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Dedicated to Angelos and Irina.

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Rezime

U ovoj tezi je predstavljena kategorijska generalizacija BF teorije na $2BF$ i $3BF$ teorije, prelaskom sa pojma gejdž grupe simetrija na pojmove gejdž 2-grupe i gejdž 3-grupe, u okviru formalizma više gejdž teorije. Razmatrana su $2BF$ dejstva sa vezama koja opisuju teoriju gravitacije i Jang-Milsovog polja i $3BF$ dejstva sa vezama koja opisuju teoriju Klajn-Gordonovog, Dirakovog, Vajlovog i Majorana polja kuplovanog sa Ajnštajn-Kartanovom gravitacijom. Ova klasična dejstva napisana su u obliku koji je prilagođen kvantizacionoj proceduri spinske pene, tj. prirodno su podeljena na dva sektora: topološki sektor i sektor sa vezama. U okviru ovih $3BF$ teorija, struktura 3-grupe dovodi do pojave nove gejdž grupe koja određuje spektar polja materije prisutne u teoriji, na sličan način kao što obična gejdž grupa određuje spektar gejdž bozona u Jang-Milsovoj teoriji. Ovakva formulacija polja materije i gravitacije nam omogućava da prepíšemo ceo Standardni Model kuplovan sa gravitacijom kao $3BF$ dejstvo sa vezama i dovodi nas korak bliže konstruisanju unifikovanog opisa i neperturbativne kvantizacije polja gravitacije i materije. U nastavku je određena kompletna gejdž grupa simetrije topološkog $3BF$ dejstva sprovođenjem kompletne Hamiltonove analize $3BF$ dejstva za proizvoljnu semistriktnu Lijevu 3-grupu, koristeći Dirakovu proceduru. Određivanje ukupne grupe simetrija je važan korak u kanonskoj kvantizaciji teorije kompletnog Standardnog Modela elementarnih čestica kuplovanog sa Ajnštajn-Kartanovom gravitacijom, formulisanog kao $3BF$ dejstvo sa vezama. Rezultujuća gejdž grupa simetrije se sastoji od pet vrsta transformacija: G -, H -, L -, M - i N -gejdž transformacija. Pokazuje se da je razmatrana teorija invarijantna na difeomorfizme, jer je $3BF$ dejstvo sa vezama manifestno kovarijantno, a grupa lokalnih translacija dobijena je kao podgrupa direktnog proizvoda ukupne gejdž simetrije i Eno-Tajtelboim transformacija. Kao važan korak u kovarijantnoj kvantizacionoj proceduri spinske pene, razmatrana je topološka suma po stanjima Z , konstruisana za klasično $3BF$ dejstvo za generalnu 3-grupu i 4-dimenzionalnu prostorvremensku mnogostrukost $\mathcal{M}_4$. Konstruisana suma po stanjima specijalan je slučaj Porterove topološke kvantne teorije polja za $d = 4$ i $n = 3$ i topološka je invarijanta 4-dimenzionalne mnogostrukosti, što je utvrđeno ispitivanjem njene invarijantnosti pri Pahnerovim potezima. Ova suma po stanjima je generalizacija sume po stanjima koju su formulisali Žireli, Pfajfer i Popesku za slučaj $2BF$ dejstva sa odgovarajućom strukturom 2-grupe.

Ključne reči: BF teorija, teorija kategorija, više gradijentne teorije, $3BF$ teorija.

Naučna oblast: Fizika

Uža naučna oblast: Teorijska fizika visokih energija

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Abstract

In this thesis we study the categorical generalizations of a BF theory to $2BF$ and $3BF$ theories, by passing from the notion of a gauge group to the notions of a gauge 2-group and a gauge 3-group in the framework of higher gauge theory. In particular, we construct the constrained $2BF$ actions describing the dynamics of the gravitational and the Yang-Mills fields, and the constrained $3BF$ actions describing the dynamics of Klein-Gordon, Dirac, Weyl and Majorana fields coupled to Einstein-Cartan gravity. The action is naturally split into a topological sector and a sector with simplicity constraints, adapted to the spinfoam quantization program. In addition, the structure of the 3-group gives rise to a novel gauge group which specifies the spectrum of matter fields present in the theory, just like the ordinary gauge group specifies the spectrum of gauge bosons in Yang-Mills theory. This allows us to rewrite the whole Standard Model coupled to gravity as a constrained $3BF$ action, facilitating the nonperturbative quantization of both gravity and matter fields. We determine the full gauge symmetry of the $3BF$ action by carrying out the complete Hamiltonian analysis of the $3BF$ action for an arbitrary semistrict Lie 3-group, using the Dirac procedure. This analysis is an important step in the canonical quantization of the complete Standard Model of elementary particles coupled to Einstein-Cartan gravity, formulated as a $3BF$ action with suitable simplicity constraints. We show that the resulting gauge symmetry group consists of the already familiar G -, H -, and L -gauge transformations, as well as additional M - and N -gauge transformations, which have not been discussed in the existing literature. As expected, since the $3BF$ action is formulated in a manifestly covariant way, we establish that diffeomorphisms are a symmetry of the theory, and are obtained as a subgroup of the direct product of the full gauge symmetry group and the Henneaux-Teitelboim transformations. As an important step in the covariant spinfoam quantization of the theory, we construct a triangulation independent topological state sum Z , based on the classical $3BF$ action for a general 3-group and a 4-dimensional spacetime manifold $\mathcal{M}_4$. The obtained state sum coincides with Porter's TQFT for $d = 4$ and $n = 3$. In order to verify that the constructed state sum is a topological invariant of the underlying 4-dimensional manifold, we analyze its behavior under Pachner moves, and we obtain that the state sum Z remains the same. The constructed state sum is a generalization of the work done by Girelli, Pfeiffer, and Popescu for the case of state sum based on the topological $2BF$ action with the underlying 2-group structure.

Key words: BF theory, category theory, higher gauge theory, $3BF$ theory.

Scientific field: Physics

Research area: Theoretical high energy physics

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Part I

Classical Theory

Chapter 1

Introduction

Within the framework of *Loop Quantum Gravity*, the nonperturbative quantization of gravity can be studied using either a covariant or a canonical approach (for a detailed introduction to the approaches to quantum gravity within this framework, see [1], [2], [3], [4]). The primary aim of the *covariant approach* is to define the path integral of the gravitational field,

$$Z = \int \mathcal{D}g \, e^{iS[g]}, \quad (1.1)$$

by considering a triangulation $\mathcal{T}(\mathcal{M}_D)$ of a spacetime manifold $\mathcal{M}_D$ and defining the integral as a discretized *state sum* over gravitational-field configurations on the simplices comprising the triangulation. The state sum associated with a triangulation $\mathcal{T}(\mathcal{M}_D)$ of a D -dimensional manifold $\mathcal{M}_D$ is defined as

$$Z = \sum_{\{\phi\}} \prod_{v \in \mathcal{T}} \mathcal{A}_v(\phi) \prod_{\epsilon \in \mathcal{T}} \mathcal{A}_\epsilon(\phi) \cdots \prod_{\sigma_D \in \mathcal{T}} \mathcal{A}_{\sigma_D}(\phi), \quad (1.2)$$

where the products run over all vertices v , edges ϵ , triangles Δ , tetrahedra τ , and, successively, all the way up to the D -simplices σ_D contained in the triangulation. Each cell is colored by labels ϕ describing the fundamental variables of the model, and an amplitude $\mathcal{A}$ describing the dynamics of the variables ϕ is assigned to every cell. In this way, one can define the *path integral* (1.1). Note that the amplitudes $\mathcal{A}_v(\phi)$, $\mathcal{A}_\epsilon(\phi)$, $\dots$, $\mathcal{A}_{\sigma_D}(\phi)$ enter partly into the definition of the path-integral measure and partly into the definition of the action $S[\phi]$.

This quantization technique is known as the *spin-foam quantization method*. The *spin-foam formalism* and its generalization within category theory, the *spin-cube formalism*, implement a *covariant approach* to quantum gravity in which the path integral is defined in the same manner as in *Feynman's path-integral formulation*, while being motivated by *canonical loop quantization*. This approach can be divided into three main steps.

1. First, one defines a classical action $S[g]$ consisting of two sectors: a topological BF -theory sector and a constraint sector.
2. Next, using the algebraic structure—namely, the Lie group G associated with the topological sector of the theory—one defines a state sum Z_{BF} that is independent of the triangulation $\mathcal{T}(\mathcal{M}_4)$.
3. Finally, imposing constraints on the topological state sum Z_{BF} yields a state sum corresponding to the physical theory of interest.

This method of quantizing gravity has been successfully applied to various choices of the action $S[g]$, the Lie group G , and the spacetime dimension D .

Topological BF theory is specified by a gauge-invariant BF action defined for a Lie group G , or equivalently for its corresponding Lie algebra $\mathfrak{g}$:

$$S_{BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}}. \quad (1.3)$$

Here, $\mathcal{F} \equiv d\alpha + \alpha \wedge \alpha$ is a $\mathfrak{g}$ -valued 2-form and the curvature of the connection 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$, defined on a principal G -bundle over a four-dimensional spacetime manifold $\mathcal{M}_4$. The Lagrange multiplier $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$ is a 2-form, while $\langle _, _ \rangle_{\mathfrak{g}}$ denotes a G -invariant, symmetric, nondegenerate bilinear form.

A brief review of classical BF theory is given in Chapter 4. We carry out the complete Hamiltonian analysis of the topological BF action and then count the degrees of freedom of the theory. As expected, the BF action is shown to describe a theory with no local propagating degrees of freedom. Using the Castellani procedure, we compute the generator of gauge symmetries in BF theory and then determine the variations of all fields in the theory and of their conjugate momenta. From these results, we obtain two types of gauge-symmetry transformations in BF theory—the G -gauge and M -gauge transformations, both already known in the literature—as well as the commutation relations of the full gauge-symmetry group $\mathcal{G}_{BF}$ of the BF action. We demonstrate that BF theory is invariant under diffeomorphisms. We then consider two physically relevant BF models obtained by adding suitable terms, referred to as *constraints*, to the BF action: *Yang–Mills theory* for the group $SU(N)$ in Minkowski spacetime and the *Plebanski model* for General Relativity.

Within the BF -theory framework, three-dimensional quantum gravity was first formulated in the *Ponzano–Regge model* for a triangulated three-dimensional manifold and the group $SU(2)$ [5]. An irreducible representation of $SU(2)$ is assigned to each edge of the triangulation, and the amplitudes are summed over all possible configurations of these spins. A variety of models have been defined in four dimensions, among which the *Barrett–Crane model* [6], [7] and the *Ooguri model* [8] are some of the best known. Finally, the most sophisticated construction, the so-called *EPRL/FK model*, is a model of four-dimensional quantum gravity independently developed by J. Engle, R. Pereira, E. Livine, and C. Rovelli [9], and by L. Freidel and K. Krasnov [10], within the *spin-foam formalism*. The particular choice of fields in the *EPRL/FK model* is motivated by a result from canonical loop quantization [1], where a state of the gravitational field is described by a *spin network* whose edges are colored by half-integers $i, j \in \mathbb{N}/2$. These models represent independent attempts to define a quantum theory of gravity, with varying degrees of success, but all focus on pure gravity without matter. Attempts to include matter in the theory have had limited success [11], mainly because mass terms cannot be expressed within this framework. The reason is that tetrad fields are absent from the topological sector of BF theory.

The covariant quantization of gravity within BF theory—that is, the construction of the topological BF state sum in the three- and four-dimensional cases by means of the standard spin-foam quantization procedure—is discussed in Chapter 7. In the three-dimensional case, starting from topological BF theory, we construct the state sum describing the *Ponzano–Regge model*; this is possible because the classical theory of three-dimensional gravity has no local propagating degrees of freedom. In the four-dimensional case, we then construct the topological BF state sum corresponding to the *Ooguri model*. For a physical four-dimensional spacetime manifold, however, the situation is more complicated. Four-dimensional gravity is not a topological theory, and therefore the resulting state sum does not correspond to the physical theory. A quantum theory of gravity can be obtained only by modifying the amplitudes of the topological state sum. The final, third step of the spin-foam quantization procedure will not be addressed in this dissertation.

To overcome the problem of coupling matter in BF models of quantum gravity, a new approach has been developed within *category theory*. It employs a categorical generalization of

the BF action in the context of *higher gauge theory* [12]. A brief introduction to the formalism of higher category theory is presented in Chapter 2. In category theory, a group is defined as a category with a single object in which every morphism is invertible. The notion of a category can be generalized to so-called *higher categories*, which contain not only objects and morphisms but also 2-morphisms (morphisms between morphisms), and so forth. A 2-group can then be defined as a 2-category with a single object in which all morphisms and 2-morphisms are invertible. More specifically, we use the idea of the *categorical ladder*, shown in Table 1.1. The categorical generalization of a BF action defined for a Lie group leads to a $2BF$ action defined for a suitable 2-group. Every strict 2-group has been shown to be equivalent to a *crossed module* $(H \xrightarrow{\partial} G, \triangleright)$, where G and H are Lie groups, $\partial : H \rightarrow G$ is a homomorphism from H to G , and $\triangleright : G \times H \rightarrow H$ is an action of G on H .

categorical structure	algebraic structure	linearized structure	topological action	degrees of freedom
Lie group	Lie group	Lie algebra	BF theory	gauge fields
Lie 2-group	Lie crossed module	differential Lie crossed module	$2BF$ theory	tetrads
Lie 3-group	Lie 2-crossed module	differential Lie 2-crossed module	$3BF$ theory	scalar and fermionic fields

Table 1.1: The categorical ladder.

Just as a connection α on a principal G -bundle over a four-dimensional spacetime manifold $\mathcal{M}_4$, and subsequently a BF action, arise naturally from a Lie group G , a 2-group structure naturally gives rise to an ordered pair, the 2-connection (α, β) . Here, $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$ is the usual $\mathfrak{g}$ -valued connection 1-form, while $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h})$ is a new $\mathfrak{h}$ -valued connection 2-form, where $\mathfrak{h}$ is the Lie algebra of the Lie group H . For a 2-connection, the so-called *fake 2-curvature*, an ordered pair $(\mathcal{F}, \mathcal{G})$, is defined as follows [12]:

$$\mathcal{F} = d\alpha + \alpha \wedge \alpha - \partial\beta, \quad \mathcal{G} = d\beta + \alpha \wedge^\triangleright \beta. \quad (1.4)$$

Here, $\alpha \wedge^\triangleright \beta$ indicates that α and β are multiplied both as differential forms, using the wedge product, and as algebra elements, through the action $\triangleright$ of $\mathfrak{g}$ on $\mathfrak{h}$. The ordered pair of curvatures $(\mathcal{F}, \mathcal{G})$ is called fake because of the term $\partial\beta$ in the definition of $\mathcal{F}$; see Chapter 5 for details.

Using these variables, one can define a $2BF$ action invariant under 2-gauge transformations [13], [14],

$$S_{2BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}}, \quad (1.5)$$

where the 2-form $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$ and the 1-form $C \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ are Lagrange multipliers. Furthermore, $\langle _, _ \rangle_{\mathfrak{g}}$ and $\langle _, _ \rangle_{\mathfrak{h}}$ denote G -invariant, symmetric, nondegenerate bilinear forms on $\mathfrak{g}$ and $\mathfrak{h}$, respectively.

For a suitable choice, the 2-group structure introduces tetrad fields into the topological $2BF$ action, as demonstrated in [15]. In that work, the action for General Relativity was written as a constrained $2BF$ action for a particular choice of 2-group, so that the tetrad fields are already present in the topological sector of the theory. This result opened the possibility of coupling matter to gravity in a straightforward way.

In Chapter 5, we analyze classical $2BF$ theory. As in the case of the BF action, a Hamiltonian analysis and a count of the degrees of freedom show that the $2BF$ action is also topological,

describing a theory with no local propagating degrees of freedom. We obtain the finite symmetry transformations of the $2BF$ action—the G -, H -, M -, and N -gauge transformations—and the commutation relations of the full gauge-symmetry group $\mathcal{G}_{2BF}$. We demonstrate the diffeomorphism invariance of $2BF$ theory. For a suitable choice of 2-group, we then present *General Relativity* as a constrained $2BF$ theory, following [15]. Finally, the last section of Chapter 5 is devoted to the *Einstein–Yang–Mills theory*, namely, the theory of gravity and gauge fields formulated as a constrained $2BF$ theory [16].

In Chapter 8, we carry out the second step of the covariant spin-foam quantization procedure for $2BF$ theory [13]. We demonstrate how to construct a triangulation-independent state sum Z , starting from the classical $2BF$ action for a general strict 2-group and an arbitrary triangulation of any smooth d -dimensional spacetime manifold, for $d \in \{3, 4\}$. For $d = 3$, the resulting state sum is precisely Yetter’s model, whereas for $d = 4$ it coincides with Porter’s TQFT for $d = 4$ and $n = 2$. We analyze the behavior of the constructed state sum under Pachner moves, which are local changes of triangulation that preserve topology; any two triangulations of the same manifold are related by a finite sequence of Pachner moves. In three dimensions there are four Pachner moves—the $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ moves and their inverses—whereas in four dimensions there are five distinct Pachner moves—the $3 \leftrightarrow 3$, $4 \leftrightarrow 2$, and $5 \leftrightarrow 1$ moves and their inverses. The setup for the analysis of the behavior of the state sum under these Pachner moves is presented in Section 8.3, while the details of the calculation are given in Appendix E.1. We find that the $2BF$ state sum remains unchanged under these transformations of the triangulation, proving that it is a *topological invariant* of the manifold.

Nevertheless, although a 2-group structure can naturally describe gravitational and vector fields, matter fields cannot be naturally expressed within the algebraic structure of a 2-group. In other words, the matter sector of the action cannot be written as a topological term plus constraints. An action written in this way is therefore only partially adapted to the spin-foam quantization procedure, and its second and third steps cannot be carried out for the full theory of gravity and matter. This is precisely the problem addressed in this dissertation.

To construct a unified description of gravity and matter, a further step in the categorical ladder was proposed: the generalization of the algebraic structure from a 2-group to a 3-group [16]. The levels of the categorical ladder are shown in Table 1.1. As it turns out, the 3-group structure provides a successful description of all fields present in the Standard Model and of their interaction with gravity. Moreover, this structure introduces a new gauge group associated with the scalar and fermionic fields of the theory [16]. This is a new and unexpected result that has the potential to open a new direction of research and to offer an explanation for the structure of the matter sector of the Standard Model and beyond.

Within category theory, a 3-group is defined as a 3-category with a single object in which all morphisms, 2-morphisms, and 3-morphisms are invertible. Just as a strict 2-group is equivalent to a crossed module, every semistrict 3-group has been shown to be equivalent to a 2-crossed module [17]. A Lie 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$ is an algebraic structure specified by three Lie groups G , H , and L , together with homomorphisms $\delta : L \rightarrow H$ and $\partial : H \rightarrow G$, an action $\triangleright$ of G on all three groups, and a G -equivariant map

$$\{_, _\}_{\text{pf}} : H \times H \rightarrow L,$$

called the Peiffer lifting.

Analogously to the construction of the topological BF and $2BF$ actions, one can define a topological $3BF$ action on a manifold $\mathcal{M}_4$ for a 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$. Given the Lie algebras $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$ associated with the Lie groups G , H , and L , one naturally introduces an ordered triple, the 3-connection (α, β, γ) , whose components are algebra-valued differential forms $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$, $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h})$, and $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l})$. The corresponding *fake*

3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is

$$\mathcal{F} = d\alpha + \alpha \wedge \alpha - \partial\beta, \quad \mathcal{G} = d\beta + \alpha \wedge^\triangleright \beta - \delta\gamma, \quad \mathcal{H} = d\gamma + \alpha \wedge^\triangleright \gamma + \beta \wedge^{\{\cdot\}} \beta. \quad (1.6)$$

See [17], [18] for details. One can now define the $3BF$ action

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}, \quad (1.7)$$

where $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$, $C \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, and $D \in \mathcal{A}^0(\mathcal{M}_4, \mathfrak{l})$ are Lagrange multipliers. The forms $\langle _, _ \rangle_{\mathfrak{g}}$, $\langle _, _ \rangle_{\mathfrak{h}}$, and $\langle _, _ \rangle_{\mathfrak{l}}$ are G -invariant, symmetric, nondegenerate bilinear forms on $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$, respectively.

For a theory of gravity and matter formulated as a constrained $3BF$ action for a particular 3-group, both the covariant and the canonical quantization procedures must be developed. For the time being, we focus on their initial steps.

Canonical quantization procedure. The first step toward the canonical quantization of the theory is the Hamiltonian analysis, which yields the algebra of first- and second-class constraints. The first-class constraints become conditions on the physical states that define the Hilbert space, while the Hamiltonian constraint determines the dynamics. With this goal in mind, as the first step in the canonical quantization of $3BF$ theory, we focused on determining the complete gauge-symmetry group of the topological $3BF$ action [19]. The complete Hamiltonian analysis of the $3BF$ action for a general semistrict Lie 3-group, performed using the Dirac procedure, is presented in Chapter 6. It generalizes the Hamiltonian analysis of the $2BF$ action carried out in [20], [21], [22], [23], as well as the Hamiltonian analysis of the particular 2-crossed module corresponding to scalar electrodynamics performed in [24]. The analysis of the Hamiltonian structure yields the algebra of the first- and second-class constraints present in the theory. As usual, the first-class constraints generate gauge transformations, which leave the physical state of the system unchanged. After computing the first-class constraints, we use the Castellani procedure to construct the generator of gauge transformations on a spatial hypersurface, and then generalize the resulting transformations to the entire spacetime. We obtain the complete gauge symmetry of topological $3BF$ theory, consisting of five types of finite gauge transformations: the G -, H -, and L -gauge transformations already known in the literature, and the additional M - and N -gauge transformations, which are among our principal results. Using this result, we analyze the structure of the complete gauge-symmetry group $\mathcal{G}_{3BF}$. The resulting analysis establishes a relationship between the gauge-symmetry group of the $3BF$ action and the 3-group structure on which the action is based. We demonstrate the diffeomorphism invariance of $3BF$ theory. This analysis is an important step toward studying the gauge-symmetry group of gravity coupled to matter, formulated as a constrained $3BF$ action [16], and toward its canonical quantization.

First step of the spin-foam quantization procedure. Once the appropriate 3-groups have been identified and the corresponding $3BF$ actions constructed, suitable constraints must be imposed on the degrees of freedom in the topological sector of the $3BF$ action in order to obtain the desired classical dynamics of matter and gravitational fields. By imposing suitable constraints on the action (1.7), one can obtain actions for matter fields interacting with gravity. In addition to the previously defined constrained $2BF$ actions for the *Yang–Mills* and *Proca vector fields*, in Chapter 6 we construct the corresponding constrained $3BF$ actions for the *Klein–Gordon*, *Dirac*, *Weyl*, and *Majorana fields* interacting with Einstein–Cartan gravity. This construction leads to an unexpected new result. As we shall see, scalar and fermionic fields are *naturally associated with a new gauge group*, thereby generalizing the notion of the gauge group in Yang–Mills theory. The new gauge group opens the possibility of classifying matter fields and describing structures such as the families of quarks and leptons. However, because

of the complexity of the algebraic properties of 3-groups within this new category-theoretic formalism, at this stage of our research we have focused only on reconstructing already known theories, such as the Standard Model. Any potential explanation of the matter-field spectrum of the Standard Model is therefore left for future work.

Second step of the spin-foam quantization procedure. Completing the first step of the spin-foam quantization procedure suggests that its second and third steps may also be realized. The action is written in terms of differential forms, allowing it to be adapted to a piecewise-flat spacetime manifold using Regge calculus [25]. In particular, all fields and field strengths appearing in the $3BF$ action can be naturally assigned to a d -dimensional simplex of a four-dimensional triangulation: a 0-form is assigned to a vertex, a zero-dimensional object; a 1-form to an edge; and so on. This correspondence is summarized in Table 1.2.

d	triangulation	dual triangulation	form	fields	field strengths
0	vertex $\bullet$	4-complex	0-form	$\phi, \psi_{\tilde{\alpha}}, \bar{\psi}^{\tilde{\alpha}}$	
1	edge $/$	3-polyhedron 	1-form	ω^{ab}, A^I, e^a	
2	triangle $\triangle$	polygon 	2-form	β^a, B^{ab}	R^{ab}, F^I, T^a
3	tetrahedron 	edge $/$	3-form	$\gamma, \gamma_{\tilde{\alpha}}, \bar{\gamma}^{\tilde{\alpha}}$	$\mathcal{G}^a$
4	4-simplex 	vertex $\bullet$	4-form		$\mathcal{H}, \mathcal{H}_{\tilde{\alpha}}, \bar{\mathcal{H}}^{\tilde{\alpha}}$

Table 1.2: Correspondence between fields and field strengths and the elements of the triangulation $\mathcal{T}(\mathcal{M}_4)$.

Once the classical Regge-discretized $3BF$ action has been constructed, the next step of the quantization procedure is to define a state sum Z giving the path integral of the theory. The topological nature of the $3BF$ action, together with the gauge 3-group structure, ensures that such a state sum is a topological invariant and hence independent of the triangulation. In Chapter 9, we formulate a triangulation-independent topological state sum Z for the classical $3BF$ action associated with a general semistrict 3-group on a four-dimensional spacetime manifold $\mathcal{M}_4$. This state sum coincides with Porter's abstract definition of a topological quantum field theory (TQFT) [26] for $d = 4$ and $n = 3$, where d denotes the dimension of the manifold $\mathcal{M}$ and n the level of the categorical ladder. This definition generalizes Yetter's state-sum construction. To verify that the constructed state sum is indeed topological, we analyze its behavior under Pachner moves [27]. The analysis shows that the state sum Z remains unchanged and is therefore a topological invariant of the four-dimensional manifold. The setup of the invariance proof is presented in Chapter 9, while the detailed analysis of the Pachner moves is given in Appendix E.

Unfortunately, a complete and precise implementation of this step requires generalizations of the Peter–Weyl and Plancherel theorems to 2-groups and 3-groups. The Peter–Weyl theorem provides a decomposition of functions on a group into a sum over suitable irreducible representations, which allows one to specify the spectrum of labels assigned to the d -simplices of a triangulation and thereby determine the domain of the fields living on each d -simplex. For 2-groups and 3-groups, representation theory is not yet sufficiently developed to provide

such a construction, and analogues of the Peter–Weyl and Plancherel theorems remain open mathematical results. While awaiting such formulations, one may attempt to *infer* the appropriate structure of irreducible representations of 2-groups or 3-groups, as was done in [15] in the construction of the *spin-cube model* of quantum gravity.

Third step of the spin-foam quantization procedure. Finally, because the physical theory of interest is not topological but has local propagating degrees of freedom, our ultimate goal is not to construct a topological invariant—the state sum Z —but rather a state sum with nontrivial dynamics. To achieve this, constraints must be imposed on the topological state sum Z , completing the third step of the spin-foam quantization procedure. With this in mind, the principal motivation for the first part of our research in [16] was to rewrite the action for gravity and matter so that the topological part is explicitly separated from the constraints, making their imposition straightforward. The result of this third and final step would be a state sum describing a theory of quantum gravity coupled to matter.

Notation and conventions

Local Lorentz indices are denoted by lowercase Latin letters $a, b, c, \dots$, taking the values $0, 1, 2, 3$. They are raised and lowered using the Minkowski metric η_{ab} with signature $(-, +, +, +)$. Spacetime indices are denoted by Greek letters $\mu, \nu, \dots$ and are raised and lowered using the spacetime metric $g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu$, where e^a_μ are the tetrad fields. Spatial indices are denoted by lowercase Latin letters from the middle of the alphabet, $i, j, \dots$, while the temporal component is denoted by 0. Indices labeling the generators of the groups G , H , and L are denoted, respectively, by lowercase Greek letters from the beginning of the alphabet $\alpha, \beta, \dots$, lowercase Latin letters from the beginning of the alphabet $a, b, c, \dots$, and uppercase Latin letters $A, B, C, \dots$. Antisymmetrization of a tensor over two indices is denoted by

$$A_{[a_1|a_2\dots a_{n-1}|a_n]} = \frac{1}{2} (A_{a_1 a_2 \dots a_{n-1} a_n} - A_{a_n a_2 \dots a_{n-1} a_1}) ,$$

while total antisymmetrization over all indices is denoted by

$$A_{[a_1 \dots a_n]} = \frac{1}{n!} \sum_{\sigma \in S_n} (-1)^{\text{sgn}(\sigma)} A_{a_{\sigma(1)} \dots a_{\sigma(n)}} .$$

Similarly, symmetrization of a tensor over two indices is denoted by

$$A_{(a_1|a_2\dots a_{n-1}|a_n)} = \frac{1}{2} (A_{a_1 a_2 \dots a_{n-1} a_n} + A_{a_n a_2 \dots a_{n-1} a_1}) ,$$

while total symmetrization over all indices is denoted by

$$A_{(a_1 \dots a_n)} = \frac{1}{n!} \sum_{\sigma \in S_n} A_{a_{\sigma(1)} \dots a_{\sigma(n)}} .$$

We use natural units, $c = \hbar = 1$ and $G = l_p^2$, where l_p is the Planck length.

If G is a finite group, $\int_G dg = 1/|G| \sum_{g \in G}$ denotes the normalized sum over all elements of the group, while δ_G denotes the corresponding delta distribution on G . For each element $g \in G$, this distribution is defined by $\delta_G(g) = |G|$ if g is the identity element, $g = e$, and by $\delta_G(g) = 0$ otherwise, i.e., if $g \neq e$. If G is a Lie group, $\int_G dg$ and δ_G denote the Haar measure and the delta distribution on G , respectively. The set of all k -simplices, where $0 < k < d$, is denoted by Λ_k . The set of all vertices Λ_0 is finite and ordered, and each k -simplex is labeled by a $(k+1)$ -tuple of vertices $(i_0 \dots i_k)$, where $i_0, \dots, i_k \in \Lambda_0$ and $i_0 < \dots < i_k$. The number of k -simplices is denoted by $|\Lambda_k|$.

For every triangulation $\mathcal{T}(\mathcal{M}_d)$ of a d -dimensional manifold $\mathcal{M}_d$, one can define its dual triangulation $\mathcal{T}(\mathcal{M}_d)^*$. The *dual triangulation*, or *dual lattice*, is defined as follows [1]. A vertex of the dual lattice is placed at the center of each d -simplex of the triangulation. Connecting the vertices corresponding to adjacent d -simplices gives the edges of the dual triangulation. Each such edge corresponds to a $(d - 1)$ -simplex shared by two adjacent d -simplices. A polygon, or 2-complex, obtained by connecting the appropriate edges $\epsilon \in \mathcal{T}(\mathcal{M}_d)^*$ corresponds to a $(d - 2)$ -simplex of $\mathcal{T}(\mathcal{M}_d)$. The remaining elements of the dual triangulation are assigned in the same way, each dual to the corresponding element of the original triangulation, up to the d -complex dual to a vertex $v \in \mathcal{T}(\mathcal{M}_d)$. The set of all k -complexes of the dual triangulation, where $0 < k < d$, is denoted by Λ_k^* . The total number of k -complexes in the dual lattice is denoted by $|\Lambda_k^*|$.

All additional notation and conventions are defined explicitly in the text where they first appear.

Chapter 2

Higher gauge theories

Ordinary gauge theory describes how 0-dimensional particles transform under parallel transport along a 1-dimensional path in the base of a fibre bundle¹, with an element of a group naturally assigned to each path in the manifold. Similarly, 2-gauge theory defines parallel transport of 1-dimensional objects by introducing a 2-connection on a fibre 2-bundle². A fibre 2-bundle generalizes a fibre bundle within 2-gauge theory: its fibres are not manifolds, but categories endowed with an appropriate smooth structure.

A category consists of elements called *objects* and *morphisms*, namely maps between those objects. A group may therefore be regarded as a category with a single object in which all morphisms are invertible. In a higher gauge theory, the symmetry group that specifies an ordinary gauge theory is replaced by a higher-categorical generalization of a group—a 2-group.

2.1 Gauge theory

A *gauge theory* is a field theory whose Lagrangian, which determines the dynamics of the system, remains invariant under *local transformations*³, called *gauge transformations*. Gauge theories contain redundant degrees of freedom: in addition to the variables that describe the physical content of the system, their actions contain nonphysical variables. They possess a *gauge symmetry*, which acts through nonphysical transformations of the dynamical variables. Dynamical systems of this type are also called singular systems, and their treatment requires the *Hamiltonian analysis* of constrained systems, discussed in greater detail in the next chapter. In the Hamiltonian formalism, these theories are characterized by the presence of constraints in the action.

Gauge transformations form a Lie group known as the *symmetry group*, or the *gauge group*, of the theory. Each generator of this group is associated with a field called a *gauge field*.

Mathematically, gauge theories are formulated in the language of differential geometry, namely on fibre bundles. General relativity is likewise defined on a fibre bundle, specifically

¹A *fibre bundle* $E(B, F, \pi)$ is a manifold E equipped with a smooth map π (the *projection*) onto a manifold B (the *base*), such that, for each point b of the base, the *fibre over* b , $\pi^{-1}(b) \equiv \{e \in E \mid \pi(e) = b\}$, is diffeomorphic to a manifold F (the *typical fibre*), and there exists a neighbourhood U_b for which $\pi^{-1}(U_b)$ is diffeomorphic to the direct product $U_b \times F$.

²A *fibre 2-bundle*.

³We distinguish between *global* and *local* symmetries of a physical system. When the Lagrangian is invariant under transformations that have the same form at every spacetime point at which physical processes occur, i.e. when the transformation parameter does not depend on the spacetime point, the theory is said to possess a *global symmetry*. A *local symmetry* is a symmetry under transformations whose parameter depends on the spacetime coordinates.

the *tangent bundle*⁴, whereas gauge theories are formulated on a *principal bundle*⁵.

In a gauge theory we define a connection 1-form⁶ $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$, valued in the algebra $\mathfrak{g}$ and also called the *gauge potential* in physics. A connection is specified by a differential operator that measures the departure of tensor fields at a point from the tensor fields obtained by *parallel transport* to neighbouring points, namely by the *covariant derivative* ∇ .

The gauge *curvature* describes how the gauge potential changes when transported around a closed loop in the manifold. For a connection α , the curvature 2-form is defined by

$$F = d\alpha + \alpha \wedge \alpha. \quad (2.1)$$

Gauge curvature is a fundamental quantity in gauge theory and plays a central role in determining both the dynamics and the interactions of particles with gauge fields.

Gauge theories have proved remarkably successful in describing and predicting the behaviour of elementary particles and their interactions. The *Standard Model*, whose gauge symmetry is $SU(3) \times SU(2) \times U(1)$, has been tested and confirmed by numerous experiments, including those performed at high-energy particle accelerators such as the Large Hadron Collider (LHC). Nevertheless, important questions remain open, among them the unified description of all fundamental interactions, which continues to be an active area of research in theoretical physics.

2.2 2-gauge theory

A 2-category, the generalization of a category, consists of objects, morphisms, and morphisms between morphisms—2-morphisms [12].

The higher-categorical generalization of gauge theory, namely 2-gauge theory, is a theory whose fundamental symmetries are described by a 2-group. In addition to assigning group elements $g \in G$ to paths, as in ordinary gauge theory, a 2-gauge theory assigns elements $h \in H$ to surfaces. These labels of the elementary constituents of the manifold cannot be chosen arbitrarily; they must satisfy the following conditions.

1. For every surface labelled by $h \in H$, the labels g_1 and g_2 of its source and target paths satisfy $\partial(h) = g_2 g_1^{-1}$.
2. For every volume, the surface holonomy around it is trivial.

In this section we examine the composition of labelled paths and surfaces, so as to construct composites of arbitrary size from elementary constituents of the manifold.

2.2.1 2-group

In category theory, a 2-group is a 2-category with a single object in which all morphisms and 2-morphisms are invertible. Every strict 2-group is known to be equivalent to a crossed module $(H \xrightarrow{\partial} G, \triangleright)$.

Definition 2.2.1 (Pre-crossed module and crossed module) A *pre-crossed module* $(H \xrightarrow{\partial} G, \triangleright)$ consists of:

⁴To each point m of a D -dimensional manifold $\mathcal{M}$ one may associate a D -dimensional tangent space $T_m(\mathcal{M})$, consisting of tangent vectors obtained from the various curves through m . Assigning the vector space $T_m(\mathcal{M})$ to every point m of $\mathcal{M}$ yields the *tangent bundle* $T(\mathcal{M})$, i.e. a vector bundle with base $\mathcal{M}$, fibre $\mathbb{R}^D$, and projection $\pi : T_m(\mathcal{M}) \rightarrow m$. All tangent spaces are mutually isomorphic, but establishing a correspondence between vectors in the tangent spaces associated with different points requires the introduction of a *connection*.

⁵When the typical fibre is a Lie group, $F = G$, the bundle is called a *principal bundle* P with structure group G .

⁶A *connection* assigns to each path γ joining points x and y of a manifold $\mathcal{M}$ a map $\rho(\gamma) : T_x(\mathcal{M}) \rightarrow T_y(\mathcal{M})$.

- a group G whose elements are morphisms, with composition as the group operation,

$$\bullet \xleftarrow{g_1} \bullet \xleftarrow{g_2} \bullet = \bullet \xleftarrow{g_1 g_2} \bullet ;$$

- a group H consisting of all 2-morphisms whose source is the identity,

$$\begin{array}{c} 1 \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow h \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial h \end{array} ,$$

where horizontal composition is the group operation,

$$\begin{array}{c} 1 \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow h \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial h \end{array} \begin{array}{c} 1 \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow h' \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial h' \end{array} = \begin{array}{c} 1 \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow hh' \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial(hh') \end{array} ;$$

- a group homomorphism $\partial : H \rightarrow G$ mapping each 2-morphism $h \in H$ to its target $\partial h \in G$,

$$\begin{array}{c} 1 \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow h \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial h \end{array} ,$$

so that

$$\partial(hh') = \partial(h)\partial(h') ; \quad (2.2)$$

- an action $\triangleright$ of G on both groups, such that

- G acts on itself by horizontal conjugation,

$$\bullet \xleftarrow{g_0} \bullet \xleftarrow{g} \bullet \xleftarrow{g_0^{-1}} \bullet = \bullet \xleftarrow{g_0 g g_0^{-1}} \bullet ,$$

or, formally, for all $g_0, g \in G$,

$$g_0 \triangleright g = g_0 g g_0^{-1} , \quad (2.3)$$

- an element $g \in G$ acts on an element $h \in H$ by horizontal conjugation with its identity 2-morphism 1_g , producing the 2-morphism $g \triangleright h \in H$,

$$\begin{array}{c} g \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow 1_g \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ g \end{array} \begin{array}{c} 1 \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow h \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial h \end{array} \begin{array}{c} g^{-1} \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow 1_g^{-1} \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ g^{-1} \end{array} = \begin{array}{c} 1 \\ \swarrow \quad \searrow \\ \bullet \quad \quad \bullet \\ \downarrow g \triangleright h \\ \bullet \quad \quad \bullet \\ \nwarrow \quad \nearrow \\ \partial(g \triangleright h) \end{array} ,$$

that is, for all $g \in G$ and $h \in H$,

$$g \partial h g^{-1} = \partial(g \triangleright h) , \quad (2.4)$$

- the action $\triangleright$ is G -equivariant, i.e. for all $h, h' \in H$ and $g, g' \in G$,

$$(gg') \triangleright h = g \triangleright (g' \triangleright h) , \quad g \triangleright (hh') = (g \triangleright h)(g \triangleright h') . \quad (2.5)$$

In a pre-crossed module, the **Piesser commutator** $\langle h_1, h_2 \rangle_{\text{pf}}$ is defined for all $h_1, h_2 \in H$ by

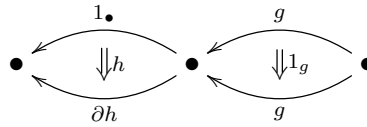
$$\langle h_1, h_2 \rangle_{\text{pf}} = h_1 h_2 h_1^{-1} \partial(h_1) \triangleright h_2^{-1}. \quad (2.6)$$

A pre-crossed module in which all Piesser commutators are trivial is called a **crossed module**. Equivalently, the **Piesser identity** holds for all $h, h' \in H$:

$$(\partial h) \triangleright h' = h h' h^{-1}. \quad (2.7)$$

For each element $g \in G$, the operation $\triangleright$ defines an automorphism of H . Since the composition of two automorphisms is again an automorphism, the automorphisms of H form a group $\text{Aut}(H)$, and $\triangleright$ maps every $g \in G$ to an element of this group, i.e. $\triangleright : G \rightarrow \text{Aut}(H)$.

The notions of a 2-group and a crossed module are equivalent. Thus, to every 2-morphism $\alpha \in \mathcal{G}$, an element of the 2-group $\mathcal{G}$ specified by a pair (h, g) ,



there corresponds the following element of the 2-group, namely the 2-morphism

$$g \rightarrow (\partial h) g. \quad (2.8)$$

2.2.2 Lie 2-algebra

In analogy with pre-crossed and crossed modules, the language of Lie 2-algebras gives rise to the notions of a *differential pre-crossed module* and a *differential crossed module*.

Definition 2.2.2 (Differential pre-crossed module and differential crossed module) A *differential pre-crossed module* $(\mathfrak{h} \xrightarrow{\partial} \mathfrak{g}, \triangleright)$ is specified by the algebras $\mathfrak{g}$ and $\mathfrak{h}$, a map $\partial : \mathfrak{h} \rightarrow \mathfrak{g}$, and an action $\triangleright$ of $\mathfrak{g}$ on the algebras $\mathfrak{g}$ and $\mathfrak{h}$.

For a differential pre-crossed module, the **Piesser commutator** of two elements $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$ is defined by

$$\langle \underline{h}_1, \underline{h}_2 \rangle_{\text{pf}} = [\underline{h}_1, \underline{h}_2] - \partial(\underline{h}_1) \triangleright \underline{h}_2. \quad (2.9)$$

A *differential pre-crossed module* is a **differential crossed module** when all of its Piesser commutators vanish, or equivalently when the **Piesser identity** holds for all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$:

$$\partial(\underline{h}_1) \triangleright \underline{h}_2 = [\underline{h}_1, \underline{h}_2]. \quad (2.10)$$

The action $\triangleright$ of elements of $\mathfrak{g}$ on elements of $\mathfrak{h}$ and $\mathfrak{g}$ is defined by the action of the generators of $\mathfrak{g}$ on the generators of the corresponding algebras:

$$\tau_\alpha \triangleright \tau_\beta = \triangleright_{\alpha\beta}{}^\gamma \tau_\gamma, \quad \tau_\alpha \triangleright t_a = \triangleright_{\alpha a}{}^b t_b, \quad (2.11)$$

The components of a differential crossed module have the following properties.

1. By definition, the action $\triangleright$ of elements of $\mathfrak{g}$ on the same algebra is the adjoint action. Thus, for all $\underline{g}_0, \underline{g} \in \mathfrak{g}$,

$$\underline{g}_0 \triangleright \underline{g} = [\underline{g}_0, \underline{g}]. \quad (2.12)$$

or, in a basis,

$$\triangleright_{\alpha\beta}{}^\gamma = f_{\alpha\beta}{}^\gamma. \quad (2.13)$$

2. The map $\partial : H \rightarrow G$ is $\mathfrak{g}$ -equivariant, so for all $\underline{h} \in \mathfrak{h}$ and $\underline{g} \in \mathfrak{g}$,

$$\partial(\underline{g} \triangleright \underline{h}) = [\underline{g}, \partial(\underline{h})], \quad (2.14)$$

or, in a basis,

$$\partial_a^\beta f_{\alpha\beta}^\gamma = \triangleright_{\alpha a}^b \partial_b^\gamma. \quad (2.15)$$

3. The action $\triangleright$ of $\mathfrak{g}$ on $\mathfrak{h}$ is a $\mathfrak{g}$ -equivariant map. Hence, for all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$,

$$\underline{g} \triangleright [\underline{h}_1, \underline{h}_2] = [\underline{g} \triangleright \underline{h}_1, \underline{h}_2] + [\underline{h}_1, \underline{g} \triangleright \underline{h}_2], \quad (2.16)$$

or, in a basis,

$$f_{ab}^c \triangleright_{\alpha c}^d = 2 \triangleright_{\alpha[a}^c f_{c|b]}^d. \quad (2.17)$$

4. The Peiffer commutator $(\underline{h}_1, \underline{h}_2) \in \mathfrak{h} \times \mathfrak{h} \mapsto \langle \underline{h}_1, \underline{h}_2 \rangle_{\text{pf}} \in \mathfrak{h}$ is a bilinear $\mathfrak{g}$ -equivariant map. Thus all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$ and $\underline{g} \in \mathfrak{g}$ satisfy

$$\underline{g} \triangleright \langle \underline{h}_1, \underline{h}_2 \rangle_{\text{pf}} = \langle \underline{g} \triangleright \underline{h}_1, \underline{h}_2 \rangle_{\text{pf}} + \langle \underline{h}_1, \underline{g} \triangleright \underline{h}_2 \rangle_{\text{pf}}, \quad (2.18)$$

or, in a basis,

$$\triangleright_{\alpha c}^d (f_{ab}^c - \partial_a^\beta \triangleright_{\beta b}^c) = \triangleright_{\alpha a}^c (f_{cb}^d - \partial_c^\beta \triangleright_{\beta b}^d) + \triangleright_{\alpha b}^c (f_{ac}^d - \partial_a^\beta \triangleright_{\beta c}^d). \quad (2.19)$$

5. In the language of algebras, the *Peiffer identity* defined in (2.10) becomes

$$f_{ab}^c = \partial_a^\alpha \triangleright_{\alpha b}^c. \quad (2.20)$$

2.2.3 Composition of morphisms

In 2-gauge theory, geometric objects are labelled at two levels: curves are labelled by elements $g \in G$, and surfaces by elements $h \in H$. The composition and reversal of orientation of curves are defined just as in ordinary gauge theory.

To discuss the composition of morphisms, it is convenient to introduce the following notation. We denote the source and target of a morphism g by $\partial_1^-(g)$ and $\partial_1^+(g)$, respectively.

- The horizontal composition of morphisms g_1 and g_2 , when they are composable, i.e. when $\partial_1^-(g_1) = \partial_1^+(g_2)$, is the morphism $g_1 \#_1 g_2$:

$$\begin{array}{c} \xrightarrow{g_1} \quad \xrightarrow{g_2} \\ z \bullet \quad \bullet y \quad \bullet x \end{array} = \begin{array}{c} \xrightarrow{g_1 g_2} \\ z \bullet \quad \bullet x \end{array},$$

or, formally,

$$g_1 \#_1 g_2 = g_1 g_2. \quad (2.21)$$

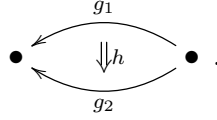
- Composition of morphisms is associative. Thus, for morphisms g_1, g_2 , and g_3 satisfying $\partial_1^-(g_1) = \partial_1^+(g_2)$ and $\partial_1^-(g_2) = \partial_1^+(g_3)$, one has

$$\begin{array}{c} \xrightarrow{g_1 g_2} \quad \xrightarrow{g_3} \\ z \bullet \quad \bullet y \quad \bullet x \end{array} = \begin{array}{c} \xrightarrow{g_1} \quad \xrightarrow{g_2 g_3} \\ z \bullet \quad \bullet y' \quad \bullet x \end{array}.$$

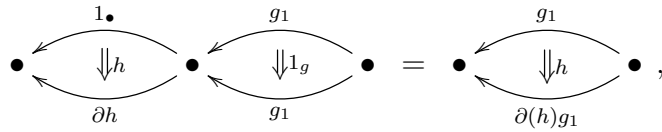
2.2.4 Composition of 2-morphisms

In this section we describe how the composition of elementary surfaces is defined in 2-gauge theory. Surfaces are labelled by elements $h \in H$, and both their *vertical* and *horizontal compositions* may be defined. We also define the *left whiskering* and *right whiskering* of a 2-morphism by a 1-morphism.

For each surface, we may choose two reference points on its boundary and divide that boundary into two curves: a source curve labelled by $g_1 \in G$ and a target curve labelled by $g_2 \in G$, as shown in the diagram



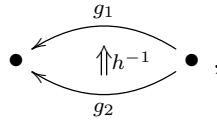
The 2-morphism $h \in H$ maps the curve $g_1 \in G$ to the curve $\partial(h)g_1 \in G$,



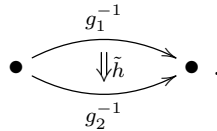
so that $h \in H$ satisfies

$$\partial(h) = g_2 g_1^{-1}. \quad (2.22)$$

The orientation of a surface may be reversed, in which case the surface is labelled by the inverse element,



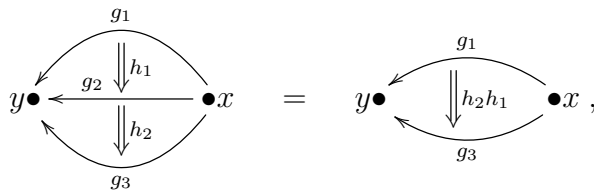
whereas reversing the orientation of the curves yields a surface element labelled by $\tilde{h} = g_1^{-1} \triangleright h^{-1}$:



To discuss the composability of two 2-morphisms, we introduce the following notation. The source and target of the k -arrow ($k = 1, 2$) of a 2-morphism h are denoted by $\partial_k^-(h)$ and $\partial_k^+(h)$, respectively. We can now specify the possible compositions of elementary surfaces and the conditions under which each composition is defined.

Vertical composition of 2-morphisms

- The vertical composition of the 2-morphisms (g_1, h_1) and (g_2, h_2) , when they are composable, i.e. when $\partial_2^+(h_1) = \partial_2^-(h_2)$, gives the 2-morphism $(g_1, h_2 h_1)$,



or, for the pair (g_1, h_1) and (g_2, h_2) ,

$$(g_2, h_2) \#_2 (g_1, h_1) = (g_1, h_2 h_1). \quad (2.23)$$

- Vertical composition is associative. Thus, for the 2-morphisms (g_1, h_1) , (g_2, h_2) , and (g_3, h_3) , when $\partial_1^+(h_1) = \partial_1^-(h_2)$ and $\partial_1^+(h_2) = \partial_1^-(h_3)$,

$$\begin{array}{c}
 \begin{array}{ccc}
 & g_1 & \\
 \curvearrowright & \Downarrow h_2 h_1 & \curvearrowleft \\
 y \bullet & \xleftarrow{g_3} & \bullet x \\
 \curvearrowleft & \Downarrow h_3 & \curvearrowright \\
 & g_4 &
 \end{array}
 =
 \begin{array}{ccc}
 & g_1 & \\
 \curvearrowright & \Downarrow h_1 & \curvearrowleft \\
 y \bullet & \xleftarrow{g_2} & \bullet x \\
 \curvearrowleft & \Downarrow h_3 h_2 & \curvearrowright \\
 & g_4 &
 \end{array}
 ,
 \end{array}$$

one has

$$(g_3, h_3) \#_2 (g_1, h_2 h_1) = (g_2, h_3 h_2) \#_2 (g_1, h_1). \quad (2.24)$$

- For every morphism g , there exists a 2-morphism $(g, 1_g)$ that is the identity for vertical composition of 2-morphisms:

$$\begin{array}{ccc}
 & g & \\
 \curvearrowright & \Downarrow 1_g & \curvearrowleft \\
 y \bullet & & \bullet x \\
 \curvearrowleft & g &
 \end{array}
 .$$

Horizontal composition of 2-morphisms

- The horizontal composition of the 2-morphisms (g_1, h_1) and (g_2, h_2) , when they are composable, i.e. when $\partial_1^-(h_1) = \partial_1^+(h_2)$, gives the 2-morphism $(g_1 g_2, h_1 g_1 \triangleright h_2)$:

$$\begin{array}{ccc}
 \begin{array}{ccc}
 & g_1 & \\
 \curvearrowright & \Downarrow h_1 & \curvearrowleft \\
 z \bullet & & y \bullet \\
 \curvearrowleft & g'_1 &
 \end{array}
 \begin{array}{ccc}
 & g_2 & \\
 \curvearrowright & \Downarrow h_2 & \curvearrowleft \\
 y \bullet & & \bullet x \\
 \curvearrowleft & g'_2 &
 \end{array}
 =
 \begin{array}{ccc}
 & g_1 g_2 & \\
 \curvearrowright & \Downarrow h_1 g_1 \triangleright h_2 & \curvearrowleft \\
 z \bullet & & \bullet x \\
 \curvearrowleft & g'_1 g'_2 &
 \end{array}
 .
 \end{array}$$

Horizontal composition of 2-morphisms is the group operation in $G \times H$:

$$\begin{aligned}
 & (g_1, h_1) \#_1 (g_2, h_2) = \\
 & \begin{array}{ccccccc}
 & 1 \bullet & & g_1 & & 1 \bullet & & g_2 \\
 \curvearrowright & \Downarrow h_1 & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_1} & \curvearrowleft & \curvearrowright & \Downarrow h_2 & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_2} & \curvearrowleft \\
 \bullet & & \bullet & & \bullet & & \bullet & & \bullet & & \bullet \\
 \curvearrowleft & \partial(h_1) & \curvearrowright & g_1 & \curvearrowleft & \partial(h_2) & \curvearrowright & g_2 &
 \end{array} = \\
 & \begin{array}{ccccccccccc}
 & 1 \bullet & & g_1 & & 1 \bullet & & g_1^{-1} & & g_1 & & g_2 \\
 \curvearrowright & \Downarrow h_1 & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_1} & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_1}^{-1} & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_1} & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_2} & \curvearrowleft \\
 \bullet & & \bullet & & \bullet & & \bullet & & \bullet & & \bullet & & \bullet & & \bullet \\
 \curvearrowleft & \partial(h_1) & \curvearrowright & g_1 & \curvearrowleft & \partial(h_2) & \curvearrowright & g_1^{-1} & \curvearrowleft & g_1 & \curvearrowleft & g_2 &
 \end{array} = \\
 & \begin{array}{ccccc}
 & 1 \bullet & & 1 \bullet & & g_1 g_2 \\
 \curvearrowright & \Downarrow h_1 & \curvearrowleft & \curvearrowright & \Downarrow g_1 \triangleright h_2 & \curvearrowleft & \curvearrowright & \Downarrow 1_{g_1 g_2} & \curvearrowleft \\
 \bullet & & \bullet & & \bullet & & \bullet & & \bullet \\
 \curvearrowleft & \partial(h_1) & \curvearrowright & \partial(g_1 \triangleright h_2) & \curvearrowright & g_1 g_2 &
 \end{array} = \\
 & \begin{array}{ccc}
 & 1 \bullet & \\
 \curvearrowright & \Downarrow h_1 g_1 \triangleright h_2 & \curvearrowleft \\
 \bullet & & \bullet \\
 \curvearrowleft & \partial(h_1 g_1 \triangleright h_2) & \curvearrowright \\
 & & \begin{array}{ccc}
 & g_1 g_2 & \\
 \curvearrowright & \Downarrow 1_{g_1 g_2} & \curvearrowleft \\
 \bullet & & \bullet \\
 \curvearrowleft & g_1 g_2 &
 \end{array}
 \end{array} = (g_1 g_2, h_1 g_1 \triangleright h_2) .
 \end{aligned}$$

- Horizontal composition of 2-morphisms is associative. Thus, for 2-morphisms (g_1, h_1) , (g_2, h_2) , and (g_3, h_3) satisfying $\partial_1^-(h_1) = \partial_1^+(h_2)$ and $\partial_1^-(h_2) = \partial_1^+(h_3)$, one has

$$\begin{array}{c}
 \bullet \xleftarrow{g_1 g_2} \bullet \xleftarrow{g_3} \bullet \\
 \Downarrow h_1 g_1 \triangleright h_2 \quad \Downarrow h_3 \\
 \bullet \xleftarrow{g'_1 g'_2} \bullet \xleftarrow{g'_3} \bullet
 \end{array}
 =
 \begin{array}{c}
 \bullet \xleftarrow{g_1} \bullet \xleftarrow{g_2 g_3} \bullet \\
 \Downarrow h_1 \quad \Downarrow h_2 g_2 \triangleright h_3 \\
 \bullet \xleftarrow{g'_1} \bullet \xleftarrow{g'_2 g'_3} \bullet
 \end{array},$$

or, formally,

$$(g_1 g_2, h_1 g_1 \triangleright h_2) \#_1 (g_3, h_3) = (g_1, h_1) \#_1 (g_2 g_3, h_2 g_2 \triangleright h_3). \quad (2.25)$$

Proof. Let us verify this equality. The left-hand side gives

$$(g_1 g_2, h_1 g_1 \triangleright h_2) \#_1 (g_3, h_3) = (g_1 g_2 g_3, h_1 g_1 \triangleright h_2 (g_1 g_2) \triangleright h_3). \quad (2.26)$$

The right-hand side likewise gives

$$\begin{aligned}
 (g_1, h_1) \#_1 (g_2 g_3, h_2 g_2 \triangleright h_3) &= (g_1 g_2 g_3, h_1 g_1 \triangleright (h_2 g_2 \triangleright h_3)) \\
 &= (g_1 g_2 g_3, h_1 g_1 \triangleright h_2 g_1 \triangleright (g_2 \triangleright h_3)) \\
 &= (g_1 g_2 g_3, h_1 g_1 \triangleright h_2 (g_1 g_2) \triangleright h_3),
 \end{aligned} \quad (2.27)$$

where the identity (2.5) has been used. ■

- There exists a 2-morphism that serves as the identity for horizontal composition of 2-morphisms:

$$\begin{array}{c}
 x \bullet \xleftarrow{1_x} \bullet x \\
 \Downarrow 1_{1_x} \\
 x \bullet \xleftarrow{1_x} \bullet x
 \end{array}$$

- The composition of morphisms is independent of the order in which horizontal and vertical compositions are performed; explicitly,

$$\left((g'_1, h'_1) \#_2 (g_1, h_1) \right) \#_1 \left((g'_2, h'_2) \#_2 (g_2, h_2) \right) = \left((g'_1, h'_1) \#_1 (g'_2, h'_2) \right) \#_2 \left((g_1, h_1) \#_1 (g_2, h_2) \right), \quad (2.28)$$

This is conveniently illustrated in diagrammatic notation, in which a diagram of the form

$$\begin{array}{ccccc}
 & g_1 & & g_2 & \\
 x \bullet & \xleftarrow{g'_1} \bullet & y \bullet & \xleftarrow{g'_2} \bullet & z \\
 & \Downarrow h_1 & & \Downarrow h_2 & \\
 & g'_1 & & g'_2 & \\
 & \Downarrow h'_1 & & \Downarrow h'_2 & \\
 & g''_1 & & g''_2 &
 \end{array}$$

uniquely determines a 2-morphism. Equation (2.28) is called the *interchange law*.

Proof. We now prove (2.28). Its left-hand side is

$$\begin{aligned}
 ((g'_1, h'_1) \#_2 (g_1, h_1)) \#_1 ((g'_2, h'_2) \#_2 (g_2, h_2)) &= (g_1, h'_1 h_1) \#_1 (g_2, h'_2 h_2) \\
 &= (g_1 g_2, h'_1 h_1 g_1 \triangleright (h'_2 h_2)).
 \end{aligned} \quad (2.29)$$

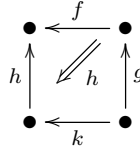
The right-hand side of (2.28) becomes

$$\begin{aligned}
((g'_1, h'_1) \#_1 (g'_2, h'_2)) \#_2 ((g_1, h_1) \#_1 (g_2, h_2)) &= (g'_1 g'_2, h'_1 g'_1 \triangleright h'_2) \#_2 (g_1 g_2, h_1 g_1 \triangleright h_2) \\
&= (g_1 g_2, h'_1 g'_1 \triangleright h'_2 h_1 g_1 \triangleright h_2) \\
&= (g_1 g_2, h'_1 (\partial(h_1) g_1) \triangleright h'_2 h_1 g_1 \triangleright h_2) \\
&= (g_1 g_2, h'_1 \partial(h_1) \triangleright (g_1 \triangleright h'_2) h_1 g_1 \triangleright h_2) \\
&= (g_1 g_2, h'_1 h_1 (g_1 \triangleright h'_2) h_1^{-1} h_1 g_1 \triangleright h_2) \\
&= (g_1 g_2, h'_1 h_1 g_1 \triangleright h'_2 g_1 \triangleright h_2) \\
&= (g_1 g_2, h'_1 h_1 g_1 \triangleright (h'_2 h_2)).
\end{aligned} \tag{2.30}$$

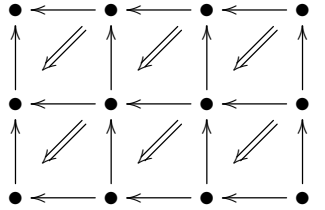
In this calculation we used the identities (2.5) and (2.7), thereby proving (2.28). ■

Composition of a 2-morphism and a 1-morphism

Let us calculate the holonomy associated with a surface in the presence of a 1-form $\alpha \in \mathfrak{g}$ and a 2-form $\beta \in \mathfrak{h}$. We interpret the surface

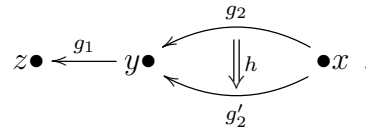


as a 2-morphism $h : fg \rightarrow hk$. A larger surface can be obtained by whiskering:



Using *whiskering*, we may determine the holonomy $\text{hol}(\Sigma)$ associated with a surface Σ . The problem is reduced to finding the holonomies of the small squares into which the surface is subdivided, composing them as defined above, and then taking the limit in which the squares become infinitesimally small.

- *Left whiskering* composes a 2-morphism h with a 1-morphism g_1 lying on its left, when $\partial_1^-(g_1) = \partial_1^+(h)$:



The resulting 2-morphism, denoted by $g_1 \#_1 (g_2, h)$, is the 2-morphism h *left-whiskered* by the 1-morphism g_1 . This composition is the horizontal composition of the two 2-morphisms $(g_1, 1_{g_1}) \#_1 (g_2, h)$,

$$\begin{array}{c}
\begin{array}{ccccc}
& g_1 & & g_2 & \\
z \bullet & \xleftarrow{\quad} & y \bullet & \xleftarrow{\quad} & \bullet x \\
& \Downarrow 1_{g_1} & & \Downarrow h & \\
& g_1 & & g'_2 &
\end{array}
& = &
\begin{array}{ccc}
& g_1 g_2 & \\
z \bullet & \xleftarrow{\quad} & \bullet x \\
& \Downarrow g_1 \triangleright h & \\
& g_1 g'_2 &
\end{array}
\end{array} ,$$

or, formally,

$$g_1 \#_1 (g_2, h) = (g_1, 1_{g_1}) \#_1 (g_2, h) = (g_1 g_2, g_1 \triangleright h). \tag{2.31}$$

- Similarly, we can form the *right whiskering* of a 2-morphism h by a 1-morphism g_2 , when $\partial_1^-(h) = \partial_1^+(g_2)$. We denote this composition by $(g_1, h) \#_1 g_2$:

$$\begin{array}{ccccc} & & g_1 & & \\ & \swarrow & \Downarrow h & \searrow & \\ z \bullet & & & & \bullet y \xleftarrow{g_2} x \bullet \\ & \nwarrow & \Downarrow g'_1 & \swarrow & \end{array} .$$

The resulting 2-morphism is the 2-morphism h *right-whiskered* by the 1-morphism g_2 . It is obtained as the horizontal composition of the two 2-morphisms $(g_1, h) \#_1 (g_2, 1_{g_2})$,

$$\begin{array}{ccccc} & & g_1 & & g_2 & & & & g_1 g_2 & & \\ & \swarrow & \Downarrow h & \nwarrow & \Downarrow 1_{g_2} & \swarrow & & & \Downarrow h & \nwarrow & \\ z \bullet & & & & y \bullet & & \bullet x & = & z \bullet & & \bullet x \\ & \nwarrow & \Downarrow g'_1 & \swarrow & \Downarrow g_2 & \nwarrow & & & \Downarrow g'_1 g_2 & \nwarrow & \end{array} ,$$

or, formally,

$$(g_1, h) \#_1 g_2 = (g_1, h) \#_1 (g_2, 1_{g_2}) = (g_1 g_2, h g_1 \triangleright 1_{g_2}) = (g_1 g_2, h) . \quad (2.32)$$

This construction allows us to combine two 2-morphisms as illustrated in the following example.

Example 2.2.1 *Let us determine the holonomy associated with a surface composed of two squares:*

$$\begin{array}{ccccc} n \bullet & \xleftarrow{g_{mn}} & m \bullet & \xleftarrow{g_{lm}} & l \bullet \\ \uparrow g_{kn} & \swarrow h_{jmn} & \uparrow g_{jm} & \swarrow h_{ilm} & \uparrow g_{il} \\ k \bullet & \xleftarrow{g_{jk}} & j \bullet & \xleftarrow{g_{ij}} & i \bullet \end{array} .$$

The calculation proceeds in two steps. First, we compose the 2-morphism h_{ilm} with the 1-morphism g_{mn} acting from the left, obtaining the 2-morphism

$$g_{mn} \#_1 (g_{lm} g_{il}, h_{ilm}) : g_{mn} g_{lm} g_{il} \rightarrow g_{mn} g_{jm} g_{ij} , \quad (2.33)$$

$$\begin{array}{ccccc} n \bullet & \xleftarrow{g_{mn}} & m \bullet & \xleftarrow{g_{lm}} & l \bullet \\ & \swarrow h_{jmn} & \uparrow g_{jm} & \swarrow h_{ilm} & \uparrow g_{il} \\ & & j \bullet & \xleftarrow{g_{ij}} & i \bullet \end{array} .$$

Next, we form $(g_{mn} g_{jm}, h_{jmn}) \#_1 g_{ij}$, the composition of the 2-morphism h_{jmn} with the 1-morphism g_{ij} acting from the right:

$$(g_{mn} g_{jm}, h_{jmn}) \#_1 g_{ij} : g_{mn} g_{jm} g_{ij} \rightarrow g_{kn} g_{jk} g_{ij} , \quad (2.34)$$

$$\begin{array}{ccccc} n \bullet & \xleftarrow{g_{mn}} & m \bullet & & \\ \uparrow g_{kn} & \swarrow h_{jmn} & \uparrow g_{jm} & & \\ k \bullet & \xleftarrow{g_{jk}} & j \bullet & \xleftarrow{g_{ij}} & i \bullet \end{array} .$$

Finally, the vertical composition of these two 2-morphisms gives

$$((g_{mn} g_{jm}, h_{jmn}) \#_1 g_{ij}) \#_2 (g_{mn} \#_1 (g_{lm} g_{il}, h_{ilm})) : g_{mn} g_{lm} g_{il} \rightarrow g_{kn} g_{jk} g_{ij} , \quad (2.35)$$

$$\begin{array}{ccccc} n \bullet & \xleftarrow{g_{mn}} & m \bullet & \xleftarrow{g_{lm}} & l \bullet \\ \uparrow g_{kn} & \swarrow h_{jmn} & \uparrow g_{jm} & \swarrow h_{ilm} & \uparrow g_{il} \\ k \bullet & \xleftarrow{g_{jk}} & j \bullet & \xleftarrow{g_{ij}} & i \bullet \end{array} .$$

In the same manner as in the preceding example, we shall construct the surface holonomy associated with a tetrahedron, a result of particular relevance to a *triangulation of spacetime*.

2.2.5 2-connection and 2-curvature

Just as a Lie group G gives rise to a connection α , used in the formulation of BF theory, a 2-group gives rise to a 2-connection: an ordered pair (α, β) consisting of a $\mathfrak{g}$ -valued 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$ and a $\mathfrak{h}$ -valued 2-form $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h})$, where $\mathfrak{h}$ is the Lie algebra corresponding to the Lie group H . The associated *fake 2-curvature* is the ordered pair $(\mathcal{F}, \mathcal{G})$ defined by

$$\mathcal{F} = d\alpha + \alpha \wedge \alpha - \partial\beta, \quad \mathcal{G} = d\beta + \alpha \wedge^\triangleright \beta. \quad (2.36)$$

Here $\alpha \wedge^\triangleright \beta$ denotes both the wedge product of the differential forms α and β and the product of their algebra-valued components induced by the action $\triangleright$; see [16]. The 2-curvature $(\mathcal{F}, \mathcal{G})$ is called *fake* because of the additional term $\partial\beta$ in the definition of $\mathcal{F}$ [12].

The curvatures may be expanded in bases of the corresponding algebras and differential forms:

$$\mathcal{F} = \frac{1}{2} \mathcal{F}^\alpha_{\mu\nu} \tau_\alpha dx^\mu \wedge dx^\nu, \quad \mathcal{G} = \frac{1}{3!} \mathcal{G}^a_{\mu\nu\rho} t_a dx^\mu \wedge dx^\nu \wedge dx^\rho,$$

with coefficients

$$\begin{aligned} \mathcal{F}^\alpha_{\mu\nu} &= \partial_\mu \alpha^\alpha_\nu - \partial_\nu \alpha^\alpha_\mu + f_{\beta\gamma}^\alpha \alpha^\beta_\mu \alpha^\gamma_\nu - \beta^a_{\mu\nu} \partial_a^\alpha, \\ \mathcal{G}^a_{\mu\nu\rho} &= \partial_\mu \beta^a_{\nu\rho} + \partial_\nu \beta^a_{\rho\mu} + \partial_\rho \beta^a_{\mu\nu} + \alpha^\alpha_\mu \beta^b_{\nu\rho} \triangleright_{\alpha b}^a + \alpha^\alpha_\nu \beta^b_{\rho\mu} \triangleright_{\alpha b}^a + \alpha^\alpha_\rho \beta^b_{\mu\nu} \triangleright_{\alpha b}^a. \end{aligned} \quad (2.37)$$

Fake curvature

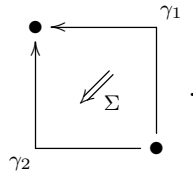
According to Baez [28], the adjective *fake* originates in the work of Breen and Messing and distinguishes two types of curvature: the *2-curvature* and the *fake 2-curvature*. The ordinary 2-curvature, the ordered pair (F, G) associated with a 2-connection (α, β) , where $\alpha \in \mathcal{A}^1(\mathfrak{g}, \mathcal{M}_4)$ and $\beta \in \mathcal{A}^2(\mathfrak{h}, \mathcal{M}_4)$, is defined in the standard way:

$$\begin{aligned} F &= d\alpha + \alpha \wedge \alpha, \\ G &= d\beta + \alpha \wedge^\triangleright \beta. \end{aligned} \quad (2.38)$$

The *fake 2-curvature*, the ordered pair $(\mathcal{F}, \mathcal{G})$, contains an additional term:

$$\begin{aligned} \mathcal{F} &= F - \partial\beta, \\ \mathcal{G} &= G. \end{aligned} \quad (2.39)$$

The notion of fake 2-curvature is introduced for the following reason. Consider a surface Σ :



This square represents a 2-morphism $\Sigma : \gamma_1 \rightarrow \gamma_2$, where γ_1 and γ_2 are the 1-morphisms shown in the diagram. We wish to calculate

$$\text{hol}(\Sigma) : \text{hol}(\gamma_1) \rightarrow \text{hol}(\gamma_2). \quad (2.40)$$

The holonomy $\text{hol}(\Sigma)$ depends on the connection 2-form β , whereas its source and target depend only on the 1-form α . Requiring this 2-morphism to have a well-defined source and target leads to a relation between the connections α and β , which we now derive.

In terms of the 1-connection α , one has

$$\text{hol}(\gamma_1) = \mathcal{P}\exp\left(\int_{\gamma_1} \alpha\right), \quad \text{hol}(\gamma_2) = \mathcal{P}\exp\left(\int_{\gamma_2} \alpha\right). \quad (2.41)$$

Moreover, a 2-morphism $h : g_1 \rightarrow g_2$ is specified by an element $h \in H$, with $g_2 = \partial(h)g_1$. Hence the 2-morphism $\text{hol}(\Sigma) : \text{hol}(\gamma_1) \rightarrow \text{hol}(\gamma_2)$ is specified by an element $h \in H$ such that

$$\mathcal{P}\exp\left(\int_{\gamma_2} \alpha\right) = \partial(h) \mathcal{P}\exp\left(\int_{\gamma_1} \alpha\right), \quad (2.42)$$

or $\partial(h) = \mathcal{P}\exp\left(\int_{\partial\Sigma} \alpha\right)$. Here the integration is performed along the boundary $\partial\Sigma = \gamma_2\gamma_1^{-1}$. Since the square is small, Stokes' theorem gives

$$\partial(h) = \mathcal{P}\exp\left(\int_{\partial\Sigma} \alpha\right) \approx \exp\left(\int_{\Sigma} F\right). \quad (2.43)$$

On the other hand, in terms of the 2-form β , the 2-morphism h is

$$h \approx \exp\left(\int_{\Sigma} \beta\right). \quad (2.44)$$

Substituting this expression for h into equation (2.43) gives the relation between the 1-connection α and the 2-connection β ,

$$\partial(\exp\left(\int_{\Sigma} \beta\right)) \approx \exp\left(\int_{\Sigma} F\right), \quad (2.45)$$

and hence the identity

$$\partial(\beta) = F = d\alpha + \alpha \wedge \alpha. \quad (2.46)$$

We may now introduce the fake curvature $\mathcal{F} = F - \partial(h)$.

Transformations of the 2-connection and 2-curvature

In category theory, 2-gauge transformations are generated by elements of the groups G and H . Under a G -gauge transformation, the 2-connection transforms as

$$\alpha' = g\alpha g^{-1} + gdg^{-1}, \quad \beta' = g \triangleright \beta, \quad (2.47)$$

where the transformation parameter $g : \mathcal{M}_4 \rightarrow G$ is an element of the principal G -bundle over $\mathcal{M}_4$. Under an H -gauge transformation generated by the parameter $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, the 2-connection transforms as

$$\alpha' = \alpha + \partial\eta, \quad \beta' = \beta + d\eta + \alpha' \wedge^{\triangleright} \eta - \eta \wedge \eta. \quad (2.48)$$

Theorem 1 *The composition of a G -gauge transformation and an H -gauge transformation gives the following transformation of the 2-connection:*

$$\begin{aligned} \alpha'' &= g\alpha g^{-1} + gdg^{-1} + \partial(\eta), \\ \beta'' &= g \triangleright \beta + d\eta + \alpha'' \wedge^{\triangleright} \eta - \eta \wedge \eta, \end{aligned} \quad (2.49)$$

where $g : \mathcal{M}_4 \rightarrow G$ and $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ are the parameters of the G - and H -gauge transformations, respectively.

Using the definition of the 2-curvature (1.4) and the transformation rules above, we obtain the transformation law for the 2-curvature.

Theorem 2 *Under a G -gauge transformation, the 2-curvature $(\mathcal{F}, \mathcal{G})$ transforms as*

$$\mathcal{F} \rightarrow g\mathcal{F}g^{-1}, \quad \mathcal{G} \rightarrow g \triangleright \mathcal{G}, \quad (2.50)$$

whereas under an H -gauge transformation it transforms according to

$$\mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{G} \rightarrow \mathcal{G} + \mathcal{F} \wedge^{\triangleright} \eta, \quad (2.51)$$

where $g : \mathcal{M}_4 \rightarrow G$ and $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ are the parameters of the G - and H -gauge transformations, respectively.

For further details on the transformation of the 2-curvature under 2-gauge transformations, see [29] and Appendix A.

2.3 3-gauge theory

In category theory, the higher-categorical generalization of a 2-category is a 3-category. It consists of objects, morphisms, morphisms between morphisms called 2-morphisms, and morphisms between 2-morphisms called 3-morphisms. See [16] for further details.

In 3-gauge theory, the elementary constituents of a manifold are labelled at three levels: curves by elements $g \in G$, surfaces by elements $h \in H$, and volumes by elements $l \in L$. For a 3-gauge-theory configuration to be well defined, these labels cannot be arbitrary; they must satisfy the following conditions.

1. For every surface labelled by $h \in H$, the labels g_1 and g_2 of its source and target curves satisfy $\partial(h) = g_2 g_1^{-1}$.
2. For every volume, the label $l \in L$ satisfies $\delta(l) = h_2 h_1^{-1}$, where h_1 and h_2 are its source and target surfaces, respectively.
3. For every 4-dimensional hypersurface of the manifold, the volume holonomy around it is trivial.

In this section we consider the various operations used to combine labelled paths, surfaces, and volumes, thereby constructing structures of arbitrary size from elementary constituents of the manifold. The resulting configurations may be viewed as classical configurations of 3-gauge theory; later, in the quantum theory, they will be the configurations summed over in the state sum.

2.3.1 3-group

By analogy with groups and 2-groups in category theory, a 3-group is defined as a 3-category with a single object in which all morphisms, 2-morphisms, and 3-morphisms are invertible. Just as a strict 2-group is equivalent to a crossed module, a semistrict 3-group—a Gray group—is equivalent to a 2-crossed module [17], [30].

Definition 2.3.1 (2-crossed module) A 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{-, -\})$ consists of:

- a group G whose elements are morphisms, with composition as the group operation,

$$\bullet \xleftarrow{g_1} \bullet \xleftarrow{g_2} \bullet = \bullet \xleftarrow{g_1 g_2} \bullet ;$$

- a group H consisting of all 2-morphisms whose source is the identity,

$$\begin{array}{c} 1. \bullet \\ \curvearrowright \\ \bullet \quad \Downarrow h \quad \bullet \\ \curvearrowleft \\ \partial h \end{array} ,$$

where horizontal composition is the group operation,

$$\begin{array}{c} 1. \bullet \\ \curvearrowright \\ \bullet \quad \Downarrow h_1 \quad \bullet \\ \curvearrowleft \\ \partial h_1 \end{array} \quad \begin{array}{c} 1. \bullet \\ \curvearrowright \\ \bullet \quad \Downarrow h_2 \quad \bullet \\ \curvearrowleft \\ \partial h_2 \end{array} = \begin{array}{c} 1. \bullet \\ \curvearrowright \\ \bullet \quad \Downarrow h_1 h_2 \quad \bullet \\ \curvearrowleft \\ \partial(h_1 h_2) \end{array} ;$$

- a group L consisting of the 3-morphisms whose source is the identity,

$$\begin{array}{ccc} \begin{array}{c} \bullet \\ \downarrow \scriptstyle 1_{1_\bullet} \\ \bullet \\ \uparrow \scriptstyle 1_\bullet \end{array} & \xRightarrow{l} & \begin{array}{c} \bullet \\ \downarrow \scriptstyle \delta l \\ \bullet \\ \uparrow \scriptstyle \partial \delta l = 1_\bullet \end{array} \end{array};$$

- a group homomorphism $\partial : H \rightarrow G$ mapping each 2-morphism $h \in H$ to its target $\partial h \in G$,

$$\begin{array}{c} \bullet \\ \downarrow \scriptstyle h \\ \bullet \\ \uparrow \scriptstyle \partial h \end{array},$$

so that, for all $h_1, h_2 \in H$,

$$\partial(h_1 h_2) = \partial(h_1) \partial(h_2);$$

- a group homomorphism $\delta : L \rightarrow H$ mapping each 3-morphism $l \in L$ to its target $\delta l \in H$:

$$\begin{array}{ccc} \begin{array}{c} \bullet \\ \downarrow \scriptstyle 1_{1_\bullet} \\ \bullet \\ \uparrow \scriptstyle 1_\bullet \end{array} & \xRightarrow{l} & \begin{array}{c} \bullet \\ \downarrow \scriptstyle \delta l \\ \bullet \\ \uparrow \scriptstyle \partial \delta l = 1_\bullet \end{array} \end{array},$$

where, for every $l \in L$, the element $\partial(\delta l) = e$ is the identity of G ;

- an action $\triangleright$ of G on all three groups, such that
 - the action of G on itself is given by horizontal conjugation,

$$\begin{array}{ccccccc} \bullet & \xleftarrow{g_0} & \bullet & \xleftarrow{g} & \bullet & \xleftarrow{g_0^{-1}} & \bullet \\ & & & & = & & \\ \bullet & \xleftarrow{g_0 g g_0^{-1}} & \bullet & & & & \end{array},$$

or, formally, for all $g_0, g \in G$,

$$g_0 \triangleright g = g_0 g g_0^{-1},$$

- an element of G acts on an element of H by horizontal conjugation with its identity 2-morphism 1_g , yielding the 2-morphism $g \triangleright h \in H$,

$$\begin{array}{ccccccc} \bullet & \xleftarrow{g} & \bullet & \xleftarrow{1_\bullet} & \bullet & \xleftarrow{g^{-1}} & \bullet \\ & \downarrow \scriptstyle 1_g & & \downarrow \scriptstyle h & & \downarrow \scriptstyle 1_{g^{-1}} & \\ \bullet & \xleftarrow{g} & \bullet & \xleftarrow{\partial h} & \bullet & \xleftarrow{g^{-1}} & \bullet \\ & & & & = & & \\ \bullet & \xleftarrow{1} & \bullet & \xleftarrow{\partial(g \triangleright h)} & \bullet & & \end{array},$$

that is, for all $g \in G$ and $h \in H$,

$$g \partial h g^{-1} = \partial(g \triangleright h),$$

- an element of G acts on an element of L by $\triangleright$, yielding the 3-morphism $g \triangleright l \in L$,

$$\begin{array}{ccccccc} \bullet & \xleftarrow{g_1} & \begin{array}{c} \bullet \\ \downarrow \scriptstyle h_1 \\ \bullet \\ \uparrow \scriptstyle g'_2 \end{array} & \xRightarrow{l} & \bullet & \xleftarrow{g_1} & \begin{array}{c} \bullet \\ \downarrow \scriptstyle h_2 \\ \bullet \\ \uparrow \scriptstyle g'_2 \end{array} \\ & & & & = & & \\ \bullet & \xleftarrow{g_1} & \begin{array}{c} \bullet \\ \downarrow \scriptstyle g_1 \triangleright h_1 \\ \bullet \\ \uparrow \scriptstyle g_1 g'_2 \end{array} & \xRightarrow{g_1 \triangleright l} & \bullet & \xleftarrow{g_1} & \begin{array}{c} \bullet \\ \downarrow \scriptstyle g_1 \triangleright h_2 \\ \bullet \\ \uparrow \scriptstyle g_1 g'_2 \end{array} \end{array};$$

- a G -equivariant map called the **Peiffer lifting**,

$$\{-, -\}_{\text{pf}} : H \times H \rightarrow L.$$

The components of a 2-crossed module have the following properties.

1. The homomorphisms ∂ and δ are G -equivariant. Thus, for every $g \in G$ and $h \in H$,

$$g \triangleright \partial(h) = \partial(g \triangleright h), \quad g \triangleright \delta(l) = \delta(g \triangleright l), \quad (2.52)$$

The action of G on H , respectively on L , is a smooth left action and defines an automorphism of H , respectively of L , i.e. $\triangleright : G \rightarrow \text{Aut}(H)$ and $\triangleright : G \rightarrow \text{Aut}(L)$. For all $g, g_1, g_2 \in G$, $h_1, h_2 \in H$, $l_1, l_2 \in L$, and $e \in H, L$,

$$g_1 \triangleright (g_2 \triangleright e) = (g_1 g_2) \triangleright e, \quad g \triangleright (h_1 h_2) = (g \triangleright h_1)(g \triangleright h_2), \quad g \triangleright (l_1 l_2) = (g \triangleright l_1)(g \triangleright l_2). \quad (2.53)$$

The Peiffer lifting is a G -equivariant bilinear map. Thus, for all $h_1, h_2 \in H$ and $g \in G$,

$$g \triangleright \{h_1, h_2\}_{\text{pf}} = \{g \triangleright h_1, g \triangleright h_2\}_{\text{pf}}. \quad (2.54)$$

2. For all $h_1, h_2 \in H$, the Peiffer commutator $\langle h_1, h_2 \rangle_{\text{pf}}$ defined in (2.6) satisfies

$$\delta(\{h_1, h_2\}_{\text{pf}}) = \langle h_1, h_2 \rangle_{\text{pf}}. \quad (2.55)$$

3. For all $l_1, l_2 \in L$, the following identity holds:

$$[l_1, l_2] = \{\delta(l_1), \delta(l_2)\}_{\text{pf}}, \quad (2.56)$$

where $[l, k] = lkl^{-1}k^{-1}$.

4. The Peiffer lifting satisfies the following identities for all $h_1, h_2, h_3 \in H$:

$$\{h_1 h_2, h_3\}_{\text{pf}} = \{h_1, h_2 h_3 h_2^{-1}\}_{\text{pf}} \partial(h_1) \triangleright \{h_2, h_3\}_{\text{pf}}, \quad (2.57)$$

and

$$\{h_1, h_2 h_3\}_{\text{pf}} = \{h_1, h_2\}_{\text{pf}} \{h_1, h_3\}_{\text{pf}} \{ \langle h_1, h_3 \rangle_{\text{pf}}^{-1}, \partial(h_1) \triangleright h_2 \}_{\text{pf}}. \quad (2.58)$$

5. For all $h \in H$ and $l \in L$, one has

$$\{\delta(l), h\}_{\text{pf}} \{h, \delta(l)\}_{\text{pf}} = l(\partial(h) \triangleright l^{-1}). \quad (2.59)$$

Crossed-module substructure

Theorem 3 The substructure $(L \xrightarrow{\delta} H, \triangleright')$ of a 2-crossed module forms a crossed module, where $\triangleright'$ is the action of H on L defined, for every $h \in H$ and $l \in L$, by

$$h \triangleright' l = l \{ \delta(l)^{-1}, h \}_{\text{pf}}. \quad (2.60)$$

We divide the proof of this theorem into four parts. See [30] for further details.

Lemma 1 For all $h_1, h_2 \in H$ and $l \in L$,

$$(h_1 h_2) \triangleright' l = h_1 \triangleright' (h_2 \triangleright l).$$

Proof. For all $h_1, h_2 \in H$ and $l \in L$,

$$\begin{aligned} (h_1 h_2) \triangleright' l &= l \{ \delta(l)^{-1}, h_1 h_2 \}_{\text{pf}} \\ &= l \{ \delta(l)^{-1}, h_1 \}_{\text{pf}} \{ \delta(l)^{-1}, h_2 \}_{\text{pf}} \{ \langle \delta(l)^{-1}, h_2 \rangle_{\text{pf}}^{-1}, \partial(\delta(l))^{-1} \triangleright h_1 \}_{\text{pf}} \\ &= l \{ \delta(l)^{-1}, h_1 \}_{\text{pf}} \{ \delta(l)^{-1}, h_2 \}_{\text{pf}} \{ \langle \delta(l)^{-1}, h_2 \rangle_{\text{pf}}^{-1}, h_1 \}_{\text{pf}}, \end{aligned}$$

where in the first line we used the definition of $\triangleright'$ from Theorem (3), in the second line the identity (2.58), and in the third line $\partial(\delta) = 1$. On the other hand,

$$\begin{aligned} h_1 \triangleright' (h_2 \triangleright' l) &= h_1 \triangleright' (l \{ \delta(l)^{-1}, h_2 \}_{\text{pf}}) \\ &= (h_1 \triangleright' l) (h_1 \triangleright' \{ \delta(l)^{-1}, h_2 \}_{\text{pf}}) \\ &= l \{ \delta(l)^{-1}, h_1 \}_{\text{pf}} \{ \delta(l)^{-1}, h_2 \}_{\text{pf}} \{ \delta(\{ \delta(l)^{-1}, h_2 \}_{\text{pf}})^{-1}, h_1 \}_{\text{pf}} \\ &= l \{ \delta(l)^{-1}, h_1 \}_{\text{pf}} \{ \delta(l)^{-1}, h_2 \}_{\text{pf}} \{ \langle \delta(l)^{-1}, h_2 \rangle_{\text{pf}}^{-1}, h_1 \}_{\text{pf}}, \end{aligned}$$

where the identity (2.56) was used in the final step. ■

Lemma 2 For every $h \in \mathfrak{h}$ and $l_1, l_2 \in \mathfrak{l}$,

$$h \triangleright' (l_1 l_2) = (h \triangleright' l_1) (h \triangleright' l_2).$$

Proof. For every $h \in \mathfrak{h}$ and $l_1, l_2 \in \mathfrak{l}$,

$$\begin{aligned} h \triangleright' (l_1 l_2) &= (l_1 l_2) \{ \delta(l_1 l_2)^{-1}, h \}_{\text{pf}} \\ &= (l_1 l_2) \{ \delta(l_2)^{-1} \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 \{ \delta(l_2)^{-1}, \delta(l_1)^{-1} h \delta(l_1) \}_{\text{pf}} \partial(\delta(l_2)^{-1}) \triangleright \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 \{ \delta(l_2)^{-1}, \delta(l_1)^{-1} h \delta(l_1) \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 \{ \delta(l_2)^{-1}, [\delta(l_1)^{-1}, h] h \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 \{ \delta(l_2)^{-1}, \delta(\{ \delta(l_1)^{-1}, h \}_{\text{pf}}) h \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 \{ \delta(l_2)^{-1}, \delta(\{ \delta(l_1)^{-1}, h \}_{\text{pf}}) \}_{\text{pf}} \{ \delta(l_2)^{-1}, h \}_{\text{pf}} \{ \langle \delta(l_2)^{-1}, h \rangle_{\text{pf}}^{-1}, \partial(\delta(l_2)^{-1}) \triangleright \delta(\{ \delta(l_1)^{-1}, h \}_{\text{pf}}) \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 [l_2^{-1}, \{ \delta(l_1)^{-1}, h \}_{\text{pf}}] \{ \delta(l_2)^{-1}, h \}_{\text{pf}} \{ \delta(\{ \delta(l_2)^{-1}, h \}_{\text{pf}})^{-1}, \delta(\{ \delta(l_1)^{-1}, h \}_{\text{pf}}) \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 l_2 l_2^{-1} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} l_2 \{ \delta(l_1)^{-1}, h \}_{\text{pf}}^{-1} \{ \delta(l_2)^{-1}, h \}_{\text{pf}} \{ \delta(l_2)^{-1}, h \}_{\text{pf}}^{-1} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 \{ \delta(l_1)^{-1}, h \}_{\text{pf}} l_2 \{ \delta(l_1)^{-1}, h \}_{\text{pf}}^{-1} \{ \delta(l_2)^{-1}, h \}_{\text{pf}} \{ \delta(l_2)^{-1}, h \}_{\text{pf}}^{-1} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \{ \delta(l_2)^{-1}, h \}_{\text{pf}} \{ \delta(l_1)^{-1}, h \}_{\text{pf}}^{-1} \{ \delta(l_1)^{-1}, h \}_{\text{pf}} \\ &= l_1 \{ \delta(l_1)^{-1}, h \}_{\text{pf}} l_2 \{ \delta(l_2)^{-1}, h \}_{\text{pf}} \\ &= (h \triangleright' l_1) (h \triangleright' l_2), \end{aligned}$$

where we used (2.57) in the third line, $\partial(\delta) = 1$ in the fourth and eighth lines, (2.56) in the sixth and ninth lines, and (2.58) in the seventh line. ■

Lemma 3 The *Peiffer identity* holds for all $l_1, l_2 \in L$:

$$\delta(l_1) \triangleright' l_2 = l_1 l_2 l_1^{-1}.$$

Proof. For all $l_1, l_2 \in L$,

$$\delta(l_1) \triangleright' l_2 = l_2 \{ \delta(l_2)^{-1}, \delta(l_1) \}_{\text{pf}} = l_2 l_2^{-1} l_1 l_2 l_1^{-1} = l_1 l_2 l_1^{-1},$$

where the first equality uses the definition of $\triangleright'$ from Theorem (3) and the second uses (2.56). ■

Lemma 4 For all $h \in H$ and $l \in L$,

$$\delta(h \triangleright' l) = h \triangleright' \delta(l).$$

Proof. For all $h \in H$ and $l \in L$,

$$\begin{aligned}
 \delta(h \triangleright' l) &= \delta(l\{\delta(l)^{-1}, h\}_{\text{pf}}) \\
 &= \delta(l)\delta(\{\delta(l)^{-1}, h\}_{\text{pf}}) \\
 &= \delta(l)\langle \delta(l)^{-1}, h \rangle_{\text{pf}} \\
 &= \delta(l)\delta(l)^{-1}h\delta(l)\partial(\delta(l))^{-1} \triangleright' h^{-1} \\
 &= h\delta(l)h^{-1} \\
 &= h \triangleright' \delta(l),
 \end{aligned}$$

where we used the definition (3) of $\triangleright'$ in the first line, (2.55) in the third, the definition of the Peiffer commutator (2.6) in the fourth, $\partial(\delta) = 1$ in the fifth, and the fact that H acts on itself by conjugation in the last line. ■

By contrast, the substructure $(H \xrightarrow{\partial} G, \triangleright)$ is in general only a pre-crossed module, since the Peiffer identity need not hold. If ∂ is trivial and H is Abelian, however, all Peiffer commutators are trivial and the Peiffer identity holds for all $h_1, h_2 \in H$:

$$\partial(h_1) \triangleright h_2 = h_1 h_2 h_1^{-1}. \quad (2.61)$$

Useful identities

The elements $h_1, h_2, h_3 \in H$ satisfy the following identities [17]:

$$\{h_1 h_2, h_3\}_{\text{pf}} = (h_1 \triangleright' \{h_2, h_3\}_{\text{pf}}) \{h_1, \partial(h_2) \triangleright h_3\}_{\text{pf}}, \quad (2.62)$$

$$\{h_1, h_2 h_3\}_{\text{pf}} = \{h_1, h_2\}_{\text{pf}} (\partial(h_1) \triangleright h_2) \triangleright' \{h_1, h_3\}_{\text{pf}}. \quad (2.63)$$

Using the fifth axiom of a 2-crossed module, we obtain, for every $h \in H$ and $l \in L$,

$$\{h, \delta(l)^{-1}\}_{\text{pf}} = (h \triangleright' l^{-1}) (\partial(h) \triangleright l), \quad (2.64)$$

and hence the following identities for all $h_1, h_2 \in H$:

$$\{h_1, h_2\}_{\text{pf}}^{-1} = h_1 \triangleright' \{h_1^{-1}, \partial(h_1) \triangleright h_2\}_{\text{pf}}, \quad (2.65)$$

$$\{h_1, h_2\}_{\text{pf}}^{-1} = \partial(h_1) \triangleright \{h_1^{-1}, h_1 h_2 h_1^{-1}\}_{\text{pf}}, \quad (2.66)$$

$$\{h_1, h_2\}_{\text{pf}}^{-1} = (h_1 h_2 h_1^{-1}) \triangleright' \{h_1, h_2^{-1}\}_{\text{pf}}, \quad (2.67)$$

$$\{h_1, h_2\}_{\text{pf}}^{-1} = (\partial(h_1) \triangleright h_2) \triangleright' \{h_1, h_2^{-1}\}_{\text{pf}}. \quad (2.68)$$

2.3.2 Lie 3-algebra

In analogy with a 2-crossed module, the language of Lie 3-algebras gives rise to a *differential 2-crossed module*.

Definition 2.3.2 (Differential 2-crossed module) A differential 2-crossed module is specified by Lie algebras $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$, together with maps $\partial : \mathfrak{h} \rightarrow \mathfrak{g}$ and $\delta : \mathfrak{l} \rightarrow \mathfrak{h}$,

$$\mathfrak{l} \xrightarrow{\delta} \mathfrak{h} \xrightarrow{\partial} \mathfrak{g}, \quad (2.69)$$

an action $\triangleright$ of $\mathfrak{g}$ on all three algebras, and a $\mathfrak{g}$ -equivariant bilinear map called the **Peiffer lifting**:

$$\{-, -\}_{\text{pf}} : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathfrak{l}. \quad (2.70)$$

Choosing bases $T_A \in \mathfrak{l}$, $t_a \in \mathfrak{h}$, and $\tau_\alpha \in \mathfrak{g}$,

$$[T_A, T_B] = f_{AB}{}^C T_C, \quad [t_a, t_b] = f_{ab}{}^c t_c, \quad [\tau_\alpha, \tau_\beta] = f_{\alpha\beta}{}^\gamma \tau_\gamma, \quad (2.71)$$

the maps ∂ and δ may be expressed in these bases as

$$\partial(t_a) = \partial_a{}^\alpha \tau_\alpha, \quad \delta(T_A) = \delta_A{}^a t_a. \quad (2.72)$$

with the identity

$$\delta_A{}^a \partial_a{}^\alpha = 0. \quad (2.73)$$

The action $\triangleright$ of elements of $\mathfrak{g}$ on elements of $\mathfrak{l}$, $\mathfrak{h}$, and $\mathfrak{g}$ is defined through the action of the generators of $\mathfrak{g}$ on the generators of the respective algebras:

$$\tau_\alpha \triangleright T_A = \triangleright_{\alpha A}{}^B T_B, \quad \tau_\alpha \triangleright t_a = \triangleright_{\alpha a}{}^b t_b, \quad \tau_\alpha \triangleright \tau_\beta = \triangleright_{\alpha\beta}{}^\gamma \tau_\gamma. \quad (2.74)$$

The coefficients $X_{ab}{}^A$ specifying the **Peiffer lifting** are defined by

$$\{t_a, t_b\}_{\text{pf}} = X_{ab}{}^A T_A. \quad (2.75)$$

The components of a differential 2-crossed module have the following properties.

1. The action of $\mathfrak{g}$ on itself is the adjoint action. Thus, for all $\underline{g}_0, \underline{g} \in \mathfrak{g}$,

$$\underline{g}_0 \triangleright \underline{g} = [\underline{g}_0, \underline{g}], \quad (2.76)$$

or, in a basis,

$$\triangleright_{\alpha\beta}{}^\gamma = f_{\alpha\beta}{}^\gamma. \quad (2.77)$$

2. The maps $\partial : H \rightarrow G$ and $\delta : L \rightarrow H$ are $\mathfrak{g}$ -equivariant. Thus, for all $\underline{l} \in \mathfrak{l}$, $\underline{h} \in \mathfrak{h}$, and $\underline{g} \in \mathfrak{g}$,

$$\partial(\underline{g} \triangleright \underline{h}) = [\underline{g}, \partial(\underline{h})], \quad \delta(\underline{g} \triangleright \underline{l}) = \underline{g} \triangleright \delta(\underline{l}), \quad (2.78)$$

or, in a basis,

$$\partial_a{}^\beta f_{\alpha\beta}{}^\gamma = \triangleright_{\alpha a}{}^b \partial_b{}^\gamma, \quad \partial_B{}^a \triangleright_{\alpha A}{}^B = \triangleright_{\alpha b}{}^a \delta_A{}^b. \quad (2.79)$$

3. The action $\triangleright$ of $\mathfrak{g}$ on $\mathfrak{h}$ and $\mathfrak{l}$ is $\mathfrak{g}$ -equivariant. Hence, for all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$, $\underline{l}_1, \underline{l}_2$, and $\underline{g} \in \mathfrak{g}$, one has

$$\underline{g} \triangleright [\underline{h}_1, \underline{h}_2] = [\underline{g} \triangleright \underline{h}_1, \underline{h}_2] + [\underline{h}_1, \underline{g} \triangleright \underline{h}_2], \quad (2.80)$$

$$\underline{g} \triangleright [\underline{l}_1, \underline{l}_2] = [\underline{g} \triangleright \underline{l}_1, \underline{l}_2] + [\underline{l}_1, \underline{g} \triangleright \underline{l}_2], \quad (2.81)$$

or, in a basis,

$$f_{ab}{}^c \triangleright_{\alpha c}{}^d = 2 \triangleright_{\alpha[a}{}^c f_{c|b]}{}^d, \quad (2.82)$$

$$f_{AB}{}^C \triangleright_{\alpha C}{}^D = 2 \triangleright_{\alpha[A}{}^C f_{C|B]}{}^D. \quad (2.83)$$

4. The Peiffer lifting is a $\mathfrak{g}$ -equivariant map. Thus, for all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$ and $\underline{g} \in \mathfrak{g}$,

$$\underline{g} \triangleright \{\underline{h}_1, \underline{h}_2\}_{\text{pf}} = \{\underline{g} \triangleright \underline{h}_1, \underline{h}_2\}_{\text{pf}} + \{\underline{h}_1, \underline{g} \triangleright \underline{h}_2\}_{\text{pf}}, \quad (2.84)$$

or, in terms of the corresponding coefficients in a chosen basis,

$$X_{ab}{}^B \triangleright_{\alpha B}{}^A = \triangleright_{\alpha a}{}^c X_{cb}{}^A + \triangleright_{\alpha b}{}^c X_{ac}{}^A. \quad (2.85)$$

5. For all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$, the Peiffer commutator defined in (2.9) becomes, in the language of algebras,

$$\delta(\{\underline{h}_1, \underline{h}_2\}_{\text{pf}}) = -\langle \underline{h}_1, \underline{h}_2 \rangle_{\text{pf}}, \quad (2.86)$$

or, in a basis,

$$X_{ab}{}^A \delta_A{}^c = f_{ab}{}^c - \partial_a{}^\alpha \triangleright_{\alpha b}{}^c. \quad (2.87)$$

6. For all $\underline{l}_1, \underline{l}_2 \in \mathfrak{l}$, the following identity holds:

$$[\underline{l}_1, \underline{l}_2] = \{\delta(\underline{l}_1), \delta(\underline{l}_2)\}_{\text{pf}}, \quad (2.88)$$

or, equivalently,

$$f_{AB}{}^C = \delta_A{}^a \delta_B{}^b X_{ab}{}^C. \quad (2.89)$$

7. For all $\underline{h}_1, \underline{h}_2, \underline{h}_3 \in \mathfrak{h}$, the Peiffer lifting satisfies the following identities:

$$\{[\underline{h}_1, \underline{h}_2], \underline{h}_3\}_{\text{pf}} = \partial(\underline{h}_1) \triangleright \{\underline{h}_2, \underline{h}_3\}_{\text{pf}} + \{\underline{h}_1, [\underline{h}_2, \underline{h}_3]\}_{\text{pf}} - \partial(\underline{h}_2) \triangleright \{\underline{h}_1, \underline{h}_3\}_{\text{pf}} - \{\underline{h}_2, [\underline{h}_1, \underline{h}_3]\}_{\text{pf}}, \quad (2.90)$$

$$\{[\underline{h}_1, \underline{h}_2], \underline{h}_3\}_{\text{pf}} = \{\partial(\underline{h}_1) \triangleright \underline{h}_2, \underline{h}_3\}_{\text{pf}} - \{\partial(\underline{h}_2) \triangleright \underline{h}_1, \underline{h}_3\}_{\text{pf}} - \{\underline{h}_1, \delta\{\underline{h}_2, \underline{h}_3\}_{\text{pf}}\}_{\text{pf}} + \{\underline{h}_2, \delta\{\underline{h}_1, \underline{h}_3\}_{\text{pf}}\}_{\text{pf}}. \quad (2.91)$$

In a basis, this identity becomes

$$f_{ab}{}^d X_{dc}{}^B = \partial_a{}^\alpha X_{bc}{}^A \triangleright_{\alpha A}{}^B + X_{ad}{}^B f_{bc}{}^d - \partial_b{}^\alpha \triangleright_{\alpha A}{}^B X_{ac}{}^A - X_{bd}{}^B f_{ac}{}^d. \quad (2.92)$$

In addition, for all $\underline{h}_1, \underline{h}_2, \underline{h}_3 \in \mathfrak{h}$,

$$\{\underline{h}_1, [\underline{h}_2, \underline{h}_3]\}_{\text{pf}} = \{\delta\{\underline{h}_1, \underline{h}_2\}_{\text{pf}}, \underline{h}_3\}_{\text{pf}} - \{\delta\{\underline{h}_1, \underline{h}_3\}_{\text{pf}}, \underline{h}_2\}_{\text{pf}}, \quad (2.93)$$

or, in a basis,

$$X_{ad}{}^A f_{bc}{}^d = X_{ab}{}^B \delta_B{}^d X_{dc}{}^A - X_{ac}{}^B \delta_B{}^d X_{db}{}^A. \quad (2.94)$$

8. Every $\underline{h} \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$ satisfy

$$\{\delta(\underline{l}), \underline{h}\} + \{\underline{h}, \delta(\underline{l})\} = -\partial(\underline{h}) \triangleright \underline{l}, \quad (2.95)$$

or, equivalently,

$$\delta_A{}^a X_{ab}{}^B + \delta_A{}^a X_{ba}{}^B = -\partial_b{}^\alpha \triangleright_{\alpha A}{}^B. \quad (2.96)$$

Differential crossed-module substructure

Theorem 4 A substructure of a differential 2-crossed module forms a differential crossed module $(L \xrightarrow{\delta} H, \triangleright')$, where the action $\triangleright'$ of $\mathfrak{h}$ on $\mathfrak{l}$ is defined, for all $\underline{h} \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$, by

$$\underline{h} \triangleright' \underline{l} = -\{\delta(\underline{l}), \underline{h}\}_{\text{pf}}, \quad (2.97)$$

or, in a basis,

$$t_a \triangleright' T_A = \triangleright'_{aA}{}^B T_B, \quad \triangleright'_{aA}{}^B = -\delta_A{}^b X_{ba}{}^B. \quad (2.98)$$

We divide the proof of this theorem into four parts.

Lemma 5 For all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$,

$$[\underline{h}_1, \underline{h}_2] \triangleright' \underline{l} = \underline{h}_1 \triangleright' (\underline{h}_2 \triangleright' \underline{l}) - \underline{h}_2 \triangleright' (\underline{h}_1 \triangleright' \underline{l}).$$

Proof. For all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$,

$$\begin{aligned} [\underline{h}_1, \underline{h}_2] \triangleright' \underline{l} &= -\{\delta(\underline{l}), [\underline{h}_1, \underline{h}_2]\}_{\text{pf}} \\ &= -\{\delta(\{\delta(\underline{l}), \underline{h}_1\}_{\text{pf}}), \underline{h}_2\}_{\text{pf}} + \{\delta(\{\delta(\underline{l}), \underline{h}_2\}_{\text{pf}}), \underline{h}_1\}_{\text{pf}} \\ &= -\underline{h}_2 \triangleright' (\underline{h}_1 \triangleright' \underline{l}) + \underline{h}_1 \triangleright' (\underline{h}_2 \triangleright' \underline{l}), \end{aligned}$$

where we used the definition (2.97) in the first and third lines and the identity (2.93) in the second line. ■

Lemma 6 For all $\underline{l}_1, \underline{l}_2 \in \mathfrak{l}$ and $\underline{h} \in \mathfrak{h}$,

$$\underline{h} \triangleright' [\underline{l}_1, \underline{l}_2] = [\underline{h} \triangleright' \underline{l}_1, \underline{l}_2] + [\underline{l}_1, \underline{h} \triangleright' \underline{l}_2].$$

Proof. For all $\underline{l}_1, \underline{l}_2 \in \mathfrak{l}$ and $\underline{h} \in \mathfrak{h}$,

$$\begin{aligned} \underline{h} \triangleright' [\underline{l}_1, \underline{l}_2] &= -\{\delta([\underline{l}_1, \underline{l}_2]), \underline{h}\}_{\text{pf}} \\ &= -\{[\delta(\underline{l}_1), \delta(\underline{l}_2)], \underline{h}\}_{\text{pf}} \\ &= -\partial(\delta(\underline{l}_1)) \triangleright \{\delta(\underline{l}_2), \underline{h}\}_{\text{pf}} - \{\delta(\underline{l}_1), [\delta(\underline{l}_2), \underline{h}]\}_{\text{pf}} \\ &\quad + \partial(\delta(\underline{l}_2)) \triangleright \{\delta(\underline{l}_1), \underline{h}\}_{\text{pf}} + \{\delta(\underline{l}_2), [\delta(\underline{l}_1), \underline{h}]\}_{\text{pf}} \\ &= -\{\delta(\underline{l}_1), [\delta(\underline{l}_2), \underline{h}]\}_{\text{pf}} + \{\delta(\underline{l}_2), [\delta(\underline{l}_1), \underline{h}]\}_{\text{pf}} \\ &= -\{\delta(\{\delta(\underline{l}_1), \delta(\underline{l}_2)\}_{\text{pf}}), \underline{h}\}_{\text{pf}} + \{\delta(\{\delta(\underline{l}_1), \underline{h}\}_{\text{pf}}), \delta(\underline{l}_2)\}_{\text{pf}} \\ &\quad + \{\delta(\{\delta(\underline{l}_2), \delta(\underline{l}_1)\}_{\text{pf}}), \underline{h}\}_{\text{pf}} - \{\delta(\{\delta(\underline{l}_2), \underline{h}\}_{\text{pf}}), \delta(\underline{l}_1)\}_{\text{pf}} \\ &= -\{\delta([\underline{l}_1, \underline{l}_2], \underline{h})\}_{\text{pf}} + [\delta(\underline{l}_1), \underline{h}]_{\text{pf}, \underline{l}_2} + \{\delta([\underline{l}_2, \underline{l}_1], \underline{h})\}_{\text{pf}} - [\delta(\underline{l}_2), \underline{h}]_{\text{pf}, \underline{l}_1} \\ &= +\underline{h} \triangleright' [\underline{l}_1, \underline{l}_2] - [\underline{h} \triangleright' \underline{l}_1, \underline{l}_2] - \underline{h} \triangleright' [\underline{l}_2, \underline{l}_1] + [\underline{h} \triangleright' \underline{l}_2, \underline{l}_1], \end{aligned}$$

from which (6) follows. Here we used (2.90) in the third line, $\partial(\delta) = 0$ in the fourth, (2.93) in the fifth, (2.88) in the sixth, and the definition (2.97) of $\triangleright'$ in the seventh line. ■

Lemma 7 For all $\underline{l}_1, \underline{l}_2 \in \mathfrak{l}$, the *Peiffer identity* holds:

$$\delta(\underline{l}_1) \triangleright' \underline{l}_2 = [\underline{l}_1, \underline{l}_2].$$

Proof. For all $\underline{l}_1, \underline{l}_2 \in \mathfrak{l}$,

$$\delta(\underline{l}_1) \triangleright' \underline{l}_2 = -\{\delta(\underline{l}_2), \delta(\underline{l}_1)\}_{\text{pf}} = -[\underline{l}_2, \underline{l}_1],$$

which proves (7); the first equality uses the definition (2.97), and the second uses (2.88). ■

Lemma 8 For all $\underline{h} \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$,

$$\delta(\underline{h} \triangleright' \underline{l}) = [\underline{h}, \delta(\underline{l})].$$

Proof. For all $\underline{h} \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$,

$$\delta(\underline{h} \triangleright' \underline{l}) = -\delta(\{\delta(\underline{l}), \underline{h}\}_{\text{pf}}) = -\langle \delta(\underline{l}), \underline{h} \rangle_{\text{pf}} = -[\delta(\underline{l}), \underline{h}] + \partial(\delta(\underline{l})) \triangleright \underline{h},$$

Using $\partial\delta = 0$, this proves (8). We used the definition (2.97) in the first equality, (2.86) in the second, and the definition of the Peiffer commutator (2.9) in the third. ■

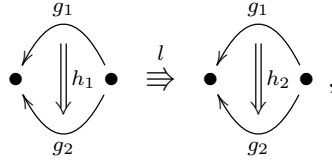
Property 5 shows that a trivial map δ , or a trivial Peiffer lifting, implies that L must be Abelian. Conversely, if L is Abelian, at least one of these two maps, δ or the Peiffer lifting, must be trivial. This indicates that the components defining a 3-group are not independent: the groups G , H , and L and the maps ∂ , δ , and $\{_, _ \}_{\text{pf}}$ must be chosen carefully so that they indeed form a 3-group structure.

For a more detailed analysis of the 3-group structure, see [18].

2.3.3 Composition of 3-morphisms

In this section we describe the composition of elementary paths, surfaces, and volumes in 3-gauge theory. Geometric objects are labelled at three levels. Curves are labelled by elements $g \in G$, and their composition and reversal of orientation are defined as in ordinary gauge theory. Surfaces are labelled by elements $h \in H$, and their vertical composition is defined exactly as in the 2-gauge theory discussed above. Left and right whiskering of a 2-morphism by a 1-morphism are likewise unchanged, whereas horizontal composition is different: in 3-gauge theory it gives rise to an *interchange 3-morphism*. Volumes are labelled by elements $l \in L$.

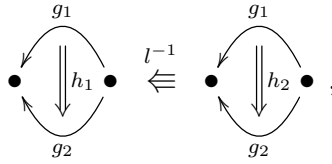
For each volume, we divide its boundary into two surfaces: a source surface labelled by $\partial_3^-(l) = h_1$ and a target surface labelled by $\partial_3^+(l) = h_2$. On their common boundary we choose two reference points and divide the boundary into two curves, a source curve labelled by $\partial_2^-(l) = g_1$ and a target curve labelled by $\partial_2^+(l) = g_2$, as shown below,



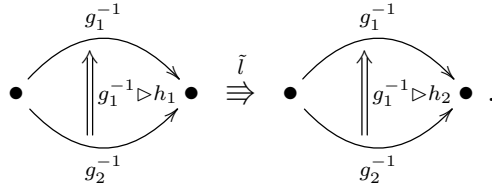
so that the element $l \in L$ satisfies

$$\delta(l) = h_2 h_1^{-1}. \quad (2.99)$$

The orientation of the volume may be reversed, in which case the oppositely oriented volume is labelled by the inverse element l^{-1} ,



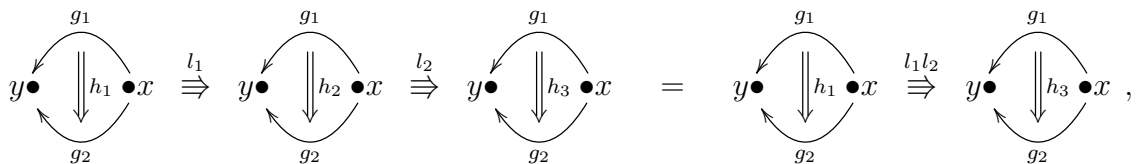
whereas reversing the orientations of the curves and surfaces yields a volume element labelled by $\tilde{l} = g_1^{-1} \triangleright l$:



To specify when elementary volumes are composable, we denote the source and target of the k -arrow ($k = 1, 2, 3$) of a 3-morphism l by $\partial_k^-(l)$ and $\partial_k^+(l)$, respectively.

Upward composition of 3-morphisms

- The *upward composition* of two composable 3-morphisms, i.e. when $\partial_3^+(l_1) = \partial_3^-(l_2)$, gives the 3-morphism



or, for the two 3-morphisms (g_1, h_1, l_1) and (g_1, h_2, l_2) ,

$$(g_1, h_2, l_2) \#_3 (g_1, h_1, l_1) = (g_1, h_1, l_2 l_1). \quad (2.100)$$

- Upward composition of 3-morphisms is associative. Thus, for $l_1, l_2, l_3 \in L$ satisfying $\partial_3^-(l_3) = \partial_3^+(l_2)$ and $\partial_3^-(l_3) = \partial_3^+(l_2)$,

$$\begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_1 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{l_2 l_1} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_3 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{l_3} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_4 \\ \bullet \xleftarrow{g_2} \bullet \end{array} = \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_1 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{l_1} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_2 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{l_3 l_2} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_4 \\ \bullet \xleftarrow{g_2} \bullet \end{array} ,$$

or, formally,

$$(g_1, h_3, l_3) \#_3 (g_1, h_1, l_2 l_1) = (g_1, h_2, l_3 l_2) \#_3 (g_1, h_1, l_1) . \quad (2.101)$$

- For every element $h \in H$, there exists a 3-morphism that is the identity for upward composition:

$$\begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{1_h} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h \\ \bullet \xleftarrow{g_2} \bullet \end{array} .$$

Vertical composition of 3-morphisms

- The vertical composition of two composable 3-morphisms, i.e. when $\partial_2^+(l_1) = \partial_2^-(l_2)$,

$$\begin{array}{c} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_1 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{l_1} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h'_1 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \\ \begin{array}{c} \bullet \xleftarrow{g_2} \bullet \\ \Downarrow h_2 \\ \bullet \xleftarrow{g_3} \bullet \end{array} \xRightarrow{l_2} \begin{array}{c} \bullet \xleftarrow{g_2} \bullet \\ \Downarrow h'_2 \\ \bullet \xleftarrow{g_3} \bullet \end{array} , \end{array}$$

gives the 3-morphism

$$\begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_2 h_1 \\ \bullet \xleftarrow{g_3} \bullet \end{array} \xRightarrow{l_2 h_2 \triangleright' l_1} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow \delta(l_2 h_2 \triangleright' l_1) h_2 h_1 \\ \bullet \xleftarrow{g_3} \bullet \end{array} .$$

Formally, the vertical composition of (g_1, h_1, l_1) and (g_2, h_2, l_2) gives

$$(g_2, h_2, l_2) \#_2 (g_1, h_1, l_1) = (g_1, h_2 h_1, l_2 h_2 \triangleright' l_1) . \quad (2.102)$$

- Vertical composition is associative. Thus, for $l_1, l_2, l_3 \in L$ satisfying $\partial_2^+(l_1) = \partial_2^-(l_2)$ and $\partial_2^+(l_2) = \partial_2^-(l_3)$,

$$\begin{array}{c} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h_1 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \xRightarrow{l_1} \begin{array}{c} \bullet \xrightarrow{g_1} \bullet \\ \Downarrow h'_1 \\ \bullet \xleftarrow{g_2} \bullet \end{array} \\ \begin{array}{c} \bullet \xleftarrow{g_2} \bullet \\ \Downarrow h_2 \\ \bullet \xleftarrow{g_3} \bullet \end{array} \xRightarrow{l_2} \begin{array}{c} \bullet \xleftarrow{g_2} \bullet \\ \Downarrow h'_2 \\ \bullet \xleftarrow{g_3} \bullet \end{array} \\ \begin{array}{c} \bullet \xleftarrow{g_3} \bullet \\ \Downarrow h_3 \\ \bullet \xleftarrow{g_1} \bullet \end{array} \xRightarrow{l_3} \begin{array}{c} \bullet \xleftarrow{g_3} \bullet \\ \Downarrow h'_3 \\ \bullet \xleftarrow{g_1} \bullet \end{array} , \end{array}$$

one has

$$(g_3, h_3, l_3) \#_2 (g_1, h_2 h_1, l_2 h_2 \triangleright' l_1) = (g_2, h_3 h_2, l_3 h_3 \triangleright' l_2) \#_2 (g_1, h_1, l_1). \quad (2.103)$$

Proof. We prove this equality in the same way as (2.25). The left-hand side of (2.103) gives

$$(g_3, h_3, l_3) \#_2 (g_1, h_2 h_1, l_2 h_2 \triangleright' l_1) = (g_1, h_3 h_2 h_1, l_3 h_3 \triangleright' (l_2 h_2 \triangleright' l_1)). \quad (2.104)$$

The right-hand side of (2.103) likewise gives

$$\begin{aligned} (g_2, h_3 h_2, l_3 h_3 \triangleright' l_2) \#_2 (g_1, h_1, l_1) &= (g_1, h_3 h_2 h_1, l_3 h_3 \triangleright' l_2 (h_3 h_2) \triangleright' l_1) \\ &= (g_1, h_3 h_2 h_1, l_3 h_3 \triangleright' l_2 h_3 \triangleright' (h_2 \triangleright' l_1)) \\ &= (g_1, h_3 h_2 h_1, l_3 h_3 \triangleright' (l_2 h_2 \triangleright' l_1)), \end{aligned} \quad (2.105)$$

where Lemma 2 has been used. This proves (2.103). ■

- Composition of 3-morphisms is independent of the order in which vertical and upward compositions are performed; explicitly,

$$((g_2, h'_2, l'_2) \#_3 (g_2, h_2, l_2)) \#_2 ((g_1, h'_1, l'_1) \#_3 (g_1, h_1, l_1)) = ((g_2, h'_2, l'_2) \#_2 (g_1, h'_1, l'_1)) \#_3 ((g_2, h_2, l_2) \#_2 (g_1, h_1, l_1)), \quad (2.106)$$

This is conveniently seen in diagrammatic notation, where a diagram of the form

$$\begin{array}{ccccc} \begin{array}{c} \begin{array}{ccc} y \bullet & \xleftarrow{g_2} & \bullet x \\ \downarrow h_1 \\ g_1 \end{array} & \xRightarrow{l_1} & \begin{array}{ccc} y \bullet & \xleftarrow{g_2} & \bullet x \\ \downarrow h'_1 \\ g_1 \end{array} & \xRightarrow{l'_1} & \begin{array}{ccc} y \bullet & \xleftarrow{g_2} & \bullet x \\ \downarrow h'_1 \\ g_1 \end{array} \\ \begin{array}{ccc} y \bullet & \xleftarrow{g_2} & \bullet x \\ \downarrow h_2 \\ g_3 \end{array} & \xRightarrow{l_2} & \begin{array}{ccc} y \bullet & \xleftarrow{g_2} & \bullet x \\ \downarrow h'_2 \\ g_3 \end{array} & \xRightarrow{l'_2} & \begin{array}{ccc} y \bullet & \xleftarrow{g_2} & \bullet x \\ \downarrow h'_2 \\ g_3 \end{array} \end{array}$$

uniquely determines a 3-morphism.

Proof. Let us prove (2.106). Its left-hand side is

$$\begin{aligned} ((g_2, h'_2, l'_2) \#_3 (g_2, h_2, l_2)) \#_2 ((g_1, h'_1, l'_1) \#_3 (g_1, h_1, l_1)) &= (g_2, h_2, l'_2 l_2) \#_2 (g_1, h_1, l'_1 l_1) \\ &= (g_1, h_2 h_1, l'_2 l_2 h_2 \triangleright' (l'_1 l_1)). \end{aligned} \quad (2.107)$$

The right-hand side of (2.106) is

$$\begin{aligned} ((g_2, h'_2, l'_2) \#_2 (g_1, h'_1, l'_1)) \#_3 ((g_2, h_2, l_2) \#_2 (g_1, h_1, l_1)) &= (g_1, h'_2 h'_1, l'_2 h'_2 \triangleright' l'_1) \#_3 (g_1, h_2 h_1, l_2 h_2 \triangleright' l_1) \\ &= (g_1, h_2 h_1, l'_2 h'_2 \triangleright' l'_1 l_2 h_2 \triangleright' l_1) & (l'_2 = \delta(l_2) h_2) \\ &= (g_1, h_2 h_1, l'_2 (\delta(l_2) h_2) \triangleright' l'_1 l_2 h_2 \triangleright' l_1) & (\text{Lema 1}) \\ &= (g_1, h_2 h_1, l'_2 \delta(l_2) \triangleright' (h_2 \triangleright' l'_1) l_2 h_2 \triangleright' l_1) & (\text{Pajferov id.}) \\ &= (g_1, h_2 h_1, l'_2 l_2 (h_2 \triangleright' l'_1) l_2^{-1} l_2 h_2 \triangleright' l_1) & (l_2^{-1} l_2 = e) \\ &= (g_1, h_2 h_1, l'_2 l_2 h_2 \triangleright' l'_1 h_2 \triangleright' l_1) & (\text{Lema 2}) \\ &= (g_1, h_2 h_1, l'_2 l_2 h_2 \triangleright' (l'_1 l_1)). \end{aligned} \quad (2.108)$$

This proves (2.106). ■

Composition of a 3-morphism and a 1-morphism

- *Left whiskering* composes a 3-morphism l with a 1-morphism g_1 lying on its left, when $\partial_1^+(l) = \partial_1^-(g_1)$:

$$\begin{array}{c}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \xleftarrow{g_1} & y \bullet & \begin{array}{c} \xrightarrow{g_2} \\ \parallel h_1 \\ \xleftarrow{g'_2} \end{array} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 \xRightarrow{l}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \xleftarrow{g_1} & y \bullet & \begin{array}{c} \xrightarrow{g_2} \\ \parallel h_2 \\ \xleftarrow{g'_2} \end{array} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 .
 \end{array}$$

The resulting 3-morphism, denoted by $g_1 \#_1 l$, is the 3-morphism l *left-whiskered* by the 1-morphism g_1 . This composition is formed as $(g_1, 1_{g_1}) \#_1 (g_2, h_1, l)$,

$$\begin{array}{c}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \xleftarrow{g_1} & y \bullet & \begin{array}{c} \xrightarrow{g_2} \\ \parallel h_1 \\ \xleftarrow{g'_2} \end{array} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 \xRightarrow{l}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \xleftarrow{g_1} & y \bullet & \begin{array}{c} \xrightarrow{g_2} \\ \parallel h_2 \\ \xleftarrow{g'_2} \end{array} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 = \\
 \\
 \begin{array}{c}
 \begin{array}{ccc}
 z \bullet & \xleftarrow{g_1 g_2} & \bullet x \\
 & \parallel g_1 \triangleright h_1 \\
 & \xleftarrow{g_1 g'_2}
 \end{array}
 \end{array}
 \xRightarrow{g_1 \triangleright l}
 \begin{array}{c}
 \begin{array}{ccc}
 z \bullet & \xleftarrow{g_1 g_2} & \bullet x \\
 & \parallel g_1 \triangleright h_2 \\
 & \xleftarrow{g_1 g'_2}
 \end{array}
 \end{array}
 ,
 \end{array}$$

or, formally,

$$(g_1, 1_{g_1}) \#_1 (g_2, h_1, l) = (g_1 g_2, g_1 \triangleright h, g_1 \triangleright l) . \quad (2.109)$$

- Similarly, we may right-whisker the 3-morphism l by a 1-morphism g_2 , when $\partial_1^-(l) = \partial_1^+(g_2)$,

$$\begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \begin{array}{c} \xleftarrow{g_1} \\ \parallel h_1 \\ \xleftarrow{g'_1} \end{array} & y \bullet & \xleftarrow{g_2} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 \xRightarrow{l}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \begin{array}{c} \xleftarrow{g_1} \\ \parallel h_2 \\ \xleftarrow{g'_1} \end{array} & y \bullet & \xleftarrow{g_2} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 ,$$

which is formed as $(g_1, h_1, l) \#_1 (g_2, 1_{g_2})$,

$$\begin{array}{c}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \begin{array}{c} \xleftarrow{g_1} \\ \parallel h_1 \\ \xleftarrow{g'_1} \end{array} & y \bullet & \begin{array}{c} \xleftarrow{g_2} \\ \parallel 1_{g_2} \\ \xleftarrow{g_2} \end{array} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 \xRightarrow{l}
 \begin{array}{c}
 \begin{array}{ccccc}
 z \bullet & \begin{array}{c} \xleftarrow{g_1} \\ \parallel h_2 \\ \xleftarrow{g'_1} \end{array} & y \bullet & \begin{array}{c} \xleftarrow{g_2} \\ \parallel 1_{g_2} \\ \xleftarrow{g_2} \end{array} & \bullet x \\
 & & & & \\
 & & & &
 \end{array}
 \end{array}
 = \\
 \\
 \begin{array}{c}
 \begin{array}{ccc}
 z \bullet & \xleftarrow{g_1 g_2} & \bullet x \\
 & \parallel h_1 \\
 & \xleftarrow{g'_1 g_2}
 \end{array}
 \end{array}
 \xRightarrow{l}
 \begin{array}{c}
 \begin{array}{ccc}
 z \bullet & \xleftarrow{g_1 g_2} & \bullet x \\
 & \parallel h_2 \\
 & \xleftarrow{g'_1 g_2}
 \end{array}
 \end{array}
 ,
 \end{array}$$

or, formally,

$$(g_1, h_1, l) \#_1 (g_2, 1_{g_2}) = (g_1 g_2, h_1, l) . \quad (2.110)$$

Composition of a 3-morphism and a 2-morphism

- *Upper whiskering* of a 3-morphism l by a 2-morphism h_2 , when they are composable, i.e. when $\partial_2^+(l) = \partial_2^-(h_2)$, is defined as the vertical composition of (g_1, h_1, l) and $(g_2, h_2, 1_{h_2})$,

$$\begin{array}{ccc}
 \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array} & \xRightarrow{l} & \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array} \\
 \begin{array}{c} y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowright \\ g_3 \end{array} & \xRightarrow{1_{h_2}} & \begin{array}{c} y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowright \\ g_3 \end{array} ,
 \end{array}$$

which gives the 3-morphism

$$\begin{array}{ccc}
 \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array} & \xRightarrow{h_2 \triangleright' l} & \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array}
 \end{array}$$

Formally,

$$(g_1, h_1, l) \#_2 (g_2, h_2, 1_{h_2}) = (g_1, h_2 h_1, h_2 \triangleright' l). \quad (2.111)$$

- *Lower whiskering* of a 3-morphism l by a 2-morphism h_1 , when they are composable, i.e. when $\partial_2^-(l) = \partial_2^+(h_1)$, is defined as the vertical composition of $(g_1, h_1, 1_{h_1})$ and (g_2, h_2, l) ,

$$\begin{array}{ccc}
 \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array} & \xRightarrow{1_{h_1}} & \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array} \\
 \begin{array}{c} y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowright \\ g_3 \end{array} & \xRightarrow{l} & \begin{array}{c} y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowright \\ g_3 \end{array} ,
 \end{array}$$

which gives the 3-morphism

$$\begin{array}{ccc}
 \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array} & \xRightarrow{l} & \begin{array}{c} g_1 \\ \curvearrowright \\ y \bullet \xleftarrow{g_2} \bullet x \\ \curvearrowleft \\ g_3 \end{array}
 \end{array}$$

Formally,

$$(g_1, h_1, 1_{h_1}) \#_2 (g_2, h_2, l) = (g_1, h_2 h_1, l). \quad (2.112)$$

2.3.4 Horizontal composition of 2-morphisms—the interchange 3-morphism

- The *horizontal composition* of the 2-morphisms h_1 and h_2 , when $\partial_1^-(h_1) = \partial_1^+(h_2)$, gives an *interchange 3-morphism*.

$$\begin{array}{ccccc}
 & g_1 & & g_2 & \\
 & \curvearrowright & & \curvearrowright & \\
 z \bullet & \xleftarrow{\quad} & y \bullet & \xleftarrow{\quad} & \bullet x \\
 & \curvearrowleft & & \curvearrowleft & \\
 & g'_1 & & g'_2 &
 \end{array}$$

The composition gives a 3-morphism whose source is the 2-morphism

$$\partial_3^-(l) = ((g_1, h_1) \#_1 g'_2) \#_2 (g_1 \#_1 (g_2, h_2)),$$

and whose target is the 2-morphism

$$\partial_3^+(l) = (g'_1 \#_1 (g_2, h_2)) \#_2 ((g_1, h_1) \#_1 g_2) ,$$

$$\begin{array}{c} \begin{array}{ccccc} & g_1 & & g_2 & \\ z \bullet & \xleftarrow{\quad} & y \bullet & \xleftarrow{\quad} & \bullet x \\ & \Downarrow h_1 & & \Downarrow h_2 & \\ & g'_1 & & g'_2 & \end{array} & = & \begin{array}{ccc} & g_1 g_2 & \\ z \bullet & \xleftarrow{\quad} & \bullet x \\ & \Downarrow h_1 g_1 \triangleright h_2 & \\ & g'_1 g'_2 & \end{array} & \xRightarrow{l} & \begin{array}{ccc} & g_1 g_2 & \\ z \bullet & \xleftarrow{\quad} & \bullet x \\ & \Downarrow g'_1 \triangleright h_2 h_1 & \\ & g'_1 g'_2 & \end{array} \end{array}$$

Formally,

$$(g_1, h_1) \#_1 (g_2, h_2) = (g_1 g_2, h_1 g_1 \triangleright h_2, l) . \quad (2.113)$$

In equation (2.113), the 3-morphism l is the inverse Peiffer lifting $\{h_1, g_1 \triangleright h_2\}_{\text{pf}}^{-1}$. Using (2.99), we obtain

$$(\partial(h_1)g_1) \triangleright h_2 h_1 = \delta(l)h_1(g_1 \triangleright h_2) . \quad (2.114)$$

Using the definition of the Peiffer commutator, namely (2.55), and the first property of a 2-crossed module, $\delta(\{h_1, h_2\}_{\text{p}}) = \langle h_1, h_2 \rangle_{\text{pf}}$, we find that the 3-morphism l satisfies

$$\delta(l)^{-1} = h_1 g_1 \triangleright h_2 h_1^{-1} (\partial(h_1)g_1) \triangleright h_2^{-1} = \langle h_1, g_1 \triangleright h_2 \rangle_{\text{pf}} = \delta(\{h_1, g_1 \triangleright h_2\}_{\text{p}}) . \quad (2.115)$$

- Consider the horizontal composition of the vertical composite of the 2-morphisms h_1 and h'_1 with the 2-morphism h_2 on the right. When these are composable, i.e. when $\partial_1^-(h_1) = \partial_1^-(h'_1) = \partial_1^+(h_2)$ and $\partial_2^+(h_1) = \partial_2^-(h'_1)$,

$$\begin{array}{ccccc} & g_1 & & g_2 & \\ z \bullet & \xleftarrow{\quad} & y \bullet & \xleftarrow{\quad} & \bullet x \\ & \Downarrow h_1 & & \Downarrow h_2 & \\ & g'_1 & & g'_2 & \\ & \Downarrow h'_1 & & & \\ & g''_1 & & & \end{array}$$

the result is the interchange 3-morphism $(g_1, h'_1 h_1) \#_1 (g_2, h_2)$,

$$\begin{array}{ccccc} & g_1 & & g_2 & \\ z \bullet & \xleftarrow{\quad} & y \bullet & \xleftarrow{\quad} & \bullet x \\ & \Downarrow h'_1 h_1 & & \Downarrow h_2 & \\ & g''_1 & & g'_2 & \end{array} = (g_1, h'_1 h_1) \#_1 (g_2, h_2) = (g_1 g_2, h'_1 h_1 g_1 \triangleright h_2, l) .$$

In the preceding formula, $l = \{h'_1 h_1, g_1 \triangleright h_2\}_{\text{pf}}^{-1}$, with source surface $h'_1 h_1 g_1 \triangleright h_2$ and target surface $g''_1 \triangleright h_2 h'_1 h_1$.

- Consider the horizontal composition of the vertical composite of the 2-morphisms h_2 and h'_2 with the 2-morphism h_1 on the left. When these are composable, i.e. when $\partial_1^+(h_2) = \partial_1^+(h'_2) = \partial_1^-(h_1)$ and $\partial_2^+(h_2) = \partial_2^-(h'_2)$,

$$\begin{array}{ccccc} & g_1 & & g_2 & \\ z \bullet & \xleftarrow{\quad} & y \bullet & \xleftarrow{\quad} & \bullet x \\ & \Downarrow h_1 & & \Downarrow h_2 & \\ & g'_1 & & g'_2 & \\ & & & \Downarrow h'_2 & \\ & & & g''_2 & \end{array}$$

the result is the interchange 3-morphism $(g_1, h_1) \#_1 (g_2, h'_2 h_2)$,

$$\begin{array}{c}
 \begin{array}{ccccc}
 & & g_1 & & \\
 & \swarrow & & \searrow & \\
 z \bullet & & y \bullet & & x \bullet \\
 & \nwarrow & & \nearrow & \\
 & & g'_1 & & \\
 & & & & g'_2
 \end{array}
 \end{array}
 = (g_1, h_1) \#_1 (g_2, h'_2 h_2) = (g_1 g_2, h_1 g_1 \triangleright (h'_2 h_2), l) .$$

In the preceding formula, $l = \{h_1, g_1 \triangleright (h'_2 h_2)\}_{\text{pf}}^{-1}$, with source surface $h_1 g_1 \triangleright (h'_2 h_2)$ and target surface $g'_1 \triangleright (h'_2 h_2) h_1$.

2.3.5 3-connection and 3-curvature

Let $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$ be the Lie algebras corresponding to the groups G , H , and L . A 3-connection is an ordered triple (α, β, γ) of algebra-valued differential forms: $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$, $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h})$, and $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l})$. The corresponding *fake*⁷ 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is defined by

$$\begin{aligned}
 \mathcal{F} &= d\alpha + \alpha \wedge \alpha - \partial\beta, & \mathcal{G} &= d\beta + \alpha \wedge^\triangleright \beta - \delta\gamma, \\
 \mathcal{H} &= d\gamma + \alpha \wedge^\triangleright \gamma + \beta \wedge^{\{\cdot\}} \beta.
 \end{aligned} \tag{2.118}$$

The connection γ is a differential 3-form, and its corresponding field strength $\mathcal{H}$ is therefore a differential 4-form. Consequently, the spacetime manifold $\mathcal{M}$ on which the $3BF$ action is defined must be at least four-dimensional. Comparing the definition of the 3-curvature in 3-gauge theory with that of the 2-curvature in 2-gauge theory, we see that $\mathcal{G}$ contains an additional term⁸.

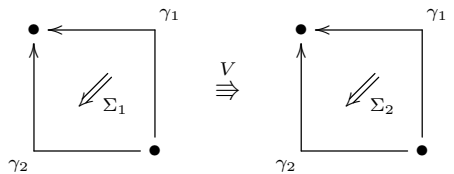
⁷We distinguish between the 3-curvature and the *fake* 3-curvature. The ordinary 3-curvature, the ordered triple (F, G, H) associated with the connection forms $\alpha \in \mathcal{A}^1(\mathfrak{g}, \mathcal{M}_4)$, $\beta \in \mathcal{A}^2(\mathfrak{h}, \mathcal{M}_4)$, and $\gamma \in \mathcal{A}^3(\mathfrak{l}, \mathcal{M}_4)$, is defined in the standard way:

$$\begin{aligned}
 F &= d\alpha + \alpha \wedge \alpha, \\
 G &= d\beta + \alpha \wedge^\triangleright \beta, \\
 H &= d\gamma + \alpha \wedge^\triangleright \gamma.
 \end{aligned} \tag{2.116}$$

The *fake* 3-curvature, the ordered triple $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, contains additional terms and is defined by

$$\begin{aligned}
 \mathcal{F} &= F - \partial\beta, \\
 \mathcal{G} &= G - \delta\gamma, \\
 \mathcal{H} &= H + \beta \wedge^{\{\cdot\}} \beta.
 \end{aligned} \tag{2.117}$$

⁸Consider a volume V :



This cube represents the 3-morphism

$$V : \Sigma_1 \rightarrow \Sigma_2, \tag{2.119}$$

where Σ_1 and Σ_2 are the 2-morphisms shown in the diagram. We wish to calculate

$$\text{hol}(V) : \text{hol}(\Sigma_1) \rightarrow \text{hol}(\Sigma_2). \tag{2.120}$$

The holonomy $\text{hol}(V)$ depends on the 3-connection γ , whereas its source and target depend only on the 2-form β . A relation between the 2-connection β and the 3-connection γ ensures that the source and target of the 3-morphism V are well defined. On the one hand,

$$\text{hol}(\Sigma_1) = \mathcal{P}\exp\left(\int_{\Sigma_1} \beta\right), \quad \text{hol}(\Sigma_2) = \mathcal{P}\exp\left(\int_{\Sigma_2} \beta\right). \tag{2.121}$$

The curvatures may be expanded in bases of the corresponding algebras and differential forms:

$$\mathcal{F} = \frac{1}{2} \mathcal{F}^\alpha_{\mu\nu} \tau_\alpha dx^\mu \wedge dx^\nu, \quad \mathcal{G} = \frac{1}{3!} \mathcal{G}^a_{\mu\nu\rho} t_a dx^\mu \wedge dx^\nu \wedge dx^\rho, \quad \mathcal{H} = \frac{1}{4!} \mathcal{H}^A_{\mu\nu\rho\sigma} T_A dx^\mu \wedge dx^\nu \wedge dx^\rho \wedge dx^\sigma,$$

with coefficients

$$\begin{aligned} \mathcal{F}^\alpha_{\mu\nu} &= \partial_\mu \alpha^\alpha_\nu - \partial_\nu \alpha^\alpha_\mu + f_{\beta\gamma}^\alpha \alpha^\beta_\mu \alpha^\gamma_\nu - \beta^a_{\mu\nu} \partial_a^\alpha, \\ \mathcal{G}^a_{\mu\nu\rho} &= \partial_\mu \beta^a_{\nu\rho} + \partial_\nu \beta^a_{\rho\mu} + \partial_\rho \beta^a_{\mu\nu} + \alpha^\alpha_\mu \beta^b_{\nu\rho} \triangleright_{\alpha b}^a + \alpha^\alpha_\nu \beta^b_{\rho\mu} \triangleright_{\alpha b}^a + \alpha^\alpha_\rho \beta^b_{\mu\nu} \triangleright_{\alpha b}^a - \gamma^A_{\mu\nu\rho} \delta_A^a, \\ \mathcal{H}^A_{\mu\nu\rho\sigma} &= \partial_\mu \gamma^A_{\nu\rho\sigma} - \partial_\nu \gamma^A_{\rho\sigma\mu} + \partial_\rho \gamma^A_{\sigma\mu\nu} - \partial_\sigma \gamma^A_{\mu\nu\rho} \\ &\quad + 2\beta^a_{\mu\nu} \beta^b_{\rho\sigma} X_{\{ab\}}^A - 2\beta^a_{\mu\rho} \beta^b_{\nu\sigma} X_{\{ab\}}^A + 2\beta^a_{\mu\sigma} \beta^b_{\nu\rho} X_{\{ab\}}^A \\ &\quad + \alpha^\alpha_\mu \gamma^B_{\nu\rho\sigma} \triangleright_{\alpha B}^A - \alpha^\alpha_\nu \gamma^B_{\rho\sigma\mu} \triangleright_{\alpha B}^A + \alpha^\alpha_\rho \gamma^B_{\sigma\mu\nu} \triangleright_{\alpha B}^A - \alpha^\alpha_\sigma \gamma^B_{\mu\nu\rho} \triangleright_{\alpha B}^A. \end{aligned}$$

see [17], [18] for further details.

Transformations of the 3-connection and 3-curvature

In category theory, a 3-group gives rise to three types of gauge transformations, generated by the groups G , H , and L . Under a G -gauge transformation, the 3-connection transforms as

$$\alpha' = g\alpha g^{-1} + gdg^{-1}, \quad \beta' = g \triangleright \beta, \quad \gamma' = g \triangleright \gamma, \quad (2.128)$$

where $g : \mathcal{M}_4 \rightarrow G$ is an element of the principal G -bundle over $\mathcal{M}_4$. Under an H -gauge transformation generated by $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, the 3-connection transforms according to

$$\alpha' = \alpha + \partial\eta, \quad \beta' = \beta + d\eta + \alpha' \wedge^\triangleright \eta - \eta \wedge \eta, \quad \gamma' = \gamma - \beta' \wedge^{\{\cdot\}} \eta - \eta \wedge^{\{\cdot\}} \beta. \quad (2.129)$$

Finally, under an L -gauge transformation generated by $\theta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$, the 3-connection transforms as

$$\alpha' = \alpha, \quad \beta' = \beta - \delta\theta, \quad \gamma' = \gamma - d\theta - \alpha \wedge^\triangleright \theta. \quad (2.130)$$

Moreover, a 3-morphism $l : h_1 \rightarrow h_2$ is specified by an element $l \in L$ such that $h_2 = \delta(l)h_1$. Hence the 3-morphism $\text{hol}(V) : \text{hol}(\Sigma_1) \rightarrow \text{hol}(\Sigma_2)$ is specified by an element $l \in L$ satisfying

$$\mathcal{P}\exp\left(\int_{\Sigma_2} \beta\right) = \delta(l) \mathcal{P}\exp\left(\int_{\Sigma_1} \beta\right), \quad (2.122)$$

or

$$\delta(l) = \mathcal{P}\exp\left(\int_{\partial V} \beta\right), \quad (2.123)$$

where the boundary surface of the volume is $\partial V = \Sigma_2 \Sigma_1^{-1}$. For a sufficiently small volume, Stokes' theorem gives

$$\mathcal{P}\exp\left(\int_{\partial V} \beta\right) \approx \exp\left(\int_V G\right). \quad (2.124)$$

On the other hand, the 3-morphism l expressed in terms of the 3-form γ is

$$l \approx \exp\left(\int_V \gamma\right). \quad (2.125)$$

For these expressions to agree, one must have

$$\delta(\exp\left(\int_V \gamma\right)) \approx \exp\left(\int_V G\right), \quad (2.126)$$

and therefore

$$\delta(\gamma) = G = d\beta + \alpha \wedge \beta. \quad (2.127)$$

Theorem 5 *The composition of G -, H -, and L -gauge transformations gives the following transformation of the 3-connection:*

$$\begin{aligned}\tilde{\alpha} &= g\alpha g^{-1} + g\mathrm{d}g^{-1} + \partial(\eta), \\ \tilde{\beta} &= g \triangleright \beta + \mathrm{d}\eta + \tilde{\alpha} \wedge^{\triangleright} \eta - \eta \wedge \eta - \delta(\theta), \\ \tilde{\gamma} &= g \triangleright \gamma - \mathrm{d}\theta - \tilde{\alpha} \wedge \theta - \tilde{\beta} \wedge^{\{\cdot\}} \eta - \eta \wedge^{\{\cdot\}} (g \triangleright \beta) + \eta \wedge^{\triangleright'} \theta,\end{aligned}\tag{2.131}$$

where $g : \mathcal{M}_4 \rightarrow G$, $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, and $\theta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$ are the parameters of the G -, H -, and L -gauge transformations, respectively.

From the transformations of the 3-connection, we obtain the transformation law under 3-gauge transformations for the 3-curvature defined in (2.118).

Theorem 6 *Under a G -gauge transformation, the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ transforms as*

$$\mathcal{F} \rightarrow g\mathcal{F}g^{-1}, \quad \mathcal{G} \rightarrow g \triangleright \mathcal{G}, \quad \mathcal{H} \rightarrow g \triangleright \mathcal{H},\tag{2.132}$$

under an H -gauge transformation as

$$\mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{G} \rightarrow \mathcal{G} + \mathcal{F} \wedge^{\triangleright} \eta, \quad \mathcal{H} \rightarrow \mathcal{H} - \mathcal{G}' \wedge^{\{\cdot\}} \eta + \eta \wedge^{\{\cdot\}} \mathcal{G},\tag{2.133}$$

and under an L -gauge transformation as

$$\mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{G} \rightarrow \mathcal{G}, \quad \mathcal{H} \rightarrow \mathcal{H} - \mathcal{F} \wedge^{\triangleright} \theta,\tag{2.134}$$

where $g : \mathcal{M}_4 \rightarrow G$, $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, and $\theta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$ are the parameters of the G -, H -, and L -gauge transformations, respectively.

For further details and proofs of these theorems, see [18] and Appendix A.

Chapter 3

Hamiltonian analysis

The Hamiltonian analysis of a theory is the necessary first step in the canonical quantization procedure formulated by Paul Dirac for constrained Hamiltonian systems [31]. This procedure allows us to formulate a quantum theory for systems possessing gauge symmetry. The approach may be divided into two main steps.

1. First, one must perform the Hamiltonian analysis of the system. Its outcome is the algebra of first- and second-class constraints present in the theory. First-class constraints generate unphysical transformations of the dynamical variables, namely gauge transformations, which do not change the physical state of the system. The first-class constraints F and second-class constraints S together define a subspace Γ^* of the phase space Γ , whose dimension is

$$2n = 2N - (2F + S), \quad (3.1)$$

and in which the dynamics of the independent degrees of freedom takes place.

This chapter presents the basic ideas of Dirac's method. It then gives a brief overview of a systematic procedure for constructing the *generator of gauge transformations* from a known Hamiltonian structure, namely the *Castellani procedure*.

2. The transition from the classical theory to a quantum theory with gauge degrees of freedom in the coordinate representation proceeds as follows:

- The dynamical coordinate variables and their canonical momenta become operators:

$$q(x) \rightarrow \hat{q}(x),$$

$$\pi(x) \rightarrow \hat{\pi}(x);$$

- The Poisson bracket is then used to introduce the Dirac bracket $\{_, _\}_D$, thereby eliminating the second-class constraints, if any, from the theory;
- The Dirac bracket of the operators q and π , i.e. the *Heisenberg algebra*, becomes a commutator:

$$\{q(x), \pi(x)\} \rightarrow -i[\hat{q}(x), \hat{\pi}(x)];$$

- The first-class constraints become conditions on physical states:

$$\hat{\Phi} |\psi\rangle = 0.$$

These conditions are analogous to the Gupta–Bleuler conditions in quantum electrodynamics.

3.1 Lagrangian and Hamiltonian formalisms

This section provides a brief review of the Lagrangian and Hamiltonian formalisms. Let n be the *number of degrees of freedom* of a physical system, that is, the minimum number of quantities required to describe completely the positions of all particles, or equivalently the configuration of the system. The Lagrangian function, or *Lagrangian*, is a function of the form

$$L(q_1, \dots, q_N, \dot{q}_1, \dots, \dot{q}_N, t), \quad (3.2)$$

where q_i and $\dot{q}_i$, $i = 1, \dots, N$, are the generalized coordinates and their time derivatives. The *Hamiltonian action* is defined by

$$S = \int_{t_1}^{t_2} dt L(q_1, \dots, q_N, \dot{q}_1, \dots, \dot{q}_N, t). \quad (3.3)$$

If only potential forces act on the physical system under consideration and the Lagrangian contains only first time derivatives of the generalized coordinates, the *Euler–Lagrange equations of motion* take the form

$$\frac{\delta S}{\delta q_i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \quad i = 1, \dots, N, \quad (3.4)$$

where $\frac{\delta S}{\delta q_i}$ is the functional derivative of the action with respect to the generalized coordinate q_i . The equations of motion follow from *Hamilton's principle of least action*. The Euler–Lagrange equations may be written as

$$\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \ddot{q}_j = \frac{\partial L}{\partial q_i} - \frac{\partial^2 L}{\partial \dot{q}_i \partial q_j} \dot{q}_j, \quad i = 1, \dots, N, \quad (3.5)$$

where the matrix $\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j}$ is the Hessian of the Lagrangian L . If the Hessian is invertible, i.e.

$$\det H^{ij} = \det \left(\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \right) \neq 0, \quad (3.6)$$

then, provided that the conditions for the existence and uniqueness of solutions to the system of differential equations are satisfied and initial conditions are specified, equations (3.5) can be solved for the accelerations $\ddot{q}_i$. Such theories are called *nonsingular*. A theory is said to be *singular* if condition (3.6) is not satisfied, that is, if

$$\det H^{ij} = 0. \quad (3.7)$$

The *canonical momenta* are defined by

$$p^i = \pi(q_i) = \frac{\delta L}{\delta \dot{q}_i}, \quad i = 1, \dots, N. \quad (3.8)$$

We can now reformulate the definition of a nonsingular theory. Since the Hessian may be expressed as $\frac{\partial p^i}{\partial \dot{q}_j}$, condition (3.6) can be understood as the requirement that the velocities $\dot{q}_i$ be expressible as functions of the generalized coordinates and their canonical momenta.

For a nonsingular theory, we define the *Hamiltonian*,

$$H = \sum_i p^i \dot{q}_i - L. \quad (3.9)$$

The equations of motion in the Hamiltonian formalism are

$$\dot{p}^i = \frac{\partial H}{\partial q_i}, \quad \dot{q}_i = -\frac{\partial H}{\partial p^i}, \quad i = 1, \dots, N. \quad (3.10)$$

The *Poisson bracket* of the functions $f = f(q_1, \dots, q_N, \dot{q}_1, \dots, \dot{q}_N, t)$ and $g = g(q_1, \dots, q_N, \dot{q}_1, \dots, \dot{q}_N, t)$ is defined by

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p^i} - \frac{\partial g}{\partial p^i} \frac{\partial f}{\partial q_i} \right). \quad (3.11)$$

Using the Poisson bracket, we define the time evolution of an arbitrary function f on phase space by

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}, \quad (3.12)$$

so that Hamilton's equations of motion may be written as

$$\dot{p}^i = \{p^i, H\}, \quad \dot{q}_i = \{q_i, H\}, \quad i = 1, \dots, N. \quad (3.13)$$

The definitions given in this section generalize directly to a theory with infinitely many degrees of freedom. For a theory describing a field ϕ^α on spacetime with coordinates x^μ , the *Lagrangian density* $\mathcal{L}$ is a function of the form

$$L = \int_{\Sigma_3} d^3x \mathcal{L}(\phi^\alpha, \partial_\mu \phi^\alpha), \quad (3.14)$$

and the *action* is defined by

$$S = \int d^4x \mathcal{L}(\phi^\alpha, \partial_\mu \phi^\alpha). \quad (3.15)$$

In this case, the *canonical momenta* are functions of the coordinates,

$$p^\alpha(x) = \pi(\phi_\alpha) = \frac{\delta L}{\delta \dot{\phi}_\alpha(x)}, \quad (3.16)$$

while the *Hamiltonian* is

$$H = \int_{\Sigma_3} d^3x \left[\sum_\alpha p^\alpha(x) \dot{\phi}_\alpha(x) \right] - L. \quad (3.17)$$

This section has provided a brief review of the Lagrangian and Hamiltonian formalisms, with our attention restricted entirely to nonsingular theories, i.e. those satisfying condition (3.6). In the following section we consider *singular theories*.

3.2 Constrained systems

Singular theories are characterized by redundant degrees of freedom in the physical system, namely by the presence of additional, unphysical variables in the action, so that the total number of variables in the theory exceeds the number required to describe the physics of the system. The analysis of such physical systems requires the *Hamiltonian analysis of constrained systems*, or the *Dirac procedure*.

3.2.1 Dirac theory

Primary constraints

If relations (3.8) cannot be inverted to express the velocities as functions of the generalized coordinates and their conjugate momenta, i.e. if condition (3.7) is satisfied, the theory is singular. All physically relevant theories, including the Standard Model of elementary particles and General Relativity, belong to this class. *Singular theories* possess a set of relations among the variables and momenta, called *primary constraints*,

$$P_m(q, p) \approx 0, \quad m = 1, \dots, P, \quad (3.18)$$

where P is the total number of primary constraints in the theory and the symbol $\approx$ denotes weak equality¹. Equation (3.8) yields the primary constraints in the form

$$P(q_i) \equiv \pi(q_i) - \frac{\delta L}{\delta \dot{q}_i}. \quad (3.19)$$

The primary constraints determine the subspace Γ_1 of the full phase space Γ in which the dynamics of the system takes place. The primary constraints P_m need not all be mutually independent. If the number of independent primary constraints is $P' \leq P$, then the dimension of the phase-space subspace is $2N - P'$. The Hessian $\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j}$ is then a matrix of rank $N - P'$.

Canonical and total Hamiltonians

We define the *canonical Hamiltonian*

$$H_c = \sum_{i=1}^N p^i \dot{q}_i - L, \quad (3.20)$$

which is well defined on the surface determined by the primary constraints. For a singular theory, we define the *total Hamiltonian* on the entire phase space Γ by

$$H_T = H_c + \sum_m \lambda^m P_m, \quad (3.21)$$

where $\lambda^m(q)$ are arbitrary Lagrange multipliers. The equations of motion for the canonical variables and their momenta obtained from the total Hamiltonian are

$$\dot{p}_i = -\frac{\partial H_T}{\partial q_i}, \quad \dot{q}_i = \frac{\partial H_T}{\partial p_i}, \quad i = 1, \dots, N, \quad (3.22)$$

which, upon using equation (3.21), become

$$\dot{p}_i = -\frac{\partial H_c}{\partial q_i} - \lambda^m \frac{\partial P_m}{\partial q_i}, \quad \dot{q}_i = \frac{\partial H_c}{\partial p_i} + \lambda^m \frac{\partial P_m}{\partial p_i}, \quad i = 1, \dots, N. \quad (3.23)$$

¹Let $F(q, p)$ be a function defined and differentiable in a neighbourhood $\mathcal{O} \subseteq \Gamma$ containing the subspace Γ_1 . If $F(q, p)$ vanishes on Γ_1 , then F is said to be *weakly equal* to zero:

$$F(q, p) \approx 0 \quad \Leftrightarrow \quad F(q, p)|_{\Gamma_1} = 0.$$

If F and all its first derivatives vanish on Γ_1 , then it is said to be *strongly equal* to zero:

$$F(q, p) = 0 \quad \Leftrightarrow \quad F(q, p)|_{\Gamma_1} = \frac{\partial F(q, p)}{\partial q}|_{\Gamma_1} = \frac{\partial F(q, p)}{\partial p}|_{\Gamma_1} = 0.$$

If $A(q, p)$ is an arbitrary dynamical variable in the theory, its equation of motion is

$$\dot{A} = \frac{\partial A}{\partial q_i} \dot{q}_i + \frac{\partial A}{\partial p_i} \dot{p}_i, \quad (3.24)$$

and equations (3.22) therefore give

$$\dot{A} = \frac{\partial A}{\partial q_i} \frac{\partial H_T}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial H_T}{\partial q_i} = \{A, H_T\}. \quad (3.25)$$

Consistency conditions and secondary constraints

The *consistency conditions* for the primary constraints require them to be preserved under the dynamical evolution of the system:

$$\dot{P}(q_i) \equiv \{P(q_i), H_T\} \approx 0. \quad (3.26)$$

The consistency conditions lead to one of three possible outcomes.

1. The Poisson bracket in (3.26) is a linear combination of primary constraints and is therefore weakly equal to zero; the consistency condition is then satisfied automatically.
2. The consistency conditions yield *secondary constraints* in the theory:

$$\mathcal{S}(q_i) = \{P(q_i), H_T\} \approx 0. \quad (3.27)$$

The resulting secondary constraints must themselves satisfy *consistency conditions*:

$$\dot{\mathcal{S}}(q_i) \equiv \{\mathcal{S}(q_i), H_T\} \approx 0. \quad (3.28)$$

The consistency conditions for the secondary constraints may produce new constraints in the theory, namely *tertiary constraints*. One must then require that the tertiary constraints also be preserved, i.e. that they satisfy their consistency conditions, again leading to one of the outcomes listed here. The process continues until the consistency conditions cease to generate new constraints.

3. Finally, the consistency conditions may determine some of the Lagrange multipliers λ^m .

Together, the P primary constraints $P(q, p)$ and the K new constraints $\mathcal{S}(q, p)$ (secondary, tertiary, and so forth) obtained from the consistency conditions constitute the complete set of constraints in the theory:

$$\phi_m(q, p) \approx 0, \quad m = 1, \dots, P, P+1, \dots, P+K. \quad (3.29)$$

It can be shown that a function $F(q, p)$ on phase space is weakly equal to zero if and only if it is a linear combination of constraints:

$$F \approx 0 \quad \Leftrightarrow \quad F = \sum_m \lambda^m \phi_m.$$

Using the definition of the total Hamiltonian, the consistency conditions for all constraints in the theory become

$$\{\phi_m, H_c\} + \sum_{m'} \lambda^{m'} \{\phi_m, \phi_{m'}\} \approx 0. \quad (3.30)$$

Let L be the number of consistency conditions that are not satisfied automatically, i.e. those imposing restrictions on the Lagrange multipliers. The resulting system of L differential equations (3.30) can then be solved for λ^m :

$$\lambda^m = U^m + \sum_a v^a(t) V_a^m, \quad (3.31)$$

where $v^a(t)$ are arbitrary functions of time, V_a^m form the homogeneous part,

$$\sum_{m'} \sum_a v^a(t) V_a^{m'} \{ \phi_m, \phi_{m'} \} \approx 0, \quad (3.32)$$

and U^m is a particular solution of the system of differential equations. We therefore conclude that the total Hamiltonian contains arbitrary functions of time. Finally, because the functions $v^a(t)$ are arbitrary, the following relation holds:

$$\sum_{m'} V_a^{m'} \{ \phi_m, \phi_{m'} \} \approx 0. \quad (3.33)$$

First- and second-class constraints

A dynamical variable $R(q, p)$ is *first class* if its Poisson brackets with all constraints in the theory vanish weakly:

$$\{R(q, p), \phi_m(q, p)\} \approx 0.$$

Like any function that is weakly equal to zero, this Poisson bracket is a linear combination of constraints. It follows from this definition that the total Hamiltonian H_T is a first-class quantity. If a variable $R(q, p)$ is not first class, it is *second class*. Although the distinction between primary and secondary constraints is of little significance for the Hamiltonian analysis, the distinction between first- and second-class constraints is essential to the dynamical interpretation of constraints within Hamiltonian theory. First-class constraints are defined by

$$\{\Phi(q, p), \phi_m(q, p)\} \approx 0, \quad (3.34)$$

whereas second-class constraints satisfy

$$\{\chi(q, p), \phi_m(q, p)\} \not\approx 0, \quad (3.35)$$

where $\phi_m(q, p)$ denotes all constraints in the theory, primary and secondary (and tertiary, if present, and so forth).

At the end of the preceding subsection, we concluded that the total Hamiltonian contains arbitrary functions of time $v^a(t)$. The presence of these arbitrary functions in the Hamiltonian, and hence in the equations of motion and their solutions, means that the variables $(q(t), p(t))$ cannot be determined uniquely from given initial conditions $(q(t=0), p(t=0))$. If the variables $(q(t), p(t))$ cannot be determined uniquely at each instant, they have no direct physical interpretation and are therefore not physical variables of the theory. Let f be a variable in the theory. Its dynamical evolution is

$$\begin{aligned} f(\delta t) &= f(t=0) + \delta t \dot{f} \\ &= f(t=0) + \delta t \{f, H_T\} \\ &= f(t=0) + \delta t \{f, H_c\} + \delta t \sum_m U^m \{f, \phi_m\} + \delta t \sum_m \sum_a v^a(t) V_a^m \{f, \phi_m\} \\ &= f(t=0) + \delta t \{f, H'\} + \delta t \sum_a v^a(t) \{f, \phi_a\}, \end{aligned} \quad (3.36)$$

where $H' = H_c + \sum_m U^m \phi_m$ and $\phi_a = \sum_m V_a^m \phi_m$. Since the total Hamiltonian is first class and relation (3.33) holds, the quantities H' and ϕ_a are also first class, i.e. their Poisson brackets with all constraints in the theory vanish weakly. The last term in (3.36) contains arbitrary functions of time. Consequently, the function

$$f_1(\delta t) = f(t=0) + \delta t \{f, H'\},$$

and the function

$$f_2(\delta t) = f(t=0) + \delta t \{f, H'\} + \delta t \sum_a v^a(t) \{f, \phi_a\},$$

represent the same physical state. The transformation

$$\delta f(\delta t) = \delta t \sum_a v^a(t) \{f, \phi_a\} = \sum_a \epsilon^a(t) \{f, \phi_a\}, \quad (3.37)$$

is unphysical. We therefore conclude that the primary first-class constraints ϕ_a generate unphysical transformations of the dynamical variables which leave the physical state of the system unchanged, namely *gauge transformations*.

Since both the successive application and the difference of two unphysical transformations are again unphysical transformations,

$$\delta_1(\delta_2 f) - \delta_2(\delta_1 f) = \sum_{a,b} \epsilon_1^b(t) \epsilon_2^a(t) (\{f, \phi_a\}, \phi_b) - \{f, \phi_b\}, \phi_a) = \sum_{a,b} \epsilon_1^b(t) \epsilon_2^a(t) \{f, \{\phi_a, \phi_b\}\},$$

we conclude that the Poisson bracket of two primary first-class constraints, $\{\phi_a, \phi_b\}$, also generates unphysical transformations. This Poisson bracket must be a linear combination of constraints and may therefore be expected to contain secondary constraints as well. These secondary constraints are first class, because the Poisson bracket of first-class constraints is itself a first-class constraint. Hence some secondary first-class constraints are expected to generate unphysical transformations in the theory; in the next subsection we demonstrate that this is true of all first-class constraints.

The physical information about the system at an arbitrary instant t can be obtained from functions defined on the subspace Γ' of the full phase space Γ determined by the constraints of the theory, that is, from the *physical observables* of the theory.

Second-class constraints and physical observables in the theory

Let Φ_f , $f = 1, \dots, F$, be the first-class constraints of the theory and χ_s , $s = 1, \dots, S$, its second-class constraints. Since the constraints χ_s are second class, the matrix $\Delta_{rs} = \{\chi_r, \chi_s\}$ is nonsingular and therefore invertible. Every odd-dimensional antisymmetric matrix has vanishing determinant; since Δ_{rs} is both antisymmetric and invertible, the number of second-class constraints must be even.

We can define a new bracket, the *Dirac bracket*,

$$\{f, g\}_D = \{f, g\} - \{f, \chi_r\} \Delta_{rs}^{-1} \{\chi_s, g\}, \quad (3.38)$$

which, like the Poisson bracket, is an antisymmetric bilinear map. It follows directly from the definition that the Dirac bracket of any second-class constraint with an arbitrary variable vanishes:

$$\{\chi_p, g\}_D = \{\chi_p, g\} - \{\chi_p, \chi_r\} \Delta_{rs}^{-1} \{\chi_s, g\} = \{\chi_p, g\} - \Delta_{pr} \Delta_{rs}^{-1} \{\chi_s, g\} = \{\chi_p, g\} - \delta_{ps} \{\chi_s, g\} = 0. \quad (3.39)$$

Thus, after the Dirac bracket has been introduced, the second-class constraints become strongly equal to zero, and the equation of motion for an arbitrary variable g in the theory is

$$\dot{g} \approx \{g, H_T\}_D. \quad (3.40)$$

The distinction between first- and second-class constraints was defined using the Poisson bracket in equations (3.34) and (3.35). Once the Dirac bracket is introduced, the Poisson bracket is no longer required: the second-class constraints become strong equalities and the entire theory may be formulated in terms of Dirac brackets. As we have seen, the presence of second-class constraints means that the theory contains dynamical degrees of freedom that are not relevant. These variables are eliminated by defining the Dirac bracket solely in terms of the physically relevant degrees of freedom. In practice, however, this construction can be extremely difficult in theories with a large number of second-class constraints.

Number of degrees of freedom

In the general case of N initial fields, F independent first-class constraints, and S independent second-class constraints, the *number of local propagating degrees of freedom* is

$$n = N - F - \frac{S}{2}. \quad (3.41)$$

Equation (3.41) follows from the fact that the presence of S second-class constraints is equivalent to the elimination of $S/2$ canonical coordinates and $S/2$ of their momenta. The presence of F first-class constraints in the theory is equivalent to the elimination of F canonical coordinates; moreover, because first-class constraints generate gauge symmetries, one may impose F gauge-fixing conditions for the corresponding F canonical momenta. There are therefore $2N - 2F - S$ independent canonical coordinates and momenta, so that $2n = 2N - 2F - S$, which gives equation (3.41).

3.2.2 Generator of gauge transformations

In this subsection we present an algorithm for constructing the generator of gauge symmetries in a theory, known as the *Castellani procedure*.

Consider a system specified by the total Hamiltonian H_T and the complete set of constraints ϕ_m , where $m = 1, \dots, M, M+1, \dots, M+K$. Let $(q(t), p(t))$ be a trajectory in phase space parameterized by t , whose initial point $(q(t=0), p(t=0))$ lies on the hypersurface determined by the constraints. The equations of motion obtained from the total Hamiltonian, (3.22), do not have the same form as those obtained using (3.10), but their difference is unphysical. Using $H_T = H' + \sum_a v^a(t)\phi_a$, one obtains

$$\dot{p}_i = -\frac{\partial H'}{\partial q_i} - v^a(t)\frac{\partial \phi_a}{\partial q_i}, \quad \dot{q}_i = \frac{\partial H'}{\partial p_i} + v^a(t)\frac{\partial \phi_a}{\partial p_i}, \quad i = 1, \dots, N, \quad (3.42)$$

whereas the equations of motion for $v^a(t)$ are

$$\phi_a(q, p) = 0. \quad (3.43)$$

Now consider a new trajectory beginning at the same point $(q(t=0), p(t=0))$ on the hypersurface determined by the constraints, such that $(q(t) + \delta_0 q(t), p(t) + \delta_0 p(t))$ also satisfies the equations of motion for the function $v^a(t) + \delta_0 v^a(t)$:

$$\begin{aligned} \delta_0 \dot{p}_i &= -\sum_{j=1}^N \left(\delta_0 q_j \frac{\partial}{\partial q_j} + \delta_0 p_j \frac{\partial}{\partial p_j} \right) \frac{\partial H_T}{\partial q_i} - \delta_0 v^a(t) \frac{\partial \phi_a}{\partial q_i}, \\ \delta_0 \dot{q}_i &= \sum_{j=1}^N \left(\delta_0 q_j \frac{\partial}{\partial q_j} + \delta_0 p_j \frac{\partial}{\partial p_j} \right) \frac{\partial H_T}{\partial p_i} + \delta_0 v^a(t) \frac{\partial \phi_a}{\partial p_i}, \\ \sum_{j=1}^N \left(\delta_0 q_j \frac{\partial}{\partial q_j} + \delta_0 p_j \frac{\partial}{\partial p_j} \right) \phi_a(q, p) &= 0. \end{aligned} \quad (3.44)$$

The difference between these two trajectories at a fixed instant t is an unphysical transformation.

On the other hand, the variations of the forms under gauge transformations are determined by the generator:

$$\begin{aligned}\delta_0 q_i &= \epsilon(t) \{ q_i, G \} = \epsilon(t) \frac{\partial G}{\partial p_i}, \\ \delta_0 p_i &= \epsilon(t) \{ p_i, G \} = -\epsilon(t) \frac{\partial G}{\partial q_i},\end{aligned}\tag{3.45}$$

where the definition of the Poisson bracket has been used. It follows that

$$\begin{aligned}\delta_0 \dot{q}_i &= \dot{\epsilon}(t) \frac{\partial G}{\partial p_i} + \epsilon(t) \left\{ \frac{\partial G}{\partial p_i}, H_T \right\}, \\ \delta_0 \dot{p}_i &= -\dot{\epsilon}(t) \frac{\partial G}{\partial q_i} - \epsilon(t) \left\{ \frac{\partial G}{\partial q_i}, H_T \right\}.\end{aligned}\tag{3.46}$$

Comparing equations (3.44) and (3.46), we obtain

$$\begin{aligned}\sum_{j=1}^N \left(\delta_0 q_j \frac{\partial}{\partial q_j} + \delta_0 p_j \frac{\partial}{\partial p_j} \right) \frac{\partial H_T}{\partial p_i} + \delta_0 v^a(t) \frac{\partial \phi_a}{\partial p_i} &= \dot{\epsilon}(t) \frac{\partial G}{\partial p_i} + \epsilon(t) \left\{ \frac{\partial G}{\partial p_i}, H_T \right\}, \\ \sum_{j=1}^N \left(\delta_0 q_j \frac{\partial}{\partial q_j} + \delta_0 p_j \frac{\partial}{\partial p_j} \right) \frac{\partial H_T}{\partial q_i} + \delta_0 v^a(t) \frac{\partial \phi_a}{\partial q_i} &= \dot{\epsilon}(t) \frac{\partial G}{\partial q_i} + \epsilon(t) \left\{ \frac{\partial G}{\partial q_i}, H_T \right\},\end{aligned}\tag{3.47}$$

that is,

$$\begin{aligned}\epsilon(t) \left\{ \frac{\partial H_T}{\partial p_i}, G \right\} + \delta_0 v^a(t) \frac{\partial \phi_a}{\partial p_i} &= \dot{\epsilon}(t) \frac{\partial G}{\partial p_i} + \epsilon(t) \left\{ \frac{\partial G}{\partial p_i}, H_T \right\}, \\ \epsilon(t) \left\{ \frac{\partial H_T}{\partial q_i}, G \right\} + \delta_0 v^a(t) \frac{\partial \phi_a}{\partial q_i} &= \dot{\epsilon}(t) \frac{\partial G}{\partial q_i} + \epsilon(t) \left\{ \frac{\partial G}{\partial q_i}, H_T \right\}, \\ \epsilon(t) \{ \phi_a(q, p), G \} &= 0.\end{aligned}\tag{3.48}$$

These equations may be rewritten as

$$\begin{aligned}\frac{\partial}{\partial p_i} \left(\dot{\epsilon}(t) G + \epsilon(t) \{ H_T, G \} - v^a(t) \phi_a \right) &= 0, \\ \frac{\partial}{\partial q_i} \left(\dot{\epsilon}(t) G + \epsilon(t) \{ H_T, G \} - v^a(t) \phi_a \right) &= 0, \\ \epsilon(t) \{ \phi_a(q, p), G \} &= 0,\end{aligned}\tag{3.49}$$

and hence, for any dynamical variable $F(q, p)$,

$$\{ F(q, p), \dot{\epsilon}(t) G + \epsilon(t) \{ H_T, G \} - v^a(t) \phi_a \} = 0.\tag{3.50}$$

It follows that $\dot{\epsilon}(t) G + \epsilon(t) \{ H_T, G \} - v^a(t) \phi_a$ is a trivial function, equal to zero or to some power of the constraints, and consequently

$$G = \alpha^a \phi_a, \quad \{ G, H_T \} = 0.\tag{3.51}$$

In the general case, the generators of gauge transformations take the form

$$G = \sum_{n=0}^k \epsilon^{(n)} G_n,\tag{3.52}$$

where $\epsilon^{(n)}$ denotes the n th time derivative of the parameter ϵ , $\epsilon^{(n)} = \frac{\partial^n \epsilon}{\partial t^n}$. The generators G_n , where $n = 1, \dots, k$, are determined recursively by the *Castellani procedure* and must satisfy the relations

$$\begin{aligned} G_k &= C_{PFC}, \\ &\vdots \\ G_1 + \{G_2, H_T\} &= C_{PFC}, \\ G_0 + \{G_1, H_T\} &= C_{PFC}, \\ \{G_0, H_T\} &= C_{PFC}, \end{aligned} \tag{3.53}$$

where C_{PFC} denotes a primary first-class constraint. The total number k of generators is equal to the number of generations of secondary constraints. If the theory contains only primary and secondary constraints, then $k = 1$, and the generator takes the form

$$G = \epsilon(t)G_0 + \dot{\epsilon}(t)G_1. \tag{3.54}$$

In this case, the recursive algorithm of the Castellani procedure described by relations (3.53) reduces to

$$\begin{aligned} G_1 &= C_{PFC}, \\ G_0 + \{G_1, H_T\} &= C_{PFC}, \\ \{G_0, H_T\} &= C_{PFC}, \end{aligned} \tag{3.55}$$

where C_{PFC} denotes a primary first-class constraint.

The variation of an arbitrary variable A defined on phase space under a symmetry transformation is calculated using

$$\delta_0 A = \{A, G\}. \tag{3.56}$$

Chapter 4

BF theory

This chapter presents BF theory for a general Lie group on an n -dimensional manifold. As a topological theory, BF theory has no local propagating degrees of freedom. Physically relevant systems are therefore obtained by imposing suitable constraints on the topological BF action, thereby defining a constrained BF theory. The best-known example of this construction is the Plebanski action [32]. Alternatively, the symmetry of the BF action may be broken by adding a symmetry-breaking term, as in the MacDowell–Mansouri theory [33].

The covariant quantization procedure applied to different classical BF actions yields *topological quantum field theories* in the sense that they satisfy the Atiyah axioms. It has produced numerous *spin-foam models* of quantum gravity, ranging from the Ponzano–Regge model of three-dimensional gravity [5] to the currently most sophisticated *EPRL/FK* model in four spacetime dimensions [9], [10]. A limitation shared by these spin-foam models is that they describe pure gravity without matter fields. This is a consequence of the fact that the tetrad fields do not appear explicitly in the BF action; instead, they emerge from the classical equations of motion and are therefore on-shell variables. Consequently, matter cannot be introduced at the quantum level within this BF framework. For further information on BF theory, see [32], [34], [35].

The chapter is organized as follows. Section 4.1 gives a brief review of topological BF theory. Subsection 4.1.1 presents its Hamiltonian analysis and resulting canonical structure, including the first- and second-class constraints and their algebra. The same subsection derives the Bianchi identity obeyed by the first-class constraints, which reduces the number of independent first-class constraints, and uses these results to count the physical degrees of freedom. The gauge generator and the variations of all variables and their canonical momenta are also presented in Subsection 4.1.1; the Castellani procedure used to obtain the generator is provided in Appendix D.1. These results are then used in Subsection 4.1.2 to determine all gauge symmetries of BF theory, namely the groups of G - and M -gauge transformations. We assemble these results into the complete symmetry-group structure, including its Lie algebra, and exhibit a particular choice of parameters that yields diffeomorphism transformations. Finally, Section 4.2 discusses Yang–Mills theory in flat spacetime as a constrained topological BF theory, while Section 4.3 presents the well-known Plebanski action for general relativity.

4.1 Topological BF theory

Given a Lie group G , its corresponding Lie algebra $\mathfrak{g}$, and a four-dimensional spacetime manifold $\mathcal{M}_4$, the topological BF action is defined by

$$S_{BF} = \int_{\mathcal{M}_4} \langle B \wedge F \rangle_{\mathfrak{g}}. \quad (4.1)$$

Here F is the curvature 2-form of the $\mathfrak{g}$ -valued connection 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$, as defined in equation (2.1); $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$ is a Lagrange-multiplier 2-form; and $\langle _, _ \rangle_{\mathfrak{g}}$ denotes a G -invariant symmetric nondegenerate bilinear form. The action (4.1) is manifestly diffeomorphism invariant and invariant under gauge transformations generated by G . Varying the action (4.1) with respect to B^β and α^β gives the equations of motion

$$F^\beta = 0, \quad \nabla B^\beta \equiv dB^\beta + f_{\gamma\delta}{}^\beta \alpha^\gamma \wedge B^\delta = 0. \quad (4.2)$$

Here β is a G -group index labelling the generators of $\mathfrak{g}$, and $f_{\gamma\delta}{}^\beta$ are the structure constants of G .

The first equation in (4.2) implies that the connection α is flat, i.e. $\alpha = 0$ up to gauge transformations. Substituting this result into the second equation shows that the Lagrange multiplier B is constant. Thus the action (4.1) describes a theory without local propagating degrees of freedom, namely a *topological theory*¹.

The number of degrees of freedom is obtained formally through the Hamiltonian analysis of the action, carried out in Section 4.1.1.

4.1.1 Hamiltonian analysis of topological BF theory

Topological BF theory is defined by the action (4.1). Expanding it in spacetime indices $\mu, \nu, \dots$ and group indices $\alpha, \beta, \dots$ of G gives

$$S_{BF} = \int_{\mathcal{M}_4} d^4x \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} B^\alpha{}_{\mu\nu} F^\beta{}_{\rho\sigma} g_{\alpha\beta}. \quad (4.7)$$

Assuming that spacetime is globally hyperbolic, we may foliate it by spatial hypersurfaces Σ_3 and write the Lagrangian of BF theory as

$$L_{3BF} = \int_{\Sigma_3} d^3\vec{x} \frac{1}{4} \epsilon^{\mu\nu\rho\sigma} B^\alpha{}_{\mu\nu} F^\beta{}_{\rho\sigma} g_{\alpha\beta}. \quad (4.8)$$

¹A *topological quantum field theory* is a quantum field theory concerned with the calculation of topological invariants. In a topological theory, the path integral is

$$Z^I = \int \mathcal{D}\phi \exp[iS_{TOP}[\phi]], \quad (4.3)$$

where the action is a constant $\chi(\mathcal{M}_D)$ depending only on the topology:

$$S_{TOP}[\phi] = \int_{\mathcal{M}_D} d^Dx \mathcal{L}^{TOP}(\phi, \partial\phi) = \chi(\mathcal{M}_D). \quad (4.4)$$

Therefore the constant $\exp[i\chi]$ factors out of the path integral, giving

$$Z^I = \exp[i\chi(\mathcal{M}_D)] \int \mathcal{D}\phi = \text{const} \cdot \exp[i\chi(\mathcal{M}_D)]. \quad (4.5)$$

We call this path integral the first quantization. The second quantization, i.e. the path integral Z^{II} of the discretized theory, is obtained by summing over all triangulations of the manifold. The discretized theory is *topological* if the partition function Z^I is independent of the triangulation. In that case the first and second quantizations coincide:

$$Z^{II} = \sum_{T(\mathcal{M}_D)} Z^I = Z^I \sum_{T(\mathcal{M}_D)} 1 = \text{const} \cdot Z^I, \quad (4.6)$$

Here Z^I factors out of the sum because it is triangulation independent. Since the triangulation fixes the number of degrees of freedom of the discretized theory while physical observables do not depend on it, the theory has no physical degrees of freedom.

The *canonical momenta*, defined in (3.16), conjugate to the variables $B^\alpha_{\mu\nu}$ and α^α are obtained by varying the Lagrangian with respect to their time derivatives:

$$\begin{aligned}\pi(B)_\alpha^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 B^\alpha_{\mu\nu}} = 0, \\ \pi(\alpha)_\alpha^\mu &= \frac{\delta L}{\delta \partial_0 \alpha^\alpha_\mu} = \frac{1}{2} \epsilon^{0\mu\nu\rho} B_{\alpha\nu\rho}.\end{aligned}\tag{4.9}$$

Since these equations cannot be solved for the time derivatives, they define the *primary constraints* (3.19):

$$\begin{aligned}P(B)_\alpha^{\mu\nu} &\equiv \pi(B)_\alpha^{\mu\nu} \approx 0, \\ P(\alpha)_\alpha^\mu &\equiv \pi(\alpha)_\alpha^\mu - \frac{1}{2} \epsilon^{0\mu\nu\rho} B_{\alpha\nu\rho} \approx 0.\end{aligned}\tag{4.10}$$

Using the fundamental Poisson brackets

$$\begin{aligned}\{B^\alpha_{\mu\nu}(\vec{x}), \pi(B)_{\beta}^{\rho\sigma}(\vec{y})\} &= 2\delta^\alpha_\beta \delta^\rho_{[\mu} \delta^\sigma_{\nu]} \delta^{(3)}(\vec{x} - \vec{y}), \\ \{\alpha^\alpha_\mu(\vec{x}), \pi(\alpha)_\beta^\nu(\vec{y})\} &= \delta^\alpha_\beta \delta^\nu_\mu \delta^{(3)}(\vec{x} - \vec{y}),\end{aligned}\tag{4.11}$$

we calculate the Poisson bracket between the primary constraints:

$$\{P(B)_\alpha^{jk}(\vec{x}), P(\alpha)_\beta^i(\vec{y})\} = \epsilon^{0ijk} g_{\alpha\beta} \delta^{(3)}(\vec{x} - \vec{y}).\tag{4.12}$$

The *canonical Hamiltonian*, defined in (3.20), is

$$H_c = \int_{\Sigma_3} d^3\vec{x} \left[\frac{1}{2} \pi(B)_\alpha^{\mu\nu} \partial_0 B^\alpha_{\mu\nu} + \pi(\alpha)_\alpha^\mu \partial_0 \alpha^\alpha_\mu \right] - L.\tag{4.13}$$

Using the definition of the curvature $F^\alpha_{\mu\nu}$, we may rewrite (4.13) so that the time derivatives multiply primary constraints and therefore vanish identically on shell:

$$H_c = - \int_{\Sigma_3} d^3\vec{x} \epsilon^{0ijk} \left[\frac{1}{2} B_{\alpha 0i} F^\alpha_{jk} + \frac{1}{2} \alpha^\alpha_0 \nabla_i B_{\alpha jk} \right].\tag{4.14}$$

In this form, the canonical Hamiltonian does not depend on the canonical momenta and contains only the fields and their time derivatives. Introducing Lagrange multipliers λ for the two primary constraints, we define the off-shell *total Hamiltonian* of (3.21):

$$H_T = H_c + \int_{\Sigma_3} d^3\vec{x} \left[\frac{1}{2} \lambda(B)_\alpha^{\mu\nu} P(B)_\alpha^{\mu\nu} + \lambda(\alpha)_\alpha^\mu P(\alpha)_\alpha^\mu \right].\tag{4.15}$$

Because the primary constraints must be preserved in time, the *consistency conditions* (3.26) must hold for all of them. For the primary constraints $P(B)_\alpha^{0i}$ and $P(\alpha)_\alpha^0$, these conditions produce the secondary constraints (3.27) of the theory, $\mathcal{S}$:

$$\begin{aligned}\mathcal{S}(F)_\alpha^i &\equiv \frac{1}{2} \epsilon^{0ijk} F_{\alpha jk} \approx 0, \\ \mathcal{S}(\nabla B)_\alpha &\equiv \frac{1}{2} \epsilon^{0ijk} \nabla_{[i} B_{\alpha j]k} \approx 0,\end{aligned}\tag{4.16}$$

whereas for the primary constraints $P(\alpha)_\alpha{}^k$ and $P(B)_\alpha{}^{jk}$, the consistency conditions determine the Lagrange multipliers:

$$\begin{aligned}\lambda(B)_{\alpha ij} &\approx \nabla_i B_{\alpha 0j} - \nabla_j B_{\alpha 0i} + g_{\alpha\gamma}{}^\beta \alpha_{\beta 0} B^\gamma{}_{ij}, \\ \lambda(\alpha)^\alpha{}_i &\approx \nabla_i \alpha^\alpha{}_0.\end{aligned}\tag{4.17}$$

The Lagrange multipliers

$$\lambda(B)^\alpha{}_{0i}, \quad \lambda(\alpha)^\alpha{}_0,\tag{4.18}$$

remain undetermined. The *consistency conditions for the secondary constraints* (3.28) produce no further constraints:

$$\begin{aligned}\{\mathcal{S}(F)^{\alpha i}, H_T\} &= f_{\beta\gamma}{}^\alpha \mathcal{S}(F)^{\beta i} \alpha^\gamma{}_0 \approx 0, \\ \{\mathcal{S}(\nabla B)_\alpha, H_T\} &= -f_{\beta\gamma\alpha} B^\gamma{}_{0k} \mathcal{S}(F)^{\beta k} + f_{\beta\alpha}{}^\gamma \alpha^\beta{}_0 \mathcal{S}(\nabla B)_\gamma \approx 0.\end{aligned}\tag{4.19}$$

Substituting the expressions for the Lagrange multipliers, the total Hamiltonian reduces to

$$H_T = \int_{\Sigma_3} d^3\vec{x} \left[\lambda(B)_{\alpha 0i} \Phi(B)^{\alpha i} + \lambda(\alpha)_\alpha \Phi(\alpha)^\alpha - B_{\alpha 0i} \Phi(F)^{\alpha i} - \alpha_{\alpha 0} \Phi(\nabla B)^\alpha \right],\tag{4.20}$$

where the *first-class constraints* (3.34) are

$$\begin{aligned}\Phi(B)^{\alpha i} &= P(B)^{\alpha 0i}, \\ \Phi(\alpha)^\alpha &= P(\alpha)^\alpha{}_0, \\ \Phi(F)^{\alpha i} &= \mathcal{S}(F)^{\alpha i} - \nabla_j P(B)^{\alpha ij}, \\ \Phi(\nabla B)^\alpha &= \mathcal{S}(\nabla B)^\alpha + \nabla_i P(\alpha)^{\alpha i} - \frac{1}{2} f_{\gamma\beta}{}^\alpha B^\beta{}_{ij} P(B)^{\gamma ij},\end{aligned}\tag{4.21}$$

and the *second-class constraints* (3.35) are

$$\chi(B)_\alpha{}^{jk} = P(B)_\alpha{}^{jk} \quad \chi(\alpha)_\alpha{}^i = P(\alpha)_\alpha{}^i.\tag{4.22}$$

The nonvanishing Poisson brackets among the first-class constraints are

$$\begin{aligned}\{\Phi(F)^\alpha{}_i(\vec{x}), \Phi(\nabla B)_\beta(\vec{y})\} &= f_{\beta\gamma}{}^\alpha \Phi(F)^\gamma{}_i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{\Phi(\nabla B)_\alpha(\vec{x}), \Phi(\nabla B)_\beta(\vec{y})\} &= f_{\alpha\beta}{}^\gamma \Phi(\nabla B)_\gamma(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}),\end{aligned}\tag{4.23}$$

while the Poisson brackets between first- and second-class constraints give

$$\begin{aligned}\{\Phi(F)^{\alpha i}(\vec{x}), \chi(\alpha)_\beta{}^j(\vec{y})\} &= -f_{\beta\gamma}{}^\alpha \chi(B)^{\gamma ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{\Phi(\nabla B)^\alpha(\vec{x}), \chi(\alpha)_\beta{}^i(\vec{y})\} &= f_{\beta\gamma}{}^\alpha \chi(\alpha)^{\gamma i}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{\Phi(\nabla B)^\alpha(\vec{x}), \chi(B)_\beta{}^{ij}(\vec{y})\} &= -f_{\beta\gamma}{}^\alpha \chi(B)^{\gamma ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}).\end{aligned}\tag{4.24}$$

The Poisson brackets of the first-class constraints with the Hamiltonian are

$$\begin{aligned}
\{\Phi(B)_\alpha{}^i, H_T\} &= \Phi(F)_\alpha{}^i, \\
\{\Phi(\alpha)_\alpha, H_T\} &= \Phi(\nabla B)_\alpha, \\
\{\Phi(F)^\alpha{}_i, H_T\} &= -\alpha^\beta{}_0 f_{\beta\gamma}{}^\alpha \Phi(F)^\gamma{}_i, \\
\{\Phi(\nabla B)_\alpha, H_T\} &= B_{\beta 0 i} f_{\alpha\gamma}{}^\beta \Phi(F)^{\gamma i} - \alpha^\beta{}_0 f_{\alpha\beta}{}^\gamma \Phi(\nabla B)_\gamma,
\end{aligned} \tag{4.25}$$

These results will later be used to construct the generator of the gauge symmetries of BF theory and determine its full symmetry group.

Degrees of freedom of topological BF theory

Determining the number of independent components of the primary constraints requires the Bianchi identities.

Theorem 7 (BI for BF theory) *The Bianchi identity (BI) for the connection 1-form α and its curvature 2-form*

$$F^\alpha = d\alpha^\alpha + f_{\beta\gamma}{}^\alpha \alpha^\beta \wedge \alpha^\gamma, \tag{4.26}$$

is

$$\epsilon^{\lambda\mu\nu\rho} \nabla_\mu F^\alpha{}_{\nu\rho} = 0. \tag{4.27}$$

$\alpha^\alpha{}_\mu$	$B^\alpha{}_{\mu\nu}$
$4p$	$6p$

Table 4.1: Total number of initial fields in BF theory.

Counting the initial fields shown in Table 4.1, we find $N = 10p$, where p is the dimension of G . Similarly, Table 4.2 gives the number of second-class constraints, $S = 6p$. The first-class

$\chi(B)_\alpha{}^{jk}$	$\chi(\alpha)_\alpha{}^i$
$3p$	$3p$

Table 4.2: Total number of second-class constraints in BF theory.

constraints are not all independent; they satisfy

$$\nabla_i \Phi(F)_\alpha{}^i = \epsilon^{ijk} \nabla_i F_{\alpha jk}, \tag{4.28}$$

which becomes

$$\nabla_i \Phi(F)_\alpha{}^i = 0, \tag{4.29}$$

because the right-hand side of (4.28), $\epsilon^{ijk} \nabla_i F_{jk}^a = 0$, is the $\lambda = 0$ component of the Bianchi identity (4.27).

Taking this relation among the first-class constraints into account, their number of independent components is read from Table 4.3. Thus the number of independent first-class constraints is

$$F = 8p - p = 7p.$$

$\Phi(B)_\alpha^i$	$\Phi(\alpha)_\alpha$	$\Phi(F)^{\alpha i}$	$\Phi(\nabla B)^\alpha$
$3p$	p	$3p - p$	p

Table 4.3: Total number of first-class constraints in BF theory.

Here the total number of first-class constraints listed in Table 4.3 has been reduced by the p relations (4.29). Therefore the number of degrees of freedom in BF theory, defined by (3.41), is

$$n = 10p - 7p - \frac{6p}{2} = 0, \quad (4.30)$$

so BF theory has no local propagating degrees of freedom.

Gauge generator for BF theory

The gauge generator of BF theory is

$$G = \int \nabla_0 \epsilon^\alpha_i \Phi(B)_\alpha^i - \epsilon^\alpha_i \Phi(F)_\alpha^i + \nabla_0 \epsilon^\alpha \Phi(\alpha)_\alpha + \epsilon^\alpha (f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} - \Phi(\nabla B)_\alpha). \quad (4.31)$$

Its derivation by the Castellani procedure is given in Appendix D.1. The variations of the form variables and their conjugate momenta follow from equation (3.56):

$$\begin{aligned} \delta_0 B^\alpha_{0i} &= \nabla_0 \epsilon^\alpha_i - f_{\beta\gamma}{}^\alpha \epsilon^\beta B^\gamma_{0i}, & \delta_0 \pi(B)_\alpha^{0i} &= -f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{0i}, \\ \delta_0 B^\alpha_{ij} &= 2\nabla_{[i} \epsilon^\alpha_{j]} - f_{\beta\gamma}{}^\alpha \epsilon^\beta B^\gamma_{ij}, & \delta_0 \pi(B)_\alpha^{ij} &= -f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{ij}, \\ \delta_0 \alpha^\alpha_0 &= \nabla_0 \epsilon^\alpha, & \delta_0 \pi(\alpha)_\alpha^0 &= -f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{0i} - f_{\alpha\beta}{}^\gamma \epsilon^\gamma \pi(\alpha)_\gamma, \\ \delta_0 \alpha^\alpha_i &= \nabla_i \epsilon^\alpha, & \delta_0 \pi(\alpha)_\alpha^i &= -f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{ij} - f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(\alpha)_\gamma^i + \epsilon^{0ijk} \nabla_j \epsilon_{\alpha k}. \end{aligned} \quad (4.32)$$

4.1.2 Symmetries of the BF action

The G symmetry group

Inspecting the variations (4.32), we see that some terms correspond to the transformation defined in Theorem 8,

$$\delta_0 \alpha^\alpha_\mu = \partial_\mu \epsilon_\mathfrak{g}^\alpha + f_{\beta\gamma}{}^\alpha \epsilon^\beta_\mu \epsilon_\mathfrak{g}^\gamma, \quad \delta_0 B^\alpha_{\mu\nu} = -f_{\beta\gamma}{}^\alpha \epsilon_\mathfrak{g}^\beta B^\gamma_{\mu\nu}, \quad (4.33)$$

where the free parameter is $\epsilon_\mathfrak{g}^\alpha = \epsilon^\alpha$.

Theorem 8 (G -gauge transformations) *In BF theory based on an arbitrary Lie group G , the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \text{Ad}_g \alpha + g dg^{-1}, & B &\rightarrow B' = g B g^{-1}, \\ \beta &\rightarrow \beta' = g \triangleright \beta, & C &\rightarrow C, \end{aligned} \quad (4.34)$$

where $g = \exp(\epsilon_\mathfrak{g} \cdot \hat{G}) = \exp(\epsilon_\mathfrak{g} \hat{G}^\alpha) \in G$, and $\epsilon_\mathfrak{g} : \mathcal{M}_4 \rightarrow \mathfrak{g}$ is the transformation parameter.

Proof. Under the G -gauge transformations of Theorem 8, with parameter $g \in G$, the curvature F transforms as

$$\mathcal{F} \rightarrow \mathcal{F}' = g \mathcal{F} g^{-1}. \quad (4.35)$$

The theorem follows by direct calculation:

$$\begin{aligned} S_{BF} \rightarrow S'_{BF} &= \frac{1}{4} \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} (B^\alpha_{\mu\nu} - f_{\gamma\delta}{}^\alpha \epsilon^\gamma_{\mathfrak{g}} B^\delta_{\mu\nu}) (F^\beta_{\rho\sigma} - f_{\epsilon\tau}{}^\beta \epsilon^\epsilon_{\mathfrak{g}} F^\tau_{\rho\sigma}) g_{\alpha\beta} \\ &= S_{BF} - \frac{1}{4} \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} (f_{\gamma\delta}{}^\alpha \epsilon^\gamma_{\mathfrak{g}} B^\delta_{\mu\nu} F^\beta_{\rho\sigma} + f_{\epsilon\tau}{}^\beta \epsilon^\epsilon_{\mathfrak{g}} B^\alpha_{\mu\nu} F^\tau_{\rho\sigma}) g_{\alpha\beta}, \end{aligned} \quad (4.36)$$

where the second term vanishes. Equivalently, for a finite transformation,

$$\langle B, F \rangle_{\mathfrak{g}} \rightarrow \langle g^{-1}Bg, g^{-1}Fg \rangle_{\mathfrak{g}} = \langle B, F \rangle_{\mathfrak{g}}, \quad (4.37)$$

because the Killing form $\langle _, _ \rangle_{\mathfrak{g}}$ is G -invariant. ■

Consider two infinitesimal G -gauge transformations with parameters $\epsilon_{\mathfrak{g}}^{\alpha_1}$ and $\epsilon_{\mathfrak{g}}^{\beta_2}$. To calculate the commutator of their generators, we use the Baker–Campbell–Hausdorff (BCH) formula for small transformation parameters:

$$e^{\epsilon_{\mathfrak{g}}^{\alpha_1} \hat{G}_\alpha} e^{\epsilon_{\mathfrak{g}}^{\beta_2} \hat{G}_\beta} = e^{\epsilon_{\mathfrak{g}}^{\alpha_1} \hat{G}_\alpha + \epsilon_{\mathfrak{g}}^{\beta_2} \hat{G}_\beta + \frac{1}{2} \epsilon_{\mathfrak{g}}^{\alpha_1} \epsilon_{\mathfrak{g}}^{\beta_2} [\hat{G}_\alpha, \hat{G}_\beta] + O(\epsilon_{\mathfrak{g}}^3)}, \quad (4.38)$$

which implies

$$e^{\epsilon_{\mathfrak{g}}^{\alpha_1} \hat{G}_\alpha} e^{\epsilon_{\mathfrak{g}}^{\beta_2} \hat{G}_\beta} - e^{\epsilon_{\mathfrak{g}}^{\beta_2} \hat{G}_\beta} e^{\epsilon_{\mathfrak{g}}^{\alpha_1} \hat{G}_\alpha} = \epsilon_{\mathfrak{g}}^{\alpha_1} \epsilon_{\mathfrak{g}}^{\beta_2} [\hat{G}_\alpha, \hat{G}_\beta] + O(\epsilon_{\mathfrak{g}}^3). \quad (4.39)$$

Using (4.39), we find that the generators of the G -gauge transformations defined in Theorem 8 satisfy

$$[\hat{G}_\alpha, \hat{G}_\beta] = f_{\alpha\beta}{}^\gamma \hat{G}_\gamma, \quad (4.40)$$

where $f_{\alpha\beta}{}^\gamma$ are the structure constants of $\mathfrak{g}$. Since the generators are isomorphic, $\hat{G}_\alpha \cong \tau_\alpha$, the G -gauge-transformation group of Theorem 8 is precisely the group G associated with the BF action.

The $\tilde{M}$ symmetry group

The remaining terms in (4.32), of the form $\delta_0 B^\alpha_{0i} = \nabla_0 \epsilon_i$ and $\delta_0 B^\alpha_{ij} = 2\nabla_{[i} \epsilon_{j]}$, correspond to the transformation defined in Theorem 9:

$$\delta_0 \alpha^\alpha_\mu = 0, \quad \delta_0 B^\alpha_{\mu\nu} = 2\nabla_{[\mu} \epsilon_{\nu]}^\alpha, \quad (4.41)$$

where the free parameter is $\epsilon_{\mathfrak{m}\mu}^\alpha = \epsilon_\mu^\alpha$.

Theorem 9 (M -gauge transformations) *In BF theory based on an arbitrary Lie group G , the following transformation is a symmetry:*

$$\alpha \rightarrow \alpha' = \alpha, \quad B \rightarrow B' = B + \nabla \epsilon_{\mathfrak{m}}, \quad (4.42)$$

where $\epsilon_{\mathfrak{m}\mu}^\alpha$ is an arbitrary $\mathfrak{g}$ -valued 1-form, and ∇ is the covariant exterior derivative,

$$\nabla \epsilon_{\mathfrak{m}} = d\epsilon_{\mathfrak{m}} + [\alpha \wedge \epsilon_{\mathfrak{m}}]. \quad (4.43)$$

Proof. The theorem follows by direct calculation:

$$\begin{aligned} S_{BF} \rightarrow S'_{BF} &= \frac{1}{4} \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} (B^\alpha_{\mu\nu} + 2\partial_{[\mu} \epsilon_{\nu]}^\alpha + 2f_{\gamma\delta}{}^\alpha \alpha^\gamma_{[\mu} \epsilon_{\nu]}^\delta) F^\beta_{\rho\sigma} g_{\alpha\beta} \\ &= S_{BF} - \frac{1}{2} \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\rho\sigma\nu} (\partial_\mu F^\tau_{\rho\sigma} + f_{\gamma\beta}{}^\tau \alpha^\gamma_\mu F^\beta_{\rho\sigma}) \epsilon_{\mathfrak{m}\nu}^\delta g_{\tau\delta} + \text{boundary term}, \end{aligned} \quad (4.44)$$

where the second term vanishes by the Bianchi identity (7), since $\epsilon^{\mu\rho\sigma\nu} (\partial_\mu F^\tau_{\rho\sigma} + f_{\gamma\beta}^\tau \alpha^\gamma_\mu F^\beta_{\rho\sigma}) = 0$. ■ The transformations of Theorem 9 are linear, and two successive M -gauge transformations give a single M -gauge transformation with parameter $\epsilon_{\mathfrak{m}1} + \epsilon_{\mathfrak{m}2}$. Denoting their generators by $\hat{M}_\alpha^\mu$,

$$e^{\epsilon_{\mathfrak{m}1} \cdot \hat{M}} e^{\epsilon_{\mathfrak{m}2} \cdot \hat{M}} = e^{(\epsilon_{\mathfrak{m}1} + \epsilon_{\mathfrak{m}2}) \cdot \hat{M}}, \quad (4.45)$$

where $\epsilon_{\mathfrak{m}} \cdot \hat{M} = \epsilon_{\mathfrak{m}\mu}^\alpha \hat{M}_\alpha^\mu$, it follows that their commutator is trivial:

$$[\hat{M}_\alpha^\mu, \hat{M}_\beta^\nu] = 0. \quad (4.46)$$

Thus the M -gauge transformations form an Abelian group, denoted below by $\tilde{M}$. The index structure of its parameters and generators shows that it is isomorphic to $\mathbb{R}^{4p}$, where p is the dimension of G :

$$\tilde{M} \cong \mathbb{R}^{4p}. \quad (4.47)$$

We next examine the relation between the M -gauge transformations and the G -gauge transformations defined above²,

$$[\epsilon_{\mathfrak{g}} \cdot \hat{G}, \epsilon_{\mathfrak{m}} \cdot \hat{M}] = (\epsilon_{\mathfrak{g}} \triangleright \epsilon_{\mathfrak{m}}) \cdot \hat{M}, \quad (4.48)$$

which gives the commutator

$$[\hat{G}_\alpha, \hat{M}_\beta^\mu] = f_{\alpha\beta}^\gamma \hat{M}_\gamma^\mu. \quad (4.49)$$

This completes the calculation of the algebra of gauge-transformation generators in BF theory.

Full gauge symmetry group of the BF action

Combining the preceding results, the algebra of generators of the full gauge symmetry group is as follows.

- The algebra $\mathfrak{g}$ of G is given by

$$[\hat{G}_\alpha, \hat{G}_\beta] = f_{\alpha\beta}^\gamma \hat{G}_\gamma. \quad (4.50)$$

- The algebra of the M -gauge generators is

$$[\hat{M}_\alpha^\mu, \hat{M}_\beta^\nu] = 0. \quad (4.51)$$

- The action of the generators of G on the M -gauge generators is

$$[\hat{G}_\alpha, \hat{M}_\beta^\mu] = f_{\alpha\beta}^\gamma \hat{M}_\gamma^\mu. \quad (4.52)$$

The commutation relations (4.51) show that $\tilde{M}$ is an *invariant subgroup*³ of the full symmetry group $\mathcal{G}_{BF}$. The relations (4.52) then show that the semidirect product of G and $\tilde{M}$ is the full symmetry group of the BF action:

$$\mathcal{G}_{BF} = G \ltimes \tilde{M}. \quad (4.53)$$

²The action of $\epsilon_{\mathfrak{g}}$ on $\epsilon_{\mathfrak{m}}$ is defined by $\epsilon_{\mathfrak{g}} \triangleright \epsilon_{\mathfrak{m}} \equiv \triangleright_{\alpha\beta}^\gamma \epsilon_{\mathfrak{g}}^\alpha \epsilon_{\mathfrak{m}}^\beta dx^\mu$, where $\triangleright_{\alpha\beta}^\gamma = f_{\alpha\beta}^\gamma$. Hence

$$(\epsilon_{\mathfrak{g}} \triangleright \epsilon_{\mathfrak{m}}) \cdot \hat{M} = f_{\alpha\beta}^\gamma \epsilon_{\mathfrak{g}}^\alpha \epsilon_{\mathfrak{m}}^\beta \hat{M}_\gamma^\mu.$$

³A subgroup is an *invariant subgroup*, equivalently a *normal subgroup*, if it is invariant under conjugation by elements of the full group. Thus H is a normal subgroup of G if $H \leq G$ and, for every $h \in H$ and $g \in G$, there exists $h' \in H$ such that $ghg^{-1} = h'$. At the algebraic level, the corresponding notion is an *ideal*. An algebra A is a subalgebra of L if $[A, A] \subset A$, and a subalgebra A is an ideal of L if $[A, L] \subset A$.

Diffeomorphisms

Another important question is the diffeomorphism invariance of the theory. Since the BF action is written in a manifestly covariant form using differential forms, diffeomorphisms are clearly symmetries. From the structure of the gauge group $\mathcal{G}_{BF}$, however, it is not immediately apparent whether $\text{Diff}(\mathcal{M}_4, \mathbb{R})$ is one of its subgroups.

Consider a diffeomorphism

$$x^\mu \rightarrow x'^\mu = x^\mu + \xi^\mu(x), \quad (4.54)$$

where $\xi^\mu(x)$ is an arbitrary function of the coordinates, and denote the gauge-symmetry parameters by $\epsilon_i(x)$. If diffeomorphisms are symmetries of the action, then for every field $\phi(x)$ and every diffeomorphism parameter $\xi^\mu(x)$ there exists a choice of gauge parameters $\epsilon_i(x)$ and Henneaux–Teitelboim parameters⁴ $\epsilon^{\text{HT}}(x)$ such that

$$(\delta_0^{\text{gauge}} + \delta_0^{\text{HT}} + \delta_0^{\text{diff}}) \phi = 0. \quad (4.57)$$

In other words, if diffeomorphisms are symmetries of the theory, their form variations can be expressed as the sum of the gauge and HT form variations:

$$\delta_0^{\text{diff}} \phi = -\delta_0^{\text{gauge}} \phi - \delta_0^{\text{HT}} \phi. \quad (4.58)$$

Specifically, the BF action depends on $\alpha^\alpha{}_\mu$ and $B^\alpha{}_{\mu\nu}$. The HT parameters $\epsilon^{\text{HT}\alpha\beta}{}_{\mu\nu\rho}$ are defined by (4.55):

$$\delta_0^{\text{HT}} \alpha^\alpha{}_\mu = \frac{1}{2} \epsilon^{\text{HT}\alpha\beta}{}_{\mu\nu\rho} \frac{\delta S}{\delta B^\beta{}_{\nu\rho}}, \quad \delta_0^{\text{HT}} B^\alpha{}_{\mu\nu} = -\epsilon^{\text{HT}\alpha\beta}{}_{\rho\mu\nu} \frac{\delta S}{\delta \alpha^\beta{}_\rho}, \quad (4.59)$$

whereas the gauge parameters ϵ_g^α and $\epsilon_m^\alpha{}_\mu$ are defined in Theorems 8 and 9. A choice of these parameters exists for which (4.57) holds for every field. In particular, choose

$$\epsilon_g^\alpha = -\xi^\lambda \alpha^\alpha{}_\lambda, \quad \epsilon_m^\alpha{}_\mu = \xi^\lambda B^\alpha{}_{\mu\lambda}, \quad (4.60)$$

and the HT parameters as

$$\epsilon^{\text{HT}\alpha\beta}{}_{\mu\nu\rho} = \xi^\lambda g^{\alpha\beta} \epsilon_{\mu\nu\rho\lambda}, \quad (4.61)$$

Then equation (4.58) gives precisely the standard form variations under diffeomorphisms:

$$\begin{aligned} \delta_0^{\text{diff}} \alpha^\alpha{}_\mu &= -\partial_\mu \xi^\lambda \alpha^\alpha{}_\lambda - \xi^\lambda \partial_\lambda \alpha^\alpha{}_\mu, \\ \delta_0^{\text{diff}} B^\alpha{}_{\mu\nu} &= -\partial_\mu \xi^\lambda B^\alpha{}_{\lambda\nu} - \partial_\nu \xi^\lambda B^\alpha{}_{\mu\lambda} - \xi^\lambda \partial_\lambda B^\alpha{}_{\mu\nu}. \end{aligned} \quad (4.62)$$

This establishes that diffeomorphisms are indeed symmetries of the theory. Although they are not contained in the full gauge symmetry group $\mathcal{G}_{BF}$ itself, they are contained in the direct product of the full gauge group and the HT symmetry group.

⁴Any action depending on at least two fields $\phi_1(x)$ and $\phi_2(x)$ is invariant under the Henneaux–Teitelboim (HT) transformation [36] with parameter ϵ^{HT} ,

$$\delta_0^{\text{HT}} \phi_1 = \epsilon^{\text{HT}}(x) \frac{\delta S}{\delta \phi_2}, \quad \delta_0^{\text{HT}} \phi_2 = -\epsilon^{\text{HT}}(x) \frac{\delta S}{\delta \phi_1}, \quad (4.55)$$

as follows directly by varying the action:

$$\delta^{\text{HT}} S[\phi_1, \phi_2] = \frac{\delta S}{\delta \phi_1} \delta_0^{\text{HT}} \phi_1 + \frac{\delta S}{\delta \phi_2} \delta_0^{\text{HT}} \phi_2 = 0. \quad (4.56)$$

This symmetry is present even in theories without gauge symmetry, but it does not appear in the gauge generator (4.31) obtained by the Castellani procedure. The reason is that the HT variations of ϕ_1 and ϕ_2 always vanish on shell: they are linear combinations of constraints and hence vanish weakly.

4.2 Yang–Mills theory

In physics we are usually interested in nontopological theories, namely theories with local propagating degrees of freedom. A topological BF action can be converted into such a theory by adding a further term that imposes a *simplicity constraint*. The resulting action defines a *constrained BF theory*.

One example is *Yang–Mills theory* with gauge group $SU(N)$ in Minkowski spacetime, which may be written as the following constrained BF action:

$$S = \int B_I \wedge F^I + \lambda^I \wedge \left(B_I - \frac{12}{g} M_{abI} \delta^a \wedge \delta^b \right) + \zeta^{abI} \left(M_{abI} \varepsilon_{cdef} \delta^c \wedge \delta^d \wedge \delta^e \wedge \delta^f - g_{IJ} F^J \wedge \delta_a \wedge \delta_b \right). \quad (4.63)$$

Here $F \equiv dA + A \wedge A$ is the curvature 2-form of the connection 1-form $A \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{su}(N))$, while $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{su}(N))$ is a Lagrange-multiplier 2-form. The Killing form $g_{IJ} \equiv \langle \tau_I, \tau_J \rangle_{\mathfrak{su}(N)} \propto f_{IK}{}^L f_{JL}{}^K$ raises and lowers the indices $I, J, \dots$ labelling the generators of $SU(N)$, and $f_{IJ}{}^K$ are the structure constants of $\mathfrak{su}(N)$. The action (4.63) is obtained by imposing constraints on the topological action (4.1), through two additional terms consisting of the products of Lagrange multipliers—the 2-form λ^I and the 0-form ζ^{abI} —with their corresponding constraints. The function, or 0-form, M_{abI} is also a Lagrange multiplier, and g is the Yang–Mills coupling constant. Finally, δ^a is a nondynamical 1-form for which there exists a global coordinate system in which its components equal the Kronecker delta $\delta^a{}_\mu$. It is therefore a background field that defines the global spacetime metric

$$\eta_{\mu\nu} = \eta_{ab} \delta^a{}_\mu \delta^b{}_\nu, \quad (4.64)$$

where $\eta_{ab} \equiv \text{diag}(-1, +1, +1, +1)$ is the Minkowski metric. Hence the spacetime manifold $\mathcal{M}_4$ is flat. The indices $a, b, \dots$ are local Lorentz indices taking values $0, \dots, 3$. The field δ^a has all the properties of the tetrad 1-form e^a in flat Minkowski spacetime. The action (4.63) is manifestly diffeomorphism invariant and invariant under $SU(N)$ gauge transformations, but it is not background independent because of the field δ^a . Varying the action (4.63) with respect to ζ^{abI} , M_{abI} , A^I , B_I , and λ^I gives

$$M_{abI} \varepsilon_{cdef} \delta^c \wedge \delta^d \wedge \delta^e \wedge \delta^f - F_I \wedge \delta_a \wedge \delta_b = 0, \quad (4.65)$$

$$-\frac{12}{g} \lambda^I \wedge \delta^a \wedge \delta^b + \zeta^{abI} \varepsilon_{cdef} \delta^c \wedge \delta^d \wedge \delta^e \wedge \delta^f = 0, \quad (4.66)$$

$$-dB_I + f_{JI}{}^K B_K \wedge A^J + d(\zeta^{abI} \delta_a \wedge \delta_b) - f_{JI}{}^K \zeta^{abI} \delta_a \wedge \delta_b \wedge A^J = 0, \quad (4.67)$$

$$F_I + \lambda_I = 0, \quad (4.68)$$

$$B_I - \frac{12}{g} M_{abI} \delta^a \wedge \delta^b = 0. \quad (4.69)$$

The background field δ^a is not varied. Equations (4.65), (4.66), (4.68), and (4.69) express the Lagrange multipliers as algebraic functions of the field strength $F^I{}_{\mu\nu}$ of the dynamical field A^I :

$$\begin{aligned} M_{abI} &= \frac{1}{48} \varepsilon_{abcd} F^I{}^{cd}, & \zeta^{abI} &= \frac{1}{4g} \varepsilon^{abcd} F^I{}_{cd}, \\ \lambda_{Iab} &= F_{Iab}, & B_{Iab} &= \frac{1}{2g} \varepsilon_{abcd} F^I{}^{cd}. \end{aligned} \quad (4.70)$$

Here $F_{Iab} = F_{I\mu\nu}\delta_a^\mu\delta_b^\nu$, with analogous notation for all other variables, and δ^a_μ is assumed to be invertible. Equations (4.70) together with the differential equation (4.67) yield the equation of motion for the gauge field A^I_μ :

$$\nabla_\rho F^{I\rho\mu} \equiv \partial_\rho F^{I\rho\mu} + f_{JK}^I A^J_\rho F^{K\rho\mu} = 0. \quad (4.71)$$

This is precisely the classical equation of motion of a free Yang–Mills field. The action (4.63) can be modified to describe a *massive vector field*, governed by the Proca equation, by adding the mass term

$$-\frac{1}{4!}m^2 A_{I\mu} A^I_\nu \eta^{\mu\nu} \varepsilon_{abcd} \delta^a \wedge \delta^b \wedge \delta^c \wedge \delta^d. \quad (4.72)$$

This term explicitly breaks the $SU(N)$ gauge symmetry of the action.

4.3 The Plebanski action for general relativity

The best-known constrained BF theory is the *Plebanski action for general relativity* [32], [34]. In 1977, Plebanski showed that the gravitational field may be described using B , rather than the metric, as the fundamental variable.

Choosing the Lorentz group $G = SO(3, 1)$ as the gauge group, the BF action is

$$S = \int_{\mathcal{M}_4} B_{ab} \wedge R^{ab}. \quad (4.73)$$

Here R^{ab} is the curvature 2-form of the spin connection ω^{ab} , B_{ab} is a Lagrange-multiplier 2-form, and the Killing form is

$$g_{ab,cd} = \frac{1}{2}(\eta_{ac}\eta_{bd} - \eta_{ad}\eta_{bc}),$$

where η_{ab} is the Minkowski metric. The antisymmetric pair of indices labels the generators J_{ab} of the Lorentz group $SO(3, 1)$. Adding the appropriate constraints gives the constrained BF action known as the *Plebanski action*:

$$S = \int_{\mathcal{M}_4} B_{ab} \wedge R^{ab} + \frac{1}{2}\phi_{abcd} B^{ab} \wedge B^{cd} + \mu\phi_{abcd} (a_1 g^{ab,cd} + a_2 \varepsilon^{abcd}). \quad (4.74)$$

Here ϕ_{abcd} is a Lagrange-multiplier 0-form imposing the constraint on $B^{ab} \wedge B^{cd}$, μ is a Lagrange-multiplier 4-form, $a_1, a_2 \in \mathbb{R}$ are real parameters, and $g^{ab,cd}$ is the inverse Killing form. Varying this action with respect to B_{ab} , ω^{ab} , ϕ_{abcd} , and μ gives

$$R_{ab} - \phi_{abcd} B^{cd} = 0, \quad (4.75)$$

$$\nabla B_{ab} = 0, \quad (4.76)$$

$$\frac{1}{2}B^{ab} \wedge B^{cd} + \mu (a_1 g^{ab,cd} + a_2 \varepsilon^{abcd}) = 0, \quad (4.77)$$

$$\phi_{abcd} (a_1 g^{ab,cd} + a_2 \varepsilon^{abcd}) = 0. \quad (4.78)$$

Solving equation (4.77) shows that the Lagrange multiplier has the form

$$B_{ab} = \frac{\alpha}{2} \varepsilon_{abcd} e^c \wedge e^d + \beta e_a \wedge e_b, \quad (4.79)$$

where α and β satisfy

$$a_2 \alpha \beta = \frac{a_1}{4} (\alpha^2 - \beta^2).$$

For the standard choice $a_1 = 0$ and $a_2 = 1$ used in the modern literature, there are two possible solutions:

$$B_{ab} = \frac{\alpha}{2} \varepsilon_{abcd} e^c \wedge e^d, \quad \beta = 0, \quad (4.80)$$

$$B_{ab} = \beta e_a \wedge e_b, \quad \alpha = 0. \quad (4.81)$$

Both solutions therefore imply that the differential form B_{ab} is *simple*⁵. Accordingly, the constraint

$$\frac{1}{2} B^{ab} \wedge B^{cd} + \mu (a_1 g^{ab,cd} + a_2 \epsilon^{abcd}) ,$$

leading to these solutions is called the *simplicity constraint*.

Solving the system (4.75)–(4.78) shows that equation (4.75) is equivalent to the vacuum Einstein equation of general relativity:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0. \quad (4.82)$$

In this model, the tetrad fields are not explicitly present; they appear only as solutions of the equations of motion. The tetrads are therefore on-shell objects and cannot themselves be quantized. This makes the Plebanski model unsuitable for coupling matter fields to gravity [11], [15], [37]. Nevertheless, the model has been successfully quantized as a formulation of general relativity within the spin-foam approach [1], [2], [9], [10].

⁵A differential 2-form is *simple*, or *decomposable*, if it can be written as the exterior product of two 1-forms.

Chapter 5

$2BF$ theory

To address the difficulty of quantizing the Standard Model matter fields coupled to gravity within the Plebanski model, a new research direction has been developed: generalized BF models in the context of category theory [12]; see [15], [22], [23], [37], [38]. The first step in this categorical generalization, the so-called *categorical ladder*, is the generalization of the notion of a group to that of a 2-group. This approach is based on the idea that gauge symmetries in physics may be described not only by Lie groups but also by other objects. Generalizing a BF theory defined for a Lie group to a theory defined for a general semistrict 2-group yields a $2BF$ theory, also known as a *BFCG theory* [12], [13], [17], [39].

In the spin-foam quantization framework, higher category theory has been successfully applied to formulate a quantum-gravity model based on the Poincaré 2-group [39] and the corresponding $2BF$ action, the so-called *spincube model of quantum gravity*. Since tetrads appear in the topological sector of the spincube model as fundamental fields of the $2BF$ action, this model could in principle be extended to include all matter fields of the Standard Model. At the quantum level, however, the actions describing matter fields must likewise be written in a form adapted to spin-foam quantization. As it turns out, this requires one further step of categorical generalization: the formulation of a $3BF$ theory, discussed in the next chapter.

In this chapter we first define and analyze the symmetries of the topological $2BF$ action in Section 5.1, following a presentation similar to Chapter 4. Section 5.1 gives a brief overview of topological $2BF$ theory. Subsection 5.1.1 contains its Hamiltonian analysis and the resulting canonical structure, an analysis of the Bianchi identities obeyed by the first-class constraints and reducing their number of independent components, and a count of the physical degrees of freedom. As expected, the analysis shows that $2BF$ theory is topological, with no local propagating degrees of freedom. The end of Subsection 5.1.1 gives the final gauge generator and the form variations of all variables and their canonical momenta, while the Castellani procedure used to obtain the generator is presented in Appendix D.2.2.

In Subsection 5.1.2, the resulting form variations are used to derive the finite transformations of all gauge symmetries of $2BF$ theory. Subsection 5.1.2 is divided into four parts. We first discuss the gauge group G and the corresponding G -gauge transformations. The second part presents the group $\tilde{M}$, or the M -gauge transformations; the third analyzes the group $\tilde{H}$ formed by the H -gauge transformations already known in the literature; and the fourth analyzes the group $\tilde{N}$ and the N -gauge transformations that also arise in the theory. The commutators of their generators are stated here, with calculational details in Appendix D.2.3. Combining these results gives the complete structure of the gauge-symmetry group of the $2BF$ action, including its Lie algebra, as well as a particular choice of parameters that reproduces diffeomorphism transformations in $2BF$ theory.

Finally, Sections 5.2 and 5.3 modify the topological $2BF$ action by adding suitable constraints, thereby obtaining constrained $2BF$ actions with nontrivial dynamics: *general relativ-*

ity and *Einstein–Yang–Mills theory*. We show how gravity and a Yang–Mills field in curved spacetime can be written as constrained $2BF$ actions. This brings us one step closer to expressing the total action for all Standard Model matter and gravity in a form adapted to covariant spin-foam quantization.

5.1 Topological $2BF$ theory

Using the definitions of a 2-group, equivalently a crossed module, a 2-connection, and a 2-curvature, one defines the following generalization of the BF action, called the $2BF$ action [13], [17]:

$$S_{2BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}}. \quad (5.1)$$

Here the 2-form $\mathcal{F} \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$ and the 3-form $\mathcal{G} \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{h})$ are the components of the 2-curvature defined in equation (2.36); the 2-form $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$ and the 1-form $C \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ are Lagrange multipliers; and $\langle _, _ \rangle_{\mathfrak{g}}$ and $\langle _, _ \rangle_{\mathfrak{h}}$ denote G -invariant, symmetric, nondegenerate bilinear forms on $\mathfrak{g}$ and $\mathfrak{h}$. As a consequence of the crossed-module structure (see [16]), the bilinear form $\langle _, _ \rangle_{\mathfrak{h}}$ is also H -invariant. See [13], [17] for a detailed review and further references.

Like the BF action, the $2BF$ action is topological, as can be seen from its equations of motion. Varying the action (5.1) with respect to B^α and C^a gives

$$\mathcal{F}^\alpha = 0, \quad \mathcal{G}^a = 0, \quad (5.2)$$

where α labels the generators of G , and a labels those of H . Variation with respect to the 2-connection variables α^α and β^a gives the equations of motion for the Lagrange multipliers:

$$dB_\alpha - f_{\alpha\beta}{}^\gamma B_\gamma \wedge \alpha^\beta - \triangleright_{\alpha a}{}^b C_b \wedge \beta^a = 0, \quad (5.3)$$

$$dC_a - \partial_a{}^\alpha B_\alpha + \triangleright_{\alpha a}{}^b C_b \wedge \alpha^\alpha = 0. \quad (5.4)$$

These are first-order differential equations describing a theory without propagating degrees of freedom. This statement is established rigorously by Hamiltonian analysis, as in [22], [23]. The Hamiltonian analysis therefore shows that $2BF$ theory is a *topological theory*.

5.1.1 Hamiltonian analysis of topological $2BF$ theory

The topological $2BF$ action (5.1) gives the Lagrangian

$$L_{2BF} = \int_{\Sigma_3} d^3\vec{x} \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} B^\alpha{}_{\mu\nu} \mathcal{F}^\beta{}_{\rho\sigma} g_{\alpha\beta} + \frac{1}{3!} C^a{}_\mu \mathcal{G}^b{}_{\nu\rho\sigma} g_{ab} \right). \quad (5.5)$$

Canonical momenta for the variables $B^\alpha{}_{\mu\nu}$, $\alpha^\alpha{}_\mu$, $C^a{}_\mu$, and $\beta^a{}_{\mu\nu}$ are obtained by varying the action with respect to their time derivatives:

$$\begin{aligned} \pi(B)_\alpha{}^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 B^\alpha{}_{\mu\nu}} = 0, \\ \pi(\alpha)_\alpha{}^\mu &= \frac{\delta L}{\delta \partial_0 \alpha^\alpha{}_\mu} = \frac{1}{2} \epsilon^{0\mu\nu\rho} B_{\alpha\nu\rho}, \\ \pi(C)_a{}^\mu &= \frac{\delta L}{\delta \partial_0 C^a{}_\mu} = 0, \\ \pi(\beta)_a{}^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 \beta^a{}_{\mu\nu}} = -\epsilon^{0\mu\nu\rho} C_{a\rho}. \end{aligned} \quad (5.6)$$

Thus, the *primary constraints* of the theory are obtained using equation (3.19):

$$\begin{aligned}
P(B)_\alpha{}^{\mu\nu} &\equiv \pi(B)_\alpha{}^{\mu\nu} \approx 0, \\
P(\alpha)_\alpha{}^\mu &\equiv \pi(\alpha)_\alpha{}^\mu - \frac{1}{2}\epsilon^{0\mu\nu\rho} B_{\alpha\nu\rho} \approx 0, \\
P(C)_a{}^\mu &\equiv \pi(C)_a{}^\mu \approx 0, \\
P(\beta)_a{}^{\mu\nu} &\equiv \pi(\beta)_a{}^{\mu\nu} + \epsilon^{0\mu\nu\rho} C_{a\rho} \approx 0.
\end{aligned} \tag{5.7}$$

The fundamental Poisson brackets of the variables and their canonical momenta are defined by

$$\begin{aligned}
\{B^\alpha{}_{\mu\nu}(\vec{x}), \pi(B)_\beta{}^{\rho\sigma}(\vec{y})\} &= 2\delta_\beta^\alpha \delta_{[\mu}^\rho \delta_{\nu]}^\sigma \delta^{(3)}(\vec{x} - \vec{y}), \\
\{\alpha^\alpha{}_\mu(\vec{x}), \pi(\alpha)_\beta{}^\nu(\vec{y})\} &= \delta_\beta^\alpha \delta_\mu^\nu \delta^{(3)}(\vec{x} - \vec{y}), \\
\{C^a{}_\mu(\vec{x}), \pi(C)_b{}^\nu(\vec{y})\} &= \delta_b^a \delta_\mu^\nu \delta^{(3)}(\vec{x} - \vec{y}), \\
\{\beta^a{}_{\mu\nu}(\vec{x}), \pi(\beta)_b{}^{\rho\sigma}(\vec{y})\} &= 2\delta_b^a \delta_{[\mu}^\rho \delta_{\nu]}^\sigma \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{5.8}$$

From this definition we obtain the algebra of primary constraints,

$$\begin{aligned}
\{P(B)_\alpha{}^{jk}(\vec{x}), P(\alpha)_\beta{}^i(\vec{y})\} &= \epsilon^{0ijk} g_{\alpha\beta}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{P(C)_a{}^k(\vec{x}), P(\beta)_b{}^{ij}(\vec{y})\} &= -\epsilon^{0ijk} g_{ab}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}),
\end{aligned} \tag{5.9}$$

while all remaining Poisson brackets vanish. The canonical *on-shell* Hamiltonian is defined by equation (3.20),

$$H_c = \int_{\Sigma_3} d^3\vec{x} \left[\frac{1}{2} \pi(B)_\alpha{}^{\mu\nu} \partial_0 B^\alpha{}_{\mu\nu} + \pi(\alpha)_\alpha{}^\mu \partial_0 \alpha^\alpha{}_\mu + \pi(C)_a{}^\mu \partial_0 C^a{}_\mu + \frac{1}{2} \pi(\beta)_a{}^{\mu\nu} \partial_0 \beta^a{}_{\mu\nu} \right] - L, \tag{5.10}$$

that is,

$$\begin{aligned}
H_c \approx & - \int_{\Sigma_3} d^3\vec{x} \epsilon^{0ijk} \left[\frac{1}{2} B_{\alpha 0i} \mathcal{F}^\alpha{}_{jk} + \frac{1}{6} C_{a0} \mathcal{G}^a{}_{ijk} \right. \\
& \left. + \beta^a{}_{0i} \left(\nabla_j C_{ak} - \frac{1}{2} \partial_a{}^\alpha B_{\alpha jk} \right) + \frac{1}{2} \alpha^\alpha{}_0 \left(\nabla_i B_{\alpha jk} - C_{ai} \triangleright_{ab}{}^a \beta^b{}_{jk} \right) \right].
\end{aligned} \tag{5.11}$$

In the preceding expression we used the weak vanishing of the primary constraints. Adding products of Lagrange multipliers λ and primary constraints defines the *total off-shell Hamiltonian* of equation (3.21):

$$H_T = H_c + \int_{\Sigma_3} d^3\vec{x} \left[\frac{1}{2} \lambda(B)_\alpha{}^{\mu\nu} P(B)_\alpha{}^{\mu\nu} + \lambda(\alpha)_\alpha{}^\mu P(\alpha)_\alpha{}^\mu + \lambda(C)_a{}^\mu P(C)_a{}^\mu + \frac{1}{2} \lambda(\beta)_a{}^{\mu\nu} P(\beta)_a{}^{\mu\nu} \right]. \tag{5.12}$$

The *consistency conditions* (3.26) must hold for the primary constraints. For $P(B)_\alpha{}^{0i}$, $P(\alpha)_\alpha{}^0$, $P(C)_a{}^0$, and $P(\beta)_a{}^{0i}$, they give rise to the *secondary constraints* $\mathcal{S}$,

$$\begin{aligned}
\mathcal{S}(\mathcal{F})_\alpha{}^i &\equiv \frac{1}{2} \epsilon^{0ijk} \mathcal{F}_{\alpha jk} \approx 0, \\
\mathcal{S}(\nabla B)_\alpha &\equiv \frac{1}{2} \epsilon^{0ijk} (\nabla_{[i} B_{\alpha j]k} - C_{a[i} \triangleright_{ab}{}^a \beta^b{}_{j]k}) \approx 0, \\
\mathcal{S}(\mathcal{G})_a &\equiv \frac{1}{6} \epsilon^{0ijk} \mathcal{G}_{aijk} \approx 0, \\
\mathcal{S}(\nabla C)_a{}^i &\equiv \epsilon^{0ijk} (\nabla_{[j} C_{ak]} - \frac{1}{2} \partial_a{}^\alpha B_{\alpha jk}) \approx 0,
\end{aligned} \tag{5.13}$$

whereas for the primary constraints $P(\alpha)_\alpha^k$, $P(B)_\alpha^{jk}$, $P(\beta)_a^{jk}$, and $P(C)_a^k$, the consistency conditions determine the Lagrange multipliers:

$$\begin{aligned}
\lambda(B)_{\alpha ij} &\approx \nabla_i B_{\alpha 0j} - \nabla_j B_{\alpha 0i} + C_{a0} \beta_{ij}^b \triangleright_{\alpha b}^a + C_{bi} \triangleright_{\alpha}^b \beta_a^{\alpha}{}_{0j} \\
&\quad - C_{bj} \triangleright_{\alpha}^b \beta_a^{\alpha}{}_{0i} + g_{\beta\gamma} \alpha^\beta{}_0 B^\gamma{}_{ij}, \\
\lambda(\alpha)^\alpha{}_i &\approx \nabla_i \alpha^\alpha{}_0 + \partial_a^\alpha \beta^a{}_{0i}, \\
\lambda(C)^a{}_i &\approx \nabla_i C^a{}_0 + C_{ij}^b \triangleright_{\alpha}^a \beta_b^\alpha \alpha^\alpha{}_0 + B_{\alpha 0i} \partial^a \alpha, \\
\lambda(\beta)^a{}_{ij} &\approx \nabla_i \beta^a{}_{0j} - \nabla_j \beta^a{}_{0i} - \beta_{ij}^b \triangleright_{\alpha b}^a \alpha^\alpha{}_0.
\end{aligned} \tag{5.14}$$

The remaining Lagrange multipliers,

$$\lambda(B)^\alpha{}_{0i}, \quad \lambda(\alpha)^\alpha{}_0, \quad \lambda(C)^a{}_0, \quad \lambda(\beta)^a{}_{0i}, \tag{5.15}$$

remain undetermined. The consistency conditions for the secondary constraints produce no tertiary constraints; explicitly,

$$\begin{aligned}
\{\mathcal{S}(\mathcal{F})^{\alpha i}, H_T\} &= f_{\beta\gamma} \alpha^\beta \mathcal{S}(\mathcal{F})^{\beta i} \alpha^\gamma{}_0, \\
\{\mathcal{S}(\nabla B)_\alpha, H_T\} &= f_{\beta\gamma\alpha} B^\gamma{}_{0k} \mathcal{S}(\mathcal{F})^{\beta k} + f_{\beta\alpha} \gamma \alpha^\beta{}_0 \mathcal{S}(\nabla B)_\gamma + C_{a0} \triangleright_{\alpha b}^a \mathcal{S}(\mathcal{G})^b \\
&\quad - \triangleright_{\alpha a}^b \beta^a{}_{0k} \mathcal{S}(\nabla C)_b{}^k, \\
\{\mathcal{S}(\mathcal{G})^a, H_T\} &= \triangleright_{\alpha b}^a \beta^b{}_{0k} \mathcal{S}(\mathcal{F})^{\alpha k} - \alpha^\alpha{}_0 \triangleright_{\alpha b}^a \mathcal{S}(\mathcal{G})^b, \\
\{\mathcal{S}(\nabla C)_a{}^i, H_T\} &= C_{b0} \triangleright_{\alpha}^b \beta_a^\alpha \mathcal{S}(\mathcal{F})^{\alpha i} + \triangleright_{\alpha a}^b \alpha^\alpha{}_0 \mathcal{S}(\nabla C)_b{}^i,
\end{aligned} \tag{5.16}$$

The total Hamiltonian can be written as

$$\begin{aligned}
H_T = \int_{\Sigma_3} d^3\vec{x} \Big[&\lambda(B)^\alpha{}_{0i} \Phi(B)_\alpha{}^i + \lambda(\alpha)^\alpha \Phi(\alpha)_\alpha + \lambda(C)^a{}_0 \Phi(C)_a + \lambda(\beta)^a{}_{0i} \Phi(\beta)_a{}^i \\
&- B_{\alpha 0i} \Phi(\mathcal{F})^{ai} - \alpha_{\alpha 0} \Phi(\nabla B)^\alpha - C_{a0} \Phi(\mathcal{G})^a - \beta_{a0i} \Phi(\nabla C)^{ai} \Big],
\end{aligned} \tag{5.17}$$

where

$$\begin{aligned}
\Phi(B)_\alpha{}^i &= P(B)_\alpha{}^{0i}, \\
\Phi(\alpha)_\alpha &= P(\alpha)_\alpha{}^0, \\
\Phi(C)_a &= P(C)_a{}^0, \\
\Phi(\beta)_a{}^i &= P(\beta)_a{}^{0i}, \\
\Phi(\mathcal{F})^{\alpha i} &= \mathcal{S}(\mathcal{F})^{\alpha i} - \nabla_j P(B)^{\alpha ij} - P(C)_a{}^i \partial^{a\alpha}, \\
\Phi(\mathcal{G})_a &= \mathcal{S}(\mathcal{G})_a + \nabla_i P(C)_a{}^i - \frac{1}{2} \beta_{bij} \triangleright_{\alpha}^b \beta_a^\alpha P(B)^{\alpha ij}, \\
\Phi(\nabla C)_a{}^i &= \mathcal{S}(\nabla C)_a{}^i - \nabla_j P(\beta)_a{}^{ij} + C_{bj} \triangleright_{\alpha}^b \beta_a^\alpha P(B)^{\alpha ij} - \partial_a^\alpha P(\alpha)_\alpha{}^i, \\
\Phi(\nabla B)_\alpha &= \mathcal{S}(\nabla B)_\alpha + \nabla_i P(\alpha)_\alpha{}^i - \frac{1}{2} B_{bij} f_{\alpha\gamma}^\beta P(B)^{\gamma ij} \\
&\quad - C_{bi} \triangleright_{\alpha a}^b P(C)^{ai} - \frac{1}{2} \beta_{bij} \triangleright_{\alpha a}^b P(\beta)^{aij},
\end{aligned} \tag{5.18}$$

are the first-class constraints, while the second-class constraints are

$$\chi(B)_\alpha{}^{jk} = P(B)_\alpha{}^{jk}, \quad \chi(C)_a{}^i = P(C)_a{}^i, \quad \chi(\alpha)_\alpha{}^i = P(\alpha)_\alpha{}^i, \quad \chi(\beta)_a{}^{ij} = P(\beta)_a{}^{ij}. \quad (5.19)$$

The Poisson algebra of the first-class constraints is

$$\begin{aligned} \{ \Phi(\mathcal{G})^a(\vec{x}), \Phi(\nabla C)_b{}^i(\vec{y}) \} &= -\triangleright_{ab}{}^a \Phi(\mathcal{F})^{\alpha i}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\mathcal{G})^a(\vec{x}), \Phi(\nabla B)_\alpha(\vec{y}) \} &= \triangleright_{ab}{}^a \Phi(\mathcal{G})^b(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla C)_a{}^i(\vec{x}), \Phi(\nabla B)_\alpha(\vec{y}) \} &= \triangleright_{\alpha a}{}^b \Phi(\nabla C)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\mathcal{F})^\alpha{}_i(\vec{x}), \Phi(\nabla B)_\beta(\vec{y}) \} &= f_{\beta\gamma}{}^\alpha \Phi(\mathcal{F})^\gamma{}_i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla B)_\alpha(\vec{x}), \Phi(\nabla B)_\beta(\vec{y}) \} &= f_{\alpha\beta}{}^\gamma \Phi(\nabla B)_\gamma(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \end{aligned} \quad (5.20)$$

and the Poisson algebra of the first- and second-class constraints is

$$\begin{aligned} \{ \Phi(\mathcal{F})^{\alpha i}(\vec{x}), \chi(\alpha)_\beta{}^j(\vec{y}) \} &= -f_{\beta\gamma}{}^\alpha \chi(B)^{\gamma ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\mathcal{G})^a(\vec{x}), \chi(\alpha)_\alpha{}^i(\vec{y}) \} &= -\triangleright_{ab}{}^a \chi(C)^{bi}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\mathcal{G})^a(\vec{x}), \chi(\beta)_b{}^{ij}(\vec{y}) \} &= \triangleright_{ab}{}^a \chi(B)^{\alpha ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla C)^{ai}(\vec{x}), \chi(\alpha)_\alpha{}^j(\vec{y}) \} &= -\triangleright_{ab}{}^a \chi(\beta)^{bij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla C)^{ai}(\vec{x}), \chi(C)_b{}^j(\vec{y}) \} &= \triangleright_{ab}{}^a \chi(B)^{\alpha ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(\alpha)_\beta{}^i(\vec{y}) \} &= f_{\beta\gamma}{}^\alpha \chi(\alpha)^{\gamma i}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(\beta)_a{}^{ij}(\vec{y}) \} &= \triangleright_{\alpha a}{}^b \chi(\beta)_b{}^{ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(B)_\beta{}^{ij}(\vec{y}) \} &= -f_{\beta\gamma}{}^\alpha \chi(B)^{\gamma ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(C)_a{}^i(\vec{y}) \} &= -\triangleright^{\alpha b}{}_a \chi(C)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}). \end{aligned} \quad (5.21)$$

The Poisson brackets of the first-class constraints with the Hamiltonian are

$$\begin{aligned} \{ \Phi(B)_\alpha{}^i, H_T \} &= \Phi(F)_\alpha{}^i, \\ \{ \Phi(\alpha)_\alpha, H_T \} &= \Phi(\nabla B)_\alpha, \\ \{ \Phi(F)^\alpha{}_i, H_T \} &= -\alpha^\beta{}_0 f_{\beta\gamma}{}^\alpha \Phi(F)^\gamma{}_i, \\ \{ \Phi(\nabla B)_\alpha, H_T \} &= -B_{\beta 0i} f_{\alpha\gamma}{}^\beta \Phi(F)^{\gamma i} - \alpha^\beta{}_0 f_{\alpha\beta}{}^\gamma \Phi(\nabla B)_\gamma, \\ \{ \Phi(C)_a, H_T \} &= \Phi(\mathcal{G})_a, \\ \{ \Phi(\beta)_a{}^i, H_T \} &= \Phi(\nabla C)_a{}^i, \\ \{ \Phi(\mathcal{G})_a, H_T \} &= \alpha^\alpha{}_0 \triangleright_{\alpha a}{}^b \Phi(\mathcal{G})_b - \beta_{b 0i} \triangleright_{\alpha a}{}^b \Phi(F)^{\alpha i}, \\ \{ \Phi(\nabla C)_a{}^i, H_T \} &= \alpha^\alpha{}_0 \triangleright_{\alpha a}{}^b \Phi(\nabla C)_b{}^i - C_{b 0} \triangleright_{\alpha a}{}^b \Phi(F)^{\alpha i}. \end{aligned} \quad (5.22)$$

Number of degrees of freedom of topological $2BF$ theory

To determine the number of degrees of freedom of topological $2BF$ theory, we use the following Bianchi identities (BI).

Lemma 9 (BI for the 1-forms α and C .) *The corresponding curvature 2-forms*

$$F^\alpha = d\alpha^\alpha + f_{\beta\gamma}{}^\alpha \alpha^\beta \wedge \alpha^\gamma, \quad T^a = dC^a + \triangleright_{ab}{}^a \alpha^\alpha \wedge C^b, \quad (5.23)$$

satisfy the Bianchi identities

$$\epsilon^{\lambda\mu\nu\rho} \nabla_\mu F^\alpha{}_{\nu\rho} = 0, \quad (5.24)$$

$$\epsilon^{\lambda\mu\nu\rho} (\nabla_\mu T^a{}_{\nu\rho} - \triangleright_{ab}{}^a F^\alpha{}_{\mu\nu} C^b{}_\rho) = 0. \quad (5.25)$$

Lemma 10 (BI for the 2-forms B and β .) *The corresponding 3-curvatures are*

$$S^\alpha = dB^\alpha + f_{\beta\gamma}{}^\alpha \alpha^\beta \wedge B^\gamma, \quad G^a = d\beta^a + \triangleright_{ab}{}^a \alpha^\alpha \wedge \beta^b, \quad (5.26)$$

and satisfy the Bianchi identities

$$\epsilon^{\lambda\mu\nu\rho} \left(\frac{2}{3} \nabla_\lambda S^\alpha{}_{\mu\nu\rho} - f_{\beta\gamma}{}^\alpha F^\beta{}_{\lambda\mu} B^\gamma{}_{\nu\rho} \right) = 0, \quad (5.27)$$

$$\epsilon^{\lambda\mu\nu\rho} \left(\frac{2}{3} \nabla_\lambda G^a{}_{\mu\nu\rho} - \triangleright_{ab}{}^a F^\alpha{}_{\lambda\mu} \beta^b{}_{\nu\rho} \right) = 0. \quad (5.28)$$

For $2BF$ theory, the initial number of field components N can be determined from Table (5.1). Here p is the dimension of the Lie group G , and q is the dimension of the Lie group H .

$\alpha^\alpha{}_\mu$	$\beta^a{}_{\mu\nu}$	$B^\alpha{}_{\mu\nu}$	$C^a{}_\mu$
$4p$	$6q$	$6p$	$4q$

Table 5.1: Number of initial field components in $2BF$ theory.

Counting the fields in Table (5.1) gives $N = 10(p+q)$. The number of independent components of the second-class constraints is found by counting the constraints in Table (5.2), yielding $S = 6(p+q)$.

$\chi(B)_\alpha{}^{jk}$	$\chi(C)_a{}^i$	$\chi(\alpha)_\alpha{}^i$	$\chi(\beta)_a{}^{ij}$
$3p$	$3q$	$3p$	$3q$

Table 5.2: Second-class constraints in $2BF$ theory.

The first-class constraints are not all independent and satisfy

$$\nabla_i \Phi(\mathcal{F})_\alpha{}^i + \frac{1}{2} \partial_{a\alpha} \Phi(\mathcal{G})^a - \frac{1}{2} \partial^a{}_\alpha \nabla_i \chi(C)_a{}^i - \frac{1}{2} f_{\beta\gamma\alpha} \partial_a{}^\beta \beta^a{}_{ij} \chi(B)^{\gamma ij} = \epsilon^{ijk} \nabla_i F_{\alpha jk}, \quad (5.29)$$

Using $\epsilon^{ijk} \nabla_i F_{jk}^a = 0$, the $\lambda = 0$ component of BI (5.24), this reduces to

$$\nabla_i \Phi(\mathcal{F})_\alpha{}^i + \frac{1}{2} \partial_{a\alpha} \Phi(\mathcal{G})^a - \frac{1}{2} \partial^a{}_\alpha \nabla_i \chi(C)_a{}^i - \frac{1}{2} f_{\beta\gamma\alpha} \partial_a{}^\beta \beta^a{}_{ij} \chi(B)^{\gamma ij} = 0. \quad (5.30)$$

Similarly, the first-class constraints satisfy

$$\begin{aligned}
& \nabla_i \Phi(\nabla C)_a^i - \frac{1}{2} C_{bi} \triangleright_{\alpha a} {}^b \Phi(\mathcal{F})^{\alpha i} + \partial_{a\alpha} S(\nabla B)^\alpha + \\
& + \frac{1}{2} F^\beta_{ij} \triangleright_{\beta c} {}^b \chi(\beta)^{cij} + T^b_{jk} \triangleright_{\alpha a} {}^b \chi(B)^{\alpha jk} - \partial_{a\alpha} \nabla_i \chi(\alpha)^{\alpha i} \\
& = \epsilon^{ijk} (\nabla_i T_{ajk} - \triangleright_{ab} {}^a F_{jk}^\alpha C_i^b),
\end{aligned} \tag{5.31}$$

Since the right-hand side is the $\lambda = 0$ component of (5.25), equation (5.31) yields the relation

$$\begin{aligned}
& \nabla_i \Phi(\nabla C)_a^i - \frac{1}{2} C_{bi} \triangleright_{\alpha a} {}^b \Phi(\mathcal{F})^{\alpha i} + \partial_{a\alpha} S(\nabla B)^\alpha + \\
& + \frac{1}{2} F^\beta_{ij} \triangleright_{\beta c} {}^b \chi(\beta)^{cij} + T^b_{jk} \triangleright_{\alpha a} {}^b \chi(B)^{\alpha jk} - \partial_{a\alpha} \nabla_i \chi(\alpha)^{\alpha i} = 0.
\end{aligned} \tag{5.32}$$

The number of first-class constraints follows by counting the constraints in Table (5.3). It is

$$F = 8(p + q) - (p + q) = 7(p + q),$$

where we subtracted the p relations (5.30) and the q relations (5.32). Thus, from the definition (3.41),

$$n = 10(p + q) - 7(p + q) - \frac{6(p + q)}{2} = 0. \tag{5.33}$$

We conclude that 2BF theory has no local propagating degrees of freedom and is therefore a *topological theory*.

$\Phi(B)_\alpha^i$	$\Phi(C)_a$	$\Phi(\alpha)_\alpha$	$\Phi(\beta)_a^i$	$\Phi(\mathcal{F})^{\alpha i}$	$\Phi(\mathcal{G})^a$	$\Phi(\nabla C)^{\alpha i}$	$\Phi(\nabla B)^\alpha$
$3p$	q	p	$3q$	$3p - p$	q	$3q - q$	p

Table 5.3: First-class constraints in 2BF theory.

Gauge-transformation generator for 2BF theory

The generator of gauge transformations in 2BF theory is

$$\begin{aligned}
G = \int_{\Sigma_3} d^3 \vec{x} & \left((\nabla_0 \epsilon^\alpha_i) \Phi(B)_\alpha^i - \epsilon^\alpha_i \Phi(\mathcal{F})_\alpha^i + (\nabla_0 \epsilon^\alpha) \Phi(\alpha)_\alpha \right. \\
& + \epsilon^\alpha (f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} + C_{a0} \triangleright_{ab} {}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{ab} {}^a \Phi(\beta)^{b0i} - \Phi(\nabla B)_\alpha) \\
& + (\nabla_0 \epsilon^a) \Phi(C)_a - \epsilon^a (\beta_{b0i} \triangleright_{\alpha a} {}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a) \\
& \left. + (\nabla_0 \epsilon^a_i) \Phi(\beta)_a^i - \epsilon^a_i (C_{b0} \triangleright_{\alpha a} {}^b \Phi(B)^{\alpha i} + \Phi(\nabla C)_a^i) \right).
\end{aligned} \tag{5.34}$$

Here ϵ^{ab}_i , ϵ^{ab} , ϵ_i , ϵ^a , and ϵ^a_i are independent gauge parameters. The derivation of the generator (5.34) is presented in Appendix D.2.

The form variations of the variables and their conjugate momenta are calculated using

(3.56):

$$\begin{aligned}
\delta_0 B^\alpha_{0i} &= \nabla_0 \epsilon^\alpha_i - f_{\beta\gamma}{}^\alpha \epsilon^\beta B^\gamma_{0i} & \delta_0 \pi(B)_\alpha^{0i} &= f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{0i}, \\
&\quad - \epsilon^a \triangleright_{\alpha a}{}^b \beta_{b0i} - \epsilon^a_i \triangleright_{\alpha a}{}^b C_{b0}, & & \\
\delta_0 B^\alpha_{ij} &= 2\nabla_{[i} \epsilon^\alpha_{j]} - f_{\beta\gamma}{}^\alpha \epsilon^\beta B^\gamma_{ij} & \delta_0 \pi(B)_\alpha^{ij} &= f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{ij} - \epsilon^a_k \epsilon^{0ijk} \partial_{a\alpha}, \\
&\quad - \epsilon^a \triangleright_{\alpha a}{}^b \beta_{bij} - 2\epsilon^a_{[j} \triangleright_{\alpha a}{}^b C_{b|ij]}, & & \\
\delta_0 \alpha^a_0 &= \nabla_0 \epsilon^a, & \delta_0 \pi(\alpha)_\alpha^0 &= -f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{0i} - f_{\alpha\beta}{}^\gamma \epsilon^\gamma \pi(\alpha)_\gamma^0 \\
& & &\quad - \epsilon^b \triangleright_{\alpha b}{}^a \pi(C)_a - \epsilon^b_i \triangleright_{\alpha b}{}^a \pi(\beta)_a^i, \\
\delta_0 \alpha^a_i &= \nabla_i \epsilon^a + \partial_a^\alpha \epsilon^a_i, & \delta_0 \pi(\alpha)_\alpha^i &= -f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(B)_\gamma^{ij} - f_{\alpha\beta}{}^\gamma \epsilon^\beta \pi(\alpha)_\gamma^i, \tag{5.35} \\
& & & \\
\delta_0 C^a_0 &= \nabla_0 \epsilon^a - \epsilon^\alpha \triangleright_{\alpha b}{}^a C^b_0, & \delta_0 \pi(C)_a^0 &= \epsilon^\alpha \triangleright_{\alpha a}{}^b \pi(C)_b^0 + \epsilon_{bi} \triangleright_{\alpha a}{}^b \pi(B)^{\alpha 0i}, \\
\delta_0 C^a_i &= \nabla_i \epsilon^a - \epsilon^\alpha \triangleright_{\alpha b}{}^a C^b_i, & \delta_0 \pi(C)_a^i &= \epsilon^\alpha \triangleright_{\alpha a}{}^b \pi(C)_b^i + \epsilon_{bj} \triangleright_{\alpha a}{}^b \pi(B)^{\alpha ij}, \\
\delta_0 \beta^a_{0i} &= \nabla_0 \epsilon^a_i - \epsilon^\alpha \triangleright_{\alpha b}{}^a \beta^b_{0i}, & \delta_0 \pi(\beta)_a^{0i} &= \epsilon^\alpha \triangleright_{\alpha a}{}^b \pi(\beta)_b^{0i} + \epsilon_b \triangleright_{\alpha a}{}^b \pi(B)^{\alpha 0i}, \\
\delta_0 \beta^a_{ij} &= 2\nabla_{[i} \epsilon^a_{j]} - \epsilon^\alpha \triangleright_{\alpha b}{}^a \beta^b_{ij}, & \delta_0 \pi(\beta)_a^{ij} &= \epsilon^\alpha \triangleright_{\alpha a}{}^b \pi(\beta)_b^{ij} + \epsilon_b \triangleright_{\alpha a}{}^b \pi(B)^{\alpha ij} \\
& & &\quad - \epsilon^\alpha_k \epsilon^{0ijk} \partial_{a\alpha}.
\end{aligned}$$

5.1.2 Symmetries of the $2BF$ action

The G symmetry group

The action (5.1) has additional symmetries beyond those defined for the BF action in Theorems 8 and 9.

The transformations generated by the gauge parameter ϵ_g^α , inferred from the form variations (5.35), are

$$\begin{aligned}
\delta_0 \alpha^\alpha_\mu &= -\partial_\mu \epsilon_g^\alpha - f_{\beta\gamma}{}^\alpha \alpha^\beta_\mu \epsilon_g^\gamma, & \delta_0 B^\alpha_{\mu\nu} &= f_{\beta\gamma}{}^\alpha \epsilon_g^\beta B^\gamma_{\mu\nu}, \\
\delta_0 \beta^a_{\mu\nu} &= \triangleright_{\alpha b}{}^a \epsilon_g^\alpha \beta^b_{\mu\nu}, & \delta_0 C^a_\mu &= \triangleright_{\alpha b}{}^a \epsilon_g^\alpha C^b_\mu,
\end{aligned} \tag{5.36}$$

which can equivalently be written as

$$\begin{aligned}
\alpha &\rightarrow \alpha' = \alpha - \nabla \epsilon_g, & B &\rightarrow B' = B - [B, \epsilon_g], \\
\beta &\rightarrow \beta' = \beta + \epsilon_g \triangleright \beta, & C &\rightarrow C' = C + \epsilon_g \triangleright C,
\end{aligned} \tag{5.37}$$

These infinitesimal transformations lead to the finite transformations defined in Theorem 10.

Theorem 10 (G -gauge transformations) *In a $2BF$ theory constructed for an arbitrary crossed module $(H \xrightarrow{\partial} G, \triangleright)$, the following is a symmetry transformation:*

$$\begin{aligned}
\alpha &\rightarrow \alpha' = \text{Ad}_g \alpha + g dg^{-1}, & B &\rightarrow B' = g B g^{-1}, \\
\beta &\rightarrow \beta' = g \triangleright \beta, & C &\rightarrow C' = g \triangleright C,
\end{aligned} \tag{5.38}$$

where $g = \exp(\epsilon_g \cdot \hat{G}) = \exp(\epsilon_{g\alpha} \hat{G}^\alpha) \in G$, and $\epsilon_g : \mathcal{M}_4 \rightarrow \mathfrak{g}$ is the transformation parameter.

Proof. Under these transformations, the 2-curvature transforms as

$$\mathcal{F} \rightarrow \mathcal{F}' = g \mathcal{F} g^{-1}, \quad \mathcal{G} \rightarrow \mathcal{G}' = g \triangleright \mathcal{G}. \tag{5.39}$$

Invariance of the 2BF action follows from the G -invariance of the bilinear forms $\langle _, _ \rangle_{\mathfrak{g}}$ and $\langle _, _ \rangle_{\mathfrak{h}}$:

$$S_{2BF} = \int_{\mathcal{M}_4} \left(\langle B, \mathcal{F} \rangle_{\mathfrak{g}} + \langle C, G \rangle_{\mathfrak{h}} \right) \rightarrow S'_{2BF} = \int_{\mathcal{M}_4} \left(\langle g^{-1}Bg, g^{-1}\mathcal{F}g \rangle_{\mathfrak{g}} + \langle g^{-1}\triangleright C, g^{-1}\triangleright G \rangle_{\mathfrak{h}} \right), \quad (5.40)$$

and hence the 2BF action is invariant. This can also be shown as in Theorem 8. ■

The preceding theorem generalizes Theorem 8 to 2BF theory.

The $\tilde{M}$ symmetry group

Inspecting the field transformations, the terms $\delta_0 B^\alpha_{0i} = \nabla_0 \epsilon_i$ and $\delta_0 B^\alpha_{ij} = 2\nabla_{[i} \epsilon_{j]}$ are seen to correspond to the transformation defined in Theorem 11, as in BF theory.

Theorem 11 (M -gauge transformations) *In a 2BF theory over an arbitrary crossed module $(H \xrightarrow{\partial} G, \triangleright)$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha, & B &\rightarrow B' = B - \nabla \epsilon_{\mathfrak{m}}, \\ \beta &\rightarrow \beta' = \beta, & C^a &\rightarrow C'^a = C^a - \partial^a_{\alpha} \epsilon_{\mathfrak{m}}^{\alpha}, \end{aligned} \quad (5.41)$$

where $\epsilon_{\mathfrak{m}}$ is an arbitrary $\mathfrak{g}$ -valued 1-form and ∇ is the covariant exterior derivative defined in the standard way,

$$\nabla \epsilon_{\mathfrak{m}} = d\epsilon_{\mathfrak{m}} + [\alpha \wedge \epsilon_{\mathfrak{m}}]. \quad (5.42)$$

Proof. Under an M -gauge transformation, the 2BF action becomes

$$S'_{2BF} = S_{2BF} + \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \left(-\frac{1}{2} (\nabla_{\mu} \epsilon_{\mathfrak{m}}^{\alpha}{}_{\nu}) \mathcal{F}_{\alpha\rho\sigma} - \frac{1}{3!} \partial^a_{\alpha} \epsilon_{\mathfrak{m}}^{\alpha}{}_{\mu} G_{a\nu\rho\sigma} \right). \quad (5.43)$$

Using the definition of the 2-curvature (2.37), we obtain

$$S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \left(-\frac{1}{2} (\nabla_{\mu} \epsilon_{\mathfrak{m}}^{\alpha}{}_{\nu}) (F_{\alpha\rho\sigma} - \partial^a_{\alpha} \beta_{a\rho\sigma}) - \frac{1}{3!} \partial^a_{\alpha} \epsilon_{\mathfrak{m}}^{\alpha}{}_{\mu} 3\nabla_{\nu} \beta_{a\rho\sigma} \right). \quad (5.44)$$

The second and third terms in parentheses cancel, leaving

$$S'_{3BF} = S_{3BF} - \frac{1}{2} \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \epsilon_{\mathfrak{m}}^{\alpha}{}_{\mu} \nabla_{\nu} F_{\alpha\rho\sigma}. \quad (5.45)$$

The term $\epsilon^{\mu\nu\rho\sigma} \nabla_{\nu} F_{\alpha\rho\sigma} = 0$ is BI (5.24). We conclude that the 2BF action S_{2BF} is invariant under the M -gauge transformations of Theorem 11. ■

This theorem generalizes Theorem 9 to 2BF theory. The commutators among the generators of G -gauge transformations, among those of M -gauge transformations, and between the G - and M -gauge generators are calculated as in BF theory, giving the same results, namely equations (4.40), (4.46), and (4.49). As for the symmetries of the BF action, the generators obey the isomorphism $\hat{G}_{\alpha} \cong \tau_{\alpha}$. Thus the group of G -gauge transformations in Theorem 10 is precisely the group G of the crossed module $(H \xrightarrow{\partial} G, \triangleright)$. This important result will not hold for the remaining symmetry transformations of the 2BF action.

The $\tilde{H}$ symmetry group

In the form variations (5.35), consider the variations on the spatial hypersurface Σ_3 associated with the parameter ϵ^a_i . They allow us to extend the variations to a parameter ϵ^a_μ and define the H -gauge transformations in Theorem 12:

$$\begin{aligned} \delta_0 \alpha^\alpha_\mu &= \partial_a^\alpha \epsilon_\mathfrak{h}^a_\mu, & \delta_0 B^\alpha_{\mu\nu} &= -2C_{a[\mu} \epsilon_\mathfrak{h}^b_{\nu]} \triangleright_{\beta b}^a g^{\alpha\beta}, \\ \delta_0 \beta^a_{\mu\nu} &= 2\nabla_{[\mu} \epsilon_\mathfrak{h}^a_{\nu]}, & \delta_0 C^a_\mu &= 0, \end{aligned} \quad (5.46)$$

where we identify $\epsilon_\mathfrak{h}^a_i = \epsilon^a_i$ and $\epsilon_\mathfrak{h}^a_0 = 0$.

Theorem 12 (H -gauge transformations) *In a 2BF theory over an arbitrary crossed module $(H \xrightarrow{\partial} G, \triangleright)$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha - \partial \epsilon_\mathfrak{h}, & B &\rightarrow B' = B - C' \wedge^\mathcal{T} \epsilon_\mathfrak{h}, \\ \beta &\rightarrow \beta' = \beta - \nabla' \epsilon_\mathfrak{h} - \epsilon_\mathfrak{h} \wedge \epsilon_\mathfrak{h}, & C &\rightarrow C' = C, \end{aligned} \quad (5.47)$$

where $\epsilon_\mathfrak{h}$ is an arbitrary $\mathfrak{h}$ -valued 1-form, and ∇' denotes the covariant derivative with connection α' . The map τ is defined in Appendix A by equation (A.13) and solves

$$\langle C \wedge^\mathcal{T} \epsilon_\mathfrak{h}, \mathcal{F} \rangle_\mathfrak{g} + \langle C, \mathcal{F} \wedge^\triangleright \epsilon_\mathfrak{h} \rangle_\mathfrak{h} = 0.$$

Proof. Invariance follows by direct verification. The 3-curvature transforms as

$$\begin{aligned} \mathcal{F} &\rightarrow \mathcal{F}' = \mathcal{F}, \\ \mathcal{G} &\rightarrow \mathcal{G}' = \mathcal{G} - \mathcal{F} \wedge^\triangleright \epsilon_\mathfrak{h}. \end{aligned} \quad (5.48)$$

Under the 3-curvature transformations (5.48) and the transformations of the Lagrange multipliers, the action S_{3BF} becomes

$$S'_{2BF} = S_{2BF} + \int_{\mathcal{M}_4} \left(-\langle C' \wedge^\mathcal{T} \epsilon_\mathfrak{h}, \mathcal{F} \rangle_\mathfrak{g} - \langle C', \mathcal{F} \wedge^\triangleright \epsilon_\mathfrak{h} \rangle_\mathfrak{h} \right). \quad (5.49)$$

The definition of the map $\mathcal{T}$ in equation (A.13) ensures that the terms in parentheses cancel, so the action remains invariant. ■

Denote the generators of the H -gauge transformations of Theorem 12 by $\hat{H}_a^\mu$. Applying the same procedure used for the G - and M -gauge commutators gives the H -gauge commutators. Two successive infinitesimal H -gauge transformations with parameters $\epsilon_{\mathfrak{h}1}$ and $\epsilon_{\mathfrak{h}2}$ give

$$e^{\epsilon_{\mathfrak{h}1} \cdot \hat{H}} e^{\epsilon_{\mathfrak{h}2} \cdot \hat{H}} - e^{\epsilon_{\mathfrak{h}2} \cdot \hat{H}} e^{\epsilon_{\mathfrak{h}1} \cdot \hat{H}} = 0, \quad (5.50)$$

where $\epsilon_\mathfrak{h} \cdot \hat{H} = \epsilon_\mathfrak{h}^a_\mu \hat{H}_a^\mu$. Hence the H -gauge commutator is

$$[\hat{H}_a^\mu, \hat{H}_b^\nu] = 0. \quad (5.51)$$

This commutator is calculated in Appendix D.2.3. Thus the H -gauge transformations form an Abelian group, denoted by $\tilde{H}$. The index structure of its parameters and generators shows that it is isomorphic to $\mathbb{R}^{4q}$, where q is the dimension of H :

$$\tilde{H} \cong \mathbb{R}^{4q}. \quad (5.52)$$

The commutators of the generators of G and $\tilde{M}$ with the H -gauge generators are

$$[\hat{G}_\alpha, \hat{H}_a^\mu] = \triangleright_{\alpha a}^b \hat{H}_b^\mu, \quad [\hat{H}_a, \hat{M}_\alpha^\mu] = 0. \quad (5.53)$$

The $\tilde{N}$ symmetry group

Theorem 13 (N -gauge transformations) *In a 2BF theory over an arbitrary crossed module $(H \xrightarrow{\partial} G, \triangleright)$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha, & B &\rightarrow B' = B - \beta \wedge^T \epsilon_n, \\ \beta &\rightarrow \beta' = \beta, & C &\rightarrow C' = C - \nabla \epsilon_n, \end{aligned} \quad (5.54)$$

where ϵ_n is an arbitrary $\mathfrak{h}$ -valued 0-form.

Proof. The proof is direct. Under the transformations defined in Theorem 13, the 2BF action transforms as

$$\begin{aligned} S_{2BF} &\rightarrow S'_{2BF} = \int_{\mathcal{M}_4} dx^4 \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} (B_{\alpha\mu\nu} - \beta_{b\mu\nu} \triangleright_{\alpha a} {}^b \epsilon_n^a) \mathcal{F}^\alpha_{\rho\sigma} + \frac{1}{3!} (C^a_\mu + \nabla_\mu \epsilon_n^a) G_{a\nu\rho\sigma} \right) \\ &= S_{2BF} + \int_{\mathcal{M}_4} dx^4 \epsilon^{\mu\nu\rho\sigma} \left(-\frac{1}{4} \beta_{b\mu\nu} \triangleright_{\alpha a} {}^b \epsilon_n^a \mathcal{F}^\alpha_{\rho\sigma} - \frac{1}{2} \nabla_\nu \nabla_\mu \epsilon_n^a \beta_{a\rho\sigma} \right) \\ &= S_{2BF} + \int_{\mathcal{M}_4} dx^4 \epsilon^{\mu\nu\rho\sigma} \left(-\frac{1}{4} \beta_{b\mu\nu} \triangleright_{\alpha a} {}^b \epsilon_n^a \mathcal{F}^\alpha_{\rho\sigma} + \frac{1}{4} \triangleright_{ab} {}^a F^\alpha_{\mu\nu} \epsilon_n^b \beta_{a\rho\sigma} \right). \end{aligned} \quad (5.55)$$

Here we used the fact that the term

$$\epsilon^{\mu\nu\rho\sigma} \triangleright_{\alpha a} {}^b \epsilon_n^a \beta_{b\mu\nu} \partial_c^\alpha \beta^c_{\rho\sigma} = \epsilon^{\mu\nu\rho\sigma} f_{ca} {}^b \epsilon_n^a \beta_{b\mu\nu} \beta^c_{\rho\sigma} = 0,$$

vanishes identically by the Peiffer identity (2.20) and the antisymmetry of the structure constants. ■

Denote the group of N -gauge transformations defined in Theorem 13 by $\tilde{N}$. These transformations are linear, and the composition of two of them is an N -gauge transformation with parameter $\epsilon_{n1} + \epsilon_{n2}$. Denoting the generators of $\tilde{N}$ by $\hat{N}_a$, we obtain

$$e^{\epsilon_{n1} \cdot \hat{N}} e^{\epsilon_{n2} \cdot \hat{N}} = e^{(\epsilon_{n1} + \epsilon_{n2}) \cdot \hat{N}}, \quad (5.56)$$

where $\epsilon_n \cdot \hat{N} = \epsilon_n^a \hat{N}_a$; hence the gauge generators commute:

$$[\hat{N}_a, \hat{N}_b] = 0. \quad (5.57)$$

Thus $\tilde{N}$ is Abelian, and the index structure of its parameters and generators shows that it is isomorphic to $\mathbb{R}^q$, where q is the dimension of H :

$$\tilde{N} \cong \mathbb{R}^q. \quad (5.58)$$

We may next examine the commutators of N -gauge transformations with the G -, H -, and M -gauge transformations. For the G -gauge transformations we obtain¹

$$[\epsilon_g \cdot \hat{G}, \epsilon_n \cdot \hat{N}] = (\epsilon_g \triangleright \epsilon_n) \cdot \hat{N}, \quad (5.59)$$

so the commutator of the G - and N -gauge transformations is

$$[\hat{G}_\alpha, \hat{N}_a] = \triangleright_{\alpha a} {}^b \hat{N}_b. \quad (5.60)$$

¹The action of ϵ_g on ϵ_n , denoted $\epsilon_g \triangleright \epsilon_n$, is defined by $\epsilon_g \triangleright \epsilon_n \equiv \triangleright_{\alpha a} {}^b \epsilon_g^\alpha \epsilon_n^a$. It follows that

$$(\epsilon_g \triangleright \epsilon_n) \cdot \hat{N} = \triangleright_{\alpha a} {}^b \epsilon_g^\alpha \epsilon_n^a \hat{N}_b.$$

Examining the relation between the N - and H -gauge transformations gives

$$e^{\epsilon_{\mathfrak{h}} \cdot \hat{H}} e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} - e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} e^{\epsilon_{\mathfrak{h}} \cdot \hat{H}} = -(\epsilon_{\mathfrak{n}} \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}}) \cdot \hat{M}, \quad (5.61)$$

whose proof is given in Appendix D.2.3. The commutator of an H -gauge generator with an N -gauge generator is therefore a linear combination of M -gauge generators:

$$[\hat{H}_a^\mu, \hat{N}^b] = \triangleright_{\alpha a}^b \hat{M}^{\alpha\mu}. \quad (5.62)$$

Similarly, one verifies that

$$e^{\epsilon_{\mathfrak{m}} \cdot \hat{M}} e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} = e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} e^{\epsilon_{\mathfrak{m}} \cdot \hat{M}}, \quad (5.63)$$

which shows that the M - and N -gauge generators commute:

$$[\hat{M}_\alpha^\mu, \hat{N}_a] = 0. \quad (5.64)$$

This completes the calculation of the algebra of gauge generators in $2BF$ theory.

Complete gauge-symmetry group of the $2BF$ action

Combining the preceding results, the algebra of the generators of the complete gauge-symmetry group is as follows.

- The algebra $\mathfrak{g}$ of the group G in the 2-crossed module $(H \xrightarrow{\partial} G, \triangleright,)$ satisfies

$$[\hat{G}_\alpha, \hat{G}_\beta] = f_{\alpha\beta}{}^\gamma \hat{G}_\gamma. \quad (5.65)$$

- The algebra of $\tilde{H}$, consisting of the H -gauge generators, is

$$[\hat{H}_a^\mu, \hat{H}_b^\nu] = 0, \quad (5.66)$$

- The algebra of the M -gauge generators is

$$[\hat{M}_\alpha^\mu, \hat{M}_\beta^\nu] = 0. \quad (5.67)$$

- The algebra of the N -gauge generators is

$$[\hat{N}_a, \hat{N}_b] = 0. \quad (5.68)$$

- The commutators between the generators of $\tilde{M}$ and $\tilde{N}$ are

$$[\hat{M}_\alpha^\mu, \hat{N}_a] = 0. \quad (5.69)$$

- The action of the generators of $\tilde{H}$ on the M - and N -gauge generators is

$$\begin{aligned} [\hat{H}_a^\mu, \hat{N}^b] &= \triangleright_{\alpha a}^b \hat{M}^{\alpha\mu}, \\ [\hat{H}_a^\mu, \hat{M}_\alpha^\nu] &= 0, \end{aligned} \quad (5.70)$$

Diffeomorphisms

As in BF theory, if diffeomorphisms are symmetries, their form variations can be expressed as the sum of the variations under gauge and HT transformations:

$$\delta_0^{\text{diff}} \phi = -\delta_0^{\text{gauge}} \phi - \delta_0^{\text{HT}} \phi. \quad (5.73)$$

Specifically, the $2BF$ action depends on α^α_μ , $\beta^a_{\mu\nu}$, $B^\alpha_{\mu\nu}$, and C^a_μ . The HT-transformation parameters $\epsilon^{\text{HT}\alpha\beta}_{\mu\nu\rho}$ and $\epsilon^{\text{HT}ab}_{\mu\nu\rho}$ are defined by (4.55),

$$\begin{aligned} \delta_0^{\text{HT}} \alpha^\alpha_\mu &= \frac{1}{2} \epsilon^{\text{HT}\alpha\beta}_{\mu\nu\rho} \frac{\delta S}{\delta B^\beta_{\nu\rho}}, & \delta_0^{\text{HT}} B^\alpha_{\mu\nu} &= -\epsilon^{\text{HT}\alpha\beta}_{\rho\mu\nu} \frac{\delta S}{\delta \alpha^\beta_\rho}, \\ \delta_0^{\text{HT}} \beta^a_{\mu\nu} &= \epsilon^{\text{HT}ab}_{\mu\nu\rho} \frac{\delta S}{\delta C^b_\rho}, & \delta_0^{\text{HT}} C^a_\mu &= -\frac{1}{2} \epsilon^{\text{HT}ab}_{\nu\rho\mu} \frac{\delta S}{\delta \beta^b_{\nu\rho}}, \end{aligned} \quad (5.74)$$

while the gauge parameters ϵ_g^α , $\epsilon_h^a_\mu$, $\epsilon_m^\alpha_\mu$, and ϵ_n^a are defined in Theorems 10–13. A choice of these parameters exists for which equation (4.57) holds for every field. In particular, choose the gauge parameters

$$\epsilon_g^\alpha = -\xi^\lambda \alpha^\alpha_\lambda, \quad \epsilon_h^a_\mu = \xi^\lambda \beta^a_{\mu\lambda}, \quad \epsilon_m^\alpha_\mu = \xi^\lambda B^\alpha_{\mu\lambda}, \quad \epsilon_n^a = -\xi^\lambda C^a_\lambda, \quad (5.75)$$

and the HT parameters

$$\epsilon^{\text{HT}\alpha\beta}_{\mu\nu\rho} = \xi^\lambda g^{\alpha\beta} \epsilon_{\mu\nu\rho\lambda}, \quad \epsilon^{\text{HT}ab}_{\mu\nu\rho} = \xi^\lambda g^{ab} \epsilon_{\lambda\mu\nu\rho}, \quad (5.76)$$

then equation (5.73) gives precisely the standard form variations corresponding to diffeomorphisms:

$$\begin{aligned} \delta_0^{\text{diff}} \alpha^\alpha_\mu &= -\partial_\mu \xi^\lambda \alpha^\alpha_\lambda - \xi^\lambda \partial_\lambda \alpha^\alpha_\mu, \\ \delta_0^{\text{diff}} \beta^a_{\mu\nu} &= -\partial_\mu \xi^\lambda \beta^a_{\lambda\nu} - \partial_\nu \xi^\lambda \beta^a_{\mu\lambda} - \xi^\lambda \partial_\lambda \beta^a_{\mu\nu}, \\ \delta_0^{\text{diff}} B^\alpha_{\mu\nu} &= -\partial_\mu \xi^\lambda B^\alpha_{\lambda\nu} - \partial_\nu \xi^\lambda B^\alpha_{\mu\lambda} - \xi^\lambda \partial_\lambda B^\alpha_{\mu\nu}, \\ \delta_0^{\text{diff}} C^a_\mu &= -\partial_\mu \xi^\lambda C^a_\lambda - \xi^\lambda \partial_\lambda C^a_\mu. \end{aligned} \quad (5.77)$$

This establishes that diffeomorphisms are indeed symmetries of the theory, although they are not contained in the complete gauge-symmetry group $\mathcal{G}_{2BF}$, but in the direct product of that group with the HT symmetry group.

5.2 General relativity

An important example of a crossed-module structure consists of a vector space V and its isometry group O . The vector space V may be regarded as an Abelian Lie group whose operation is vector addition, so a representation of O on V becomes an action $\triangleright$ of O on the group V . To define the crossed module $(V \xrightarrow{\partial} O, \triangleright)$, one also introduces a trivial homomorphism $\partial : V \rightarrow O$, mapping every element of V to the identity of O . The *Poincaré 2-group*, or its equivalent crossed module, is constructed in this way. The Lie groups are

$$G = SO(3, 1), \quad H = \mathbb{R}^4, \quad (5.78)$$

the map ∂ is trivial, and $\triangleright$ is the natural action of $SO(3, 1)$ on $\mathbb{R}^4$, defined by

$$M_{ab} \triangleright P_c = \eta_{[bc} P_{a]}, \quad (5.79)$$

where M_{ab} and P_a denote the generators of $SO(3, 1)$ and $\mathbb{R}^4$. By the crossed-module axioms, the action of $SO(3, 1)$ on itself is conjugation. At the algebra level this is the adjoint representation, so it is specified by the standard commutation relations for the generators of $SO(3, 1)$:

$$M_{ab} \triangleright M_{cd} = [M_{ab}, M_{cd}] \equiv \eta_{ad} M_{bc} - \eta_{ac} M_{bd} + \eta_{bc} M_{ad} - \eta_{bd} M_{ac}. \quad (5.80)$$

The 2-connection (α, β) is then a pair of algebra-valued differential forms: a 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{so}(3, 1))$ and a 2-form $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{R}^4)$,

$$\alpha = \omega^{ab} M_{ab}, \quad \beta = \beta^a P_a, \quad (5.81)$$

where ω^{ab} is the spin connection. The corresponding 2-curvature, the ordered pair $(\mathcal{F}, \mathcal{G})$, is

$$\begin{aligned} \mathcal{F} &= (d\omega^{ab} + \omega^a_c \wedge \omega^{cb}) M_{ab} \equiv R^{ab} M_{ab}, \\ \mathcal{G} &= (d\beta^a + \omega^a_b \wedge \beta^b) P_a \equiv \nabla \beta^a P_a \equiv G^a P_a, \end{aligned} \quad (5.82)$$

Because the homomorphism ∂ is trivial, the fake curvature equals the ordinary curvature. Defining the bilinear forms

$$\langle M_{ab}, M_{cd} \rangle_{\mathfrak{g}} = \eta_{a[c} \eta_{b]d}, \quad \langle P_a, P_b \rangle_{\mathfrak{h}} = \eta_{ab}, \quad (5.83)$$

one finds that the 1-form C^a transforms under Lorentz transformations and diffeomorphisms in the same way as the tetrad 1-form e^a ; thus C^a may be identified with the tetrad. The $2BF$ action (5.1) for the Poincaré 2-group is therefore

$$S_{2BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a. \quad (5.84)$$

The identification $C^a \equiv e^a$ is crucial, since it makes the tetrad fields explicit in the $2BF$ action for the Poincaré group. To introduce the degrees of freedom of general relativity into the action (5.84), one adds the term

$$S = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right). \quad (5.85)$$

Here λ_{ab} is a Lagrange-multiplier 2-form and l_p is the Planck length. Varying the action (5.85) with respect to B_{ab} , e_a , ω_{ab} , β_a , and λ_{ab} gives

$$R_{ab} - \lambda_{ab} = 0, \quad (5.86)$$

$$\nabla \beta_a + \frac{1}{8\pi l_p^2} \varepsilon_{abcd} \lambda^{bc} \wedge e^d = 0, \quad (5.87)$$

$$\nabla B_{ab} - e_{[a} \wedge \beta_{b]} = 0, \quad (5.88)$$

$$\nabla e_a = 0, \quad (5.89)$$

$$B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d = 0. \quad (5.90)$$

Equations (5.89) and (5.90) imply $\nabla B^{ab} = 0$, and equation (5.88) then gives $e_{[a} \wedge \beta_{b]} = 0$. Assuming invertible tetrads, $e \equiv \det(e^a_\mu) \neq 0$, this condition is equivalent to $\beta^a = 0$ [15]. Equations (5.86), (5.88), (5.89), and (5.90) further give

$$\lambda^{ab}_{\mu\nu} = R^{ab}_{\mu\nu}, \quad \beta^a_{\mu\nu} = 0, \quad B_{ab\mu\nu} = \frac{1}{8\pi l_p^2} \varepsilon_{abcd} e^c_\mu e^d_\nu, \quad \omega^{ab}_\mu = \Delta^{ab}_\mu. \quad (5.91)$$

Here the Ricci rotation coefficients are

$$\Delta^{ab}_\mu \equiv \frac{1}{2} (c^{abc} - c^{cab} + c^{bca}) e_{c\mu}, \quad (5.92)$$

where

$$c^{abc} = e^\mu_b e^\nu_c (\partial_\mu e^a_\nu - \partial_\nu e^a_\mu) . \quad (5.93)$$

The last equation expresses the spin connection ω^{ab} as a function of the tetrads, and hence the 2-form R^{ab} can also be written in this form. The remaining equation (5.87) is the tetrad equation of motion:

$$\varepsilon_{abcd} R^{bc} \wedge e^d = 0 . \quad (5.94)$$

This is precisely Einstein's equation written in the language of differential forms:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0 .$$

We conclude that the action (5.85) is classically equivalent to *general relativity*.

5.3 Einstein–Yang–Mills theory

As noted above, the principal advantage of the theory (5.85) over the Plebanski model is the explicit presence of tetrad fields in its topological sector. This permits matter fields to be coupled to gravity, as in [15]. A Yang–Mills matter field can also be coupled within the 2-group formalism [16]. The Poincaré 2-group can be extended to include $SU(N)$ gauge fields. We choose the following Lie groups for the crossed module:

$$G = SO(3, 1) \times SU(N) , \quad H = \mathbb{R}^4 , \quad (5.95)$$

The action $\triangleright$ of G is chosen as follows. As for the Poincaré 2-group, G acts on itself by conjugation. In the action of G on H , the subgroup $SO(3, 1)$ acts on $\mathbb{R}^4$ through the vector representation (5.79), whereas the action of the subgroup $SU(N)$ on H is trivial:

$$\tau_I \triangleright P_a = 0 , \quad (5.96)$$

where τ_I are the generators of $SU(N)$. The map $\partial : H \rightarrow G$ remains trivial. The form of the 2-connection (α, β) reflects the structure of G : the direct product decomposes α into two terms, one for each subgroup,

$$\alpha = \omega^{ab} M_{ab} + A^I \tau_I , \quad \beta = \beta^a P_a , \quad (5.97)$$

where A^I is the gauge-connection 1-form. The component $\mathcal{F}$ of the 2-curvature pair $\mathcal{F}, \mathcal{G}$ becomes

$$\mathcal{F} = R^{ab} M_{ab} + F^I \tau_I , \quad F^I \equiv dA^I + f_{JK}^I A^J \wedge A^K , \quad (5.98)$$

while $\mathcal{G}$ remains the same as for the Poincaré 2-group, by (5.96). Finally, the direct-product structure of G means that the Killing form $\langle _, _ \rangle_{\mathfrak{g}}$ splits into the Killing forms of $SO(3, 1)$ and $SU(N)$:

$$\langle M_{ab}, M_{cd} \rangle = \eta_{a[c} \eta_{b]d} , \quad \langle \tau_I, \tau_J \rangle = \delta_{IJ} , \quad \langle M_{ab}, \tau_I \rangle_{\mathfrak{g}} = 0 . \quad (5.99)$$

The crossed-module axioms imply that this Killing form is invariant under both G - and H -gauge transformations. For this crossed module, the topological $2BF$ action (5.1) is

$$S_{2BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + B^I \wedge F_I + e_a \wedge \nabla \beta^a , \quad (5.100)$$

where $B^I \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{su}(N))$ is a new Lagrange multiplier. The action (5.100) is topological, so appropriate constraints must be added to obtain nontrivial dynamics. The constraint producing general relativity is given in (5.85); the constraint producing the gauge-field dynamics is the

one in (4.63), with $\delta^a \rightarrow e^a$. Thus the action for a Yang–Mills field coupled to Einstein–Cartan gravity is

$$\begin{aligned} S = \int_{\mathcal{M}_4} & B^{ab} \wedge R_{ab} + B^I \wedge F_I + e_a \wedge \nabla \beta^a \\ & - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right) + \lambda^I \wedge \left(B_I - \frac{12}{g} M_{abI} e^a \wedge e^b \right) \\ & + \zeta^{abI} \left(M_{abI} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f - g_{IJ} F^J \wedge e_a \wedge e_b \right). \end{aligned} \quad (5.101)$$

The constraint that produces a Yang–Mills field in curved spacetime in (5.101) is obtained from the Minkowski-space Yang–Mills action (4.63) by replacing the nondynamical background field δ^a appearing in the action (4.63) with the dynamical tetrad e^a . The relation between these fields is already suggested by equation (4.64), which relates δ^a to the flat Minkowski metric $\eta_{\mu\nu}$. After the replacement, the field becomes dynamical through the gravitational sector, while (4.64) becomes the standard relation between the tetrad and spacetime metric,

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu. \quad (5.102)$$

This replacement makes the action (5.101) background independent, as required in general relativity. Once again, the construction is possible because tetrads are present in the topological sector of (5.85).

Varying the action (5.101) with respect to B_{ab} , ω_{ab} , β_a , λ_{ab} , ζ^{abI} , M_{abI} , B_I , λ^I , A^I , and e^a gives the equations of motion

$$R^{ab} - \lambda^{ab} = 0, \quad (5.103)$$

$$\nabla B^{ab} - e^{[a} \wedge \beta^{b]} = 0, \quad (5.104)$$

$$\nabla e^a = 0, \quad (5.105)$$

$$B_{ab} - \frac{1}{16\pi l_p^2} \varepsilon_{abcd} e^c \wedge e^d = 0, \quad (5.106)$$

$$M_{abI} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f - F_I \wedge e_a \wedge e_b = 0, \quad (5.107)$$

$$-\frac{12}{g} \lambda^I \wedge e^a \wedge e^b + \zeta^{abI} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f = 0, \quad (5.108)$$

$$F_I + \lambda_I = 0, \quad (5.109)$$

$$B_I - \frac{12}{g} M_{abI} e^a \wedge e^b = 0, \quad (5.110)$$

$$-dB_I + B_K \wedge f_{JI}{}^K A^J + d(\zeta_I^{ab} e_a \wedge e_b) - \zeta_K^{ab} e_a \wedge e_b \wedge f_{JI}{}^K A^J = 0, \quad (5.111)$$

$$\nabla \beta_a + \frac{1}{8\pi l_p^2} \varepsilon_{abcd} \lambda^{bc} \wedge e^d - \frac{24}{g} M_{abI} \lambda^I \wedge e^b + 4\zeta^{efI} M_{efI} \varepsilon_{abcd} e^b \wedge e^c \wedge e^d - 2\zeta_{ab}{}^I F_I \wedge e^b = 0. \quad (5.112)$$

This system describes two dynamical fields, e_a and A^I , while all other variables can be expressed

in terms of them and their derivatives, as follows from equations (5.103)–(5.110):

$$\begin{aligned} \lambda_{ab\mu\nu} &= R_{ab\mu\nu}, \quad \beta_{a\mu\nu} = 0, \quad \omega_{ab\mu} = \triangle_{ab\mu}, \quad \lambda_{abI} = F_{abI}, \quad B_{\mu\nu I} = -\frac{e}{2g}\varepsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}{}_I, \\ B_{ab\mu\nu} &= \frac{1}{8\pi l_p^2}\varepsilon_{abcd}e^c{}_\mu e^d{}_\nu, \quad M_{abI} = -\frac{1}{4eg}\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}{}^I e^a{}_\rho e^b{}_\sigma, \quad \zeta^{abI} = \frac{1}{4eg}\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}{}^I e^a{}_\rho e^b{}_\sigma. \end{aligned} \quad (5.113)$$

Substituting these expressions into the equations of motion (5.111) and (5.112) gives the equation of motion for A^I ,

$$\nabla_\rho F^{I\rho\mu} \equiv \partial_\rho F^{I\rho\mu} + \Gamma^\rho{}_{\lambda\rho} F^{I\lambda\mu} + f_{JK}{}^I A^J{}_\rho F^{K\rho\mu} = 0, \quad (5.114)$$

where $\Gamma^\lambda{}_{\mu\nu}$ denotes the Levi-Civita connection, and the equation of motion for e^a ,

$$R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R = 8\pi l_p^2 T^{\mu\nu}, \quad (5.115)$$

where

$$T^{\mu\nu} \equiv -\frac{1}{4g} \left(F_{\rho\sigma}{}^I F^{\rho\sigma}{}_I g^{\mu\nu} + 4F^{\mu\rho}{}_I F^{\nu I}{}_\rho \right). \quad (5.116)$$

The system (5.113)–(5.116) is equivalent to the system (5.103)–(5.112). Notice that once again $\beta^a = 0$, as in pure gravity.

The choice of gauge group for Yang–Mills theory generalizes straightforwardly from $SU(N)$ to a more complex case such as $SU(3) \times SU(2) \times U(1)$.

Chapter 6

$3BF$ theory

Although the 2-group structure successfully describes gravitational and gauge fields, it is insufficient for the remaining matter fields, such as scalar and fermionic fields. Describing these fields requires one further step along the categorical ladder, generalizing the algebraic structure of a 2-group to that of a 3-group. We shall see that the 3-group structure successfully accommodates all Standard Model fields coupled to gravity. Moreover, it possesses a third type of gauge transformation, absent from the 2-group structure, which corresponds to the choice of scalar and fermionic fields in the theory. This unexpected and intriguing result is analyzed in detail in [16].

The structure of this chapter follows Chapters 4 and 5, which treated BF and $2BF$ theory. We first define and analyze the symmetries of the topological $3BF$ action in Section 6.1. Subsection 6.1.1 presents the Hamiltonian analysis of $3BF$ theory, yielding its complete canonical structure, its first- and second-class constraints, and their algebra. Using these results, the remainder of Subsection 6.1.1 analyzes the Bianchi identities obeyed by the first-class constraints, which reduce the number of independent first-class constraints. Combining these results gives the number of local propagating degrees of freedom and shows that $3BF$ theory is topological. Finally, the subsection constructs the gauge-symmetry generator by the Castellani procedure, whose calculational details are given in Appendix D.3.2, and computes the form variations of the variables and their conjugate momenta.

Subsection 6.1.2 contains the principal results of our work and analyzes the gauge symmetries of the $3BF$ action. The variations of all variables and their canonical momenta obtained in the preceding subsection are infinitesimal gauge transformations on a spatial hypersurface Σ_3 , with the time component of the transformation parameter set to zero. From them we can infer the infinitesimal transformations throughout spacetime. Their finite forms then yield five types of gauge transformations: the known G -, H -, and L -gauge transformations, together with the M - and N -gauge transformations obtained here as new results. Computing the commutators of their generators reveals an important difference from $2BF$ theory: in $3BF$ theory the H -gauge transformations do not form a group. Instead, the gauge group $\tilde{H}_L$, formed jointly by the H - and L -gauge transformations, plays an essential role. The commutator calculations are detailed in Appendix D.3.3, and the subsection concludes by assembling the complete gauge-symmetry-group structure.

Adding suitable constraints to the topological $3BF$ action produces *constrained $3BF$ actions* with nontrivial dynamics. Sections 6.2 and 6.3 show how the *Klein–Gordon* and *Dirac fields* in curved spacetime can be written in this form. For completeness, Section 6.4 also analyzes *Weyl* and *Majorana fields* interacting with Einstein–Cartan gravity. Section 6.5 applies these results to construct a constrained $3BF$ action describing all Standard Model matter coupled to gravity. The chapter ends with a simple model of scalar electrodynamics as a constrained $3BF$ theory, whose complete Hamiltonian analysis is given in Appendix C.

We conclude that gravity, gauge fields, and matter fields are all successfully incorporated into the 3-group formalism. The classical theory is thereby written in a form adapted to covariant quantization.

6.1 Topological 3BF theory

As for the BF and $2BF$ actions, we define a gauge-invariant topological $3BF$ action on a manifold $\mathcal{M}_4$ for the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \})$:

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}. \quad (6.1)$$

In this equation, the 2-form $\mathcal{F} \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$, 3-form $\mathcal{G} \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{h})$, and 4-form $\mathcal{H} \in \mathcal{A}^4(\mathcal{M}_4, \mathfrak{l})$ are the components of the 3-curvature defined in (2.118). In addition to the Lagrange multipliers $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$ and $C \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ already present in $2BF$ theory, $3BF$ theory contains the Lagrange multiplier $D \in \mathcal{A}^0(\mathcal{M}_4, \mathfrak{l})$, whose interesting physical interpretation is discussed below. The brackets $\langle _, _ \rangle_{\mathfrak{g}}$, $\langle _, _ \rangle_{\mathfrak{h}}$, and $\langle _, _ \rangle_{\mathfrak{l}}$ denote G -invariant symmetric nondegenerate bilinear forms on $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$.

Varying the topological $3BF$ action (6.1) with respect to B^α , C^a , and D^A (where A labels the generators of L) gives

$$\mathcal{F}^\alpha = 0, \quad \mathcal{G}^a = 0, \quad \mathcal{H}^A = 0, \quad (6.2)$$

while variation with respect to α^α , β^a , and γ^A gives

$$dB_\alpha - f_{\alpha\beta}^\gamma B_\gamma \wedge \alpha^\beta - \triangleright_{\alpha a}^b C_b \wedge \beta^a + \triangleright_{\alpha B}^A D_A \wedge \gamma^B = 0, \quad (6.3)$$

$$dC_a - \partial_a^\alpha B_\alpha + \triangleright_{\alpha a}^b C_b \wedge \alpha^\alpha + 2X_{\{ab\}}^A D_A \wedge \beta^b = 0, \quad (6.4)$$

$$dD_A - \triangleright_{\alpha A}^B D_B \wedge \alpha^\alpha + \delta_A^a C_a = 0. \quad (6.5)$$

6.1.1 Hamiltonian analysis of topological 3BF theory

This section presents the complete Hamiltonian analysis of topological $3BF$ theory [19]. Assuming that the spacetime manifold $\mathcal{M}_4$ is globally hyperbolic, the Lagrangian of the $3BF$ action can be defined on a spatial foliation Σ_3 :

$$L_{3BF} = \int_{\Sigma_3} d^3\vec{x} \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} B_{\mu\nu}^\alpha \mathcal{F}_{\rho\sigma}^\beta g_{\alpha\beta} + \frac{1}{3!} C_\mu^a \mathcal{G}_{\nu\rho\sigma}^b g_{ab} + \frac{1}{4!} D^A \mathcal{H}_{\mu\nu\rho\sigma}^B g_{AB} \right). \quad (6.6)$$

For the Lagrangian (6.6), the conjugate momenta of $B_{\mu\nu}^\alpha$, α_μ^α , C_μ^a , $\beta_{\mu\nu}^a$, D^A , and $\gamma_{\mu\nu\rho}^A$, obtained by varying with respect to the time derivatives, are

$$\begin{aligned} \pi(B)_\alpha^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 B_{\mu\nu}^\alpha} = 0, \\ \pi(\alpha)_\alpha^\mu &= \frac{\delta L}{\delta \partial_0 \alpha_\mu^\alpha} = \frac{1}{2} \epsilon^{0\mu\nu\rho} B_{\alpha\nu\rho}, \\ \pi(C)_a^\mu &= \frac{\delta L}{\delta \partial_0 C_\mu^a} = 0, \\ \pi(\beta)_a^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 \beta_{\mu\nu}^a} = -\epsilon^{0\mu\nu\rho} C_{a\rho}, \\ \pi(D)_A &= \frac{\delta L}{\delta \partial_0 D^A} = 0, \\ \pi(\gamma)_A^{\mu\nu\rho} &= \frac{\delta L}{\delta \partial_0 \gamma_{\mu\nu\rho}^A} = \epsilon^{0\mu\nu\rho} D_A. \end{aligned} \quad (6.7)$$

Since the relations (6.7) cannot be inverted for the time derivatives, the theory has the following primary constraints:

$$\begin{aligned}
P(B)_\alpha{}^{\mu\nu} &\equiv \pi(B)_\alpha{}^{\mu\nu} \approx 0, \\
P(\alpha)_\alpha{}^\mu &\equiv \pi(\alpha)_\alpha{}^\mu - \frac{1}{2}\epsilon^{0\mu\nu\rho} B_{\alpha\nu\rho} \approx 0, \\
P(C)_a{}^\mu &\equiv \pi(C)_a{}^\mu \approx 0, \\
P(\beta)_a{}^{\mu\nu} &\equiv \pi(\beta)_a{}^{\mu\nu} + \epsilon^{0\mu\nu\rho} C_{a\rho} \approx 0, \\
P(D)_A &\equiv \pi(D)_A \approx 0, \\
P(\gamma)_A{}^{\mu\nu\rho} &\equiv \pi(\gamma)_A{}^{\mu\nu\rho} - \epsilon^{0\mu\nu\rho} D_A \approx 0.
\end{aligned} \tag{6.8}$$

We use the fundamental Poisson brackets

$$\begin{aligned}
\{B^\alpha{}_{\mu\nu}(\vec{x}), \pi(B)_\beta{}^{\rho\sigma}(\vec{y})\} &= 2\delta_\beta^\alpha \delta_{[\mu}^\rho \delta_{\nu]}^\sigma \delta^{(3)}(\vec{x} - \vec{y}), \\
\{\alpha^\alpha{}_\mu(\vec{x}), \pi(\alpha)_\beta{}^\nu(\vec{y})\} &= \delta_\beta^\alpha \delta_\mu^\nu \delta^{(3)}(\vec{x} - \vec{y}), \\
\{C^a{}_\mu(\vec{x}), \pi(C)_b{}^\nu(\vec{y})\} &= \delta_b^a \delta_\mu^\nu \delta^{(3)}(\vec{x} - \vec{y}), \\
\{\beta^a{}_{\mu\nu}(\vec{x}), \pi(\beta)_b{}^{\rho\sigma}(\vec{y})\} &= 2\delta_b^a \delta_{[\mu}^\rho \delta_{\nu]}^\sigma \delta^{(3)}(\vec{x} - \vec{y}), \\
\{D^A(\vec{x}), \pi(D)_B(\vec{y})\} &= \delta_B^A \delta^{(3)}(\vec{x} - \vec{y}), \\
\{\gamma^A{}_{\mu\nu\rho}(\vec{x}), \pi(\gamma)_B{}^{\sigma\tau\xi}(\vec{y})\} &= 3!\delta_B^A \delta_{[\mu}^\sigma \delta_\nu^\tau \delta_{\rho]}^\xi \delta^{(3)}(\vec{x} - \vec{y}),
\end{aligned} \tag{6.9}$$

to calculate the *algebra of primary constraints*:

$$\begin{aligned}
\{P(B)_\alpha{}^{jk}(\vec{x}), P(\alpha)_\beta{}^i(\vec{y})\} &= \epsilon^{0ijk} g_{\alpha\beta}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{P(C)_a{}^k(\vec{x}), P(\beta)_b{}^{ij}(\vec{y})\} &= -\epsilon^{0ijk} g_{ab}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{P(D)_A(\vec{x}), P(\gamma)_B{}^{ijk}(\vec{y})\} &= \epsilon^{0ijk} g_{AB}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{6.10}$$

All remaining Poisson brackets vanish. The *canonical on-shell Hamiltonian* is

$$\begin{aligned}
H_c = \int_\Sigma d^3\vec{x} \left[\frac{1}{2} \pi(B)_\alpha{}^{\mu\nu} \partial_0 B^\alpha{}_{\mu\nu} + \pi(\alpha)_\alpha{}^\mu \partial_0 \alpha^\alpha{}_\mu + \pi(C)_a{}^\mu \partial_0 C^a{}_\mu \right. \\
\left. + \frac{1}{2} \pi(\beta)_a{}^{\mu\nu} \partial_0 \beta^a{}_{\mu\nu} + \pi(D)_A \partial_0 D^A + \frac{1}{3!} \pi(\gamma)_A{}^{\mu\nu\rho} \partial_0 \gamma^A{}_{\mu\nu\rho} \right] - L.
\end{aligned} \tag{6.11}$$

Expanding the 3-curvature using (2.3.5), the Hamiltonian (6.11) separates into terms proportional to products of primary constraints and time derivatives, and a remainder. Since the primary constraints vanish on shell, the canonical Hamiltonian becomes

$$\begin{aligned}
H_c \approx - \int_\Sigma d^3\vec{x} \epsilon^{0ijk} \left[\frac{1}{2} B_{\alpha 0i} \mathcal{F}^\alpha{}_{jk} + \frac{1}{6} C_{a0} \mathcal{G}^a{}_{ijk} + \beta^a{}_{0i} \left(\nabla_j C_{ak} - \frac{1}{2} \partial_a{}^\alpha B_{\alpha jk} + \beta^b{}_{jk} D_A X_{\{ab\}}{}^A \right) \right. \\
\left. + \frac{1}{2} \alpha^\alpha{}_0 \left(\nabla_i B_{\alpha jk} - C_{ai} \triangleright_{\alpha b}{}^a \beta^b{}_{jk} + \frac{1}{3} D_A \triangleright_{\alpha B}{}^A \gamma^B{}_{ijk} \right) + \frac{1}{2} \gamma^A{}_{0ij} \left(\nabla_k D_A + C_{ak} \delta_A{}^a \right) \right].
\end{aligned} \tag{6.12}$$

Adding products of Lagrange multipliers λ and primary constraints gives the off-shell total Hamiltonian:

$$H_T = H_c + \int d^3\vec{x} \left[\frac{1}{2} \lambda(B)^\alpha{}_{\mu\nu} P(B)_{\alpha}{}^{\mu\nu} + \lambda(\alpha)^\alpha{}_\mu P(\alpha)_\alpha{}^\mu + \lambda(C)^a{}_\mu P(C)_a{}^\mu + \frac{1}{2} \lambda(\beta)^a{}_{\mu\nu} P(\beta)_a{}^{\mu\nu} + \lambda(D)^A P(D)_A + \frac{1}{3!} \lambda(\gamma)^A{}_{\mu\nu\rho} P(\gamma)_A{}^{\mu\nu\rho} \right]. \quad (6.13)$$

For the primary constraints to be preserved during evolution, they must satisfy the consistency conditions

$$\dot{P} \equiv \{P, H_T\} \approx 0, \quad (6.14)$$

for every primary constraint P . Applying (6.14) to the primary constraints $P(B)_\alpha{}^{0i}$, $P(\alpha)_\alpha{}^0$, $P(C)_a{}^0$, $P(\beta)_a{}^{0i}$, and $P(\gamma)_A{}^{0ij}$ gives the secondary constraints $\mathcal{S}$:

$$\begin{aligned} \mathcal{S}(\mathcal{F})_\alpha{}^i &\equiv \frac{1}{2} \epsilon^{0ijk} \mathcal{F}_{\alpha jk} \approx 0, \\ \mathcal{S}(\nabla B)_\alpha &\equiv \frac{1}{2} \epsilon^{0ijk} (\nabla_{[i} B_{\alpha j]k} - C_{a[i} \triangleright_{\alpha b}{}^a \beta^b{}_{j]k} + \frac{1}{3} D_A \triangleright_{\alpha B}{}^A \gamma^B{}_{ijk}) \approx 0, \\ \mathcal{S}(\mathcal{G})_a &\equiv \frac{1}{6} \epsilon^{0ijk} \mathcal{G}_{aijk} \approx 0, \\ \mathcal{S}(\nabla C)_a{}^i &\equiv \epsilon^{0ijk} (\nabla_{[j} C_{ak]} - \frac{1}{2} \partial_a{}^\alpha B_{\alpha jk} + \beta^b{}_{jk} D_A X_{\{ab\}}{}^A) \approx 0, \\ \mathcal{S}(\nabla D)_A{}^{ij} &\equiv \epsilon^{0ijk} (\nabla_k D_A + C_{ak} \delta_A{}^a) \approx 0. \end{aligned} \quad (6.15)$$

whereas for the primary constraints $P(\alpha)_\alpha{}^k$, $P(B)_\alpha{}^{jk}$, $P(\beta)_a{}^{jk}$, $P(C)_a{}^k$, $P(\gamma)_A{}^{ijk}$, and $P(D)_A$, the consistency conditions determine the Lagrange multipliers:

$$\begin{aligned} \lambda(B)_{\alpha ij} &\approx \nabla_i B_{\alpha 0j} - \nabla_j B_{\alpha 0i} + C_{a0} \beta^b{}_{ij} \triangleright_{\alpha b}{}^a + C_{bi} \triangleright_{\alpha}{}^b{}_a \beta^a{}_{0j} \\ &\quad - C_{bj} \triangleright_{\alpha}{}^b{}_a \beta^a{}_{0i} + f_{\beta\gamma}{}^\alpha \alpha^\beta{}_0 B^\gamma{}_{ij} + D_B \gamma^A{}_{0ij} \triangleright_{\alpha}{}^B{}_A, \\ \lambda(\alpha)^\alpha{}_i &\approx \nabla_i \alpha^\alpha{}_0 + \partial_a{}^\alpha \beta^a{}_{0i}, \\ \lambda(C)^a{}_i &\approx \nabla_i C^a{}_0 + C^b{}_i \triangleright_{\alpha}{}^a{}_b \alpha^\alpha{}_0 - 2\beta_{b0i} D_A X^{\{ba\}}{}^A + B_{\alpha 0i} \partial^{a\alpha}, \\ \lambda(\beta)^a{}_{ij} &\approx \nabla_i \beta^a{}_{0j} - \nabla_j \beta^a{}_{0i} - \beta^b{}_{ij} \triangleright_{\alpha b}{}^a \alpha^\alpha{}_0 + \gamma^A{}_{0ij} \delta_A{}^a, \\ \lambda(D)_A &\approx \alpha^\alpha{}_0 D_B \triangleright_{\alpha A}{}^B - C_{a0} \delta_A{}^a, \\ \lambda(\gamma)^A{}_{ijk} &\approx -2\beta^a{}_{0i} \beta^b{}_{jk} X_{\{ab\}}{}^A + 2\beta^a{}_{0j} \beta^b{}_{ik} X_{\{ab\}}{}^A - 2\beta^a{}_{0k} \beta^b{}_{ij} X_{\{ab\}}{}^A \\ &\quad - \alpha^\alpha{}_0 \triangleright_{\alpha B}{}^A \gamma^B{}_{ijk} + \nabla_i \gamma^A{}_{0jk} - \nabla_j \gamma^A{}_{0ik} + \nabla_k \gamma^A{}_{0ij}. \end{aligned} \quad (6.16)$$

The remaining Lagrange multipliers,

$$\lambda(B)^\alpha{}_{0i}, \quad \lambda(\alpha)^\alpha{}_0, \quad \lambda(C)^a{}_0, \quad \lambda(\beta)^a{}_{0i}, \quad \lambda(\gamma)^A{}_{0ij} \quad (6.17)$$

remain undetermined by the consistency conditions for the primary constraints. The secondary constraints must also be preserved. Their consistency conditions produce no further constraints:

$$\begin{aligned}
\{\mathcal{S}(\mathcal{F})^{\alpha i}, H_T\} &= f_{\beta\gamma}{}^{\alpha}\mathcal{S}(\mathcal{F})^{\beta i}\alpha^{\gamma}_0, \\
\{\mathcal{S}(\nabla B)_{\alpha}, H_T\} &= f_{\beta\gamma\alpha}B^{\gamma}_{0k}\mathcal{S}(\mathcal{F})^{\beta k} + f_{\beta\alpha}{}^{\gamma}\alpha^{\beta}_0\mathcal{S}(\nabla B)_{\gamma} + C_{a0}\triangleright_{\alpha b}{}^a\mathcal{S}(\mathcal{G})^b \\
&\quad - \triangleright_{\alpha a}{}^b\beta^a_{0k}\mathcal{S}(\nabla C)_b{}^k + \frac{1}{2}\triangleright_{\alpha}{}^B{}_A\gamma^A{}_{0jk}\mathcal{S}(\nabla D)_B{}^{jk}, \\
\{\mathcal{S}(\mathcal{G})^a, H_T\} &= \triangleright_{\alpha b}{}^a\beta^b_{0k}\mathcal{S}(\mathcal{F})^{\alpha k} - \alpha^{\alpha}_0\triangleright_{\alpha b}{}^a\mathcal{S}(\mathcal{G})^b, \\
\{\mathcal{S}(\nabla C)_a{}^i, H_T\} &= C_{b0}\triangleright_{\alpha}{}^b{}_a\mathcal{S}(\mathcal{F})^{\alpha i} + \triangleright_{\alpha a}{}^b\alpha^{\alpha}_0\mathcal{S}(\nabla C)_b{}^i + 2X_{\{ab\}}{}^A\beta^b_{0j}\mathcal{S}(\nabla D)_A{}^{ij}, \\
\{\mathcal{S}(\nabla D)_A{}^{ij}, H_T\} &= \alpha^{\alpha}_0\triangleright_{\alpha A}{}^B\mathcal{S}(\nabla D)_B{}^{ij}.
\end{aligned} \tag{6.18}$$

Finally, the total Hamiltonian can be written as

$$\begin{aligned}
H_T = \int_{\Sigma_3} d^3\vec{x} \Big[&\lambda(B)^{\alpha}_{0i}\Phi(B)_{\alpha}{}^i + \lambda(\alpha)^{\alpha}\Phi(\alpha)_{\alpha} + \lambda(C)^a{}_0\Phi(C)_a + \lambda(\beta)^a{}_{0i}\Phi(\beta)_a{}^i + \frac{1}{2}\lambda(\gamma)^A{}_{0ij}\Phi(\gamma)_A{}^{ij} \\
&- B_{\alpha 0i}\Phi(\mathcal{F})^{\alpha i} - \alpha_{\alpha 0}\Phi(\nabla B)^{\alpha} - C_{a0}\Phi(\mathcal{G})^a - \beta_{a0i}\Phi(\nabla C)^{\alpha i} - \frac{1}{2}\gamma_{A0ij}\Phi(\nabla D)^A{}^{ij} \Big],
\end{aligned} \tag{6.19}$$

where

$$\begin{aligned}
\Phi(B)_{\alpha}{}^i &= P(B)_{\alpha}{}^{0i}, \\
\Phi(\alpha)_{\alpha} &= P(\alpha)_{\alpha}{}^0, \\
\Phi(C)_a &= P(C)_a{}^0, \\
\Phi(\beta)_a{}^i &= P(\beta)_a{}^{0i}, \\
\Phi(\gamma)_A{}^{ij} &= P(\gamma)_A{}^{0ij}, \\
\Phi(\mathcal{F})^{\alpha i} &= \mathcal{S}(\mathcal{F})^{\alpha i} - \nabla_j P(B)^{\alpha ij} - P(C)_a{}^i\partial^{a\alpha}, \\
\Phi(\mathcal{G})_a &= \mathcal{S}(\mathcal{G})_a + \nabla_i P(C)_a{}^i - \frac{1}{2}\beta_{bij}\triangleright_{\alpha}{}^b{}_a P(B)^{\alpha ij} + P(D)^A\delta_{Aa}, \\
\Phi(\nabla C)_a{}^i &= \mathcal{S}(\nabla C)_a{}^i - \nabla_j P(\beta)_a{}^{ij} + C_{bj}\triangleright_{\alpha}{}^b{}_a P(B)^{\alpha ij} \\
&\quad - \partial_a{}^{\alpha}P(\alpha)_{\alpha}{}^i + 2D_A X_{\{ab\}}{}^A P(C)^{bi} + \beta^b_{jk}X_{\{ab\}}{}^A P(\gamma)_A{}^{ijk}, \\
\Phi(\nabla B)_{\alpha} &= \mathcal{S}(\nabla B)_{\alpha} + \nabla_i P(\alpha)_{\alpha}{}^i - \frac{1}{2}f_{\alpha\gamma}{}^{\beta}B_{\beta ij}P(B)^{\gamma ij} \\
&\quad - C_{bi}\triangleright_{\alpha a}{}^b P(C)^{ai} - \frac{1}{2}\beta_{bij}\triangleright_{\alpha a}{}^b P(\beta)^{aij} \\
&\quad - P(D)^A D_B \triangleright_{\alpha A}{}^B + \frac{1}{3!}P(\gamma)_A{}^{ijk}\gamma^B{}_{ijk}\triangleright_{\alpha B}{}^A, \\
\Phi(\nabla D)_A{}^{ij} &= \mathcal{S}(\nabla D)_A{}^{ij} + \nabla_k P(\gamma)_A{}^{ijk} - P(\beta)_a{}^{ij}\delta_A{}^a - P(B)^{\alpha ij}\triangleright_{\alpha}{}^B{}_A D_B,
\end{aligned} \tag{6.20}$$

are the first-class constraints, while

$$\begin{aligned}
\chi(B)_{\alpha}{}^{jk} &= P(B)_{\alpha}{}^{jk}, & \chi(C)_a{}^i &= P(C)_a{}^i, & \chi(D)_A &= P(D)_A, \\
\chi(\alpha)_{\alpha}{}^i &= P(\alpha)_{\alpha}{}^i, & \chi(\beta)_a{}^{ij} &= P(\beta)_a{}^{ij}, & \chi(\gamma)_A{}^{ijk} &= P(\gamma)_A{}^{ijk},
\end{aligned} \tag{6.21}$$

are the second-class constraints.

The Poisson algebra of the first-class constraints is

$$\begin{aligned}
\{ \Phi(\mathcal{G})^a(\vec{x}), \Phi(\nabla C)_b{}^i(\vec{y}) \} &= -\triangleright_{ab}{}^a \Phi(\mathcal{F})^{\alpha i}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla C)_a{}^i(\vec{x}), \Phi(\nabla C)_b{}^j(\vec{y}) \} &= -2X_{\{ab\}}{}^A \Phi(\nabla D)_A{}^{ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{G})^a(\vec{x}), \Phi(\nabla B)_\alpha(\vec{y}) \} &= \triangleright_{ab}{}^a \Phi(\mathcal{G})^b(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla C)_a{}^i(\vec{x}), \Phi(\nabla B)_\alpha(\vec{y}) \} &= \triangleright_{\alpha a}{}^b \Phi(\nabla C)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{F})^\alpha{}_i(\vec{x}), \Phi(\nabla B)_\beta(\vec{y}) \} &= f_{\beta\gamma}{}^\alpha \Phi(\mathcal{F})^\gamma{}_i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)_\alpha(\vec{x}), \Phi(\nabla B)_\beta(\vec{y}) \} &= f_{\alpha\beta}{}^\gamma \Phi(\nabla B)_\gamma(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)_\alpha(\vec{x}), \Phi(\nabla D)_A{}^{ij}(\vec{y}) \} &= \triangleright_{\alpha A}{}^B \Phi(\nabla D)_B{}^{ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{6.22}$$

The Poisson brackets of the first-class constraints with the second-class constraints are

$$\begin{aligned}
\{ \Phi(\mathcal{F})^{\alpha i}(\vec{x}), \chi(\alpha)_\beta{}^j(\vec{y}) \} &= -f_{\beta\gamma}{}^\alpha \chi(B)^{\gamma ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{G})^a(\vec{x}), \chi(\alpha)_\alpha{}^i(\vec{y}) \} &= -\triangleright_{ab}{}^a \chi(C)^{bi}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{G})^a(\vec{x}), \chi(\beta)_b{}^{ij}(\vec{y}) \} &= \triangleright_{ab}{}^a \chi(B)^{\alpha ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ai}(\vec{x}), \chi(\alpha)_\alpha{}^j(\vec{y}) \} &= -\triangleright_{ab}{}^a \chi(\beta)^{bij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ai}(\vec{x}), \chi(\beta)_b{}^{jk}(\vec{y}) \} &= -2X^{\{ac\}}{}^A g_{bc} \chi(\gamma)_A{}^{ijk}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla C)^{ai}(\vec{x}), \chi(C)_b{}^j(\vec{y}) \} &= \triangleright_{ab}{}^a \chi(B)^{\alpha ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ai}(\vec{x}), \chi(D)_A(\vec{y}) \} &= X^{\{ab\}}{}_A \chi(C)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(\alpha)_\beta{}^i(\vec{y}) \} &= f_{\beta\gamma}{}^\alpha \chi(\alpha)^{\gamma i}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(\beta)_a{}^{ij}(\vec{y}) \} &= \triangleright_{\alpha a}{}^b \chi(\beta)_b{}^{ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(\gamma)_A{}^{ijk}(\vec{y}) \} &= -\triangleright_{\alpha A}{}^B \chi(\gamma)_B{}^{ijk}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(B)_\beta{}^{ij}(\vec{y}) \} &= -f_{\beta\gamma}{}^\alpha \chi(B)^{\gamma ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(C)_a{}^i(\vec{y}) \} &= -\triangleright_{ab}{}^a \chi(C)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^\alpha(\vec{x}), \chi(D)_A(\vec{y}) \} &= -\triangleright_{\alpha A}{}^B \chi(D)_B(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla D)^{Aij}(\vec{x}), \chi(\alpha)_\alpha{}^k \} &= -\triangleright_{\alpha B}{}^A \chi(\gamma)^{Bijk}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla D)^{Aij}(\vec{x}), \chi(D)_B \} &= \frac{1}{2} \triangleright_{\alpha B}{}^A \chi(B)^{\alpha ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{6.23}$$

Finally, it is useful to calculate the Poisson brackets between the first-class constraints and the

Hamiltonian:

$$\begin{aligned}
\{\Phi(\mathcal{F})^{\alpha i}, H_T\} &= f_{\beta\gamma}{}^{\alpha}\Phi(\mathcal{F})^{\beta i}\alpha^{\gamma}_0, \\
\{\Phi(\nabla B)_{\alpha}, H_T\} &= f_{\beta\gamma\alpha}B^{\gamma}_{0k}\Phi(\mathcal{F})^{\beta k} + f_{\beta\alpha}{}^{\gamma}\alpha^{\beta}_0\Phi(\nabla B)_{\gamma} + C_{a0}\triangleright_{\alpha b}{}^a\Phi(\mathcal{G})^b \\
&\quad - \triangleright_{\alpha a}{}^b\beta^a_{0k}\Phi(\nabla C)_b{}^k + \frac{1}{2}\triangleright_{\alpha}{}^B{}_A\gamma^A{}_{0jk}\Phi(\nabla D)_B{}^{jk}, \\
\{\Phi(\mathcal{G})^a, H_T\} &= \triangleright_{\alpha b}{}^a\beta^b_{0k}\Phi(\mathcal{F})^{\alpha k} - \alpha^{\alpha}_0\triangleright_{\alpha b}{}^a\Phi(\mathcal{G})^b, \\
\{\Phi(\nabla C)_a{}^i, H_T\} &= C_{b0}\triangleright_{\alpha}{}^b{}_a\Phi(\mathcal{F})^{\alpha i} + \triangleright_{\alpha a}{}^b\alpha^{\alpha}_0\Phi(\nabla C)_b{}^i + 2X_{\{ab\}}{}^A\beta^b_{0j}\Phi(\nabla D)_A{}^{ij}, \\
\{\Phi(\nabla D)_A{}^{ij}, H_T\} &= \alpha^{\alpha}_0\triangleright_{\alpha A}{}^B\Phi(\nabla D)_B{}^{ij}.
\end{aligned} \tag{6.24}$$

Number of degrees of freedom of topological 3BF theory

The Bianchi identities (BI) for the 1-forms α and C and the 2-forms β and B are the same as in 2BF theory, equations (5.24), (5.25), (5.26), and (5.27). In addition, 3BF theory has a Bianchi identity associated with the 0-form D .

Lemma 11 (BI for the 0-form D) *Defining*

$$Q^A = dD^A + \triangleright_{\alpha B}{}^A\alpha^{\alpha} \wedge D^B, \tag{6.25}$$

one obtains the identity

$$\epsilon^{\lambda\mu\nu\rho} (\nabla_{\nu}Q^A{}_{\rho} - \triangleright_{\alpha B}{}^A F^{\alpha}{}_{\nu\rho} D^B) = 0. \tag{6.26}$$

Bianchi identities play an important role in determining the number of degrees of freedom of the action.

Let p , q , and r denote the dimensions of G , H , and L , respectively. The initial number of field components in 3BF theory follows by counting the fields in Table 6.1.

$\alpha^{\alpha}{}_{\mu}$	$\beta^a{}_{\mu\nu}$	$\gamma^A{}_{\mu\nu\rho}$	$B^{\alpha}{}_{\mu\nu}$	$C^a{}_{\mu}$	D^A
$4p$	$6q$	$4r$	$6p$	$4q$	r

Table 6.1: Initial fields in 3BF theory.

This gives $N = 10(p + q) + 5r$. Similarly, the number of independent components of the second-class constraints follows from Table (6.2).

$\chi(B)_{\alpha}{}^{jk}$	$\chi(C)_a{}^i$	$\chi(D)_A$	$\chi(\alpha)_{\alpha}{}^i$	$\chi(\beta)_a{}^{ij}$	$\chi(\gamma)_A{}^{ijk}$
$3p$	$3q$	r	$3p$	$3q$	r

Table 6.2: Second-class constraints in 3BF theory.

Thus $S = 6(p + q) + 2r$.

The first-class constraints are not all independent. Besides the identities (5.30) and (5.32), also satisfied in 2BF theory, a new identity appears in 3BF theory. Namely, the first-class constraints satisfy

$$\begin{aligned}
\nabla_i\Phi(\nabla D)_A{}^{ij} + \frac{1}{2}\delta_A{}^a\mathcal{S}(\nabla C)_a{}^j - \nabla_i\nabla_k\chi(\gamma)_A{}^{ijk} + \frac{1}{2}\chi(\beta)_a{}^{ij}\delta_A{}^a - \frac{1}{2}\triangleright_{\alpha}{}^B{}_AD_B\Phi(\mathcal{F})_{\alpha}{}^j \\
+ \frac{1}{2}\triangleright_{\alpha}{}^B{}_AD_B\partial_a{}^{\alpha}\chi(C)^{aj} = \frac{1}{2}\epsilon^{0ijk}(\nabla_iQ_{Ak} + \triangleright_{\alpha A}{}^BF^{\alpha}{}_{ik}D_B),
\end{aligned} \tag{6.27}$$

Since the right-hand side is precisely the $\lambda = 0$ component of the Bianchi identity (6.26), one obtains

$$\begin{aligned} \nabla_i \Phi(\nabla D)_A{}^{ij} + \frac{1}{2} \delta_A{}^a \mathcal{S}(\nabla C)_a{}^j - \nabla_i \nabla_k \chi(\gamma)_A{}^{ijk} + \frac{1}{2} \chi(\beta)_a{}^{ij} \delta_A{}^a \\ - \frac{1}{2} \triangleright_\alpha{}^B{}_A D_B \Phi(\mathcal{F})_\alpha{}^j + \frac{1}{2} \triangleright_\alpha{}^B{}_A D_B \partial_a{}^\alpha \chi(C)^{aj} = 0. \end{aligned} \quad (6.28)$$

The number of independent components of the first-class constraints is obtained by counting the constraints in Table 6.3 and subtracting the number of independent Bianchi identities.

$\Phi(B)_\alpha{}^i$	$\Phi(C)_a$	$\Phi(\alpha)_\alpha$	$\Phi(\beta)_a{}^i$	$\Phi(\gamma)_A{}^{ij}$	$\Phi(\mathcal{F})^{\alpha i}$	$\Phi(\mathcal{G})^a$	$\Phi(\nabla C)^{ai}$	$\Phi(\nabla B)^\alpha$	$\Phi(\nabla D)_A{}^{ij}$
$3p$	q	p	$3q$	$3r$	$3p - p$	q	$3q - q$	p	$3r - 2r$

Table 6.3: Number of first-class constraints in 3BF theory.

The number of independent first-class-constraint components is

$$F = 8(p + q) + 6r - p - q - 2r = 7(p + q) + 4r,$$

where, from the number of first-class-constraint components listed in Table 6.3, we have subtracted the p relations (5.30), the q relations (5.32), and $2r$ independent¹ relations (6.28). Using the degree-of-freedom formula (3.41), the number of degrees of freedom in 3BF theory is

$$n = 10(p + q) + 5r - 7(p + q) - 4r - \frac{6(p + q) + 2r}{2} = 0, \quad (6.29)$$

so 3BF theory has no propagating degrees of freedom.

Gauge-transformation generator for 3BF theory

The gauge-transformation generator in 3BF theory is

$$\begin{aligned} G = \int_{\Sigma_3} d^3\vec{x} \Big(& (\nabla_0 \epsilon^\alpha{}_i) \Phi(B)_\alpha{}^i - \epsilon^\alpha{}_i \Phi(\mathcal{F})_\alpha{}^i + (\nabla_0 \epsilon^\alpha) \Phi(\alpha)_\alpha + \epsilon^\alpha (f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} \\ & + C_{a0} \triangleright_{\alpha b}{}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{\alpha b}{}^a \Phi(\beta)^{b0i} - \frac{1}{2} \gamma^A{}_{0ij} \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} - \Phi(\nabla B)_\alpha) \\ & + (\nabla_0 \epsilon^a) \Phi(C)_a - \epsilon^a (\beta_{b0i} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a) \\ & + (\nabla_0 \epsilon^a{}_i) \Phi(\beta)_a{}^i - \epsilon^a{}_i (C_{b0} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} - 2\beta^b{}_{0j} X_{\{ab\}}{}^A \Phi(\gamma)_A{}^{ij} + \Phi(\nabla C)_a{}^i) \\ & + \frac{1}{2} (\nabla_0 \epsilon^A{}_{ij}) \Phi(\gamma)_A{}^{ij} - \frac{1}{2} \epsilon^A{}_{ij} \Phi(\nabla D)_A{}^{ij} \Big). \end{aligned} \quad (6.30)$$

Here $\epsilon^\alpha{}_i$, ϵ^α , ϵ^a , $\epsilon^a{}_i$, and $\epsilon^A{}_{ij}$ are independent gauge parameters. The derivation of the generator (6.30) is given in Appendix D.3.

The form variations of the variables and their conjugate momenta are calculated using

¹Equation (6.28) contains $3r$ identities, but only $2r$ are mutually independent. Taking the divergence of (6.28) gives zero identically by the Bianchi identity (5.24). Subtracting these r divergence relations, which impose no new identities on the constraints, from the total number of relations (6.28) leaves $2r$ independent identities (6.28).

(3.56):

$$\begin{aligned}
\delta_0 B^{\alpha}_{0i} &= \nabla_0 \epsilon^{\alpha}_i - f_{\beta\gamma}{}^{\alpha} \epsilon^{\beta} B^{\gamma}_{0i} \\
&\quad - \epsilon^a \triangleright_{\alpha a}{}^b \beta_{b0i} - \epsilon^a{}_i \triangleright_{\alpha a}{}^b C_{b0}, \\
\delta_0 B^{\alpha}_{ij} &= 2\nabla_{[i} \epsilon^{\alpha}_{j]} - f_{\beta\gamma}{}^{\alpha} \epsilon^{\beta} B^{\gamma}_{ij} - \epsilon^A{}_{ij} \triangleright_{\alpha A}{}^B D_B \\
&\quad - \epsilon^a \triangleright_{\alpha a}{}^b \beta_{bij} - 2\epsilon^a{}_{[j} \triangleright_{\alpha a}{}^b C_{b| i]}, \\
\delta_0 \alpha^{\alpha}_0 &= \nabla_0 \epsilon^{\alpha}, \\
\delta_0 \alpha^{\alpha}_i &= \nabla_i \epsilon^{\alpha} + \partial_a \epsilon^{\alpha} \epsilon^a_i, \\
\delta_0 C^a_0 &= \nabla_0 \epsilon^a - \epsilon^{\alpha} \triangleright_{\alpha b}{}^a C^b_0, \\
\delta_0 C^a_i &= \nabla_i \epsilon^a - \epsilon^{\alpha} \triangleright_{\alpha b}{}^a C^b_i \\
&\quad + \epsilon^{\alpha}_i \partial_a \epsilon^{\alpha} - 2\epsilon^b{}_i D_A X_{\{bc\}}{}^A g^{ac}, \\
\delta_0 \pi(B)_{\alpha}{}^{0i} &= -f_{\alpha\beta}{}^{\gamma} \epsilon^{\beta} \pi(B)_{\gamma}{}^{0i}, \\
\delta_0 \pi(B)_{\alpha}{}^{ij} &= -f_{\alpha\beta}{}^{\gamma} \epsilon^{\beta} \pi(B)_{\gamma}{}^{ij}, \\
\delta_0 \pi(\alpha)_{\alpha}{}^0 &= -f_{\alpha\beta}{}^{\gamma} \epsilon^{\beta} \pi(B)_{\gamma}{}^{0i} - f_{\alpha\beta}{}^{\gamma} \epsilon^{\beta} \pi(\alpha)_{\gamma}{}^0 \\
&\quad - \triangleright_{\alpha b}{}^a \epsilon^b \pi(C)_a{}^0 - \triangleright_{\alpha b}{}^a \epsilon^b{}_i \pi(\beta)_a{}^i \\
&\quad - \frac{1}{2} \triangleright_{\alpha B}{}^A \epsilon^B{}_{ij} \pi(\gamma)_A{}^{0ij}, \\
\delta_0 \pi(\alpha)_{\alpha}{}^i &= -f_{\alpha\beta}{}^{\gamma} \epsilon^{\beta} \pi(B)_{\gamma}{}^{ij} - f_{\alpha\beta}{}^{\gamma} \epsilon^{\beta} \pi(\alpha)_{\gamma}{}^i \\
&\quad - \triangleright_{\alpha b}{}^a \epsilon^b \pi(C)_a{}^i - \triangleright_{\alpha b}{}^a \epsilon^b{}_j \pi(\beta)_a{}^j \\
&\quad - \frac{1}{2} \triangleright_{\alpha B}{}^A \epsilon^B{}_{jk} \pi(\gamma)_A{}^{ijk} + \epsilon^{0ijk} \nabla_j \epsilon_{\alpha k}, \\
&\quad + \frac{1}{2} \epsilon^{0ijk} \epsilon^a \triangleright_{\alpha b}{}^a \beta^b{}_{jk}, \\
\delta_0 \pi(C)_a{}^0 &= \epsilon^{\alpha} \triangleright_{\alpha a}{}^b \pi(C)_b{}^0 - \epsilon_{bi} \triangleright_{\alpha a}{}^b \pi(B)^{\alpha 0i}, \\
\delta_0 \pi(C)_a{}^i &= \epsilon^{\alpha} \triangleright_{\alpha a}{}^b \pi(C)_b{}^i - \epsilon_{bj} \triangleright_{\alpha a}{}^b \pi(B)^{\alpha ij},
\end{aligned} \tag{6.31}$$

$$\begin{aligned}
\delta_0 \beta^a_{0i} &= \nabla_0 \epsilon^a_i - \epsilon^{\alpha} \triangleright_{\alpha b}{}^a \beta_{b0i}, \\
\delta_0 \beta^a_{ij} &= 2\nabla_{[i} \epsilon^a_{j]} - \epsilon^{\alpha} \triangleright_{\alpha b}{}^a \beta^b_{ij} + \epsilon^A{}_{ij} \delta_A{}^a, \\
\delta_0 \gamma^A{}_{0ij} &= -\epsilon^{\alpha} \gamma^B{}_{0ij} \triangleright_{\alpha B}{}^A + \nabla_0 \epsilon^A{}_{ij} \\
&\quad + 4\epsilon^a{}_{[i} \beta^b{}_{0| j]} X_{\{ab\}}{}^A, \\
\delta_0 \gamma^A{}_{ijk} &= -\epsilon^{\alpha} \gamma^B{}_{ijk} \triangleright_{\alpha B}{}^A + \nabla_i \epsilon^A{}_{jk} \\
&\quad - \nabla_j \epsilon^A{}_{ki} + \nabla_k \epsilon^A{}_{ij} - 3! \epsilon^a{}_{[i} \beta^b{}_{jk]} X_{\{ab\}}{}^A, \\
\delta_0 D^A &= -\epsilon^a \delta_a{}^A - \epsilon^{\alpha} D^B \triangleright_{\alpha B}{}^A, \\
\delta_0 \pi(\beta)_a{}^{0i} &= \epsilon^{\alpha} \triangleright_{\alpha a}{}^b \pi(\beta)_b{}^{0i} - \epsilon_b \triangleright_{\alpha a}{}^b \pi(B)^{\alpha 0i} \\
&\quad + 2\epsilon^b{}_j X_{\{ab\}}{}^A \pi(\gamma)_A{}^{0ij}, \\
\delta_0 \pi(\beta)_a{}^{ij} &= \epsilon^{\alpha} \triangleright_{\alpha a}{}^b \pi(\beta)_b{}^{ij} - \epsilon_b \triangleright_{\alpha a}{}^b \pi(B)^{\alpha ij} \\
&\quad + 2\epsilon^b{}_k X_{\{ab\}}{}^A \pi(\gamma)_A{}^{ijk} \\
&\quad - \epsilon^{0ijk} \nabla_k \epsilon_a - \epsilon^{0ijk} \epsilon^{\alpha}_k \partial_{\alpha a}, \\
\delta_0 \pi(\gamma)_A{}^{0ij} &= \epsilon^{\alpha} \triangleright_{\alpha A}{}^B \pi(\gamma)_B{}^{0ij}, \\
\delta_0 \pi(\gamma)_A{}^{ijk} &= \epsilon^{\alpha} \triangleright_{\alpha A}{}^B \pi(\gamma)_B{}^{ijk} - \epsilon^{0ijk} \delta_{aA} \epsilon^a, \\
\delta_0 \pi(D)_A &= 2\epsilon^a{}_i X_{\{ab\}}{}^A \pi(C)^{bi} \\
&\quad - \frac{1}{2} \epsilon_B{}^{ij} \triangleright_{\alpha A}{}^B \pi(B)^{\alpha}_{0ij} \\
&\quad + \epsilon^{\alpha} \triangleright_{\alpha A}{}^B \pi(D)_B.
\end{aligned}$$

6.1.2 Symmetries of the 3BF action

The action (6.1) has additional symmetries beyond those defined for the 2BF action in Theorems 10, 11, and 12. Specifically, the following theorems hold [19].

The G group

First consider the infinitesimal transformations with parameter $\epsilon_{\mathfrak{g}}^\alpha$, given by the form variations

$$\begin{aligned} \delta_0 \alpha^\alpha{}_\mu &= -\partial_\mu \epsilon_{\mathfrak{g}}^\alpha - f_{\beta\gamma}{}^\alpha \alpha^\beta{}_\mu \epsilon_{\mathfrak{g}}^\gamma, & \delta_0 B^\alpha{}_{\mu\nu} &= f_{\beta\gamma}{}^\alpha \epsilon_{\mathfrak{g}}^\beta B^\gamma{}_{\mu\nu}, \\ \delta_0 \beta^a{}_{\mu\nu} &= \triangleright_{\alpha b}{}^a \epsilon_{\mathfrak{g}}^\alpha \beta^b{}_{\mu\nu}, & \delta_0 C^a{}_\mu &= \triangleright_{\alpha b}{}^a \epsilon_{\mathfrak{g}}^\alpha C^b{}_\mu, \\ \delta_0 \gamma^A{}_{\mu\nu\rho} &= \triangleright_{\alpha B}{}^A \epsilon_{\mathfrak{g}}^\alpha \gamma^B{}_{\mu\nu\rho}, & \delta_0 D^A &= \triangleright_{\alpha B}{}^A \epsilon_{\mathfrak{g}}^\alpha D^B, \end{aligned} \quad (6.32)$$

which can equivalently be written as

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha + \nabla \epsilon_{\mathfrak{g}}, & B &\rightarrow B' = B + [B, \epsilon_{\mathfrak{g}}], \\ \beta &\rightarrow \beta' = \beta - \epsilon_{\mathfrak{g}} \triangleright \beta, & C &\rightarrow C' = C - \epsilon_{\mathfrak{g}} \triangleright C, \\ \gamma &\rightarrow \gamma' = \gamma - \epsilon_{\mathfrak{g}} \triangleright \gamma, & D &\rightarrow D' = D - \epsilon_{\mathfrak{g}} \triangleright D, \end{aligned} \quad (6.33)$$

From these infinitesimal transformations we infer the finite transformation defined in Theorem 14.

Theorem 14 (G -gauge transformations) *In a 3BF theory over an arbitrary 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \text{Ad}_g \alpha + g dg^{-1}, & B &\rightarrow B' = g B g^{-1}, \\ \beta &\rightarrow \beta' = g \triangleright \beta, & C &\rightarrow C' = g \triangleright C, \\ \gamma &\rightarrow \gamma' = g \triangleright \gamma, & D &\rightarrow D' = g \triangleright D, \end{aligned} \quad (6.34)$$

where $g = \exp(\epsilon_{\mathfrak{g}} \cdot \hat{G}) = \exp(\epsilon_{\mathfrak{g}\alpha} \hat{G}^\alpha) \in G$, and $\epsilon_{\mathfrak{g}} : \mathcal{M}_4 \rightarrow \mathfrak{g}$ is the transformation parameter.

Proof. The transformation of the 3-connection defined in Theorem 14 induces the following transformation of the 3-curvature:

$$\mathcal{F} \rightarrow \mathcal{F}' = g \mathcal{F} g^{-1}, \quad \mathcal{G} \rightarrow \mathcal{G}' = g \triangleright \mathcal{G}, \quad \mathcal{H} \rightarrow \mathcal{H}' = g \triangleright \mathcal{H}, \quad (6.35)$$

Under this transformation, the 3BF action becomes

$$\begin{aligned} S_{3BF} &= \int_{\mathcal{M}_4} \left(\langle B, \mathcal{F} \rangle_{\mathfrak{g}} + \langle C, \mathcal{G} \rangle_{\mathfrak{h}} + \langle D, \mathcal{H} \rangle_{\mathfrak{l}} \right) \\ &\rightarrow S'_{BF} = \int_{\mathcal{M}_4} \left(\langle g B g^{-1}, g \mathcal{F} g^{-1} \rangle_{\mathfrak{g}} + \langle g \triangleright C, g \triangleright \mathcal{G} \rangle_{\mathfrak{h}} + \langle g \triangleright D, g \triangleright \mathcal{H} \rangle_{\mathfrak{l}} \right). \end{aligned} \quad (6.36)$$

The G -invariance of the bilinear forms $\langle _, _ \rangle_{\mathfrak{g}}$, $\langle _, _ \rangle_{\mathfrak{h}}$, and $\langle _, _ \rangle_{\mathfrak{l}}$ implies that the 3BF action is invariant. Invariance can also be shown as in Theorem 8. ■

Considering two successive infinitesimal G -gauge transformations with small parameters $\epsilon_{\mathfrak{g}}^{\alpha_1}$ and $\epsilon_{\mathfrak{g}}^{\beta_2}$, we calculate the commutator of their generators as in BF theory. The generators of the G -gauge transformations defined in Theorem 14 satisfy

$$[\hat{G}_\alpha, \hat{G}_\beta] = f_{\alpha\beta}{}^\gamma \hat{G}_\gamma, \quad (6.37)$$

where $f_{\alpha\beta}{}^\gamma$ are the structure constants of $\mathfrak{g}$. As for the BF and $2BF$ transformations, there is an isomorphism $\hat{G}_\alpha \cong \tau_\alpha$. Thus the group of G -gauge transformations in Theorem 14 is precisely the group G of the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$. This important result will not hold for the remaining symmetry transformations.

The gauge group $\tilde{H}_L$

Consider the form variations associated with the parameter $\epsilon_{\mathfrak{h}}^a{}_i$. For example, equations (6.31) give the variations of $\alpha^\alpha{}_0$ and $\alpha^\alpha{}_i$ as

$$\delta_0 \alpha^\alpha{}_0 = 0, \quad \delta_0 \alpha^\alpha{}_i = -\partial_a^\alpha \epsilon_{\mathfrak{h}}^a{}_i. \quad (6.38)$$

The generator (6.30) yields symmetry transformations on a spatial hypersurface Σ_3 with vanishing time component $\epsilon_{\mathfrak{h}}^a{}_0 = 0$. This allows the transformation to be extended to a spacetime gauge parameter $\epsilon_{\mathfrak{h}}^a{}_\mu$. The variation of $\alpha^\alpha{}_\mu$ is then

$$\delta_0 \alpha^\alpha{}_\mu = -\partial_a^\alpha \epsilon_{\mathfrak{h}}^a{}_\mu, \quad (6.39)$$

and the remaining variables follow analogously. The spacetime infinitesimal transformation associated with $\epsilon_{\mathfrak{h}}^a{}_\mu$ is therefore

$$\begin{aligned} \delta_0 \alpha^\alpha{}_\mu &= -\partial_a^\alpha \epsilon_{\mathfrak{h}}^a{}_\mu, & \delta_0 B^\alpha{}_{\mu\nu} &= 2C_{a[\mu} \epsilon_{\mathfrak{h}}^b{}_{\nu]} \triangleright_{\beta b}{}^a g^{\alpha\beta}, \\ \delta_0 \beta^a{}_{\mu\nu} &= -2\nabla_{[\mu} \epsilon_{\mathfrak{h}}^a{}_{\nu]}, & \delta_0 C^a{}_\mu &= 2D_A X_{(ab)}^A \epsilon_{\mathfrak{h}}^b{}_\mu, \\ \delta_0 \gamma^A{}_{\mu\nu\rho} &= 3! \beta^a{}_{[\mu\nu} \epsilon_{\mathfrak{h}}^b{}_{\rho]} X_{(ab)}^A, & \delta_0 D &= 0. \end{aligned} \quad (6.40)$$

These infinitesimal transformations lead to the finite symmetry transformations of Theorem 15.

Theorem 15 (*H*-gauge transformations) *In a 3BF theory over an arbitrary 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha - \partial \epsilon_{\mathfrak{h}}, & B &\rightarrow B' = B - C' \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}} - \epsilon_{\mathfrak{h}} \wedge^{\mathcal{D}} \epsilon_{\mathfrak{h}} \wedge^{\mathcal{D}} D, \\ \beta &\rightarrow \beta' = \beta - \nabla' \epsilon_{\mathfrak{h}} - \epsilon_{\mathfrak{h}} \wedge \epsilon_{\mathfrak{h}}, & C &\rightarrow C' = C - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}}, \\ \gamma &\rightarrow \gamma' = \gamma + \{\beta', \epsilon_{\mathfrak{h}}\}_{\text{pf}} + \{\epsilon_{\mathfrak{h}}, \beta\}_{\text{pf}}, & D &\rightarrow D' = D, \end{aligned} \quad (6.41)$$

where $\epsilon_{\mathfrak{h}} \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ is an arbitrary $\mathfrak{h}$ -valued 1-form, and ∇' is the covariant derivative with connection α' . The maps $\mathcal{T}$, $\mathcal{D}$, $\mathcal{X}_1$, and $\mathcal{X}_2$ are defined in Appendix A, equations (A.13), (A.51), (A.47), and (A.48).

Proof. Under the symmetry transformations of Theorem 15, the 3-curvature transforms as

$$\begin{aligned} \mathcal{F} &\rightarrow \mathcal{F}' = \mathcal{F}, \\ \mathcal{G} &\rightarrow \mathcal{G}' = \mathcal{G} - \mathcal{F} \wedge^{\triangleright} \epsilon_{\mathfrak{h}}, \\ \mathcal{H} &\rightarrow \mathcal{H}' = \mathcal{H} + \{\mathcal{G}', \epsilon_{\mathfrak{h}}\}_{\text{pf}} - \{\epsilon_{\mathfrak{h}}, \mathcal{G}\}_{\text{pf}}. \end{aligned} \quad (6.42)$$

Using the 3-curvature transformations (6.42) and those of the Lagrange multipliers, the transformed action is

$$\begin{aligned} S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} &\left(-\langle C' \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}}, \mathcal{F} \rangle_{\mathfrak{g}} - \langle \epsilon_{\mathfrak{h}} \wedge^{\mathcal{D}} \epsilon_{\mathfrak{h}} \wedge^{\mathcal{D}} D, \mathcal{F} \rangle_{\mathfrak{g}} - \langle C', \mathcal{F} \wedge^{\triangleright} \epsilon_{\mathfrak{h}} \rangle_{\mathfrak{h}} - \langle D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}}, \mathcal{G} \rangle_{\mathfrak{h}} \right. \\ &\left. - \langle D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}}, \mathcal{G} \rangle_{\mathfrak{h}} + \langle D, \{\mathcal{G}, \epsilon_{\mathfrak{h}}\}_{\text{pf}} \rangle_{\mathfrak{l}} - \langle D, \{\mathcal{F} \wedge^{\triangleright} \epsilon_{\mathfrak{h}}, \epsilon_{\mathfrak{h}}\}_{\text{pf}} \rangle_{\mathfrak{l}} - \langle D, \{\epsilon_{\mathfrak{h}}, \mathcal{G}\}_{\text{pf}} \rangle_{\mathfrak{l}} \right). \end{aligned} \quad (6.43)$$

Using the definitions of $\mathcal{T}$, $\mathcal{D}$, $\mathcal{X}_1$, and $\mathcal{X}_2$ given in Appendix A, the remaining terms cancel, and the action is invariant: $S'_{3BF} = S_{3BF}$. Specifically, the first term cancels the third, the second the seventh, the fourth the eighth, and the fifth the sixth. ■

The *H*-gauge transformations do not form a group: two successive transformations are not generally of the same type, so closure fails. An analogous situation occurs for the Lorentz

group, where boosts alone are not closed and do not form a group; both rotations and boosts are required. For the H -gauge transformations, one must additionally include transformations associated with the parameter $\epsilon_{\mathfrak{l}}^A{}_{ij}$. Equations (6.31) give the corresponding form variations on Σ_3 . As for the H -gauge transformations, these can be extended to spacetime variations with parameter $\epsilon_{\mathfrak{l}}^A{}_{\mu\nu}$:

$$\begin{aligned} \delta_0 \alpha^\alpha{}_\mu &= 0, & \delta_0 B^\alpha{}_{\mu\nu} &= -D_A \triangleright_{\beta B}{}^A \epsilon_{\mathfrak{l}}^B{}_{\mu\nu} g^{\alpha\beta}, \\ \delta_0 \beta^a{}_{\mu\nu} &= \delta_A{}^a \epsilon_{\mathfrak{l}}^A{}_{\mu\nu}, & \delta_0 C^a{}_\mu &= 0, \\ \delta_0 \gamma^A{}_{\mu\nu\rho} &= \nabla_\mu \epsilon_{\mathfrak{l}}^A{}_{\nu\rho} - \nabla_\nu \epsilon_{\mathfrak{l}}^A{}_{\mu\rho} + \nabla_\rho \epsilon_{\mathfrak{l}}^A{}_{\mu\nu}, & \delta_0 D^A &= 0. \end{aligned} \quad (6.44)$$

These infinitesimal transformations correspond to the finite symmetry transformations defined in Theorem 16.

Theorem 16 (L -gauge transformations) *In a 3BF theory over an arbitrary 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha, & B &\rightarrow B' = B + D \wedge^{\mathcal{S}} \epsilon_{\mathfrak{l}}, \\ \beta &\rightarrow \beta' = \beta + \delta \epsilon_{\mathfrak{l}}, & C &\rightarrow C' = C, \\ \gamma &\rightarrow \gamma' = \gamma + \nabla \epsilon_{\mathfrak{l}}, & D &\rightarrow D' = D, \end{aligned} \quad (6.45)$$

where $\epsilon_{\mathfrak{l}} \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$ is an arbitrary $\mathfrak{l}$ -valued 2-form, and the map $\mathcal{S}$ is defined in Appendix A by equation (A.43).

Proof. Under the transformations of Theorem 16, the 3-curvature transforms as

$$\begin{aligned} \mathcal{F} &\rightarrow \mathcal{F}' = \mathcal{F}, \\ \mathcal{G} &\rightarrow \mathcal{G}' = \mathcal{G}, \\ \mathcal{H} &\rightarrow \mathcal{H}' = \mathcal{H} + \mathcal{F} \wedge^{\triangleright} \epsilon_{\mathfrak{l}}. \end{aligned} \quad (6.46)$$

Taking into account the 3-curvature transformations (6.46) and those of the Lagrange multipliers, the 3BF action becomes

$$S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} \left(\langle D \wedge^{\mathcal{S}} \epsilon_{\mathfrak{l}}, \mathcal{F} \rangle_{\mathfrak{g}} + \langle D, \mathcal{F} \wedge^{\triangleright} \epsilon_{\mathfrak{l}} \rangle_{\mathfrak{l}} \right). \quad (6.47)$$

By the definition of $\mathcal{S}$ in Appendix A, the terms in parentheses cancel. ■

Denote the generators of the H -gauge transformations in Theorem 15 by $\hat{H}_a{}^\mu$ and those of the L -gauge transformations in Theorem 16 by $\hat{L}_A{}^{\mu\nu}$. We can now test whether the H -gauge transformations defined in Theorem 15 form a group. Two successive H -gauge transformations with parameters $\epsilon_{\mathfrak{h}1}$ and $\epsilon_{\mathfrak{h}2}$ give

$$e^{\epsilon_{\mathfrak{h}1} \cdot \hat{H}} e^{\epsilon_{\mathfrak{h}2} \cdot \hat{H}} - e^{\epsilon_{\mathfrak{h}2} \cdot \hat{H}} e^{\epsilon_{\mathfrak{h}1} \cdot \hat{H}} = 2(\{\epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}2}\}_{\text{pf}} - \{\epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}1}\}_{\text{pf}}) \cdot \hat{L}, \quad (6.48)$$

where $\epsilon_{\mathfrak{h}} \cdot \hat{H} = \epsilon_{\mathfrak{h}}{}^a{}_\mu \hat{H}_a{}^\mu$ and $\epsilon_{\mathfrak{l}} \cdot \hat{L} = \frac{1}{2} \epsilon_{\mathfrak{l}}^A{}_{\mu\nu} \hat{L}_A{}^{\mu\nu}$. Using the analogue of the BCH formula (4.38), the commutator of two H -gauge generators is an L -gauge generator (details are given in Appendix D):

$$[\hat{H}_a{}^\mu, \hat{H}_b{}^\nu] = 2X_{(ab)}{}^A \hat{L}_A{}^{\mu\nu}. \quad (6.49)$$

The transformations in Theorem 16 are linear, and two successive L -gauge transformations yield an L -gauge transformation with parameter $\epsilon_{\mathfrak{l}1} + \epsilon_{\mathfrak{l}2}$:

$$e^{\epsilon_{\mathfrak{l}1} \cdot \hat{L}} e^{\epsilon_{\mathfrak{l}2} \cdot \hat{L}} = e^{(\epsilon_{\mathfrak{l}1} + \epsilon_{\mathfrak{l}2}) \cdot \hat{L}}, \quad (6.50)$$

It follows that the L -gauge transformations commute:

$$[\hat{L}_A^{\mu\nu}, \hat{L}_B^{\rho\sigma}] = 0. \quad (6.51)$$

Thus the L -gauge transformations form an Abelian group $\tilde{L}$. The index structure of the parameters and generators shows that $\tilde{L}$ is isomorphic to $\mathbb{R}^{6r}$, where r is the dimension of L :

$$\tilde{L} \cong \mathbb{R}^{6r}. \quad (6.52)$$

The Abelian group $\tilde{L}$ should not be confused with the non-Abelian group L appearing in the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$.

Consider now the relation between the H - and L -gauge transformations. From

$$e^{\epsilon_{\mathfrak{h}} \cdot \hat{H}} e^{\epsilon_{\mathfrak{l}} \cdot \hat{L}} = e^{\epsilon_{\mathfrak{l}} \cdot \hat{L}} e^{\epsilon_{\mathfrak{h}} \cdot \hat{H}}, \quad (6.53)$$

we conclude that their generators commute:

$$[\hat{H}_a^\mu, \hat{L}_A^{\nu\rho}] = 0. \quad (6.54)$$

Relations (6.49), (6.51), and (6.54) show that the H - and L -gauge transformations together form a group, denoted by $\tilde{H}_L$. Finally, the action of G on the H - and L -gauge transformations follows from

$$[\epsilon_{\mathfrak{g}} \cdot \hat{G}, \epsilon_{\mathfrak{h}} \cdot \hat{H}] = (\epsilon_{\mathfrak{g}} \triangleright \epsilon_{\mathfrak{h}}) \cdot \hat{H}, \quad [\epsilon_{\mathfrak{g}} \cdot \hat{G}, \epsilon_{\mathfrak{l}} \cdot \hat{L}] = (\epsilon_{\mathfrak{g}} \triangleright \epsilon_{\mathfrak{l}}) \cdot \hat{L}, \quad (6.55)$$

which gives the commutation relations

$$\begin{aligned} [\hat{G}_\alpha, \hat{H}_a^\mu] &= \triangleright_{\alpha a}{}^b \hat{H}_b^\mu, \\ [\hat{G}_\alpha, \hat{L}_A^{\mu\nu}] &= \triangleright_{\alpha A}{}^B \hat{L}_B^{\mu\nu}. \end{aligned} \quad (6.56)$$

Theorems 14, 15, and 16 describe the G -, H -, and L -gauge transformations; see [18], [40] for further information.

The groups $\tilde{M}$ and $\tilde{N}$

Next consider the infinitesimal transformation with parameter $\epsilon_{\mathfrak{m}}^\alpha$ in (6.31). As above, the Hamiltonian variations on a spatial hypersurface Σ_3 determine the transformations throughout spacetime. Since the time component vanishes on the hypersurface, $\epsilon_{\mathfrak{m}}^\alpha = 0$, we extend them to spacetime form variations with parameter $\epsilon_{\mathfrak{m}}^\alpha$:

$$\begin{aligned} \delta_0 \alpha^\alpha{}_\mu &= 0, & \delta_0 B^\alpha{}_{\mu\nu} &= -2\nabla_{[\mu} \epsilon_{\mathfrak{m}}^\alpha{}_{\nu]}, \\ \delta_0 \beta^a{}_{\mu\nu} &= 0, & \delta_0 C^a{}_\mu &= -\partial^a{}_\alpha \epsilon_{\mathfrak{m}}^\alpha{}_\mu, \\ \delta_0 \gamma^A{}_{\mu\nu\rho} &= 0, & \delta_0 D^A &= 0. \end{aligned} \quad (6.57)$$

This result gives the finite spacetime symmetry transformation defined in Theorem 17, called an M -gauge transformation.

Theorem 17 (M -gauge transformations) *In a 3BF theory over an arbitrary 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha, & B &\rightarrow B' = B - \nabla \epsilon_{\mathfrak{m}}, \\ \beta &\rightarrow \beta' = \beta, & C^a &\rightarrow C'^a = C^a - \partial^a{}_\alpha \epsilon_{\mathfrak{m}}^\alpha, \\ \gamma &\rightarrow \gamma' = \gamma, & D &\rightarrow D' = D, \end{aligned} \quad (6.58)$$

where the transformation parameter $\epsilon_{\mathfrak{m}} \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$ is an arbitrary $\mathfrak{g}$ -valued 1-form.

Proof. Consider the transformation of the $3BF$ action under the transformations defined in Theorem 17. We obtain

$$S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \left(-\frac{1}{2} (\nabla_\mu \epsilon_m^\alpha{}_\nu) \mathcal{F}_{\alpha\rho\sigma} - \frac{1}{3!} \partial^a{}_\alpha \epsilon_m^\alpha{}_\mu \mathcal{G}_{a\nu\rho\sigma} \right). \quad (6.59)$$

Using the definition of the 3-curvature (2.3.5) gives

$$S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \left(-\frac{1}{2} (\nabla_\mu \epsilon_m^\alpha{}_\nu) (F_{\alpha\rho\sigma} - \partial^a{}_\alpha \beta_{a\rho\sigma}) - \frac{1}{3!} \partial^a{}_\alpha \epsilon_m^\alpha{}_\mu (3\nabla_\nu \beta_{a\rho\sigma} - \delta^A_a \gamma_{A\nu\rho\sigma}) \right). \quad (6.60)$$

In the preceding expression, the second and third terms cancel, while the last term vanishes by the identity (2.73); hence

$$S'_{3BF} = S_{3BF} - \frac{1}{2} \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \epsilon_m^\alpha{}_\mu \nabla_\nu F_{\alpha\rho\sigma}. \quad (6.61)$$

Finally, the relation $\epsilon^{\mu\nu\rho\sigma} \nabla_\nu F_{\alpha\rho\sigma} = 0$ is precisely the Bianchi identity (5.24). We conclude that S_{3BF} is invariant under the transformations defined in Theorem 17. ■

The transformations defined in Theorem 17 are linear: two successive M -gauge transformations give a single M -gauge transformation with parameter $\epsilon_{m1} + \epsilon_{m2}$. Denoting the generators of M -gauge transformations by $\hat{M}_\alpha{}^\mu$, we may write

$$e^{\epsilon_{m1} \cdot \hat{M}} e^{\epsilon_{m2} \cdot \hat{M}} = e^{(\epsilon_{m1} + \epsilon_{m2}) \cdot \hat{M}}, \quad (6.62)$$

where $\epsilon_m \cdot \hat{M} = \epsilon_m^\alpha{}_\mu \hat{M}_\alpha{}^\mu$. This yields the commutation relation

$$[\hat{M}_\alpha{}^\mu, \hat{M}_\beta{}^\nu] = 0. \quad (6.63)$$

Thus, the M -gauge transformations form an Abelian group $\tilde{M}$. From the index structure of its parameters and generators, this group is isomorphic to $\mathbb{R}^{4p}$, where p is the dimension of G :

$$\tilde{M} \cong \mathbb{R}^{4p}. \quad (6.64)$$

We can next examine the relation between the M -gauge transformations and the G -, H -, and L -gauge transformations defined above. In particular, the generators of the G -gauge transformations satisfy

$$[\epsilon_g \cdot \hat{G}, \epsilon_m \cdot \hat{M}] = (\epsilon_g \triangleright \epsilon_m) \cdot \hat{M}, \quad (6.65)$$

which gives the commutator

$$[\hat{G}_\alpha, \hat{M}_\beta{}^\mu] = f_{\alpha\beta}{}^\gamma \hat{M}_\gamma{}^\mu. \quad (6.66)$$

For the generators of the H - and L -gauge transformations, we obtain

$$\begin{aligned} e^{\epsilon_h \cdot \hat{H}} e^{\epsilon_m \cdot \hat{M}} &= e^{\epsilon_m \cdot \hat{M}} e^{\epsilon_h \cdot \hat{H}}, \\ e^{\epsilon_l \cdot \hat{L}} e^{\epsilon_m \cdot \hat{M}} &= e^{\epsilon_m \cdot \hat{M}} e^{\epsilon_l \cdot \hat{L}}, \end{aligned} \quad (6.67)$$

and therefore the generators of the M -gauge transformations commute with both the generators of the H -gauge transformations and those of the L -gauge transformations:

$$[\hat{H}_a, \hat{M}_\alpha{}^\mu] = 0, \quad [\hat{L}_A{}^{\mu\nu}, \hat{M}_\alpha{}^\rho] = 0. \quad (6.68)$$

The final type of transformation follows from the form variations in (6.31) associated with the parameter ϵ_n^a :

$$\begin{aligned} \delta_0 \alpha^\alpha{}_\mu &= 0, & \delta_0 B^\alpha{}_{\mu\nu} &= \beta_{b\mu\nu} \triangleright_{\alpha'a}{}^b \epsilon_n^a g^{\alpha\alpha'}, \\ \delta_0 \beta^a{}_{\mu\nu} &= 0, & \delta_0 C^a{}_\mu &= -\nabla_\mu \epsilon_n^a, \\ \delta_0 \gamma^A{}_{\mu\nu\rho} &= 0, & \delta_0 D^A &= \delta^A_a \epsilon_n^a. \end{aligned} \quad (6.69)$$

These N -gauge transformations are defined in Theorem 18. They are already spacetime transformations because their parameter carries no spacetime index.

Theorem 18 (N -gauge transformations) *In a 3BF theory over an arbitrary 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$, the following transformation is a symmetry:*

$$\begin{aligned} \alpha &\rightarrow \alpha' = \alpha, & B &\rightarrow B' = B - \beta \wedge^T \epsilon_n, \\ \beta &\rightarrow \beta' = \beta, & C &\rightarrow C' = C - \nabla \epsilon_n, \\ \gamma &\rightarrow \gamma' = \gamma, & D^A &\rightarrow D'^A = D^A + \delta^A_a \epsilon_n^a, \end{aligned} \quad (6.70)$$

where $\epsilon_n : \mathcal{M}_4 \rightarrow \mathfrak{h}$ is an arbitrary $\mathfrak{h}$ -valued 0-form.

Proof. Under the transformations of Theorem 18, the 3BF action becomes

$$S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} dx^4 \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} \beta_{b\mu\nu} \triangleright_{\alpha a} {}^b \epsilon_n^a \mathcal{F}^\alpha_{\rho\sigma} - \frac{1}{3!} (\nabla_\mu \epsilon_n^a) \mathcal{G}_{a\nu\rho\sigma} + \frac{1}{4!} \delta^A_a \epsilon_n^a \mathcal{H}_{A\mu\nu\rho\sigma} \right). \quad (6.71)$$

Using the definition of the 3-curvature (2.118), we obtain

$$\begin{aligned} S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} dx^4 \epsilon^{\mu\nu\rho\sigma} & \left(\frac{1}{4} \beta_{b\mu\nu} \triangleright_{\alpha a} {}^b \epsilon_n^a (F^\alpha_{\rho\sigma} - \partial_c {}^\alpha \beta^c_{\rho\sigma}) - \frac{1}{3!} (\nabla_\mu \epsilon_n^a) (3 \nabla_\nu \beta_{a\rho\sigma} - \delta^A_a \gamma_{A\nu\rho\sigma}) \right. \\ & \left. + \frac{1}{4!} \delta^A_a \epsilon_n^a (4 \nabla_\mu \gamma_{A\nu\rho\sigma} + 6 X_{(bc)A} \beta^b_{\mu\nu} \beta^c_{\rho\sigma}) \right). \end{aligned} \quad (6.72)$$

After integration by parts, the last term in the first line of (6.72) cancels the first term in the second line. In addition, using the identity

$$\frac{1}{2} \epsilon^{\mu\nu\rho\sigma} (\nabla_\nu \nabla_\mu \epsilon_n^a) \beta_{a\rho\sigma} = -\frac{1}{4} \epsilon^{\mu\nu\rho\sigma} \beta_{b\rho\sigma} \triangleright_{\alpha a} {}^b \epsilon_n^a F^\alpha_{\mu\nu}, \quad (6.73)$$

the first and third terms in the first line also cancel. Finally,

$$S'_{3BF} = S_{3BF} + \int_{\mathcal{M}_4} dx^4 \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} \epsilon_{na} \triangleright_{\alpha(b|} {}^a \partial_{|c)} {}^\alpha \beta^b_{\mu\nu} \beta^c_{\rho\sigma} + \frac{1}{4} \epsilon_{na} \delta_A^a X_{(bc)}^A \beta^b_{\mu\nu} \beta^c_{\rho\sigma} \right). \quad (6.74)$$

The sum of the two remaining terms vanishes by the symmetrized form of identity (2.87),

$$\triangleright_{\alpha(b|} {}^a \partial_{|c)} {}^\alpha + \delta_A^a X_{(bc)}^A = f_{(bc)}^a = 0,$$

because the structure constants are antisymmetric. Thus the action S_{3BF} is invariant under the transformations of Theorem 18. ■

The N -gauge transformations of Theorem 18 form a group denoted by $\tilde{N}$. They are linear, and two successive transformations give an N -gauge transformation with parameter $\epsilon_{n1} + \epsilon_{n2}$. Denoting the generators of $\tilde{N}$ by $\hat{N}_a$, we write

$$e^{\epsilon_{n1} \cdot \hat{N}} e^{\epsilon_{n2} \cdot \hat{N}} = e^{(\epsilon_{n1} + \epsilon_{n2}) \cdot \hat{N}}, \quad (6.75)$$

where $\epsilon_n \cdot \hat{N} = \epsilon_n^a \hat{N}_a$. Hence the commutator of two N -gauge generators is

$$[\hat{N}_a, \hat{N}_b] = 0, \quad (6.76)$$

so $\tilde{N}$ is Abelian. The index structure of its parameters and generators shows that $\tilde{N}$ is isomorphic to $\mathbb{R}^q$, where q is the dimension of H :

$$\tilde{N} \cong \mathbb{R}^q. \quad (6.77)$$

We can next examine the relation between the N -gauge transformations and the G -, H -, and L -gauge transformations defined above. For the G -gauge generators,

$$[\epsilon_g \cdot \hat{G}, \epsilon_n \cdot \hat{N}] = (\epsilon_g \triangleright \epsilon_n) \cdot \hat{N}, \quad (6.78)$$

so the commutator of the G - and N -gauge transformations is

$$[\hat{G}_\alpha, \hat{N}_a] = \triangleright_{\alpha a}{}^b \hat{N}_b. \quad (6.79)$$

The relation between the N - and H -gauge transformations follows from

$$e^{\epsilon_{\mathfrak{h}} \cdot \hat{H}} e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} - e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} e^{\epsilon_{\mathfrak{h}} \cdot \hat{H}} = -(\epsilon_{\mathfrak{n}} \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}}) \cdot \hat{M}. \quad (6.80)$$

The proof is given in Appendix D. The commutator of the H - and N -gauge transformations is a linear combination of M -gauge generators:

$$[\hat{H}_a{}^\mu, \hat{N}^b] = \triangleright_{\alpha a}{}^b \hat{M}^{\alpha\mu}. \quad (6.81)$$

The same procedure gives

$$e^{\epsilon_{\mathfrak{l}} \cdot \hat{L}} e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} = e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} e^{\epsilon_{\mathfrak{l}} \cdot \hat{L}}, \quad e^{\epsilon_{\mathfrak{m}} \cdot \hat{M}} e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} = e^{\epsilon_{\mathfrak{n}} \cdot \hat{N}} e^{\epsilon_{\mathfrak{m}} \cdot \hat{M}}, \quad (6.82)$$

showing that the L - and M -gauge generators commute with the N -gauge generators:

$$[\hat{M}_\alpha{}^\mu, \hat{N}_a] = 0, \quad [\hat{L}_A{}^{\mu\nu}, \hat{N}_a] = 0. \quad (6.83)$$

Complete gauge-symmetry group of the $3BF$ action

Combining the preceding results, the complete algebra of gauge-symmetry generators is as follows.

- The algebra $\mathfrak{g}$ corresponding to the group G of the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$ is given by

$$[\hat{G}_\alpha, \hat{G}_\beta] = f_{\alpha\beta}{}^\gamma \hat{G}_\gamma. \quad (6.84)$$

- The algebra corresponding to $\tilde{H}_L$ consists of the H - and L -gauge generators satisfying

$$[\hat{H}_a{}^\mu, \hat{H}_b{}^\nu] = 2X_{(ab)}{}^A \hat{L}_A{}^{\mu\nu}, \quad [\hat{L}_A{}^{\mu\nu}, \hat{L}_B{}^{\rho\sigma}] = 0, \quad [\hat{H}_a{}^\mu, \hat{L}_A{}^{\nu\rho}] = 0. \quad (6.85)$$

- The algebra of the M -gauge generators is

$$[\hat{M}_\alpha{}^\mu, \hat{M}_\beta{}^\nu] = 0. \quad (6.86)$$

- The algebra of the N -gauge generators is

$$[\hat{N}_a, \hat{N}_b] = 0. \quad (6.87)$$

- The commutators between generators of $\tilde{M}$ and $\tilde{N}$ are

$$[\hat{M}_\alpha{}^\mu, \hat{N}_a] = 0. \quad (6.88)$$

- The action of the generators of $\tilde{H}_L$ on the M - and N -gauge generators is

$$\begin{aligned}
[\hat{H}_a^\mu, \hat{N}^b] &= \triangleright_{\alpha a}{}^b \hat{M}^{\alpha\mu}, \\
[\hat{H}_a^\mu, \hat{M}_\alpha^\nu] &= 0, \\
[\hat{L}_A^{\nu\rho}, \hat{M}_\alpha^\mu] &= 0, \\
[\hat{L}_A^{\mu\nu}, \hat{N}_a] &= 0.
\end{aligned} \tag{6.89}$$

- The action of the generators of G on the H -, L -, M -, and N -gauge generators is

$$\begin{aligned}
[\hat{G}_\alpha, \hat{H}_a^\mu] &= \triangleright_{\alpha a}{}^b \hat{H}_b^\mu, \\
[\hat{G}_\alpha, \hat{L}_A^{\mu\nu}] &= \triangleright_{\alpha A}{}^B \hat{L}_B^{\mu\nu}, \\
[\hat{G}_\alpha, \hat{M}_\beta^\mu] &= f_{\alpha\beta}{}^\gamma \hat{M}_\gamma^\mu, \\
[\hat{G}_\alpha, \hat{N}_a] &= \triangleright_{\alpha a}{}^b \hat{N}_b.
\end{aligned} \tag{6.90}$$

Equations (6.84)–(6.90) determine the structure of the complete gauge-symmetry group. The Hasse-type diagram in Figure 6.1 shows all relevant subgroups of the complete symmetry group $\mathcal{G}_{3BF}$, with *normal subgroups* boxed.

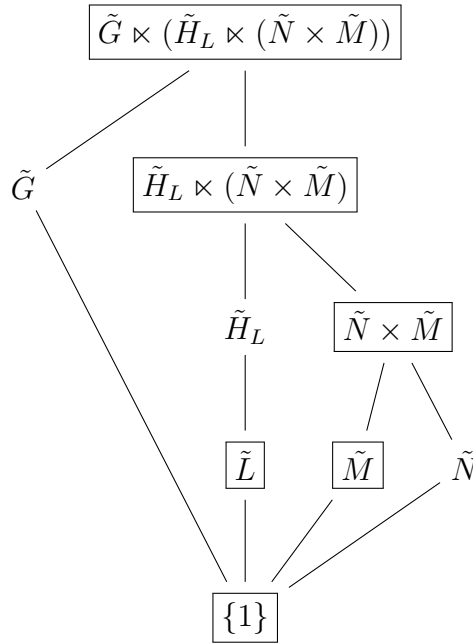


Figure 6.1: Relevant subgroups of the complete symmetry group $\mathcal{G}_{3BF}$. Normal subgroups are boxed.

The commutation relations show that $\tilde{L}$, $\tilde{M}$, and $\tilde{N}$ are subgroups of the complete symmetry group $\mathcal{G}_{3BF}$. The groups $\tilde{L}$ and $\tilde{M}$ are normal because their only nontrivial commutators, with generators of G , are linear combinations of generators of $\tilde{L}$ and $\tilde{M}$, respectively. The group $\tilde{N}$ is not normal because commutators of $\hat{N}_a$ with $\hat{H}_a^\mu$ are linear combinations of $\hat{M}_\alpha^\mu$. Nevertheless, the generators of $\tilde{N}$ and $\tilde{M}$ commute, so $\tilde{N}$ is normal in the direct product $\tilde{N} \times \tilde{M}$. This direct product is a normal subgroup of the complete symmetry group.

The preceding discussion showed that the H - and L -gauge generators together form $\tilde{H}_L$. The commutator of $\hat{H}_a^\mu$ with $\hat{N}_b$ shows that this group is not normal in $\mathcal{G}_{3BF}$. The subgroups $\tilde{N} \times \tilde{M}$ and $\tilde{H}_L$ therefore form the semidirect product $\tilde{H}_L \ltimes (\tilde{N} \times \tilde{M})$. It is semidirect because $\tilde{H}_L$ is not normal in this product, while $\tilde{N} \times \tilde{M}$ is. The resulting group is a normal subgroup of the complete symmetry group $\mathcal{G}_{3BF}$.

Finally, including the G -gauge generators and using (6.90) gives the complete gauge-symmetry group $\mathcal{G}_{3BF}$:

$$\mathcal{G}_{3BF} = G \ltimes (\tilde{H}_L \ltimes (\tilde{N} \times \tilde{M})). \quad (6.91)$$

Diffeomorphisms

Another important issue is the diffeomorphism invariance of 3BF theory. As in 2BF theory, if diffeomorphisms are symmetries, their form variations can be expressed as the sum of gauge and HT variations:

$$\delta_0^{\text{diff}} \phi = -\delta_0^{\text{gauge}} \phi - \delta_0^{\text{HT}} \phi. \quad (6.92)$$

The 3BF action depends on α^α_μ , $\beta^a_{\mu\nu}$, $\gamma^A_{\mu\nu\rho}$, $B^\alpha_{\mu\nu}$, C^a_μ , and D^A . The HT parameters $\epsilon^{\text{HT}\alpha\beta}_{\mu\nu\rho}$, $\epsilon^{\text{HT}ab}_{\mu\nu\rho}$, and $\epsilon^{\text{HT}AB}_{\mu\nu\rho}$ are defined by (4.55),

$$\begin{aligned} \delta_0^{\text{HT}} \alpha^\alpha_\mu &= \frac{1}{2} \epsilon^{\text{HT}\alpha\beta}_{\mu\nu\rho} \frac{\delta S}{\delta B^\beta_{\nu\rho}}, & \delta_0^{\text{HT}} B^\alpha_{\mu\nu} &= -\epsilon^{\text{HT}\alpha\beta}_{\rho\mu\nu} \frac{\delta S}{\delta \alpha^\beta_\rho}, \\ \delta_0^{\text{HT}} \beta^a_{\mu\nu} &= \epsilon^{\text{HT}ab}_{\mu\nu\rho} \frac{\delta S}{\delta C^b_\rho}, & \delta_0^{\text{HT}} C^a_\mu &= -\frac{1}{2} \epsilon^{\text{HT}ab}_{\nu\rho\mu} \frac{\delta S}{\delta \beta^b_{\nu\rho}}, \\ \delta_0^{\text{HT}} \gamma^A_{\mu\nu\rho} &= \epsilon^{\text{HT}AB}_{\mu\nu\rho} \frac{\delta S}{\delta D^B}, & \delta_0^{\text{HT}} D^A &= -\frac{1}{3!} \epsilon^{\text{HT}AB}_{\mu\nu\rho} \frac{\delta S}{\delta \gamma^B_{\mu\nu\rho}}, \end{aligned} \quad (6.93)$$

while the gauge parameters ϵ_g^α , $\epsilon_h^a_\mu$, $\epsilon_l^A_{\mu\nu}$, $\epsilon_m^\alpha_\mu$, and ϵ_n^a are defined in Theorems 14–18. A choice exists for which equation (4.57) holds for every field. In particular, choose

$$\epsilon_g^\alpha = -\xi^\lambda \alpha^\alpha_\lambda, \quad \epsilon_h^a_\mu = \xi^\lambda \beta^a_{\mu\lambda}, \quad \epsilon_l^A_{\mu\nu} = \xi^\lambda \gamma^A_{\mu\nu\lambda}, \quad \epsilon_m^\alpha_\mu = \xi^\lambda B^\alpha_{\mu\lambda}, \quad \epsilon_n^a = -\xi^\lambda C^a_\lambda, \quad (6.94)$$

and the HT parameters

$$\epsilon^{\text{HT}\alpha\beta}_{\mu\nu\rho} = \xi^\lambda g^{\alpha\beta} \epsilon_{\mu\nu\rho\lambda}, \quad \epsilon^{\text{HT}ab}_{\mu\nu\rho} = \xi^\lambda g^{ab} \epsilon_{\lambda\mu\nu\rho}, \quad \epsilon^{\text{HT}AB}_{\mu\nu\rho} = \xi^\lambda g^{AB} \epsilon_{\mu\nu\rho\lambda}, \quad (6.95)$$

then equation (6.92) gives the standard form variations corresponding to diffeomorphisms:

$$\begin{aligned} \delta_0^{\text{diff}} \alpha^\alpha_\mu &= -\partial_\mu \xi^\lambda \alpha^\alpha_\lambda - \xi^\lambda \partial_\lambda \alpha^\alpha_\mu, \\ \delta_0^{\text{diff}} \beta^a_{\mu\nu} &= -\partial_\mu \xi^\lambda \beta^a_{\lambda\nu} - \partial_\nu \xi^\lambda \beta^a_{\mu\lambda} - \xi^\lambda \partial_\lambda \beta^a_{\mu\nu}, \\ \delta_0^{\text{diff}} \gamma^A_{\mu\nu\rho} &= -\partial_\mu \xi^\lambda \gamma^A_{\lambda\nu\rho} - \partial_\nu \xi^\lambda \gamma^A_{\mu\lambda\rho} - \partial_\rho \xi^\lambda \gamma^A_{\mu\nu\lambda} - \xi^\lambda \partial_\lambda \gamma^A_{\mu\nu\rho}, \\ \delta_0^{\text{diff}} B^\alpha_{\mu\nu} &= -\partial_\mu \xi^\lambda B^\alpha_{\lambda\nu} - \partial_\nu \xi^\lambda B^\alpha_{\mu\lambda} - \xi^\lambda \partial_\lambda B^\alpha_{\mu\nu}, \\ \delta_0^{\text{diff}} C^a_\mu &= -\partial_\mu \xi^\lambda C^a_\lambda - \xi^\lambda \partial_\lambda C^a_\mu, \\ \delta_0^{\text{diff}} D^A &= -\xi^\lambda \partial_\lambda D^A. \end{aligned} \quad (6.96)$$

Thus diffeomorphisms are symmetries of the theory, although they are not contained in the complete gauge-symmetry group $\mathcal{G}_{3BF}$, but in its direct product with the HT symmetry group.

6.2 Klein–Gordon theory

This section shows how the 3-group structure, and hence the corresponding 3BF theory, can describe a Klein–Gordon field interacting with gravity [16]. We first specify the 2-crossed module on which the 3BF theory is defined and then obtain the required dynamics by adding

suitable *constraints* to the topological $3BF$ action. Define the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$ using the Lie groups

$$G = SO(3, 1), \quad H = \mathbb{R}^4, \quad L = \mathbb{R}. \quad (6.97)$$

The group G acts on itself by conjugation and on H through the vector representation, while its action on L is trivial. This defines $\triangleright$. The map ∂ is trivial, as in pure gravity. The map δ is also trivial, sending every element of L to the identity of H . Finally, the Peiffer lifting is trivial, sending every ordered pair of elements of H to the identity of L . This choice of 2-crossed module describes a scalar field interacting with gravity, as shown below.

For this 2-crossed module, the 3-connection (α, β, γ) is

$$\alpha = \omega^{ab} M_{ab}, \quad \beta = \beta^a P_a, \quad \gamma = \gamma \mathbb{I}, \quad (6.98)$$

where $\mathbb{I}$ denotes the generator of the Lie group $\mathbb{R}$. Since ∂ , δ , and the Peiffer lifting are trivial, the fake 3-curvature (2.118) reduces to the ordinary 3-curvature,

$$\mathcal{F} = R^{ab} M_{ab}, \quad \mathcal{G} = \nabla \beta^a P_a, \quad \mathcal{H} = d\gamma. \quad (6.99)$$

Here we used the trivial action of G on L , namely $M_{ab} \triangleright \mathbb{I} = 0$. Thus the 3-form γ transforms as a scalar under Lorentz transformations. Its Lagrange multiplier D transforms in the same way, as is also apparent from its index structure. Since D is a 0-form, it is a scalar under diffeomorphisms as well. Consequently, D transforms as a real scalar field under every symmetry; setting $D \equiv \phi$, the topological $3BF$ action (6.1) becomes

$$S_{3BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \phi d\gamma, \quad (6.100)$$

where the bilinear form on L is defined by $\langle \mathbb{I}, \mathbb{I} \rangle_{\mathbb{I}} = 1$.

To describe a scalar field of mass m , with Klein–Gordon dynamics and interacting with gravity, suitable constraints must be added to the topological action (6.100):

$$\begin{aligned} S = & \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \phi d\gamma \\ & - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right) \\ & + \lambda \wedge \left(\gamma - \frac{1}{2} H_{abc} e^a \wedge e^b \wedge e^c \right) \\ & + \Lambda^{ab} \wedge \left(H_{abc} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - d\phi \wedge e_a \wedge e_b \right) \\ & - \frac{1}{2 \cdot 4!} m^2 \phi^2 \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d. \end{aligned} \quad (6.101)$$

In this expression, the first line is the topological sector (6.100); the second is the familiar gravitational *simplicity constraint* introduced in (5.85); the third and fourth are new *simplicity constraints* containing the Lagrange-multiplier 1-forms λ and Λ^{ab} and the 0-form H_{abc} ; and the final line supplies the mass m of the scalar field ϕ . Varying (6.101) with respect to B_{ab} , ω_{ab} ,

$\beta_a, \lambda_{ab}, \Lambda_{ab}, \gamma, \lambda, H_{abc}, \phi$, and e^a gives

$$R^{ab} - \lambda^{ab} = 0, \quad (6.102)$$

$$\nabla B^{ab} - e^{[a} \wedge \beta^{b]} = 0, \quad (6.103)$$

$$\nabla e^a = 0, \quad (6.104)$$

$$B_{ab} - \frac{1}{16\pi l_p^2} \varepsilon_{abcd} e^c \wedge e^d = 0, \quad (6.105)$$

$$H_{abc} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - d\phi \wedge e_a \wedge e_b = 0, \quad (6.106)$$

$$d\phi - \lambda = 0, \quad (6.107)$$

$$\gamma - \frac{1}{2} H_{abc} e^a \wedge e^b \wedge e^c = 0, \quad (6.108)$$

$$-\frac{1}{2} \lambda \wedge e^a \wedge e^b \wedge e^c + \varepsilon^{cdef} \Lambda^{ab} \wedge e_d \wedge e_e \wedge e_f = 0, \quad (6.109)$$

$$d\gamma - d(\Lambda^{ab} \wedge e_a \wedge e_b) - \frac{1}{4!} m^2 \phi \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d = 0, \quad (6.110)$$

$$\begin{aligned} \nabla \beta_a + \frac{1}{8\pi l_p^2} \varepsilon_{abcd} \lambda^{bc} \wedge e^d + \frac{3}{2} H_{abc} \lambda \wedge e^b \wedge e^c + 3H^{def} \varepsilon_{abcd} \Lambda_{ef} \wedge e^b \wedge e^c \\ - 2\Lambda_{ab} \wedge d\phi \wedge e^b - 2\frac{1}{4!} m^2 \phi \varepsilon_{abcd} e^b \wedge e^c \wedge e^d = 0. \end{aligned} \quad (6.111)$$

This system contains two dynamical fields, the tetrad e^a and the scalar ϕ ; all other variables can be expressed in terms of them and their derivatives:

$$\begin{aligned} \lambda_{ab\mu\nu} &= R_{ab\mu\nu}, & \omega^{ab}{}_{\mu} &= \Delta^{ab}{}_{\mu}, & \gamma_{\mu\nu\rho} &= -\frac{e}{2} \varepsilon_{\mu\nu\rho\sigma} \partial^\sigma \phi, \\ \beta^a{}_{\mu\nu} &= 0, & \Lambda^{ab}{}_{\mu} &= \frac{1}{12e} g_{\mu\lambda} \varepsilon^{\lambda\nu\rho\sigma} \partial_\nu \phi e^a{}_{\rho} e^b{}_{\sigma}, & \lambda_\mu &= \partial_\mu \phi, \\ H^{abc} &= \frac{1}{6e} \varepsilon^{\mu\nu\rho\sigma} \partial_\mu \phi e^a{}_{\nu} e^b{}_{\rho} e^c{}_{\sigma}, & B_{ab\mu\nu} &= \frac{1}{8\pi l_p^2} \varepsilon_{abcd} e^c{}_{\mu} e^d{}_{\nu}. \end{aligned} \quad (6.112)$$

Unlike the equations for the Lagrange multipliers, the equations for e^a and ϕ are differential equations. The equation for ϕ , (6.110), gives the Klein–Gordon equation

$$(\nabla_\mu \nabla^\mu - m^2) \phi = 0, \quad (6.113)$$

while the tetrad equation (6.111) is

$$R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R = 8\pi l_p^2 T^{\mu\nu}, \quad (6.114)$$

where the energy–momentum tensor of the real scalar field is

$$T^{\mu\nu} \equiv \partial^\mu \phi \partial^\nu \phi - \frac{1}{2} g^{\mu\nu} (\partial_\rho \phi \partial^\rho \phi + m^2 \phi^2). \quad (6.115)$$

6.3 Einstein–Cartan–Dirac theory

To describe a Dirac field interacting with Einstein–Cartan gravity, define the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$ as follows [16]. The Lie groups are

$$G = SO(3, 1), \quad H = \mathbb{R}^4, \quad L = \mathbb{R}^8(\mathbb{G}), \quad (6.116)$$

where $\mathbb{G}$ denotes the set of Grassmann numbers. The maps ∂ and δ and the Peiffer lifting remain trivial, as for the scalar field. The group G acts on itself by conjugation, on H in the vector representation, and on L in the spinor representation. More precisely, if P_α and P^α are the eight generators of $\mathbb{R}^8(\mathbb{G})$, with $\alpha = 1, \dots, 4$, the action $\triangleright$ of G on L is

$$M_{ab} \triangleright P_\alpha = \frac{1}{2}(\sigma_{ab})^\beta{}_\alpha P_\beta, \quad M_{ab} \triangleright P^\alpha = -\frac{1}{2}(\sigma_{ab})^\alpha{}_\beta P^\beta, \quad (6.117)$$

where $\sigma_{ab} = \frac{1}{4}[\gamma_a, \gamma_b]$ and γ_a are the Dirac matrices satisfying

$$\{\gamma_a, \gamma_b\} \equiv \gamma_a \gamma_b + \gamma_b \gamma_a = -2\eta_{ab}.$$

As for the scalar field, the choice of L determines the matter fields present in the theory, while the action $\triangleright$ of G on L ensures their required transformation properties.

Having completed the definition of the 2-crossed module, we can define the corresponding $3BF$ action. For this 3-group, the 3-connection (α, β, γ) is

$$\alpha = \omega^{ab} M_{ab}, \quad \beta = \beta^a P_a, \quad \gamma = \gamma^\alpha P_\alpha + \bar{\gamma}_\alpha P^\alpha, \quad (6.118)$$

and the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is

$$\begin{aligned} \mathcal{F} &= R^{ab} M_{ab}, \quad \mathcal{G} = \nabla \beta^a P_a, \\ \mathcal{H} &= \left(d\gamma^\alpha + \frac{1}{2} \omega^{ab} (\sigma_{ab})^\alpha{}_\beta \gamma^\beta \right) P_\alpha + \left(d\bar{\gamma}_\alpha - \frac{1}{2} \omega^{ab} \bar{\gamma}_\beta (\sigma_{ab})^\beta{}_\alpha \right) P^\alpha \\ &\equiv (\vec{\nabla} \gamma)^\alpha P_\alpha + (\bar{\gamma} \overleftarrow{\nabla})_\alpha P^\alpha. \end{aligned} \quad (6.119)$$

These expressions use the action $\triangleright$ defined in (6.117). The bilinear form $\langle _, _ \rangle_{\mathfrak{l}}$ is defined on the generators of L by

$$\begin{aligned} \langle P_\alpha, P_\beta \rangle_{\mathfrak{l}} &= 0, \quad \langle P^\alpha, P^\beta \rangle_{\mathfrak{l}} = 0, \\ \langle P_\alpha, P^\beta \rangle_{\mathfrak{l}} &= -\delta_\alpha^\beta, \quad \langle P^\alpha, P_\beta \rangle_{\mathfrak{l}} = \delta_\beta^\alpha. \end{aligned} \quad (6.120)$$

Unlike the symmetric forms encountered so far, this bilinear form is antisymmetric on generators. The reason is as follows. For elements $A, B \in \mathfrak{l}$ it must be a symmetric, bilinear, nondegenerate form. Expanding A and B in an algebra basis gives

$$\langle A, B \rangle_{\mathfrak{l}} = A^I B^J g_{IJ}, \quad \langle B, A \rangle_{\mathfrak{l}} = B^J A^I g_{JI}. \quad (6.121)$$

Symmetry requires the two expressions to agree. Since the coefficients in $\mathfrak{l}$ are Grassmann numbers, $A^I B^J = -B^J A^I$, implying $g_{IJ} = -g_{JI}$. The antisymmetry in (6.120) therefore compensates for the anticommuting nature of Grassmann numbers and makes the bilinear form symmetric for arbitrary $A, B \in \mathfrak{l}$.

The action $\triangleright$ of G on L is chosen to give the Lagrange multiplier D in (6.1) its spinorial character. The group L determines the structure of D , whose components are eight independent Grassmann matter fields. Since D is a differential 0-form transforming in the spinor representation of $SO(3, 1)$, it can be identified with a Dirac bispinor:

$$D = \psi^\alpha P_\alpha + \bar{\psi}_\alpha P^\alpha. \quad (6.122)$$

As for the scalar field, this demonstrates how the matter sector can be fixed by choosing the group L and the action $\triangleright$ of G on it, both components of the 2-crossed module. The Lorentz-transformation properties of the fields are set by the choice of $\triangleright$.

For this 2-crossed module and the preceding field identifications, the corresponding 3BF action is

$$S_{3BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + (\bar{\gamma} \overleftarrow{\nabla})_\alpha \psi^\alpha + \bar{\psi}_\alpha (\overrightarrow{\nabla} \gamma)^\alpha. \quad (6.123)$$

To obtain spinor fields with the appropriate dynamics coupled to Einstein–Cartan gravity, suitable *simplicity constraints* must be added to (6.123):

$$\begin{aligned} S = & \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + (\bar{\gamma} \overleftarrow{\nabla})_\alpha \psi^\alpha + \bar{\psi}_\alpha (\overrightarrow{\nabla} \gamma)^\alpha \\ & - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right) \\ & - \lambda^\alpha \wedge \left(\bar{\gamma}_\alpha - \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c (\bar{\psi} \gamma^d)_\alpha \right) \\ & + \bar{\lambda}_\alpha \wedge \left(\gamma^\alpha + \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c (\gamma^d \psi)^\alpha \right) \\ & - \frac{1}{12} m \bar{\psi} \psi \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d + 2\pi i l_p^2 \bar{\psi} \gamma_5 \gamma^a \psi \varepsilon_{abcd} e^b \wedge e^c \wedge \beta^d. \end{aligned} \quad (6.124)$$

As in the scalar case, the first line is the topological sector (6.123), the second is the gravitational constraint, and the third and fourth are new *simplicity constraints* for the Dirac field containing the Lagrange-multiplier 1-forms λ^α and $\bar{\lambda}_\alpha$. The fifth line contains the Dirac mass term and a term ensuring the appropriate interaction between torsion and Dirac spin. Einstein–Cartan theory gives

$$T_a \equiv \nabla e_a = 2\pi l_p^2 s_a, \quad (6.125)$$

as one of the equations of motion, where

$$s_a = i \varepsilon_{abcd} e^b \wedge e^c \bar{\psi} \gamma_5 \gamma^d \psi \quad (6.126)$$

is the Dirac-spin 2-form. Other choices are possible, but we restrict attention to this one.

Varying the action (6.124) with respect to B_{ab} , λ^{ab} , $\bar{\gamma}_\alpha$, γ^α , λ^α , $\bar{\lambda}_\alpha$, $\bar{\psi}_\alpha$, ψ^α , e^a , β^a , and ω^{ab}

gives

$$R^{ab} - \lambda^{ab} = 0, \quad (6.127)$$

$$B_{ab} - \frac{1}{16\pi l_p^2} \varepsilon_{abcd} e^c \wedge e^d = 0, \quad (6.128)$$

$$(\vec{\nabla} \psi)^\alpha - \lambda^\alpha = 0, \quad (6.129)$$

$$(\bar{\psi} \overleftarrow{\nabla})_\alpha - \bar{\lambda}_\alpha = 0, \quad (6.130)$$

$$\bar{\gamma}_\alpha - \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c (\bar{\psi} \gamma^d)_\alpha = 0, \quad (6.131)$$

$$\gamma^\alpha + \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c (\gamma^d \psi)^\alpha = 0, \quad (6.132)$$

$$\begin{aligned} d\gamma^\alpha + \omega^\alpha{}_\beta \wedge \gamma^\beta + \frac{i}{6} \lambda^\beta \wedge \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \gamma^{d\alpha}{}_\beta + \frac{1}{12} m \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \psi^\alpha \\ + i2\pi l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c (\gamma_5 \gamma^d \psi)^\alpha = 0, \end{aligned} \quad (6.133)$$

$$\begin{aligned} d\bar{\gamma}_\alpha - \bar{\gamma}_\beta \wedge \omega^\beta{}_\alpha + \frac{i}{6} \bar{\lambda}_\beta \wedge \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \gamma^{d\beta}{}_\alpha - \frac{1}{12} m \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \bar{\psi}_\alpha \\ - i2\pi l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c (\bar{\psi} \gamma_5 \gamma^d)_\alpha = 0, \end{aligned} \quad (6.134)$$

$$\begin{aligned} \nabla \beta_a + 2\varepsilon_{abcd} \lambda^{bc} \wedge e^d - \frac{i}{2} \varepsilon_{abcd} \lambda^\alpha \wedge e^b \wedge e^c (\bar{\psi} \gamma^d)_\alpha + \frac{i}{2} \varepsilon_{abcd} \bar{\lambda}_\alpha \wedge e^b \wedge e^c (\gamma^d \psi)^\alpha \\ - \frac{1}{3} \varepsilon_{abcd} e^b \wedge e^c \wedge e^d m \bar{\psi} \psi - 4\pi l_p^2 i \varepsilon_{abcd} e^b \wedge \beta^c \bar{\psi} \gamma_5 \gamma^d \psi = 0, \end{aligned} \quad (6.135)$$

$$\nabla e_a - i2\pi l_p^2 \varepsilon_{abcd} e^b \wedge e^c \bar{\psi} \gamma_5 \gamma^d \psi = 0, \quad (6.136)$$

$$\nabla B_{ab} - e_{[a} \wedge \beta_{b]} + \bar{\gamma} \frac{1}{8} [\gamma_a, \gamma_b] \psi + \bar{\psi} \frac{1}{8} [\gamma_a, \gamma_b] \gamma = 0. \quad (6.137)$$

The only dynamical fields are e^a , ψ , and $\bar{\psi}$; the remaining fields can be expressed algebraically in terms of them and their derivatives:

$$\begin{aligned} B_{ab\mu\nu} &= \frac{1}{8\pi l_p^2} \varepsilon_{abcd} e^c{}_\mu e^d{}_\nu, & \lambda^\alpha{}_\mu &= (\vec{\nabla}_\mu \psi)^\alpha, & \bar{\lambda}_{\alpha\mu} &= (\bar{\psi} \overleftarrow{\nabla}_\mu)_\alpha, \\ \bar{\gamma}_{\alpha\mu\nu\rho} &= i\varepsilon_{abcd} e^a{}_\mu e^b{}_\nu e^c{}_\rho (\bar{\psi} \gamma^d)_\alpha, & \gamma^\alpha{}_{\mu\nu\rho} &= -i\varepsilon_{abcd} e^a{}_\mu e^b{}_\nu e^c{}_\rho (\gamma^d \psi)^\alpha, \\ \beta^a{}_{\mu\nu} &= 0, & \lambda_{ab\mu\nu} &= R_{ab\mu\nu}, & \omega^{ab}{}_\mu &= \Delta^{ab}{}_\mu + K^{ab}{}_\mu. \end{aligned} \quad (6.138)$$

Here $K^{ab}{}_\mu$ is the contorsion tensor, defined in the standard way from the torsion tensor. The action also produces the required torsion, since one equation of motion is precisely (6.125):

$$T_a \equiv \nabla e_a = 2\pi l_p^2 s_a. \quad (6.139)$$

Finally, the equations of motion for ψ and $\bar{\psi}$ are the standard covariant Dirac equations

$$(i\gamma^a e^\mu{}_a \vec{\nabla}_\mu - m)\psi = 0, \quad (6.140)$$

and its conjugate,

$$\bar{\psi}(i\overleftarrow{\nabla}_\mu e^\mu{}_a \gamma^a + m) = 0, \quad (6.141)$$

where e^μ_a is the inverse tetrad. The equation of motion for e^a is

$$R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R = 8\pi l_p^2 T^{\mu\nu}, \quad (6.142)$$

where the energy–momentum tensor is

$$T^{\mu\nu} \equiv \frac{i}{2}\bar{\psi}\gamma^a\overset{\leftrightarrow}{\nabla}^\nu e^\mu_a\psi - \frac{1}{2}g^{\mu\nu}\bar{\psi}\left(i\gamma^a\overset{\leftrightarrow}{\nabla}_\rho e^\rho_a - 2m\right)\psi, \quad (6.143)$$

Here the notation $\overset{\leftrightarrow}{\nabla} = \overset{\rightarrow}{\nabla} - \overset{\leftarrow}{\nabla}$ has been used. As expected, equations (6.139), (6.140), (6.141), and (6.142) are precisely the equations of motion describing a Dirac field interacting with Einstein–Cartan gravity.

6.4 Weyl and Majorana fields interacting with Einstein–Cartan gravity

As is well known, the solution space of the Dirac equation is not an irreducible representation of the Lorentz group. Dirac fermions can be rewritten by separating fields of left and right chirality, which are irreducible representations and therefore do not change chirality under Lorentz transformations. To treat these spinor representations within our approach, the theories of left- and right-handed Weyl fields must be written as constrained $3BF$ actions. For simplicity, only a left-handed field will be considered here; the theory of a right-handed field is defined analogously. Weyl and Majorana fermions can both be treated in this way, with an additional mass term required in the Majorana case.

The corresponding 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$ is defined as follows [16]. The Lie groups G , H , and L are

$$G = SO(3, 1), \quad H = \mathbb{R}^4, \quad L = \mathbb{R}^4(\mathbb{G}). \quad (6.144)$$

The maps ∂ and δ , as well as the Peiffer lifting, are trivial. The action $\triangleright$ of G on G , H , and L is defined as in the Dirac-fermion case, while the spinor representation for the left-handed field is

$$M_{ab} \triangleright P^\alpha = \frac{1}{2}(\sigma_{ab})^\alpha_\beta P^\beta, \quad M_{ab} \triangleright P_{\dot{\alpha}} = \frac{1}{2}(\bar{\sigma}_{ab})^{\dot{\beta}}_{\dot{\alpha}} P_{\dot{\beta}}, \quad (6.145)$$

where $\sigma^{ab} = -\bar{\sigma}^{ab} = \frac{1}{4}(\sigma^a\bar{\sigma}^b - \sigma^b\bar{\sigma}^a)$, with $\sigma^a = (1, \vec{\sigma})$ and $\bar{\sigma}^a = (1, -\vec{\sigma})$, and $\vec{\sigma}$ denotes the three Pauli matrices. The four generators of L are denoted by P^α and $P_{\dot{\alpha}}$, where the Weyl indices $\alpha, \dot{\alpha}$ take the values 1, 2.

For this choice of 2-crossed module, the corresponding 3-connection (α, β, γ) has the form

$$\alpha = \omega^{ab}M_{ab}, \quad \beta = \beta^a P_a, \quad \gamma = \gamma_\alpha P^\alpha + \bar{\gamma}^{\dot{\alpha}} P_{\dot{\alpha}}, \quad (6.146)$$

while the corresponding fake 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is defined by equation (2.118):

$$\begin{aligned} \mathcal{F} &= R^{ab}M_{ab}, \quad \mathcal{G} = \nabla\beta^a P_a, \\ \mathcal{H} &= (d\gamma_\alpha + \frac{1}{2}\omega_{ab}(\sigma^{ab})^\beta_\alpha\gamma_\beta)P^\alpha + (d\bar{\gamma}^{\dot{\alpha}} + \frac{1}{2}\omega_{ab}(\bar{\sigma}^{ab})^{\dot{\alpha}}_{\dot{\beta}}\bar{\gamma}^{\dot{\beta}})P_{\dot{\alpha}} \equiv (\overset{\rightarrow}{\nabla}\gamma)_\alpha P^\alpha + (\bar{\gamma}\overset{\leftarrow}{\nabla})^{\dot{\alpha}} P_{\dot{\alpha}}. \end{aligned} \quad (6.147)$$

As in the Dirac-spinor case, the Lagrange multiplier D is identified with the spinor fields ψ_α and $\bar{\psi}^{\dot{\alpha}}$:

$$D = \psi_\alpha P^\alpha + \bar{\psi}^{\dot{\alpha}} P_{\dot{\alpha}}, \quad (6.148)$$

The bilinear form $\langle _, _ \rangle_I$ on L is defined by its action on the generators:

$$\langle P^\alpha, P^\beta \rangle_I = \varepsilon^{\alpha\beta}, \quad \langle P_{\dot{\alpha}}, P_{\dot{\beta}} \rangle_I = \varepsilon_{\dot{\alpha}\dot{\beta}}, \quad \langle P^\alpha, P_{\dot{\beta}} \rangle_I = 0, \quad \langle P_{\dot{\alpha}}, P^\beta \rangle_I = 0. \quad (6.149)$$

In these equations, $\varepsilon^{\alpha\beta}$ and $\varepsilon_{\dot{\alpha}\dot{\beta}}$ are the standard notation for the two-dimensional Levi-Civita symbol. The topological $3BF$ action (6.1) for spinor fields and gravity can now be defined as

$$S_{3BF} = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \psi^\alpha \wedge (\vec{\nabla} \gamma)_\alpha + \bar{\psi}_{\dot{\alpha}} \wedge (\bar{\gamma} \vec{\nabla})^{\dot{\alpha}}. \quad (6.150)$$

To obtain a theory with the appropriate Weyl-spinor dynamics, the relevant *simplicity constraints* must be added to the action (6.150), giving the constrained $3BF$ action

$$\begin{aligned} S = & \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab} + e_a \wedge \nabla \beta^a + \psi^\alpha \wedge (\vec{\nabla} \gamma)_\alpha + \bar{\psi}_{\dot{\alpha}} \wedge (\bar{\gamma} \vec{\nabla})^{\dot{\alpha}} \\ & - \lambda_{ab} \wedge (B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d) \\ & - \lambda^\alpha \wedge (\gamma_\alpha + \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \sigma^d_{\alpha\beta} \bar{\psi}^{\dot{\beta}}) - \bar{\lambda}_{\dot{\alpha}} \wedge (\bar{\gamma}^{\dot{\alpha}} + \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \bar{\sigma}^{d\dot{\alpha}\beta} \psi_\beta) \\ & - 4\pi l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c (\bar{\psi}_{\dot{\alpha}} \bar{\sigma}^{d\dot{\alpha}\beta} \psi_\beta). \end{aligned} \quad (6.151)$$

The third line contains the new constraints and the Lagrange-multiplier 1-forms λ_α and $\bar{\lambda}^{\dot{\alpha}}$. The fourth line ensures the appropriate interaction between torsion and the spin of the Weyl field. Using the torsion–spin interaction for Dirac particles, we have defined the spin of the Weyl field as

$$s_a \equiv i \varepsilon_{abcd} e^b \wedge e^c \psi^\alpha \sigma^d_{\alpha\beta} \bar{\psi}^{\dot{\beta}}, \quad (6.152)$$

and its relation to torsion as

$$T_a = 4\pi l_p^2 s_a. \quad (6.153)$$

Majorana fields are defined analogously, with the following additional mass term in the action:

$$-\frac{1}{12} m \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d (\psi^\alpha \psi_\alpha + \bar{\psi}_{\dot{\alpha}} \bar{\psi}^{\dot{\alpha}}). \quad (6.154)$$

Varying the action (6.151) successively with respect to B_{ab} , λ^{ab} , γ_α , $\bar{\gamma}^{\dot{\alpha}}$, λ_α , $\bar{\lambda}^{\dot{\alpha}}$, ψ_α , $\bar{\psi}^{\dot{\alpha}}$, e^a , β^a , and ω^{ab} gives the equations of motion presented in Appendix B. The only dynamical degrees of freedom are the fields ψ_α , $\bar{\psi}^{\dot{\alpha}}$, and e^a ; the remaining variables can be expressed algebraically as functions of these fields and their derivatives:

$$\begin{aligned} \lambda^{ab}{}_{\mu\nu} &= R^{ab}{}_{\mu\nu}, \quad B_{ab\mu\nu} = \frac{1}{8\pi l_p^2} \varepsilon_{abcd} e^c{}_\mu e^d{}_\nu, \quad \lambda_{\alpha\mu} = \nabla_\mu \psi_\alpha, \quad \bar{\lambda}^{\dot{\alpha}}{}_\mu = \nabla_\mu \bar{\psi}^{\dot{\alpha}}, \\ \gamma_{\alpha\mu\nu\rho} &= i \varepsilon_{abcd} e^a{}_\mu e^b{}_\nu e^c{}_\rho \sigma^d_{\alpha\beta} \bar{\psi}^{\dot{\beta}}, \quad \bar{\gamma}^{\dot{\alpha}}{}_{\mu\nu\rho} = i \varepsilon_{abcd} e^a{}_\mu e^b{}_\nu e^c{}_\rho \bar{\sigma}^{d\dot{\alpha}\beta} \psi_\beta, \quad \omega_{ab\mu} = \Delta_{ab\mu} + K_{ab\mu}. \end{aligned} \quad (6.155)$$

Notice that the result $\beta = 0$ is unchanged. Finally, the equations of motion for the dynamical fields are

$$\begin{aligned} \bar{\sigma}^{a\dot{\alpha}\beta} e^\mu{}_a \nabla_\mu \psi_\beta &= 0, \quad \sigma^a{}_{\alpha\dot{\beta}} e^\mu{}_a \nabla_\mu \bar{\psi}^{\dot{\beta}} = 0, \\ R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R &= 8\pi l_p^2 T^{\mu\nu}, \\ T^{\mu\nu} &\equiv \frac{i}{2} \bar{\psi} \bar{\sigma}^b e^\nu{}_b \nabla^\mu \psi + \frac{i}{2} \psi \sigma^b e^\nu{}_b \nabla^\mu \bar{\psi} - g^{\mu\nu} \frac{1}{2} \left(i \bar{\psi} \bar{\sigma}^a e^\lambda{}_a \nabla_\lambda \psi + i \psi \sigma^a e^\lambda{}_a \nabla_\lambda \bar{\psi} \right). \end{aligned} \quad (6.156)$$

In the Majorana case, the equations of motion (6.155) remain unchanged, while the equations of motion for the fields ψ_α and $\bar{\psi}^{\dot{\alpha}}$ are

$$i\sigma^a_{\alpha\dot{\beta}}e^\mu_a\nabla_\mu\bar{\psi}^{\dot{\beta}} - m\psi_\alpha = 0, \quad i\bar{\sigma}^{a\dot{\alpha}\beta}e^\mu_a\nabla_\mu\psi_\beta - m\bar{\psi}^{\dot{\alpha}} = 0, \quad (6.157)$$

and the energy–momentum tensor has the form

$$T^{\mu\nu} \equiv \frac{i}{2}\bar{\psi}\bar{\sigma}^b e^\nu_b \nabla^\mu\psi + \frac{i}{2}\psi\sigma^b e^\nu_b \nabla^\mu\bar{\psi} - g^{\mu\nu} \frac{1}{2} \left[i\bar{\psi}\bar{\sigma}^a e^\lambda_a \nabla_\lambda\psi + i\psi\sigma^a e^\lambda_a \nabla_\lambda\bar{\psi} - \frac{1}{2}m(\psi\psi + \bar{\psi}\bar{\psi}) \right]. \quad (6.158)$$

6.5 The Standard Model

The appropriate 3-group describing all fields present in the Standard Model, with their corresponding dynamics, is obtained by choosing the Lie groups G , H , and L as follows [16]:

$$G = SO(3,1) \times SU(3) \times SU(2) \times U(1), \quad H = \mathbb{R}^4, \quad L = \mathbb{R}^4(\mathbb{C}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}), \quad (6.159)$$

where $\mathbb{C}$ denotes the set of complex numbers. The motivation for this choice of groups becomes clear from Table 6.4.

First generation of leptons	red color first generation of quarks	green color first generation of quarks	blue color first generation of quarks
$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}_L$	$\begin{pmatrix} u_r \\ d_r \end{pmatrix}_L$	$\begin{pmatrix} u_g \\ d_g \end{pmatrix}_L$	$\begin{pmatrix} u_b \\ d_b \end{pmatrix}_L$
$(\nu_e)_R$	$(u_r)_R$	$(u_g)_R$	$(u_b)_R$
$(e^-)_R$	$(d_r)_R$	$(d_g)_R$	$(d_b)_R$

Table 6.4: Matter fields present in the Standard Model of particle physics (first generation).

Counting the fields in Table 6.4, we conclude that 16 spinors are needed to describe the first generation of spinorial matter fields in the Standard Model of particle physics; hence the group L must contain $L = \mathbb{R}^{64}(\mathbb{G})$. Since there are three generations of matter, the full subgroup of L describing the fermionic fields is $L = \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G})$. To define the Higgs sector, two complex scalar fields $\begin{pmatrix} \phi^+ \\ \phi_0 \end{pmatrix}$ are required, so the subgroup of L corresponding to the scalar sector of the Standard Model is $L = \mathbb{R}^4(\mathbb{C})$.

To define a 2-crossed module whose associated 3BF action has the appropriate dynamics, the maps ∂ and δ and the Peiffer lifting must be chosen to be trivial. Moreover, by the definition of a 2-crossed module, the action of G on itself is conjugation. The subgroup $SO(3,1)$ of G acts on H in the vector representation, whereas the action of the subgroup $SU(3) \times SU(2) \times U(1)$ of G on H is trivial. The action of $SO(3,1)$ on the subgroup of L corresponding to the scalar matter sector, namely the $\mathbb{R}^4(\mathbb{C})$ subgroup of L , is trivial, while on each set of four generators describing a spinor field it is given by the spinor representation, as shown in Section 6.3. The transformation properties of the spinor fields under the subgroup $SU(3) \times SU(2) \times U(1)$ of G are specified by the action of this subgroup on L .

6.5.1 Leptons and the electroweak interaction

We demonstrate the construction of the 2-crossed module on the simple example of one lepton family and the electroweak interaction. The remainder of the Standard Model is defined analogously.

The Lie groups G , H , and L are defined as follows:

$$G = SO(3, 1) \times SU(2) \times U(1), \quad H = \mathbb{R}^4, \quad L^{\text{leptons and Higgs boson}} = \mathbb{R}^{16}(\mathbb{G}) \times \mathbb{R}^4(\mathbb{C}). \quad (6.160)$$

The corresponding 3-connection is then

$$\alpha = \omega^{ab} M_{ab} + W^I T_I + A Y, \quad \beta = \beta^a P_a, \quad \gamma = \gamma_\alpha^{\tilde{L}} P_\alpha^{\tilde{L}} + \gamma^{\dot{\alpha}}_{\tilde{L}} P_{\dot{\alpha}}^{\tilde{L}} + \gamma_\alpha^{\tilde{R}} P_\alpha^{\tilde{R}} + \gamma^{\dot{\alpha}}_{\tilde{R}} P_{\dot{\alpha}}^{\tilde{R}} + \gamma^{\tilde{a}} P_{\tilde{a}}. \quad (6.161)$$

Here the indices $I, J, \dots$ take the values 1, 2, 3 and enumerate the Pauli matrices, the generators of $SU(2)$; the indices $\tilde{L}, \tilde{L}', \dots$ take the values 1, 2 and enumerate the components of the left-handed doublet; $\tilde{R}$ denotes the right-handed singlets $(e^-)_R$ and $(\nu_e)_R$; and the indices $\tilde{a}, \tilde{b}, \dots$ take the values 1, 2 and enumerate the components of the scalar doublet. We also define the index $\tilde{i} = (\tilde{L}, \tilde{R})$, which takes the values 1, $\dots$, 4.

The action of G on L is defined as follows:

$$\begin{aligned} M_{ab} \triangleright P^\alpha_i &= \frac{1}{2}(\sigma_{ab})^\alpha_\beta P^\beta_i, & M_{ab} \triangleright P_{\dot{\alpha}i} &= \frac{1}{2}(\bar{\sigma}_{ab})^{\dot{\beta}}_{\dot{\alpha}} P_{\dot{\beta}i}, & M_{ab} \triangleright P_{\tilde{a}} &= 0, \\ T_I \triangleright P^\alpha_{\tilde{L}} &= \frac{1}{2}(\sigma_I)^{\tilde{L}'}_{\tilde{L}} P^\alpha_{\tilde{L}'}, & T_I \triangleright P_{\dot{\alpha}\tilde{L}} &= \frac{1}{2}(\sigma_I)^{\tilde{L}'}_{\tilde{L}} P_{\dot{\alpha}\tilde{L}'}, \\ T_I \triangleright P^\alpha_{\tilde{R}} &= 0, & T_I \triangleright P_{\dot{\alpha}\tilde{R}} &= 0, & T_I \triangleright P_{\tilde{a}} &= \frac{1}{2}(\sigma_I)^{\tilde{b}}_{\tilde{a}} P_{\tilde{b}}, \\ Y \triangleright P^\alpha_{\tilde{L}} &= -P^\alpha_{\tilde{L}}, & Y \triangleright P^\alpha_{\tilde{R}} &= -2P^\alpha_{\tilde{R}}, & Y \triangleright P_{\tilde{a}} &= P_{\tilde{a}}, \\ Y \triangleright P_{\dot{\alpha}\tilde{L}} &= -P_{\dot{\alpha}\tilde{L}}, & Y \triangleright P_{\dot{\alpha}\tilde{R}} &= -2P_{\dot{\alpha}\tilde{R}}. \end{aligned} \quad (6.162)$$

For this choice of 2-crossed module, the corresponding 3-curvatures are

$$\begin{aligned} \mathcal{F} &= R^{ab} M_{ab} + F^I T_I + F Y, & \mathcal{G} &= \nabla \beta^a P_a, \\ \mathcal{H} &= (\vec{\nabla} \gamma^{\tilde{L}})_\alpha P^\alpha_{\tilde{L}} + (\bar{\gamma}_{\tilde{L}}^{\leftarrow} \nabla^{\leftarrow})^{\dot{\alpha}} P_{\dot{\alpha}}^{\tilde{L}} + (\vec{\nabla} \gamma^{\tilde{R}})_\alpha P^\alpha_{\tilde{R}} + (\bar{\gamma}_{\tilde{R}}^{\leftarrow} \nabla^{\leftarrow})^{\dot{\alpha}} P_{\dot{\alpha}}^{\tilde{R}} + d\gamma^{\tilde{a}} P_{\tilde{a}}. \end{aligned} \quad (6.163)$$

The topological $3BF$ action is

$$S = \int B_{ab} R^{ab} + B_I F^I + B F + e_a \nabla \beta^a + \psi^\alpha_{\tilde{i}} (\vec{\nabla} \gamma^{\tilde{i}})_\alpha + \bar{\psi}_{\dot{\alpha}}^{\tilde{i}} (\bar{\gamma}_{\tilde{i}}^{\leftarrow} \nabla^{\leftarrow})^{\dot{\alpha}} + \phi^{\tilde{a}} d\gamma_{\tilde{a}}. \quad (6.164)$$

The notation can now be simplified by introducing indices $\hat{a}$ that enumerate the generators of G , indices $\hat{a}$ for H , and indices $\hat{A}$ for L . To introduce the appropriate degrees of freedom describing the first lepton family interacting with the electroweak gauge fields, the Higgs field, and gravity, the corresponding *simplicity constraints* must be added to the topological action

(6.164) as follows:

$$\begin{aligned}
S = & \int B_{\hat{\alpha}} \wedge \mathcal{F}^{\hat{\alpha}} + e_{\hat{a}} \wedge \mathcal{G}^{\hat{a}} + D_{\hat{A}} \wedge \mathcal{H}^{\hat{A}} \\
& + \left(B_{\hat{\alpha}} - C_{\hat{\alpha}}^{\hat{\beta}} M_{cd\hat{\beta}} e^c \wedge e^d \right) \wedge \lambda^{\hat{\alpha}} - \left(\gamma_{\hat{A}} - e^a \wedge e^b \wedge e^c C_{\hat{A}}^{\hat{B}} M_{abc\hat{B}} \right) \wedge \lambda^{\hat{A}} \\
& + \zeta^{ab}{}_{\hat{\alpha}} \wedge \left(M_{ab}{}^{\hat{\alpha}} \varepsilon^{cdef} e_c \wedge e_d \wedge e_e \wedge e_f - F^{\hat{\alpha}} \wedge e_c \wedge e_d \right) \\
& + \zeta^{ab}{}_{\hat{A}} \wedge \left(M_{abc}{}^{\hat{A}} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - F^{\hat{A}} \wedge e_a \wedge e_b \right) \\
& - \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \left(Y_{\hat{A}\hat{B}\hat{C}} D^{\hat{A}} D^{\hat{B}} D^{\hat{C}} + M_{\hat{A}\hat{B}} D^{\hat{A}} D^{\hat{B}} + L_{\hat{A}\hat{B}\hat{C}\hat{D}} D^{\hat{A}} D^{\hat{B}} D^{\hat{C}} D^{\hat{D}} \right) \\
& - 4\pi i l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c D_{\hat{A}} T^{d\hat{A}}{}_{\hat{B}} D^{\hat{B}},
\end{aligned} \tag{6.165}$$

where

$$\begin{aligned}
B_{\hat{\alpha}} &= [B_{ab} \ B_I \ B]^T, \quad \mathcal{F}^{\hat{\alpha}} = [R_{ab} \ F_I \ F]^T, \quad D_{\hat{A}} = [\psi^{\alpha}{}_{\bar{L}} \ \bar{\psi}^{\alpha}{}_{\bar{L}} \ \psi^{\alpha}{}_{\bar{R}} \ \bar{\psi}^{\alpha}{}_{\bar{R}} \ \phi_{\bar{a}}], \\
\mathcal{H}^{\hat{A}} &= [(\vec{\nabla} \gamma_{\bar{L}})_{\alpha} \ (\bar{\gamma}_{\bar{L}} \overleftarrow{\nabla})^{\alpha} \ (\vec{\nabla} \gamma_{\bar{R}})_{\alpha} \ (\bar{\gamma}_{\bar{R}} \overleftarrow{\nabla})^{\alpha} \ d\gamma_{\bar{a}}]^T, \quad \gamma_{\hat{A}} = [\gamma^{\alpha}{}_{\bar{L}} \ \bar{\gamma}^{\alpha}{}_{\bar{L}} \ \gamma^{\alpha}{}_{\bar{R}} \ \bar{\gamma}^{\alpha}{}_{\bar{R}} \ \gamma_{\bar{a}}], \\
\lambda^{\hat{\alpha}} &= [-\lambda^{ab} \ \lambda^I \ \lambda]^T, \quad M_{cd\hat{\alpha}} = [\varepsilon_{abcd} \ M_{cdI} \ M_{cd}], \\
\lambda^{\hat{A}} &= [\lambda_{\alpha L} \ \bar{\lambda}^{\alpha}{}_{\bar{L}} \ \lambda_{\alpha R} \ \bar{\lambda}^{\alpha}{}_{\bar{R}} \ \lambda_{\bar{a}}]^T, \quad \zeta^{cd}{}_{\hat{\alpha}} = [0 \ \zeta^{cd}{}_I \ \zeta^{cd}], \quad \zeta^{ab}{}_{\hat{A}} = [\zeta^{ab} \ 0 \ 0], \\
M_{abc\hat{A}} &= [\varepsilon_{abcd} \sigma^d{}_{\alpha\beta} \bar{\psi}^{\dot{\beta}}{}_L \ \varepsilon_{abcd} \bar{\sigma}^{d\dot{\alpha}\beta} \psi_{\beta L} \ \varepsilon_{abcd} \sigma^d{}_{\alpha\beta} \bar{\psi}^{\dot{\beta}}{}_R \ \varepsilon_{abcd} \bar{\sigma}^{d\dot{\alpha}\beta} \psi_{\beta R} \ M_{abc\bar{a}}].
\end{aligned}$$

The matrices $C_{\hat{\beta}}^{\hat{\alpha}}$, $C_{\hat{B}}^{\hat{A}}$, $M_{\hat{A}\hat{B}}$, $Y_{\hat{A}\hat{B}\hat{C}}$, $L_{\hat{A}\hat{B}\hat{C}\hat{D}}$, and $T^{d\hat{A}}{}_{\hat{B}}$ are constant matrices containing information about the relevant coupling constants, the Higgs-field mass, the Yukawa coupling, the mixing angles, the Higgs self-coupling, and torsion.

6.6 Scalar electrodynamics as a constrained $3BF$ theory

As a first step toward studying the Hamiltonian structure of $3BF$ theories, we consider the simplest nontrivial example: scalar electrodynamics coupled to gravity [24].

The standard formulation of scalar electrodynamics coupled to gravity is given by the action

$$S = \int d^4k \sqrt{-g} \left[-\frac{1}{16\pi l_p^2} R - \frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma} + g^{\mu\nu} \nabla_{\mu} \phi^* \nabla_{\nu} \phi - m^2 \phi^* \phi \right]. \tag{6.166}$$

Here $g_{\mu\nu}$ is the spacetime metric, $g \equiv \det(g_{\mu\nu})$ is its determinant, R is the Ricci scalar, and l_p is the Planck length. The covariant derivative ∇_{μ} of the complex scalar field ϕ is defined by $\nabla_{\mu} \phi = (\partial_{\mu} + ikA_{\mu})\phi$, where A_{μ} is the electromagnetic potential and k denotes the coupling constant, i.e. the electric charge of the field ϕ . In this section, the model is reformulated as a constrained $3BF$ theory for a particular 3-group. The Hamiltonian structure of the theory is considered as a necessary step toward its canonical quantization. For simplicity, the Hamiltonian analysis is performed here only for the topological sector, neglecting the constraint sector; see Appendix C.

To obtain scalar electrodynamics interacting with Einstein–Cartan gravity, the crossed module is chosen as follows. The Lie groups G , H , and L are

$$G = SO(3, 1) \times U(1), \quad H = \mathbb{R}^4, \quad L = \mathbb{R}^2.$$

The maps ∂ and δ are trivial. The action of the algebra $\mathfrak{g}$ on the algebras $\mathfrak{h}$ and $\mathfrak{l}$ is defined by its action on the generators:

$$\begin{aligned} M_{ab} \triangleright P_c &= \triangleright_{ab,c}{}^d P_d = \delta_{[a}{}^d \eta_{b]c} P_d = \eta_{[b|c} P_{|a]}, & T \triangleright P_a &= 0, \\ M_{ab} \triangleright P_A &= 0, & T \triangleright P_A &= \triangleright_A{}^B P_B \end{aligned} \quad (6.167)$$

where M_{ab} are the six generators of $\mathfrak{so}(3, 1)$, T is the generator of $\mathfrak{u}(1)$, P_a are the four generators of $\mathbb{R}^4$, and P_A are the two generators of $\mathbb{R}^2$. In the preceding expression, the action $\triangleright_A{}^B$ of the algebra $\mathfrak{u}(1)$ on $\mathbb{R}^2$ is defined by

$$\triangleright_A{}^B = iq \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The action of $\mathfrak{g}$ on itself is given by the adjoint representation and, for $\mathfrak{g} = \mathfrak{so}(3, 1) \times \mathfrak{u}(1)$, reads

$$\begin{aligned} M_{ab} \triangleright M_{cd} &= \triangleright_{ab,cd}{}^{ef} M_{ef} = f_{ab,cd}{}^{ef} M_{ef} = \eta_{ad} M_{bc} + \eta_{bc} M_{ad} - \eta_{ac} M_{bd} - \eta_{bd} M_{ac}, \\ M_{ab} \triangleright T &= 0, \quad T \triangleright M_{ab} = 0, \quad T \triangleright T = 0, \end{aligned} \quad (6.168)$$

as a consequence of the direct-product structure and the fact that $U(1)$ is Abelian. The Peiffer lifting

$$\{_, _ \}_{\text{pf}} : H \times H \rightarrow L,$$

is also trivial, i.e. all coefficients $X_{ab}{}^A$ vanish:

$$\{P_a, P_b\}_{\text{pf}} \equiv X_{ab}{}^A T_A = 0. \quad (6.169)$$

The 3-curvature corresponding to this choice of 2-crossed module is obtained by first defining the connection (α, β, γ) and then applying formula (2.118). Owing to the direct-product structure, the connection $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$ can be written as $\alpha = \omega + A$, where $\omega \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{so}(3, 1))$ and $A \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{u}(1))$ are differential 1-forms valued in the corresponding algebras. We also define the connections $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathbb{R}^4)$ and $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathbb{R}^2)$. The corresponding 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ now follows from formula (2.118):

$$\begin{aligned} \mathcal{F} &= R^{ab} M_{ab} + FT = (d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}) M_{ab} + dA T, \\ \mathcal{G} &= \mathcal{G}^a P_a = (d\beta^a + \omega^a{}_b \wedge \beta^b) P_a, \\ \mathcal{H} &= \mathcal{H}^A P_A = (d\gamma^A + \triangleright_B{}^A A \wedge \gamma^B) P_A. \end{aligned} \quad (6.170)$$

Notice that the connection ω^{ab} is absent from the last expression, as follows from the definitions of the action $\triangleright$ and the Peiffer lifting $\{_, _ \}_{\text{pf}}$; see (6.167) and (6.169):

$$\begin{aligned} \mathcal{H} &= d\gamma + \alpha \wedge^\triangleright \gamma + \{\beta \wedge \beta\} \\ &= d\gamma^A P_A + (\omega^{ab} M_{ab} + AT) \wedge^\triangleright (\gamma^A P_A) \\ &= d\gamma^A P_A + \omega^{ab} \wedge \gamma^A M_{ab} \triangleright P_A + A \wedge \gamma^A T \triangleright P_A \\ &= d\gamma^A P_A + A \wedge \gamma^A \triangleright_A{}^B P_B \\ &= (d\gamma^A + \triangleright_B{}^A A \wedge \gamma^B) P_A. \end{aligned} \quad (6.171)$$

The components of the differential 2-forms F and R^{ab} , the 3-form $\mathcal{G}$, and the 4-form $\mathcal{H}$ are

$$\begin{aligned}
F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu, \\
R^{ab}{}_{\mu\nu} &= \partial_\mu \omega^{ab}{}_\nu - \partial_\nu \omega^{ab}{}_\mu + \omega^a{}_{c\mu} \omega^{cb}{}_\nu - \omega^a{}_{c\nu} \omega^{cb}{}_\mu, \\
\mathcal{G}^a{}_{\mu\nu\rho} &= \partial_\mu \beta^a{}_{\nu\rho} + \partial_\nu \beta^a{}_{\rho\mu} + \partial_\rho \beta^a{}_{\mu\nu} + \omega^a{}_{b\mu} \beta^b{}_{\nu\rho} + \omega^a{}_{b\nu} \beta^b{}_{\rho\mu} + \omega^a{}_{b\rho} \beta^b{}_{\mu\nu}, \\
\mathcal{H}^A{}_{\mu\nu\rho\sigma} &= \partial_\mu \gamma^A{}_{\nu\rho\sigma} - \partial_\nu \gamma^A{}_{\rho\sigma\mu} + \partial_\rho \gamma^A{}_{\sigma\mu\nu} - \partial_\sigma \gamma^A{}_{\mu\nu\rho} \\
&\quad + \triangleright_B{}^A A_\mu \gamma^B{}_{\nu\rho\sigma} - \triangleright_B{}^A A_\nu \gamma^B{}_{\rho\sigma\mu} + \triangleright_B{}^A A_\rho \gamma^B{}_{\sigma\mu\nu} - \triangleright_B{}^A A_\sigma \gamma^B{}_{\mu\nu\rho}.
\end{aligned} \tag{6.172}$$

We can now define the 3BF action:

$$S_{3BF} = \int_{\mathcal{M}_4} \left(\langle B, \mathcal{F} \rangle_{\mathfrak{g}} + \langle C, \mathcal{G} \rangle_{\mathfrak{h}} + \langle D, \mathcal{H} \rangle_{\mathfrak{l}} \right), \tag{6.173}$$

where $B \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{so}(3, 1))$, $C \in \mathcal{A}^1(\mathcal{M}_4, \mathbb{R}^4)$, and $D \in \mathcal{A}^0(\mathcal{M}_4, \mathbb{R}^2)$ are Lagrange multipliers. The forms $\langle _, _ \rangle_{\mathfrak{g}}$, $\langle _, _ \rangle_{\mathfrak{h}}$, and $\langle _, _ \rangle_{\mathfrak{l}}$ are G -invariant symmetric nondegenerate bilinear forms on $\mathfrak{g}$, $\mathfrak{h}$, and $\mathfrak{l}$, respectively, defined by their action on the generators:

$$\langle M_{ab}, M_{cd} \rangle_{\mathfrak{g}} = g_{ab, cd}, \quad \langle T, T \rangle_{\mathfrak{g}} = 1, \quad \langle M_{ab}, T \rangle_{\mathfrak{g}} = 0, \quad \langle P_a, P_b \rangle_{\mathfrak{h}} = \eta_{ab}, \quad \langle P_A, P_B \rangle_{\mathfrak{l}} = g_{AB},$$

where

$$g_{ab, cd} = \eta_{a[c} \eta_{b]d}, \quad \eta_{ab} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad g_{AB} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Identifying the Lagrange multiplier C^a with the tetrad e^a and the Lagrange multiplier D^A with the scalar-field doublet ϕ^A ,

$$\phi = \phi^A P_A = \phi P_1 + \phi^* P_2,$$

on the basis of their transformation properties, as discussed in Section (6.2), the action (6.173) can be written as

$$S_{3BF} = \int_{\mathcal{M}_4} d^4x \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} B^{ab}{}_{\mu\nu} R^{cd}{}_{\rho\sigma} g_{ab, cd} + \frac{1}{4} B_{\mu\nu} F_{\rho\sigma} + \frac{1}{3!} e^a{}_\mu \mathcal{G}^b{}_{\nu\rho\sigma} \eta_{ab} + \frac{1}{4!} \phi^A \mathcal{H}^B{}_{\mu\nu\rho\sigma} g_{AB} \right). \tag{6.174}$$

Varying the action (6.174) gives the equations of motion

$$\begin{aligned}
\delta B^{ab} &: & 2R_{ab} &= 0, \\
\delta \omega^{ab} &: & \nabla B_{ab} - e_{[a} \wedge \beta_{b]} &= 0, \\
\delta B &: & F &= 0, \\
\delta A &: & dB + \phi_A \triangleright_B{}^A \gamma^B &= 0, \\
\delta e^a &: & \mathcal{G}_a &= 0, \\
\delta \beta^a &: & \nabla e_a &= 0, \\
\delta \phi^A &: & \nabla \gamma_A &= 0, \\
\delta \gamma^A &: & \nabla \phi_A &= 0.
\end{aligned} \tag{6.175}$$

To obtain a theory describing a doublet of scalar fields ϕ^A of mass m and charge q , minimally coupled to gravity and the electromagnetic field, the appropriate constraints must be added to the action (6.174) so that its equations of motion are equivalent to those obtained by varying the action (6.166):

$$\begin{aligned}
S = \int_{\mathcal{M}_4} & B^{ab} \wedge R_{ab} + B \wedge F + e_a \wedge \nabla \beta^a + \phi_A \nabla \gamma^A \\
& - \lambda_{ab} \wedge \left(B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d \right) \\
& + \lambda^A \wedge \left(\gamma_A - \frac{1}{2} H_{abcA} e^a \wedge e^b \wedge e^c \right) + \Lambda^{abA} \wedge \left(H_{abcA} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - \nabla \phi_A A \wedge e_a \wedge e_b \right) \\
& + \lambda \wedge \left(B - \frac{12}{q} M_{ab} e^a \wedge e^b \right) + \zeta^{ab} \left(M_{ab} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f - F \wedge e_a \wedge e_b \right) \\
& - \frac{1}{2 \cdot 4!} m^2 \phi_A \phi^A \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d.
\end{aligned} \tag{6.176}$$

Varying the action (6.176) successively with respect to B_{ab} , B , ω_{ab} , β_a , λ_{ab} , Λ^{abA} , γ^A , λ^A , H_{abcA} , ζ^{ab} , M_{ab} , λ , A , ϕ^A , and e^a gives the equations of motion

$$R^{ab} - \lambda^{ab} = 0, \tag{6.177}$$

$$F + \lambda = 0, \tag{6.178}$$

$$\nabla B^{ab} - e^{[a} \wedge \beta^{b]} = 0, \tag{6.179}$$

$$\nabla e^a = 0, \tag{6.180}$$

$$B^{ab} - \frac{1}{16\pi l_p^2} \varepsilon^{abcd} e_c \wedge e_d = 0, \tag{6.181}$$

$$H_{abcA} \varepsilon^{cdef} e_d \wedge e_e \wedge e_f - \nabla \phi_A \wedge e_a \wedge e_b = 0, \tag{6.182}$$

$$\nabla \phi_A - \lambda_A = 0, \tag{6.183}$$

$$\gamma_A - \frac{1}{2} H_{abcA} e^a \wedge e^b \wedge e^c = 0, \tag{6.184}$$

$$-\frac{1}{2} \lambda^A \wedge e^a \wedge e^b \wedge e^c + \varepsilon^{cdef} \Lambda^{abA} \wedge e_d \wedge e_e \wedge e_f = 0, \tag{6.185}$$

$$M_{ab} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f - F \wedge e_a \wedge e_b = 0, \tag{6.186}$$

$$-\frac{12}{q} \lambda \wedge e^a \wedge e^b + \zeta^{ab} \varepsilon_{cdef} e^c \wedge e^d \wedge e^e \wedge e^f = 0, \tag{6.187}$$

$$B - \frac{12}{g} M_{ab} e^a \wedge e^b = 0, \tag{6.188}$$

$$-dB + d(\zeta^{ab} e_a \wedge e_b) - \phi_A \triangleright_B^A \gamma^B - \Lambda^{abA} \triangleright_B^A \phi_B \wedge e_a \wedge e_b = 0, \tag{6.189}$$

$$\nabla \gamma_A - \nabla(\Lambda^{ab}{}_A \wedge e_a \wedge e_b) - \frac{1}{4!} m^2 \phi_A \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d = 0, \tag{6.190}$$

$$\tag{6.191}$$

$$\begin{aligned}
& \nabla \beta_a + \frac{1}{8\pi l_p^2} \varepsilon_{abcd} \lambda^{bc} \wedge e^d + \frac{3}{2} H_{abcA} \lambda^A \wedge e^b \wedge e^c + 3H^{defA} \varepsilon_{abcd} \Lambda_{efA} \wedge e^b \wedge e^c \\
& - 2\Lambda_{abA} \wedge \nabla \phi^A \wedge e^b - 2\frac{1}{4!} m^2 \phi_A \phi^A \varepsilon_{abcd} e^b \wedge e^c \wedge e^d \\
& - \frac{24}{q} M_{ab} \lambda \wedge e^b + 4\zeta^{ef} M_{ef} \varepsilon_{abcd} e^b \wedge e^c \wedge e^d - 2\zeta_{ab} F \wedge e^b = 0.
\end{aligned} \tag{6.192}$$

The dynamical degrees of freedom are the tetrad e^a , the scalar field ϕ^A , and the electromagnetic potential A , while the remaining variables can be determined as functions of the dynamical variables and their derivatives. Equations (6.177)–(6.188) give the following expressions for the nondynamical variables:

$$\begin{aligned}
& \lambda_{ab\mu\nu} = R_{ab\mu\nu}, \quad \omega^{ab}{}_\mu = \Delta^{ab}{}_\mu, \quad \gamma^A{}_{\mu\nu\rho} = -\frac{1}{2e} \varepsilon_{\mu\nu\rho\sigma} \nabla^\sigma \phi^A, \\
& \Lambda^{abA}{}_\mu = \frac{1}{12e} g_{\mu\lambda} \varepsilon^{\lambda\nu\rho\sigma} \nabla_\nu \phi^A e^a{}_\rho e^b{}_\sigma, \quad \beta^a{}_{\mu\nu} = 0, \quad B_{ab\mu\nu} = \frac{1}{8\pi l_p^2} \varepsilon_{abcd} e^c{}_\mu e^d{}_\nu, \\
& H^{abcA} = \frac{1}{6e} \varepsilon^{\mu\nu\rho\sigma} \nabla_\mu \phi^A e^a{}_\nu e^b{}_\rho e^c{}_\sigma, \quad \lambda^A{}_\mu = \nabla_\mu \phi^A, \\
& \lambda_{\mu\nu} = F_{\mu\nu}, \quad B_{\mu\nu} = -\frac{1}{2eq} \varepsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}, \\
& M^{ab} = -\frac{1}{4e} \varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu} e^a{}_\rho e^b{}_\sigma, \quad \zeta^{ab} = \frac{1}{4eq} \varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu} e^a{}_\rho e^b{}_\sigma.
\end{aligned} \tag{6.193}$$

Notice that equations (6.179), (6.180), and (6.181) imply that the connection $\beta^a = 0$, as in pure gravity. The equation of motion (6.190) gives the covariant Klein–Gordon equation for the scalar field coupled to the electromagnetic potential A :

$$(\nabla_\mu \nabla^\mu - m^2) \phi_A = 0. \tag{6.194}$$

Equation (6.189) gives the differential equation of motion for the electromagnetic potential A :

$$\nabla_\mu F^{\mu\nu} = j^\nu, \quad j^\mu \equiv \frac{1}{2} \left(\nabla^\nu \phi^A \triangleright^B{}_A \phi_B - \phi_A \triangleright_B{}^A \nabla^\nu \phi^B \right) = iq \left(\nabla \phi^* \phi - \phi^* \nabla \phi \right). \tag{6.195}$$

Finally, after simplification, the equation of motion (6.191) for e^a gives

$$\begin{aligned}
& R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R = 8\pi l_p^2 T^{\mu\nu}, \\
& T^{\mu\nu} \equiv \nabla^\mu \phi_A \nabla^\nu \phi^A - \frac{1}{2} g^{\mu\nu} (\nabla_\rho \phi_A \nabla^\rho \phi^A + m^2 \phi_A \phi^A) - \frac{1}{4q} (F_{\rho\sigma} F^{\rho\sigma} g^{\mu\nu} + 4F^{\mu\rho} F_\rho{}^\nu).
\end{aligned} \tag{6.196}$$

The system of equations (6.177)–(6.191) is equivalent to the system (6.193)–(6.196).

The complete Hamiltonian analysis of the topological sector of scalar electrodynamics is given in Appendix C.

Part II

Quantum theory

Chapter 7

Spin-foam models: BF theory

Einstein's general theory of relativity led to our understanding of the nature of space and time as manifestations of the gravitational field. Just as other physical fields exhibit quantum properties at a certain scale, it is natural to expect the gravitational field, and therefore spacetime itself, to possess quantum properties. Our understanding of the nature of space and time must therefore be modified to take these quantum features into account. The problem is that the present theories, general relativity and quantum field theory, cannot describe the quantum behavior of the gravitational field. A *quantum theory of gravity* is needed, with the predictive power to describe phenomena in which both gravity and quantum theory play a role, such as black holes, the early universe, and physics at short distances.

In 1936, Bronstein repeated the Bohr–Rosenfeld analysis of the electromagnetic field for the gravitational field and showed that quantum theory forbids determining the field in an arbitrarily small region of spacetime. If we measure the field at a point x , which we wish to determine with precision L , the *Heisenberg uncertainty relation* between a particle's position and momentum implies that the momentum uncertainty must satisfy $\Delta p \geq \hbar/L$. In the ultra-relativistic limit, the energy is $E \sim cp$, showing that sharp localization requires high energy. General relativity tells us that energy curves space, and that the curvature grows as the energy becomes more concentrated, until a black hole forms when a mass $M \sim E/c^2$ is concentrated within a radius $R \sim GM/c^2$. By demanding increasingly precise localization, we reach the scale $L_{Plank} = R$, below which it is impossible to proceed because the localized region would be hidden behind a black-hole horizon¹. The preceding relations give

$$L_{Plank} = \sqrt{\frac{\hbar G}{c^3}} \sim 10^{-33} \text{cm}.$$

At scales larger than the *Planck length* L_{Plank} , spacetime can be regarded as a smooth manifold, while below this scale quantum fluctuations of spacetime become non-negligible and the notion of length ceases to be meaningful.

The preceding analysis suggests that quantum field theory, a formalism in which quantum fields are defined on a spacetime manifold, is not an adequate picture of the world in quantum gravity. A *quantum description of geometry* is required, in which geometry is described by quantum states and spacetime is a semiclassical approximation to such a quantum configuration. One possible description of the quantum states of geometry, i.e. of the gravitational field, is provided by the formalism of *loop quantum gravity*².

¹Here it is assumed that General Relativity remains valid without modification at scales below the Planck scale, i.e. that a solution of Einstein's equations describing such a small black hole exists.

²English: *Loop Quantum Gravity*.

Loop quantum gravity

Loop quantum gravity is an approach to quantizing gravity that is more than thirty-five years old, originating with Ashtekar's work in 1986. As a theory of pure gravity, it does not seek to solve the *unification problem*, i.e. to unify the interactions and reduce the number of degrees of freedom of the *Standard Model*. Quantization within the *canonical quantization procedure* requires choosing an algebra of fields that become quantum operators; in this case, the algebra is based on the *holonomies of the gravitational connection*. A holonomy becomes an operator that creates a *loop state*. The theory is background independent, so a loop state is meaningful only in relation to other loops, and an infinitesimal displacement of a loop does not produce a new state but one equivalent up to a gauge transformation. The state space of the theory is a separable Hilbert space with a basis of loop states, where finite linear combinations of loop states are called *spin networks*.

Quantum properties manifest themselves in the discrete spectra of the eigenvalues of operators corresponding to quantities that describe local properties of the gravitational field, such as the operator associated with each link of the boundary graph of the dual triangulation,

$$\vec{E}_l = 8\pi\gamma\hbar G \vec{L}_l, \quad (7.1)$$

whose values are normals to the boundary triangles of the triangulation, to which the area operator corresponds:

$$\hat{A}_l = 8\pi\gamma\hbar G |\vec{L}|^2. \quad (7.2)$$

This operator has the discrete spectrum

$$A = 8\pi\gamma\hbar G \sqrt{j(j+1)}, \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

where γ is the *Barbero–Immirzi parameter*, a dimensionless constant in the theory whose value is fixed so that the number of microstates in LQG agrees with the semiclassically calculated black-hole entropy [41], [42]. Another observable in this theory is the oriented volume of a tetrahedron,

$$V^2 = \frac{2}{9} (\vec{E}_1 \cdot \vec{E}_2) \times \vec{E}_3 = \frac{2}{9} \epsilon_{ijk} E_1^i E_2^j E_3^k, \quad (7.3)$$

or the volume operator $\hat{V}$,

$$\hat{V} = \frac{\sqrt{2}}{3} (8\pi\gamma\hbar G)^{\frac{3}{2}} \sqrt{(\vec{L}_1 \cdot \vec{L}_2) \times \vec{L}_3} \quad (7.4)$$

which also has a discrete spectrum:

$$\hat{V} |i_v\rangle = V |i_v\rangle. \quad (7.5)$$

This means that space can be regarded as being composed of *cells*, i.e. tetrahedra whose volumes are determined by this spectrum.

The four area operators $\hat{A}_a$ and the volume operator $\hat{V}$ form a maximal commuting set of operators describing the quantum state of a tetrahedron, so they can be diagonalized simultaneously, and the quantum states of the tetrahedron's geometry are uniquely determined by their eigenvalues $|j_a, V\rangle$. Since six numbers, for example the six edge lengths, are required for a unique classical description of a tetrahedron, we see that, as expected in quantum theory, the quantum state of a tetrahedron possesses a certain quantum uncertainty at the Planck scale.

This qualitative picture of the quantum structure of spacetime is preserved in *covariant quantization*, i.e. the *spin-foam quantization procedure*. Within the covariant quantization procedure, *spin foam* refers to a *functional approach* to quantizing gravity in which the

configuration-space integral is defined in the same way as in the *Feynman definition of the path integral*.

According to Michael Atiyah's definition, an $(n + 1)$ -dimensional *topological quantum field theory* (TQFT) is a functorial assignment of a finite-dimensional Hilbert space $\mathcal{H}_\Sigma$ to every closed oriented n -manifold Σ , and of a vector $\mathcal{Z}_\mathcal{M} \in \mathcal{H}_\Sigma$ to every oriented $(n + 1)$ -manifold $\mathcal{M}$ having Σ as its boundary. If we consider a quantum state of space formed from $|\Lambda_3|$ tetrahedra in LQG, it can be represented as a graph in the dual triangulation, where vertices correspond to tetrahedra and the links between them to the triangles separating adjacent tetrahedra. In analogy with Atiyah's general definition, a *spin-foam model* assigns to every oriented graph³ Γ a Hilbert space $\mathcal{H}_\Gamma$, and to every foam⁴ $\mathcal{C}$ having the graph Γ as its boundary⁵, a vector $\mathcal{Z}_\mathcal{C} \in \mathcal{H}_\Gamma$. The following axioms are satisfied [43]:

1. (multiplicativity) $\mathcal{H}_{\Gamma_1 \cup \Gamma_2} = \mathcal{H}_{\Gamma_1} \otimes \mathcal{H}_{\Gamma_2}$,
2. (duality) $\mathcal{H}_{\bar{\Gamma}} = \mathcal{H}_\Gamma^*$, $\mathcal{Z}_{\bar{\Gamma}} = \mathcal{Z}_\Gamma^\dagger$,
3. (functoriality)⁶ $\mathcal{Z}_{\mathcal{C}_1 \cup_\Gamma \mathcal{C}_2} = \langle \mathcal{Z}_{\bar{\mathcal{C}}_2} | \mathcal{Z}_{\mathcal{C}_1} \rangle_{\mathcal{H}_\Gamma} = \langle \mathcal{Z}_{\bar{\mathcal{C}}_1} | \mathcal{Z}_{\mathcal{C}_2} \rangle_{\mathcal{H}_{\bar{\Gamma}}}$,
4. $\mathcal{H}_\emptyset = \mathbb{C}$,
5. $\mathcal{Z}_{\mathbb{1}_\Gamma} = \text{id}_{\mathcal{H}_\Gamma}$.

The dynamics of loop quantum gravity fits this definition, with the Hilbert space associated with a graph Γ given by the lattice $SU(2)$ Yang–Mills space $L^2(SU(2))^{|L_\Gamma|}/SU(2)^{|N_\Gamma|}$. Each vertex v in the dual triangulation of spacetime is colored by i_v , corresponding to an eigenvalue of the volume operator V for that cell, while each link ϵ in the dual triangulation is colored by j_ϵ , corresponding to an eigenvalue of the area A joining the two cells of space. Such a quantum state of space is called a *spin network*. Furthermore, if we consider the *evolution of a spin network* in time, the label i_v that colored a vertex of the spin network now colors an edge of the *spin foam*, while j_ϵ now colors a face. The foam amplitudes $\mathcal{Z}_\mathcal{C}$ are defined by summing the spin-foam amplitudes $\mathcal{Z}_\mathcal{C}(\sigma)$ over the colors of the faces of the triangulation $\sigma = \{j_f\}$, i.e. over the irreducible representations j_f of the group $SU(2)$,

$$\mathcal{Z}_\mathcal{C} = \sum_\sigma \mathcal{Z}_\mathcal{C}(\sigma),$$

as shown in the following sections.

³A *graph* is an ordered pair $\Gamma = (N_\Gamma, L_\Gamma)$, where N_Γ is a finite set of nodes and L_Γ is a set of ordered pairs of nodes, i.e. the links of the graph Γ . The nodes n and n' of a link $l = (n, n')$ are called the source and target of the link l and are denoted by $\partial^-(l)$ and $\partial^+(l)$, respectively. The inverse link is defined by $l^{-1} \equiv (n', n)$, while $\bar{\Gamma}$ is the inverse graph of Γ , obtained by reversing every link of Γ .

⁴A *foam* is an ordered triple $\mathcal{C} = (V_\mathcal{C}, E_\mathcal{C}, F_\mathcal{C})$, where $V_\mathcal{C}$ is a finite set of vertices, $E_\mathcal{C}$ is a set of ordered pairs of vertices, i.e. edges $e = (v, v')$, and $F_\mathcal{C}$ is a finite set of faces. A face is a finite sequence of edges $f = (e_1, \dots, e_{n_f})$, where $\partial^+(e_n) = \partial^-(e_{n+1})$ and $\partial^+(e_{n_f}) = \partial^-(e_1)$. Note that any subset F of the set of faces $F_\mathcal{C}$ naturally defines a subcomplex of the foam $\mathcal{C}$, consisting of the vertices, edges, and faces occurring in F . The foam obtained from $\mathcal{C}$ by reversing all its edges and faces is denoted by $\bar{\mathcal{C}}$.

⁵Edges $E_\mathcal{C}$ that occur exactly once and belong to only one face of the foam are called its *links*, while the other edges are *internal edges*. Analogously, vertices $V_\mathcal{C}$ that occur only once in internal edges are called *nodes*, while the remaining ones are called *internal vertices*. In general, the sets of nodes and links of a foam $\mathcal{C}$ do not form a graph, but when they do, and when the orientation of each link agrees with that induced by the unique face passing through it, such a foam is called a *proper foam*. In that case, we define the *boundary of the foam* $\partial\mathcal{C}$ as the subcomplex of $\mathcal{C}$ containing all faces of $\mathcal{C}$ that contain at least one link. The graph corresponding to $\partial\mathcal{C}$ is the *boundary graph* of $\mathcal{C}$.

⁶The *composition of two proper foams* $\mathcal{C}_1$ and $\mathcal{C}_2$ along an oriented graph Γ , denoted by $\mathcal{C}_1 \cup_\Gamma \mathcal{C}_2$, is defined when Γ is a connected component of the boundary graphs of $\mathcal{C}_1$ and $\bar{\mathcal{C}}_2$. The composition is obtained by removing Γ and joining, for each node, the unique edges $e_1 \in \mathcal{C}_1$ and $e_2 \in \mathcal{C}_2$ containing that node into a single edge, and, for each link l , the unique faces $f_1 \in \mathcal{C}_1$ and $f_2 \in \mathcal{C}_2$ into a single face.

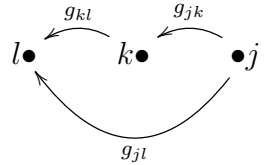
In quantum electrodynamics, a perturbative theory, a better approximation is obtained by summing higher-order Feynman diagrams, while in quantum chromodynamics, a theory defined on a spacetime lattice, a better approximation is obtained by refining the lattice. It can be shown that in loop quantum gravity these two approaches are the same, which is intuitively clear once one recognizes that the lattice points are themselves quanta of space.

In the following sections, we focus on constructing topological BF state sums for three- and four-dimensional manifolds using the standard spin-foam quantization procedure. In the three-dimensional case, the state sum presented in Section 7.2.1 gives a quantum theory of three-dimensional gravity—the *Ponzano–Regge model*—because the corresponding classical theory has no local propagating degrees of freedom. As we know, this is not the case in the physically relevant four-dimensional setting, so a quantum theory of gravity must be obtained by modifying the amplitudes of the topological state sum of the *Ooguri model* presented in Section 7.2.2. This construction is beyond the scope of our discussion; see [2] for a pedagogical treatment of the covariant quantization procedure that produces a state sum describing gravity in four dimensions.

7.1 Gauge-invariant objects

The classical equations of motion of BF theory impose the condition that the gauge connection be flat, or, in the language of holonomies, that every null-homotopic curve correspond to the identity element of the gauge group. Lemma 12 considers the boundary curve of a triangle and formulates the flatness condition for the gauge connection on this element of the manifold's triangulation.

Lemma 12 *Consider a triangle (jkl) . Its edges (jk) , $j < k$ are labeled by group elements $g_{jk} \in G$. Consider diagram (7.6).*



$$\begin{array}{c} \begin{array}{ccc} l \bullet & k \bullet & j \bullet \\ & \nwarrow \quad \nearrow & \\ & g_{jl} & \end{array} \end{array} \quad (7.6)$$

The curve $\gamma_1 = g_{kl}g_{jk}$ is equal to the curve $\gamma_2 = g_{jl}$, i.e. the following identity holds:

$$g_{jl} = g_{kl}g_{jk} . \quad (7.7)$$

7.2 Quantization of the topological BF action

This section presents the quantization of the BF action using the standard heuristic spin-foam quantization procedure. First, the path integral of the topological state sum is

$$Z = \int \mathcal{D}\alpha \mathcal{D}B \exp \left(i \int_{M_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} \right) . \quad (7.8)$$

Formal integration over the Lagrange multiplier B gives

$$Z = \mathcal{N} \int \mathcal{D}\alpha \delta(\mathcal{F}) . \quad (7.9)$$

Next, the connection 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$ is discretized by coloring the edges of the triangulation $\epsilon = (jk) \in \Lambda_1$ with group elements $g_\epsilon \in G$. The measure of the path integral (7.8) is

discretized by the substitution

$$\int \mathcal{D}\alpha \quad \mapsto \quad \prod_{(jk) \in \Lambda_1} \int_G dg_{jk}, \quad (7.10)$$

where dg_{jk} denotes integration with respect to the Haar measure on the group G .

The vanishing-curvature condition is discretized on every triangle $(jkl) \in \Lambda_2$ by the δ -function $\delta(\mathcal{F})$. In passing from the smooth manifold to its triangulation, the δ -function is defined on the set of triangles of the triangulation,

$$\delta(\mathcal{F}) = \prod_{(jkl) \in \Lambda_2} \delta_G(g_{jkl}), \quad (7.11)$$

where, for every triangle $(jkl) \in \Lambda_2$, the corresponding δ -function $\delta_G(g_{jkl})$ is given by

$$\delta_G(g_{jkl}) = \delta_G(g_{kl} g_{jk} g_{jl}^{-1}). \quad (7.12)$$

Identity (7.12) follows from equation (7.7) in Lemma 12.

Substituting the previously defined discretized measure (7.10) and the δ -function (7.11) into equation (7.9) gives the state sum⁷:

$$Z = \mathcal{N} \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \left(\prod_{(jkl) \in \Lambda_2} \delta_G(g_{jkl}) \right). \quad (7.18)$$

Next, substituting expression (7.12) into (7.18) gives an explicit expression for the state sum of a given triangulation of the spacetime manifold $\mathcal{M}$. With an appropriate choice of the constant

⁷A similar procedure can be carried out in the dual triangulation. First, discretize the action:

$$S_{BF}[B, \alpha] \rightarrow S_{BF}^{disc}[B, \alpha] \equiv \sum_{f \in \Lambda_2^*} \text{tr} [B_f g_f], \quad (7.13)$$

where the integral of the curvature tensor has been replaced by a sum of holonomies g_f over all polygons f dual to the hinges of the triangulation—in the three-dimensional case, the edges $(jk) \in \Lambda_1$ —on which the curvature is nonzero, while the multiplier B has been replaced by its value B_f on each polygon f . The holonomy g_f around the polygon f can be written as the ordered product of the holonomies g_l along all edges $l \in \Lambda_2^*$ of the given polygon $f \in \Lambda_2^*$ [38]:

$$g_f = \prod_{l \in f} g_l. \quad (7.14)$$

Then, integrating over all multipliers and holonomies,

$$\begin{aligned} Z \rightarrow Z^{disc} &= \int \left(\prod_{f \in \Lambda_2^*} \mathcal{D}B_f \right) \int \left(\prod_{l \in \Lambda_1^*} \mathcal{D}g_l \right) \exp \left(i \sum_{f \in \Lambda_2^*} \text{tr} [B_f \prod_{l \in f} g_l] \right) \\ &= \int \left(\prod_{l \in \Lambda_1^*} \mathcal{D}g_l \right) \prod_{f \in \Lambda_2^*} \left(\int \mathcal{D}B_f \exp \left(i \text{tr} [B_f \prod_{l \in f} g_l] \right) \right), \end{aligned} \quad (7.15)$$

where we recognize that

$$\int \mathcal{D}B_f \exp \left(i \text{tr} [B_f \prod_{l \in f} g_l] \right) = \mathcal{N}' \delta \left(\prod_{l \in f} g_l \right) = \mathcal{N}' \delta(g_f), \quad (7.16)$$

gives the state sum

$$Z^{disc} = \mathcal{N} \left(\prod_{l \in \Lambda_1^*} \int \mathcal{D}g_l \right) \left(\prod_{f \in \Lambda_2^*} \delta(g_f) \right). \quad (7.17)$$

The resulting state sum is equivalent to equation (7.18), but is expressed in terms of elements of the dual triangulation.

$\mathcal{N}$ in front of the integral, obtained after integrating over the Lagrange multiplier B , this sum becomes independent of the triangulation, i.e. invariant under Pachner moves. Requiring this invariance for all Pachner moves determines the appropriate choice of the constant $\mathcal{N}$, given in Definition 7.2.1.

Definition 7.2.1 *Let $\mathcal{M}_d$ be a compact and oriented combinatorial d -manifold, $d \in \{3, 4\}$. The state sum of topological gauge theory is defined by*

$$Z = |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} \left(\prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \right) \left(\prod_{(jkl) \in \Lambda_2} \delta_G(g_{kl} g_{jk} g_{jl}^{-1}) \right) \quad (7.19)$$

In the preceding definition, integration is performed over the elements $g_{jk} \in G$ for every edge $(jk) \in \Lambda_1$. The δ -function in the integrand imposes the following condition.

- For every triangle $(jkl) \in \Lambda_2$, the condition $g_{kl} g_{jk} = g_{jl}$ holds (see Lemma 12).

Theorem 19 *Let $\mathcal{M}_d$ be a closed and oriented combinatorial d -manifold for $d \in \{3, 4\}$. The state sum (7.19) is invariant under Pachner moves.*

For now, we leave Theorem 19 without proof. Chapters 8 and 9 considered categorical generalizations of this theorem and their proofs, so the proof of the preceding theorem can be obtained straightforwardly by simplifying those more general cases.

7.2.1 $d = 3$: Ponzano–Regge model

Three-dimensional quantum gravity can be defined in several ways. The first successful approach was the *Ponzano–Regge model* of quantum gravity on a discretized three-dimensional manifold, obtained by forming the BF state sum for the choice of gauge group $G = SU(2)$.

After choosing a particular gauge group G , the topological state sum obtained in the preceding section is further transformed using the Peter–Weyl or Plancherel theorem⁸. A natural choice of G is the isometry group of the given space, which in the three-dimensional case is $SO(3)$, or equivalently $SU(2)$. Thus, for the group $SU(2)$ we may write

$$\delta(g_f) = \sum_{j_f} \left(\dim j_f \right) \text{tr} \left[D^{(j_f)}(g_f) \right], \quad j_f \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}. \quad (7.20)$$

Substituting the preceding expression for $\delta(g_f)$ into the state sum in the dual triangulation (7.17), we obtain

$$\begin{aligned} Z^{\text{disc}} &= \mathcal{N} \int \left(\prod_{l \in \Lambda_1^*} \mathcal{D}g_l \right) \prod_{f \in \Lambda_2^*} \left(\sum_{j_f \in \mathcal{N}_0/2} (2j_f + 1) \text{tr} \left[\prod_{l \in f} D^{(j_f)}(g_l) \right] \right) \\ &= \mathcal{N} \sum_{\{j_f\}} \left(\prod_{f \in \Lambda_2^*} (2j_f + 1) \right) \int \left(\prod_{l \in \Lambda_1^*} \mathcal{D}g_l \right) \left(\dots D^{(j_f)}_{\alpha}(g_l) \dots D^{(j_f^*)}_{\alpha^*}(g_{l^*}) \dots \right), \end{aligned} \quad (7.21)$$

⁸The Peter–Weyl and Plancherel theorems provide a decomposition of functions on a group into a sum over the corresponding irreducible representations of the group. The result is called the *Peter–Weyl theorem* when the group G is compact, while Plancherel proved the analogous theorem for a noncompact group G .

Theorem 20 (Peter–Weyl and Plancherel theorem) *For a δ -function whose argument is a group element $g_f \in G$, one has*

$$\delta(g_f) = \sum_{\Lambda_f} \dim(\Lambda_f) \chi(g_f, \Lambda_f),$$

where Λ_f are the unitary irreducible representations of the Lie group G , $\chi(g_f, \Lambda_f)$ is the trace of the element g_f in the representation Λ_f , and $\dim(\Lambda_f)$ is the dimension of the representation.

where all indices in the last line are contracted. Rearranging the factors in parentheses in the preceding expression so as to group the representations of the element g_l , we obtain

$$Z^{disc} = \mathcal{N} \sum_{\{j_f\}} \left(\prod_{f \in \Lambda_2^*} (2j_f + 1) \right) \text{tr} \left[\prod_{l \in \Lambda_1^*} \left(\int \mathcal{D}g_l D^{(j_1)_\alpha^a}(g_l) D^{(j_2)_\beta^b}(g_l) D^{(j_3)_\gamma^c}(g_l) \right) \right]. \quad (7.22)$$

There are clearly three such representations, corresponding to the three faces f labeled by the numbers j_f that share the edge l , as shown in Figure 7.1. This is because an edge l of the dual lattice corresponds to a triangle, which has three edges, and these in turn correspond to the faces f . For the group $SU(2)$ and $n = 3$ representations of a group element, there is only one

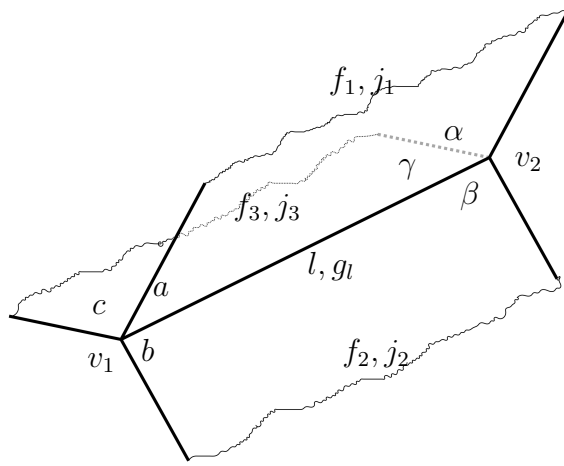


Figure 7.1: An edge l of the dual lattice and the faces f_1 , f_2 , and f_3 that share it.

intertwiner⁹ It is the $\{3j\}$ symbol, for which

$$\int \mathcal{D}g_l D^{(j_1)_\alpha^a}(g_l) D^{(j_2)_\beta^b}(g_l) D^{(j_3)_\gamma^c}(g_l) = i^{abc} i_{\alpha\beta\gamma} = \begin{pmatrix} j_1 & j_2 & j_3 \\ a & b & c \end{pmatrix}_{SU(2)} \begin{pmatrix} j_1 & j_2 & j_3 \\ \alpha & \beta & \gamma \end{pmatrix}_{SU(2)}. \quad (7.24)$$

Using (7.24), equation (7.22) becomes

$$Z^{disc} = \mathcal{N} \sum_{\{j_f\}} \left(\prod_{f \in \Lambda_2^*} (2j_f + 1) \right) \text{tr} \left[\prod_{l \in \Lambda_1^*} \begin{pmatrix} j_1 & j_2 & j_3 \\ a & b & c \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ \alpha & \beta & \gamma \end{pmatrix} \right]. \quad (7.25)$$

Further grouping the intertwiners by their common vertex, as shown in Figure 7.2, gives

$$Z^{disc} = \mathcal{N} \sum_{\{j_f\}} \left(\prod_{f \in \Lambda_2^*} (2j_f + 1) \right) \prod_{v \in \Lambda_0^*} \left(\begin{pmatrix} j_1 & j_4 & j_5 \\ a & d & e \end{pmatrix} \begin{pmatrix} j_2 & j_4 & j_6 \\ b & d & f \end{pmatrix} \begin{pmatrix} j_3 & j_5 & j_6 \\ c & e & f \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ a & b & c \end{pmatrix} \right). \quad (7.26)$$

⁹For every group element $g \in G$ and representations Λ_i of the group, there are one or more objects called *intertwiners* [1], which satisfy

$$D^{(\Lambda_1)_\alpha^a}(g) D^{(\Lambda_2)_\beta^b}(g) D^{(\Lambda_3)_\gamma^c}(g) \dots D^{(\Lambda_n)_\nu^n}(g) i_{abc\dots n}^{(\Lambda_1 \Lambda_2 \dots \Lambda_n)} = i_{\alpha\beta\gamma\dots\nu}^{(\Lambda_1 \Lambda_2 \dots \Lambda_n)}. \quad (7.23)$$

Intertwiners depend on n , the group G , and its representations, but not on the group element $g \in G$.

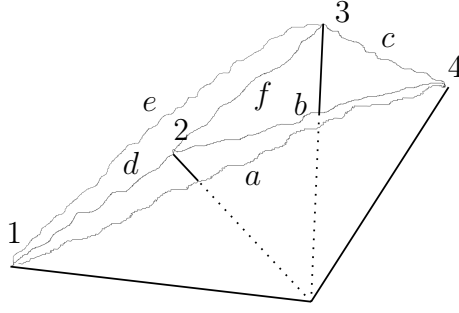


Figure 7.2: Four edges and six faces meeting at a single vertex of the dual triangulation of a 3D manifold.

A single vertex corresponds to six¹⁰ faces and four edges in the dual triangulation. The four $\{3j\}$ symbols grouped in parentheses form a $\{6j\}$ symbol, a function of six numbers $f(j_1, \dots, j_6)$ ¹¹; hence we may write

$$Z^{disc} = \mathcal{N} \sum_{j_1} \cdots \sum_{j_{|\Lambda_2^*|}} \left(\prod_{f=1}^{|\Lambda_2^*|} (2j_f + 1) \right) \left(\prod_{v=1}^{|\Lambda_0^*|} \{6j\}_{SU(2)}^v \right). \quad (7.27)$$

Expressed in terms of the elements of the triangulation, the state sum of the *Ponzano–Regge model* is

$$Z^{disc} = \mathcal{N} \sum_{j_1} \cdots \sum_{j_{|\Lambda_1|}} \left(\prod_{\epsilon=1}^{|\Lambda_1|} (2j_\epsilon + 1) \right) \left(\prod_{\tau=1}^{|\Lambda_3|} \{6j\}_{SU(2)}^\tau \right). \quad (7.28)$$

The constructed BF state sum gives a quantum theory of gravity for a three-dimensional spacetime manifold. The classical theory must therefore be recovered in the classical limit of this quantum theory. In quantum mechanics, the classical limit is obtained in the limit of large quantum numbers, where quantum discreteness becomes negligible. Ponzano and Regge observed, and Roberts formally proved, that in the limit of large quantum numbers j ,

$$\{6j\} \underset{j \rightarrow \infty}{\sim} \frac{1}{\sqrt{12\pi V}} \cos\left(S + \frac{\pi}{4}\right), \quad (7.29)$$

where V is the volume of the tetrahedron and S is the classical Regge action of the tetrahedron. Expanding the cosine and using the angle-addition formulas gives

$$\{6j\} \underset{j \rightarrow \infty}{\sim} \frac{1}{2\sqrt{-12i\pi V}} \exp(iS) + \frac{1}{2\sqrt{12i\pi V}} \exp(-iS). \quad (7.30)$$

In the limit of large quantum numbers, the sum over spins in equation (7.28) can be approximated by an integral over lengths in Regge geometry. Equation (7.30) shows that the integrand has the form of the exponential of the action, revealing that the Einstein–Hilbert action is encoded in the $\{6j\}$ symbol. Thus, in the classical limit, we obtain¹² the path integral for the Einstein–Hilbert action:

$$Z \sim \int \mathcal{D}g e^{\frac{i}{\hbar} \int \sqrt{-g} R}. \quad (7.31)$$

¹⁰A vertex of the dual triangulation corresponds to a tetrahedron of the ordinary triangulation of the 3D manifold; the four edges meeting at a vertex of the dual triangulation correspond to the triangles of $\mathcal{T}(\mathcal{M}_3)$; the six faces of the dual triangulation correspond to the six edges of $\mathcal{T}(\mathcal{M}_3)$; and so on.

¹¹Four edges meet at a single vertex, and for every pair there is a face containing both. Intertwiners of the form i^{pqr} carry information about the three faces p , q , and r that share the edge associated with that intertwiner, i.e. the three numbers j_f labeling those faces.

¹²Note that expression (7.30) contains two terms with opposite signs of the action in their exponents, so the resulting expression is slightly more complicated than equation (7.31).

The four-dimensional case is somewhat more complicated, as expected, because in four dimensions gravity is not a topological theory without local propagating degrees of freedom. The construction of the BF state sum for a four-dimensional manifold is presented in the following section.

7.2.2 $d = 4$: Ouguri model

For four-dimensional spacetime, the topological BF action is defined in the same way as in the three-dimensional case,

$$S_{BF}[B, \omega] = \int_{\mathcal{M}_4} B^{ab} \wedge R_{ab}, \quad (7.32)$$

where the curvature $R_{ab} = d\omega_{ab} + \omega_{ac} \wedge \omega^c_b$ and the Lagrange multiplier B^{ab} are now 2-forms. Repeating the procedure of Section 7.2.1 for the BF action defined on a three-dimensional spacetime manifold leads to equation (7.17). Applying Theorem 20 to the group $G = SO(4)$ shows that the state sum has the form (7.21), where the index j_f now labels the irreducible representations of the group $SO(4)$.

Next, the product in parentheses in the state sum (7.17) is rearranged so as to group the representations of the group element g_l , keeping in mind that there are now four such representations, i.e. four numbers j_f . Figure 7.3 shows that in 4D there are four faces f sharing an edge l , because an edge l of the dual lattice corresponds to a tetrahedron, which has four triangles corresponding to the four faces f .

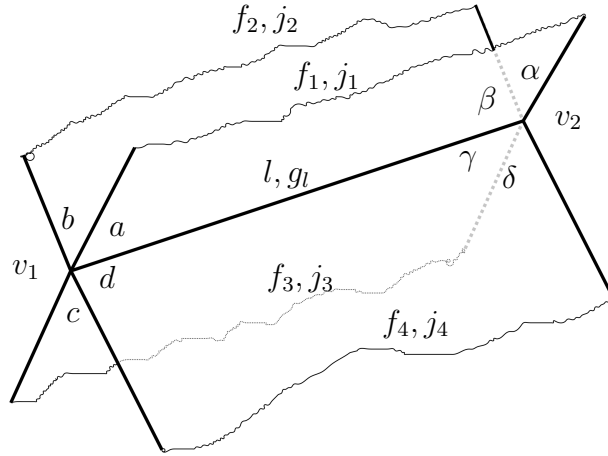


Figure 7.3: An edge l of the dual lattice and the faces f_1 , f_2 , f_3 , and f_4 that share it.

Using the definition of an intertwiner for the group $SO(4)$, one can prove the following relation:

$$\int \mathcal{D}g_l D^{(j_1)_a}_{\alpha}(g_l) D^{(j_2)_b}_{\beta}(g_l) D^{(j_3)_c}_{\gamma}(g_l) D^{(j_4)_d}_{\delta}(g_l) = \sum_{I_l} i_{I_l}^{abcd} i_{\alpha\beta\gamma\delta}^{I_l}. \quad (7.33)$$

Unlike the group $SU(2)$, which has only one intertwiner $\{3j\}$ for a given set of representations, the group $SO(4)$ has several intertwiners satisfying the defining equation (7.23). In equation (7.33), these intertwiners are enumerated by an additional index I_l .

The state sum obtained from (7.21) by an appropriate rearrangement and application of equation (7.33) is

$$\begin{aligned}
Z^{disc} &= \sum_{\{j_f\}} \left(\prod_{f \in \Lambda_2^*} \dim j_f \right) \text{tr} \left[\prod_{l \in \Lambda_1^*} \left(\int \mathcal{D}g_l D^{(j_1)}_{\alpha}(g_l) D^{(j_2)}_{\beta}(g_l) D^{(j_3)}_{\gamma}(g_l) D^{(j_4)}_{\delta}(g_l) \right) \right] \\
&= \sum_{\{j_f\}} \left(\prod_{f \in \Lambda_2^*} \dim j_f \right) \text{tr} \left[\prod_{l \in \Lambda_1^*} \sum_{I_l} i_{I_l}^{abcd} i_{\alpha\beta\gamma\delta}^{I_l} \right] \\
&= \sum_{\{j_f\}} \sum_{\{I_l\}} \left(\prod_{f \in \Lambda_2^*} \dim j_f \right) \text{tr} \left[\prod_{l \in \Lambda_1^*} i_{I_l}^{a_l b_l c_l d_l} i_{\alpha_l \beta_l \gamma_l \delta_l}^{I_l} \right] \\
&= \sum_{\{j_f\}} \sum_{\{I_l\}} \left(\prod_{f \in \Lambda_2^*} \dim j_f \right) \prod_{v \in \Lambda_0^*} \left(i_{I_1}^{efgb} i_{I_2}^{ehic} i_{I_3}^{fhja} i_{I_4}^{gjid} i_{I_5}^{abcd} \right).
\end{aligned} \tag{7.34}$$

In the final step, we grouped the intertwiners by their common vertex, as shown in Figure 7.4. A single vertex corresponds to ten¹³ faces and five edges in the dual triangulation. The five intertwiners grouped in parentheses form the $\{15j\}_{SO(4)}$ symbol, a function of fifteen numbers $f(I_1, \dots, I_5, j_1, \dots, j_{10})$ ¹⁴. Finally, the state sum of topological BF theory for a four-

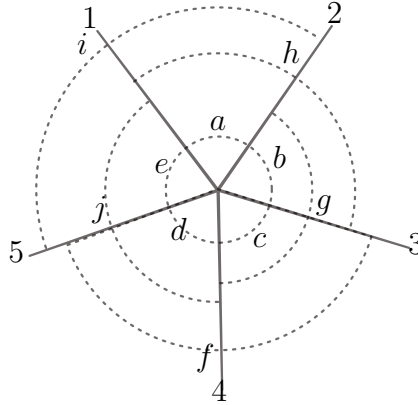


Figure 7.4: Five edges and ten faces meeting at a single vertex of the dual triangulation of a $4D$ manifold.

dimensional spacetime manifold takes the form

$$Z^{disc} = \sum_{j_1} \cdots \sum_{j_{|\Lambda_2^*|}} \sum_{I_1} \cdots \sum_{I_{|\Lambda_1^*|}} \left(\prod_{f=1}^{|\Lambda_2^*|} \dim j_f \right) \left(\prod_{v=1}^{|\Lambda_0^*|} \{15j\}_{SO(4)}^v \right), \tag{7.35}$$

or, in terms of the elements of the triangulation,

$$Z^{disc} = \sum_{j_1} \cdots \sum_{j_{|\Lambda_2|}} \sum_{I_1} \cdots \sum_{I_{|\Lambda_3|}} \left(\prod_{\Delta=1}^{|\Lambda_2|} \dim j_{\Delta} \right) \left(\prod_{\sigma=1}^{|\Lambda_4|} \{15j\}_{SO(4)}^{\sigma} \right). \tag{7.36}$$

The resulting state sum is the *Ooguri model* [4].

Unlike the three-dimensional case, where general relativity has no local propagating degrees of freedom, in four dimensions BF theory is not equivalent to general relativity. To obtain

¹³A vertex of the dual triangulation corresponds to a 4-simplex of the ordinary triangulation of the $4D$ manifold; the five edges meeting at a vertex of the dual triangulation correspond to tetrahedra of $\mathcal{T}(\mathcal{M}_4)$; the ten faces of the dual triangulation correspond to the ten triangles of $\mathcal{T}(\mathcal{M}_4)$; and so on.

¹⁴Intertwiners of the form $i_{I_l}^{pqrs}$ carry information about the edge l and the four faces p, q, r , and s that share it, i.e. about the number I_l carried by the edge and the numbers j_f carried by the faces.

a theory describing GR, the BF action must be modified by adding appropriate constraints, i.e. by formulating the Plebanski action (4.74) at the classical level, or by deforming the topological state sum (7.36) into a nontopological state sum at the quantum level. The state sum is deformed by choosing different amplitudes in the model, through the procedure described in [9][10]. This procedure yields the *EPRL/FK spin-foam model*, which represents one possible quantization of *General Relativity*.

The quantization of constrained BF theory and the construction of a quantum theory of gravity lie beyond the scope of our discussion, so the interested reader is referred to [9][10].

Chapter 8

Construction of the topological state sum: $2BF$ theory

In this chapter we focus on the second step of the covariant spin-foam quantization procedure for $2BF$ theory. We demonstrate how to construct a triangulation-independent state sum Z from the classical $2BF$ action for a general strict 2-group and an arbitrary triangulation of any smooth d -dimensional spacetime manifold, for $d \in \{3, 4\}$. For $d = 3$, the constructed state sum is precisely Yetter’s model, while for $d = 4$ it coincides with Porter’s TQFT for $d = 4$ and $n = 2$.

To verify that the constructed state sum is topological, we analyze its behavior under Pachner moves, local changes of triangulation that preserve topology, such that any two triangulations of the same manifold are related by a finite number of Pachner moves. In three dimensions there are four Pachner moves—the $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ moves and their inverses—while in 4 dimensions there are five distinct Pachner moves: the $3 \leftrightarrow 3$, $4 \leftrightarrow 2$, and $5 \leftrightarrow 1$ moves and their inverses. The setup for analyzing the behavior of the constructed state sum under these Pachner moves is presented in Section 8.3, while the calculation details are given in Appendix E.1. The state sum is found to be unchanged under these transformations of the manifold, proving that it is a *topological invariant* of the manifold. Since it is independent of the triangulation, the state sum remains unchanged under arbitrary refinement of the triangulation and therefore defines a continuum theory on a smooth manifold.

See the work of Girelli, Pfeiffer, and Popescu for further information [13].

8.1 Gauge-invariant objects

The action of classical BF theory is chosen to be independent of any background metric, i.e. to depend only on the spacetime manifold. The classical equations of motion impose the condition that the gauge connection be flat or, in the language of holonomies, that every null-homotopic curve correspond to the identity element of the gauge group. In higher gauge theories, specifically 2-gauge theory and the corresponding $2BF$ theory, this condition generalizes to the requirement that the surface holonomy of the boundary 2-sphere of every 3-ball be trivial.

In Section 2.2, we introduced a collection of operations that define compositions of arbitrary paths and surfaces in the language of 2-gauge theory, including arbitrarily large ones. In this section, these compositions are used to construct gauge-invariant quantities corresponding to closed paths and surfaces. In Lemmas 13 and 14, this procedure is applied to the boundary path of a triangle and the boundary surface of a tetrahedron, as derived in [13].

Lemma 13 *Consider a triangle (jkl) . Its edges (jk) , $j < k$ are labeled by group elements $g_{jk} \in G$, and the triangle (jkl) , $j < k < \ell$ is labeled by an element $h_{jkl} \in H$. Consider the*

$$\begin{array}{c}
\begin{array}{ccccc}
l & \bullet & k & \bullet & j \\
\swarrow g_{kl} & & \swarrow g_{jk} & & \\
\searrow g_{jl} & & \searrow h_{jkl} & & \\
\end{array}
=
\begin{array}{ccccc}
l & \bullet & k & \bullet & j \\
\swarrow g_{kl} & & \swarrow g_{jk} & & \\
\searrow \partial(h_{jkl}) & & \searrow 1_{g_{kl}g_{jk}} & & \\
\end{array}
=
\begin{array}{ccccc}
l & \bullet & k & \bullet & j \\
\swarrow g_{kl} & & \swarrow g_{jk} & & \\
\searrow \partial(h_{jkl})g_{kl}g_{jk} & & \searrow h_{jkl} & & \\
\end{array}
.
\end{array}
\tag{8.1}$$
$$g_{j\ell} = \partial(h_{jkl})g_{kl}g_{jk}. \quad (8.2)$$

First analyze the composition shown in diagram (8.3). We initially move the curve from $g_{kl}g_{jk}$ to $g_{j\ell}$. At this stage, the result is not vertically composable with the triangle $(j\ell m)$; we must first add $g_{\ell m}$ on its left. The two morphisms are then vertically composable, and the resulting 2-morphism deforms the curve to g_{jm} . The resulting 2-morphism is

Consider diagram (8.4). First deform the curve $g_{\ell m} g_{k \ell}$ to the curve g_{km} . To make vertical composition of the result with the triangle (jkm) possible, first add the curve g_{jk} on its right. The two morphisms are now vertically composable, and the resulting 2-morphism deforms the curve to g_{jm} . This gives the following 2-morphism:

$$h_{ikm}h_{k\ell m} = h_{j\ell m}(g_{\ell m} \triangleright h_{ik\ell}). \quad (8.6)$$

8.2 Quantization of the topological $2BF$ action

This section presents a combinatorial description of the construction of the state sum for $2BF$ theory on a triangulation of a manifold of dimension $d = \{3, 4\}$.

The classical $2BF$ action given by equation (8) is quantized in the same way as in BF theories, using the usual heuristic spin-foam quantization procedure. First, the path integral of the topological state sum is

$$Z = \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}B \mathcal{D}C \exp \left(i \int_{M_d} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} \right). \quad (8.7)$$

Formally integrating over the Lagrange multipliers B and C gives

$$Z = \mathcal{N} \int \mathcal{D}\alpha \mathcal{D}\beta \delta(\mathcal{F}) \delta(\mathcal{G}). \quad (8.8)$$

As in ordinary gauge theory, the connection 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_d, \mathfrak{g})$ is discretized by labeling the edges $\epsilon = (jk) \in \Lambda_1$ of the triangulation with group elements $g_\epsilon \in G$, while the connection 2-form $\beta \in \mathcal{A}^2(\mathcal{M}_d, \mathfrak{h})$ is discretized by labeling the triangles $\Delta = (jkl) \in \Lambda_2$ with elements $h_\Delta \in H$. The path-integral measures in (8.7) are discretized by the substitutions

$$\int \mathcal{D}\alpha \quad \mapsto \quad \prod_{(jk) \in \Lambda_1} \int_G dg_{jk}, \quad (8.9)$$

$$\int \mathcal{D}\beta \quad \mapsto \quad \prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl}, \quad (8.10)$$

where dg_{jk} and dh_{jkl} denote integration with respect to the Haar measure on the groups G and H . The condition that the fake curvature vanish is imposed on every triangle $(jkl) \in \Lambda_2$ by discretizing $\delta(\mathcal{F})$. When passing from the smooth manifold to its triangulation, the δ distribution is defined on the set of simplices of the triangulation:

$$\delta(\mathcal{F}) = \prod_{(jkl) \in \Lambda_2} \delta_G(g_{jkl}), \quad (8.11)$$

where, for every triangle $(jkl) \in \Lambda_2$, the corresponding δ function $\delta_G(g_{jkl})$ is given by (see equation (8.2) in Lemma 13)

$$\delta_G(g_{jkl}) = \delta_G(\partial(h_{jkl}) g_{kl} g_{jk} g_{jl}^{-1}). \quad (8.12)$$

The condition that the surface holonomy of the boundary 2-sphere of every 3-ball be trivial, $\delta(\mathcal{G})$, becomes, after discretization on the simplices of the triangulated manifold,

$$\delta(\mathcal{G}) = \prod_{(jklm) \in \Lambda_3} \delta_H(h_{jklm}), \quad (8.13)$$

where, for every tetrahedron $(jklm) \in \Lambda_3$,

$$\delta_H(h_{jklm}) = \delta_H(h_{jlm} (g_{lm} \triangleright h_{jkl}) h_{klm}^{-1} h_{jkm}^{-1}). \quad (8.14)$$

Expression (8.14) follows from equation (8.6), derived in Lemma 14.

Substituting the discretized measures (8.9) and (8.10) and the δ functions (8.11) and (8.13) defined above into equation (8.8) gives the state sum

$$Z = \mathcal{N} \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \left(\prod_{(jklm) \in \Lambda_3} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in \Lambda_3} \delta_H(h_{jklm}) \right). \quad (8.15)$$

Next, substituting (8.12) and (8.14) into (8.15) yields an explicit expression for the state sum associated with a given triangulation of the manifold $\mathcal{M}_d$. With an appropriate choice of the prefactor $\mathcal{N}$ obtained after integrating over the Lagrange multipliers B and C , this sum becomes independent of the triangulation, i.e. invariant under Pachner moves¹. Requiring this invariance for all Pachner moves determines the appropriate choice of the constant $\mathcal{N}$ given in Definition 8.2.1.

Definition 8.2.1 *Let $\mathcal{M}_d$ be a compact oriented combinatorial d -manifold for $d = \{3, 4\}$, and let $(H \xrightarrow{\partial} G, \triangleright)$ be a crossed module. The state sum of topological higher gauge theory is defined by*

$$\begin{aligned} Z = & |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} |H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|} \left(\prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \right) \left(\prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \right) \\ & \times \left(\prod_{(jkl) \in \Lambda_2} \delta_G(\partial(h_{jkl}) g_{kl} g_{jk} g_{jl}^{-1}) \right) \left(\prod_{(jklm) \in \Lambda_3} \delta_H(h_{jlm} (g_{lm} \triangleright h_{jkl}) h_{klm}^{-1} h_{jkm}^{-1}) \right) \end{aligned} \quad (8.16)$$

In the preceding definition, integration is performed over elements $g_{jk} \in G$ for every edge $(jk) \in \Lambda_1$ and elements $h_{jkl} \in H$ for every triangle $(jkl) \in \Lambda_2$. The δ functions in the integrand impose the following conditions.

1. For every triangle $(jkl) \in \Lambda_2$ labeled by a group element h_{jkl} , the condition $\partial(h_{jkl}) g_{kl} g_{jk} = g_{jl}$ requires every triangle h_{jkl} to have the appropriate source and target (see Lemma 13);
2. For every tetrahedron $(jklm) \in \Lambda_3$, the condition that its surface holonomy be trivial reduces to the requirement that $h_{jlm} (g_{lm} \triangleright h_{jkl}) h_{klm}^{-1} h_{jkm}^{-1}$ equal the identity element of the group H (see Lemma 14).

Theorem 22 *Let $\mathcal{M}_d$ be a closed oriented combinatorial d -manifold for $d = \{3, 4\}$, and let $(H \xrightarrow{\partial} G, \triangleright)$ be a crossed module. The state sum (8.16) is invariant under Pachner moves.*

8.3 Pachner moves

8.3.1 $d = 3$

In three dimensions, to verify invariance of the state sum (8.16), it is enough to show that it is unchanged under the four Pachner moves $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ and their inverses. The setup of the proof of invariance of the state sum (8.16) under Pachner moves is given in this section, while the calculation details are presented in Appendix E.1 [13].

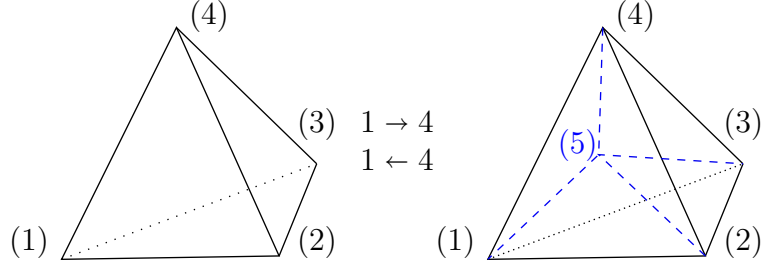
The $1 \leftrightarrow 4$ Pachner move

Label the vertices on the left-hand side of the $1 \leftrightarrow 4$ Pachner move by (1234). Adding the vertex (5) on the right-hand side of the Pachner move gives four tetrahedra:

$$M_3 = \{(2345), (1235), (1345), (1245)\}. \quad (8.17)$$

¹By Pachner's theorem, to prove that the state sum is a topological invariant, i.e. invariant under changes of triangulation, it is enough to establish invariance under the Pachner moves introduced by Udo Pachner in 1991 [44]. In three-dimensional space, the only Pachner moves are $2 \rightarrow 3$, $2 \leftarrow 3$, $1 \rightarrow 4$, and $1 \leftarrow 4$.

Theorem 21 (Pachner's theorem) *For a given piecewise-smooth manifold $\mathcal{M}_D$, every triangulation $T_1(\mathcal{M}_D)$ is related to a triangulation $T_2(\mathcal{M}_D)$ of a homeomorphic (topologically isomorphic) manifold by a finite number of Pachner moves.*



The additional triangles on the right-hand side are

$$M_2 = \{(125), (135), (145), (235), (245), (345)\}, \quad (8.18)$$

and the additional edges are

$$M_1 = \{(15), (25), (35), (45)\}. \quad (8.19)$$

On the left-hand side, the state sum is

$$\mathcal{Z}_{\text{levo}}^{1 \leftrightarrow 4} = |G|^{-2} |H|^1 \delta_H(h_{1234}) \mathcal{Z}_{\text{ostatak}}, \quad (8.20)$$

while on the right-hand side it is

$$\begin{aligned} \mathcal{Z}_{\text{desno}}^{1 \leftrightarrow 4} = & |G|^{-5} |H|^1 \int_{G^4} dg_{15} dg_{25} dg_{35} dg_{45} \int_{H^6} dh_{125} dh_{135} dh_{145} dh_{235} dh_{245} dh_{345} \\ & \left(\prod_{(jkl) \in M_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3} \delta_H(h_{jklm}) \right) \mathcal{Z}_{\text{ostatak}}. \end{aligned} \quad (8.21)$$

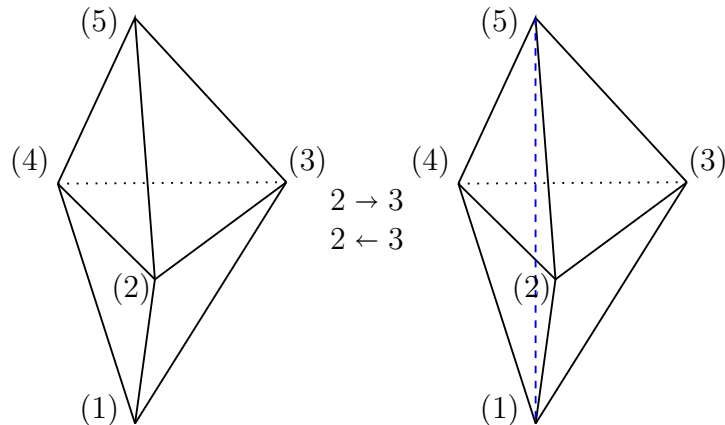
where M_2 and M_3 are given in (8.17) and (8.18). The numbers of k -simplices on the two sides of the $1 \leftrightarrow 4$ move (not counting the remainder of the triangulation) are given in Table 8.1.

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $
l.s.	4	6	4	1
d.s.	5	10	10	4

Table 8.1: Numbers of vertices $|\Lambda_0|$, edges $|\Lambda_1|$, triangles $|\Lambda_2|$, and tetrahedra $|\Lambda_3|$ on the left- and right-hand sides of the $1 \leftrightarrow 4$ Pachner move.

The proof of invariance reduces to showing that expressions (8.20) and (8.21) are equal, where the factor $\mathcal{Z}_{\text{ostatak}}$ denotes the part of the state sum that remains unchanged by the definition of the move.

The $2 \leftrightarrow 3$ Pachner move



Let the tetrahedra on the left-hand side of the move be $M_3^{\text{levo}} = \{(1234), (2345)\}$; they share the triangle $M_2^{\text{levo}} = \{(234)\}$. On the right-hand side, the tetrahedra $M_3^{\text{desno}} = \{(1235), (1245), (1345)\}$ share the edge $M_1^{\text{desno}} = \{(15)\}$, and each pair shares one of the triangles $M_2^{\text{desno}} = \{(125), (135), (145)\}$.

On the left-hand side of the $2 \leftrightarrow 3$ move, the state sum is

$$\mathcal{Z}_{\text{levo}}^{2 \leftrightarrow 3} = |G|^{-3} |H|^1 \int_H dh_{234} \delta_G(g_{234}) \delta_H(h_{1234}) \delta_H(h_{2345}) \mathcal{Z}_{\text{ostatak}}, \quad (8.22)$$

while on the right-hand side it is

$$\begin{aligned} \mathcal{Z}_{\text{desno}}^{2 \leftrightarrow 3} = & |G|^{-4} |H|^1 \int_G dg_{15} \int_{H^3} dh_{125} dh_{135} dh_{145} \\ & \delta_G(g_{125}) \delta_G(g_{135}) \delta_G(g_{145}) \delta_H(h_{1235}) \delta_H(h_{1245}) \delta_H(h_{1345}) \mathcal{Z}_{\text{ostatak}}. \end{aligned} \quad (8.23)$$

The numbers of k -simplices on the two sides of the $2 \leftrightarrow 3$ move are shown in Table 8.2.

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $
l.s.	5	9	7	2
d.s.	5	10	9	3

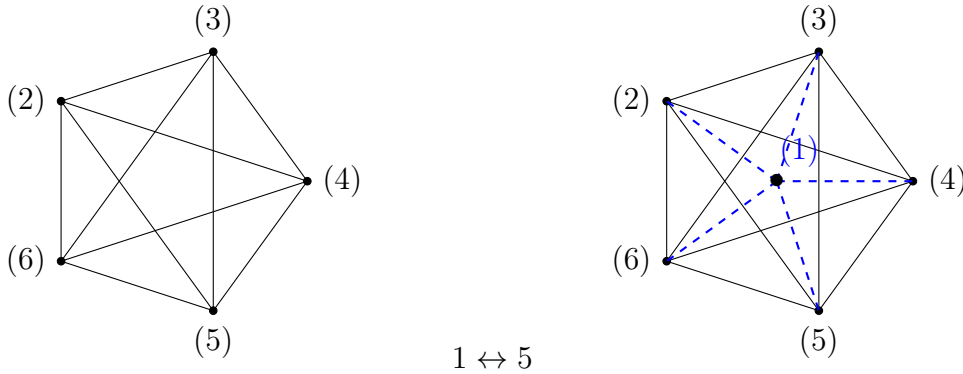
Table 8.2: Numbers of vertices $|\Lambda_0|$, edges $|\Lambda_1|$, triangles $|\Lambda_2|$, and tetrahedra $|\Lambda_3|$ on the left- and right-hand sides of the $2 \leftrightarrow 3$ Pachner move.

The proof of invariance reduces to showing that expressions (8.22) and (8.23) are equal, where the factor $\mathcal{Z}_{\text{ostatak}}$ denotes the part of the state sum that remains unchanged by the definition of the move.

8.3.2 $d = 4$

In four dimensions, to verify invariance of the state sum (8.16), it is enough to show that it is unchanged under the five Pachner moves $1 \leftrightarrow 5$, $2 \leftrightarrow 4$, and $3 \leftrightarrow 3$ and their inverses. The setup of the proof of invariance of the state sum (8.16) under Pachner moves is given in this section, while the calculation details are presented in Appendix E.1 [13].

The $1 \leftrightarrow 5$ Pachner move



Label the vertices of the 4-simplex on the left-hand side of the $1 \leftrightarrow 5$ Pachner move by (23456) . Adding the vertex (1) on the right-hand side of the Pachner move gives five 4-simplices,

$$M_4 = \{(13456), (12456), (12356), (12346), (12345)\}.$$

The right-hand side contains the additional tetrahedra

$$M_3 = \{(1234), (1235), (1236), (1245), (1246), (1256), (1345), (1346), (1356), (1456)\},$$

the additional triangles

$$(jkl) \in M_2 = \{(123), (124), (125), (126), (134), (135), (136), (145), (146), (156)\},$$

the additional edges $(jk) \in M_1 = \{(12), (13), (14), (15), (16)\}$, and the additional vertex $(j) \in M_0 = \{(1)\}$. All other simplices are present on both sides of the move.

Invariance of the state sum (8.16) under the $1 \leftrightarrow 5$ Pachner move means that the integral on the right-hand side,

$$\begin{aligned} Z_{\text{desno}}^{1 \leftrightarrow 5} &= |G|^{-11} |H|^{-4} \int_{G^5} \prod_{(jk) \in M_1} dg_{jk} \int_{H^{10}} \prod_{(jkl) \in M_2} dh_{jkl} \\ &\cdot \left(\prod_{(jkl) \in M_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3} \delta_H(h_{jklm}) \right) Z_{\text{ostatak}}, \end{aligned} \quad (8.24)$$

equals the state sum on the left-hand side,

$$Z_{\text{levo}}^{1 \leftrightarrow 5} = |G|^{-5} |H|^0 Z_{\text{ostatak}}. \quad (8.25)$$

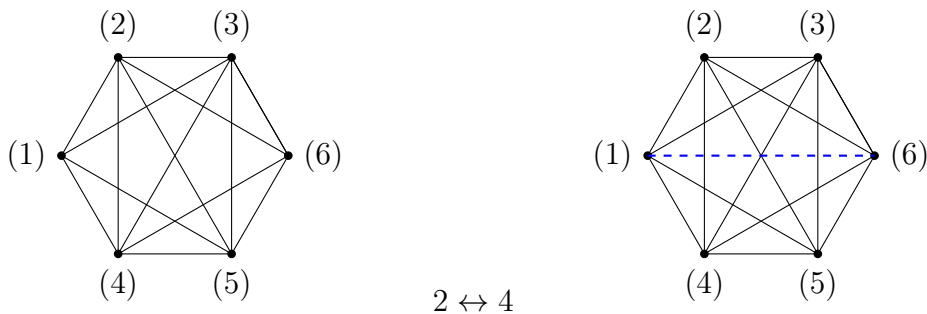
The prefactors of the state-sum integrals on the left- and right-hand sides of the move are calculated from equation (8.16), using $|G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|}$ and $|H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|}$, where $|\Lambda_0|$, $|\Lambda_1|$, $|\Lambda_2|$, and $|\Lambda_3|$ are respectively the numbers of vertices, edges, triangles, and tetrahedra in the triangulation. From the data in Table 8.3, the factor on the right-hand side is $|G|^{-11}|H|^{-4}$, while that on the left-hand side is $|G|^{-5}|H|^0$.

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $	$ \Lambda_4 $
l.s.	5	10	10	5	1
d.s.	6	15	20	15	5

Table 8.3: Numbers of vertices $|\Lambda_0|$, edges $|\Lambda_1|$, triangles $|\Lambda_2|$, tetrahedra $|\Lambda_3|$, and 4-simplices $|\Lambda_4|$ on the left- and right-hand sides of the $1 \leftrightarrow 5$ Pachner move.

The proof of invariance reduces to showing that expressions (8.24) and (8.25) are equal, where the factor Z_{ostatak} denotes the part of the state sum that remains unchanged by the definition of the move.

The $2 \leftrightarrow 4$ Pachner move



To verify invariance of the state sum (8.16) under the $2 \leftrightarrow 4$ Pachner move, arrange the vertices so that there are two 4-simplices on the left-hand side,

$$M_4^{\text{levo}} = \{(23456), (12345)\}$$

, and four 4-simplices on the right-hand side,

$$M_4^{\text{desno}} = \{(12346), (12356), (12456), (13456)\}.$$

The left-hand side then contains one tetrahedron,

$$M_3^{\text{levo}} = \{(2345)\},$$

while the right-hand side contains six tetrahedra,

$$M_3^{\text{desno}} = \{(1236), (1246), (1256), (1346), (1356), (1456)\}.$$

All other tetrahedra are present on both sides of the move. In addition, the right-hand side contains the triangles $M_2^{\text{desno}} = \{(126), (136), (146), (156)\}$ and one edge $M_1^{\text{desno}} = \{(16)\}$, while all remaining triangles and edges are present on both sides. All vertices are likewise present on both sides of the move.

On the left-hand side of the move, the integral is

$$Z_{\text{levo}}^{2 \leftrightarrow 4} = |G|^{-8} |H|^{-1} \delta_H(h_{2345}) Z_{\text{ostatak}}, \quad (8.26)$$

while on the right-hand side it is

$$Z_{\text{desno}}^{2 \leftrightarrow 4} = |G|^{-11} |H|^{-3} \int_G dg_{16} \int_{H^4} dh_{126} dh_{136} dh_{146} dh_{156} \left(\prod_{(jkl) \in M_2^{\text{desno}}} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3^{\text{desno}}} \delta_H(h_{jklm}) \right) Z_{\text{ostatak}}. \quad (8.27)$$

Counting the k -simplices on the two sides of the $2 \leftrightarrow 4$ move (see Table 8.4) gives the prefactors of the integrals: $|G|^{-8} |H|^{-1}$ on the left-hand side and $|G|^{-11} |H|^{-3}$ on the right-hand side.

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $	$ \Lambda_4 $
l.s.	6	14	16	9	2
d.s.	6	15	20	14	4

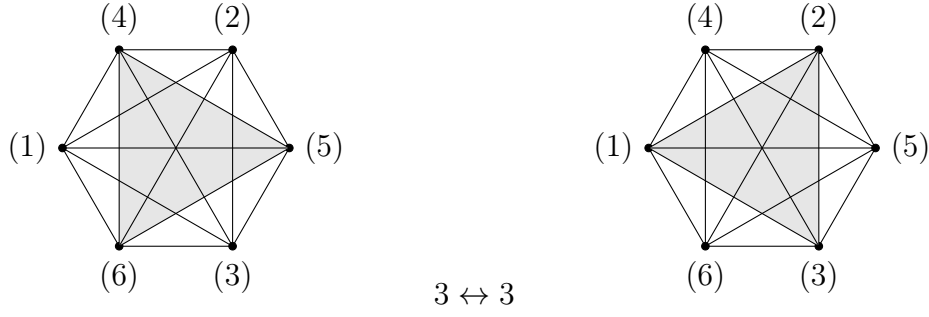
Table 8.4: Numbers of vertices $|\Lambda_0|$, edges $|\Lambda_1|$, triangles $|\Lambda_2|$, tetrahedra $|\Lambda_3|$, and 4-simplices $|\Lambda_4|$ on the two sides of the $2 \leftrightarrow 4$ move.

The proof of invariance reduces to showing that expressions (8.26) and (8.27) are equal, where the factor Z_{ostatak} denotes the part of the state sum that remains unchanged by the definition of the move.

The $3 \leftrightarrow 3$ Pachner move

Label the vertices so that the left-hand side of the $3 \leftrightarrow 3$ Pachner move contains three 4-simplices,

$$M_4^{\text{levo}} = \{(23456), (13456), (12456)\},$$



while the right-hand side contains the 4-simplices

$$M_4^{\text{desno}} = \{(12356), (12346), (12345)\}.$$

The tetrahedra $M_3^{\text{levo}} = \{(1456), (2456), (3456)\}$ occur on the left-hand side, while $M_3^{\text{desno}} = \{(1234), (1235), (1236)\}$ occur on the right-hand side. The two sides share six tetrahedra, while each side contains three tetrahedra shared by two 4-simplices. Furthermore, the left-hand side contains the triangle $M_2^{\text{levo}} = \{(456)\}$ and the right-hand side the triangle $M_2^{\text{desno}} = \{(123)\}$. All other triangles, edges, and vertices occur on both sides of the move.

Thus, on the left-hand side of the move, the integral is

$$Z_{\text{levo}}^{3 \leftrightarrow 3} = \int_H dh_{456} \delta_G(g_{456}) \delta_H(h_{3456}) \delta_H(h_{2456}) \delta_H(h_{1456}) Z_{\text{ostatak}}, \quad (8.28)$$

while on the right-hand side it is

$$Z_{\text{desno}}^{3 \leftrightarrow 3} = \int_H dh_{123} \delta_G(g_{123}) \delta_H(h_{1234}) \delta_H(h_{1235}) \delta_H(h_{1236}) Z_{\text{ostatak}}. \quad (8.29)$$

The proof of invariance reduces to showing that expressions (8.28) and (8.29) are equal, where the factor Z_{ostatak} denotes the part of the state sum that remains unchanged by the definition of the move.

Chapter 9

Construction of the topological state sum: $3BF$ theory

In this chapter we focus on the second step of the covariant spin-foam quantization procedure for $3BF$ theory. Analogously to the procedure in the preceding chapter for the $2BF$ state sum, we demonstrate how to construct a triangulation-independent state sum Z from the classical $3BF$ action for a general semistrict 3-group and an arbitrary triangulation of any smooth 4-dimensional spacetime manifold, as done in [27]. A $3BF$ theory can be formulated only when the dimension of the spacetime manifold is $d \geq 4$, so the three-dimensional case is not considered. The constructed state sum generalizes the work of Girelli, Pfeiffer, and Popescu on the $2BF$ state sum presented in the preceding chapter, i.e. it generalizes Yetter's model, and coincides with Porter's TQFT for $d = 4$ and $n = 3$.

As in the case of the $2BF$ state sum, to verify that the constructed state sum is topological we analyze its behavior under Pachner moves. Only the four-dimensional case is considered, namely invariance under the $3 \leftrightarrow 3$, $4 \leftrightarrow 2$, and $5 \leftrightarrow 1$ Pachner moves and their inverses. The setup for analyzing the behavior of the constructed state sum under these Pachner moves is presented in Section 9.3, while the calculation details are given in Appendix E.2. The state sum is found to be invariant under Pachner moves, proving that it is a *topological invariant* of the manifold [27]. We conclude that the state sum remains unchanged under arbitrary refinement of the triangulation and therefore defines a continuum theory on a smooth manifold.

However, completing the second step of the covariant spin-foam quantization procedure requires generalizations of the Peter–Weyl and Plancherel theorems to 2-groups and 3-groups, mathematical results that remain open problems. These theorems should provide a decomposition of functions on a 3-group into a sum over the corresponding irreducible representations of the 3-group. This determines the spectrum of labels assigned to the simplices of the triangulation, i.e. the value domains of the fields living on those simplices, as in the case of the BF state sum. Current attempts to complete the second quantization step for generalized BF theories within higher gauge theory rely on conjecturing the irreducible representations of 2-groups, as done for example in the spincube model of quantum gravity [15], or on other techniques; see, for example, [45], [46], [47].

This result opens the way to the third and final step of the covariant quantization procedure and to formulating a quantum theory of gravity and Standard Model matter by imposing appropriate constraints on the model variables through a modification of the state-sum amplitudes.

9.1 Gauge-invariant objects

In addition to the preceding conditions that also hold in 2BF theory, the equations of motion of 3BF theory impose a higher flatness condition on the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, further requiring the volume holonomy around the boundary 3-sphere of every 4-ball to be trivial.

In Section 2.3, we introduced a collection of operations that define compositions of paths, surfaces, and volumes in the language of 3-gauge theory. These rules can be used to calculate compositions of elementary paths, surfaces, and volumes, including arbitrarily large ones. In this section, these compositions are used to construct gauge-invariant quantities corresponding to closed paths, surfaces, and volumes. In Lemmas 13, 15, and 16, this procedure is applied to the boundary path of a triangle, the boundary surface of a tetrahedron, and the boundary volume of a 4-simplex. The result of Lemma 13 was derived for 2-groups and remains unchanged in 3-gauge theory; see [13]. The flatness condition for the boundary surface of a tetrahedron derived in Lemma 14 is generalized to 3-groups in Lemma 15. One of the main results is Lemma 16, in which the flatness condition for the boundary volume of a 4-simplex is derived.

Lemma 15 *Consider a tetrahedron $(jklm)$. Its edges (jk) , $j < k$ are labeled by group elements $g_{jk} \in G$, its triangles (jkl) , $j < k < l$ by group elements $h_{jkl} \in H$, and the tetrahedra of the manifold $(jklm)$, $j < k < l < m$ by group elements $l_{jklm} \in L$. The triangles (jkl) are oriented so that the curve $g_{kl}g_{jk}$ is the source of the 2-morphism and the curve $g_{j\ell}$ its target, i.e. the condition $g_{j\ell} = \partial(h_{jkl})g_{kl}g_{jk}$ is satisfied.*

First, cut the surface of the tetrahedron along the boundary (jm) . This determines the order of vertical composition of the constituent surfaces. We must ensure that all surfaces are composable, i.e. that they have the appropriate reference points and the correct orientation for vertical composition.

First analyze the composition shown in diagram (9.1). We initially move the curve from $g_{kl}g_{jk}$ to $g_{j\ell}$. At this stage, the result is not vertically composable with the triangle $(j\ell m)$; we must first add $g_{\ell m}$ on its left. The two morphisms are then vertically composable, and the resulting 2-morphism deforms the curve to g_{jm} . The resulting 2-morphism is

$$\begin{array}{c}
 \begin{array}{ccccccc}
 m & \xleftarrow{g_{\ell m}} & \bullet & \xleftarrow{g_{kl}} & \bullet & \xleftarrow{g_{jk}} & j \\
 & \searrow & & \searrow & & \searrow & \\
 & & \bullet & & \bullet & & \\
 & \swarrow & & \swarrow & & \swarrow & \\
 & & \bullet & & \bullet & & \\
 & \nwarrow & & \nwarrow & & \nwarrow & \\
 & & m & & \ell & & k
 \end{array}
 \end{array}
 = (g_{\ell m}g_{j\ell}, h_{j\ell m}) \#_2 (g_{\ell m} \#_1 (g_{kl}g_{jk}, h_{jkl})) = (g_{\ell m}g_{kl}g_{jk}, h_{j\ell m}(g_{\ell m} \triangleright h_{jkl})).
 \tag{9.1}$$

Consider diagram (9.2). First deform the curve from $g_{\ell m}g_{kl}$ to g_{km} . To make vertical composition of the result with the triangle (jkm) possible, first add the curve g_{jk} on its right. The two morphisms are now vertically composable, and the resulting 2-morphism deforms the curve to g_{jm} . This gives the following 2-morphism:

$$\begin{array}{c}
 \begin{array}{ccccccc}
 m & \xleftarrow{g_{\ell m}} & \bullet & \xleftarrow{g_{kl}} & \bullet & \xleftarrow{g_{jk}} & j \\
 & \searrow & & \searrow & & \searrow & \\
 & & \bullet & & \bullet & & \\
 & \swarrow & & \swarrow & & \swarrow & \\
 & & \bullet & & \bullet & & \\
 & \nwarrow & & \nwarrow & & \nwarrow & \\
 & & m & & \ell & & k
 \end{array}
 \end{array}
 = (g_{km}g_{jk}, h_{jkm}) \#_2 ((g_{\ell m}g_{kl}, h_{k\ell m}) \#_1 g_{jk}) = (g_{\ell m}g_{kl}g_{jk}, h_{jkm}h_{k\ell m}).
 \tag{9.2}$$

The two surfaces $\Sigma_1 : g_{\ell m}g_{kl}g_{jk} \rightarrow g_{jm}$ and $\Sigma_2 : g_{\ell m}g_{kl}g_{jk} \rightarrow g_{jm}$ have the same source and target. The deformation of the surface shown in diagram (9.1) into the surface shown in diagram (9.2) is given by the volume morphism $\mathcal{V} : \Sigma_1 \rightarrow \Sigma_2$ labeled by the group element l_{jklm} , i.e.

$$(g_{\ell m}g_{kl}g_{jk}, h_{jkm}h_{k\ell m}) = (g_{\ell m}g_{kl}g_{jk}, \delta(l_{jklm})h_{j\ell m}(g_{\ell m} \triangleright h_{jkl})), \tag{9.3}$$

which gives the relation

$$h_{jkm}h_{k\ell m} = \delta(l_{jk\ell m})h_{j\ell m}(g_{\ell m} \triangleright h_{jkl}). \quad (9.4)$$

Lemma 16 Consider a 4-simplex, $(jklmn)$. Its edges (jk) , $j < k$ are labeled by elements $g_{jk} \in G$, its triangles (jkl) , $j < k < \ell$ by elements $h_{jkl} \in H$, and its tetrahedra $(jklm)$, $j < k < \ell < m$ by elements $l_{jk\ell m} \in L$. The triangles (jkl) are oriented so that the source of the 2-morphism is the curve $g_{kl}g_{jk}$ and its target is the curve $g_{j\ell}$, i.e. $g_{j\ell} = \partial(h_{jkl})g_{kl}g_{jk}$, while the tetrahedra $(jklm)$ are oriented so that the source of the 3-morphism is the surface $h_{j\ell m}(g_{\ell m} \triangleright h_{jkl})$ and its target is the surface $h_{jkm}h_{k\ell m}$, i.e. $h_{jkm}h_{k\ell m} = \delta(l_{jk\ell m})h_{j\ell m}(g_{\ell m} \triangleright h_{jkl})$.

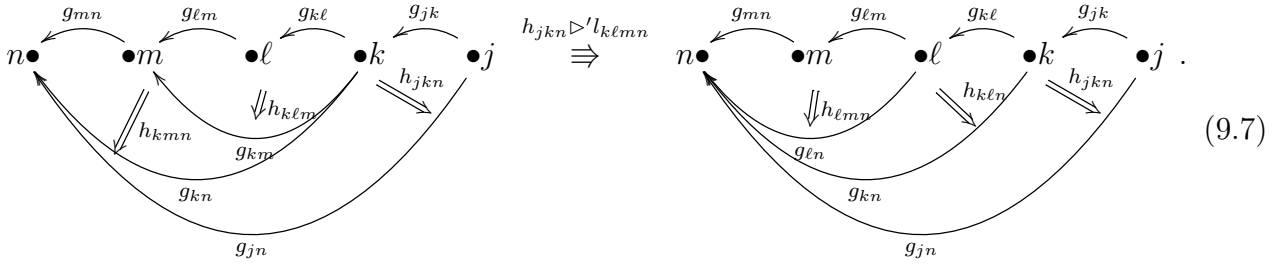
First, cut the volume of the 4-simplex along the surface $h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$. This determines the order of upward composition of the 3-morphisms. We must ensure that all volumes are composable, i.e. that they have the appropriate reference surfaces and the correct orientation for their compositions to be defined.

First consider diagram (9.5). The surface $h_{j\ell m}g_{\ell m} \triangleright h_{jkl}$ is deformed into the surface $h_{jkm}h_{k\ell m}$ by the 3-morphism $l_{jk\ell m}$. To define the composition of the resulting 3-morphism with the 2-morphism h_{jmn} , we must first add to the 3-morphism the curve g_{mn} on the left. The resulting 3-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl}), g_{mn} \triangleright l_{jk\ell m})$ can be vertically composed with the 2-morphism $(g_{mn}g_{jm}, h_{jmn})$ from below, so that the resulting 3-morphism is $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl}), h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m}))$, whose source is the surface $h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$ and whose target is the surface $h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m})$:

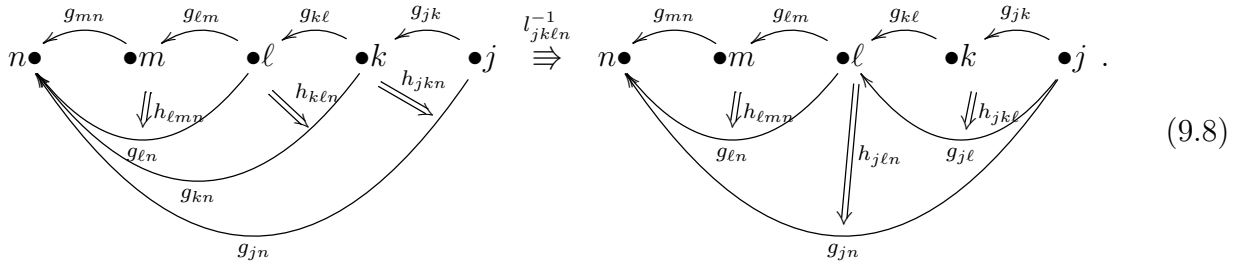
We now deform the surface to $h_{jkn}h_{kmn}g_{ml} \triangleright h_{k\ell m}$, as shown in diagram (9.6). First consider the 3-morphism $(g_{mn}g_{km}g_{jk}, h_{jmn}g_{mn} \triangleright h_{jkm}, l_{jkmn})$, whose source is the surface $h_{jmn}g_{mn} \triangleright h_{jkm}$ and whose target is the surface $h_{jkn}h_{kmn}$. This 3-morphism can be extended from above by the 2-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, g_{mn} \triangleright h_{k\ell m})$, resulting in the 3-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m}), l_{jkmn})$, whose source is the surface $h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m})$ and whose target is the surface $h_{jkn}h_{kmn}g_{mn} \triangleright h_{k\ell m}$:

The next step is to deform the surface from $h_{jkn}h_{kmn}g_{mn} \triangleright h_{k\ell m}$ to $h_{jkn}h_{k\ell n}h_{\ell mn}$, as shown in diagram (9.7). First extend the 3-morphism $(g_{mn}g_{\ell m}g_{kl}, h_{kmn}g_{mn} \triangleright h_{k\ell m}, l_{k\ell mn})$, whose source is the surface $h_{kmn}g_{mn} \triangleright h_{k\ell m}$ and whose target is the surface $h_{k\ell n}h_{\ell mn}$, by the curve g_{jk} on the right, resulting in the 3-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{kmn}g_{mn} \triangleright h_{k\ell m}, l_{k\ell mn})$. Then extend the resulting 3-morphism from below by the 2-morphism $(g_{kn}g_{jk}, h_{jkn})$, obtaining the 3-morphism

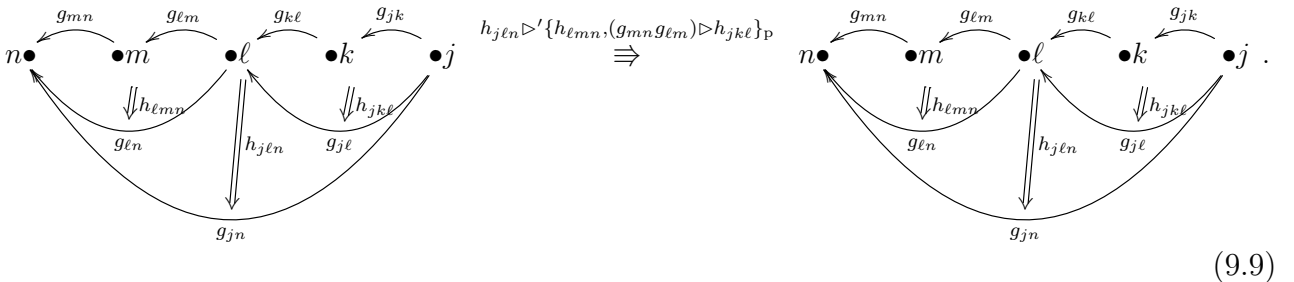
$(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}g_{mn} \triangleright h_{k\ell m}, h_{jkn} \triangleright' l_{k\ell mn})$:



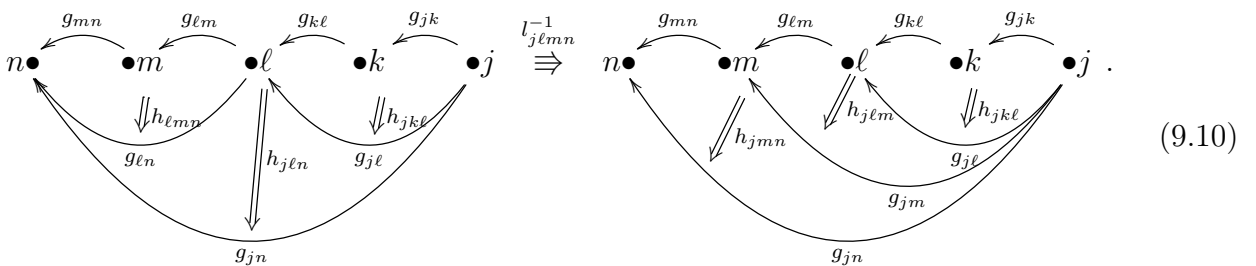
The deformation of the surface $h_{jkn}h_{k\ell n}h_{\ell mn}$ into the surface $h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$ is shown in diagram (9.8). The 3-morphism with the corresponding source and target is obtained by extending the 3-morphism $(g_{\ell n}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}, l_{jkl n}^{-1})$ from above by the 2-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{\ell mn})$. The resulting 3-morphism is $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}h_{\ell mn}, l_{jkl n}^{-1})$:



Next, we map the surface $h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$ to the surface $h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$; see diagram (9.9). This is achieved by the inverse interchange 2-morphism composition, which maps $g_{\ell n} \triangleright h_{jkl}h_{\ell mn}$ to the surface $h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$, i.e. by the 3-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, g_{\ell n} \triangleright h_{jkl}h_{\ell mn}, \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_{\text{pf}})$. Then extend the resulting 3-morphism from below by the 2-morphism $(g_{\ell n}g_{j\ell}, h_{j\ell n})$. The resulting 3-morphism with the appropriate source and target is $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{j\ell n}g_{\ell n} \triangleright h_{jkl}h_{\ell mn}, h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_{\text{pf}})$,



Finally, complete the construction with a 3-morphism that maps the surface $h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}$ to the initial surface $h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$. First deform the surface $h_{j\ell n}h_{\ell mn}$ into the surface $h_{jmn}g_{mn} \triangleright h_{j\ell m}$ by the 3-morphism $(g_{mn}g_{\ell m}g_{j\ell}, h_{j\ell n}h_{\ell mn}, l_{j\ell mn}^{-1})$. Then extend the 3-morphism $(g_{mn}g_{\ell m}g_{j\ell}, h_{j\ell n}h_{\ell mn}, l_{j\ell mn}^{-1})$ from above by the 2-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, (g_{mn}g_{\ell m}) \triangleright h_{jkl})$. The resulting 3-morphism $(g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}, l_{j\ell mn}^{-1})$ deforms the surface into the one from which we started, as shown in diagram (9.10):



We now form the upward composition of the 3-morphisms shown in diagrams (9.5)–(9.10), taking care of their order. The resulting 3-morphism is

$$\begin{aligned}
& (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{j\ell n}h_{\ell mn}(g_{mn}g_{\ell m}) \triangleright h_{jkl}, l_{j\ell mn}^{-1}) \#_3 \\
& (g_{mn}g_{\ell m}g_{kl}g_{jk}, g_{\ell n} \triangleright h_{jkl}h_{\ell mn}, h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_{\text{p}}) \#_3 \\
& (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{k\ell n}h_{\ell mn}, l_{j\ell n}^{-1}) \#_3 \\
& (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jkn}h_{k\ell mn}g_{\ell m} \triangleright h_{k\ell m}, h_{jkn} \triangleright' l_{jk\ell mn}) \#_3 \\
& (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{jkm}h_{k\ell m}), l_{jk\ell mn}) \#_3 \\
& (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl}), h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})) \\
& = (g_{mn}g_{\ell m}g_{kl}g_{jk}, h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl}), l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_{\text{pf}} \\
& \quad l_{j\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jk\ell mn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m})) .
\end{aligned} \tag{9.11}$$

The resulting 3-morphism is the identity 3-morphism, i.e. its source and target are the surface $\mathcal{V}_1 = \mathcal{V}_2 = h_{jmn}g_{mn} \triangleright (h_{j\ell m}g_{\ell m} \triangleright h_{jkl})$, so the following identity holds:

$$l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn}g_{\ell m}) \triangleright h_{jkl}\}_{\text{pf}} l_{j\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jk\ell mn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jk\ell m}) = e . \tag{9.12}$$

9.2 Quantization of the topological $3BF$ action

This section presents a combinatorial description of the construction of the $3BF$ state sum for an arbitrary triangulation of a manifold of dimension $d = 4$. The model is defined for any closed oriented combinatorial manifold $\mathcal{M}_4$ of dimension $d = 4$. It coincides with Porter's TQFT [26] for $d = 4$ and $n = 3$.

We first demonstrate how the state sum corresponding to the classical $3BF$ action (6.1) is constructed by the usual discretization procedure. We therefore consider the state sum

$$Z = \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \mathcal{D}B \mathcal{D}C \mathcal{D}D \exp \left(\int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}} \right) . \tag{9.13}$$

Formal integration over the Lagrange multipliers B , C , and D gives

$$Z = \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \delta(\mathcal{F})\delta(\mathcal{G})\delta(\mathcal{H}) . \tag{9.14}$$

As in ordinary gauge theory, the connection 1-form $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$ is discretized by labeling the edges $\epsilon = (jk) \in \Lambda_1$ of the triangulation with group elements $g_{\epsilon} \in G$. Analogously, the connection 2-form $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h})$ is represented by labeling the triangles $\Delta = (jkl) \in \Lambda_2$ with elements $h_{\Delta} \in H$, while the connection 3-form $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l})$ is represented by labeling the tetrahedra $\tau = (jk\ell m) \in \Lambda_3$ with group elements $l_{\tau} \in L$.

The measure of the state sum (9.13) is discretized by the substitutions

$$\int \mathcal{D}\alpha \quad \mapsto \quad \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} , \tag{9.15}$$

$$\int \mathcal{D}\beta \quad \mapsto \quad \prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} , \tag{9.16}$$

$$\int \mathcal{D}\gamma \quad \mapsto \quad \prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} , \tag{9.17}$$

where dg_{jk} , dh_{jkl} , and $dl_{jk\ell m}$ respectively denote integrations over the groups G , H , and L .

The condition that the fake curvature vanish, given by equation (8.2) in Lemma 13, is discretized on every triangle $(jkl) \in \Lambda_2$ by replacing $\delta(F)$ with

$$\delta_G(g_{jkl}) = \delta_G(\partial(h_{jkl}) g_{kl} g_{jk} g_{jl}^{-1}), \quad (9.18)$$

the condition that the curvature 3-form $\delta(G)$ be trivial for every tetrahedron $(jklm) \in \Lambda_3$ of the discretized manifold, given by equation (9.4) in Lemma 15, becomes

$$\delta_H(h_{jklm}) = \delta_H(\delta(l_{jklm}) h_{jlm} (g_{lm} \triangleright h_{jkl}) h_{k\ell m}^{-1} h_{jkm}^{-1}), \quad (9.19)$$

while the condition that the curvature 4-form $\delta(H)$ be trivial for every 4-simplex $(jklmn) \in \Lambda_4$ becomes identity (9.12) in Lemma 16:

$$\delta_L(l_{jklmn}) = \delta_L(l_{jkl\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} (h_{jmn} \triangleright' (g_{mn} \triangleright l_{jklm})) h_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn} g_{\ell m}) \triangleright h_{jkl}\}_p). \quad (9.20)$$

It follows that the path integral can be written as the state sum

$$Z = \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \prod_{(jklm) \in \Lambda_3} \int_L dl_{jklm} \left(\prod_{(jkl) \in \Lambda_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in \Lambda_3} \delta_H(h_{jklm}) \right) \left(\prod_{(jklmn) \in \Lambda_4} \delta_L(l_{jklmn}) \right). \quad (9.21)$$

Substituting equations (9.18), (9.19), and (9.20) into the state sum (9.21) gives an expression proportional to (9.22). For the sum in (9.21) to be independent of the triangulation of the manifold, it must be multiplied by an appropriate factor depending on the numbers of vertices, edges, triangles, tetrahedra, and 4-simplices in the triangulation, resulting in the state sum in equation (9.22).

Definition 9.2.1 Let $\mathcal{M}_d$ be a compact oriented combinatorial d -manifold, $d = 4$, and let $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_p)$ be a 2-crossed module. The state sum of topological 3-gauge theory is defined by the following expression:

$$Z = |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} |H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|} |L|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|+|\Lambda_3|-|\Lambda_4|} \left(\prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \right) \left(\prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \right) \left(\prod_{(jklm) \in \Lambda_3} \int_L dl_{jklm} \right) \left(\prod_{(jkl) \in \Lambda_2} \delta_G(\partial(h_{jkl}) g_{kl} g_{jk} g_{jl}^{-1}) \right) \left(\prod_{(jklm) \in \Lambda_3} \delta_H(\delta(l_{jklm}) h_{jlm} (g_{lm} \triangleright h_{jkl}) h_{k\ell m}^{-1} h_{jkm}^{-1}) \right) \left(\prod_{(jklmn) \in \Lambda_4} \delta_L(l_{j\ell mn}^{-1} h_{j\ell n} \triangleright' \{h_{\ell mn}, (g_{mn} g_{\ell m}) \triangleright h_{jkl}\}_p l_{jkl\ell n}^{-1} (h_{jkn} \triangleright' l_{k\ell mn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jklm})) \right). \quad (9.22)$$

In the preceding expression, we integrate over the elements $g_{jk} \in G$ for every edge $(jk) \in \Lambda_1$, over the elements $h_{jkl} \in H$ for every triangle $(jkl) \in \Lambda_2$, and over the elements l_{jklm} for every tetrahedron of the manifold $(jklm) \in \Lambda_3$, while the δ -distributions in the integrand impose the following conditions on these values. The conditions that the elements of the triangulation must satisfy were derived in Lemmas 13, 15, and 16.

1. The condition that every triangle $(jkl) \in \Lambda_2$ carrying the label h_{jkl} have the appropriate source and target is $\partial(h_{jkl}) g_{kl} g_{jk} = g_{jl}$, as shown in Lemma 13.
2. Next, the condition $h_{jkm} h_{k\ell m} = \delta(l_{jklm}) h_{jlm} (g_{lm} \triangleright h_{jkl})$ holds for every tetrahedron $(jklm) \in \Lambda_3$, i.e. every tetrahedron carrying the label l_{jklm} has a well-defined source and target; see Lemma 15.

3. Finally, the volume holonomy around every 4-simplex $(jklmn) \in \Lambda_4$ is trivial, i.e. $l_{jlmn}^{-1} h_{jln} \triangleright' \{h_{lmn}, (g_{mn} g_{lm}) \triangleright h_{jkl}\}_p l_{jklm}^{-1} (h_{jkn} \triangleright' l_{klmn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jklm})$ is the identity element of the group L for every 4-simplex $(jklmn) \in \Lambda_4$, as proved in Lemma 16.

Theorem 23 *Let $\mathcal{M}_4$ be a closed and oriented combinatorial d -manifold, $d = 4$, and let $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$ be a 2-crossed module. The state sum (9.22) is invariant under Pachner moves.*

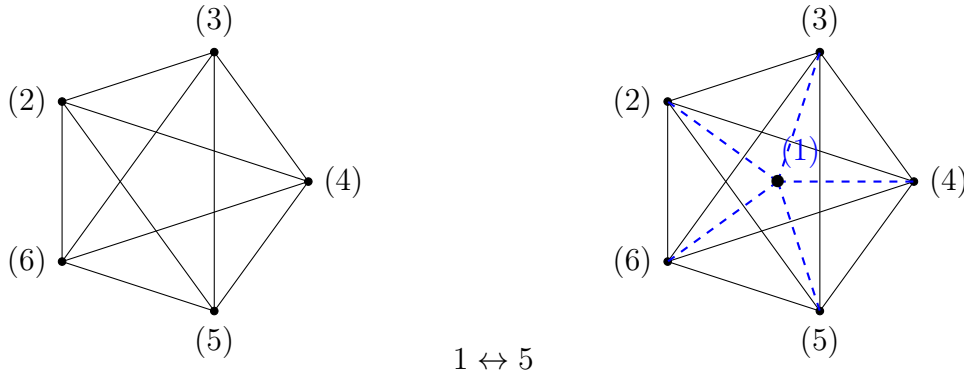
In Appendix E, we shall outline the proof of invariance of the state sum under Pachner moves [44]. In the state sum (9.22), labeling edges by elements $g_{jk} \in G$, triangles by elements $h_{jkl} \in H$, and tetrahedra by elements $l_{jklm} \in L$ is called a *coloring of the manifold*.

9.3 Pachner moves

9.3.1 $d = 4$

In the four-dimensional case, to verify the invariance of the state sum (9.22) under local changes of the triangulation of a four-dimensional manifold, it is sufficient to show that it remains unchanged under the five Pachner moves: the $1 \leftrightarrow 5$, $2 \leftrightarrow 4$, and $3 \leftrightarrow 3$ Pachner moves and their inverses. The setup of the proof of invariance of the state sum (9.22) under Pachner moves is given in this section, while the details of the calculation are presented in Appendix E.2.

The Pachner move $1 \leftrightarrow 5$



Since the state sum (9.22) does not depend on how the vertices of the triangulation are labeled, nor on their ordering, it is sufficient to establish invariance in only one case. Label the vertices of the 4-simplex on the left-hand side of the $1 \leftrightarrow 5$ Pachner move by (23456). By adding the vertex (1) on the right-hand side of the Pachner move, we obtain five new 4-simplices:

$$M_4 := \{(13456), (12456), (12356), (12346), (12345)\}.$$

On the right-hand side there are additional tetrahedra

$$M_3 := \{(1234), (1235), (1236), (1245), (1246), (1256), (1345), (1346), (1356), (1456)\},$$

additional triangles

$$(jkl) \in M_2 := \{(123), (124), (125), (126), (134), (135), (136), (145), (146), (156)\},$$

additional edges $(jk) \in M_1 := \{(12), (13), (14), (15), (16)\}$, and an additional vertex $(j) \in M_0 := \{(1)\}$. All other simplices are present on both sides of the move.

Invariance of the state sum (9.22) under the $1 \leftrightarrow 5$ Pachner move means that the integral on the right-hand side,

$$Z_{\text{desno}}^{1 \leftrightarrow 5} = |G|^{-11} |H|^{-4} |L|^{-1} \int_{G^5} \prod_{(jk) \in M_1} dg_{jk} \int_{H^{10}} \prod_{(jkl) \in M_2} dh_{jkl} \int_{L^{10}} \prod_{(jklm) \in M_3} dl_{jklm} \cdot \left(\prod_{(jkl) \in M_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3} \delta_H(h_{jklm}) \right) \left(\prod_{(jklmn) \in M_4} \delta_L(l_{jklmn}) \right) Z_{\text{ostatak}}, \quad (9.23)$$

is equal to the δ -function present on the left-hand side,

$$Z_{\text{levo}}^{1 \leftrightarrow 5} = |G|^{-5} |H|^0 |L|^{-1} \delta_L(l_{23456}) Z_{\text{ostatak}}. \quad (9.24)$$

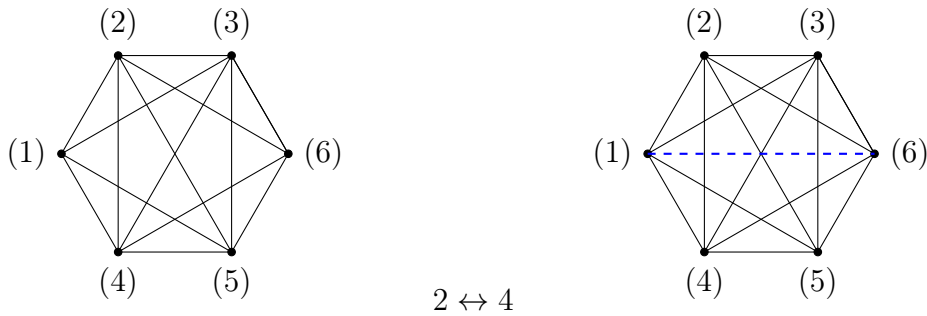
The factors in front of the state-sum integrals on the left- and right-hand sides of the move are calculated from equation (9.22); namely, we use $|G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|}$, $|H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|}$, and $|L|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|+|\Lambda_3|-|\Lambda_4|}$, where $|\Lambda_0|$, $|\Lambda_1|$, $|\Lambda_2|$, $|\Lambda_3|$, and $|\Lambda_4|$ are respectively the numbers of vertices, edges, triangles, tetrahedra, and 4-simplices in the triangulation. From the data shown in Table 9.1, the factor on the right-hand side is $|G|^{-11} |H|^{-4} |L|^{-1}$, while the factor on the left-hand side is $|G|^{-5} |H|^0 |L|^{-1}$.

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $	$ \Lambda_4 $
l.s.	5	10	10	5	1
d.s.	6	15	20	15	5

Table 9.1: Numbers of vertices $|\Lambda_0|$, edges $|\Lambda_1|$, triangles $|\Lambda_2|$, tetrahedra $|\Lambda_3|$, and 4-simplices $|\Lambda_4|$ on the left- and right-hand sides of the $1 \leftrightarrow 5$ Pachner move.

The proof of invariance of the state sum (9.22) under the $1 \leftrightarrow 5$ Pachner move reduces to proving that expressions (9.23) and (9.24) are equal, where the factor Z_{ostatak} denotes the part of the sum that remains unchanged by definition of the move. The proof that $Z_{\text{desno}}^{1 \leftrightarrow 5} = Z_{\text{levo}}^{1 \leftrightarrow 5}$ is given in Appendix E.

The Pachner move $2 \leftrightarrow 4$



To verify the invariance of the state sum (9.22) under the $2 \leftrightarrow 4$ Pachner move, order the vertices so that on the left-hand side of the move there are two 4-simplices

$$M_4^{\text{levo}} = \{(23456), (12345)\}$$

and on the right-hand side four 4-simplices

$$M_4^{\text{desno}} = \{(12346), (12356), (12456), (13456)\}.$$

Then, on the left-hand side there is one tetrahedron

$$M_3^{\text{levo}} = \{(2345)\},$$

while on the right-hand side there are six tetrahedra

$$M_3^{\text{desno}} = \{(1236), (1246), (1256), (1346), (1356), (1456)\}.$$

All other tetrahedra are present on both sides of the move. In addition, the triangles $M_2^{\text{desno}} = \{(126), (136), (146), (156)\}$ and one edge $M_1^{\text{desno}} = \{(16)\}$ are present on the right-hand side, while all remaining triangles and edges are present on both sides of the move. All vertices are likewise present on both sides of the move.

Thus, using the definition of the state sum (9.22), on the left-hand side of the move we have the integral

$$Z_{\text{levo}}^{2 \leftrightarrow 4} = |G|^{-8} |H|^{-1} |L|^{-1} \int_L dl_{2345} \delta_H(h_{2345}) \left(\prod_{(jklmn) \in M_4^{\text{levo}}} \delta_L(l_{jklmn}) \right) Z_{\text{ostatak}}, \quad (9.25)$$

while on the right-hand side the integral is

$$Z_{\text{desno}}^{2 \leftrightarrow 4} = |G|^{-11} |H|^{-3} |L|^{-1} \int_G dg_{16} \int_{H^4} dh_{126} dh_{136} dh_{146} dh_{156} \int_L dl_{1236} dl_{1246} dl_{1256} dl_{1346} dl_{1356} dl_{1456} \\ \left(\prod_{(jkl) \in M_2^{\text{desno}}} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in M_3^{\text{desno}}} \delta_H(h_{jklm}) \right) \left(\prod_{(jklmn) \in M_4^{\text{desno}}} \delta_L(l_{jklmn}) \right) Z_{\text{ostatak}}. \quad (9.26)$$

By counting the k -simplices on both sides of the $2 \leftrightarrow 4$ move (see Table 9.2), we obtain the coefficients in front of the integrals: $|G|^{-8} |H|^{-1} |L|^{-1}$ on the left-hand side of the move and $|G|^{-11} |H|^{-3} |L|^{-1}$ on the right-hand side.

	$ \Lambda_0 $	$ \Lambda_1 $	$ \Lambda_2 $	$ \Lambda_3 $	$ \Lambda_4 $
l.s.	6	14	16	9	2
d.s.	6	15	20	14	4

Table 9.2: Numbers of vertices $|\Lambda_0|$, edges $|\Lambda_1|$, triangles $|\Lambda_2|$, tetrahedra $|\Lambda_3|$, and 4-simplices $|\Lambda_4|$ on both sides of the $2 \leftrightarrow 4$ move.

The proof of invariance of the state sum (9.22) under the $2 \leftrightarrow 4$ Pachner move reduces to proving that expressions (9.25) and (9.26) are equal, where the factor Z_{ostatak} denotes the part of the sum that remains unchanged by definition of the move. The proof that $Z_{\text{desno}}^{2 \leftrightarrow 4} = Z_{\text{levo}}^{2 \leftrightarrow 4}$ is given in Appendix E.

The Pachner move $3 \leftrightarrow 3$

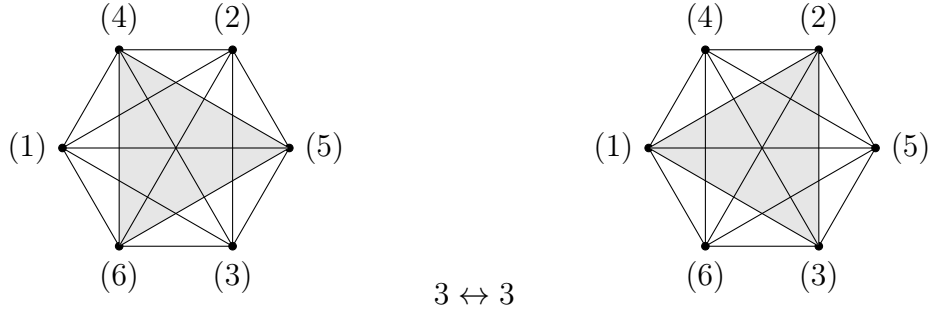
Label the vertices so that on the left-hand side of the $3 \leftrightarrow 3$ Pachner move there are three 4-simplices

$$M_4^{\text{levo}} = \{(23456), (13456), (12456)\},$$

while on the right-hand side there are the 4-simplices

$$M_4^{\text{desno}} = \{(12356), (12346), (12345)\}.$$

On the left-hand side there are the tetrahedra $M_3^{\text{levo}} = \{(1456), (2456), (3456)\}$, while on the right-hand side there are $M_3^{\text{desno}} = \{(1234), (1235), (1236)\}$. The two sides of the move share six



tetrahedra, while each side contains three tetrahedra shared by two 4-simplices. Furthermore, on the left-hand side we have the triangle $M_2^{\text{levo}} = \{(456)\}$, and on the right-hand side of the move the triangle $M_2^{\text{desno}} = \{(123)\}$. All other triangles, edges, and vertices occur on both sides of the move.

Thus, on the left-hand side of the move we have the integral

$$Z_{\text{levo}}^{3 \leftrightarrow 3} = \int_H dh_{456} \int_{L^3} dl_{1456} dl_{2456} dl_{3456} \delta_G(g_{456}) \delta_H(h_{3456}) \delta_H(h_{2456}) \delta_H(h_{1456}) \delta_L(l_{23456}) \delta_L(l_{13456}) \delta_L(l_{12456}) Z_{\text{ostatak}}, \quad (9.27)$$

while on the right-hand side we have the integral

$$Z_{\text{desno}}^{3 \leftrightarrow 3} = \int_H dh_{123} \int_{L^3} dl_{1234} dl_{1235} dl_{1236} \delta_G(g_{123}) \delta_H(h_{1234}) \delta_H(h_{1235}) \delta_H(h_{1236}) \delta_L(l_{12356}) \delta_L(l_{12346}) \delta_L(l_{12345}) Z_{\text{ostatak}}. \quad (9.28)$$

The proof of invariance of the state sum (9.22) under the $3 \leftrightarrow 3$ Pachner move reduces to proving that expressions (9.27) and (9.28) are equal, where the factor Z_{ostatak} denotes the part of the sum that remains unchanged by definition of the move.

We find that the state sum defined in (9.22) remains unchanged under all five Pachner moves, and therefore conclude that the state sum is independent of the triangulation of the 4-dimensional manifold $\mathcal{M}_4$ (see Appendix E for the proof).

Chapter 10

Conclusion

Summary

In this dissertation, we have studied the foundations of models of quantum gravity and matter within $2BF$ and $3BF$ theory. First, Chapter 2 presented a brief overview of the higher-categorical generalization of gauge theories—*higher gauge theories*—namely, 2-gauge theories whose symmetry is given by a 2-group, i.e. a crossed module $(H \xrightarrow{\partial} G, \triangleright)$, and 3-gauge theories whose symmetry is given by a 3-group, i.e. a 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$. The mathematical objects required to formulate $3BF$ theory were also introduced: a 3-connection (α, β, γ) , where the differential forms are elements of the algebras $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g})$, $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h})$, and $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l})$, and a fake 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, where $\mathcal{F} \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{g})$, $\mathcal{G} \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{h})$, and $\mathcal{H} \in \mathcal{A}^4(\mathcal{M}_4, \mathfrak{l})$. Chapter 3 then gave a brief overview of the *Hamiltonian analysis of constrained systems*, including definitions of the canonical and total Hamiltonians, primary and secondary constraints, first- and second-class constraints, the method for calculating the number of degrees of freedom in a theory, and the *Castellani procedure* for calculating the generators of gauge symmetries.

Afterwards, Chapter 4 considered BF theory and carried out a complete Hamiltonian analysis of the topological BF action. As expected, the BF action was found to describe a theory without local propagating degrees of freedom. The Castellani procedure was used to calculate the generator of gauge symmetries in BF theory, as well as the variations of all variables in the theory and their conjugate momenta. These results yielded two types of gauge-symmetry transformations in BF theory— G -gauge and M -gauge transformations, already known in the literature—together with the commutation relations of the total gauge-symmetry group of the BF action, $\mathcal{G}_{BF} = G \ltimes \tilde{M}$. Here, G is the subgroup of the total symmetry group formed by G -gauge transformations, while $\tilde{M}$ is the invariant subgroup of the total symmetry group formed by M -gauge transformations. The field variations corresponding to diffeomorphisms were represented, for a specific choice of parameters, as the sum of the field variations under gauge transformations and under HT transformations, thereby demonstrating that BF theory is diffeomorphism invariant. Two physically relevant models with local propagating degrees of freedom were then considered, obtained by adding appropriate *constraint* terms to the BF action. The first example was *Yang–Mills theory* for the $SU(N)$ group in Minkowski space, and the second was the *Plebanski model* for General Relativity.

Following the same line of presentation, Chapter 5 considered a higher-categorical generalization of BF theory— $2BF$ theory, also known in the literature as $BFCG$ theory. A complete Hamiltonian analysis of the topological $2BF$ action was carried out. As in the case of BF theory, $2BF$ theory was found to be topological, without local propagating degrees of freedom. After calculating the generator and the field variations of the variables and their conjugate momenta, the finite symmetry transformations of the $2BF$ action were obtained: G -gauge,

H -gauge, M -gauge, and N -gauge transformations. The total gauge-symmetry group was found to be $\mathcal{G}_{2BF} = G \ltimes (\tilde{H} \ltimes (\tilde{N} \times \tilde{M}))$, where the groups G and $\tilde{M}$ are defined as in the case of BF theory, $\tilde{N}$ is the group of N -gauge transformations, and $\tilde{H}$ is the group of H -gauge transformations. As in the case of BF theory, specific choices of gauge parameters and HT parameters that yield diffeomorphism transformations were obtained. It was shown how General Relativity can be represented as a constrained $2BF$ theory for a particular choice of symmetry 2-group. Finally, the last section of Chapter 5 was devoted to a discussion of *Einstein–Yang–Mills theory*, i.e. a theory of gravity and gauge fields formulated as a constrained $2BF$ theory. The advantage of this formulation of General Relativity over the formulation in terms of BF theory is that the 2-group structure introduces tetrad fields into the topological action, thereby opening the possibility of coupling matter to gravity in a straightforward way. Nevertheless, matter fields cannot be naturally expressed within the algebraic structure of a 2-group; that is, the matter sector of the action cannot be written as a sum of a topological term and a constraint. Achieving this requires one further step in the higher-categorical generalization of BF theory—the so-called $3BF$ theory.

Finally, the last chapter in the first part of the dissertation, Chapter 6, was devoted to classical $3BF$ theory. Following the Hamiltonian analysis of the theory and a procedure analogous to that used for BF and $2BF$ theory, $3BF$ theory was found to be invariant under five types of gauge transformations: G -gauge, H -gauge, L -gauge, M -gauge, and N -gauge transformations. The analysis of the symmetry transformations, i.e. the calculation of the commutators of their generators, revealed one important difference from $2BF$ theory: in $3BF$ theory, the H -gauge transformations do not form a group. The total gauge-symmetry group was found to be $\mathcal{G}_{3BF} = G \ltimes (\tilde{H}_L \ltimes (\tilde{N} \times \tilde{M}))$, where $\tilde{H}_L$ is the group formed by the H -gauge and L -gauge transformations, while the remaining groups are defined as in the case of $2BF$ theory. Constrained $3BF$ theories describing models with nontrivial dynamics, obtained by modifying the topological $3BF$ action through the addition of appropriate constraints, were then discussed. Theories describing the *Klein–Gordon* and *Dirac fields* in curved spacetime, formulated as constrained $3BF$ actions, were considered. For completeness, the *Weyl* and *Majorana fields* interacting with Einstein–Cartan gravity were also analyzed. These results were then applied to construct a constrained $3BF$ action describing all matter present in the Standard Model coupled to the gravitational field. At the end of this chapter, a simple model describing scalar electrodynamics as a constrained $3BF$ theory was presented.

The second part of the dissertation was devoted to quantum theory. Chapter 7 considered the construction of the topological BF state sum for three- and four-dimensional manifolds using the standard spin-foam quantization procedure. In the three-dimensional case, the resulting state sum yields the quantum theory of $3D$ gravity—the *Ponzano–Regge model*—as a consequence of the fact that the corresponding classical theory has no local propagating degrees of freedom. The construction of the topological BF state sum in the physically relevant case of a four-dimensional spacetime manifold—the so-called *Ooguri model*—was then presented. However, because gravity in $4D$ is not a topological theory, the resulting state sum does not represent a physical theory; a quantum theory of gravity can instead be obtained by modifying the amplitudes of the topological state sum. The final, third step of the spin-foam quantization procedure, i.e. imposing constraints on the variables present in the topological sector of the action by modifying the amplitudes of the topological state sum, lies beyond the scope of this dissertation.

In Chapter 8, the second step of the covariant spin-foam quantization procedure was carried out for $2BF$ theory. It was demonstrated how to construct a triangulation-independent state sum Z from the classical $2BF$ action for a general strict 2-group and any triangulation of any smooth d -dimensional spacetime manifold, for the cases $d \in \{3, 4\}$. For $d = 3$, the constructed state sum is precisely Yetter’s model, while for $d = 4$ it coincides with Porter’s TQFT for $d = 4$

and $n = 2$. The behavior of the constructed state sum under Pachner moves was analyzed. These are local changes of triangulation that preserve topology, such that any two triangulations of the same manifold are related by a finite number of Pachner moves. In the three-dimensional case there are four Pachner moves—the $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ moves and their inverses—while in 4 dimensions there are five distinct Pachner moves—the $3 \leftrightarrow 3$, $4 \leftrightarrow 2$, and $5 \leftrightarrow 1$ moves and their inverses. The setup for analyzing the behavior of the constructed state sum under these Pachner moves was presented in Section 8.3, while the details of the calculation were given in Appendix E.1. The state sum was found to remain unchanged under these transformations of the triangulation, proving that it is a *topological invariant* of the manifold. Because it is independent of the triangulation, the state sum is invariant under arbitrary refinement of the triangulation and therefore defines a continuum theory on a smooth manifold.

Finally, in the last chapter, Chapter 9, we focused on the second step of the covariant spin-foam quantization procedure for $3BF$ theory. By analogy with the procedure for the $2BF$ state sum in the preceding chapter, it was demonstrated how to construct a triangulation-independent state sum Z from the classical $3BF$ action for a general semistrict 3-group and any triangulation of any 4-dimensional spacetime manifold. Since $3BF$ theory can be formulated only when the dimension of the spacetime manifold is $d \geq 4$, only the case $d = 4$ was considered. The constructed state sum generalizes the work of Girelli, Pfeiffer, and Popescu on the $2BF$ state sum presented in the preceding chapter, i.e. it generalizes Yetter’s model, and it coincides with Porter’s TQFT for $d = 4$ and $n = 3$. As in the case of the $2BF$ state sum, to verify that the constructed state sum is topological, its behavior under Pachner moves for a four-dimensional manifold was analyzed, i.e. its invariance under the $3 \leftrightarrow 3$, $4 \leftrightarrow 2$, and $5 \leftrightarrow 1$ moves and their inverses. The analysis of the behavior of the constructed state sum under Pachner moves was presented in Section 9.3, with details in Appendix E.2. Once again, the state sum was found to be invariant under Pachner moves, i.e. to be a *topological invariant* of the manifold.

The appendices present the calculational details accompanying the main text.

Discussion and future research directions

The categorical generalization of the BF action to the $3BF$ action provided significant insight into the content of the matter sector. It was shown that the matter fields in the theory are determined by the choice of the gauge group L , an element of the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{Pf}})$. The group L is an entirely new structure, absent from the Standard Model, and arises wholly from the theory’s higher-categorical structure. Gauge groups corresponding to the Klein–Gordon, Dirac, Weyl, and Majorana fields were thus formulated, and the corresponding $3BF$ actions describing the dynamics of these fields coupled to gravity in the standard way were constructed. Then, by simply choosing the group L as the direct product of the groups corresponding to the individual relevant fields in the Standard Model, i.e. as $L = \mathbb{R}^4(\mathbb{C}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G}) \times \mathbb{R}^{64}(\mathbb{G})$, a group describing all matter in the Standard Model was obtained, and the theory of gravity and matter fields was formulated as a constrained $3BF$ theory. This choice of 3-group is trivial in the sense that the group L is chosen as a direct product, while the Peiffer lifting and the maps ∂ and δ are chosen to be trivial. Different choices of 3-group can be used to construct different models of gravity and matter, much as in *Grand Unified Theories*¹, where various vector-field models are constructed through different choices of the Yang–Mills gauge group. This raises the question of whether there is a better choice of 3-group corresponding to a theory of gravity with matter that could answer some of the open questions of the Standard Model. The possibility of a nontrivial unification of the fields in the theory is an interesting direction for future research and holds the greatest potential of this approach.

¹English: *Grand Unified Theory (GUT)*.

The first part of the text carried out a Hamiltonian analysis of the topological BF , $2BF$, and $3BF$ actions. However, as topological actions, they do not describe realistic physical theories containing local propagating degrees of freedom. Physical degrees of freedom are introduced by imposing constraints on the topological action. The constrained $2BF$ and $3BF$ actions formulated for nontopological theories—specifically, constrained $2BF$ actions describing the Yang–Mills field and Einstein–Cartan gravity, and constrained $3BF$ actions describing the Klein–Gordon, Dirac, Weyl, and Majorana fields coupled to gravity in the standard way—were written as the sum of a topological term and a constraint term. A natural next step in this research direction would be a Hamiltonian analysis of all such $2BF$ and $3BF$ models of gravity coupled to various vector and matter fields, together with a study of their symmetries.

The symmetry group $\mathcal{G}_{3BF}$ describing the gauge symmetries of the topological $3BF$ action was obtained. However, physical theories are described by modified $3BF$ actions obtained by adding constraints, whose symmetry groups are subgroups of the total symmetry group of the topological action. A symmetry group explicitly broken in this way can subsequently be broken spontaneously through the Higgs mechanism, which represents another interesting direction for future research.

One important result is the relation between the 2-crossed module and the symmetry group of the $3BF$ action, the so-called *duality*. The calculated commutation relations of the Lie algebra of the symmetry group $\mathcal{G}_{3BF}$ show that the structure constants depend on the choice of the groups G , H , and L of the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_{\text{pf}})$, the action $\triangleright$, and the symmetric part of the Peiffer lifting $\{_, _\}_{\text{pf}}$. However, the group $\mathcal{G}_{3BF}$ does not depend on the antisymmetric part of the Peiffer lifting, nor on the homomorphisms ∂ and δ . This result implies that several different 2-crossed modules may be *dual* to the same symmetry group $\mathcal{G}_{3BF}$; in other words, there is no one-to-one correspondence between a 2-crossed module and the symmetry group of the corresponding $3BF$ action. This result may have a practical application in constructing $3BF$ models: one could first choose a group $\mathcal{G}_{3BF}$ corresponding to the desired symmetry of the model, which automatically fixes the choice of the groups G , H , and L , the action $\triangleright$, and the symmetric part of the Peiffer lifting, and then define the remaining elements of the 2-crossed module so that all axioms in the definition of a 2-crossed module are satisfied.

In the second part of the thesis, a topological state sum Z was constructed for a general semistrict 3-group and a $4D$ spacetime manifold $\mathcal{M}_4$, and was proved to be a topological invariant of that manifold. The constructed state sum gives a combinatorial construction of a topological quantum field theory (TQFT) in the sense of Atiyah’s axioms, which can be verified explicitly. A proof that the state sum satisfies Atiyah’s axioms lies beyond the scope of this thesis and is left for future work.

Nevertheless, successfully completing the second step of the covariant spin-foam quantization procedure requires generalizations of the Peter–Weyl and Plancherel theorems to the cases of 2-groups and 3-groups, mathematical results that remain open problems. Namely, these theorems should provide a decomposition of functions on a 3-group into a sum over the corresponding irreducible representations of the 3-group. In this way, one determines the spectrum of labels of the simplices in the triangulation, i.e. the domain of values of the fields living on those simplices, as was done for the BF state sum. Current attempts to complete the second step in the quantization of generalized BF theories within higher gauge theories amount to conjecturing the irreducible representations of 2-groups, as has been done, for example, in the spin-cube model of quantum gravity.

This result opens the way to the third and final step of the covariant spin-foam quantization procedure. Since the goal is to describe a realistic physical theory, i.e. a theory containing local propagating degrees of freedom, it is necessary to construct a nontopological state sum describing a theory with nontrivial dynamics. The final, third step of the spin-foam quantization

procedure consists in imposing constraints that deform the topological theory into a physical theory. The continuation of this research aims to formulate a state sum describing a quantum theory of gravity coupled to matter. The construction of this model is left for future work.

This list is not exhaustive; in addition to the topics mentioned above, there are many other directions of research within higher-categorical generalizations of BF theories, both in physics and in mathematics.

Appendix A

Construction of gauge-invariant actions

A.1 Construction of the $2BF$ action

We denote the symmetric bilinear invariant forms on the algebras $\mathfrak{h}$ and $\mathfrak{g}$ by

$$\langle t_a, t_b \rangle_{\mathfrak{h}} = g_{ab}, \quad \langle \tau_\alpha, \tau_\beta \rangle_{\mathfrak{g}} = g_{\alpha\beta}. \quad (\text{A.1})$$

The bilinear forms have the following properties:

- $\langle _, _ \rangle_{\mathfrak{g}}$ is G -invariant:

$$\langle g\tau_\alpha g^{-1}, g\tau_\beta g^{-1} \rangle_{\mathfrak{g}} = \langle \tau_\alpha, \tau_\beta \rangle_{\mathfrak{g}}, \quad \forall g \in G; \quad (\text{A.2})$$

- $\langle _, _ \rangle_{\mathfrak{h}}$ is G -invariant,

$$\langle g \triangleright t_a, g \triangleright t_b \rangle_{\mathfrak{h}} = \langle t_a, t_b \rangle_{\mathfrak{h}}, \quad \forall g \in G, \quad (\text{A.3})$$

and also H -invariant:

$$\langle ht_a h^{-1}, ht_b h^{-1} \rangle_{\mathfrak{h}} = \langle \partial(h) \triangleright t_a, \partial(h) \triangleright t_b \rangle_{\mathfrak{h}} = \langle t_a, t_b \rangle_{\mathfrak{h}}, \quad \forall h \in H. \quad (\text{A.4})$$

A.1.1 2-gauge transformations of the 2-curvature

The groups G and H of the crossed module $(H \xrightarrow{\partial} G, \triangleright)$ generate two types of 2-gauge transformations of the 2-connection, defined by expressions (2.47) and (2.48).

Theorem 24 *The composition of G -gauge and H -gauge transformations gives the following transformation of the 2-connection:*

$$\begin{aligned} \alpha'' &= g^{-1}\alpha g + g^{-1}dg + \partial(\eta), \\ \beta'' &= g^{-1} \triangleright \beta + d\eta + \alpha'' \wedge^{\triangleright} \eta - \eta \wedge \eta, \end{aligned} \quad (\text{A.5})$$

where $g : \mathcal{M}_4 \rightarrow G$ and $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ are the parameters of the G - and H -gauge transformations, respectively.

Proof. Under G -gauge transformations, the 2-connection (α, β) transforms according to (2.47):

$$\begin{aligned} \alpha' &= g^{-1}\alpha g + g^{-1}dg, \\ \beta' &= g^{-1} \triangleright \beta. \end{aligned}$$

Subsequently transforming the 2-connection by an H -gauge transformation according to (2.48) gives

$$\begin{aligned}\alpha'' &= \alpha' + \partial(\eta) = g^{-1}\alpha g + g^{-1}dg + \partial(\eta), \\ \beta'' &= \beta' + d\eta + \alpha'' \wedge^\triangleright \eta - \eta \wedge \eta = g^{-1} \triangleright \beta + d\eta + (g^{-1}\alpha g + g^{-1}dg + \partial(\eta)) \wedge^\triangleright \eta - \eta \wedge \eta.\end{aligned}$$

This proves Theorem 24. ■

From definition (2.36) and the transformation rules (2.47) and (2.48) for the 2-connection, one obtains the transformations of the 2-curvature $(\mathcal{F}, \mathcal{G})$ under 2-gauge transformations.

Theorem 25 *Under G -gauge transformations, the 2-curvature $(\mathcal{F}, \mathcal{H})$ transforms as*

$$\mathcal{F} \rightarrow g^{-1}\mathcal{F}g, \quad \mathcal{G} \rightarrow g^{-1} \triangleright \mathcal{G}, \quad (\text{A.6})$$

where $g : \mathcal{M}_4 \rightarrow G$ is the parameter of the G -gauge transformations.

Proof. Applying the definition of the 2-curvature, we find that under G -gauge transformations the curvature $\mathcal{F}$ transforms as

$$\begin{aligned}\mathcal{F}' &= d\alpha' + \alpha' \wedge \alpha' - \partial\beta' \\ &= d(g^{-1}\alpha g + g^{-1}dg) + (g^{-1}\alpha g + g^{-1}dg) \wedge (g^{-1}\alpha g + g^{-1}dg) - \partial(g^{-1} \triangleright \beta) \\ &= dg^{-1} \wedge \alpha g + g^{-1}d\alpha g - g^{-1}\alpha \wedge dg + dg^{-1} \wedge dg \\ &\quad + g^{-1}\alpha g \wedge g^{-1}\alpha g + g^{-1}\alpha g \wedge g^{-1}dg + g^{-1}dg \wedge g^{-1}\alpha g + g^{-1}dg \wedge g^{-1}dg - g^{-1} \triangleright \partial(\beta) \\ &= dg^{-1} \wedge \alpha g + g^{-1}d\alpha g + dg^{-1} \wedge dg \\ &\quad + g^{-1}\alpha \wedge \alpha g - dg^{-1} \wedge \alpha g - dg^{-1} \wedge dg - g^{-1} \triangleright \partial(\beta) \\ &= g^{-1}d\alpha g + g^{-1}\alpha \wedge \alpha g - g^{-1} \triangleright \partial(\beta) \\ &= g^{-1}\mathcal{F}g,\end{aligned} \quad (\text{A.7})$$

where we used $g^{-1}dg = -dg^{-1}g$. Under G -gauge transformations, the curvature $\mathcal{G}$ transforms as

$$\begin{aligned}\mathcal{G}' &= d\beta' + \alpha' \wedge^\triangleright \beta' \\ &= d(g^{-1} \triangleright \beta) + (g^{-1}\alpha g + g^{-1}dg) \wedge^\triangleright (g^{-1} \triangleright \beta) \\ &= dg^{-1} \wedge^\triangleright \beta + g^{-1} \triangleright d\beta + (g^{-1}\alpha g) \wedge^\triangleright (g^{-1} \triangleright \beta) - dg^{-1} \wedge^\triangleright \beta \\ &= g^{-1} \triangleright d\beta + g^{-1} \triangleright (\alpha \wedge^\triangleright \beta) \\ &= g^{-1} \triangleright \mathcal{G}.\end{aligned} \quad (\text{A.8})$$

This proves the statement of Theorem 25. ■

Theorem 26 *Under H -gauge transformations, the 2-curvature transforms according to*

$$\mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{G} \rightarrow \mathcal{G} + \mathcal{F} \wedge^\triangleright \eta, \quad (\text{A.9})$$

where $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ is the parameter of the H -gauge transformations.

Proof. Applying the definition of the 3-curvature, we find that the curvature $\mathcal{F}$ is invariant under H -gauge transformations:

$$\begin{aligned}
\mathcal{F}'' &= d\alpha'' + \alpha'' \wedge \alpha'' - \partial\beta'' \\
&= d(\alpha + \partial\eta) + (\alpha + \partial\eta) \wedge (\alpha + \partial\eta) - \partial(\beta + d\eta + (\alpha + \partial\eta) \wedge^\triangleright \eta - \eta \wedge \eta) \\
&= d\alpha + \alpha \wedge \alpha + \alpha'' \wedge \partial\eta + \partial\eta \wedge \alpha'' - \partial\beta - \partial(\alpha'' \wedge^\triangleright \eta) \\
&= d\alpha + \alpha \wedge \alpha - \partial\beta \\
&= \mathcal{F},
\end{aligned} \tag{A.10}$$

where we used the identities

$$\begin{aligned}
d(\partial\eta) &= \partial(d\eta), \\
\partial(\eta \wedge \eta) &= \partial\eta \wedge \partial\eta, \\
\alpha'' \wedge \partial\eta + \partial\eta \wedge \alpha'' &= \partial(\alpha'' \wedge^\triangleright \eta).
\end{aligned}$$

Under H -gauge transformations, the curvature $\mathcal{G}$ transforms as

$$\begin{aligned}
\mathcal{G}'' &= d\beta'' + \alpha'' \wedge^\triangleright \beta'' \\
&= d(\beta + d\eta + \alpha'' \wedge^\triangleright \eta - \eta \wedge \eta) + (\alpha + \partial\eta) \wedge^\triangleright (\beta + d\eta + \alpha'' \wedge^\triangleright \eta - \eta \wedge \eta) \\
&= d\beta + d\alpha'' \wedge^\triangleright \eta - \alpha'' \wedge^\triangleright d\eta - d\eta \wedge \eta + \eta \wedge d\eta \\
&\quad + \alpha \wedge^\triangleright (\beta + d\eta + \alpha'' \wedge^\triangleright \eta - \eta \wedge \eta) + \partial\eta \wedge^\triangleright (\beta + d\eta + \alpha'' \wedge^\triangleright \eta - \eta \wedge \eta) \\
&= d\beta + d\alpha \wedge^\triangleright \eta + d(\partial\eta) \wedge^\triangleright \eta - \partial\eta \wedge^\triangleright d\eta - d\eta \wedge \eta + \eta \wedge d\eta \\
&\quad + \alpha \wedge^\triangleright \beta + (\alpha \wedge \alpha) \wedge^\triangleright \eta + \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \alpha \wedge^\triangleright (\eta \wedge \eta) \\
&\quad - \partial\beta \wedge^\triangleright \eta + \partial\eta \wedge^\triangleright d\eta + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \partial\eta \wedge^\triangleright (\eta \wedge \eta) \\
&= d\beta + d\alpha \wedge^\triangleright \eta + \alpha \wedge^\triangleright \beta + (\alpha \wedge \alpha) \wedge^\triangleright \eta - \partial\beta \wedge^\triangleright \eta \\
&= \mathcal{G} + \mathcal{F} \wedge \eta.
\end{aligned} \tag{A.11}$$

Here we used the following identities:

$$\begin{aligned}
d\eta \wedge \eta - \eta \wedge d\eta &= d(\partial\eta) \wedge^\triangleright \eta, \\
\alpha \wedge^\triangleright (\alpha \wedge^\triangleright \eta) &= (\alpha \wedge \alpha) \wedge^\triangleright \eta, \\
\partial\eta \wedge^\triangleright \beta &= -\partial\beta \wedge^\triangleright \eta, \\
\alpha \wedge^\triangleright (\eta \wedge \eta) &= \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta), \\
\partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \partial\eta \wedge^\triangleright (\eta \wedge \eta) &= 0.
\end{aligned}$$

This proves the statement of Theorem 26. ■

Theorem 27 Under 2-gauge transformations, the 2-curvature $(\mathcal{F}, \mathcal{G})$ transforms as

$$\begin{aligned}
\mathcal{F} &\rightarrow g^{-1} \triangleright \mathcal{F}, \\
\mathcal{G} &\rightarrow g^{-1} \triangleright \mathcal{G} + (g^{-1} \triangleright \mathcal{F}) \wedge^\triangleright \eta,
\end{aligned} \tag{A.12}$$

where $g : \mathcal{M}_4 \rightarrow G$ and $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ are the parameters of the G - and H -gauge transformations, respectively.

Proof. Successively transforming the 2-curvature first by the G -gauge transformation defined in Theorem 25, and then by the H -gauge transformation defined in Theorem 26, gives the result of Theorem 27. ■

A.1.2 2-gauge transformations of the Lagrange multipliers

Using the G -invariant symmetric nondegenerate bilinear forms on $\mathfrak{g}$ and $\mathfrak{h}$, one can define a bilinear antisymmetric map $\mathcal{T} : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathfrak{g}$ as follows [17]:

$$\langle \mathcal{T}(\underline{h}_1, \underline{h}_2), \underline{g} \rangle_{\mathfrak{g}} = -\langle \underline{h}_1, \underline{g} \triangleright \underline{h}_2 \rangle_{\mathfrak{h}}, \quad \forall \underline{h}_1, \underline{h}_2 \in \mathfrak{h}, \quad \forall \underline{g} \in \mathfrak{g}. \quad (\text{A.13})$$

In the preceding expression, underlined symbols denote elements of the algebras and non-underlined symbols denote elements of the groups; this notation will also be used below.

For every $g \in G$ and all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$, one has

$$\langle \underline{h}_1, g \triangleright \underline{h}_2 \rangle_{\mathfrak{h}} = -\langle g \triangleright \underline{h}_1, \underline{h}_2 \rangle_{\mathfrak{h}} = -\langle \underline{h}_2, g \triangleright \underline{h}_1 \rangle_{\mathfrak{h}}.$$

It can be shown that for every $g \in G$ and all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$,

$$\mathcal{T}(g \triangleright \underline{h}_1, g \triangleright \underline{h}_2) = g \mathcal{T}(\underline{h}_1, \underline{h}_2) g^{-1}. \quad (\text{A.14})$$

To show this, take $\underline{g}_0 \in \mathfrak{g}$. Using the G -invariance of $\langle _, _ \rangle_{\mathfrak{g}}$ and $\langle _, _ \rangle_{\mathfrak{h}}$, we obtain

$$\begin{aligned} \langle g^{-1} \mathcal{T}(g \triangleright \underline{h}_1, g \triangleright \underline{h}_2) g, \underline{g}_0 \rangle_{\mathfrak{g}} &= \langle \mathcal{T}(g \triangleright \underline{h}_1, g \triangleright \underline{h}_2), g \underline{g}_0 g^{-1} \rangle_{\mathfrak{g}} \\ &= -\langle g \triangleright \underline{h}_1, (g \underline{g}_0 g^{-1}) \triangleright g \triangleright \underline{h}_2 \rangle_{\mathfrak{h}} \\ &= -\langle \underline{h}_1, \underline{g}_0 \triangleright \underline{h}_2 \rangle_{\mathfrak{h}} \\ &= \langle \underline{g}_0, \mathcal{T}(\underline{h}_1, \underline{h}_2) \rangle_{\mathfrak{g}} \\ &= \langle \mathcal{T}(\underline{h}_1, \underline{h}_2), \underline{g}_0 \rangle_{\mathfrak{g}}. \end{aligned}$$

The following identity is satisfied:

$$\mathcal{T}(\underline{g}_0 \triangleright \underline{h}_1, \underline{h}_2) + \mathcal{T}(\underline{h}_1, \underline{g}_0 \triangleright \underline{h}_2) = [\underline{g}_0, \mathcal{T}(\underline{h}_1, \underline{h}_2)].$$

After fixing a basis, we can define the coefficient of the bilinear antisymmetric map $\mathcal{T} : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathfrak{g}$ by

$$\mathcal{T}(t_a, t_b) = \mathcal{T}_{ab}{}^{\alpha} \tau_{\alpha}, \quad (\text{A.15})$$

so the definition of the map $\mathcal{T}$ can be expressed in terms of this coefficient:

$$\mathcal{T}_{ab}{}^{\alpha} g_{\alpha\beta} = -\triangleright_{\alpha[b}{}^c g_{a]c}. \quad (\text{A.16})$$

The bilinear antisymmetric map $\mathcal{T}$ applied to two differential forms with values in the algebra $\mathfrak{h}$, η and ω , defines a differential form with values in the algebra $\mathfrak{g}$:

$$\omega \wedge^{\mathcal{T}} \eta = \omega^a \wedge \eta^b \mathcal{T}_{ab}{}^{\alpha} \tau_{\alpha}.$$

The map $\mathcal{T}$ plays a key role in constructing a $2BF$ action invariant under 2-gauge transformations, i.e. in defining the transformation law of the Lagrange multipliers under 2-gauge transformations.

For the action (5.1) to be gauge invariant under the curvature transformations (2.47) and (2.48), the Lagrange multipliers $B \in \mathcal{A}^2(\mathcal{M}, \mathfrak{g})$ and $C \in \mathcal{A}^1(\mathcal{M}, \mathfrak{h})$ must transform under G -gauge transformations as

$$B \rightarrow B' = g^{-1} B g, \quad C \rightarrow C' = g^{-1} \triangleright C, \quad (\text{A.17})$$

and under H -gauge transformations as

$$B \rightarrow B'' = B + C'' \wedge^{\mathcal{T}} \eta, \quad C \rightarrow C'' = C, \quad (\text{A.18})$$

where the map $\mathcal{T}$ is defined by equation (A.13).

A.2 Construction of the 3BF action

We denote the symmetric bilinear invariant forms on the algebras $\mathfrak{l}$, $\mathfrak{h}$, and $\mathfrak{g}$ by

$$\langle T_A, T_B \rangle_{\mathfrak{l}} = g_{AB}, \quad \langle t_a, t_b \rangle_{\mathfrak{h}} = g_{ab}, \quad \langle \tau_\alpha, \tau_\beta \rangle_{\mathfrak{g}} = g_{\alpha\beta}. \quad (\text{A.19})$$

The bilinear forms have the following properties:

- $\langle _, _ \rangle_{\mathfrak{g}}$ is G -invariant:

$$\langle g\tau_\alpha g^{-1}, g\tau_\beta g^{-1} \rangle_{\mathfrak{g}} = \langle \tau_\alpha, \tau_\beta \rangle_{\mathfrak{g}}, \quad \forall g \in G; \quad (\text{A.20})$$

- $\langle _, _ \rangle_{\mathfrak{h}}$ is G -invariant:

$$\langle g \triangleright t_a, g \triangleright t_b \rangle_{\mathfrak{h}} = \langle t_a, t_b \rangle_{\mathfrak{h}}, \quad \forall g \in G, \quad (\text{A.21})$$

and when $(H \xrightarrow{\partial} G, \triangleright)$ is a crossed module, it is also H -invariant:

$$\langle ht_a h^{-1}, ht_b h^{-1} \rangle_{\mathfrak{h}} = \langle \partial(h) \triangleright t_a, \partial(h) \triangleright t_b \rangle_{\mathfrak{h}} = \langle t_a, t_b \rangle_{\mathfrak{h}}, \quad \forall h \in H; \quad (\text{A.22})$$

- $\langle _, _ \rangle_{\mathfrak{l}}$ is G -invariant:

$$\langle g \triangleright T_A, g \triangleright T_B \rangle_{\mathfrak{l}} = \langle T_A, T_B \rangle_{\mathfrak{l}}, \quad \forall g \in G, \quad (\text{A.23})$$

and, in the special case in which the Peiffer lifting or the map δ is trivial, it is also H -invariant:

$$\langle h \triangleright' T_A, h \triangleright' T_B \rangle_{\mathfrak{l}} = \langle T_A - \{\delta(T_A), h\}, T_B - \{\delta(T_B), h\} \rangle_{\mathfrak{l}} = \langle T_A, T_B \rangle_{\mathfrak{l}}, \quad \forall h \in H. \quad (\text{A.24})$$

The H -invariance of $\langle _, _ \rangle_{\mathfrak{l}}$ and the properties of the crossed module $(L \xrightarrow{\delta} H, \triangleright')$ imply the L -invariance of the bilinear form:

$$\langle lT_A l^{-1}, lT_B l^{-1} \rangle_{\mathfrak{l}} = \langle \delta(l) \triangleright' T_A, \delta(l) \triangleright' T_B \rangle_{\mathfrak{l}} = \langle T_A, T_B \rangle_{\mathfrak{l}}, \quad \forall l \in L. \quad (\text{A.25})$$

A.2.1 3-gauge transformations of the 3-curvature

The structure of a 3-group, i.e. a 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _ \}_{\text{pf}})$, generates three types of gauge transformations: the G -, H -, and L -gauge transformations of the 3-connection defined by expressions (2.128), (2.129), and (2.130).

Theorem 28 *The composition of G -, H -, and L -gauge transformations gives the following transformation of the 3-connection:*

$$\begin{aligned} \tilde{\alpha} &= g^{-1} \alpha g + g^{-1} dg + \partial(\eta), \\ \tilde{\beta} &= g^{-1} \triangleright \beta + d\eta + \tilde{\alpha} \wedge^{\triangleright} \eta - \eta \wedge \eta - \delta(\theta), \\ \tilde{\gamma} &= g^{-1} \triangleright \gamma - d\theta - \tilde{\alpha} \wedge \theta - \tilde{\beta} \wedge^{\{\cdot, \cdot\}} \eta - \eta \wedge^{\{\cdot, \cdot\}} (g^{-1} \triangleright \beta) + \eta \wedge^{\triangleright'} \theta, \end{aligned} \quad (\text{A.26})$$

where $g : \mathcal{M}_4 \rightarrow G$, $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, and $\theta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$ are the parameters of the G -, H -, and L -gauge transformations, respectively.

Proof. Under G -gauge transformations, the 3-connection transforms according to (2.128):

$$\begin{aligned}\alpha' &= g^{-1}\alpha g + g^{-1}dg, \\ \beta' &= g^{-1} \triangleright \beta, \\ \gamma' &= g^{-1} \triangleright \gamma.\end{aligned}$$

Under H -gauge transformations, it transforms according to (2.129):

$$\begin{aligned}\alpha'' &= \alpha' + \partial(\eta) = g^{-1}\alpha g + g^{-1}dg + \partial(\eta), \\ \beta'' &= \beta' + d\eta + \alpha'' \wedge^\triangleright \eta - \eta \wedge \eta = g^{-1} \triangleright \beta + d\eta + (g^{-1}\alpha g + g^{-1}dg + \partial(\eta)) \wedge^\triangleright \eta - \eta \wedge \eta, \\ \gamma'' &= \gamma' - \beta'' \wedge^{\{\cdot\}} \eta - \eta \wedge^{\{\cdot\}} \beta' \\ &= g^{-1} \triangleright \gamma - (g^{-1} \triangleright \beta + d\eta + (g^{-1}\alpha g + g^{-1}dg + \partial(\eta)) \wedge^\triangleright \eta - \eta \wedge \eta) \wedge^{\{\cdot\}} \eta - \eta \wedge^{\{\cdot\}} (g^{-1} \triangleright \beta).\end{aligned}$$

Under L -gauge transformations, the connection transforms according to (2.130):

$$\begin{aligned}\tilde{\alpha} &= \alpha'' = g^{-1}\alpha g + g^{-1}dg + \partial(\eta), \\ \tilde{\beta} &= \beta'' - \delta(\theta) = g^{-1} \triangleright \beta + d\eta + (g^{-1}\alpha g + g^{-1}dg + \partial(\eta)) \wedge^\triangleright \eta - \eta \wedge \eta - \delta(\theta) \\ &= g^{-1} \triangleright \beta + d\eta + \tilde{\alpha} \wedge^\triangleright \eta - \eta \wedge \eta - \delta(\theta), \\ \tilde{\gamma} &= \gamma'' - d\theta - \tilde{\alpha} \wedge^\triangleright \theta \\ &= g^{-1} \triangleright \gamma - (g^{-1} \triangleright \beta + d\eta + (g^{-1}\alpha g + g^{-1}dg + \partial(\eta)) \wedge^\triangleright \eta - \eta \wedge \eta) \wedge^{\{\cdot\}} \eta \\ &\quad - \eta \wedge^{\{\cdot\}} (g^{-1} \triangleright \beta) - d\theta - (g^{-1}\alpha g + g^{-1}dg + \partial(\eta)) \wedge^\triangleright \theta \\ &= g^{-1} \triangleright \gamma - \tilde{\beta} \wedge^{\{\cdot\}} \eta - \eta \wedge^{\{\cdot\}} (g^{-1} \triangleright \beta) - d\theta - \tilde{\alpha} \wedge^\triangleright \theta - \eta \wedge^{\triangleright'} \theta,\end{aligned}$$

where we used the identity $\delta(\theta) \wedge^{\{\cdot\}} \eta = -\eta \wedge^{\triangleright'} \theta$. ■

From definition (2.118) and the transformation rules for the 3-connection, one obtains the transformation of the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ under 3-gauge transformations.

Theorem 29 *Under G -gauge transformations, the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ transforms as*

$$\mathcal{F} \rightarrow \mathcal{F}' = g^{-1}\mathcal{F}g, \quad \mathcal{G} \rightarrow \mathcal{G}' = g^{-1} \triangleright \mathcal{G}, \quad \mathcal{H} \rightarrow \mathcal{H}' = g^{-1} \triangleright \mathcal{H}, \quad (\text{A.27})$$

where $g : \mathcal{M}_4 \rightarrow G$ is the parameter of the G -gauge transformations.

Proof.

Applying the definition of the 3-curvature and the transformation rule for the 3-connection under G -gauge transformations, we find that under G -gauge transformations the curvature $\mathcal{F}$ transforms in the same way as in $2BF$ theory.

Under G -gauge transformations, the curvature $\mathcal{G}$ transforms as

$$\begin{aligned}\mathcal{G}' &= d\beta' + \alpha' \wedge^\triangleright \beta' - \delta(\gamma') \\ &= d(g^{-1} \triangleright \beta) + (g^{-1}\alpha g + g^{-1}dg) \wedge^\triangleright (g^{-1} \triangleright \beta) - \delta(g^{-1} \triangleright \gamma) \\ &= g^{-1} \triangleright G - g^{-1} \triangleright \delta(\gamma) \\ &= g^{-1} \triangleright \mathcal{G},\end{aligned} \quad (\text{A.28})$$

where the third line was obtained in the same way as in $2BF$ theory and by using the properties

of the map δ . Under G -gauge transformations, the curvature $\mathcal{H}$ transforms as

$$\begin{aligned}
\mathcal{H}' &= d\gamma' + \alpha' \wedge^\triangleright \gamma' + \beta' \wedge^{\{\cdot\}} \beta' \\
&= d(g^{-1} \triangleright \gamma) + (g^{-1} \alpha g + g^{-1} d g) \wedge^\triangleright (g^{-1} \triangleright \gamma) + g^{-1} \triangleright \beta \wedge^{\{\cdot\}} g^{-1} \triangleright \beta \\
&= d g^{-1} \wedge^\triangleright \gamma + g^{-1} \triangleright d\gamma + (g^{-1} \alpha g) \wedge^\triangleright (g^{-1} \triangleright \gamma) - d g^{-1} \wedge^\triangleright \gamma + g^{-1} \triangleright \beta \wedge^{\{\cdot\}} g^{-1} \triangleright \beta \\
&= g^{-1} \triangleright \gamma + g^{-1} \triangleright (\alpha \wedge \gamma) + g^{-1} \triangleright (\beta \wedge^{\{\cdot\}} \beta) \\
&= g \triangleright \mathcal{H},
\end{aligned} \tag{A.29}$$

where in the last line we used the G -equivariance of the Peiffer lifting. This proves the statement of Theorem 29. ■

Theorem 30 *Under H -gauge transformations, the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ transforms as*

$$\mathcal{F} \rightarrow \mathcal{F}' = \mathcal{F}, \quad \mathcal{G} \rightarrow \mathcal{G}' = \mathcal{G} + \mathcal{F} \wedge^\triangleright \eta, \quad \mathcal{H} \rightarrow \mathcal{H}' = \mathcal{H} - \mathcal{G}' \wedge^{\{\cdot\}} \eta + \eta \wedge^{\{\cdot\}} \mathcal{G}, \tag{A.30}$$

where $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$ is the parameter of the H -gauge transformations.

Proof. Applying the definition of the 3-curvature and the transformation rule for the 3-connection under H -gauge transformations, we find that the curvature $\mathcal{F}$ transforms in the same way as in 2BF theory. Under H -gauge transformations, the curvature $\mathcal{G}$ transforms as

$$\begin{aligned}
\mathcal{G}' &= d\beta'' + \alpha'' \wedge^\triangleright \beta'' - \delta(\gamma'') \\
&= d\beta + d\alpha \wedge^\triangleright \eta + d(\partial\eta) \wedge^\triangleright \eta - \partial\eta \wedge^\triangleright d\eta - d\eta \wedge \eta + \eta \wedge d\eta \\
&\quad + \alpha \wedge^\triangleright \beta + (\alpha \wedge \alpha) \wedge^\triangleright \eta + \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \alpha \wedge^\triangleright (\eta \wedge \eta) \\
&\quad + \partial\eta \wedge^\triangleright \beta + \partial\eta \wedge^\triangleright d\eta + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \partial\eta \wedge^\triangleright (\eta \wedge \eta) \\
&\quad - \delta(\gamma - (\beta + d\eta + (\alpha + \partial\eta) \wedge^\triangleright \eta - \eta \wedge \eta) \wedge^{\{\cdot\}} \eta - \eta \wedge^{\{\cdot\}} \beta) \\
&= \underline{d\beta} + \underline{d\alpha \wedge^\triangleright \eta} + d(\partial\eta) \wedge^\triangleright \eta - \partial\eta \wedge^\triangleright d\eta - d\eta \wedge \eta + \eta \wedge d\eta \\
&\quad + \underline{\alpha \wedge^\triangleright \beta} + \underline{(\alpha \wedge \alpha) \wedge^\triangleright \eta} + \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \alpha \wedge^\triangleright (\eta \wedge \eta) \\
&\quad + \partial\eta \wedge^\triangleright \beta + \partial\eta \wedge^\triangleright d\eta + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \partial\eta \wedge^\triangleright (\eta \wedge \eta) \\
&\quad - \underline{\delta(\gamma)} + \beta \wedge^{[\cdot]} \eta - \underline{\partial\beta \wedge^\triangleright \eta} + d\eta \wedge^{[\cdot]} \eta - \partial(d\eta) \wedge^\triangleright \eta + (\alpha \wedge^\triangleright \eta) \wedge^{[\cdot]} \eta - \partial(\alpha \wedge^\triangleright \eta) \wedge^\triangleright \eta \\
&\quad + (\partial\eta \wedge^\triangleright \eta) \wedge^{[\cdot]} \eta - \partial(\partial\eta \wedge^\triangleright \eta) \wedge^\triangleright \eta - (\eta \wedge \eta) \wedge^{[\cdot]} \eta + \partial(\eta \wedge \eta) \wedge^\triangleright \eta \\
&\quad + \eta \wedge^{[\cdot]} \beta - \partial\eta \wedge^\triangleright \beta,
\end{aligned} \tag{A.31}$$

where, recognizing that the underlined terms give $\mathcal{G} + \mathcal{F} \wedge^\triangleright \eta$ and canceling certain terms, we obtain

$$\begin{aligned}
\mathcal{G}' &= \mathcal{G} + \mathcal{F} \wedge^\triangleright \eta \\
&\quad + \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \alpha \wedge^\triangleright (\eta \wedge \eta) \\
&\quad + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) - \partial\eta \wedge^\triangleright (\eta \wedge \eta) \\
&\quad + (\alpha \wedge^\triangleright \eta) \wedge^{[\cdot]} \eta - \partial(\alpha \wedge^\triangleright \eta) \wedge^\triangleright \eta \\
&\quad + (\partial\eta \wedge^\triangleright \eta) \wedge^{[\cdot]} \eta - \partial(\partial\eta \wedge^\triangleright \eta) \wedge^\triangleright \eta - (\eta \wedge \eta) \wedge^{[\cdot]} \eta + \partial(\eta \wedge \eta) \wedge^\triangleright \eta.
\end{aligned} \tag{A.32}$$

We used $d\partial\eta = \partial d\eta$, $d\eta \wedge^{[\cdot]} \eta = d\eta \wedge \eta - \eta \wedge d\eta$, and, by definition, $\beta \wedge^{[\cdot]} \eta = -\eta \wedge^{[\cdot]} \beta$. Then, using the identities

$$\begin{aligned} (\alpha \wedge^\triangleright \eta) \wedge^{[\cdot]} \eta &= \alpha \wedge^\triangleright (\eta \wedge \eta), \\ (\partial\eta \wedge^\triangleright \eta) \wedge^{[\cdot]} \eta &= \partial\eta \wedge^\triangleright (\eta \wedge \eta), \\ (\eta \wedge \eta) \wedge^{[\cdot]} \eta &= 0, \end{aligned}$$

we obtain

$$\begin{aligned} \mathcal{G}'' &= \mathcal{G} + \mathcal{F} \wedge^\triangleright \eta + \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) \\ &\quad - \partial(\alpha \wedge^\triangleright \eta) \wedge^\triangleright \eta - \partial(\partial\eta \wedge^\triangleright \eta) \wedge^\triangleright \eta + \partial(\eta \wedge \eta) \wedge^\triangleright \eta. \end{aligned} \quad (\text{A.33})$$

Finally, using

$$\begin{aligned} \partial(\alpha \wedge^\triangleright \eta) \wedge^\triangleright \eta &= \alpha \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta) + \partial\eta \wedge^\triangleright (\alpha \wedge^\triangleright \eta), \\ \partial(\partial\eta \wedge^\triangleright \eta) \wedge^\triangleright \eta &= \partial(\eta \wedge \eta) \wedge^\triangleright \eta + \partial\eta \wedge^\triangleright (\partial\eta \wedge^\triangleright \eta), \end{aligned}$$

we find

$$\mathcal{G}'' = \mathcal{G} + \mathcal{F} \wedge^\triangleright \eta. \quad (\text{A.34})$$

Finally, applying the transformation rules for the 3-connection, we find that the transformation of the curvature $\mathcal{H}$ under H -gauge transformations is

$$\begin{aligned} \mathcal{H}'' &= d\gamma'' + \alpha'' \wedge^\triangleright \gamma'' + \beta'' \wedge^{\{\cdot\}} \beta'' \\ &= d\gamma + \alpha'' \wedge^\triangleright \gamma - d(\beta'' \wedge^{\{\cdot\}} \eta) - d(\eta \wedge^{\{\cdot\}} \beta) \\ &\quad - \alpha'' \wedge^\triangleright (\beta'' \wedge^{\{\cdot\}} \eta) - \alpha'' \wedge^\triangleright (\eta \wedge^{\{\cdot\}} \beta) + \beta'' \wedge^{\{\cdot\}} \beta'' \\ &= d\gamma + \alpha'' \wedge^\triangleright \gamma - d\beta'' \wedge^{\{\cdot\}} \eta - \beta'' \wedge^{\{\cdot\}} d\eta - d\eta \wedge^{\{\cdot\}} \beta + \eta \wedge^{\{\cdot\}} d\beta \\ &\quad - (\alpha'' \wedge^\triangleright \beta'') \wedge^{\{\cdot\}} \eta - \beta'' \wedge^{\{\cdot\}} (\alpha'' \wedge^\triangleright \eta) - (\alpha'' \wedge^\triangleright \eta) \wedge^{\{\cdot\}} \beta + \eta \wedge^{\{\cdot\}} (\alpha'' \wedge^\triangleright \beta) \\ &\quad + \beta'' \wedge^{\{\cdot\}} \beta'' \\ &= d\gamma + \alpha'' \wedge^\triangleright \gamma - (d\beta'' + \alpha'' \wedge^\triangleright \beta'') \wedge^{\{\cdot\}} \eta - \beta'' \wedge^{\{\cdot\}} (d\eta + \alpha'' \wedge^\triangleright \eta) \\ &\quad - (d\eta + \alpha'' \wedge^\triangleright \eta) \wedge^{\{\cdot\}} \beta + \eta \wedge^{\{\cdot\}} (d\beta + \alpha'' \wedge^\triangleright \beta) \\ &\quad + \beta'' \wedge^{\{\cdot\}} \beta'', \end{aligned} \quad (\text{A.35})$$

where

$$\begin{aligned} \beta'' \wedge^{\{\cdot\}} \beta'' &= \beta \wedge^{\{\cdot\}} \beta + (d\eta + \alpha'' \wedge^\triangleright \eta) \wedge^{\{\cdot\}} \beta - (\eta \wedge \eta) \wedge^{\{\cdot\}} \beta \\ &\quad + \beta'' \wedge^{\{\cdot\}} (d\eta + \alpha'' \wedge^\triangleright \eta) - \beta'' \wedge^{\{\cdot\}} (\eta \wedge \eta). \end{aligned}$$

We obtain

$$\begin{aligned} \mathcal{H}'' &= \mathcal{H} + \partial\eta \wedge^\triangleright \gamma - G'' \wedge^{\{\cdot\}} \eta + \eta \wedge^{\{\cdot\}} G + \eta \wedge^{\{\cdot\}} (\partial\eta \wedge^\triangleright \beta) \\ &\quad - \beta'' \wedge^{\{\cdot\}} (d\eta + \alpha'' \wedge^\triangleright \eta) - (d\eta + \alpha'' \wedge^\triangleright \eta) \wedge^{\{\cdot\}} \beta + (d\eta + \alpha'' \wedge^\triangleright \eta) \wedge^{\{\cdot\}} \beta \\ &\quad - (\eta \wedge \eta) \wedge^{\{\cdot\}} \beta + \beta'' \wedge^{\{\cdot\}} (d\eta + \alpha'' \wedge^\triangleright \eta) - \beta'' \wedge^{\{\cdot\}} (\eta \wedge \eta). \end{aligned} \quad (\text{A.36})$$

Certain terms cancel, and after applying the identities

$$\beta'' \wedge^{\{\cdot\}} (\eta \wedge \eta) = (\beta'' \wedge^{\langle\cdot\rangle} \eta) \wedge^{\{\cdot\}} \eta,$$

$$\begin{aligned} \partial\eta \wedge^{\triangleright} \gamma &= -\eta \wedge^{\{\cdot\}} \delta\gamma + \delta\gamma \wedge^{\{\cdot\}} \eta \\ &= -\eta \wedge^{\{\cdot\}} \delta\gamma + \delta\gamma'' \wedge^{\{\cdot\}} \eta + (\beta'' \wedge^{\langle\cdot\rangle} \eta) \wedge^{\{\cdot\}} \eta + (\eta \wedge^{\langle\cdot\rangle} \beta) \wedge^{\{\cdot\}} \eta, \end{aligned}$$

it follows that

$$\begin{aligned} \mathcal{H}'' &= \mathcal{H} - \mathcal{G}'' \wedge^{\{\cdot\}} \eta + \eta \wedge^{\{\cdot\}} \mathcal{G} \\ &\quad + \eta \wedge^{\{\cdot\}} (\partial\eta \wedge^{\triangleright} \beta) - (\eta \wedge \eta) \wedge^{\{\cdot\}} \beta + (\eta \wedge^{\langle\cdot\rangle} \beta) \wedge^{\{\cdot\}} \eta \\ &= \mathcal{H} - \mathcal{G}'' \wedge^{\{\cdot\}} \eta + \eta \wedge^{\{\cdot\}} \mathcal{G}. \end{aligned} \tag{A.37}$$

In the last line we used the identity

$$\eta \wedge^{\{\cdot\}} (\partial\eta \wedge^{\triangleright} \beta) - (\eta \wedge \eta) \wedge^{\{\cdot\}} \beta + (\eta \wedge^{\langle\cdot\rangle} \beta) \wedge^{\{\cdot\}} \eta = 0.$$

This proves the statement of Theorem 30. ■

Theorem 31 *Under L -gauge transformations, the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ transforms as*

$$\mathcal{F} \rightarrow \tilde{\mathcal{F}} = \mathcal{F}, \quad \mathcal{G} = \tilde{\mathcal{G}} \rightarrow \mathcal{G}, \quad \mathcal{H} \rightarrow \tilde{\mathcal{H}} = \mathcal{H} - \mathcal{F} \wedge^{\triangleright} \theta, \tag{A.38}$$

where $\theta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$ is the parameter of the L -gauge transformations.

Proof. Applying the definition of the 3-curvature, we find that the curvature $\mathcal{F}$ is invariant under L -gauge transformations:

$$\begin{aligned} \tilde{\mathcal{F}} &= d\tilde{\alpha} + \tilde{\alpha} \wedge \tilde{\alpha} - \partial\tilde{\beta} \\ &= d\alpha + \alpha \wedge \alpha - \partial(\beta - \delta(\theta)) \\ &= \mathcal{F}, \end{aligned} \tag{A.39}$$

where we used the identity $\partial\delta = 0$. Under L -gauge transformations, the curvature $\mathcal{G}$ transforms as

$$\begin{aligned} \tilde{\mathcal{G}} &= d\tilde{\beta} + \tilde{\alpha} \wedge^{\triangleright} \tilde{\beta} - \delta(\tilde{\gamma}) \\ &= d(\beta - \delta(\theta)) + \alpha \wedge^{\triangleright} (\beta - \delta(\theta)) - \delta(\gamma - d\theta - \alpha \wedge^{\triangleright} \theta) \\ &= \mathcal{G} - d(\delta(\theta)) - \alpha \wedge^{\triangleright} \delta(\theta) + \delta(d\theta) + \delta(\alpha \wedge^{\triangleright} \theta) \\ &= \mathcal{G}, \end{aligned} \tag{A.40}$$

where we used $d(\delta\theta) = \delta(d\theta)$. Finally, the transformation of the curvature $\mathcal{H}$ under L -gauge

transformations is

$$\begin{aligned}
 \tilde{\mathcal{H}} &= d\tilde{\gamma} + \tilde{\alpha} \wedge^\triangleright \tilde{\gamma} + \tilde{\beta} \wedge^{\{\triangleright\}} \tilde{\beta} \\
 &= d(\gamma - d\theta - \alpha \wedge^\triangleright \theta) + \alpha \wedge^\triangleright (\gamma - d\theta - \alpha \wedge^\triangleright \theta) + (\beta - \delta(\theta)) \wedge^{\{\triangleright\}} (\beta - \delta(\theta)) \\
 &= \mathcal{H} - d\alpha \wedge^\triangleright \theta + \alpha \wedge^\triangleright d\theta - \alpha \wedge^\triangleright d\theta - \alpha \wedge^\triangleright (\alpha \wedge^\triangleright \theta) \\
 &\quad - \beta \wedge^{\{\triangleright\}} \delta(\theta) - \delta(\theta) \wedge^{\{\triangleright\}} \beta + \delta(\theta) \wedge^{\{\triangleright\}} \delta(\theta) \\
 &= \mathcal{H} - d\alpha \wedge^\triangleright \theta - (\alpha \wedge \alpha) \wedge^\triangleright \theta + \partial(\beta) \wedge^\triangleright \theta \\
 &= \mathcal{H} - \mathcal{F} \wedge^\triangleright \theta,
 \end{aligned} \tag{A.41}$$

where we used the identities

$$\begin{aligned}
 \delta(\theta) \wedge^{\{\triangleright\}} \delta(\theta) &= 0, \\
 -\beta \wedge^{\{\triangleright\}} \delta(\theta) - \delta(\theta) \wedge^{\{\triangleright\}} \beta &= \partial(\beta) \wedge^\triangleright \theta.
 \end{aligned}$$

The statement of Theorem 31 is proved. ■

Theorem 32 *Under 3-gauge transformations, the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ transforms as*

$$\begin{aligned}
 \mathcal{F} &\rightarrow g^{-1} \triangleright \mathcal{F}, \\
 \mathcal{G} &\rightarrow g^{-1} \triangleright \mathcal{G} + (g^{-1} \triangleright \mathcal{F}) \wedge^\triangleright \eta, \\
 \mathcal{H} &\rightarrow g^{-1} \triangleright \mathcal{H} - (g^{-1} \triangleright \mathcal{G} + (g^{-1} \triangleright \mathcal{F}) \wedge^\triangleright \eta) \wedge^{\{\triangleright\}} \eta - \eta^{\{\triangleright\}} (g^{-1} \triangleright \mathcal{G}) - (g^{-1} \triangleright \mathcal{F}) \wedge^\triangleright \theta,
 \end{aligned} \tag{A.42}$$

where $g : \mathcal{M}_4 \rightarrow G$, $\eta \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{h})$, and $\theta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{l})$ are the parameters of the G -, H -, and L -gauge transformations, respectively.

Proof. Successively transforming the 2-curvature first by the G -gauge transformation defined in Theorem 29, then by the H -gauge transformation defined in Theorem 30, and finally by the L -gauge transformation defined in Theorem 31, gives the result of Theorem 32. ■

A.2.2 3-gauge transformations of the Lagrange multipliers

The bilinear antisymmetric map $\mathcal{T} : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathfrak{g}$ is defined in the same way as for the $2BF$ action in (A.13).

To define a gauge-invariant topological $3BF$ action, it is necessary to define the bilinear antisymmetric map $\mathcal{S} : \mathfrak{l} \times \mathfrak{l} \rightarrow \mathfrak{g}$ as follows:

$$\langle \mathcal{S}(l_1, l_2), \underline{g} \rangle_{\mathfrak{g}} = -\langle l_1, \underline{g} \triangleright l_2 \rangle_{\mathfrak{l}}, \quad \forall l_1, \forall l_2 \in \mathfrak{l}, \quad \forall \underline{g} \in \mathfrak{g}. \tag{A.43}$$

By the symmetry and nondegeneracy of the bilinear form $\langle _, _ \rangle_{\mathfrak{g}}$, we have

$$\langle l_1, \underline{g} \triangleright l_2 \rangle_{\mathfrak{l}} = -\langle \underline{g} \triangleright l_1, l_2 \rangle_{\mathfrak{l}} = -\langle l_2, \underline{g} \triangleright l_1 \rangle_{\mathfrak{l}}, \quad \forall \underline{g} \in \mathfrak{g}, \quad \forall l_1, l_2 \in \mathfrak{l}.$$

Furthermore, for every $g \in G$ and $l_1, l_2 \in \mathfrak{l}$, the following identity holds:

$$\mathcal{S}(g \triangleright l_1, g \triangleright l_2) = g \mathcal{S}(l_1, l_2) g^{-1}, \tag{A.44}$$

This is readily shown:

$$\begin{aligned}
 \langle \underline{g}, g^{-1} \mathcal{S}(g \triangleright l_1, g \triangleright l_2) g \rangle_{\mathfrak{g}} &= \langle g \underline{g} g^{-1}, \mathcal{S}(g \triangleright l_1, g \triangleright l_2) \rangle_{\mathfrak{g}} \\
 &= -\langle (g \underline{g} g^{-1}) \triangleright g \triangleright l_1, g \triangleright l_2 \rangle_{\mathfrak{l}} \\
 &= -\langle \underline{g} \triangleright l_1, l_2 \rangle_{\mathfrak{l}} = \langle \underline{g}, \mathcal{S}(l_1, l_2) \rangle_{\mathfrak{g}},
 \end{aligned}$$

Here we used the identity

$$g \triangleright (\underline{g} \triangleright \underline{l}) = (g \underline{g} g^{-1}) \triangleright g \triangleright \underline{l}.$$

The following identity is satisfied:

$$\mathcal{S}(g \triangleright \underline{l}_1, \underline{l}_2) + \mathcal{S}(\underline{l}_1, g \triangleright \underline{l}_2) = [g, \mathcal{S}(\underline{l}_1, \underline{l}_2)].$$

After fixing a basis, we can define the coefficient of the bilinear antisymmetric map $\mathcal{S} : \mathfrak{l} \times \mathfrak{l} \rightarrow \mathfrak{g}$ by

$$\mathcal{S}(T_A, T_B) = \mathcal{S}_{AB}{}^\alpha \tau_\alpha, \quad (\text{A.45})$$

so the definition of the map $\mathcal{S}$ can be expressed in terms of this coefficient as

$$\mathcal{S}_{AB}{}^\alpha g_{\alpha\beta} = -\triangleright_{\alpha[B}{}^C g_{A]C}. \quad (\text{A.46})$$

The bilinear antisymmetric map $\mathcal{S}$ applied to two differential forms with values in the algebra $\mathfrak{l}$, η and ω , defines a differential form with values in the algebra $\mathfrak{g}$:

$$\omega \wedge^{\mathcal{S}} \eta = \omega^A \wedge \eta^B \mathcal{S}_{AB}{}^\alpha \tau_\alpha.$$

We can now define the transformation laws of the Lagrange multipliers under the L -gauge transformations (A.58).

We define the transformations of the Lagrange multipliers under H -gauge transformations using the bilinear map $\mathcal{X}_1 : \mathfrak{l} \times \mathfrak{h} \rightarrow \mathfrak{h}$ defined by the identity

$$\langle \mathcal{X}_1(\underline{l}, \underline{h}_1), \underline{h}_2 \rangle_{\mathfrak{h}} = -\langle \underline{l}, \{\underline{h}_1, \underline{h}_2\} \rangle_{\mathfrak{l}}, \quad \forall \underline{h}_1, \underline{h}_2 \in \mathfrak{h}, \quad \forall \underline{l} \in \mathfrak{l}, \quad (\text{A.47})$$

and the bilinear map $\mathcal{X}_2 : \mathfrak{l} \times \mathfrak{h} \rightarrow \mathfrak{h}$ defined by

$$\langle \mathcal{X}_2(\underline{l}, \underline{h}_2), \underline{h}_1 \rangle_{\mathfrak{h}} = -\langle \underline{l}, \{\underline{h}_1, \underline{h}_2\} \rangle_{\mathfrak{l}}, \quad \forall \underline{h}_1, \underline{h}_2 \in \mathfrak{h}, \quad \forall \underline{l} \in \mathfrak{l}. \quad (\text{A.48})$$

After fixing a basis, we can define the coefficients of the bilinear maps $\mathcal{X}_1$ and $\mathcal{X}_2$:

$$\mathcal{X}_1(T_A, t_a) = \mathcal{X}_{1Aa}{}^b t_b, \quad \mathcal{X}_2(T_A, t_a) = \mathcal{X}_{2Aa}{}^b t_b. \quad (\text{A.49})$$

In a basis, the definitions of the maps $\mathcal{X}_1$ and $\mathcal{X}_2$ become

$$\mathcal{X}_{1Ab}{}^c g_{ac} = -X_{ba}{}^B g_{AB}, \quad \mathcal{X}_{2Ab}{}^c g_{ac} = -X_{ab}{}^B g_{AB}. \quad (\text{A.50})$$

For two differential forms, an algebra- $\mathfrak{l}$ -valued form ω and an algebra- $\mathfrak{h}$ -valued form η , an algebra- $\mathfrak{h}$ -valued differential form is defined by

$$\omega \wedge^{\mathcal{X}_1} \eta = \omega^A \wedge \eta^a \mathcal{X}_{1Aa}{}^b t_b, \quad \omega \wedge^{\mathcal{X}_2} \eta = \omega^A \wedge \eta^a \mathcal{X}_{2Aa}{}^b t_b.$$

For every $g \in G$, $\underline{l} \in \mathfrak{l}$, and $\underline{h} \in \mathfrak{h}$, one has

$$\mathcal{X}_1(g \triangleright \underline{l}, g^{-1} \triangleright \underline{h}) = g \triangleright \mathcal{X}_1(\underline{l}, \underline{h}), \quad \mathcal{X}_2(g \triangleright \underline{l}, g \triangleright \underline{h}) = g^{-1} \triangleright \mathcal{X}_2(\underline{l}, \underline{h}),$$

which follows from the fact that for every $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$ and $\underline{l} \in \mathfrak{l}$,

$$\begin{aligned} \langle \underline{h}_2, g^{-1} \triangleright \mathcal{X}_1(g \triangleright \underline{l}, g \triangleright \underline{h}_1) \rangle_{\mathfrak{h}} &= \langle g \triangleright \underline{h}_2, \mathcal{X}_1(g \triangleright \underline{l}, g \triangleright \underline{h}_1) \rangle_{\mathfrak{h}} \\ &= \langle g \triangleright \underline{l}, \{g \triangleright \underline{h}_1, g \triangleright \underline{h}_2\} \rangle_{\mathfrak{l}} \\ &= \langle g \triangleright \underline{l}, g \triangleright \{\underline{h}_1, \underline{h}_2\} \rangle_{\mathfrak{l}} \\ &= \langle \underline{l}, \{\underline{h}_1, \underline{h}_2\} \rangle_{\mathfrak{l}} \\ &= \langle \underline{h}_2, \mathcal{X}_1(\underline{l}, \underline{h}_1) \rangle_{\mathfrak{h}}, \end{aligned}$$

and similarly for $\mathcal{X}_2$.

Finally, it is necessary to define the trilinear map $\mathcal{D} : \mathfrak{h} \times \mathfrak{h} \times \mathfrak{l} \rightarrow \mathfrak{g}$ by

$$\langle \mathcal{D}(\underline{h}_1, \underline{h}_2, \underline{l}), \underline{g} \rangle_{\mathfrak{g}} = -\langle \underline{l}, \{ \underline{g} \triangleright \underline{h}_1, \underline{h}_2 \} \rangle_{\mathfrak{l}}, \quad \forall \underline{h}_1, \underline{h}_2 \in \mathfrak{h}, \quad \forall \underline{l} \in \mathfrak{l}, \quad \forall \underline{g} \in \mathfrak{g}, \quad (\text{A.51})$$

The coefficients of the trilinear map are defined by

$$\mathcal{D}(t_a, t_b, T_A) = \mathcal{D}_{abA}{}^{\alpha} \tau_{\alpha}, \quad (\text{A.52})$$

and the definition of the map $\mathcal{D}$ then implies that it can also be written in terms of its coefficients:

$$\mathcal{D}_{abA}{}^{\beta} g_{\alpha\beta} = -\triangleright_{\alpha a}{}^c X_{cb}{}^B g_{AB}. \quad (\text{A.53})$$

For two differential forms with values in the algebra $\mathfrak{h}$, ω and η , and a differential form ξ with values in the algebra $\mathfrak{l}$, a differential form with values in the algebra $\mathfrak{g}$ is defined by

$$\omega \wedge^{\mathcal{D}} \eta \wedge^{\mathcal{D}} \xi = \omega^a \wedge \eta^b \wedge \xi^A \mathcal{D}_{abA}{}^{\beta} \tau_{\beta}.$$

The following compatibility relations hold between the maps $\mathcal{X}_1$ and $\mathcal{D}$:

$$\langle \mathcal{D}(\underline{h}_1, \underline{h}_2, \underline{l}), \underline{g} \rangle_{\mathfrak{g}} = \langle \mathcal{X}_1(\underline{l}, \underline{g} \triangleright \underline{h}_1), \underline{h}_2 \rangle_{\mathfrak{h}}, \quad \forall \underline{h}_1, \underline{h}_2 \in \mathfrak{h}, \quad \forall \underline{l} \in \mathfrak{l}, \quad \forall \underline{g} \in \mathfrak{g}. \quad (\text{A.54})$$

It can also be shown that for every $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$, $\underline{l} \in \mathfrak{l}$, and $g \in G$,

$$\mathcal{D}(g \triangleright \underline{h}_1, g \triangleright \underline{h}_2, g \triangleright \underline{l}) = g \mathcal{D}(\underline{h}_1, \underline{h}_2, \underline{l}) g^{-1}, \quad (\text{A.55})$$

which follows because for every $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$, $\underline{l} \in \mathfrak{l}$, $\underline{g} \in \mathfrak{g}$, and $g \in G$, we may write

$$\begin{aligned} \langle g^{-1} \mathcal{D}(g \triangleright \underline{h}_1, g \triangleright \underline{h}_2, g \triangleright \underline{l}) g, \underline{g} \rangle_{\mathfrak{g}} &= \langle \mathcal{D}(g \triangleright \underline{h}_1, g \triangleright \underline{h}_2, g \triangleright \underline{l}), g \underline{g} g^{-1} \rangle_{\mathfrak{g}} \\ &= \langle \mathcal{X}_1(g \triangleright \underline{l}, g \underline{g} g^{-1} \triangleright g \triangleright \underline{h}_1), g \triangleright \underline{h}_2 \rangle_{\mathfrak{h}} \\ &= \langle \mathcal{X}_1(g \triangleright \underline{l}, g \triangleright g \triangleright \underline{h}_1), g \triangleright \underline{h}_2 \rangle_{\mathfrak{h}} \\ &= \langle g \triangleright \mathcal{X}_1(\underline{l}, \underline{g} \triangleright \underline{h}_1), g \triangleright \underline{h}_2 \rangle_{\mathfrak{h}} \\ &= \langle \mathcal{X}_1(\underline{l}, \underline{g} \triangleright \underline{h}_1), \underline{h}_2 \rangle_{\mathfrak{h}} \\ &= \langle \mathcal{D}(\underline{h}_1, \underline{h}_2, \underline{l}), \underline{g} \rangle_{\mathfrak{g}}, \end{aligned}$$

where relations (A.2.2) and the compatibility relations (A.54) were used. It follows that for all $\underline{h}_1, \underline{h}_2 \in \mathfrak{h}$, $\underline{l} \in \mathfrak{l}$, and $\underline{g} \in \mathfrak{g}$, the following identity holds:

$$\mathcal{D}(\underline{g} \triangleright \underline{h}_1, \underline{h}_2, \underline{l}) + \mathcal{D}(\underline{h}_1, \underline{g} \triangleright \underline{h}_2, \underline{l}) + \mathcal{D}(\underline{h}_1, \underline{h}_2, \underline{g} \triangleright \underline{l}) = [\underline{g}, \mathcal{D}(\underline{h}_1, \underline{h}_2, \underline{l})].$$

This defines all maps required to specify the transformation laws of the Lagrange multipliers under H -gauge transformations.

For the action (6.1) to be gauge invariant under the curvature transformations (2.128), (2.129), and (2.130), the Lagrange multipliers B , C , and D must transform under G -gauge transformations as

$$B \rightarrow g^{-1} B g, \quad C \rightarrow g^{-1} \triangleright C, \quad D \rightarrow g^{-1} \triangleright D, \quad (\text{A.56})$$

under H -gauge transformations as

$$B \rightarrow B + C' \wedge^{\mathcal{T}} \eta - \eta \wedge^{\mathcal{D}} \eta \wedge^{\mathcal{D}} D, \quad C \rightarrow C + D \wedge^{\mathcal{X}_1} \eta + D \wedge^{\mathcal{X}_2} \eta, \quad D \rightarrow D, \quad (\text{A.57})$$

and, finally, under L -gauge transformations as

$$B \rightarrow B - D \wedge^{\mathcal{S}} \theta, \quad C \rightarrow C, \quad D \rightarrow D. \quad (\text{A.58})$$

The maps $\mathcal{T}$, $\mathcal{D}$, $\mathcal{X}_1$, $\mathcal{X}_2$, and $\mathcal{S}$ are defined by equations (A.13), (A.43), (A.47), (A.48), and (A.51).

Appendix B

Equations of motion for constrained $3BF$ actions with Weyl and Majorana fields coupled to Einstein–Cartan gravity

The action for a Weyl spinor field coupled to Einstein–Cartan gravity is given by expression (6.151). Varying this action successively with respect to the variables B_{ab} , λ^{ab} , γ_α , $\bar{\gamma}^{\dot{\alpha}}$, λ_α , $\bar{\lambda}^{\dot{\alpha}}$, ψ_α , $\bar{\psi}^{\dot{\alpha}}$, e^a , β^a , and ω^{ab} yields the equations of motion:

$$\begin{aligned}
R^{ab} - \lambda^{ab} &= 0, \\
B_{ab} - \frac{1}{16\pi l_p^2} \varepsilon_{abcd} e^c \wedge e^d &= 0, \\
\nabla \psi_\alpha + \lambda_\alpha &= 0, \\
\nabla \bar{\psi}^{\dot{\alpha}} + \bar{\lambda}^{\dot{\alpha}} &= 0, \\
-\gamma_\alpha + \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \sigma^d_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\beta}} &= 0, \\
-\bar{\gamma}^{\dot{\alpha}} + \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \bar{\sigma}^{d\dot{\alpha}\beta} \psi_\beta &= 0, \\
\nabla \gamma_\alpha - \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \sigma^d_{\alpha\dot{\beta}} \bar{\lambda}^{\dot{\beta}} &= 0, \\
\nabla \bar{\gamma}^{\dot{\alpha}} - \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \bar{\sigma}^{d\dot{\alpha}\beta} \lambda_{L\beta} &= 0, \\
\nabla \beta_a + \frac{1}{8\pi l_p^2} \varepsilon_{abcd} \lambda^{bc} \wedge e^d + \frac{i}{2} \varepsilon_{abcd} e^b \wedge e^c \wedge (\bar{\lambda}_{\dot{\alpha}} \bar{\sigma}^{d\dot{\alpha}\beta} \psi_\beta + \lambda^\alpha \sigma^d_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\beta}}) - 8\pi i l_p^2 \varepsilon_{abcd} e^b \beta^c (\psi^\alpha (\sigma^d)_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\beta}}) &= 0, \\
\nabla e_a - 4\pi l_p^2 \varepsilon_{abcd} e^b \wedge e^c \wedge (\bar{\psi}_{\dot{\alpha}} \bar{\sigma}^{d\dot{\alpha}\beta} \psi_\beta) &= 0, \\
\nabla B_{ab} - e_{[a} \wedge \beta_{b]} - \frac{1}{2} \gamma \sigma^{ab}_{\alpha}{}^{\beta} \psi_\beta - \frac{1}{2} \bar{\gamma} \bar{\sigma}^{ab\dot{\alpha}}{}_{\dot{\beta}} \bar{\psi}^{\dot{\beta}} &= 0.
\end{aligned}$$

For the constrained $3BF$ action corresponding to the theory of a Majorana spinor field coupled to Einstein–Cartan gravity, we add the mass term (6.154) to the action (6.151). Varying the resulting action successively with respect to the variables B_{ab} , ψ^{ab} , γ^α , $\bar{\gamma}_{\dot{\alpha}}$, λ_α , $\bar{\lambda}^{\dot{\alpha}}$, ψ_α , $\bar{\psi}^{\dot{\alpha}}$, e^a ,

β^a , and ω_{ab} yields the equations of motion:

$$\begin{aligned}
 R^{ab} - \lambda^{ab} &= 0, \\
 B_{ab} - \frac{1}{16\pi l_p^2} \varepsilon_{abcd} e^c \wedge e^d &= 0, \\
 -\nabla \psi_\alpha + \lambda_\alpha &= 0, \\
 -\nabla \bar{\psi}^{\dot{\alpha}} + \lambda^{\dot{\alpha}} &= 0, \\
 \gamma^\alpha - \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \bar{\psi}_{\dot{\beta}} (\bar{\sigma}^d)^{\dot{\beta}\alpha} &= 0, \\
 \bar{\gamma}_{\dot{\alpha}} - \frac{i}{6} \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \psi^\beta (\sigma^d)_{\beta\dot{\alpha}} &= 0, \\
 \nabla \gamma^\alpha + \frac{i}{6} \varepsilon_{abcd} \lambda^{\dot{\beta}} \wedge e^a \wedge e^b \wedge e^c (\sigma^d)^{\alpha}_{\dot{\beta}} - \frac{1}{6} m \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \psi^\alpha \\
 - 4i\pi l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c \bar{\psi}_{\dot{\beta}} (\bar{\sigma}^d)^{\dot{\beta}\alpha} &= 0, \\
 \nabla \bar{\gamma}_{\dot{\alpha}} + \frac{i}{6} \varepsilon_{abcd} \lambda_\beta \wedge e^a \wedge e^b \wedge e^c (\bar{\sigma}^d)_{\dot{\alpha}}^\beta - \frac{1}{6} m \varepsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d \psi_{\dot{\alpha}} \\
 - 4i\pi l_p^2 \varepsilon_{abcd} e^a \wedge e^b \wedge \beta^c \psi^\beta (\sigma^d)_{\beta\dot{\alpha}} &= 0, \\
 \nabla \beta^a + \frac{1}{8\pi l_p^2} \varepsilon_{abcd} \lambda^{bc} \wedge e^d + \frac{i}{2} \varepsilon_{abcd} \lambda_\alpha \wedge e^b \wedge e^c \bar{\psi}_{\dot{\beta}} (\bar{\sigma}^d)^{\dot{\beta}\alpha} + \frac{i}{2} \varepsilon_{abcd} \lambda^{\dot{\alpha}} \wedge e^b \wedge e^c \psi^\beta (\sigma^d)_{\beta\dot{\alpha}} \\
 - \frac{1}{3} m \varepsilon_{abcd} e^b \wedge e^c \wedge e^d (\psi^\alpha \psi_\alpha + \bar{\psi}_{\dot{\alpha}} \bar{\psi}^{\dot{\alpha}}) - 8\pi i l_p^2 \varepsilon_{abcd} e^b \wedge \beta^c (\psi^\alpha (\sigma^d)_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\beta}}) &= 0, \\
 \nabla e_a - 4i\pi l_p^2 \varepsilon_{abcd} e^b \wedge e^c (\psi^\alpha (\sigma^d)_{\alpha\dot{\beta}} \bar{\psi}^{\dot{\beta}}) &= 0, \\
 \nabla B_{ab} - e_{[a} \wedge \beta_{b]} - \frac{1}{2} \psi^\alpha (\sigma^{ab})_\alpha{}^\beta \gamma_\beta - \frac{1}{2} \bar{\psi}_{\dot{\alpha}} (\bar{\sigma}^{ab})^{\dot{\alpha}}{}_{\dot{\beta}} \bar{\gamma}^{\dot{\beta}} &= 0.
 \end{aligned}$$

Appendix C

Hamiltonian analysis of scalar electrodynamics

This chapter presents the complete Hamiltonian analysis of the action (6.174).

Assuming that the spacetime manifold is globally hyperbolic, $\mathcal{M}_4 = \mathbb{R} \times \Sigma_3$, the Lagrangian of the action (6.174) takes the form:

$$L_{3BF} = \int_{\Sigma_3} d^3\vec{x} \epsilon^{\mu\nu\rho\sigma} \left(\frac{1}{4} B^{ab}{}_{\mu\nu} R^{cd}{}_{\rho\sigma} g_{ab,cd} + \frac{1}{4} B_{\mu\nu} F_{\rho\sigma} + \frac{1}{3!} e^a{}_\mu \mathcal{G}^b{}_{\nu\rho\sigma} g_{ab} + \frac{1}{4!} \phi^A \mathcal{H}^B{}_{\mu\nu\rho\sigma} g_{AB} \right). \quad (\text{C.1})$$

The *canonical momenta* of the variables $B^{ab}{}_{\mu\nu}$, $\omega^{ab}{}_\mu$, $B_{\mu\nu}$, A_μ , $e^a{}_\mu$, $\beta^a{}_{\mu\nu}$, ϕ^A , and $\gamma^A{}_{\mu\nu\rho}$ are:

$$\begin{aligned} \pi(B)_{ab}{}^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 B^{ab}{}_{\mu\nu}} = 0, & \pi(\omega)_{ab}{}^\mu &= \frac{\delta L}{\delta \partial_0 \omega^{ab}{}_\mu} = \epsilon^{0\mu\nu\rho} B_{ab\nu\rho}, \\ \pi(B)^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 B_{\mu\nu}} = 0, & \pi(A)^\mu &= \frac{\delta L}{\delta \partial_0 A_\mu} = \frac{1}{2} \epsilon^{0\mu\nu\rho} B_{\nu\rho}, \\ \pi(e)_a{}^\mu &= \frac{\delta L}{\delta \partial_0 e^a{}_\mu} = 0, & \pi(\beta)_a{}^{\mu\nu} &= \frac{\delta L}{\delta \partial_0 \beta^a{}_{\mu\nu}} = -\epsilon^{0\mu\nu\rho} e_{a\rho}, \\ \pi(\phi)_A &= \frac{\delta L}{\delta \partial_0 \phi^A} = 0, & \pi(\gamma)_A{}^{\mu\nu\rho} &= \frac{\delta L}{\delta \partial_0 \gamma^A{}_{\mu\nu\rho}} = \epsilon^{0\mu\nu\rho} \phi_A. \end{aligned} \quad (\text{C.2})$$

The *primary constraints* of the theory are:

$$\begin{aligned} P(B)_{ab}{}^{\mu\nu} &\equiv \pi(B)_{ab}{}^{\mu\nu} \approx 0, & P(\omega)_{ab}{}^\mu &\equiv \pi(\omega)_{ab}{}^\mu - \epsilon^{0\mu\nu\rho} B_{ab\nu\rho} \approx 0, \\ P(B)^{\mu\nu} &\equiv \pi(B)^{\mu\nu} \approx 0, & P(A)^\mu &= \pi(A)^\mu - \frac{1}{2} \epsilon^{0\mu\nu\rho} B_{\nu\rho} \approx 0, \\ P(e)_a{}^\mu &\equiv \pi(e)_a{}^\mu \approx 0, & P(\beta)_a{}^{\mu\nu} &\equiv \pi(\beta)_a{}^{\mu\nu} + \epsilon^{0\mu\nu\rho} e_{a\rho} \approx 0, \\ P(\phi)_A &\equiv \pi(\phi)_A \approx 0, & P(\gamma)_A{}^{\mu\nu\rho} &\equiv \pi(\gamma)_A{}^{\mu\nu\rho} - \epsilon^{0\mu\nu\rho} \phi_A \approx 0. \end{aligned} \quad (\text{C.3})$$

We define the *fundamental Poisson brackets* as follows:

$$\begin{aligned}
\{ B^{ab}{}_{\mu\nu}(x), \pi(B)_{cd}{}^{\rho\sigma}(y) \} &= 4\delta^a_{[c}\delta^b_{d]}\delta^\rho_{[\mu}\delta^\sigma_{\nu]}\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ \omega^{ab}{}_\mu(x), \pi(\omega)_{cd}{}^\nu(y) \} &= 2\delta^a_{[c}\delta^b_{d]}\delta^\nu_\mu\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ B_{\mu\nu}(x), \pi(B)^{\rho\sigma}(y) \} &= 2\delta^\rho_{[\mu}\delta^\sigma_{\nu]}\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ A_\mu(x), \pi(A)^\nu(y) \} &= \delta^\nu_\mu\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ e^a{}_\mu(x), \pi(e)_b{}^\nu(y) \} &= \delta^a_b\delta^\nu_\mu\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ \beta^a{}_{\mu\nu}(x), \pi(\beta)_b{}^{\rho\sigma}(y) \} &= 2\delta^a_b\delta^\rho_{[\mu}\delta^\sigma_{\nu]}\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ \phi^A(x), \pi(\phi)_B(y) \} &= \delta^A_B\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ \gamma^A{}_{\mu\nu\rho}(x), \pi(\gamma)_B{}^{\alpha\beta\gamma}(y) \} &= 3!\delta^A_B\delta^\alpha_{[\mu}\delta^\beta_{\nu}\delta^\gamma_{\rho]}\delta^{(3)}(\vec{x}-\vec{y}).
\end{aligned} \tag{C.4}$$

Using the fundamental Poisson brackets, we obtain the *algebra of primary constraints*,

$$\begin{aligned}
\{ P(B)^{abjk}(x), P(\omega)_{cd}{}^i(y) \} &= 4\epsilon^{0ijk}\delta^a_{[c}\delta^b_{d]}\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ P(B)^{jk}(x), P(A)^i(y) \} &= \epsilon^{0ijk}\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ P(e)^{ak}, P(\beta)_b{}^{ij}(y) \} &= -\epsilon^{0ijk}\delta^a_b(x)\delta^{(3)}(\vec{x}-\vec{y}), \\
\{ P(\phi)^A(x), P(\gamma)_B{}^{ijk}(y) \} &= \epsilon^{0ijk}\delta^A_B\delta^{(3)}(\vec{x}-\vec{y}),
\end{aligned} \tag{C.5}$$

while all other Poisson brackets vanish identically. The canonical *on-shell* Hamiltonian takes the form:

$$\begin{aligned}
H_c = \int_{\Sigma_3} d^3\vec{x} \left[\frac{1}{4}\pi(B)_{ab}{}^{\mu\nu}\partial_0 B^{ab}{}_{\mu\nu} + \frac{1}{2}\pi(\omega)_{ab}{}^\mu\partial_0\omega^{ab}{}_\mu + \frac{1}{2}\pi(B)^{\mu\nu}\partial_0 B_{\mu\nu} + \pi(A)^\mu\partial_0 A_\mu \right. \\
\left. + \pi(e)_a{}^\mu\partial_0 e^a{}_\mu + \frac{1}{2}\pi(\beta)_a{}^{\mu\nu}\partial_0\beta^a{}_{\mu\nu} + \pi(\phi)_A\partial_0 D^A + \frac{1}{3!}\pi(\gamma)_A{}^{\mu\nu\rho}\partial_0\gamma^A{}_{\mu\nu\rho} \right] - L.
\end{aligned} \tag{C.6}$$

Rewriting (C.6) so that the velocities multiply first-class constraints, which vanish *on-shell*, we obtain:

$$\begin{aligned}
H_c \approx - \int_{\Sigma_3} d^3\vec{x} \epsilon^{0ijk} \left[\frac{1}{2}B_{ab0i}R^{ab}{}_{jk} + \frac{1}{2}B_{0i}F_{jk} + \frac{1}{6}e_{a0}\mathcal{G}^a{}_{ijk} + \beta^a{}_{0i}\nabla_j e_{ak} \right. \\
\left. + \frac{1}{2}\omega^{ab}{}_0 \left(\nabla_i B_{abjk} - e_{[a|i}\beta_{|b]jk} \right) + \frac{1}{2}A_0 \left(\partial_i B_{jk} + \frac{1}{3}\phi_A \triangleright_B{}^A \gamma^B{}_{ijk} \right) + \frac{1}{2}\gamma^A{}_{0ij}\nabla_k \phi_A \right].
\end{aligned} \tag{C.7}$$

By adding a Lagrange multiplier λ for each primary constraint, we obtain the total *off-shell* Hamiltonian:

$$\begin{aligned}
H_T = H_c + \int_{\Sigma_3} d^3\vec{x} \left[\frac{1}{4}\lambda(B)^{ab}{}_{\mu\nu}P(B)_{ab}{}^{\mu\nu} + \frac{1}{2}\lambda(\omega)^{ab}{}_\mu P(\omega)_{ab}{}^\mu + \frac{1}{2}\lambda(B)_{\mu\nu}P(B)^{\mu\nu} + \lambda(A)_\mu P(A)^\mu \right. \\
\left. + \lambda(e)^a{}_\mu P(e)_a{}^\mu + \frac{1}{2}\lambda(\beta)^a{}_{\mu\nu}P(\beta)_a{}^{\mu\nu} + \lambda(\phi)^A P(\phi)_A + \frac{1}{3!}\lambda(\gamma)^A{}_{\mu\nu\rho}P(\gamma)_A{}^{\mu\nu\rho} \right].
\end{aligned} \tag{C.8}$$

The *consistency condition for the primary constraints* (3.26), applied to the primary constraints $P(B)_{ab}{}^{0i}$, $P(\omega)_{ab}{}^0$, $P(B)^{0i}$, $P(A)^0$, $P(e)_a{}^0$, $P(\beta)_a{}^{0i}$, and $P(\gamma)_A{}^{0ij}$,

$$\begin{aligned} \dot{P}(B)_{ab}{}^{0i} &\approx 0, \quad \dot{P}(\omega)_{ab}{}^0 \approx 0, \quad \dot{P}(B)^{0i} \approx 0, \quad \dot{P}(A)^0 \approx 0, \\ \dot{P}(e)_a{}^0 &\approx 0, \quad \dot{P}(\beta)_a{}^{0i} \approx 0, \quad \dot{P}(\gamma)_A{}^{0ij} \approx 0, \end{aligned} \quad (\text{C.9})$$

gives rise to the *secondary constraints* $\mathcal{S}$ in the theory,

$$\begin{aligned} \mathcal{S}(R)_{ab}{}^i &\equiv \epsilon^{0ijk} R_{abjk} && \approx 0, \\ \mathcal{S}(\nabla B)_{ab} &\equiv \epsilon^{0ijk} (\nabla_i B_{abjk} - e_{[a|i} \beta_{|b]jk}) && \approx 0, \\ \mathcal{S}(F)^i &\equiv \frac{1}{2} \epsilon^{0ijk} F_{jk} && \approx 0, \\ \mathcal{S}(\nabla B) &\equiv \frac{1}{2} \epsilon^{0ijk} (\partial_i B_{jk} + \frac{1}{3} \phi_A \triangleright_B{}^A \gamma^B{}_{ijk}) && \approx 0, \\ \mathcal{S}(\mathcal{G})_a &\equiv \frac{1}{6} \epsilon^{0ijk} \mathcal{G}_{aijk} && \approx 0, \\ \mathcal{S}(\nabla e)_a{}^i &\equiv \epsilon^{0ijk} \nabla_j e_{ak} && \approx 0, \\ \mathcal{S}(\nabla \phi)_A{}^{ij} &\equiv \epsilon^{0ijk} \nabla_k \phi_A && \approx 0, \end{aligned} \quad (\text{C.10})$$

while for the primary constraints $P(B)_{ab}{}^{jk}$, $P(\omega)_{ab}{}^k$, $P(B)^{jk}$, $P(A)^k$, $P(e)_a{}^k$, $P(\beta)_a{}^{jk}$, $P(\phi)_A$, and $P(\gamma)_A{}^{ijk}$, the consistency conditions

$$\begin{aligned} \dot{P}(B)_{ab}{}^{jk} &\approx 0, \quad \dot{P}(\omega)_{ab}{}^k \approx 0, \quad \dot{P}(B)^{jk} \approx 0, \quad \dot{P}(A)^k \approx 0, \\ \dot{P}(e)_a{}^k &\approx 0, \quad \dot{P}(\beta)_a{}^{jk} \approx 0, \quad \dot{P}(\phi)_A \approx 0, \quad \dot{P}(\gamma)_A{}^{ijk} \approx 0, \end{aligned} \quad (\text{C.11})$$

determine the Lagrange multipliers:

$$\begin{aligned} \lambda(\omega)_{ab}{}^i &\approx \nabla^i \omega_{ab0}, \\ \lambda(B)_{ab}{}^{ij} &\approx 2\nabla^{[i} B_{ab}{}^{0]j} + e_{[a|0} \beta_{|b]}{}^{ij} - 2e_{[a|}{}^{[i} \beta_{|b]}{}^{0]j} + 2\omega_{[a|}{}^c B_{|b]}{}^{cij}, \\ \lambda(A)^i &\approx \partial^i A_0, \\ \lambda(B)^{ij} &\approx 2\partial^{[i} B^{0]j} + \gamma_A{}^{0ij} \triangleright_B{}^A \phi^B, \\ \lambda(\beta)_a{}^{ij} &\approx 2\nabla^{[i} \beta_a{}^{0]j} - \omega_{ab}{}^0 \beta^{bij}, \\ \lambda(e)_a{}^i &\approx \nabla^i e_a{}^0 - \omega_a{}^{b0} e_b{}^i, \\ \lambda(\gamma)_A{}^{ijk} &\approx -A^0 \triangleright_A{}^B \gamma_B{}^{ijk} + \nabla^i \gamma_A{}^{0jk} - \nabla^j \gamma_A{}^{0ik} + \nabla^k \gamma_A{}^{0ij}, \\ \lambda(\phi)^A &\approx A^0 \triangleright_A{}^B \phi^B. \end{aligned} \quad (\text{C.12})$$

Notice that the Lagrange multipliers

$$\lambda(B)_{ab}{}^{0i}, \quad \lambda(\omega)_{ab}{}^0, \quad \lambda(B)_{0i}, \quad \lambda(A)_0, \quad \lambda(e)_a{}^0, \quad \lambda(\beta)_a{}^{0i}, \quad \lambda(\gamma)_A{}^{0ij} \quad (\text{C.13})$$

remain undetermined. The consistency conditions for the secondary constraints do not give rise to any new constraints in the theory; that is, it can be shown that:

$$\begin{aligned}
\dot{\mathcal{S}}(R)^{abi} &= \{\mathcal{S}(R)^{abi}, H_T\} = \omega^{[a|}_{c0} \mathcal{S}(R)^{c|b]i}, \\
\dot{\mathcal{S}}(\nabla B)_{ab} &= \{\mathcal{S}(\nabla B)_{ab}, H_T\} = \mathcal{S}(R)_{[a|c}{}^k B^c{}_{|b]0k} + \omega_{[a|}{}^c{}_0 \mathcal{S}(\nabla B)_{|b]c} \\
&\quad - \beta_{[a|0k} \mathcal{S}(\nabla e)_{|b]}{}^k + e_{[a|0} \mathcal{S}(\mathcal{G})_{|b]}, \\
\dot{\mathcal{S}}(F)^i &= \{\mathcal{S}(F)^i, H_T\} = 0, \\
\dot{\mathcal{S}}(\nabla B) &= \{\mathcal{S}(\nabla B), H_T\} = -\triangleright_B{}^A \gamma^B{}_{0ij} \mathcal{S}(\nabla \phi)_A{}^{ij}, \\
\dot{\mathcal{S}}(\mathcal{G})^a &= \{\mathcal{S}(\mathcal{G})^a, H_T\} = \beta_{b0k} \mathcal{S}(R)^{abk} - \omega^{ab}{}_0 \mathcal{S}(\mathcal{G})_b, \\
\dot{\mathcal{S}}(\nabla e)_a{}^i &= \{\mathcal{S}(\nabla e)_a{}^i, H_T\} = e^b{}_0 \mathcal{S}(R)_{ab}{}^i - \omega_a{}^b{}_0 \mathcal{S}(\nabla e)_b{}^i, \\
\dot{\mathcal{S}}(\nabla \phi)_A{}^{ij} &= \{\mathcal{S}(\nabla \phi)_A{}^{ij}, H_T\} = A_0 \triangleright_A{}^B \mathcal{S}(\nabla \phi)_B{}^{ij}.
\end{aligned} \tag{C.14}$$

The *total Hamiltonian* can be written in the following form:

$$\begin{aligned}
H_T = \int_{\Sigma_3} d^3\vec{x} &\left[\frac{1}{2} \lambda(B)_{ab}{}^{0i} \Phi(B)^{ab}{}_i + \frac{1}{2} \lambda(\omega)_{ab}{}^0 \Phi(\omega)^{ab} + \lambda(B)^{0i} \Phi(B)_i + \lambda(A)^0 \Phi(A) \right. \\
&+ \lambda(e)_a{}^0 \Phi(e)^a + \lambda(\beta)_a{}^{0i} \Phi(\beta)^a{}_i + \frac{1}{2} \lambda(\gamma)_A{}^{0ij} \Phi(\gamma)^A{}_{ij} \\
&- \frac{1}{2} B_{ab0i} \Phi(R)^{abi} - \frac{1}{2} \omega_{ab0} \Phi(\nabla B)^{ab} - B_{0i} \Phi(F)^i - A_0 \Phi(\nabla B) \\
&\left. - e_{a0} \Phi(\mathcal{G})^a - \beta_{a0i} \Phi(\nabla e)^{ai} - \frac{1}{2} \gamma_{A0ij} \Phi(\nabla \phi)^{Aij} \right],
\end{aligned} \tag{C.15}$$

where

$$\begin{aligned}
\Phi(B)^{ab}{}_i &= P(B)^{ab}{}_{0i}, \\
\Phi(\omega)^{ab} &= P(\omega)^{ab}{}_0, \\
\Phi(B)_i &= P(B)_{0i}, \\
\Phi(A) &= P(A)_0, \\
\Phi(e)^a &= P(e)^a{}_0, \\
\Phi(\beta)^a{}_i &= P(\beta)^a{}_{0i}, \\
\Phi(\gamma)^A{}_{ij} &= P(\gamma)^A{}_{0ij}, \\
\Phi(R)^{abi} &= \mathcal{S}(R)^{abi} - \nabla_j P(B)^{abij}, \\
\Phi(\nabla B)^{ab} &= \mathcal{S}(\nabla B)^{ab} + \nabla_i P(\omega)^{abi} + B^{[a]{}|}{}_{c|ij} P(B)^{c|b|ij} - 2e^{[a]{}|}{}_i P(e)^{|b|}{}_i - \beta^{[a]{}|}{}_{ij} P(\beta)^{|b|}{}_{ij}, \\
\Phi(F)^i &= \mathcal{S}(F)^i - \partial_j P(B)^{ij}, \\
\Phi(\nabla B) &= \mathcal{S}(\nabla B) + \partial_i P(A)^i + \frac{1}{3!} \gamma^A{}_{ijk} \triangleright_A{}^B P(\gamma)_B{}^{ijk} - \phi_A \triangleright_B{}^A P(\phi)^B, \\
\Phi(\mathcal{G})^a &= \mathcal{S}(\mathcal{G})^a + \nabla_i P(e)^{ai} - \frac{1}{4} \beta_{bij} P(B)^{abij}, \\
\Phi(\nabla e)^{ai} &= \mathcal{S}(\nabla e)^{ai} - \nabla_j P(\beta)^{aij} + \frac{1}{2} e_{bj} P(B)^{abij}, \\
\Phi(\nabla \phi)^{Aij} &= \mathcal{S}(\nabla \phi)^{Aij} + \nabla_k P(\gamma)^{Aijk} - \triangleright_B{}^A \phi^B P(B)^{ij},
\end{aligned} \tag{C.16}$$

are the *first-class constraints*, while the *second-class constraints* are:

$$\begin{aligned}
\chi(B)_{ab}{}^{jk} &= P(B)_{ab}{}^{jk}, & \chi(B)^{jk} &= P(B)^{jk}, & \chi(e)_a{}^i &= P(e)_a{}^i, & \chi(\phi)_A &= P(\phi)_A, \\
\chi(\omega)_{ab}{}^i &= P(\omega)_{ab}{}^i, & \chi(A)^i &= P(A)^i, & \chi(\beta)_a{}^{ij} &= P(\beta)_a{}^{ij}, & \chi(\gamma)_A{}^{ijk} &= P(\gamma)_A{}^{ijk}.
\end{aligned} \tag{C.17}$$

The Poisson algebra of the first-class constraints is:

$$\begin{aligned}
\{ \Phi(\mathcal{G})^a(x), \Phi(\nabla e)_b{}^i(y) \} &= -\Phi(R)^a{}_b{}^i(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{G})^a(x), \Phi(\nabla B)_{bc}(y) \} &= 2\delta^a{}_{[b]} \Phi(\mathcal{G})_{|c|}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla e)^a{}_i(x), \Phi(\nabla B)_{bc}(y) \} &= 2\delta^a{}_{[b]} \Phi(\nabla e)_{|c|i}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(R)^{abi}(x), \Phi(\nabla B)_{cd}(y) \} &= -4\delta^{[a]{}|}{}_{[c} \Phi(R)^{|b|}{}_{d]}{}^i(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ab}(x), \Phi(\nabla B)_{cd}(y) \} &= -4\delta^{[a]{}|}{}_{[c} \Phi(\nabla B)^{|b|}{}_{d]}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)(x), \Phi(\nabla \phi)_A{}^{ij}(y) \} &= -2 \triangleright_B{}^A \Phi(\nabla \phi)_B{}^{ij}(x) \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{C.18}$$

The Poisson algebra of the first- and second-class constraints is:

$$\begin{aligned}
\{ \Phi(R)^{abi}(x), \chi(\omega)_{cd}{}^j(y) \} &= 4 \delta^{[a]_{[c]} \chi(B)^{b]}_{|d]}{}^{ij}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{G})^a(x), \chi(\omega)_{cd}{}^i(y) \} &= 2 \delta^a_{[c]} \chi(e)_{|d]}{}^i(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\mathcal{G})^a(x), \chi(\beta)_c{}^{jk}(y) \} &= -\frac{1}{2} \chi(B)^a{}_c{}^{jk}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla e)^{ai}(x), \chi(\omega)_{cd}{}^j(y) \} &= -2 \delta^a_{[c]} \chi(\beta)_{|d]}{}^{ij}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla e)^{ai}(x), \chi(e)_b{}^j(y) \} &= \frac{1}{2} \chi(B)^a{}_b{}^{ij} \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ab}(x), \chi(\omega)_{cd}{}^i(y) \} &= 4 \delta^{[a]_{[c]} \chi(\omega)_{|d]}{}^{b]}{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)(x), \chi(A)^i(y) \} &= 2 \chi(A)^i \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ab}(x), \chi(\beta)_c{}^{jk}(y) \} &= -2 \delta^{[a]_{[c]} \chi(\beta)^{b]}_{|d]}{}^{jk} \delta^{(3)}(x - y), \\
\{ \Phi(\nabla B)(x), \chi(\gamma)_A{}^{ijk}(y) \} &= \triangleright_A^B \chi(\gamma)_B{}^{ijk}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ab}(x), \chi(B)_{cd}{}^{jk}(y) \} &= 4 \delta^{[a]_{[c]} \chi(B)_{|d]}{}^{b]}{}^{jk} \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)^{ab}(x), \chi(e)_a{}^i(y) \} &= -2 \delta^{[a]_{[c]} \chi(e)^{b]}_{|d]}{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla B)(x), \chi(\phi)_A(y) \} &= -\triangleright_A^B \chi(\phi)_B(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla \phi)^{Aij}(x), \chi(A)^k \} &= -\triangleright_B^A \chi(\gamma)^{Bijk}(x) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{ \Phi(\nabla \phi)^{Aij}(x), \chi(\phi)_B \} &= -\triangleright_B^A \chi(B)^{ij}(x) \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{C.19}$$

The Poisson bracket of the second-class constraints was computed in (C.5).

C.1 Bianchi identities

To calculate the number of degrees of freedom in the theory, one must use the *Bianchi identities* (BI), together with additional *generalized Bianchi identities* (GBI), which are the analogues of the ordinary BI for the additional fields present in the theory.

More specifically, the theory contains BI for the 1-form fields ω^{ab} and e^a , as well as a GBI for the 1-form A . The corresponding curvature 2-forms for the connection ω ,

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}, \quad T^a = de^a + \omega^a{}_b \wedge e^b, \quad F = dA, \tag{C.20}$$

satisfy the identities:

$$\epsilon^{\lambda\mu\nu\rho} \nabla_\mu R^{ab}{}_{\nu\rho} = 0, \tag{C.21}$$

$$\epsilon^{\lambda\mu\nu\rho} (\nabla_\mu T^a{}_{\nu\rho} - R^{ab}{}_{\mu\nu} e_{b\rho}) = 0, \tag{C.22}$$

$$\epsilon^{\lambda\mu\nu\rho} \nabla_\mu F_{\nu\rho} = 0. \tag{C.23}$$

Choosing the free index to be the time coordinate $\lambda = 0$, these identities, as the time-independent parts of the Bianchi identities, become *off-shell* restrictions in the Hamiltonian

analysis. On the other hand, choosing the free index to be a spatial coordinate gives the time-dependent parts of the Bianchi identities, which impose no restrictions and can instead be derived as consequences of Hamilton's equations of motion.

In addition, the theory contains GBI associated with the 2-form fields B^{ab} , B , and β^a . The corresponding curvature 3-forms are:

$$S^{ab} = dB^{ab} + 2\omega^{[a]c} \wedge B^{c|b]}, \quad P = dB, \quad G^a = d\beta^a + \omega^a_b \wedge \beta^b. \quad (\text{C.24})$$

Differentiating these expressions gives the following GBI:

$$\epsilon^{\lambda\mu\nu\rho} \left(\frac{1}{3} \nabla_\lambda S^{ab}_{\mu\nu\rho} - R^{[a]c}_{\lambda\mu} B^{c|b]}_{\nu\rho} \right) = 0, \quad (\text{C.25})$$

$$\epsilon^{\lambda\mu\nu\rho} \partial_\lambda P_{\mu\nu\rho} = 0, \quad (\text{C.26})$$

$$\epsilon^{\lambda\mu\nu\rho} \left(\frac{2}{3} \nabla_\lambda G^a_{\mu\nu\rho} - R^{ab}_{\lambda\mu} \beta_{b\nu\rho} \right) = 0. \quad (\text{C.27})$$

However, in four-dimensional spacetime, each of these identities is a single equation with no free spacetime indices, and therefore necessarily contains time derivatives of the fields. It follows that these identities impose no off-shell restrictions on the canonical variables.

Finally, there is a GBI for the 0-form ϕ . The corresponding curvature 1-form is

$$Q^A = d\phi^A + \triangleright_B^A A \wedge \phi^B, \quad (\text{C.28})$$

and the GBI is therefore:

$$\epsilon^{\lambda\mu\nu\rho} \left(\nabla_\nu Q^A_{\rho} - \frac{1}{2} \triangleright_B^A F_{\nu\rho} \phi^B \right) = 0. \quad (\text{C.29})$$

This GBI comprises 12 equations, obtained from six choices of the antisymmetrized pair of spacetime indices $\lambda\mu$ and two choices of the free group index A . However, these 12 identities are not independent, as can be seen by differentiating equation (C.29), which gives eight equations

$$\triangleright_B^A \epsilon^{\lambda\mu\nu\rho} \nabla_\mu F_{\nu\rho} \phi^B = 0, \quad (\text{C.30})$$

that are automatically satisfied by virtue of GBI (C.23). It follows that the theory contains only four independent identities (C.29). Fixing the index $\lambda = 0$, we obtain the time-independent components of equations (C.29) and (C.30),

$$\epsilon^{0ijk} \left(\nabla_j Q^A_k - \frac{1}{2} \triangleright_B^A F_{jk} \phi^B \right) = 0, \quad (\text{C.31})$$

$$\triangleright_B^A \epsilon^{0ijk} \partial_i F_{jk} \phi^B = 0. \quad (\text{C.32})$$

An analysis of the free indices of these equations shows that, among the six equations (C.31), the two equations (C.32) leave four independent GBI relevant to the Hamiltonian analysis.

C.2 Number of degrees of freedom

In this section, we show that the constraint structure of the theory implies the absence of local degrees of freedom (DOF) in the topological sector of the $3BF$ theory of scalar electrodynamics. In general, the number of local degrees of freedom of a theory is given by equation (3.41).

In our case, we determine the initial number of fields in the theory N by counting the field components shown in Table C.1, which gives $N = 120$. The number of independent components

$\omega^a{}_{\mu}$	A_{μ}	$\beta^a{}_{\mu\nu}$	$\gamma^A{}_{\mu\nu\rho}$	$B^a{}_{\mu\nu}$	$B_{\mu\nu}$	$e^a{}_{\mu}$	ϕ^A
24	4	24	8	36	6	16	2

Table C.1: Number of initial fields in the $3BF$ theory of scalar electrodynamics.

of the second-class constraints, $S = 70$, is shown in Table C.2.

$\chi(B)_{ab}{}^{jk}$	$\chi(B)^{jk}$	$\chi(e)_a{}^i$	$\chi(\phi)_A$	$\chi(\omega)_{ab}{}^i$	$\chi(A)^i$	$\chi(\beta)_a{}^{ij}$	$\chi(\gamma)_A{}^{ijk}$
18	3	12	2	18	3	12	2

Table C.2: Second-class constraints in the $3BF$ theory of scalar electrodynamics.

The first-class constraints are not independent because of the BI and GBI present in the theory. Differentiating the constraint $\Phi(R)^{abi}$ gives:

$$\nabla_i \Phi(R)^{abi} = \varepsilon^{0ijk} \nabla_i R^a{}_{jk} + \frac{1}{2} R^{c[a}{}_{ij} P(B)_c{}^{b]ij}. \quad (\text{C.33})$$

The first term on the right-hand side vanishes off-shell, $\varepsilon^{ijk} \nabla_i R^a{}_{jk} = 0$, as the $\lambda = 0$ component of the Bianchi identity (C.21). The second term also vanishes off-shell, as the product of two constraints,

$$R^{c[a}{}_{ij} P(B)_c{}^{b]ij} \equiv \frac{1}{2} \varepsilon_{0ijk} \mathcal{S}(R)^{c[a}{}_{ik} P(B)_c{}^{b]ij} = 0. \quad (\text{C.34})$$

It follows that we have the off-shell identity

$$\nabla_i \Phi(R)^{abi} = 0, \quad (\text{C.35})$$

that is, there are twelve independent components of $\Phi(R)^{abi}$. Similarly, differentiating the constraint $\Phi(F)^i$ gives

$$\nabla_i \Phi(F)^i = \varepsilon^{0ijk} \nabla_i F_{jk} + \frac{1}{2} F_{ij} P(B)^{ij}. \quad (\text{C.36})$$

The first term on the right-hand side vanishes, $\varepsilon^{ijk} \nabla_i F_{jk} = 0$, as the $\lambda = 0$ component of GBI (C.21). The second term again vanishes as the product of two constraints,

$$F_{ij} P(B)^{ij} \equiv \frac{1}{2} \varepsilon_{0ijk} \mathcal{S}(F)^k P(B)^{ij} = 0. \quad (\text{C.37})$$

It follows that the off-shell identity

$$\nabla_i \Phi(F)^i = 0, \quad (\text{C.38})$$

is satisfied, so there are only two independent components of the constraint $\Phi(F)^i$. Similarly, we can show that:

$$\nabla_i \Phi(\nabla e)_a{}^i - \frac{1}{2} \Phi(R)_{ab}{}^i e^b{}_i + \frac{1}{4} \varepsilon^{0ijk} \mathcal{S}(R)_{abk} P(\beta)^b{}_{ij} = \frac{1}{2} \varepsilon^{0ijk} (\nabla_i T_{ajk} - R_{abij} e^b{}_k). \quad (\text{C.39})$$

The right-hand side of expression (C.39) is the $\lambda = 0$ component of BI (C.22), so equation (C.39) becomes:

$$\nabla_i \Phi(\nabla e)_a^i - \frac{1}{2} \Phi(R)_{ab}^i e^b_i = 0. \quad (\text{C.40})$$

In the preceding relation, we used the fact that the third term on the left-hand side of expression (C.39) vanishes as the product of two constraints. Equation (C.40) implies that four components of the constraints $\Phi(\nabla e)_a^i$ and $\Phi(R)_{ab}^i$ can be expressed as linear combinations of the remaining components. Finally, it can be shown that the following relation holds:

$$\begin{aligned} \nabla_i \Phi(\nabla \phi)_A^{ij} - \frac{1}{2} \epsilon_{0ikl} \triangleright_A \mathcal{S}(F)^l \chi(\gamma)_B^{ijk} + \triangleright^B_A \phi_B \Phi(F)^j \\ + \frac{1}{2} \epsilon_{0ilm} \triangleright^B_A P(B)^{ij} \mathcal{S}(\nabla \phi)_B^{lm} = \epsilon^{0ijk} \left(\nabla_i Q_{Ak} + \frac{1}{2} \triangleright^B_A F_{ik} \phi_B \right), \end{aligned} \quad (\text{C.41})$$

from which it follows that:

$$\nabla_i \Phi(\nabla \phi)_A^{ij} + \triangleright^B_A \phi_B \Phi(F)^j = 0. \quad (\text{C.42})$$

The right-hand side of expression (C.41) is the $\lambda = 0$ component of GBI (C.29), while the terms that are products of two constraints vanish off-shell. Identity (C.42) implies that six components of the first-class constraints $\Phi(\nabla \phi)_A^{ij}$ and $\Phi(F)^j$ can be represented as linear combinations of the others. However, the analysis of the Bianchi identities in the preceding section showed that only four of these six identities are linearly independent, and hence four identities (C.42) remain.

Taking the identities (C.35), (C.38), (C.40), and (C.42) into account, we can now calculate the total number of independent first-class constraints. Table C.3 shows that the total number of first-class constraint components is $F^* = 100$. The identities (C.35), (C.38), (C.40), and (C.42) reduce the number of independent components of these constraints, as indicated explicitly in the table. Thus, subtracting from this number the six identities (C.35), the identity (C.38), the four identities (C.40), and the four identities (C.42), we obtain $F = 85$ independent first-class constraint components.

$\Phi(B)_{ab}^i$	$\Phi(B)^i$	$\Phi(e)_a$	$\Phi(\omega)_{ab}$	$\Phi(A)$	$\Phi(\beta)_a^i$	$\Phi(\gamma)_A^{ij}$	$\Phi(R)_{ab}^i$	$\Phi(F)^i$	$\Phi(\mathcal{G})_a$	$\Phi(\nabla e)_a^i$	$\Phi(\nabla B)_{ab}$	$\Phi(\nabla B)$	$\Phi(\nabla \phi)_A^{ij}$
18	3	4	6	1	12	6	$18 - 6$	$3 - 1$	4	$12 - 4$	6	1	$6 - 4$

Table C.3: First-class constraints in the $3BF$ theory of scalar electrodynamics.

Therefore, substituting all the results obtained into equation (3.41), we find

$$n = 120 - 85 - \frac{70}{2} = 0, \quad (\text{C.43})$$

that the $3BF$ theory defined by the action (6.174) has no local propagating degrees of freedom.

C.3 Gauge-symmetry generator

Using the results of the Hamiltonian analysis of the action (6.174), we can calculate the gauge-symmetry generator and the variations of the configuration variables in the topological $3BF$ theory of scalar electrodynamics. The results presented in this section generalize the calculation in [48] of the generator and the variations of the configuration variables in the topological $2BF$ action for the Poincaré 2-group, and constitute a special case of the results for the $3BF$ theory based on a general semistrict 2-group given in Chapter 6.

The Castellani procedure gives the generator:

$$\begin{aligned}
G = \int_{\Sigma_3} d^3\vec{x} & \left(\frac{1}{2}(\nabla_0 \epsilon^{ab}{}_i) \Phi(B)_{ab}{}^i - \frac{1}{2} \epsilon^{ab}{}_i \Phi(R)_{ab}{}^i + \frac{1}{2}(\nabla_0 \epsilon^{ab}) \Phi(\omega)_{ab} - \frac{1}{2} \epsilon^{ab} \Phi(\nabla B)_{ab} \right. \\
& + (\partial_0 \epsilon_i) \Phi(B)^i - \epsilon_i \Phi(F)^i + (\partial_0 \epsilon) \Phi(A) - \epsilon \Phi(\nabla B) \\
& + (\nabla_0 \epsilon^a) \Phi(e)_a - \epsilon^a \Phi(\mathcal{G})_a + (\nabla_0 \epsilon^a{}_i) \Phi(\beta)_a{}^i - \epsilon^a{}_i \Phi(\nabla e)_a{}^i \\
& + \frac{1}{2}(\nabla_0 \epsilon^A{}_{ij}) \Phi(\gamma)_A{}^{ij} - \frac{1}{2} \epsilon^A{}_{ij} \Phi(\nabla \phi)_A{}^{ij} \\
& + \epsilon^{ab} (\beta_{[a|0i} P(\beta)_{|b]}{}^i + e_{[a|0} P(e)_{|b]} + B_{[a|c0i} P(B)^c{}_{|b]}{}^i) - \epsilon \gamma_{A0ij} \triangleright_B{}^A P(\gamma)^{Bij} \\
& \left. + \epsilon^a \beta_{b0i} P(B)^{abi} + \epsilon^a{}_i e_{b0} P(B)_a{}^{bi} \right), \tag{C.44}
\end{aligned}$$

where $\epsilon^{ab}{}_i$, ϵ^{ab} , ϵ_i , ϵ , ϵ^a , $\epsilon^a{}_i$, and $\epsilon^A{}_{ij}$ are independent gauge-transformation parameters.

We can then calculate the variations of the configuration variables and their conjugate momenta in the theory:

$$\begin{aligned}
\delta_0 \omega^{ab}{}_0 &= \nabla_0 \epsilon^{ab}, & \delta_0 \pi(\omega)_{ab}{}^0 &= -2\epsilon_{[a|}{}^c{}_i \pi(B)_{c|b]}{}^{0i} - 2\epsilon_{[a|}{}^c \pi(\omega)_{c|b]}{}^0, \\
& & & + 2\epsilon_{[a|} \pi(e)_{|b]}{}^0 + 2\epsilon_{[a|i} \pi(\beta)_{|b]}{}^{0i}, \\
\delta_0 \omega^{ab}{}_i &= \nabla_i \epsilon^{ab}, & \delta_0 \pi(\omega)_{ab}{}^i &= -2\epsilon_{[a|}{}^c{}_j \pi(B)_{c|b]}{}^{ij} - 2\epsilon_{[a|}{}^c{}_i \pi(\omega)_{c|b]}{}^i \\
& & & + 2\epsilon_{[a|} \pi(e)_{|b]}{}^i + 2\epsilon_{[a|j} \pi(\beta)_{|b]}{}^{ij} \\
& & & + 2\epsilon^{0ijk} \nabla_{[j} \epsilon_{ab|k]} + \epsilon^{0ijk} \epsilon_{[a|} \beta_{|b]}{}^{jk}, \\
\delta_0 B^{ab}{}_{0i} &= \nabla_0 \epsilon^{ab}{}_i + \epsilon^{[a|}{}_i e^{b]}{}_0, & \delta_0 \pi(B)_{ab}{}^{0i} &= 2\epsilon_{[a|c} \pi(B)_{|b]}{}^{ci}, \\
& & & + 2\epsilon^{[a|c} B^{b]}{}_{c0i} + \epsilon^{[a|} \beta^{b]}{}_{0i}, \\
\delta_0 B^{ab}{}_{ij} &= 2\nabla_{[i} \epsilon^{ab}{}_{j]} + 2\epsilon^{[a|c} B^{b]}{}_{cij}, & \delta_0 \pi(B)_{ab}{}^{ij} &= 2\epsilon_{[a|c} \pi(B)_{|b]}{}^{cij}, \\
& & & + 2\epsilon^{[a|}{}_{[i} e^{b]}{}_{j]} + \epsilon^{[a|} \beta^{b]}{}_{ij}, \\
\delta_0 A_0 &= \partial_0 \epsilon, & \delta_0 \pi(A)^0 &= -\frac{1}{2} \epsilon^A{}_{ij} \triangleright_B{}^A \pi(\gamma)_B{}^{0ij}, \\
\delta_0 A_i &= \partial_i \epsilon, & \delta_0 \pi(A)^i &= \epsilon^{0ijk} \partial_j \epsilon_k - \frac{1}{2} \epsilon^A{}_{jk} \triangleright_B{}^A \pi(\gamma)_B{}^{ijk}, \\
\delta_0 B_{0i} &= \partial_0 \epsilon_i, & \delta_0 \pi(B)^{0i} &= 0,
\end{aligned} \tag{C.45}$$

$$\begin{aligned}
\delta_0 B_{ij} &= 2\partial_{[i}\epsilon_{j]} + \epsilon^A_{ij} \triangleright^B_A \phi_B, & \delta_0 \pi(B)^{ij} &= -\epsilon^{0ijk} \partial_k \epsilon, \\
\delta_0 \beta^a_{0i} &= \nabla_0 \epsilon^a_i - \epsilon^{ab} \beta_{b0i}, & \delta_0 \pi(\beta)_a^{0i} &= -\epsilon_{ab} \pi(\beta)^{b0i} + \frac{1}{2} \epsilon^b_i \pi(B)_{ab}^{0i}, \\
\delta_0 \beta^a_{ij} &= 2\nabla_{[i} \epsilon^a_{j]} - \epsilon^{ab} \beta_{bij}, & \delta_0 \pi(\beta)_a^{ij} &= -\epsilon_{ab} \pi(\beta)^{bij} + \frac{1}{2} \epsilon^b_i \pi(B)_{ab}^{ij} \\
&& &- \epsilon^{0ijk} \nabla_k \epsilon^a, \\
\delta_0 e^a_0 &= \nabla_0 \epsilon^a - \epsilon^{ab} e_{b0}, & \delta_0 \pi(e)_a^0 &= -\epsilon_{ab} \pi(e)^{b0} + \frac{1}{2} \epsilon^b_i \pi(B)_{ab}^{0i}, \\
\delta_0 e^a_i &= \nabla_i \epsilon^a - \epsilon^{ab} e_{bi}, & \delta_0 \pi(e)_a^i &= -\epsilon_{ab} \pi(e)^{bi} + \epsilon^{0ijk} \left(\nabla_{[j} \epsilon_{a|k]} + \epsilon_{ab} \beta^{bjk} \right) \\
&& &+ \frac{1}{2} \epsilon^b_j \pi(B)_{ab}^{ij}, \\
\delta_0 \gamma^A_{0ij} &= \nabla_0 \epsilon^A_{ij} - \epsilon \gamma^B_{0ij} \triangleright^A_B, & \delta_0 \pi(\gamma)_A^{0ij} &= \epsilon \triangleright^B_A \pi(\gamma)_B^{0ij}, \\
\delta_0 \gamma^A_{ijk} &= -\epsilon \gamma^B_{ijk} \triangleright^A_B + \nabla_i \epsilon^A_{jk} \\
&& - \nabla_j \epsilon^A_{ik} + \nabla_k \epsilon^A_{ij}, & \delta_0 \pi(\gamma)_A^{ijk} &= \epsilon \triangleright^B_A \left(\pi(\gamma)_B^{ijk} + \epsilon^{0ijk} \phi_B \right), \\
\delta_0 \phi^A &= \epsilon \phi^B \triangleright^A_B, & \delta_0 \pi(\phi)_A &= -\epsilon \triangleright^B_A \pi(\phi)_B + \frac{1}{3!} \epsilon \epsilon^{0ijk} \triangleright^B_A \gamma_{Bijk} \\
&& &- \frac{1}{2} \triangleright^B_A \epsilon^B_{ij} \pi(B)^{ij} - \frac{1}{2} \epsilon^{0ijk} \nabla_i \epsilon^A_{jk},
\end{aligned} \tag{C.46}$$

Appendix D

Total gauge-symmetry group

D.1 Gauge transformations in topological BF theory

The generator of gauge transformations in BF theory is

$$G = \int_{\Sigma_3} d^3\vec{x} \left((\nabla_0 \epsilon_{\mathfrak{g}}^\alpha) (\tilde{G}_1)_\alpha + \epsilon_{\mathfrak{g}}^\alpha (\tilde{G}_0)_\alpha + (\nabla_0 \epsilon_{\mathfrak{m}}^{\alpha i}) (\tilde{M}_1)_\alpha{}^i + \epsilon_{\mathfrak{m}}^{\alpha i} (\tilde{M}_0)_\alpha{}^i \right), \quad (\text{D.1})$$

where:

$$\begin{aligned} (\tilde{G}_1)_\alpha &= -\Phi(\alpha)_\alpha, \\ (\tilde{G}_0)_\alpha &= -(f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} + C_{a0} \triangleright_{\alpha b}{}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{\alpha b}{}^a \Phi(\beta)^{b0i} - \Phi(\nabla B)_\alpha), \\ (\tilde{M}_1)_\alpha{}^i &= -\Phi(B)_\alpha{}^i, \\ (\tilde{M}_0)_\alpha{}^i &= \Phi(\mathcal{F})_\alpha{}^i, \end{aligned} \quad (\text{D.2})$$

where $\epsilon_{\mathfrak{g}}^\alpha$ and $\epsilon_{\mathfrak{m}}^{\alpha i}$ are independent gauge-transformation parameters.

D.1.1 Gauge-symmetry group of the phase space of the BF action

The algebra formed by the symmetry-group generators $(M_0)_\alpha{}^i$, $(M_1)_\alpha{}^i$, $(G_0)_\alpha$, and $(G_1)_\alpha$ defined in Appendix D.1 is:

$$\begin{aligned} \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{G}_0)_\beta(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{G}_0)_\gamma \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{M}_0)_\beta{}^i(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{M}_0)_\gamma{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{M}_1)_\beta{}^i(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{M}_1)_\gamma{}^i \delta^{(3)}(\vec{x} - \vec{y}), \end{aligned} \quad (\text{D.3})$$

These commutation relations show that the gauge-symmetry group has the following structure. First, the generators $(\tilde{M}_1)_\alpha{}^i$ and $(\tilde{M}_0)_\alpha{}^i$ form the algebra $\mathfrak{a}_1$,

$$\mathfrak{a}_1 = \text{span}\{(\tilde{M}_1)_\alpha{}^i\} \oplus \text{span}\{(\tilde{M}_0)_\alpha{}^i\},$$

which generates the subgroup $\tilde{M}_1 \times \tilde{M}_0$ of the total symmetry group $\tilde{\mathcal{G}}_{\Sigma_3}$. Moreover, the subgroup $\tilde{M}_1 \times \tilde{M}_0$ is a normal subgroup of $\tilde{\mathcal{G}}_{\Sigma_3}$. The group $\tilde{G}_1 \times \tilde{G}_0$ is also a subgroup of the total symmetry group. These two subgroups, one normal and the other not, can therefore be combined by a semidirect product, yielding the total symmetry group $\tilde{\mathcal{G}}_{\Sigma_3}$:

$$\tilde{\mathcal{G}}_{\Sigma_3} = (\tilde{G}_1 \times \tilde{G}_0) \ltimes (\tilde{M}_1 \times \tilde{M}_0).$$

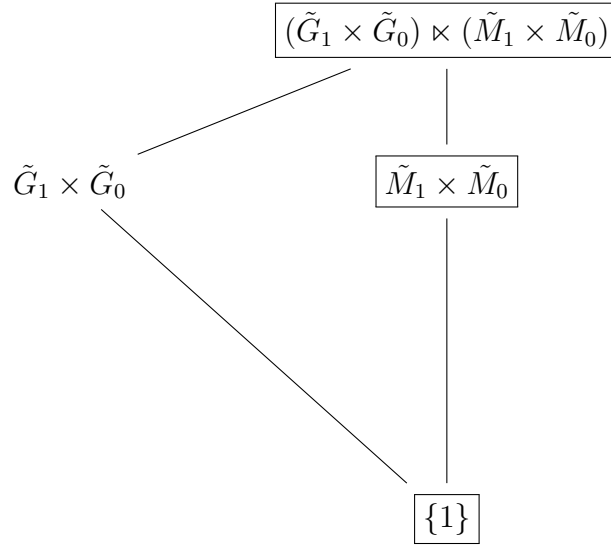


Figure D.1: The symmetry group $\mathcal{G}_{\Sigma_3}$ in phase space. Normal subgroups are boxed.

D.1.2 Construction of the symmetry generator of BF theory

Substituting the generators (D.2) into equation (D.1), we obtain the gauge-symmetry generator in BF theory in the following form:

$$G = - \int_{\Sigma_3} d^3 \vec{x} \left((\nabla_0 \epsilon_m^\alpha)_i \Phi(B)_\alpha^i - \epsilon_m^\alpha{}_i \Phi(\mathcal{F})_\alpha^i + (\nabla_0 \epsilon_g^\alpha) \Phi(\alpha)_\alpha + \epsilon_g^\alpha (f_{\alpha\gamma}{}^\beta B_{\beta 0 i} \Phi(B)^{\gamma i} - \Phi(\nabla B)_\alpha) \right), \quad (D.4)$$

where ϵ_g^α and $\epsilon_m^\alpha{}_i$ are independent gauge-transformation parameters.

The generator of gauge transformations in topological BF theory (4.31) is obtained using the *Castellani procedure* (3.53), with the relations (3.55) satisfied for each pair of generators G_0 and G_1 .

Let us first assume that the generator has the form:

$$G = \int \dot{\epsilon}^\alpha{}_i (G_1)_\alpha^i + \epsilon^\alpha{}_i (G_0)_\alpha^i + \dot{\epsilon}^\alpha (G_1)_\alpha + \epsilon^\alpha (G_0)_\alpha. \quad (D.5)$$

Choosing $(G_1)_\alpha^i = C_{PFC}$ and $(G_1)_\alpha = C_{PFC}$, where C_{PFC} denotes a primary first-class constraint, the natural choice is:

$$(G_1)_\alpha^i = \Phi(B)_\alpha^i, \quad (G_1)_\alpha = \Phi(\alpha)_\alpha. \quad (D.6)$$

It remains to determine the two generators G_0 . Castellani's second condition for the generator $(G_0)_{m\alpha}^i$ gives

$$\begin{aligned} (G_0)_{m\alpha}^i - \{\Phi(B)_\alpha^i, H_T\} &= (C_{PFC})_\alpha^i, \\ (G_0)_{m\alpha}^i - \Phi(\mathcal{F})_\alpha^i &= (C_{PFC})_\alpha^i, \end{aligned} \quad (D.7)$$

where $(G_0)_{m\alpha}^i = (C_{PFC})_\alpha^i + \Phi(\mathcal{F})_\alpha^i$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{m\alpha}^i, H_T\} &= (C_{PFC1})_\alpha^i, \\ \{(C_{PFC})_\alpha^i + \Phi(\mathcal{F})_\alpha^i, H_T\} &= (C_{PFC1})_\alpha^i, \\ \{(C_{PFC})_\alpha^i, H_T\} - f_{\beta\gamma\alpha} \alpha^\beta{}_0 \Phi(\mathcal{F})^{\gamma i} &= (C_{PFC1})_\alpha^i, \end{aligned} \quad (D.8)$$

which gives the equation

$$(C_{PFC})_\alpha{}^i = f_{\beta\gamma\alpha}\alpha^\beta{}_0\Phi(B)^{\gamma i}.$$

It follows that the generator is:

$$(G_0)_{\mathfrak{m}\alpha}{}^i = f_{\beta\gamma\alpha}\alpha^\beta{}_0\Phi(B)^{\gamma i} + \Phi(\mathcal{F})_\alpha{}^i. \quad (\text{D.9})$$

Castellani's second condition for the generator $(G_0)_{\mathfrak{g}\alpha}$ gives

$$\begin{aligned} (G_0)_{\mathfrak{g}\alpha} - \{\Phi(\alpha)_\alpha, H_T\} &= (C_{PFC})_\alpha, \\ (G_0)_{\mathfrak{g}\alpha} - \Phi(\nabla B)_\alpha &= (C_{PFC})_\alpha, \end{aligned} \quad (\text{D.10})$$

that is, $(G_0)_{\mathfrak{g}\alpha} = (C_{PFC})_\alpha + \Phi(\nabla B)_\alpha$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{g}\alpha}, H_T\} &= (C_{PFC1})_\alpha, \\ \{(C_{PFC})_\alpha + \Phi(\nabla B)_\alpha, H_T\} &= (C_{PFC1})_\alpha, \\ \{(C_{PFC})_\alpha, H_T\} + B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(\mathcal{F})^{\gamma i} - \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\nabla B)_\gamma &= (C_{PFC1})_\alpha, \end{aligned} \quad (\text{D.11})$$

that is,

$$(C_{PFC})_\alpha = -B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(B)^{\gamma i} + \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\alpha)_\gamma.$$

It follows that the generator is:

$$(G_0)_{\mathfrak{g}\alpha} = -B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(B)^{\gamma i} + \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\alpha)_\gamma + \Phi(\nabla B)_\alpha. \quad (\text{D.12})$$

At this point, it is useful to summarize the results and introduce new notation:

$$\begin{aligned} \dot{\epsilon}_{\mathfrak{m}}{}^\alpha{}_i(G_1)_{\mathfrak{m}\alpha}{}^i + \epsilon_{\mathfrak{m}}{}^\alpha{}_i(G_0)_{\mathfrak{m}\alpha}{}^i &= -\nabla_0\epsilon_{\mathfrak{m}}{}^\alpha{}_i\Phi(B)_\alpha{}^i + \epsilon_{\mathfrak{m}}{}^\alpha{}_i\Phi(\mathcal{F})_\alpha{}^i \\ &= \nabla_0\epsilon_{\mathfrak{m}}{}^\alpha{}_i(\tilde{M}_1)_\alpha{}^i + \epsilon_{\mathfrak{m}}{}^\alpha{}_i(\tilde{M}_0)_\alpha{}^i. \end{aligned} \quad (\text{D.13})$$

Notice that the time derivative of the parameter combines with other terms into a covariant derivative in the time direction.

For the second part of the total generator, we obtain:

$$\begin{aligned} \dot{\epsilon}_{\mathfrak{g}}{}^\alpha(G_1)_{\mathfrak{g}\alpha} + \epsilon_{\mathfrak{g}}{}^\alpha(G_0)_{\mathfrak{g}\alpha} &= -\dot{\epsilon}_{\mathfrak{g}}{}^\alpha\Phi(\alpha)_\alpha - \epsilon_{\mathfrak{g}}{}^\alpha(B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(B)^{\gamma i} - \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\alpha)_\gamma - \Phi(\nabla B)_\alpha) \\ &= -\nabla_0\epsilon_{\mathfrak{g}}{}^\alpha\Phi(\alpha)_\alpha - \epsilon_{\mathfrak{g}}{}^\alpha(B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(B)^{\gamma i} - \Phi(\nabla B)_\alpha) \\ &= \nabla_0\epsilon_{\mathfrak{g}}{}^\alpha(\tilde{G}_1)_\alpha + \epsilon_{\mathfrak{g}}{}^\alpha(\tilde{G}_0)_\alpha. \end{aligned} \quad (\text{D.14})$$

D.2 Gauge transformations in topological $2BF$ theory

The generator of gauge transformations in $2BF$ theory is

$$G = \int_{\Sigma_3} d^3\vec{x} \left((\nabla_0 \epsilon_{\mathfrak{g}}^\alpha) (\tilde{G}_1)_\alpha + \epsilon_{\mathfrak{g}}^\alpha (\tilde{G}_0)_\alpha + (\nabla_0 \epsilon_{\mathfrak{h}}^a{}_i) (\tilde{H}_1)_a{}^i + \epsilon_{\mathfrak{h}}^a{}_i (\tilde{H}_0)_a{}^i \right. \\ \left. + (\nabla_0 \epsilon_{\mathfrak{m}}^\alpha{}_i) (\tilde{M}_1)_\alpha{}^i + \epsilon_{\mathfrak{m}}^\alpha{}_i (\tilde{M}_0)_\alpha{}^i + (\nabla_0 \epsilon_{\mathfrak{n}}^a) (\tilde{N}_1)_a + \epsilon_{\mathfrak{n}}^a (\tilde{N}_0)_a \right), \quad (\text{D.15})$$

where:

$$\begin{aligned} (\tilde{G}_1)_\alpha &= -\Phi(\alpha)_\alpha, \\ (\tilde{G}_0)_\alpha &= -(f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} + C_{a0} \triangleright_{\alpha b}{}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{\alpha b}{}^a \Phi(\beta)^{b0i} - \Phi(\nabla B)_\alpha), \\ (\tilde{H}_1)_a{}^i &= -\Phi(\beta)_a{}^i, \\ (\tilde{H}_0)_a{}^i &= C_{b0} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\nabla C)_a{}^i, \\ (\tilde{M}_1)_\alpha{}^i &= -\Phi(B)_\alpha{}^i, \\ (\tilde{M}_0)_\alpha{}^i &= \Phi(\mathcal{F})_\alpha{}^i, \\ (\tilde{N}_1)_a &= -\Phi(C)_a, \\ (\tilde{N}_0)_a &= \beta_{b0i} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a, \end{aligned} \quad (\text{D.16})$$

where $\epsilon_{\mathfrak{g}}^\alpha$, $\epsilon_{\mathfrak{h}}^a{}_i$, $\epsilon_{\mathfrak{m}}^\alpha{}_i$, and $\epsilon_{\mathfrak{n}}^a$ are independent gauge-transformation parameters.

D.2.1 Gauge-symmetry group of the $2BF$ action

The algebra formed by the symmetry-group generators $(M_0)_\alpha{}^i$, $(M_1)_\alpha{}^i$, $(G_0)_\alpha$, $(G_1)_\alpha$, $(H_0)_a{}^i$, $(H_1)_a{}^i$, $(N_0)_a$, and $(N_1)_a$ defined in the preceding section D.2 is:

$$\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{G}_0)_\beta(\vec{y})\} = f_{\alpha\beta}{}^\gamma (\tilde{G}_0)_\gamma \delta^{(3)}(\vec{x} - \vec{y}), \quad (\text{D.17})$$

$$\begin{aligned} \{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{M}_0)^{\alpha i} \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{H}_1)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{M}_1)^{\alpha i} \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{H}_0)_a(\vec{x}), (\tilde{N}_1)^{bi}(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{M}_1)^{\alpha i} \delta^{(3)}(\vec{x} - \vec{y}), \end{aligned} \quad (\text{D.18})$$

$$\begin{aligned} \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{M}_0)_\beta{}^i(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{M}_0)_\gamma{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{M}_1)_\beta{}^i(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{M}_1)_\gamma{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{H}_1)_a{}^i(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{H}_1)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{H}_0)_a{}^i(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{H}_0)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{N}_1)_a(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{N}_1)_b(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\ \{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{N}_0)_a(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{N}_0)_b(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \end{aligned} \quad (\text{D.19})$$

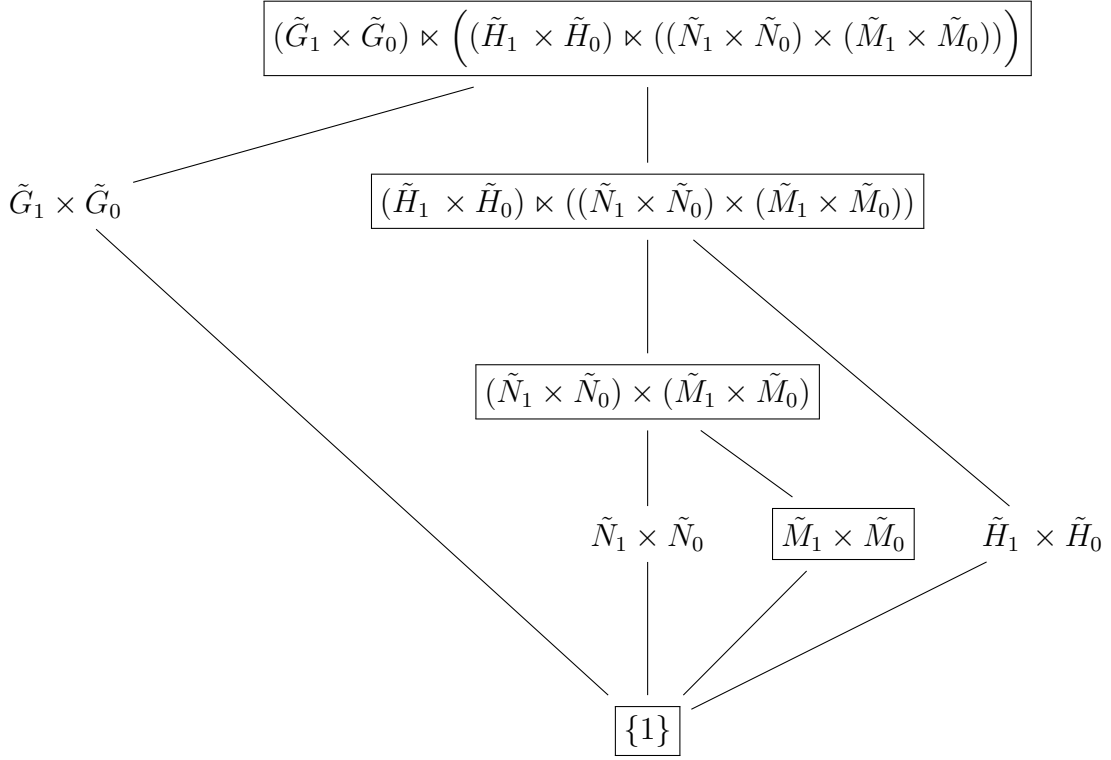


Figure D.2: The symmetry group $\mathcal{G}_{\Sigma_3}$ in phase space. Normal subgroups are boxed.

The gauge-symmetry group has the following structure. First, the groups $\tilde{M}_1 \times \tilde{M}_0$, $\tilde{N}_1 \times \tilde{N}_0$, and $\tilde{H}_1 \times \tilde{H}_0$, with the corresponding algebras $\mathfrak{a}_1$, $\mathfrak{a}_2$, and $\mathfrak{a}_3$, where

$$\begin{aligned} \mathfrak{a}_1 &= \text{span}\{(\tilde{M}_1)_\alpha{}^i\} \oplus \text{span}\{(\tilde{M}_0)_\alpha{}^i\}, & \mathfrak{a}_2 &= \text{span}\{(\tilde{N}_1)_a\} \oplus \text{span}\{(\tilde{N}_0)_a\}, \\ \mathfrak{a}_3 &= \text{span}\{(\tilde{H}_1)_a{}^i\} \oplus \text{span}\{(\tilde{H}_0)_a{}^i\} \end{aligned} \quad (\text{D.20})$$

are subgroups of the total symmetry group $\tilde{\mathcal{G}}_{\Sigma_3}$. The subgroup $\tilde{M}_1 \times \tilde{M}_0$ is a normal subgroup of the total symmetry group. The groups $\tilde{N}_1 \times \tilde{N}_0$ and $\tilde{H}_1 \times \tilde{H}_0$ are not normal subgroups of the total symmetry group, as follows from the Poisson brackets $\{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\}$ and $\{(\tilde{H}_1)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\}$, which equal linear combinations of the generators of the groups M_1 and M_0 , respectively. Since the generators of these groups commute with one another, one can form the direct product $(\tilde{N}_1 \times \tilde{N}_0) \times (\tilde{M}_1 \times \tilde{M}_0)$, and the resulting group is a normal subgroup of the total symmetry group.

The resulting group can be combined with $\tilde{H}_1 \times \tilde{H}_0$ by a semidirect product, since $(\tilde{N}_1 \times \tilde{N}_0) \times (\tilde{M}_1 \times \tilde{M}_0)$ is normal whereas $\tilde{H}_1 \times \tilde{H}_0$ is not. This gives the normal subgroup $(\tilde{H}_1 \times \tilde{H}_0) \times ((\tilde{N}_1 \times \tilde{N}_0) \times (\tilde{M}_1 \times \tilde{M}_0))$, corresponding to the algebra $\mathfrak{a}_4$:

$$\mathfrak{a}_4 = \text{span}\{(\tilde{M}_0)_\alpha{}^i, (\tilde{M}_1)_\alpha{}^i, (\tilde{H}_0)_a{}^i, (\tilde{H}_1)_a{}^i, (\tilde{N}_0)_a, (\tilde{N}_1)_a\}.$$

Finally, the resulting group can be combined with $\tilde{G}_1 \times \tilde{G}_0$ by a semidirect product, yielding the total symmetry group $\tilde{\mathcal{G}}_{\Sigma_3}$:

$$\tilde{\mathcal{G}}_{\Sigma_3} = (\tilde{G}_1 \times \tilde{G}_0) \ltimes \left((\tilde{H}_1 \times \tilde{H}_0) \ltimes \left((\tilde{N}_1 \times \tilde{N}_0) \times (\tilde{M}_1 \times \tilde{M}_0) \right) \right).$$

The complete structure of the total symmetry group is shown in Figure D.2. Normal subgroups of the total symmetry group are boxed.

D.2.2 Construction of the symmetry generator of $2BF$ theory

Substituting the generators (D.16) into equation (D.15), we obtain the gauge-symmetry generator in $2BF$ theory in the following form:

$$\begin{aligned}
G = & - \int_{\Sigma_3} d^3 \vec{x} \left((\nabla_0 \epsilon_m^\alpha)_i \Phi(B)_\alpha^i - \epsilon_m^\alpha{}_i \Phi(\mathcal{F})_\alpha^i + (\nabla_0 \epsilon_g^\alpha) \Phi(\alpha)_\alpha \right. \\
& + \epsilon_g^\alpha (f_{\alpha\gamma}{}^\beta B_{\beta 0 i} \Phi(B)^{\gamma i} + C_{a0} \triangleright_{\alpha b}{}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{\alpha b}{}^a \Phi(\beta)^{b0i} - \Phi(\nabla B)_\alpha) \\
& + (\nabla_0 \epsilon_n^a) \Phi(C)_a - \epsilon_n^a (\beta_{b0i} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a) \\
& \left. + (\nabla_0 \epsilon_h^a)_i \Phi(\beta)_a^i - \epsilon_h^a{}_i (C_{b0} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\nabla C)_a^i) \right), \tag{D.21}
\end{aligned}$$

where ϵ_g^α , $\epsilon_h^a{}_i$, $\epsilon_m^\alpha{}_i$, and ϵ_n^a are independent gauge-transformation parameters.

The gauge-symmetry generator (D.15) in $2BF$ theory (5.1) is obtained using the Castellani procedure. Let us first assume that the generator has the structure:

$$\begin{aligned}
G = & \int_{\Sigma_3} d^3 \vec{x} \left(\dot{\epsilon}_m^\alpha{}_i (G_1)_{m\alpha}^i + \epsilon_m^\alpha{}_i (G_0)_{m\alpha}^i + \dot{\epsilon}_g^\alpha (G_1)_{g\alpha} + \epsilon_g^\alpha (G_0)_{g\alpha} \right. \\
& \left. + \dot{\epsilon}_h^a{}_i (G_1)_{ha}^i + \epsilon_h^a{}_i (G_0)_{ha}^i + \dot{\epsilon}_n^a (G_1)_{na} + \epsilon_n^a (G_0)_{na} \right). \tag{D.22}
\end{aligned}$$

The first step of the Castellani procedure is to impose the conditions

$$(G_1)_{m\alpha}^i = C_{PFC}, \quad (G_1)_{g\alpha} = C_{PFC}, \quad (G_1)_{ha}^i = C_{PFC}, \quad (G_1)_{na} = C_{PFC}, \tag{D.23}$$

with the natural choice:

$$(G_1)_{m\alpha}^i = -\Phi(B)_\alpha^i, \quad (G_1)_{g\alpha} = -\Phi(\alpha)_\alpha, \quad (G_1)_{ha}^i = -\Phi(C)_\alpha^i, \quad (G_1)_{na} = -\Phi(\beta)_a. \tag{D.24}$$

It remains to determine the four generators G_0 .

Castellani's second condition for the generator $(G_0)_{m\alpha}^i$ gives

$$\begin{aligned}
(G_0)_{m\alpha}^i - \{\Phi(B)_\alpha^i, H_T\} &= (C_{PFC})_\alpha^i, \\
(G_0)_{m\alpha}^i - \Phi(\mathcal{F})_\alpha^i &= (C_{PFC})_\alpha^i, \tag{D.25}
\end{aligned}$$

where $(G_0)_{m\alpha}^i = (C_{PFC})_\alpha^i + \Phi(\mathcal{F})_\alpha^i$. Castellani's third condition then gives

$$\begin{aligned}
\{(G_0)_{m\alpha}^i, H_T\} &= (C_{PFC1})_\alpha^i, \\
\{(C_{PFC})_\alpha^i + \Phi(\mathcal{F})_\alpha^i, H_T\} &= (C_{PFC1})_\alpha^i, \\
\{(C_{PFC})_\alpha^i, H_T\} - f_{\beta\gamma\alpha} \alpha^\beta{}_0 \Phi(\mathcal{F})^{\gamma i} &= (C_{PFC1})_\alpha^i, \tag{D.26}
\end{aligned}$$

which gives the equation

$$(C_{PFC})_\alpha^i = f_{\beta\gamma\alpha} \alpha^\beta{}_0 \Phi(B)^{\gamma i}.$$

It follows that the generator is:

$$(G_0)_{m\alpha}^i = f_{\beta\gamma\alpha} \alpha^\beta{}_0 \Phi(B)^{\gamma i} + \Phi(\mathcal{F})_\alpha^i. \tag{D.27}$$

Castellani's second condition for the generator $(G_0)_{g\alpha}$ gives

$$\begin{aligned}
(G_0)_{g\alpha} - \{\Phi(\alpha)_\alpha, H_T\} &= (C_{PFC})_\alpha, \\
(G_0)_{g\alpha} - \Phi(\nabla B)_\alpha &= (C_{PFC})_\alpha, \tag{D.28}
\end{aligned}$$

that is, $(G_0)_{\mathfrak{g}\alpha} = (C_{PFC})_\alpha + \Phi(\nabla B)_\alpha$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{g}\alpha}, H_T\} &= (C_{PFC1})_\alpha, \\ \{(C_{PFC})_\alpha + \Phi(\nabla B)_\alpha, H_T\} &= (C_{PFC1})_\alpha, \\ \{(C_{PFC})_\alpha, H_T\} + B_{\beta 0i} f_{\alpha\gamma}^\beta \Phi(\mathcal{F})^{\gamma i} - \alpha^\beta{}_0 f_{\alpha\beta}^\gamma \Phi(\nabla B)_\gamma + C_{a0} \triangleright_{\alpha b} {}^a \Phi(\mathcal{G})^b + \beta_{a0i} \triangleright_{\alpha b} {}^a \Phi(\nabla C)^{bi} &= (C_{PFC1})_\alpha, \end{aligned} \quad (\text{D.29})$$

that is,

$$(C_{PFC})_\alpha = -B_{\beta 0i} f_{\alpha\gamma}^\beta \Phi(B)^{\gamma i} + \alpha^\beta{}_0 f_{\alpha\beta}^\gamma \Phi(\alpha)_\gamma - C_{a0} \triangleright_{\alpha b} {}^a \Phi(C)^b - \beta_{a0i} \triangleright_{\alpha b} {}^a \Phi(\beta)^{bi}.$$

It follows that the generator is:

$$(G_0)_{\mathfrak{g}\alpha} = -B_{\beta 0i} f_{\alpha\gamma}^\beta \Phi(B)^{\gamma i} + \alpha^\beta{}_0 f_{\alpha\beta}^\gamma \Phi(\alpha)_\gamma - C_{a0} \triangleright_{\alpha b} {}^a \Phi(C)^b - \beta_{a0i} \triangleright_{\alpha b} {}^a \Phi(\beta)^{bi} + \Phi(\nabla B)_\alpha. \quad (\text{D.30})$$

Castellani's second condition for the generator $(G_0)_{\text{na}}$ gives

$$\begin{aligned} (G_0)_{\text{na}} - \{\Phi(C)_a, H_T\} &= (C_{PFC})_a, \\ (G_0)_{\text{na}} - \Phi(\mathcal{G})_a &= (C_{PFC})_a, \end{aligned} \quad (\text{D.31})$$

where $(G_0)_{\text{na}} = (C_{PFC})_a + \Phi(\mathcal{G})_a$.

The third Castellani condition then gives

$$\begin{aligned} \{(G_0)_{\text{na}}, H_T\} &= (C_{PFC1})_a, \\ \{(C_{PFC})_a + \Phi(\mathcal{G})_a, H_T\} &= (C_{PFC1})_a, \\ \{(C_{PFC})_a, H_T\} + \alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b \Phi(\mathcal{G})_b - \beta_{b0i} \triangleright_{\alpha a} {}^b \Phi(\mathcal{F})^{\alpha i} &= (C_{PFC1})_a, \end{aligned} \quad (\text{D.32})$$

which gives:

$$(C_{PFC})_a = -\alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b \Phi(C)_b + \beta_{b0i} \triangleright_{\alpha a} {}^b \Phi(B)^{\alpha i}.$$

It follows that the generator is:

$$(G_0)_{\text{na}} = -\alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b \Phi(C)_b + \beta_{b0i} \triangleright_{\alpha a} {}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a.$$

Castellani's second condition for the generator $(G_0)_{\mathfrak{h}a}{}^i$ gives

$$\begin{aligned} (G_0)_{\mathfrak{h}a}{}^i - \{\Phi(\beta)_a{}^i, H_T\} &= (C_{PFC})_a{}^i, \\ (G_0)_{\mathfrak{h}a}{}^i - \Phi(\nabla C)_a{}^i &= (C_{PFC})_a{}^i, \end{aligned} \quad (\text{D.33})$$

that is, $(G_0)_{\mathfrak{h}a}{}^i = (C_{PFC})_a{}^i + \Phi(\nabla C)_a{}^i$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{h}a}{}^i, H_T\} &= (C_{PFC1})_a{}^i, \\ \{(C_{PFC})_a{}^i + \Phi(\nabla C)_a{}^i, H_T\} &= (C_{PFC1})_a{}^i, \\ \{(C_{PFC})_a{}^i, H_T\} + \alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b \Phi(\nabla C)_b{}^i - C_{b0} \triangleright_{\alpha a} {}^b \Phi(\mathcal{F})^{\alpha i} &= (C_{PFC1})_a{}^i, \end{aligned}$$

which gives the result:

$$(C_{PFC})_a^i = -\alpha^{\alpha}_0 \triangleright_{\alpha a} {}^b\Phi(\beta)_b^i + C_{b0} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i}.$$

It follows that the generator is:

$$(G_0)_{\mathfrak{h}a}^i = -\alpha^{\alpha}_0 \triangleright_{\alpha a} {}^b\Phi(\beta)_b^i + C_{b0} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i} + \Phi(\nabla C)_a^i.$$

At this point, it is useful to summarize the results and introduce new notation:

$$\begin{aligned} \dot{\epsilon}_{\mathfrak{m}}^{\alpha}{}_i (G_1)_{\mathfrak{m}\alpha}^i + \epsilon_{\mathfrak{m}}^{\alpha}{}_i (G_0)_{\mathfrak{m}\alpha}^i &= -\nabla_0 \epsilon_{\mathfrak{m}}^{\alpha}{}_i \Phi(B)_{\alpha}^i + \epsilon_{\mathfrak{m}}^{\alpha}{}_i \Phi(\mathcal{F})_{\alpha}^i \\ &= \nabla_0 \epsilon_{\mathfrak{m}}^{\alpha}{}_i (\tilde{M}_1)_{\alpha}^i + \epsilon_{\mathfrak{m}}^{\alpha}{}_i (\tilde{M}_0)_{\alpha}^i. \end{aligned} \quad (\text{D.34})$$

Notice that the time derivative of the parameter combines with other terms into a covariant derivative in the time direction.

For the second part of the total generator, we obtain:

$$\begin{aligned} \dot{\epsilon}_{\mathfrak{g}}^{\alpha} (G_1)_{\mathfrak{g}\alpha} + \epsilon_{\mathfrak{g}}^{\alpha} (G_0)_{\mathfrak{g}\alpha} &= -\dot{\epsilon}_{\mathfrak{g}}^{\alpha} \Phi(\alpha)_{\alpha} - \epsilon_{\mathfrak{g}}^{\alpha} (B_{\beta 0 i} f_{\alpha \gamma}^{\beta} \Phi(B)^{\gamma i} - \alpha^{\beta}_0 f_{\alpha \beta}^{\gamma} \Phi(\alpha)_{\gamma} \\ &\quad + C_{a0} \triangleright_{\alpha b} {}^a\Phi(C)^b + \beta_{a0 i} \triangleright_{\alpha b} {}^a\Phi(\beta)_b^i - \Phi(\nabla B)_{\alpha}) \\ &= -\nabla_0 \epsilon_{\mathfrak{g}}^{\alpha} \Phi(\alpha)_{\alpha} - \epsilon_{\mathfrak{g}}^{\alpha} (B_{\beta 0 i} f_{\alpha \gamma}^{\beta} \Phi(B)^{\gamma i} \\ &\quad + C_{a0} \triangleright_{\alpha b} {}^a\Phi(C)^b + \beta_{a0 i} \triangleright_{\alpha b} {}^a\Phi(\beta)_b^i - \Phi(\nabla B)_{\alpha}) \\ &= \nabla_0 \epsilon_{\mathfrak{g}}^{\alpha} (\tilde{G}_1)_{\alpha} + \epsilon_{\mathfrak{g}}^{\alpha} (\tilde{G}_0)_{\alpha}. \end{aligned} \quad (\text{D.35})$$

In addition, it follows that:

$$\begin{aligned} \dot{\epsilon}_{\mathfrak{h}}^a{}_i (G_1)_{\mathfrak{h}a}^i + \epsilon_{\mathfrak{h}}^a{}_i (G_0)_{\mathfrak{h}a}^i &= -\nabla_0 \epsilon_{\mathfrak{h}}^a{}_i \Phi(\beta)_a^i + \epsilon_{\mathfrak{h}}^a{}_i (C_{b0} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i} + \Phi(\nabla C)_a^i) \\ &= \nabla_0 \epsilon_{\mathfrak{h}}^a{}_i (\tilde{H}_1)_a^i + \epsilon_{\mathfrak{h}}^a{}_i (\tilde{H}_0)_a^i, \end{aligned} \quad (\text{D.36})$$

$$\begin{aligned} \dot{\epsilon}_{\mathfrak{n}}^a (G_1)_{\mathfrak{n}a} + \epsilon_{\mathfrak{n}}^a (G_0)_{\mathfrak{n}a} &= -\nabla_0 \epsilon_{\mathfrak{n}}^a \Phi(C)_a + \epsilon_{\mathfrak{n}}^a (\beta_{b0 i} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a) \\ &= \nabla_0 \epsilon_{\mathfrak{n}}^a (\tilde{N}_1)_a + \epsilon_{\mathfrak{n}}^a (\tilde{N}_0)_a. \end{aligned} \quad (\text{D.37})$$

D.2.3 Calculation of the symmetry algebra of the $2BF$ action

To obtain the structure of the symmetry group of the $2BF$ action, as presented in Subsection 5.1.2, we must first calculate the commutators among the generators of the G -, H -, M -, and N -gauge symmetries. This procedure is described in Section 5.1.2, while the details of the calculation are given in this section.

The commutator $[H, H]$

We now calculate the commutator of the generators of H -gauge transformations, that is, equation (5.51). After transforming the variables under an H -gauge transformation with parameter

$\epsilon_{\mathfrak{h}1}$, we obtain

$$\alpha' = \alpha - \partial\epsilon_{\mathfrak{h}1}, \quad (\text{D.38})$$

$$\beta' = \beta - \frac{\alpha - \partial\epsilon_{\mathfrak{h}1}}{\nabla} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1}, \quad (\text{D.39})$$

$$B' = B - C \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}1}, \quad (\text{D.40})$$

$$C' = C, \quad (\text{D.41})$$

Applying a further H -gauge transformation with parameter $\epsilon_{\mathfrak{h}2}$ gives:

$$\begin{aligned} \alpha'' &= \alpha - \partial\epsilon_{\mathfrak{h}1} - \partial\epsilon_{\mathfrak{h}2}, \\ \beta'' &= \beta - \frac{\alpha - \partial\epsilon_{\mathfrak{h}1}}{\nabla} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1} - \frac{\alpha - \partial\epsilon_{\mathfrak{h}1} - \partial\epsilon_{\mathfrak{h}2}}{\nabla} \epsilon_{\mathfrak{h}2} - \epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}2}, \\ B'' &= B - C \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}1} - C \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}2} \\ C'' &= C, \end{aligned} \quad (\text{D.42})$$

For the variables α^α_μ , $B^\alpha_{\mu\nu}$, and C^a_μ , we find that the transformations commute:

$$\begin{aligned} e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} \alpha^\alpha_\mu &= e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} \alpha^\alpha_\mu, \\ e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} B^\alpha_{\mu\nu} &= e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} B^\alpha_{\mu\nu}, \\ e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} C^a_\mu &= e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} C^a_\mu, \end{aligned} \quad (\text{D.43})$$

For the remaining variable $\beta^a_{\mu\nu}$, the difference between equation (D.42) and the analogous equation with $\epsilon_{\mathfrak{h}1} \leftrightarrow \epsilon_{\mathfrak{h}2}$ is:

$$\begin{aligned} (e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} - e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H}) \frac{1}{2} \beta^a_{\mu\nu} &= \partial_b^\alpha \epsilon_{\mathfrak{h}2}^b{}_{[\mu} \epsilon_{\mathfrak{h}1}^c{}_{\nu]} \triangleright_{\alpha c}^a - \partial_b^\alpha \epsilon_{\mathfrak{h}1}^b{}_{[\mu} \epsilon_{\mathfrak{h}2}^c{}_{\nu]} \triangleright_{\alpha c}^a \\ &= 0. \end{aligned} \quad (\text{D.44})$$

Taking the results (D.43) and (D.44) into account, we conclude that the H -gauge transformations commute:

$$e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} - e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} = 0. \quad (\text{D.45})$$

The commutator $[H, N]$

Let us calculate the commutator between the generators of H -gauge and N -gauge transformations, that is, derive equation (5.62). This is done by calculating the expression

$$(e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) A, \quad (\text{D.46})$$

for all variables A present in the theory. For the variables α^α_μ , $\beta^a_{\mu\nu}$, and C^a_μ , we find that the transformations commute:

$$\begin{aligned} e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} \alpha^\alpha_\mu &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} \alpha^\alpha_\mu, \\ e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} \beta^a_{\mu\nu} &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} \beta^a_{\mu\nu}, \\ e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} C^a_\mu &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} C^a_\mu. \end{aligned} \quad (\text{D.47})$$

The remaining variable $B^\alpha_{\mu\nu}$ transforms under H -gauge transformations as follows:

$$\begin{aligned} B' &= B - C \wedge^\tau \epsilon_{\mathfrak{h}}, \\ C' &= C. \end{aligned} \tag{D.48}$$

Applying a further N -gauge transformation to the variable gives:

$$\begin{aligned} B'' &= B' - \beta' \wedge^\tau \epsilon_{\mathfrak{n}} \\ &= B - C \wedge^\tau \epsilon_{\mathfrak{h}} - \left(\beta - \frac{\{\alpha^\alpha - \partial_a^\alpha \epsilon_{\mathfrak{h}}^a\}}{\nabla \epsilon_{\mathfrak{h}}} - \epsilon_{\mathfrak{h}} \wedge \epsilon_{\mathfrak{h}} \right) \wedge^\tau \epsilon_{\mathfrak{n}}, \\ C'' &= C' - \frac{\{\alpha^\alpha - \partial_a^\alpha \epsilon_{\mathfrak{h}}^a\}}{\nabla} \epsilon_{\mathfrak{n}} \\ &= C - \frac{\{\alpha^\alpha - \partial_a^\alpha \epsilon_{\mathfrak{h}}^a\}}{\nabla} \epsilon_{\mathfrak{n}}. \end{aligned} \tag{D.49}$$

Next, let us reverse the order of the transformations. First, the transformation of the variables under an N -gauge transformation is

$$\begin{aligned} B^\cdot &= B - \beta \wedge^\tau \epsilon_{\mathfrak{n}}, \\ C^\cdot &= C - \nabla \epsilon_{\mathfrak{n}}, \end{aligned} \tag{D.50}$$

while a subsequent H -gauge transformation gives:

$$\begin{aligned} B^{\cdot\cdot} &= B^\cdot - C^\cdot \wedge^\tau \epsilon_{\mathfrak{h}} \\ &= B - \beta \wedge^\tau \epsilon_{\mathfrak{n}} - (C - \nabla \epsilon_{\mathfrak{n}}) \wedge^\tau \epsilon_{\mathfrak{h}}, \\ C^{\cdot\cdot} &= C^\cdot \\ &= C - \nabla \epsilon_{\mathfrak{n}}. \end{aligned} \tag{D.51}$$

The difference between equations (D.49) and (D.51) is:

$$\begin{aligned} (e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) B^\alpha &= \nabla \epsilon_{\mathfrak{n}}^a \wedge \epsilon_{\mathfrak{h}}^b \mathcal{T}_{ab}^\alpha - \nabla \epsilon_{\mathfrak{h}}^a \wedge \epsilon_{\mathfrak{n}}^b \mathcal{T}_{ab}^\alpha + \partial_a^\beta \triangleright_{\beta c}^b \epsilon_{\mathfrak{h}}^a \wedge \epsilon_{\mathfrak{h}}^c \wedge \epsilon_{\mathfrak{n}}^d \mathcal{T}_{bd}^\alpha - \epsilon_{\mathfrak{h}}^a \wedge \epsilon_{\mathfrak{h}}^b f_{ab}^c \epsilon_{\mathfrak{n}}^d \mathcal{T}_{cd}^\alpha, \\ (e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) C^c &= -\partial_a^\beta \epsilon_{\mathfrak{h}}^a \triangleright_{\beta b}^c \epsilon_{\mathfrak{n}}^b, \end{aligned} \tag{D.52}$$

Using the definition of the map $\mathcal{T}$, the preceding equations reduce to

$$\begin{aligned} (e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) B^\alpha &= \nabla \epsilon_{\mathfrak{n}}^a \wedge \epsilon_{\mathfrak{h}}^b \mathcal{T}_{ab}^\alpha - \nabla \epsilon_{\mathfrak{h}}^a \wedge \epsilon_{\mathfrak{n}}^b \mathcal{T}_{ab}^\alpha = \nabla (\epsilon_{\mathfrak{n}} \wedge^\tau \epsilon_{\mathfrak{h}})^\alpha, \\ (e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) C^c &= \partial_\alpha^c (\epsilon_{\mathfrak{n}} \wedge^\tau \epsilon_{\mathfrak{h}})^\alpha, \end{aligned} \tag{D.53}$$

Comparing equations (D.47) and (D.53) with equation (6.58), we obtain the final result for the commutator:

$$(e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) = -(\epsilon_{\mathfrak{n}} \wedge^\tau \epsilon_{\mathfrak{h}}) \cdot M. \tag{D.54}$$

D.3 Gauge transformations in topological $3BF$ theory

The generator of gauge transformations in $3BF$ theory is:

$$\begin{aligned}
G = \int_{\Sigma_3} d^3\vec{x} & \left((\nabla_0 \epsilon_{\mathfrak{g}}^\alpha) (\tilde{G}_1)_\alpha + \epsilon_{\mathfrak{g}}^\alpha (\tilde{G}_0)_\alpha + (\nabla_0 \epsilon_{\mathfrak{h}}^a{}_i) (\tilde{H}_1)_a{}^i + \epsilon_{\mathfrak{h}}^a{}_i (\tilde{H}_0)_a{}^i \right. \\
& + \frac{1}{2} (\nabla_0 \epsilon_{\mathfrak{l}}^A{}_{ij}) (\tilde{L}_1)_A{}^{ij} + \frac{1}{2} \epsilon_{\mathfrak{l}}^A{}_{ij} (\tilde{L}_0)_A{}^{ij} \\
& \left. + (\nabla_0 \epsilon_{\mathfrak{m}}^\alpha{}_i) (\tilde{M}_1)_\alpha{}^i + \epsilon_{\mathfrak{m}}^\alpha{}_i (\tilde{M}_0)_\alpha{}^i + (\nabla_0 \epsilon_{\mathfrak{n}}^a) (\tilde{N}_1)_a + \epsilon_{\mathfrak{n}}^a (\tilde{N}_0)_a \right), \tag{D.55}
\end{aligned}$$

where

$$\begin{aligned}
(\tilde{G}_1)_\alpha &= -\Phi(\alpha)_\alpha, \\
(\tilde{G}_0)_\alpha &= -\left(f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} + C_{a0} \triangleright_{\alpha b}{}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{\alpha b}{}^a \Phi(\beta)^{b0i} \right. \\
&\quad \left. - \frac{1}{2} \gamma^A{}_{0ij} \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} - \Phi(\nabla B)_\alpha \right), \\
(\tilde{H}_1)_a{}^i &= -\Phi(\beta)_a{}^i, \\
(\tilde{H}_0)_a{}^i &= C_{b0} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} - 2\beta^b{}_{0j} X_{(ab)}{}^A \Phi(\gamma)_A{}^{ij} + \Phi(\nabla C)_a{}^i, \\
(\tilde{L}_1)_A{}^{ij} &= \Phi(\gamma)_A{}^{ij}, \\
(\tilde{L}_0)_A{}^{ij} &= -\Phi(\nabla D)_A{}^{ij}, \\
(\tilde{M}_1)_\alpha{}^i &= -\Phi(B)_\alpha{}^i, \\
(\tilde{M}_0)_\alpha{}^i &= \Phi(\mathcal{F})_\alpha{}^i, \\
(\tilde{N}_1)_a &= -\Phi(C)_a, \\
(\tilde{N}_0)_a &= \beta_{b0i} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a, \tag{D.56}
\end{aligned}$$

where $\epsilon_{\mathfrak{g}}^\alpha$, $\epsilon_{\mathfrak{h}}^a{}_i$, $\epsilon_{\mathfrak{l}}^A{}_{ij}$, $\epsilon_{\mathfrak{m}}^\alpha{}_i$, and $\epsilon_{\mathfrak{n}}^a$ are independent gauge-transformation parameters.

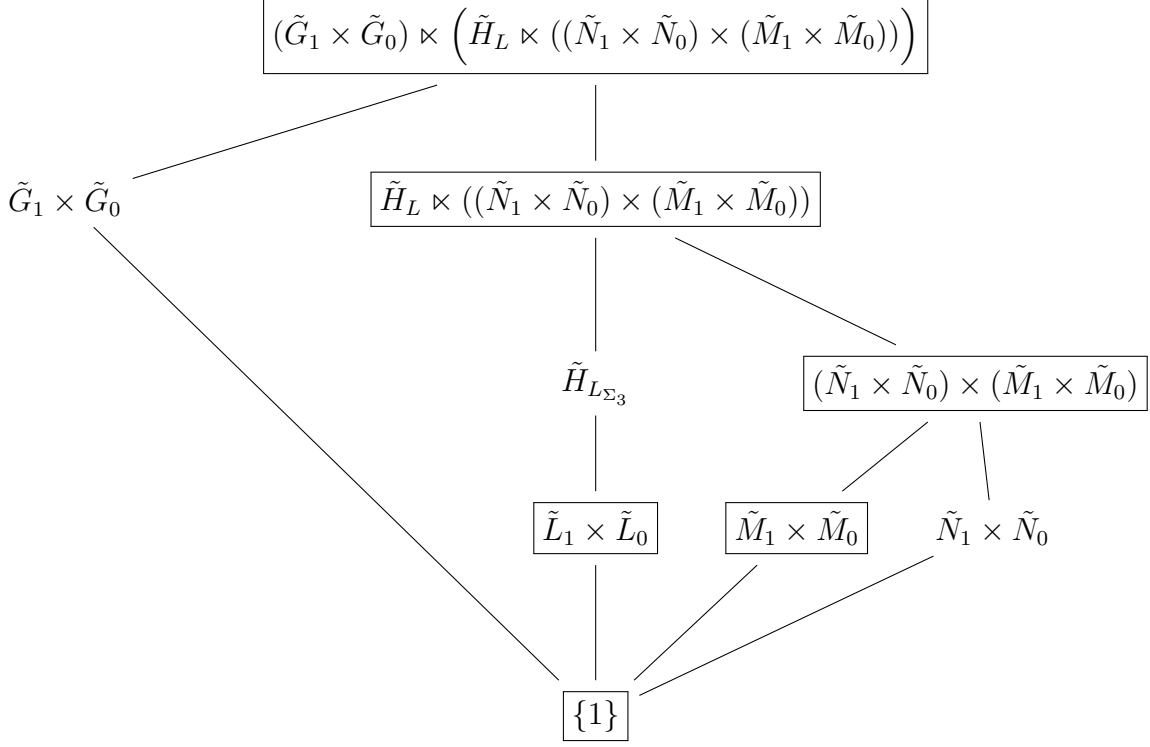
D.3.1 Gauge-symmetry group of the $3BF$ action

The algebra formed by the symmetry-group generators $(\tilde{M}_0)_\alpha{}^i$, $(\tilde{M}_1)_\alpha{}^i$, $(\tilde{G}_0)_\alpha$, $(\tilde{G}_1)_\alpha$, $(\tilde{H}_0)_a{}^i$, $(\tilde{H}_1)_a{}^i$, $(\tilde{N}_0)_a$, $(\tilde{N}_1)_a$, $(\tilde{L}_0)_A{}^{ij}$, and $(\tilde{L}_1)_A{}^{ij}$ defined in Appendix D.3 is:

$$\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{G}_0)_\beta(\vec{y})\} = f_{\alpha\beta}{}^\gamma (\tilde{G}_0)_\gamma \delta^{(3)}(\vec{x} - \vec{y}), \tag{D.57}$$

$$\begin{aligned}
\{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{H}_0)_b{}^j(\vec{y})\} &= 2X_{(ab)}{}^A (\tilde{L}_0)_A{}^{ij} \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{H}_1)_b{}^j(\vec{y})\} &= 2X_{(ab)}{}^A (\tilde{L}_1)_A{}^{ij} \delta^{(3)}(\vec{x} - \vec{y}), \tag{D.58}
\end{aligned}$$

$$\begin{aligned}
\{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{M}_0)^{\alpha i} \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{H}_1)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{M}_1)^{\alpha i} \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{H}_0)_a(\vec{x}), (\tilde{N}_1)^{bi}(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{M}_1)^{\alpha i} \delta^{(3)}(\vec{x} - \vec{y}), \tag{D.59}
\end{aligned}$$

Figure D.3: The symmetry group $\mathcal{G}_{\Sigma_3}$ in phase space. Normal subgroups are boxed.

$$\begin{aligned}
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{M}_0)_\beta{}^i(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{M}_0)_\gamma{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{M}_1)_\beta{}^i(\vec{y})\} &= f_{\alpha\beta}{}^\gamma (\tilde{M}_1)_\gamma{}^i \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{H}_1)_a{}^i(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{H}_1)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{H}_0)_a{}^i(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{H}_0)_b{}^i(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{N}_1)_a(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{N}_1)_b(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{N}_0)_a(\vec{y})\} &= \triangleright_{\alpha a}{}^b (\tilde{N}_0)_b(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}), \\
\{(\tilde{G}_0)_\alpha(\vec{x}), (\tilde{L}_0)_A{}^{ij}(\vec{y})\} &= \triangleright_{\alpha A}{}^B (\tilde{L}_0)_B{}^{ij}(\vec{x}) \delta^{(3)}(\vec{x} - \vec{y}).
\end{aligned} \tag{D.60}$$

The gauge-symmetry group has the following structure. First, the groups $\tilde{M}_1 \times \tilde{M}_0$, $\tilde{N}_1 \times \tilde{N}_0$, and $\tilde{L}_1 \times \tilde{L}_0$, with the corresponding algebras $\mathfrak{a}_1$, $\mathfrak{a}_2$, and $\mathfrak{a}_3$, where

$$\begin{aligned}
\mathfrak{a}_1 &= \text{span}\{(\tilde{M}_1)_\alpha{}^i\} \oplus \text{span}\{(\tilde{M}_0)_\alpha{}^i\}, & \mathfrak{a}_2 &= \text{span}\{(\tilde{N}_1)_a\} \oplus \text{span}\{(\tilde{N}_0)_a\}, \\
\mathfrak{a}_3 &= \text{span}\{(\tilde{L}_1)_A{}^{ij}\} \oplus \text{span}\{(\tilde{L}_0)_A{}^{ij}\},
\end{aligned} \tag{D.61}$$

are subgroups of the total symmetry group $\tilde{\mathcal{G}}_{\Sigma_3}$. In addition, the subgroups $\tilde{L}_1 \times \tilde{L}_0$ and $\tilde{M}_1 \times \tilde{M}_0$ are normal subgroups of the total symmetry group. The group $\tilde{N}_1 \times \tilde{N}_0$ is not a normal subgroup of the total symmetry group, as follows from the Poisson brackets $\{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\}$ and $\{(\tilde{H}_1)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\}$, which equal linear combinations of the generators of $\tilde{M}_1 \times \tilde{M}_0$. Since the generators of these groups commute with one another, one can form the direct product $(\tilde{N}_1 \times \tilde{N}_0) \times (\tilde{M}_1 \times \tilde{M}_0)$, and the resulting group is a normal subgroup of the total symmetry group.

Next, consider the subgroup $\tilde{H}_{L\Sigma_3}$ determined by the algebra generated by $(\tilde{L}_1)_A{}^{ij}$, $(\tilde{L}_0)_A{}^{ij}$, $(\tilde{H}_1)_a{}^i$, and $(\tilde{H}_0)_a{}^i$. This group is not a normal subgroup of the total symmetry group because of the Poisson brackets $\{(\tilde{H}_0)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\}$ and $\{(\tilde{H}_1)_a{}^i(\vec{x}), (\tilde{N}_0)^b(\vec{y})\}$, for the same reason as before. These two subgroups, one normal and the other not, can now be combined by a semidirect product, giving the normal subgroup $H_L \ltimes ((N_1 \times N_0) \times (M_1 \times M_0))$, corresponding to the algebra $\mathfrak{a}_4$:

$$\mathfrak{a}_4 = \text{span}\{(\tilde{M}_0)_\alpha{}^i, (\tilde{M}_1)_\alpha{}^i, (\tilde{H}_0)_a{}^i, (\tilde{H}_1)_a{}^i, (\tilde{N}_0)_a, (\tilde{N}_1)_a, (\tilde{L}_0)_A{}^{ij}, (\tilde{L}_1)_A{}^{ij}\}.$$

Finally, following the same line of reasoning and considering the group $\tilde{G}_1 \times \tilde{G}_0$, we obtain the total symmetry group $\tilde{\mathcal{G}}_{\Sigma_3}$:

$$\tilde{\mathcal{G}}_{\Sigma_3} = (\tilde{G}_1 \times \tilde{G}_0) \ltimes \left(\tilde{H}_L \ltimes ((\tilde{N}_1 \times \tilde{N}_0) \times (\tilde{M}_1 \times \tilde{M}_0)) \right).$$

The complete structure of the total symmetry group is shown in Figure D.3. Normal subgroups of the total symmetry group are boxed.

D.3.2 Construction of the symmetry generator of 3BF theory

Substituting the generators (D.56) into equation (6.30), we obtain the gauge-symmetry generator in 3BF theory in the following form:

$$\begin{aligned} G = & - \int_{\Sigma_3} d^3\vec{x} \left((\nabla_0 \epsilon_{\mathfrak{m}}^\alpha)_i \Phi(B)_\alpha{}^i - \epsilon_{\mathfrak{m}}^\alpha{}_i \Phi(\mathcal{F})_\alpha{}^i + (\nabla_0 \epsilon_{\mathfrak{g}}^\alpha) \Phi(\alpha)_\alpha + \epsilon_{\mathfrak{g}}^\alpha (f_{\alpha\gamma}{}^\beta B_{\beta 0i} \Phi(B)^{\gamma i} \right. \\ & + C_{a0} \triangleright_{ab}{}^a \Phi(C)^{b0} + \beta_{a0i} \triangleright_{ab}{}^a \Phi(\beta)^{b0i} - \frac{1}{2} \gamma^A{}_{0ij} \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} - \Phi(\nabla B)_\alpha \\ & + (\nabla_0 \epsilon_{\mathfrak{n}}^a) \Phi(C)_a - \epsilon_{\mathfrak{n}}^a (\beta_{b0i} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a) \\ & + (\nabla_0 \epsilon_{\mathfrak{h}}^a)_i \Phi(\beta)_a{}^i - \epsilon_{\mathfrak{h}}^a{}_i (C_{b0} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} - 2\beta^b{}_{0j} X_{(ab)}{}^A \Phi(\gamma)_A{}^{ij} + \Phi(\nabla C)_a{}^i) \\ & \left. - \frac{1}{2} (\nabla_0 \epsilon_{\mathfrak{l}}^A{}_{ij}) \Phi(\gamma)_A{}^{ij} + \frac{1}{2} \epsilon_{\mathfrak{l}}^A{}_{ij} \Phi(\nabla D)_A{}^{ij} \right), \end{aligned} \quad (\text{D.62})$$

where $\epsilon_{\mathfrak{g}}^\alpha$, $\epsilon_{\mathfrak{h}}^a{}_i$, $\epsilon_{\mathfrak{l}}^A{}_{ij}$, $\epsilon_{\mathfrak{m}}^\alpha{}_i$, and $\epsilon_{\mathfrak{n}}^a$ are independent gauge-transformation parameters.

The gauge-symmetry generator (D.55) in 3BF theory (6.1) is obtained using the Castellani procedure, with the following relations satisfied for each pair of generators G_0 and G_1 :

$$G_1 = C_{PFC}, \quad (\text{D.63})$$

$$G_0 + \{G_1, H_T\} = C_{PFC}, \quad (\text{D.64})$$

$$\{G_0, H_T\} = C_{PFC}, \quad (\text{D.65})$$

where C_{PFC} denotes a first-class constraint. Let us first assume that the generator has the structure:

$$\begin{aligned} G = & \int_{\Sigma_3} d^3\vec{x} \left(\dot{\epsilon}_{\mathfrak{m}}^\alpha{}_i (G_1)_{\mathfrak{m}\alpha}{}^i + \epsilon_{\mathfrak{m}}^\alpha{}_i (G_0)_{\mathfrak{m}\alpha}{}^i + \dot{\epsilon}_{\mathfrak{g}}^\alpha (G_1)_{\mathfrak{g}\alpha} + \epsilon_{\mathfrak{g}}^\alpha (G_0)_{\mathfrak{g}\alpha} \right. \\ & + \dot{\epsilon}_{\mathfrak{h}}^a{}_i (G_1)_{\mathfrak{h}a}{}^i + \epsilon_{\mathfrak{h}}^a{}_i (G_0)_{\mathfrak{h}a}{}^i + \dot{\epsilon}_{\mathfrak{n}}^a (G_1)_{\mathfrak{n}a} + \epsilon_{\mathfrak{n}}^a (G_0)_{\mathfrak{n}a} \\ & \left. + \frac{1}{2} \dot{\epsilon}_{\mathfrak{l}}^A{}_{ij} (G_1)_{\mathfrak{l}A}{}^{ij} + \frac{1}{2} \epsilon_{\mathfrak{l}}^A{}_{ij} (G_0)_{\mathfrak{l}A}{}^{ij} \right). \end{aligned} \quad (\text{D.66})$$

The first step of the Castellani procedure is to impose the conditions

$$\begin{aligned} (G_1)_{\mathfrak{m}\alpha}{}^i &= C_{PFC}, & (G_1)_{\mathfrak{g}\alpha} &= C_{PFC}, & (G_1)_{\mathfrak{h}a}{}^i &= C_{PFC}, \\ (G_1)_{\mathfrak{n}a} &= C_{PFC}, & (G_1)_{\mathfrak{l}A}{}^{ij} &= C_{PFC}, \end{aligned} \quad (\text{D.67})$$

with the natural choice:

$$\begin{aligned} (G_1)_{\mathfrak{m}\alpha}{}^i &= -\Phi(B)_\alpha{}^i, & (G_1)_{\mathfrak{g}\alpha} &= -\Phi(\alpha)_\alpha, & (G_1)_{\mathfrak{h}a}{}^i &= -\Phi(C)_\alpha{}^i, \\ (G_1)_{\mathfrak{n}a} &= -\Phi(\beta)_a, & (G_1)_{\mathfrak{l}A}{}^{ij} &= \Phi(\gamma)_A{}^{ij}. \end{aligned} \quad (\text{D.68})$$

It remains to determine the five generators G_0 .

Castellani's second condition for the generator $(G_0)_{\mathfrak{m}\alpha}{}^i$ gives

$$\begin{aligned} (G_0)_{\mathfrak{m}\alpha}{}^i - \{\Phi(B)_\alpha{}^i, H_T\} &= (C_{PFC})_\alpha{}^i, \\ (G_0)_{\mathfrak{m}\alpha}{}^i - \Phi(\mathcal{F})_\alpha{}^i &= (C_{PFC})_\alpha{}^i, \end{aligned} \quad (\text{D.69})$$

where $(G_0)_{\mathfrak{m}\alpha}{}^i = (C_{PFC})_\alpha{}^i + \Phi(\mathcal{F})_\alpha{}^i$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{m}\alpha}{}^i, H_T\} &= (C_{PFC1})_\alpha{}^i, \\ \{(C_{PFC})_\alpha{}^i + \Phi(\mathcal{F})_\alpha{}^i, H_T\} &= (C_{PFC1})_\alpha{}^i, \\ \{(C_{PFC})_\alpha{}^i, H_T\} - f_{\beta\gamma\alpha}\alpha^\beta{}_0\Phi(\mathcal{F})^{\gamma i} &= (C_{PFC1})_\alpha{}^i, \end{aligned} \quad (\text{D.70})$$

which gives the equation

$$(C_{PFC})_\alpha{}^i = f_{\beta\gamma\alpha}\alpha^\beta{}_0\Phi(B)^{\gamma i}.$$

It follows that the generator is:

$$(G_0)_{\mathfrak{m}\alpha}{}^i = f_{\beta\gamma\alpha}\alpha^\beta{}_0\Phi(B)^{\gamma i} + \Phi(\mathcal{F})_\alpha{}^i. \quad (\text{D.71})$$

Castellani's second condition for the generator $(G_0)_{\mathfrak{g}\alpha}$ gives

$$\begin{aligned} (G_0)_{\mathfrak{g}\alpha} - \{\Phi(\alpha)_\alpha, H_T\} &= (C_{PFC})_\alpha, \\ (G_0)_{\mathfrak{g}\alpha} - \Phi(\nabla B)_\alpha &= (C_{PFC})_\alpha, \end{aligned} \quad (\text{D.72})$$

that is, $(G_0)_{\mathfrak{g}\alpha} = (C_{PFC})_\alpha + \Phi(\nabla B)_\alpha$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{g}\alpha}, H_T\} &= (C_{PFC1})_\alpha, \\ \{(C_{PFC})_\alpha + \Phi(\nabla B)_\alpha, H_T\} &= (C_{PFC1})_\alpha, \\ \{(C_{PFC})_\alpha, H_T\} + B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(\mathcal{F})^{\gamma i} - \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\nabla B)_\gamma \\ + C_{a0}\triangleright_{\alpha b}{}^a\Phi(\mathcal{G})^b + \beta_{a0i}\triangleright_{\alpha b}{}^a\Phi(\nabla C)^{bi} - \frac{1}{2}\gamma^A{}_{0ij}\triangleright_{\alpha A}{}^B\Phi(\nabla D)_B{}^{ij} &= (C_{PFC1})_\alpha, \end{aligned} \quad (\text{D.73})$$

that is,

$$(C_{PFC})_\alpha = -B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(B)^{\gamma i} + \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\alpha)_\gamma - C_{a0}\triangleright_{\alpha b}{}^a\Phi(C)^b - \beta_{a0i}\triangleright_{\alpha b}{}^a\Phi(\beta)^{bi} + \frac{1}{2}\gamma^A{}_{0ij}\triangleright_{\alpha A}{}^B\Phi(\gamma)_B{}^{ij}.$$

It follows that the generator is:

$$\begin{aligned} (G_0)_{\mathfrak{g}\alpha} &= -B_{\beta 0i}f_{\alpha\gamma}{}^\beta\Phi(B)^{\gamma i} + \alpha^\beta{}_0f_{\alpha\beta}{}^\gamma\Phi(\alpha)_\gamma - C_{a0}\triangleright_{\alpha b}{}^a\Phi(C)^b \\ &\quad - \beta_{a0i}\triangleright_{\alpha b}{}^a\Phi(\beta)^{bi} + \frac{1}{2}\gamma^A{}_{0ij}\triangleright_{\alpha A}{}^B\Phi(\gamma)_B{}^{ij} + \Phi(\nabla B)_\alpha. \end{aligned} \quad (\text{D.74})$$

Castellani's second condition for the generator $(G_0)_{na}$ gives

$$\begin{aligned} (G_0)_{na} - \{\Phi(C)_a, H_T\} &= (C_{PFC})_a, \\ (G_0)_{na} - \Phi(\mathcal{G})_a &= (C_{PFC})_a, \end{aligned} \quad (D.75)$$

where $(G_0)_{na} = (C_{PFC})_a + \Phi(\mathcal{G})_a$.

The third Castellani condition then gives

$$\begin{aligned} \{(G_0)_{na}, H_T\} &= (C_{PFC1})_a, \\ \{(C_{PFC})_a + \Phi(\mathcal{G})_a, H_T\} &= (C_{PFC1})_a, \\ \{(C_{PFC})_a, H_T\} + \alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b\Phi(\mathcal{G})_b - \beta_{b0i} \triangleright_{\alpha a} {}^b\Phi(\mathcal{F})^{\alpha i} &= (C_{PFC1})_a, \end{aligned} \quad (D.76)$$

which gives

$$(C_{PFC})_a = -\alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b\Phi(C)_b + \beta_{b0i} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i}.$$

It follows that the generator is:

$$(G_0)_{na} = -\alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b\Phi(C)_b + \beta_{b0i} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a.$$

Castellani's second condition for the generator $(G_0)_{\mathfrak{h}a}{}^i$ gives

$$\begin{aligned} (G_0)_{\mathfrak{h}a}{}^i - \{\Phi(\beta)_a{}^i, H_T\} &= (C_{PFC})_a{}^i, \\ (G_0)_{\mathfrak{h}a}{}^i - \Phi(\nabla C)_a{}^i &= (C_{PFC})_a{}^i, \end{aligned} \quad (D.77)$$

that is, $(G_0)_{\mathfrak{h}a}{}^i = (C_{PFC})_a{}^i + \Phi(\nabla C)_a{}^i$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{h}a}{}^i, H_T\} &= (C_{PFC1})_a{}^i, \\ \{(C_{PFC})_a{}^i + \Phi(\nabla C)_a{}^i, H_T\} &= (C_{PFC1})_a{}^i, \\ \{(C_{PFC})_a{}^i, H_T\} + \alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b\Phi(\nabla C)_b{}^i - C_{b0} \triangleright_{\alpha a} {}^b\Phi(\mathcal{F})^{\alpha i} + 2\beta^b{}_{0j} X_{(ab)}{}^A \Phi(\nabla D)_A{}^{ij} &= (C_{PFC1})_a{}^i, \end{aligned}$$

which gives the result:

$$(C_{PFC})_a{}^i = -\alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b\Phi(\beta)_b{}^i + C_{b0} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i} - 2\beta^b{}_{0j} X_{(ab)}{}^A \Phi(\gamma)_A{}^{ij}.$$

It follows that the generator is:

$$(G_0)_{\mathfrak{h}a}{}^i = -\alpha^\alpha{}_0 \triangleright_{\alpha a} {}^b\Phi(\beta)_b{}^i + C_{b0} \triangleright_{\alpha a} {}^b\Phi(B)^{\alpha i} - 2\beta^b{}_{0j} X_{(ab)}{}^A \Phi(\gamma)_A{}^{ij} + \Phi(\nabla C)_a{}^i.$$

Castellani's second condition for the generator $(G_0)_{\mathfrak{l}A}{}^{ij}$ gives:

$$\begin{aligned} (G_0)_{\mathfrak{l}A}{}^{ij} + \{\Phi(\gamma)_A{}^{ij}, H_T\} &= (C_{PFC})_A{}^{ij}, \\ (G_0)_{\mathfrak{l}A}{}^{ij} + \Phi(\nabla D)_A{}^{ij} &= (C_{PFC})_A{}^{ij}, \end{aligned} \quad (D.78)$$

that is, we obtain $(G_0)_{\mathfrak{l}A}{}^{ij} = (C_{PFC})_A{}^{ij} - \Phi(\nabla D)_A{}^{ij}$. Castellani's third condition then gives

$$\begin{aligned} \{(G_0)_{\mathfrak{l}A}{}^{ij}, H_T\} &= (C_{PFC1})_A{}^{ij}, \\ \{(C_{PFC})_A{}^{ij} - \Phi(\nabla D)_A{}^{ij}, H_T\} &= (C_{PFC1})_A{}^{ij}, \\ \{(C_{PFC})_A{}^{ij}, H_T\} - \alpha^\alpha{}_0 \triangleright_{\alpha A} {}^B\Phi(\nabla D)_B{}^{ij} &= (C_{PFC1})_A{}^{ij}, \end{aligned} \quad (D.79)$$

which gives the result

$$(C_{PFC})_A{}^{ij} = \alpha^\alpha{}_0 \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij}.$$

The generator is therefore:

$$(G_0)_{IA}{}^{ij} = \alpha^\alpha{}_0 \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} - \Phi(\nabla D)_A{}^{ij}. \quad (\text{D.80})$$

At this point, it is useful to summarize the results and introduce new notation:

$$\begin{aligned} \dot{\epsilon}_m^\alpha{}_i (G_1)_{m\alpha}{}^i + \epsilon_m^\alpha{}_i (G_0)_{m\alpha}{}^i &= -\nabla_0 \epsilon_m^\alpha{}_i \Phi(B)_\alpha{}^i + \epsilon_m^\alpha{}_i \Phi(\mathcal{F})_\alpha{}^i \\ &= \nabla_0 \epsilon_m^\alpha{}_i (\tilde{M}_1)_\alpha{}^i + \epsilon_m^\alpha{}_i (\tilde{M}_0)_\alpha{}^i. \end{aligned} \quad (\text{D.81})$$

Notice that the time derivative of the parameter combines with other terms into a covariant derivative in the time direction.

For the second part of the total generator, we obtain:

$$\begin{aligned} \dot{\epsilon}_g^\alpha (G_1)_{g\alpha} + \epsilon_g^\alpha (G_0)_{g\alpha} &= -\dot{\epsilon}_g^\alpha \Phi(\alpha)_\alpha - \epsilon_g^\alpha (B_{\beta 0i} f_{\alpha\gamma}{}^\beta \Phi(B)^{\gamma i} - \alpha^\beta{}_0 f_{\alpha\beta}{}^\gamma \Phi(\alpha)_\gamma \\ &\quad + C_{a0} \triangleright_{ab}{}^a \Phi(C)^b + \beta_{a0i} \triangleright_{ab}{}^a \Phi(\beta)_b{}^i - \frac{1}{2} \gamma^A{}_{0ij} \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} - \Phi(\nabla B)_\alpha) \\ &= -\nabla_0 \epsilon_g^\alpha \Phi(\alpha)_\alpha - \epsilon_g^\alpha (B_{\beta 0i} f_{\alpha\gamma}{}^\beta \Phi(B)^{\gamma i} \\ &\quad + C_{a0} \triangleright_{ab}{}^a \Phi(C)^b + \beta_{a0i} \triangleright_{ab}{}^a \Phi(\beta)_b{}^i - \frac{1}{2} \gamma^A{}_{0ij} \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} - \Phi(\nabla B)_\alpha) \\ &= \nabla_0 \epsilon_g^\alpha (\tilde{G}_1)_\alpha + \epsilon_g^\alpha (\tilde{G}_0)_\alpha. \end{aligned} \quad (\text{D.82})$$

In addition, it follows that:

$$\begin{aligned} \dot{\epsilon}_h^a{}_i (G_1)_{h\alpha}{}^i + \epsilon_h^a{}_i (G_0)_{h\alpha}{}^i &= -\nabla_0 \epsilon_h^a{}_i \Phi(\beta)_\alpha{}^i + \epsilon_h^a{}_i (C_{b0} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} - 2\beta^b{}_{0j} X_{(ab)}{}^A \Phi(\gamma)_A{}^{ij} + \Phi(\nabla C)_a{}^i) \\ &= \nabla_0 \epsilon_h^a{}_i (\tilde{H}_1)_a{}^i + \epsilon_h^a{}_i (\tilde{H}_0)_a{}^i, \end{aligned} \quad (\text{D.83})$$

$$\begin{aligned} \dot{\epsilon}_n^a (G_1)_{na} + \epsilon_n^a (G_0)_{na} &= -\nabla_0 \epsilon_n^a \Phi(C)_a + \epsilon_n^a (\beta_{b0i} \triangleright_{\alpha a}{}^b \Phi(B)^{\alpha i} + \Phi(\mathcal{G})_a) \\ &= \nabla_0 \epsilon_n^a (\tilde{N}_1)_a + \epsilon_n^a (\tilde{N}_0)_a. \end{aligned} \quad (\text{D.84})$$

Finally, we obtain

$$\begin{aligned} \frac{1}{2} \dot{\epsilon}_l^A{}_{ij} (G_1)_{lA}{}^{ij} + \frac{1}{2} \epsilon_l^A{}_{ij} (G_0)_{lA}{}^{ij} &= \frac{1}{2} \dot{\epsilon}_l^A{}_{ij} \Phi(\gamma)_A{}^{ij} + \frac{1}{2} \epsilon_l^A{}_{ij} \alpha^\alpha{}_0 \triangleright_{\alpha A}{}^B \Phi(\gamma)_B{}^{ij} \\ &\quad - \frac{1}{2} \epsilon_l^A{}_{ij} \Phi(\nabla D)_A{}^{ij} \\ &= \frac{1}{2} \nabla_0 \epsilon_l^A{}_{ij} \Phi(\gamma)_A{}^{ij} - \frac{1}{2} \epsilon_l^A{}_{ij} \Phi(\nabla D)_A{}^{ij} \\ &= \frac{1}{2} \nabla_0 \epsilon_l^A{}_{ij} (\tilde{L}_1)_A{}^{ij} + \frac{1}{2} \epsilon_l^A{}_{ij} (\tilde{L}_0)_A{}^{ij}. \end{aligned} \quad (\text{D.85})$$

D.3.3 Calculation of the symmetry algebra of the $3BF$ action

To obtain the structure of the symmetry group of the $3BF$ action, as presented in Subsection 6.1.2, we must first calculate the commutators among the generators of the G -, H -, L -, M -, and N -gauge symmetries. This procedure is described in Section 6.1.2, while the details of the calculation are given in this section.

The commutator $[H, H]$

We now calculate the commutator of the generators of H -gauge transformations, that is, equation (6.49). After transforming the variables under an H -gauge transformation with parameter $\epsilon_{\mathfrak{h}1}$, we obtain

$$\alpha' = \alpha - \partial\epsilon_{\mathfrak{h}1}, \quad (\text{D.86})$$

$$\beta' = \beta - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1}, \quad (\text{D.87})$$

$$\gamma' = \gamma + \{\beta - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1}, \epsilon_{\mathfrak{h}1}\}_{\text{pf}} + \{\epsilon_{\mathfrak{h}1}, \beta\}_{\text{pf}}, \quad (\text{D.88})$$

$$B' = B - (C - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}1}) \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge^{\mathcal{D}} \epsilon_{\mathfrak{h}1} \wedge^{\mathcal{D}} D, \quad (\text{D.89})$$

$$C' = C - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}1}, \quad (\text{D.90})$$

$$D' = D. \quad (\text{D.91})$$

Applying a further H -gauge transformation with parameter $\epsilon_{\mathfrak{h}2}$ gives:

$$\begin{aligned} \alpha'' &= \alpha - \partial\epsilon_{\mathfrak{h}1} - \partial\epsilon_{\mathfrak{h}2}, \\ \beta'' &= \beta - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1} - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1} - \partial\epsilon_{\mathfrak{h}2}} \epsilon_{\mathfrak{h}2} - \epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}2}, \\ \gamma'' &= \gamma + \{\beta - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1}, \epsilon_{\mathfrak{h}1}\}_{\text{pf}} + \{\epsilon_{\mathfrak{h}1}, \beta\}_{\text{pf}} \\ &\quad + \{\beta - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1} - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1} - \partial\epsilon_{\mathfrak{h}2}} \epsilon_{\mathfrak{h}2} - \epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}2}, \epsilon_{\mathfrak{h}2}\}_{\text{pf}} \\ &\quad + \{\epsilon_{\mathfrak{h}2}, \beta - \nabla^{\alpha - \partial\epsilon_{\mathfrak{h}1}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}1}\}_{\text{pf}}, \\ B'' &= B - (C - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}1}) \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}1} - \epsilon_{\mathfrak{h}1} \wedge^{\mathcal{D}} \epsilon_{\mathfrak{h}1} \wedge^{\mathcal{D}} D \\ &\quad - (C - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}2} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}2}) \wedge^{\mathcal{T}} \epsilon_{\mathfrak{h}2} - \epsilon_{\mathfrak{h}2} \wedge^{\mathcal{D}} \epsilon_{\mathfrak{h}2} \wedge^{\mathcal{D}} D, \\ C'' &= C - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}1} - D \wedge^{\mathcal{X}_1} \epsilon_{\mathfrak{h}2} - D \wedge^{\mathcal{X}_2} \epsilon_{\mathfrak{h}2}, \\ D'' &= D. \end{aligned} \quad (\text{D.92})$$

For the variables $\alpha^\alpha{}_\mu$, $C^a{}_\mu$, and D^A , we find that the transformations commute:

$$\begin{aligned} e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} \alpha^\alpha{}_\mu &= e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} \alpha^\alpha{}_\mu, \\ e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} C^a{}_\mu &= e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} C^a{}_\mu, \\ e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} D^A &= e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} D^A. \end{aligned} \quad (\text{D.93})$$

For the remaining variables, $\beta_{\mu\nu}^a$, $\gamma_{\mu\nu\rho}^A$, and $B_{\mu\nu}^\alpha$, the difference between equation (D.3.3) and the analogous equation with $\epsilon_{\mathfrak{h}1} \leftrightarrow \epsilon_{\mathfrak{h}2}$ is:

$$\begin{aligned}
(e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} - e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H}) \frac{1}{2} \beta_{\mu\nu}^a &= \partial_b^\alpha \epsilon_{\mathfrak{h}2}{}^b{}_{[\mu} \epsilon_{\mathfrak{h}1}{}^c{}_{\nu]} \triangleright_{\alpha c}{}^a - \partial_b^\alpha \epsilon_{\mathfrak{h}1}{}^b{}_{[\mu} \epsilon_{\mathfrak{h}2}{}^c{}_{\nu]} \triangleright_{\alpha c}{}^a \\
&= 2\delta_A{}^a X_{(bc)}{}^A \epsilon_{\mathfrak{h}1}{}^b{}_{[\mu} \epsilon_{\mathfrak{h}2}{}^c{}_{\nu]} \\
&= \delta_A{}^a (\{\epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}2}\}_{\text{pf}} - \{\epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}1}\}_{\text{pf}})^A{}_{\mu\nu}, \\
(e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} - e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H}) \frac{1}{3!} \gamma_{\mu\nu\rho}^A &= 2(\partial_{[\mu} \epsilon_{\mathfrak{h}1}{}^a{}_{\nu]} \epsilon_{\mathfrak{h}2}{}^b{}_{\rho]} X_{(ab)}{}^A + 2\epsilon_{\mathfrak{h}1}{}^a{}_{[\nu} (\partial_{\mu} \epsilon_{\mathfrak{h}2}{}^b{}_{\rho]}) X_{(ab)}{}^A \\
&\quad + 2\alpha_{[\mu} \epsilon_{\mathfrak{h}1}{}^a{}_{\nu} \epsilon_{\mathfrak{h}2}{}^b{}_{\rho]} X_{(ab)}{}^B \triangleright_{\alpha B}{}^A \\
&= \nabla_{[\mu} (\{\epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}2}\}_{\text{pf}} - \{\epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}1}\}_{\text{pf}})^A{}_{\nu\rho]}, \\
(e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} - e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H}) \frac{1}{2} B_{\mu\nu}^\alpha &= D^A \epsilon_{\mathfrak{h}2}{}^a{}_{[\mu} \epsilon_{\mathfrak{h}1}{}^b{}_{\nu]} (X_{1Aa}{}^c + X_{2Aa}{}^c) \mathcal{T}_{cb}{}^\alpha \\
&\quad - D^A \epsilon_{\mathfrak{h}1}{}^b{}_{[\mu} \epsilon_{\mathfrak{h}2}{}^a{}_{\nu]} (X_{1Ab}{}^c + X_{2Ab}{}^c) \mathcal{T}_{ca}{}^\alpha \\
&= -2D_A \epsilon_{\mathfrak{h}1}{}^a{}_{[\mu} \epsilon_{\mathfrak{h}2}{}^b{}_{\nu]} (X_{(ac)}{}^A \triangleright_{ab}{}^c + X_{(bc)}{}^A \triangleright_{ac}{}^c) \\
&= -2D_A \epsilon_{\mathfrak{h}1}{}^a{}_{[\mu} \epsilon_{\mathfrak{h}2}{}^b{}_{\nu]} X_{(ab)}{}^B \triangleright_{\alpha B}{}^A \\
&= (D \wedge^S (\{\epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}2}\}_{\text{pf}} - \{\epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}1}\}_{\text{pf}}))^\alpha{}_{\mu\nu}.
\end{aligned} \tag{D.94}$$

Comparing equations (D.93) and (D.94) with equations (6.45), we conclude that the commutator of two H -gauge transformations is an L -gauge transformation with parameter $\epsilon_{\mathfrak{f}}^A{}_{\mu\nu} = 4\epsilon_{\mathfrak{h}1}{}^a{}_{[\mu} \epsilon_{\mathfrak{h}2}{}^b{}_{\nu]} X_{(ac)}{}^A$:

$$e^{\epsilon_{\mathfrak{h}1} \cdot H} e^{\epsilon_{\mathfrak{h}2} \cdot H} - e^{\epsilon_{\mathfrak{h}2} \cdot H} e^{\epsilon_{\mathfrak{h}1} \cdot H} = 2(\{\epsilon_{\mathfrak{h}1} \wedge \epsilon_{\mathfrak{h}2}\}_{\text{pf}} - \{\epsilon_{\mathfrak{h}2} \wedge \epsilon_{\mathfrak{h}1}\}_{\text{pf}}) \cdot \hat{L}. \tag{D.95}$$

The commutator $[H, N]$

Let us calculate the commutator between the generators of H -gauge and N -gauge transformations, that is, derive equation (6.81). This is done by calculating the expression

$$(e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} - e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H}) A, \tag{D.96}$$

for all variables A present in the theory. For the variables α_μ^α , $\beta_{\mu\nu}^a$, $\gamma_{\mu\nu\rho}^A$, and D^A , we find that the transformations commute:

$$\begin{aligned}
e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} \alpha_\mu^\alpha &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} \alpha_\mu^\alpha, \\
e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} \beta_{\mu\nu}^a &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} \beta_{\mu\nu}^a, \\
e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} \gamma_{\mu\nu\rho}^A &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} \gamma_{\mu\nu\rho}^A, \\
e^{\epsilon_{\mathfrak{h}} \cdot H} e^{\epsilon_{\mathfrak{n}} \cdot N} D^A &= e^{\epsilon_{\mathfrak{n}} \cdot N} e^{\epsilon_{\mathfrak{h}} \cdot H} D^A.
\end{aligned} \tag{D.97}$$

The remaining variables $B_{\mu\nu}^\alpha$ and C_μ^a transform under H -gauge transformations as follows:

$$\begin{aligned}
B' &= B - (C - D \wedge^{\chi_1} \epsilon_{\mathfrak{h}} - D \wedge^{\chi_2} \epsilon_{\mathfrak{h}}) \wedge^\tau \epsilon_{\mathfrak{h}} - \epsilon_{\mathfrak{h}} \wedge^{\mathcal{D}} \epsilon_{\mathfrak{h}} \wedge^{\mathcal{D}} D, \\
C' &= C - D \wedge^{\chi_1} \epsilon_{\mathfrak{h}} - D \wedge^{\chi_2} \epsilon_{\mathfrak{h}}.
\end{aligned} \tag{D.98}$$

Applying a further N -gauge transformation to these variables gives:

$$\begin{aligned}
B'' &= B' - \beta' \wedge^{\mathcal{T}} \epsilon_n \\
&= B - (C - D \wedge^{\chi_1} \epsilon_h - D \wedge^{\chi_2} \epsilon_h) \wedge^{\mathcal{T}} \epsilon_h - \epsilon_h \wedge^{\mathcal{D}} \epsilon_h \wedge^{\mathcal{D}} D - (\beta - \frac{\{\alpha^\alpha - \partial_a^\alpha \epsilon_h^a\}}{\nabla \epsilon_h} - \epsilon_h \wedge \epsilon_h) \wedge^{\mathcal{T}} \epsilon_n, \\
C'' &= C' - \frac{\{\alpha^\alpha - \partial_a^\alpha \epsilon_h^a\}}{\nabla} \epsilon_n \\
&= C - D \wedge^{\chi_1} \epsilon_h - D \wedge^{\chi_2} \epsilon_h - \frac{\{\alpha^\alpha - \partial_a^\alpha \epsilon_h^a\}}{\nabla} \epsilon_n.
\end{aligned} \tag{D.99}$$

Next, let us reverse the order of the transformations. First, the transformation of the variables under an N -gauge transformation is

$$\begin{aligned}
B^\cdot &= B - \beta \wedge^{\mathcal{T}} \epsilon_n, \\
C^\cdot &= C - \nabla \epsilon_n,
\end{aligned} \tag{D.100}$$

while a subsequent H -gauge transformation gives:

$$\begin{aligned}
B^{\cdot\cdot} &= B^\cdot - (C^\cdot - D^\cdot \wedge^{\chi_1} \epsilon_h - D^\cdot \wedge^{\chi_2} \epsilon_h) \wedge^{\mathcal{T}} \epsilon_h - \epsilon_h \wedge^{\mathcal{D}} \epsilon_h \wedge^{\mathcal{D}} D^\cdot \\
&= B - \beta \wedge^{\mathcal{T}} \epsilon_n - (C - \nabla \epsilon_n - (D + \delta \epsilon_n) \wedge^{\chi_1} \epsilon_h - (D + \delta \epsilon_n) \wedge^{\chi_2} \epsilon_h) \wedge^{\mathcal{T}} \epsilon_h - \epsilon_h \wedge^{\mathcal{D}} \epsilon_h \wedge^{\mathcal{D}} (D + \delta \epsilon_n), \\
C^{\cdot\cdot} &= C^\cdot - D^\cdot \wedge^{\chi_1} \epsilon_h - D^\cdot \wedge^{\chi_2} \epsilon_h \\
&= C - \nabla \epsilon_n - (D + \delta \epsilon_n) \wedge^{\chi_1} \epsilon_h - (D + \delta \epsilon_n) \wedge^{\chi_2} \epsilon_h.
\end{aligned} \tag{D.101}$$

The difference between equations (D.99) and (D.101) is:

$$\begin{aligned}
(e^{\epsilon_h \cdot H} e^{\epsilon_n \cdot N} - e^{\epsilon_n \cdot N} e^{\epsilon_h \cdot H}) B^\alpha &= \nabla \epsilon_n^a \wedge \epsilon_h^b \mathcal{T}_{ab}^\alpha + \delta^A_a \epsilon_n^a \epsilon_h^b \wedge \epsilon_h^d X_{1Ab}^c \mathcal{T}_{cd}^\alpha \\
&\quad + \delta^A_a \epsilon_n^a \epsilon_h^b \wedge \epsilon_h^d X_{2Ab}^c \mathcal{T}_{cd}^\alpha - \epsilon_h^a \wedge \epsilon_h^b \delta_A^c \epsilon_n^c D_{Aab}^\alpha, \\
&\quad \nabla \epsilon_h^a \wedge \epsilon_n^b \mathcal{T}_{ab}^\alpha + \partial_a^\beta \epsilon_h^a \triangleright_{\beta c}^b \epsilon_h^c \epsilon_n^d \mathcal{T}_{bd}^\alpha - \epsilon_h^a \wedge \epsilon_h^b f_{ab}^c \epsilon_n^d \mathcal{T}_{cd}^\alpha, \\
(e^{\epsilon_h \cdot H} e^{\epsilon_n \cdot N} - e^{\epsilon_n \cdot N} e^{\epsilon_h \cdot H}) C^c &= -(\delta^A_a \epsilon_n^a) \wedge \epsilon_h^b X_{1Ab}^c - (\delta^A_a \epsilon_n^a) \wedge \epsilon_h^b X_{2Ab}^c - \partial_a^\beta \epsilon_h^a \triangleright_{\beta b}^c \epsilon_n^b,
\end{aligned} \tag{D.102}$$

Using the definitions of the maps $\mathcal{T}$, $\mathcal{D}$, χ_1 , and χ_2 , the preceding equations reduce to

$$\begin{aligned}
(e^{\epsilon_h \cdot H} e^{\epsilon_n \cdot N} - e^{\epsilon_n \cdot N} e^{\epsilon_h \cdot H}) B^\alpha &= \nabla \epsilon_n^a \wedge \epsilon_h^b \mathcal{T}_{ab}^\alpha - \nabla \epsilon_h^a \wedge \epsilon_n^b \mathcal{T}_{ab}^\alpha = \nabla (\epsilon_n \wedge^{\mathcal{T}} \epsilon_h)^\alpha, \\
(e^{\epsilon_h \cdot H} e^{\epsilon_n \cdot N} - e^{\epsilon_n \cdot N} e^{\epsilon_h \cdot H}) C^c &= \partial_\alpha^c (\epsilon_n \wedge^{\mathcal{T}} \epsilon_h)^\alpha,
\end{aligned} \tag{D.103}$$

Comparing equations (D.97) and (D.103) with equation (6.58), we obtain the final result for the commutator:

$$(e^{\epsilon_h \cdot H} e^{\epsilon_n \cdot N} - e^{\epsilon_n \cdot N} e^{\epsilon_h \cdot H}) = -(\epsilon_n \wedge^{\mathcal{T}} \epsilon_h) \cdot M. \tag{D.104}$$

Appendix E

Pachner-move invariance of the state sum

E.1 Invariance of the $2BF$ state sum under Pachner moves

E.1.1 $n = 3$

The Pachner move $1 \leftrightarrow 4$

The *left-hand side* of the $1 \leftrightarrow 4$ Pachner move is given by expression (8.20) and cannot be simplified:

$$l.s. = \delta_H(h_{134} g_{34} \triangleright h_{123} h_{234}^{-1} h_{124}^{-1}). \quad (\text{E.1})$$

Let us analyze the *right-hand side of the move* $1 \leftrightarrow 4$, given by expression (8.21). We integrate over g_{15} using $\delta_G(g_{125})$, over g_{25} using $\delta_G(g_{235})$, and over g_{35} using $\delta_G(g_{345})$:

$$\begin{aligned} g_{15} &= \partial(h_{125}) g_{25} g_{12}, \\ g_{25} &= \partial(h_{235}) g_{35} g_{23}, \\ g_{35} &= \partial(h_{345}) g_{45} g_{34}. \end{aligned} \quad (\text{E.2})$$

Next, we integrate over h_{135} using $\delta_H(h_{1345})$, over h_{125} using $\delta_H(h_{1245})$, and over h_{235} using $\delta_H(h_{2345})$:

$$\begin{aligned} h_{135} &= h_{145} (g_{45} \triangleright h_{134}) h_{345}^{-1}, \\ h_{125} &= h_{145} (g_{45} \triangleright h_{124}) h_{245}^{-1}, \\ h_{235} &= h_{245} (g_{45} \triangleright h_{234}) h_{345}^{-1}. \end{aligned} \quad (\text{E.3})$$

The remaining δ -functions on the group G reduce to $\delta_G(e)^3$. Let us show this. Using equations (E.3) and identity (8.2) for the triangle (234):

$$\begin{aligned} \delta_G(g_{245}) &= \delta_G(\partial(h_{245}) g_{45} g_{24} g_{25}^{-1}) \\ &= \delta_G(\partial(h_{245}) g_{45} g_{24} g_{23}^{-1} g_{35}^{-1} \partial(h_{235})^{-1}) \\ &= \delta_G(\partial(h_{245}) g_{45} g_{24} g_{23}^{-1} g_{35}^{-1} \partial(h_{345}) g_{45} \triangleright \partial(h_{234})^{-1} \partial(h_{245})^{-1}) \\ &= \delta_G(g_{45} g_{24} g_{23}^{-1} g_{34}^{-1} g_{45}^{-1} \partial(h_{345})^{-1} \partial(h_{345}) g_{45} g_{34} g_{23} g_{24}^{-1} g_{45}^{-1}) \\ &= \delta_G(e). \end{aligned} \quad (\text{E.4})$$

For the remaining two δ -functions on the group G , we find that they are equivalent to the first one, and therefore

$$\begin{aligned}
\delta_G(g_{135}) &= \delta_G(\partial(h_{135}) g_{35} g_{13} g_{15}^{-1}) \\
&= \delta_G(\partial(h_{145}) g_{45} \triangleright \partial(h_{134}) \partial(h_{345})^{-1} \partial(h_{345}) g_{45} g_{34} g_{13} g_{12}^{-1} g_{25}^{-1} \partial(h_{125})^{-1}) \\
&= \delta_G(\partial(h_{145}) g_{45} \partial(h_{134}) g_{45}^{-1} g_{45} g_{34} g_{13} g_{12}^{-1} g_{25}^{-1} \partial(h_{245}) g_{45} \triangleright \partial(h_{124})^{-1} \partial(h_{145})^{-1}) \\
&= \delta_G(g_{45} g_{14} g_{13}^{-1} g_{34}^{-1} g_{34} g_{13} g_{12}^{-1} g_{25}^{-1} \partial(h_{245}) g_{45} g_{24} g_{12} g_{14}^{-1} g_{45}^{-1}) \\
&= \delta_G(g_{25}^{-1} \partial(h_{245}) g_{45} g_{24}) \\
&= \delta_G(g_{245}) \\
&= \delta_G(e),
\end{aligned} \tag{E.5}$$

$$\begin{aligned}
\delta_G(g_{145}) &= \delta_G(\partial(h_{145}) g_{45} g_{14} g_{15}^{-1}) \\
&= \delta_G(\partial(h_{145}) g_{45} g_{14} g_{12}^{-1} g_{25}^{-1} \partial(h_{125})^{-1}) \\
&= \delta_G(\partial(h_{145}) g_{45} g_{14} g_{12}^{-1} g_{25}^{-1} \partial(h_{245}) g_{45} \triangleright \partial(h_{124})^{-1} \partial(h_{145})^{-1}) \\
&= \delta_G(g_{45} g_{14} g_{12}^{-1} g_{25}^{-1} \partial(h_{245}) g_{45} \partial(h_{124})^{-1} g_{45}^{-1}) \\
&= \delta_G(g_{14} g_{12}^{-1} g_{25}^{-1} \partial(h_{245}) g_{45} g_{24} g_{12} g_{14}^{-1}) \\
&= \delta_G(g_{25}^{-1} \partial(h_{245}) g_{45} g_{24}) \\
&= \delta_G(g_{245}) \\
&= \delta_G(e).
\end{aligned} \tag{E.6}$$

We used equations (E.2) and (E.3), together with identity (8.2) for the triangles (134) and (124). Analyzing $\delta_H(h_{1235})$ then gives:

$$\begin{aligned}
\delta_H(h_{1235}) &= \delta_H(h_{135} (g_{35} \triangleright h_{123}) h_{235}^{-1} h_{125}^{-1}) \\
&= \delta_H(h_{145} (g_{45} \triangleright h_{134}) h_{345}^{-1} (g_{35} \triangleright h_{123}) h_{235}^{-1} h_{245} g_{45} \triangleright h_{124}^{-1} h_{145}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{134} h_{345}^{-1} (g_{35} \triangleright h_{123}) (h_{245} (g_{45} \triangleright h_{234}) h_{345}^{-1})^{-1} h_{245} g_{45} \triangleright h_{124}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{134} h_{345}^{-1} (g_{35} \triangleright h_{123}) h_{345} g_{45} \triangleright h_{234}^{-1} g_{45} \triangleright h_{124}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{134} (\partial(h_{345})^{-1} g_{35}) \triangleright h_{123} g_{45} \triangleright h_{234}^{-1} g_{45} \triangleright h_{124}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{134} (g_{45} g_{34}) \triangleright h_{123} g_{45} \triangleright h_{234}^{-1} g_{45} \triangleright h_{124}^{-1}) \\
&= \delta_H(h_{134} g_{34} \triangleright h_{123} h_{234}^{-1} h_{124}^{-1}).
\end{aligned} \tag{E.7}$$

We therefore conclude that the *right-hand side of the move* is:

$$d.s. = \delta_G(e)^3 \delta_H(h_{134} g_{34} \triangleright h_{123} h_{234}^{-1} h_{124}^{-1}). \tag{E.8}$$

The constants in front of the integrals are $|G|^{-2}|H|$ on the left-hand side of the move and $|G|^{-5}|H|$ on its right-hand side, which compensates for the difference in the factors $|G|$ in expressions (E.1) and (E.8). We conclude that the state sum (8.16) is invariant under the $1 \leftrightarrow 4$ Pachner move.

The Pachner move $2 \leftrightarrow 3$

On the *left-hand side* of the $2 \leftrightarrow 3$ Pachner move, we have the integral:

$$\int dh_{234} \delta_G(h_{234}) \delta_H(h_{1234}) \delta_H(h_{2345}). \quad (\text{E.9})$$

First, we integrate over h_{234} using $\delta_H(h_{2345})$. We obtain

$$h_{234} = g_{45}^{-1} \triangleright h_{245}^{-1} g_{45}^{-1} \triangleright h_{235} g_{45}^{-1} \triangleright h_{345}, \quad (\text{E.10})$$

Substituting this result into the expression for the remaining δ -function on the group G and using identity (8.2) for the triangles (245), (235), and (345) gives:

$$\begin{aligned} \delta_G(g_{234}) &= \delta_G(\partial(h_{234}) g_{34} g_{23} g_{24}^{-1}) \\ &= \delta_G(g_{45}^{-1} \triangleright (\partial(h_{245})^{-1} \partial(h_{235}) \partial(h_{345})) g_{34} g_{23} g_{24}^{-1}) \\ &= \delta_G(g_{45}^{-1} g_{45} g_{24} g_{25} g_{25}^{-1} g_{23}^{-1} g_{35}^{-1} g_{35}^{-1} g_{34}^{-1} g_{45}^{-1} g_{45} g_{34} g_{23} g_{24}^{-1}) \\ &= \delta_G(e). \end{aligned} \quad (\text{E.11})$$

The remaining δ -function on the group H is:

$$\begin{aligned} \delta_H(h_{1234}) &= \delta_H(h_{134} (g_{34} \triangleright h_{123}) h_{234}^{-1} h_{124}^{-1}) \\ &= \delta_H(h_{134} (g_{34} \triangleright h_{123}) g_{45}^{-1} \triangleright (h_{345}^{-1} h_{235}^{-1} h_{245}) h_{124}^{-1}). \end{aligned} \quad (\text{E.12})$$

We conclude that the *left-hand side of the move* is:

$$l.s. = |G| \delta_H(h_{134} (g_{34} \triangleright h_{123}) g_{45}^{-1} \triangleright (h_{345}^{-1} h_{235}^{-1} h_{245}) h_{124}^{-1}). \quad (\text{E.13})$$

On the *right-hand side* of the Pachner move, we have the integral:

$$\int dg_{15} dh_{125} dh_{135} dh_{145} \delta_G(g_{125}) \delta_G(g_{135}) \delta_G(g_{145}) \delta_H(h_{1235}) \delta_H(h_{1245}) \delta_H(h_{1345}). \quad (\text{E.14})$$

First, we integrate over g_{15} using $\delta_G(g_{135})$,

$$g_{15} = \partial(h_{135}) g_{35} g_{13}, \quad (\text{E.15})$$

and then over h_{125} using $\delta_H(h_{1235})$ and over h_{135} using $\delta_H(h_{1345})$:

$$\begin{aligned} h_{125} &= h_{135} (g_{35} \triangleright h_{123}) h_{235}^{-1}, \\ h_{135} &= h_{145} (g_{45} \triangleright h_{134}) h_{345}^{-1}. \end{aligned} \quad (\text{E.16})$$

Using these results in the expressions for the remaining two δ -functions on the group G , we find that they reduce to $\delta_G(e)^2$:

$$\begin{aligned}
\delta_G(g_{125}) &= \delta_G(\partial(h_{125}) g_{25} g_{12} g_{15}^{-1}) \\
&= \delta_G(\partial(h_{135}) g_{35} \triangleright \partial(h_{123}) \partial(h_{235})^{-1} g_{25} g_{12} g_{13}^{-1} g_{35}^{-1} \partial(h_{135})^{-1}) \\
&= \delta_G(g_{35} \triangleright \partial(h_{123}) \partial(h_{235})^{-1} g_{25} g_{12} g_{13}^{-1} g_{35}^{-1}) \\
&= \delta_G(g_{35} g_{13} g_{12}^{-1} g_{23}^{-1} g_{35}^{-1} g_{35} g_{23} g_{25}^{-1} g_{25} g_{12} g_{13}^{-1} g_{35}^{-1}) \\
&= \delta_G(e),
\end{aligned} \tag{E.17}$$

$$\begin{aligned}
\delta_G(g_{145}) &= \delta_G(\partial(h_{145}) g_{45} g_{14} g_{15}^{-1}) \\
&= \delta_G(\partial(h_{145}) g_{45} g_{14} g_{13}^{-1} g_{35}^{-1} \partial(h_{135})^{-1}) \\
&= \delta_G(\partial(h_{145}) g_{45} g_{14} g_{13}^{-1} g_{35}^{-1} \partial(h_{345}) g_{45} \triangleright \partial(h_{134})^{-1} \partial(h_{145})^{-1}) \\
&= \delta_G(g_{45} g_{14} g_{13}^{-1} g_{35}^{-1} g_{35} g_{34}^{-1} g_{45}^{-1} g_{45} g_{34} g_{13} g_{14}^{-1} g_{45}^{-1}) \\
&= \delta_G(e).
\end{aligned} \tag{E.18}$$

The remaining δ -function on the group H is:

$$\begin{aligned}
\delta_H(h_{1245}) &= \delta_H(h_{145} (g_{45} \triangleright h_{124}) h_{245}^{-1} h_{125}^{-1}) \\
&= \delta_H(h_{145} (g_{45} \triangleright h_{124}) h_{245}^{-1} h_{235} g_{35} \triangleright h_{123}^{-1} h_{135}^{-1}) \\
&= \delta_H(h_{145} g_{45} \triangleright h_{124} h_{245}^{-1} h_{235} g_{35} \triangleright h_{123}^{-1} h_{345} g_{45} \triangleright h_{134}^{-1} h_{145}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{124} h_{245}^{-1} h_{235} g_{35} \triangleright h_{123}^{-1} h_{345} g_{45} \triangleright h_{134}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{124} h_{245}^{-1} h_{235} h_{345} (\partial(h_{345})^{-1} g_{35}) \triangleright h_{123}^{-1} g_{45} \triangleright h_{134}^{-1}) \\
&= \delta_H(g_{45} \triangleright h_{124} h_{245}^{-1} h_{235} h_{345} (g_{45} g_{34}) \triangleright h_{123}^{-1} g_{45} \triangleright h_{134}^{-1}) \\
&= \delta_H(h_{124} g_{45}^{-1} \triangleright (h_{245}^{-1} h_{235} h_{345}) g_{34} \triangleright h_{123}^{-1} h_{134}^{-1}) \\
&= \delta_H(h_{134} (g_{34} \triangleright h_{123}) g_{45}^{-1} \triangleright (h_{345}^{-1} h_{235}^{-1} h_{245}) h_{124}^{-1}).
\end{aligned} \tag{E.19}$$

We see that expressions (E.12) and (E.19) are equal. The remaining integration over the element h_{145} is trivial, so the *right-hand side of the move* is:

$$\begin{aligned}
d.s. &= \delta_G(e)^2 \delta_H(h_{134} (g_{34} \triangleright h_{123}) g_{45}^{-1} \triangleright (h_{345}^{-1} h_{235}^{-1} h_{245}) h_{124}^{-1}) \\
&= |G|^2 \delta_H(h_{134} (g_{34} \triangleright h_{123}) g_{45}^{-1} \triangleright (h_{345}^{-1} h_{235}^{-1} h_{245}) h_{124}^{-1}).
\end{aligned} \tag{E.20}$$

The difference in the factors $|G|$ in expressions (E.13) and (E.20) is compensated by the constants in front of the integrals—the factor $|G|^{-4}|H|^1$ on the right-hand side and the factor $|G|^{-3}|H|^1$ on the left-hand side of the move. We conclude that the sum (8.16) is invariant under the $2 \leftrightarrow 3$ Pachner move.

E.1.2 $n = 4$

The Pachner move $1 \leftrightarrow 5$

The *left-hand side* of the $1 \leftrightarrow 5$ Pachner move is given by expression (8.25) and cannot be simplified. Let us examine the *right-hand side of the move*, given by equation (8.24). First, we integrate over the variable g_{12} using the δ -function $\delta_G(g_{123})$, then over the variable g_{13} using $\delta_G(g_{134})$, over g_{14} using $\delta_G(g_{145})$, and over the variable g_{15} using $\delta_G(g_{156})$:

$$\begin{aligned} g_{12} &= g_{23}^{-1} \partial(h_{123})^{-1} g_{13}, \\ g_{13} &= g_{34}^{-1} \partial(h_{134})^{-1} g_{14}, \\ g_{14} &= g_{45}^{-1} \partial(h_{145})^{-1} g_{15}, \\ g_{15} &= g_{56}^{-1} \partial(h_{156})^{-1} g_{16}. \end{aligned} \tag{E.21}$$

Next, we integrate over the variable h_{123} using $\delta_H(h_{1234})$, over h_{124} using $\delta_H(h_{1245})$, over h_{125} using $\delta_H(h_{1256})$, over h_{134} using $\delta_H(h_{1345})$, over h_{135} using $\delta_H(h_{1356})$, and over h_{145} using $\delta_H(h_{1456})$:

$$\begin{aligned} h_{123} &= g_{34}^{-1} \triangleright h_{134}^{-1} g_{34}^{-1} \triangleright h_{124} g_{34}^{-1} \triangleright h_{234}, \\ h_{124} &= g_{45}^{-1} \triangleright h_{145}^{-1} g_{45}^{-1} \triangleright h_{125} g_{45}^{-1} \triangleright h_{245}, \\ h_{125} &= g_{56}^{-1} \triangleright h_{156}^{-1} g_{56}^{-1} \triangleright h_{126} g_{56}^{-1} \triangleright h_{256}, \\ h_{134} &= g_{45}^{-1} \triangleright h_{145}^{-1} g_{45}^{-1} \triangleright h_{135} g_{45}^{-1} \triangleright h_{345}, \\ h_{135} &= g_{56}^{-1} \triangleright h_{156}^{-1} g_{56}^{-1} \triangleright h_{136} g_{56}^{-1} \triangleright h_{356}, \\ h_{145} &= g_{56}^{-1} \triangleright h_{156}^{-1} g_{56}^{-1} \triangleright h_{146} g_{56}^{-1} \triangleright h_{456}. \end{aligned} \tag{E.22}$$

After these integrations, the six δ -functions on the group G present on the right-hand side of the move reduce to $\delta_G(e)^6$. We obtain:

$$\delta_G(g_{124}) = \delta_G(g_{125}) = \delta_G(g_{126}) = \delta_G(g_{135}) = \delta_G(g_{136}) = \delta_G(g_{146}) = \delta_G(e).$$

The details of the calculation are the same as for the $1 \leftrightarrow 5$ Pachner move of the $3BF$ state sum and are given in the next section. Let us now show that the remaining δ -functions on the group H reduce to $\delta_H(e)^4$. First, a straightforward calculation gives, for $\delta_H(h_{1235})$:

$$\begin{aligned} \delta_H(h_{1235}) &= \delta_H(h_{135} (g_{35} \triangleright h_{123}) h_{235}^{-1} h_{125}^{-1}) \\ &= \delta_H(h_{135} (g_{35} g_{34}^{-1}) \triangleright (h_{134}^{-1} h_{124} h_{234}) h_{235}^{-1} h_{125}^{-1}) \\ &= \delta_H(h_{135} (g_{35} g_{34}^{-1}) \triangleright ((g_{45}^{-1} \triangleright (h_{345}^{-1} h_{135}^{-1} h_{145} h_{145}^{-1} h_{125} h_{245}) h_{234}) h_{235}^{-1} h_{125}^{-1}) \\ &= \delta_H(h_{135} h_{345} h_{345}^{-1} h_{135}^{-1} h_{125} h_{245} h_{345}^{-1} (g_{35} g_{34}^{-1}) \triangleright h_{234} h_{235}^{-1} h_{125}^{-1}) \\ &= \delta_H(h_{245} h_{345}^{-1} (g_{35} g_{34}^{-1}) \triangleright h_{234} h_{235}^{-1}) \\ &= \delta_H(h_{245} h_{345}^{-1} h_{345} (\partial(h_{345})^{-1} g_{35} g_{34}^{-1}) \triangleright h_{234} h_{345}^{-1} h_{235}^{-1}) \\ &= \delta_H(h_{245} g_{45} \triangleright h_{234} h_{345}^{-1} h_{235}^{-1}) \\ &= \delta_H(e). \end{aligned} \tag{E.23}$$

In the preceding calculation, we used the identity $g_{45} = \partial(h_{345})^{-1} g_{35} g_{34}^{-1}$ for the triangle (345), together with the expression $h_{245} g_{45} \triangleright h_{234} h_{345}^{-1} h_{235}^{-1} = e$ for the tetrahedron (2345). Applying

an analogous procedure to the value of the δ -function $\delta_H(h_{1236})$, we obtain:

$$\begin{aligned}
\delta_H(h_{1236}) &= \delta_H(h_{136} (g_{36} \triangleright h_{123}) h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} (g_{46} g_{34}) \triangleright h_{123} h_{346}^{-1} h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} g_{46} \triangleright (h_{134}^{-1} h_{124} h_{234}) h_{346}^{-1} h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} g_{46} \triangleright (h_{134}^{-1} h_{124} h_{234}) g_{46} \triangleright h_{234}^{-1} h_{246}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} (g_{46} g_{45}^{-1}) \triangleright (h_{345}^{-1} h_{135}^{-1} h_{125} h_{245}) h_{246}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} (g_{46} g_{45}^{-1}) \triangleright (h_{345}^{-1} g_{56}^{-1} \triangleright (h_{356}^{-1} h_{136}^{-1} h_{126} h_{256}) h_{245}) h_{246}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} (g_{46} g_{45}^{-1}) \triangleright h_{345}^{-1} h_{456} h_{356}^{-1} h_{136}^{-1} h_{126} h_{256} g_{56} \triangleright h_{245} h_{456}^{-1} h_{246}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} h_{456} g_{56} \triangleright h_{345}^{-1} h_{356}^{-1} h_{136}^{-1} h_{126} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{136}^{-1} h_{126} h_{126}^{-1}) \\
&= \delta_H(e).
\end{aligned} \tag{E.24}$$

We used identity (8.6) for the tetrahedra (2456) and (3456), together with identity (8.2) for the triangle (456).

By a similar procedure, we find that the δ -functions satisfy $\delta_H(h_{1246}) = \delta_H(h_{1346}) = \delta_H(e)$. We thus obtain the following value for the *right-hand side* of the move:

$$d.s. = \delta_G(e)^6 \delta_H(e)^4 = |G|^6 |H|^4. \tag{E.25}$$

The factors in expression (E.25) are compensated by the constants in front of the integrals—the factor $|G|^{-11} |H|^{-4}$ on the right-hand side and the factor $|G|^{-5} |H|^0$ on the left-hand side of the move. We conclude that the sum (8.16) is invariant under the $1 \leftrightarrow 5$ Pachner move.

The Pachner move $2 \leftrightarrow 4$

The *left-hand side* of the move is equal to the δ -function

$$\delta_H(h_{2345}) = h_{235} h_{345} (g_{45} \triangleright h_{234}^{-1}) h_{245}^{-1}. \tag{E.26}$$

To determine the left-hand side of the move, we shall use Lemma 17.

Lemma 17 *Suppose that, for a given 4-simplex $(jklmn)$, identity (8.6) is satisfied for the four tetrahedra $(klmn)$, $(jlmn)$, $(jkmn)$, and $(jkl n)$, and identity (8.2) is satisfied for all triangles on their boundary. Then identity (8.6) is also satisfied for the fifth tetrahedron $(jklm)$.*

Using this general result, we can show that the left-hand side of the move reduces to

$$\delta_H(h_{2345}) = \delta_H(e) = |H|. \tag{E.27}$$

Let us now examine the *right-hand side* of the move, that is, the integral (8.27):

$$\int dg_{16} \int dh_{126} dh_{136} dh_{146} dh_{156} \delta_G(g_{126}) \delta_G(g_{136}) \delta_G(g_{146}) \delta_G(g_{156}) \delta_H(h_{1236}) \delta_H(h_{1246}) \delta_H(h_{1256}) \delta_H(h_{1346}) \delta_H(h_{1356}) \delta_H(h_{1456}). \tag{E.28}$$

First, we integrate over g_{16} using $\delta_G(g_{126})$,

$$g_{16} = \partial(h_{126}) g_{26} g_{12}. \tag{E.29}$$

Next, we integrate over h_{126} using $\delta_H(h_{1236})$, over h_{136} using $\delta_H(h_{1346})$, and over h_{146} using $\delta_H(h_{1456})$, which gives

$$\begin{aligned} h_{126} &= h_{136} (g_{36} \triangleright h_{123}) h_{236}^{-1}, \\ h_{136} &= h_{146} (g_{46} \triangleright h_{134}) h_{346}^{-1}, \\ h_{146} &= h_{156} (g_{56} \triangleright h_{145}) h_{456}^{-1}. \end{aligned} \tag{E.30}$$

The remaining three δ -functions on the group G reduce to $\delta_G(e)^3$; that is, one finds $\delta_G(g_{136}) = \delta_G(g_{146}) = \delta_G(g_{156}) = \delta_G(e)$. The proof is the same as in the $3BF$ case; see the next section for details.

The remaining three δ -functions on the group H reduce to $\delta_H(e)^3$ by a procedure similar to that for the $1 \leftrightarrow 5$ Pachner move; that is, we obtain $\delta_H(h_{1356}) = \delta_H(h_{1246}) = \delta_H(h_{1256}) = \delta_H(e)$. First, let us show that $\delta_H(h_{1356}) = \delta_H(e)$:

$$\begin{aligned} \delta_H(h_{1356}) &= \delta_H(h_{156} (g_{56} \triangleright h_{135}) h_{356}^{-1} h_{136}^{-1}) \\ &= \delta_H(h_{156} (g_{56} \triangleright h_{135}) h_{356}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{146}^{-1}) \\ &= \delta_H(g_{56} \triangleright h_{135} h_{356}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{456} g_{56} \triangleright h_{145}^{-1}) \\ &= \delta_H(g_{56} \triangleright (g_{45} \triangleright h_{134} h_{345}) h_{356}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{456}) \\ &= \delta_H(g_{46} \triangleright h_{134} h_{456} g_{56} \triangleright h_{345} h_{356}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1}) \\ &= \delta_H(h_{456} g_{56} \triangleright h_{345} h_{356}^{-1} h_{346}) \\ &= \delta_H(e). \end{aligned} \tag{E.31}$$

Next, the δ -function $\delta_H(h_{1246})$ is:

$$\begin{aligned} \delta_H(h_{1246}) &= \delta_H(h_{146} (g_{46} \triangleright h_{124}) h_{246}^{-1} h_{126}^{-1}) \\ &= \delta_H(h_{146} (g_{46} \triangleright h_{124}) h_{246}^{-1} h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{146}^{-1}) \\ &= \delta_H(g_{46} \triangleright h_{124} h_{246}^{-1} h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1}) \\ &= \delta_H(g_{46} \triangleright h_{124} (\partial(h_{246})^{-1} \partial(h_{236}) g_{36}) \triangleright h_{123}^{-1} h_{246}^{-1} h_{236} h_{346} g_{46} \triangleright h_{134}^{-1}) \\ &= \delta_H(g_{46} \triangleright h_{124} (g_{46} g_{24} g_{23}^{-1}) \triangleright h_{123}^{-1} g_{46} \triangleright h_{234} g_{46} \triangleright h_{134}^{-1}) \\ &= \delta_H(g_{46} \triangleright h_{124} g_{46} \triangleright h_{234} (g_{46} g_{34}) \triangleright h_{123}^{-1} g_{46} \triangleright h_{134}^{-1}) \\ &= \delta_H(e). \end{aligned} \tag{E.32}$$

The remaining δ -function $\delta_H(h_{1256})$ is equal to:

$$\begin{aligned}
\delta_H(h_{1256}) &= \delta_H(h_{156} (g_{56} \triangleright h_{125}) h_{256}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{156} (g_{56} \triangleright h_{125}) h_{256}^{-1} (h_{236} g_{36} \triangleright h_{123}^{-1} h_{136}^{-1})) \\
&= \delta_H(h_{156} (g_{56} \triangleright h_{125}) h_{256}^{-1} h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{146}^{-1}) \\
&= \delta_H(h_{156} (g_{56} \triangleright h_{125}) h_{256}^{-1} h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{456} g_{56} \triangleright h_{145}^{-1} h_{156}^{-1}) \\
&= \delta_H(g_{56} \triangleright h_{125} h_{256}^{-1} h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{456} g_{56} \triangleright h_{145}^{-1}) \\
&= \delta_H(h_{256}^{-1} h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} h_{456} g_{56} \triangleright (g_{45} \triangleright h_{124} h_{245}^{-1})) \\
&= \delta_H(h_{236} g_{36} \triangleright h_{123}^{-1} h_{346} g_{46} \triangleright h_{134}^{-1} g_{46} \triangleright h_{124} h_{246}^{-1}) \\
&= \delta_H(h_{236} h_{346} (g_{46} g_{34}^{-1}) \triangleright h_{123}^{-1} g_{46} \triangleright h_{134}^{-1} g_{46} \triangleright h_{124} h_{246}^{-1}) \\
&= \delta_H(g_{46} \triangleright h_{234} (g_{46} g_{34}^{-1}) \triangleright h_{123}^{-1} g_{46} \triangleright h_{134}^{-1} g_{46} \triangleright h_{124}) \\
&= \delta_H(e).
\end{aligned} \tag{E.33}$$

The remaining integration—over the element h_{156} of the group H —is trivial, and hence the *right-hand side* finally reduces to:

$$d.s. = \delta_G(e)^3 \delta_H(e)^3 = |G|^3 |H|^3. \tag{E.34}$$

The constants in front of the integrals are $|G|^{-8} |H|^{-1}$ on the left-hand side of the move and $|G|^{-11} |H|^{-3}$ on its right-hand side, which compensates for the difference between expressions (E.27) and (E.34). We therefore conclude that the state sum (8.16) is invariant under the $2 \leftrightarrow 4$ Pachner move.

The Pachner move $3 \leftrightarrow 3$

Let us first analyze the right-hand side of the move, that is, the integral:

$$\int_H dh_{123} \delta_G(g_{123}) \delta_H(h_{1234}) \delta_H(h_{1235}) \delta_H(h_{1236}). \tag{E.35}$$

We integrate over h_{123} using $\delta_H(h_{1234})$ and obtain:

$$h_{123} = g_{34}^{-1} \triangleright h_{134}^{-1} g_{34}^{-1} \triangleright h_{124} g_{34}^{-1} \triangleright h_{234}. \tag{E.36}$$

It can be shown that the remaining δ -function on the group G is the δ -function $\delta_G(g_{123}) = \delta_G(e)$; see the proof of invariance of the $3BF$ state sum under the $3 \leftrightarrow 3$ Pachner move.

For the δ -function $\delta_H(h_{1235})$, applying identity (8.2) for the triangle (345) and identity (8.6)

for the tetrahedra (1345), (1245), and (2345) gives:

$$\begin{aligned}
\delta_H(h_{1235}) &= \delta_H(h_{135} (g_{35} \triangleright h_{123}) h_{235}^{-1} h_{125}^{-1}) \\
&= \delta_H(h_{135} (g_{35} g_{34}^{-1}) \triangleright (h_{134}^{-1} h_{124} h_{234}) h_{235}^{-1} h_{125}^{-1}) \\
&= \delta_H(h_{135} h_{345} g_{45} \triangleright (h_{134}^{-1} h_{124} h_{234}) h_{345}^{-1} h_{235}^{-1} h_{125}^{-1}) \\
&= \delta_H(h_{145} g_{45} \triangleright (h_{124} h_{234}) h_{345}^{-1} h_{235}^{-1} h_{125}^{-1}) \\
&= \delta_H(h_{245} g_{45} \triangleright h_{234} h_{345}^{-1} h_{235}^{-1}) \\
&= \delta_H(e).
\end{aligned} \tag{E.37}$$

For the δ -function $\delta_H(h_{1236})$, applying identity (8.2) for the triangle (346) and identity (8.6) for the tetrahedra (1346), (1246), and (2346) gives:

$$\begin{aligned}
\delta_H(h_{1236}) &= \delta_H(h_{136} (g_{36} \triangleright h_{123}) h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} (g_{36} g_{34}^{-1}) \triangleright (h_{134}^{-1} h_{124} h_{234}) h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{136} h_{346} g_{46} \triangleright (h_{134}^{-1} h_{124} h_{234}) h_{346}^{-1} h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{146} g_{46} \triangleright (h_{124} h_{234}) h_{346}^{-1} h_{236}^{-1} h_{126}^{-1}) \\
&= \delta_H(h_{246} g_{46} \triangleright h_{234} h_{346}^{-1} h_{236}^{-1}) \\
&= \delta_H(e).
\end{aligned} \tag{E.38}$$

We conclude that the expression on the right-hand side of the Pachner move reduces to

$$d.s. = \delta_G(e) \delta_H(e)^2 = |G| |H|^2. \tag{E.39}$$

Let us now analyze the *left-hand side of the move*, that is, the integral:

$$\int_H dh_{456} \delta_G(g_{456}) \delta_H(h_{3456}) \delta_H(h_{2456}) \delta_H(h_{1456}). \tag{E.40}$$

We integrate over h_{456} using the δ -function $\delta_H(h_{3456})$. Substituting the resulting expression for h_{456} throughout, we find that the δ -function $\delta_G(g_{456})$, by virtue of identity (8.2) for the triangles (346), (356), and (345), reduces to:

$$\delta_G(g_{456}) = \delta_G(e). \tag{E.41}$$

By a procedure similar to that used on the right-hand side of the move, we find that the δ -functions $\delta_H(h_{1456})$ and $\delta_H(h_{2456})$ are equal to $\delta_H(e)^2$. Finally, the left-hand side of the move is:

$$l.s. = \delta_G(e) \delta_H(e)^2 = |G| |H|^2. \tag{E.42}$$

The number of k -simplices on the two sides of the $3 \leftrightarrow 3$ move is the same for every k ; hence, the coefficients in front of the integrals are equal in this case and are therefore immaterial.

E.2 Invariance of the $3BF$ state sum under Pachner moves

This appendix proves that the state sum (9.22) is independent of the triangulation of the manifold, that is, invariant under Pachner moves.

E.2.1 $n = 4$

The Pachner move $1 \leftrightarrow 5$

Using identity (2.90), the δ -function on the *left-hand side of the move*, $\delta_L(l_{23456})$, is:

$$\delta_L(l_{23456}) = \delta_L(l_{2346}^{-1}(h_{236} \triangleright' l_{3456})l_{2356}h_{256} \triangleright' (g_{56} \triangleright l_{2345})l_{2456}^{-1}h_{246} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{234}\}_{\text{pf}}). \quad (\text{E.43})$$

Let us examine the *right-hand side of the move*, given after integration by equation (9.23). First, we integrate over the variable g_{12} using the δ -function $\delta_G(g_{123})$, then over the variable g_{13} using $\delta_G(g_{134})$, over g_{14} using $\delta_G(g_{145})$, and over the variable g_{15} using $\delta_G(g_{156})$:

$$\begin{aligned} g_{12} &= g_{23}^{-1} \partial(h_{123})^{-1} g_{13}, \\ g_{13} &= g_{34}^{-1} \partial(h_{134})^{-1} g_{14}, \\ g_{14} &= g_{45}^{-1} \partial(h_{145})^{-1} g_{15}, \\ g_{15} &= g_{56}^{-1} \partial(h_{156})^{-1} g_{16}. \end{aligned} \quad (\text{E.44})$$

Next, we integrate over the variable h_{123} using $\delta_H(h_{1234})$, over h_{124} using $\delta_H(h_{1245})$, over h_{125} using $\delta_H(h_{1256})$, over h_{134} using $\delta_H(h_{1345})$, over h_{135} using $\delta_H(h_{1356})$, and over h_{145} using $\delta_H(h_{1456})$:

$$\begin{aligned} h_{123} &= g_{34}^{-1} \triangleright h_{134}^{-1} g_{34}^{-1} \triangleright \delta(l_{1234})^{-1} g_{34}^{-1} \triangleright h_{124} g_{34}^{-1} \triangleright h_{234}, \\ h_{124} &= g_{45}^{-1} \triangleright h_{145}^{-1} g_{45}^{-1} \triangleright \delta(l_{1245})^{-1} g_{45}^{-1} \triangleright h_{125} g_{45}^{-1} \triangleright h_{245}, \\ h_{125} &= g_{56}^{-1} \triangleright h_{156}^{-1} g_{56}^{-1} \triangleright \delta(l_{1256})^{-1} g_{56}^{-1} \triangleright h_{126} g_{56}^{-1} \triangleright h_{256}, \\ h_{134} &= g_{45}^{-1} \triangleright h_{145}^{-1} g_{45}^{-1} \triangleright \delta(l_{1345})^{-1} g_{45}^{-1} \triangleright h_{135} g_{45}^{-1} \triangleright h_{345}, \\ h_{135} &= g_{56}^{-1} \triangleright h_{156}^{-1} g_{56}^{-1} \triangleright \delta(l_{1356})^{-1} g_{56}^{-1} \triangleright h_{136} g_{56}^{-1} \triangleright h_{356}, \\ h_{145} &= g_{56}^{-1} \triangleright h_{156}^{-1} g_{56}^{-1} \triangleright \delta(l_{1456})^{-1} g_{56}^{-1} \triangleright h_{146} g_{56}^{-1} \triangleright h_{456}. \end{aligned} \quad (\text{E.45})$$

After these integrations, the six δ -functions on the group G present on the right-hand side of the move reduce to $\delta_G(e)$ ⁶. Let us outline the proof. First,

$$\begin{aligned} \delta_G(g_{124}) &= \delta_G(\partial(h_{124})g_{24}g_{12}g_{14}^{-1}) \\ &= \delta_G(\partial(h_{124})g_{24}g_{23}^{-1}\partial(h_{123})^{-1}g_{13}g_{14}^{-1}) \\ &= \delta_G(\partial(h_{124})g_{24}g_{23}^{-1}g_{34}^{-1}\partial(h_{234})^{-1}\partial(h_{124})^{-1}\partial(h_{134})g_{34}g_{13}g_{14}^{-1}) \\ &= \delta_G(\partial(h_{124})g_{24}g_{23}^{-1}g_{34}^{-1}(g_{34}g_{23}^{-1}g_{24}^{-1})\partial(h_{124})^{-1}e) \\ &= \delta_G(e). \end{aligned} \quad (\text{E.46})$$

Next, we obtain

$$\begin{aligned}
\delta_G(g_{125}) &= \delta_G(\partial(h_{125})g_{25}g_{12}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{125})g_{25}g_{23}^{-1}\partial(h_{123})^{-1}g_{13}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{125})g_{25}g_{23}^{-1}g_{34}^{-1}\partial(h_{234})^{-1}\partial(h_{124})^{-1}\partial(h_{134})g_{34}g_{13}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{125})g_{25}g_{23}^{-1}g_{34}^{-1}\partial(h_{234})^{-1}g_{45}^{-1}(\partial(h_{245})^{-1}\partial(h_{125})^{-1}\partial(h_{145}))g_{45}g_{14}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{125})g_{25}g_{23}^{-1}g_{34}^{-1}(g_{34}g_{23}^{-1}g_{24}^{-1})g_{45}^{-1}(g_{45}g_{24}^{-1}g_{25}^{-1})\partial(h_{125})^{-1}e) \\
&= \delta_G(e).
\end{aligned} \tag{E.47}$$

Similarly,

$$\begin{aligned}
\delta_G(g_{126}) &= \delta_G(\partial(h_{126})g_{26}g_{12}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{126})g_{26}g_{23}^{-1}\partial(h_{123})^{-1}g_{13}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{126})g_{26}g_{23}^{-1}g_{34}^{-1}\partial(h_{234})^{-1}\partial(h_{124})^{-1}\partial(h_{134})g_{34}g_{13}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{126})g_{26}g_{23}^{-1}g_{34}^{-1}\partial(h_{234})^{-1}g_{45}^{-1}\partial(h_{245})^{-1}\partial(h_{125})^{-1}\partial(h_{145})g_{45}\partial(h_{134})g_{34}g_{13}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{126})g_{26}g_{23}^{-1}g_{34}^{-1}\partial(h_{234})^{-1}g_{45}^{-1}\partial(h_{245})^{-1}g_{56}^{-1}\partial(h_{256})^{-1}\partial(h_{126})^{-1}\partial(h_{156})g_{56} \\
&\quad \partial(h_{145})g_{45}g_{14}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{126})g_{26}g_{23}^{-1}g_{34}^{-1}(g_{34}g_{23}^{-1}g_{24}^{-1})g_{45}^{-1}(g_{45}g_{24}^{-1}g_{25}^{-1})g_{56}^{-1}(g_{56}g_{25}^{-1}g_{26}^{-1})\partial(h_{126})^{-1} \\
&\quad (g_{16}g_{15}^{-1}g_{56}^{-1})g_{56}g_{15}g_{16}^{-1}) \\
&= \delta_G(e).
\end{aligned} \tag{E.48}$$

For $\delta_G(g_{135})$, we likewise obtain

$$\begin{aligned}
\delta_G(g_{135}) &= \delta_G(\partial(h_{135})g_{35}g_{13}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{135})g_{35}g_{34}^{-1}\partial(h_{134})^{-1}g_{14}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{135})g_{35}g_{34}^{-1}g_{45}^{-1}\partial(h_{345})^{-1}\partial(h_{135})^{-1}\partial(h_{145})g_{45}g_{14}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{135})g_{35}g_{34}^{-1}g_{45}^{-1}\partial(h_{345})^{-1}\partial(h_{135})^{-1}\partial(h_{145})g_{45}g_{45}^{-1}\partial(h_{145})^{-1}g_{15}g_{15}^{-1}) \\
&= \delta_G(\partial(h_{135})g_{35}g_{34}^{-1}g_{45}^{-1}(g_{45}g_{34}^{-1}g_{35}^{-1})\partial(h_{135})^{-1}) \\
&= \delta_G(e),
\end{aligned} \tag{E.49}$$

and similarly for $\delta_G(g_{136})$:

$$\begin{aligned}
\delta_G(g_{136}) &= \delta_G(\partial(h_{136})g_{36}g_{13}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{136})g_{36}g_{34}^{-1}\partial(h_{134})^{-1}g_{14}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{136})g_{36}g_{34}^{-1}g_{45}^{-1}\partial(h_{345})^{-1}\partial(h_{135})^{-1}\partial(h_{145})g_{45}g_{14}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{136})g_{36}g_{34}^{-1}g_{45}^{-1}\partial(h_{345})^{-1}g_{56}^{-1}\partial(h_{356})^{-1}\partial(h_{136})^{-1}\partial(h_{156})g_{56}\partial(h_{145})g_{45}g_{14}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{136})g_{36}g_{34}^{-1}g_{45}^{-1}(g_{45}g_{34}^{-1}g_{35}^{-1})g_{56}^{-1}(g_{56}g_{35}^{-1}g_{36}^{-1})\partial(h_{136})^{-1}e) \\
&= \delta_H(e).
\end{aligned} \tag{E.50}$$

Finally, the remaining δ -function $\delta_G(g_{146})$ on the group G becomes

$$\begin{aligned}
\delta_G(g_{146}) &= \delta_G(\partial(h_{146})g_{46}g_{14}g_{16}^{-1}) \\
&= \delta_G(\partial(h_{146})g_{46}(g_{45}^{-1}\partial(h_{145})^{-1}g_{15})g_{16}^{-1}) \\
&= \delta_G(\partial(h_{146})g_{46}g_{45}^{-1}\partial(h_{145})^{-1}(g_{56}^{-1}\partial(h_{156})^{-1}g_{16})g_{16}^{-1}) \\
&= \delta_G(\partial(h_{146})g_{46}g_{45}^{-1}g_{56}^{-1}\partial(h_{456})^{-1}\partial(h_{146})^{-1}\partial(h_{156})g_{56}(g_{56}^{-1}\partial(h_{156})^{-1}g_{16})g_{16}^{-1}) \\
&= \delta_G(e).
\end{aligned} \tag{E.51}$$

Next, we integrate over l_{1235} using $\delta_L(l_{12345})$, over l_{1236} using $\delta_L(l_{12346})$, over l_{1246} using $\delta_L(l_{12456})$, and over l_{1346} using $\delta_L(l_{13456})$,

$$l_{1235} = (h_{125} \triangleright' l_{2345})l_{1245}h_{145} \triangleright' (g_{45} \triangleright l_{1234})l_{1345}^{-1}h_{135} \triangleright' \{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}, \tag{E.52}$$

$$l_{1236} = (h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1234})l_{1346}^{-1}h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}, \tag{E.53}$$

$$l_{1246} = (h_{126} \triangleright' l_{2456})l_{1256}h_{156} \triangleright' (g_{56} \triangleright l_{1245})l_{1456}^{-1}h_{146} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}, \tag{E.54}$$

$$l_{1346} = (h_{136} \triangleright' l_{3456})l_{1356}h_{156} \triangleright' (g_{56} \triangleright l_{1345})l_{1456}^{-1}h_{146} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}. \tag{E.55}$$

Let us now show that the remaining δ -functions on the group H reduce to $\delta_H(e)^4$. First, a straightforward calculation gives, for $\delta_H(h_{1235})$:

$$\begin{aligned}
\delta_H(h_{1235}) &= \delta_H(\delta(l_{1235})h_{135}(g_{35} \triangleright h_{123})h_{235}^{-1}h_{125}^{-1}) \\
&= \delta_H\left(\delta((h_{125} \triangleright' l_{2345})l_{1245}h_{145} \triangleright' (g_{45} \triangleright l_{1234})l_{1345}^{-1}h_{135} \triangleright' \{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}})h_{135} \right. \\
&\quad \left. (g_{35} \triangleright h_{123})h_{235}^{-1}h_{125}^{-1}\right) \\
&= \delta_H\left((h_{125}\delta(l_{2345})h_{125}^{-1}\delta(l_{1245})h_{145}(g_{45} \triangleright \delta(l_{1234}))h_{145}^{-1}\delta(l_{1345})^{-1}h_{135}\delta(\{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}})h_{135}^{-1}) \right. \\
&\quad \left. h_{135}(g_{35} \triangleright h_{123})h_{235}^{-1}h_{125}^{-1}\right) \\
&= \delta_H\left(h_{235}h_{345}(g_{45} \triangleright h_{234}^{-1}h_{245}^{-1}h_{125}^{-1}h_{125}h_{245}(g_{45} \triangleright h_{124}^{-1}h_{145}^{-1}h_{145}(g_{45} \triangleright (h_{124}h_{234}(g_{34} \triangleright h_{123}^{-1})h_{134}^{-1})) \right. \\
&\quad \left. h_{145}^{-1}(h_{145}(g_{45} \triangleright h_{134})h_{345}^{-1}h_{135}^{-1})h_{135}\delta(\{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}})h_{135}^{-1}h_{135}(g_{35} \triangleright h_{123})h_{235}^{-1}\right) \\
&= \delta_H(h_{345}((g_{45}g_{34}) \triangleright h_{123}^{-1})h_{345}^{-1}\delta(\{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}})(g_{35} \triangleright h_{123})).
\end{aligned} \tag{E.56}$$

Using the identity

$$\delta\{h_1, h_2\}_{\text{pf}}(\partial(h_1) \triangleright h_2) h_1 h_2^{-1} h_1^{-1} = e, \quad (\text{E.57})$$

and the equation $g_{35} = \partial(h_{345})g_{45}g_{34}$, where for the elements h_1 and h_2 we choose $h_1 = h_{345}$ and $h_2 = (g_{45}g_{34}) \triangleright h_{123}$, we obtain

$$\delta_H(h_{1235}) = \delta_H(e). \quad (\text{E.58})$$

Applying an analogous procedure to the value of the δ -function $\delta_H(h_{1236})$, we obtain:

$$\begin{aligned} \delta_H(h_{1236}) &= \delta_H(\delta(l_{1236})h_{136}(g_{36} \triangleright h_{123})h_{236}^{-1}h_{126}^{-1}) \\ &= \delta_H\left(\delta((h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1236}l_{1346}^{-1}h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}))h_{136}(g_{36} \triangleright h_{123})h_{236}^{-1}h_{126}^{-1}\right) \\ &= \delta_H\left(h_{126}\delta(l_{2346})h_{126}^{-1}\delta(l_{1246})h_{146}(g_{46} \triangleright \delta(l_{1234}))h_{146}^{-1}\delta(l_{1346})^{-1}h_{136}\delta(\{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}})h_{136}^{-1}\right. \\ &\quad \left.h_{136}(g_{36} \triangleright h_{123})h_{236}^{-1}h_{126}^{-1}\right) \\ &= \delta_H\left(h_{236}h_{346}(g_{46} \triangleright h_{234}^{-1})h_{246}^{-1}h_{126}^{-1}h_{126}h_{246}(g_{46} \triangleright h_{124}^{-1})h_{146}^{-1}h_{146}(g_{46} \triangleright (h_{124}h_{234}(g_{34} \triangleright h_{123}^{-1})h_{134}^{-1}))\right. \\ &\quad \left.h_{146}^{-1}(h_{146}(g_{46} \triangleright h_{134})h_{346}^{-1}h_{136}^{-1})h_{136}\delta(\{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}})h_{136}^{-1}h_{136}(g_{36} \triangleright h_{123})h_{236}^{-1}\right) \\ &= \delta_H(h_{346}((g_{46}g_{34}) \triangleright h_{123}^{-1})h_{346}^{-1}\delta(\{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}})(g_{36} \triangleright h_{123})). \end{aligned} \quad (\text{E.59})$$

Using the equation $g_{36} = \partial(h_{346})g_{46}g_{34}$ and identity (2.6) with $h_1 = h_{346}$ and $h_2 = (g_{46}g_{34}) \triangleright h_{123}$, we obtain:

$$\delta_H(h_{1236}) = \delta_H(e). \quad (\text{E.60})$$

By a similar procedure, we find that the δ -functions satisfy $\delta_H(h_{1246}) = \delta_H(h_{1346}) = \delta_H(e)$. The remaining δ -function on the group L is $\delta_L(l_{12356})$,

$$\delta_L(l_{12356}) = \delta_L(l_{1236}^{-1}(h_{126} \triangleright' l_{2356})l_{1256}h_{156} \triangleright' (g_{56} \triangleright l_{1235})l_{1356}^{-1}h_{136} \triangleright' \{h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}). \quad (\text{E.61})$$

Using equations (E.52), (E.53), (E.54), and (E.55), we obtain:

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L\left(h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1}(h_{136} \triangleright' l_{3456})l_{1356}h_{156} \triangleright' (g_{56} \triangleright l_{1345})l_{1456}^{-1}\right. \\ &\quad \left.h_{146} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}h_{146} \triangleright' (g_{46} \triangleright l_{1234})^{-1}h_{146} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1}l_{1456}\right. \\ &\quad \left.h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1}l_{1256}^{-1}(h_{126} \triangleright' l_{2456})^{-1}(h_{126} \triangleright' l_{2346}^{-1})(h_{126} \triangleright' l_{2356})l_{1256}\right. \\ &\quad \left.h_{156} \triangleright' (g_{56} \triangleright ((h_{125} \triangleright' l_{2345})l_{1245}h_{145} \triangleright' (g_{45} \triangleright l_{1234})l_{1345}^{-1}h_{135} \triangleright' \{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}))\right. \\ &\quad \left.l_{1356}^{-1}h_{136} \triangleright' \{h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}\right). \end{aligned} \quad (\text{E.62})$$

Using identity (2.62), the δ -function $\delta_L(l_{12356})$ reduces to:

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L\left((h_{136} \triangleright' l_{3456})l_{1356}h_{156} \triangleright' (g_{56} \triangleright l_{1345})l_{1456}^{-1}\right. \\ &\quad \left.h_{146} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}h_{146} \triangleright' (g_{46} \triangleright l_{1234})^{-1}h_{146} \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1}l_{1456}\right. \\ &\quad \left.\delta(h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1}) \triangleright' \left((\delta(l_{1256})^{-1}h_{126}) \triangleright' (l_{2456}^{-1}l_{2346}^{-1}l_{2356})h_{156} \triangleright' (g_{56} \triangleright (h_{125} \triangleright' l_{2345}))\right)\right. \\ &\quad \left.h_{156} \triangleright' (g_{56} \triangleright (h_{145} \triangleright' (g_{45} \triangleright l_{1234})l_{1345}^{-1}))l_{1356}^{-1}(h_{136}h_{346}) \triangleright' \{h_{346}^{-1}h_{356}g_{56} \triangleright h_{345}, (g_{56}g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}\right). \end{aligned} \quad (\text{E.63})$$

After commuting the elements, the δ -function reduces to

$$\begin{aligned}
\delta_L(l_{12356}) &= \delta_L \left((h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1245})^{-1}) \delta(l_{1256})^{-1} h_{126}) \triangleright' (l_{2456}^{-1} l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345})) \right. \\
&\quad h_{156} \triangleright' (g_{56} \triangleright (h_{145} \triangleright' (g_{45} \triangleright l_{1234}) l_{1345}^{-1})) l_{1356}^{-1} (h_{136} h_{346}) \triangleright' \{h_{346}^{-1} h_{356} g_{56} \triangleright h_{345}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} \\
&\quad h_{136} \triangleright' l_{3456} h_{156} \triangleright' (g_{56} \triangleright l_{1345}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' (\{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}}) \\
&\quad \left. (\delta(l_{1456})^{-1} h_{146}) \triangleright' ((g_{46} \triangleright l_{1234})^{-1}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1} \right).
\end{aligned} \tag{E.64}$$

The tetrahedron (3456) is part of the integrand on both sides of the move, so we may use identity (9.19) for $\delta_H(h_{3456})$, namely the equality $h_{346}^{-1} h_{356} g_{56} \triangleright h_{345} = h_{346}^{-1} \triangleright' \delta(l_{3456})^{-1} h_{456}$. Applying identity (2.62) then gives

$$\begin{aligned}
\{h_{346}^{-1} h_{356} g_{56} \triangleright h_{345}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} &= \{h_{346}^{-1} \triangleright' \delta(l_{3456})^{-1} h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} \\
&= (h_{346}^{-1} \triangleright' \delta(l_{3456})^{-1}) \triangleright' \{h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} \\
&\quad \{h_{346}^{-1} \triangleright' \delta(l_{3456})^{-1}, (g_{46} g_{34}) \triangleright h_{123}\}_{\text{pf}} \\
&= h_{346}^{-1} \triangleright' l_{3456}^{-1} \{h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} \\
&\quad ((g_{46} g_{34}) \triangleright h_{123} h_{346}^{-1}) \triangleright' l_{3456},
\end{aligned} \tag{E.65}$$

where, in the last line, we used the definition of the action $\triangleright'$ of the group H on the group L . Substituting equality (E.65) into equation (E.64), we obtain:

$$\begin{aligned}
\delta_L(l_{12356}) &= \delta_L \left((h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1245})^{-1}) \delta(l_{1256})^{-1} h_{126} \delta(l_{2456})^{-1}) \triangleright' (l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}) l_{2456}^{-1}) \right. \\
&\quad h_{156} \triangleright' (g_{56} \triangleright (h_{145} \triangleright' (g_{45} \triangleright l_{1234}))) (h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1345})^{-1}) \delta(l_{1356})^{-1} h_{136} \delta(l_{3456})^{-1} h_{346}) \triangleright' \\
&\quad (\{h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} ((g_{46} g_{34}) \triangleright h_{123}) \triangleright' l_{3456}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' (\{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}}) \\
&\quad \left. (\delta(l_{1456})^{-1} h_{146}) \triangleright' ((g_{46} \triangleright l_{1234})^{-1}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1} \right).
\end{aligned} \tag{E.66}$$

Commuting the element l_{3456} to the end of the expression gives

$$\begin{aligned}
\delta_L(l_{12356}) &= \delta_L \left((h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1245})^{-1}) \delta(l_{1256})^{-1} h_{126} \delta(l_{2456})^{-1}) \triangleright' (l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}) l_{2456}^{-1}) \right. \\
&\quad h_{156} \triangleright' (g_{56} \triangleright (h_{145} \triangleright' (g_{45} \triangleright l_{1234}))) (h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1345})^{-1}) \delta(l_{1356})^{-1} h_{136} \delta(l_{3456})^{-1} h_{346}) \triangleright' \\
&\quad (\{h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' (\{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}}) \\
&\quad (\delta(l_{1456})^{-1} h_{146}) \triangleright' ((g_{46} \triangleright l_{1234})^{-1}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1} \\
&\quad \left. (h_{156} g_{56} \triangleright h_{145} h_{246} g_{46} \triangleright h_{234} h_{346}^{-1}) \triangleright' l_{3456} \right).
\end{aligned} \tag{E.67}$$

Acting on the entire argument of the δ -function with $(h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1245})^{-1}) \delta(l_{1256})^{-1} h_{126} \delta(l_{2456})^{-1})^{-1} \triangleright'$, we obtain

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L(l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}) l_{2456}^{-1} (h_{246} h_{456} (g_{56} g_{45}) \triangleright h_{124}^{-1}) \triangleright' \\ &\quad \left((g_{56} g_{45}) \triangleright l_{1234} ((g_{56} g_{45}) \triangleright h_{134} h_{456}^{-1}) \triangleright' \{h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} \right. \\ &\quad \left. h_{456}^{-1} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}} h_{456}^{-1} \triangleright g_{46} \triangleright l_{1234}^{-1} (h_{456}^{-1} g_{46} \triangleright h_{124}) \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}^{-1}\}_{\text{pf}} \right) \\ &\quad (h_{246} g_{46} \triangleright h_{234} h_{346}^{-1}) \triangleright' l_{3456}. \end{aligned} \quad (\text{E.68})$$

After using identity (2.63) for $\{h_{456}, (g_{56} g_{45}) \triangleright (h_{134} g_{34} \triangleright h_{123})\}_{\text{pf}}$, that is, the equality

$$\{h_{456}, (g_{56} g_{45}) \triangleright (h_{134} g_{34} \triangleright h_{123})\}_{\text{pf}} = \{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}} (g_{46} \triangleright h_{134}) \triangleright' \{h_{456}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}, \quad (\text{E.69})$$

the expression reduces to:

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L(l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}) l_{2456}^{-1} \\ &\quad h_{246} \triangleright' \left((h_{456} (g_{56} g_{45}) \triangleright h_{124}^{-1}) \triangleright' \left((g_{56} g_{45}) \triangleright l_{1234} h_{456}^{-1} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright (h_{134} g_{34} \triangleright h_{123})\}_{\text{pf}} \right. \right. \\ &\quad \left. \left. h_{456}^{-1} \triangleright g_{46} \triangleright l_{1234}^{-1} \right) \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}^{-1}\}_{\text{pf}} \right) (h_{246} g_{46} \triangleright h_{234} h_{346}^{-1}) \triangleright' l_{3456}. \end{aligned} \quad (\text{E.70})$$

Next, applying identity (2.63) to the term $\{h_{456}, (g_{56} g_{45}) \triangleright (h_{124}^{-1} \delta(l_{1234}) h_{134} g_{34} \triangleright h_{123})\}_{\text{pf}}$, we see that the terms containing l_{1234} cancel, and we obtain the expression

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L(l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}) l_{2456}^{-1} \\ &\quad h_{246} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright (h_{124}^{-1} \delta(l_{1234}) h_{134} g_{34} \triangleright h_{123})\}_{\text{pf}} (h_{246} g_{46} \triangleright h_{234} h_{346}^{-1}) \triangleright' l_{3456} \\ &= \delta_L(l_{2346}^{-1} l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}) l_{2456}^{-1} h_{246} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{234}\}_{\text{pf}} (\delta(l_{2346})^{-1} h_{236}) \triangleright' l_{3456}) \\ &= \delta_L(l_{23456}). \end{aligned} \quad (\text{E.71})$$

We conclude that the remaining δ -function $\delta_L(l_{12356})$ on the right-hand side of the move reduces to the delta function $\delta_L(l_{23456})$ on its left-hand side. The integrations over the elements l_{1234} , l_{1245} , l_{1256} , l_{1345} , l_{1356} , and l_{1456} are trivial, and the final expression for the integral on the right-hand side is:

$$d.s. = \delta_G(e)^6 \delta_H(e)^4 \delta_L(l_{23456}) = |G|^6 |H|^4 \delta_L(l_{23456}). \quad (\text{E.72})$$

The difference in the factors in expressions (E.43) and (E.72) is compensated by the constants in front of the integrals—the factor $|G|^{-11} |H|^{-4} |L|^{-1}$ on the right-hand side and the factor $|G|^{-5} |H|^0 |L|^{-1}$ on the left-hand side. We conclude that the sum (9.22) is invariant under the $1 \leftrightarrow 5$ Pachner move.

The Pachner move $2 \leftrightarrow 4$

Let us first analyze the *left-hand side of the move*, the expression

$$\int_L dl_{2345} \delta_H(h_{2345}) \delta_L(l_{23456}) \delta_L(l_{12345}). \quad (\text{E.73})$$

First, we integrate over l_{2345} using the δ -function $\delta_L(l_{12345})$,

$$l_{2345} = h_{125}^{-1} \triangleright' (l_{1235} h_{135} \triangleright' \{h_{345}, (g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1} l_{1345} h_{145} \triangleright' (g_{45} \triangleright l_{1234})^{-1} l_{1245}^{-1}). \quad (\text{E.74})$$

The remaining δ -function on the group H , $\delta_H(h_{2345})$, now becomes

$$\begin{aligned}\delta_H(h_{2345}) &= \delta_H(\delta(l_{2345})h_{245}(g_{45}\triangleright h_{234})h_{345}^{-1}h_{235}^{-1}) \\ &= \delta_H\left(h_{125}^{-1}\delta(l_{1235})h_{135}\delta(\{h_{345},(g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1})h_{135}^{-1}\delta(l_{1345})h_{145}(g_{45}\triangleright\delta(l_{1234}))^{-1}h_{145}^{-1}\right. \\ &\quad \left.\delta(l_{1245})^{-1}h_{125}h_{245}(g_{45}\triangleright h_{234})h_{345}^{-1}h_{235}^{-1}\right).\end{aligned}\tag{E.75}$$

Applying identity (9.19) to the tetrahedra (1235), (1345), (1234), and (1245), expression (E.75) reduces to:

$$\begin{aligned}\delta_H(h_{2345}) &= \delta_H\left(h_{125}^{-1}h_{125}h_{235}(g_{35}\triangleright h_{123}^{-1})h_{135}^{-1}h_{135}\delta(\{h_{345},(g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1})h_{135}^{-1}h_{135}h_{345}(g_{45}\triangleright h_{134}^{-1})\right. \\ &\quad \left.h_{145}^{-1}h_{145}g_{45}\triangleright(h_{134}(g_{34}\triangleright h_{123})h_{234}^{-1}h_{124}^{-1})h_{145}^{-1}h_{145}(g_{45}\triangleright h_{124})h_{245}^{-1}h_{125}h_{125}h_{245}(g_{45}\triangleright h_{234})h_{345}^{-1}h_{235}^{-1}\right) \\ &= \delta_H\left((g_{35}\triangleright h_{123}^{-1})\delta(\{h_{345},(g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1})h_{345}(g_{45}g_{34})\triangleright h_{123}h_{345}^{-1}\right).\end{aligned}\tag{E.76}$$

Then, applying identity (2.6) for $h_1 = h_{345}$ and $h_2 = (g_{45}g_{34})\triangleright h_{123}$, together with the identity $g_{35} = \partial(h_{345})g_{45}g_{34}$, we obtain:

$$\delta_H(h_{2345}) = \delta_H(e).\tag{E.77}$$

The remaining δ -function on the group L , $\delta_L(l_{23456})$, is

$$\delta_L(l_{23456}) = \delta_L(l_{2346}^{-1}(h_{236}\triangleright' l_{3456})l_{2356}h_{256}\triangleright'(g_{56}\triangleright l_{2345})l_{2456}^{-1}h_{246}\triangleright'\{h_{456},(g_{56}g_{45})\triangleright h_{234}\}_{\text{pf}}).\tag{E.78}$$

Applying equation (E.75), we obtain:

$$\begin{aligned}\delta_L(l_{23456}) &= \delta_L\left(l_{2346}^{-1}(h_{236}\triangleright' l_{3456})l_{2356}h_{256}\triangleright'\left(g_{56}\triangleright(h_{125}^{-1}\triangleright'(l_{1235}h_{135}\triangleright'\{h_{345},(g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1}\right.\right. \\ &\quad \left.\left.l_{1345}h_{145}\triangleright'(g_{45}\triangleright l_{1234})^{-1}l_{1245}^{-1})\right)l_{2456}^{-1}h_{246}\triangleright'\{h_{456},(g_{56}g_{45})\triangleright h_{234}\}_{\text{pf}}\right).\end{aligned}\tag{E.79}$$

Commuting the elements gives the expression:

$$\begin{aligned}\delta_L(l_{23456}) &= \delta_L\left(l_{2456}^{-1}l_{2346}^{-1}l_{2356}(h_{256}g_{56}\triangleright h_{125}^{-1})\triangleright'g_{56}\triangleright l_{1235}(h_{256}g_{56}\triangleright h_{125}^{-1}g_{56}\triangleright h_{135})\triangleright'\right. \\ &\quad \left.\left((g_{35}\triangleright h_{123}h_{356}^{-1})\triangleright' l_{3456}\{g_{56}\triangleright h_{345},(g_{56}g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1}(g_{56}\triangleright h_{345}(g_{56}g_{45})\triangleright(h_{123}h_{234}^{-1})h_{456}^{-1})\triangleright'\right.\right. \\ &\quad \left.\left.\{h_{456},(g_{56}g_{45})\triangleright h_{234}\}_{\text{pf}}\right)(h_{256}g_{56}\triangleright h_{125}^{-1})\triangleright'g_{56}\triangleright l_{1345}\right. \\ &\quad \left.(h_{256}g_{56}\triangleright h_{125}^{-1}g_{56}\triangleright h_{145})\triangleright'((g_{56}g_{45})\triangleright l_{1234})^{-1}(h_{256}g_{56}\triangleright h_{125}^{-1})\triangleright'g_{56}\triangleright l_{1245}^{-1}\right).\end{aligned}\tag{E.80}$$

Finally, the left-hand side of the move is:

$$l.s. = \delta_H(e)\delta_L(l_{23456}) = |H|\delta_L(l_{23456}).\tag{E.81}$$

Let us now determine the *right-hand side of the move*, i.e., the integral (9.26). First, we integrate over g_{16} using $\delta_G(g_{126})$,

$$g_{16} = \partial(h_{126})g_{26}g_{12}.\tag{E.82}$$

Then, we integrate over h_{126} using $\delta_H(h_{1236})$, over h_{136} using $\delta_H(h_{1346})$, and over h_{146} using $\delta_H(h_{1456})$, which gives

$$\begin{aligned}h_{126} &= \delta(l_{1236})h_{136}(g_{36}\triangleright h_{123})h_{236}^{-1}, \\ h_{136} &= \delta(l_{1346})h_{146}(g_{46}\triangleright h_{134})h_{346}^{-1}, \\ h_{146} &= \delta(l_{1456})h_{156}(g_{56}\triangleright h_{145})h_{456}^{-1}.\end{aligned}\tag{E.83}$$

The remaining three δ -functions on the group G reduce to $\delta_G(e)^3$. It is straightforward to show that the δ -function $\delta_G(g_{136})$,

$$\delta_G(g_{136}) = \delta_G(\partial(h_{136}) g_{36} g_{13} g_{16}^{-1}), \quad (\text{E.84})$$

after substituting equation (E.82), becomes:

$$\delta_G(g_{136}) = \delta_G(\partial(h_{136}) g_{36} g_{13} g_{12}^{-1} g_{26}^{-1} \partial(h_{126})^{-1}). \quad (\text{E.85})$$

Using identity (E.83) for the elements h_{126} , h_{136} , and h_{146} and the fact that $\partial(\delta l) = 0$ for every element $l \in L$, and then applying identity (9.18) to the triangles (156), (145), (456), (134), (346), (236), and (123), the δ -function $\delta_G(g_{136})$ becomes $\delta_G(e)$. Similarly, one shows that $\delta_G(g_{146}) = \delta_G(g_{156}) = \delta_G(e)$. Next, we integrate over l_{1236} using $\delta_L(l_{12346})$ and obtain

$$l_{1236} = (h_{126} \triangleright' l_{2346}) l_{1246} h_{146} \triangleright' (g_{46} \triangleright l_{1234}) l_{1346}^{-1} h_{136} \triangleright' \{h_{346}, (g_{46} g_{34}) \triangleright h_{123}\}_{\text{pf}}, \quad (\text{E.86})$$

over l_{1246} using $\delta_L(l_{12456})$,

$$l_{1246} = (h_{126} \triangleright' l_{2456}) l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1245}) l_{1456}^{-1} h_{146} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}\}_{\text{pf}}, \quad (\text{E.87})$$

and over l_{1346} using $\delta_L(l_{13456})$,

$$l_{1346} = (h_{136} \triangleright' l_{3456}) l_{1356} h_{156} \triangleright' (g_{56} \triangleright l_{1345}) l_{1456}^{-1} h_{146} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}}. \quad (\text{E.88})$$

The remaining δ -functions on the group H reduce to $\delta_H(e)^3$ by a procedure similar to that used for the $1 \leftrightarrow 5$ Pachner move; that is, we obtain $\delta_H(h_{1256}) = \delta_H(h_{1356}) = \delta_H(h_{1456}) = \delta_H(e)$. The remaining δ -function on the group L , namely $\delta_L(l_{12356})$, is:

$$\delta_L(l_{12356}) = \delta_L(l_{1236}^{-1} (h_{126} \triangleright' l_{2356}) l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1235}) l_{1356}^{-1} h_{136} \triangleright' \{h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}}). \quad (\text{E.89})$$

After substituting equations (E.86), (E.87), and (E.88), we obtain

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L \left((h_{136} \triangleright' \{h_{346}, (g_{46} g_{34}) \triangleright h_{123}\}_{\text{pf}})^{-1} l_{1346} h_{146} \triangleright' (g_{46} \triangleright l_{1234})^{-1} l_{1246}^{-1} (h_{126} \triangleright' l_{2346})^{-1} \right. \\ &\quad \left. (h_{126} \triangleright' l_{2356}) l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1235}) l_{1356}^{-1} h_{136} \triangleright' \{h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}} \right) \\ &= \delta_L \left((h_{126} \triangleright' l_{2456})^{-1} (h_{126} \triangleright' l_{2346})^{-1} (h_{126} \triangleright' l_{2356}) (h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' l_{1235} \right. \\ &\quad \left. \delta(l_{1256}) \triangleright' \left(\delta(l_{1356})^{-1} \triangleright' (h_{136} \triangleright' \{h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}} (h_{136} h_{346}) \triangleright' \{h_{346}^{-1}, g_{36} \triangleright h_{123}\}_{\text{pf}} \right. \right. \\ &\quad \left. \left. (h_{136} \triangleright' l_{3456}) \right) h_{156} \triangleright' (g_{56} \triangleright l_{1345}) l_{1456}^{-1} h_{146} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}} h_{146} \triangleright' (g_{46} \triangleright l_{1234})^{-1} \right. \\ &\quad \left. \left. h_{146} \triangleright' \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1} l_{1456} h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1} \right) \right). \end{aligned} \quad (\text{E.90})$$

Commuting the elements so that their order agrees with the order on the left-hand side of the move, i.e., with the δ -function (E.80), and applying identity (2.62), namely,

$$\{h_{346}^{-1} h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}} = h_{346}^{-1} \triangleright' \{h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}} \{h_{346}^{-1}, g_{36} \triangleright h_{123}\}_{\text{pf}}, \quad (\text{E.91})$$

we obtain:

$$\begin{aligned} \delta_L(l_{12356}) &= \delta_L \left((h_{126} \triangleright' l_{2456})^{-1} (h_{126} \triangleright' l_{2346})^{-1} (h_{126} \triangleright' l_{2356}) (h_{126} h_{256} g_{56} \triangleright h_{125}^{-1}) \triangleright' l_{1235} \right. \\ &\quad \left. \delta(l_{1256}) \triangleright' \left(\delta(l_{1356})^{-1} \triangleright' ((h_{136} h_{346}) \triangleright' \{h_{346}^{-1} h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}} (h_{136} \triangleright' l_{3456})) \right. \right. \\ &\quad \left. \left. h_{156} \triangleright' (g_{56} \triangleright l_{1345}) (\delta(l_{1456})^{-1} h_{146}) \triangleright' (\{h_{456}, (g_{56} g_{45}) \triangleright h_{134}\}_{\text{pf}} (g_{46} \triangleright l_{1234})^{-1} \right. \right. \\ &\quad \left. \left. \{h_{456}, (g_{56} g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1}) h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1} \right) \right). \end{aligned} \quad (\text{E.92})$$

Applying identity (2.62) once again,

$$\begin{aligned}
(h_{136}h_{346}) \triangleright' \{h_{346}^{-1}h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}(h_{136} \triangleright' l_{3456}) &= \\
(h_{136}h_{346}) \triangleright' \{h_{346}^{-1} \triangleright' \delta(l_{3456})^{-1}h_{456}g_{56} \triangleright h_{345}^{-1}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}(h_{136} \triangleright' l_{3456}) &= \\
(h_{136} \triangleright' \delta(l_{3456})^{-1}h_{136}h_{346}) \triangleright' (\{h_{456}g_{56} \triangleright h_{345}^{-1}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}((g_{46}g_{34}) \triangleright h_{123}h_{346}^{-1}) \triangleright' l_{3456}^{-1}), & \\
\end{aligned} \tag{E.93}$$

and substituting this expression into (E.92), we obtain:

$$\begin{aligned}
\delta_L(l_{12356}) &= \delta_L \left((h_{126} \triangleright' l_{2456})^{-1} (h_{126} \triangleright' l_{2346})^{-1} (h_{126} \triangleright' l_{2356}) (h_{126}h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' l_{1235} \right. \\
&\quad \delta(l_{1256}) \triangleright' \left((\delta(l_{1356})^{-1}h_{136} \triangleright' \delta(l_{3456})^{-1}h_{136}h_{346}) \triangleright' \right. \\
&\quad \left. \left(\{h_{456}g_{56} \triangleright h_{345}^{-1}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}((g_{46}g_{34}) \triangleright h_{123}h_{346}^{-1}) \triangleright' l_{3456} \right) \right. \\
&\quad \left. (h_{156}g_{56} \triangleright h_{135}g_{56} \triangleright (h_{345}g_{45} \triangleright h_{134}^{-1})h_{456}^{-1}) \triangleright' (\{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}(g_{46} \triangleright l_{1234})^{-1} \right. \\
&\quad \left. \left. \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1} \right) \right) (h_{126}h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' (h_{156} \triangleright' (g_{56} \triangleright l_{1345})(g_{56} \triangleright l_{1245})^{-1}) \Big). \\
\end{aligned} \tag{E.94}$$

Commuting the elements l_{3456} and $\{h_{456}g_{56} \triangleright h_{345}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}$, and then using identity (2.62), we obtain:

$$\begin{aligned}
\delta_L(l_{12356}) &= \delta_L \left((h_{126} \triangleright' l_{2456})^{-1} (h_{126} \triangleright' l_{2346})^{-1} (h_{126} \triangleright' l_{2356}) (h_{126}h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' l_{1235} \right. \\
&\quad (h_{126}h_{256}g_{56} \triangleright h_{125}^{-1}h_{135}(g_{56}g_{35}) \triangleright h_{123}g_{56} \triangleright h_{356}^{-1}) \triangleright' g_{56} \triangleright l_{3456} \\
&\quad (h_{126}h_{256}g_{56} \triangleright h_{125}^{-1}g_{56} \triangleright h_{135}g_{56} \triangleright h_{345}) \triangleright' \left(\{g_{56} \triangleright h_{345}^{-1}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}} \right. \\
&\quad \left. h_{456}^{-1} \triangleright' \{h_{456}, (g_{56}g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}((g_{56}g_{45}) \triangleright h_{134}^{-1})h_{456}^{-1} \triangleright' (\{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}(g_{46} \triangleright l_{1234})^{-1} \right. \\
&\quad \left. \left. \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}^{-1} \right) \right) (h_{126}h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' (h_{156} \triangleright' (g_{56} \triangleright l_{1345})(g_{56} \triangleright l_{1245})^{-1}) \Big). \\
\end{aligned} \tag{E.95}$$

After transformations similar to those used for the $1 \leftrightarrow 5$ Pachner move, commuting the elements l_{1234} so that their order is the same as in (E.80), and then acting on the entire expression with h_{126}^{-1} , we obtain:

$$\begin{aligned}
\delta_L(l_{12356}) &= \delta_L \left(l_{2456}^{-1}l_{2346}^{-1}l_{2356}(h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1235} (h_{256}g_{56} \triangleright h_{125}^{-1}g_{56} \triangleright h_{135}) \triangleright' \right. \\
&\quad \left((g_{35} \triangleright h_{123}h_{356}^{-1}) \triangleright' l_{3456} \{g_{56} \triangleright h_{345}, (g_{56}g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1} (g_{56} \triangleright h_{345}(g_{56}g_{45}) \triangleright (h_{123}h_{234}^{-1})h_{456}^{-1}) \triangleright' \right. \\
&\quad \left. \{h_{456}, (g_{56}g_{45}) \triangleright h_{234}\}_{\text{pf}} \right) (h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1345} \\
&\quad \left. (h_{256}g_{56} \triangleright h_{125}^{-1}g_{56} \triangleright h_{145}) \triangleright' ((g_{56}g_{45}) \triangleright l_{1234})^{-1} (h_{256}g_{56} \triangleright h_{125}^{-1}) \triangleright' g_{56} \triangleright l_{1245}^{-1} \right). \\
\end{aligned} \tag{E.96}$$

This expression is identical to the expression in equation (E.80). The remaining integrations—over the element h_{156} of the group H and the three elements of the group L , l_{1246} , l_{1256} , and l_{1356} —are trivial; hence, the right-hand side finally reduces to:

$$d.s. = \delta_G(e)^3 \delta_H(e)^3 \delta_L(l_{12356}) = |G|^3 |H|^3 \delta_L(l_{12356}). \tag{E.97}$$

The constants in front of the integrals are $|G|^{-8}|H|^{-1}|L|^{-1}$ on the left-hand side of the move and $|G|^{-11}|H|^{-3}|L|^{-1}$ on its right-hand side. This compensates for the difference between expressions (E.81) and (E.97), and we therefore conclude that the state sum (9.22) is invariant under the $2 \leftrightarrow 4$ Pachner move.

The Pachner move $3 \leftrightarrow 3$

Let us first consider the right-hand side of the move. First, we integrate over l_{1235} using $\delta_L(l_{12345})$

$$l_{1235} = (h_{125} \triangleright' l_{2345}) l_{1245} h_{145} \triangleright' (g_{45} \triangleright l_{1234}) l_{1345}^{-1} h_{135} \triangleright' \{h_{345}, (g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}} \quad (\text{E.98})$$

and over l_{1236} using $\delta_L(l_{12356})$,

$$l_{1236} = (h_{126} \triangleright' l_{2356}) l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1235}) l_{1356}^{-1} h'_{136} \triangleright \{h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}}. \quad (\text{E.99})$$

Then, we integrate over h_{123} using $\delta_H(l_{1234})$ and obtain:

$$h_{123} = g_{34}^{-1} \triangleright h_{134}^{-1} g_{34}^{-1} \triangleright \delta(l_{1234})^{-1} g_{34}^{-1} \triangleright h_{124} g_{34}^{-1} \triangleright h_{234}. \quad (\text{E.100})$$

The remaining δ -function on the group G , the δ -function $\delta_G(g_{123})$, becomes, after applying equation (E.100),

$$\delta_G(g_{123}) = \delta_G(g_{34}^{-1} \triangleright \partial(h_{134})^{-1} g_{34}^{-1} \triangleright \partial(\delta(l_{1234}))^{-1} g_{34}^{-1} \triangleright \partial(h_{124}) g_{34}^{-1} \triangleright \partial(h_{234}) g_{23} g_{12} g_{13}^{-1}), \quad (\text{E.101})$$

or, after applying the identity $\partial\delta = 0$:

$$\delta_G(g_{123}) = \delta_G(\partial(h_{134})^{-1} \partial(h_{124}) \partial(h_{234}) g_{34}^{-1} g_{23} g_{12} g_{13}^{-1} g_{34}). \quad (\text{E.102})$$

Then, applying identity (9.18) to the triangles (134), (124), and (234), we obtain:

$$\delta_G(g_{123}) = \delta_G(e). \quad (\text{E.103})$$

For the δ -function $\delta_H(h_{1235})$, after applying equation (E.98), we obtain:

$$\begin{aligned} \delta_H(h_{1235}) &= \delta_H\left((h_{125} \delta(l_{2345}) h_{125}^{-1}) \delta(l_{1245}) (h_{145} (g_{45} \triangleright \delta(l_{1234})) h_{145}^{-1}) \delta(l_{1345})^{-1} h_{135} \triangleright' \{h_{345}, g_{35} \triangleright h_{123}\}_{\text{pf}} h_{135} \right. \\ &\quad \left. ((g_{35} g_{34}^{-1}) \triangleright (h_{134}^{-1} \delta(l_{1234})^{-1} h_{124} h_{234})) h_{235}^{-1} h_{125}^{-1}\right). \end{aligned} \quad (\text{E.104})$$

Then, using conditions (9.20) for the δ -functions $\delta_L(h_{2345})$, $\delta_L(h_{1245})$, and $\delta_L(h_{1345})$, which appear on both sides of the move,

$$\begin{aligned} \delta(l_{2345}) &= h_{235} h_{345} (g_{45} \triangleright h_{234}^{-1}) h_{245}^{-1}, \\ \delta(l_{1245}) &= h_{125} h_{245} (g_{45} \triangleright h_{124}^{-1}) h_{145}^{-1}, \\ \delta(l_{1345})^{-1} &= h_{145} (g_{45} \triangleright h_{134}) h_{345}^{-1} h_{135}^{-1}, \end{aligned} \quad (\text{E.105})$$

we obtain:

$$\begin{aligned} \delta_H(h_{1235}) &= \delta_H\left(h_{125} h_{235} h_{345} (g_{45} \triangleright h_{234}^{-1}) h_{245}^{-1} h_{125}^{-1} h_{125} h_{245} (g_{45} \triangleright h_{124}^{-1}) h_{145}^{-1} h_{145} (g_{45} \triangleright \delta(l_{1234})) h_{145}^{-1} \right. \\ &\quad h_{145} (g_{45} \triangleright h_{134}) h_{345}^{-1} h_{135}^{-1} h_{135} \triangleright \delta(\{h_{345}, (g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}) \\ &\quad \left. h_{135} ((g_{35} g_{34}^{-1}) \triangleright (h_{134}^{-1} \delta(l_{1234})^{-1} h_{124} h_{234})) h_{235}^{-1} h_{125}^{-1}\right) \\ &= \delta_H\left(h_{345} (g_{45} g_{34}) \triangleright h_{123}^{-1} h_{345}^{-1} \delta(\{h_{345}, (g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}) (g_{35} \triangleright h_{123})\right). \end{aligned} \quad (\text{E.106})$$

Applying identity (2.6) for $h_1 = h_{345}$ and $h_2 = (g_{45}g_{34}) \triangleright h_{123}$, together with the equality $g_{35} = \partial(h_{345})g_{45}g_{34}$, we obtain:

$$\delta_H(h_{1235}) = \delta_H(e). \quad (\text{E.107})$$

By a similar procedure, one obtains $\delta_H(h_{1236}) = \delta_H(e)$. The remaining δ -function $\delta_H(l_{12346})$ is:

$$\delta_L(l_{12346}) = \delta_L(l_{1236}^{-1}(h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1234})l_{1346}^{-1}h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}). \quad (\text{E.108})$$

Substituting expression (E.99), followed by expression (E.98), we obtain:

$$\begin{aligned} \delta_L(l_{12346}) &= \delta_L(h_{136} \triangleright' \{h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}^{-1} l_{1356} h_{156} \triangleright' (g_{56} \triangleright l_{1235})^{-1} l_{1256}^{-1} h_{126} \triangleright' l_{2356}^{-1} \\ &\quad (h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1234})l_{1346}^{-1}h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}) \\ &= \delta_L(h_{136} \triangleright' \{h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}^{-1} l_{1356} \\ &\quad h_{156} \triangleright' (g_{56} \triangleright ((h_{125} \triangleright' l_{2345})l_{1245}h_{145} \triangleright' (g_{45} \triangleright l_{1234})l_{1345}^{-1}h_{135} \triangleright' \{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}))^{-1} \\ &\quad l_{1256}^{-1} h_{126} \triangleright' l_{2356}^{-1} (h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1234})l_{1346}^{-1}h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}). \end{aligned} \quad (\text{E.109})$$

Commuting the elements, i.e., using the Peiffer identity for the crossed module $(L \xrightarrow{\delta} H, \triangleright')$, we obtain

$$\begin{aligned} \delta_L(l_{12346}) &= \delta_L(h_{136} \triangleright' \{h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}^{-1} \\ &\quad (\delta(l_{1356})h_{156}g_{56} \triangleright h_{135}) \triangleright' g_{56} \triangleright \{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1} l_{1356} h_{156} \triangleright' (g_{56} \triangleright l_{1345}) \\ &\quad (h_{156}g_{56} \triangleright h_{145}) \triangleright' ((g_{56}g_{45}) \triangleright l_{1234})^{-1} h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1} (h_{156}g_{56} \triangleright h_{125}) \triangleright' (g_{56} \triangleright l_{2345}^{-1}) l_{1256}^{-1} \\ &\quad h_{126} \triangleright' l_{2356}^{-1} (h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1234})l_{1346}^{-1}h_{136} \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}}) \\ &= \delta_L((\delta(l_{1346})^{-1}h_{136}) \triangleright' \{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}} (\delta(l_{1346})^{-1}h_{136}) \triangleright' \{h_{356}, (g_{56}g_{35}) \triangleright h_{123}\}_{\text{pf}}^{-1} \\ &\quad ((\delta(l_{1346})^{-1}\delta(l_{1356})h_{156}g_{56} \triangleright h_{135}) \triangleright' g_{56} \triangleright \{h_{345}, (g_{45}g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1} \\ &\quad (\delta(l_{1346})^{-1}\delta(l_{1356})h_{156} \triangleright' (g_{56} \triangleright \delta(l_{1345}))h_{156}g_{56} \triangleright h_{145}) \triangleright' ((g_{56}g_{45}) \triangleright l_{1234})^{-1} l_{1346}^{-1} l_{1356} h_{156} \triangleright' (g_{56} \triangleright l_{1345}) \\ &\quad h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1} (h_{156}g_{56} \triangleright h_{125}) \triangleright' (g_{56} \triangleright l_{2345}^{-1}) l_{1256}^{-1} h_{126} \triangleright' l_{2356}^{-1} (h_{126} \triangleright' l_{2346})l_{1246}h_{146} \triangleright' (g_{46} \triangleright l_{1234})). \end{aligned} \quad (\text{E.110})$$

Applying identity (2.65), namely,

$$\{h_{346}, (g_{46}g_{34}) \triangleright h_{123}\}_{\text{pf}} = h_{346} \triangleright' \{h_{346}^{-1}, g_{36} \triangleright h_{123}\}_{\text{pf}}^{-1}, \quad (\text{E.111})$$

and then the variant of identity (2.62), namely,

$$\{h_1 h_2 h_3, h_4\}_{\text{pf}}^{-1} = \{h_1, \partial(h_2 h_3) \triangleright h_4\}_{\text{pf}}^{-1} h_1 \triangleright' \{h_2, \partial(h_2) \triangleright h_4\}_{\text{pf}}^{-1} (h_1 h_2) \triangleright' \{h_3, h_4\}_{\text{pf}}^{-1}, \quad (\text{E.112})$$

we obtain:

$$\begin{aligned} \{h_{346}^{-1} h_{356} (g_{56} \triangleright h_{345}), (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1} &= \{h_{346}^{-1}, (g_{46} g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1} h_{346}^{-1} \triangleright' \{h_{356}, (g_{56} g_{35}) \triangleright h_{123}\}_{\text{pf}}^{-1} \\ &\quad (h_{346}^{-1} h_{356}) \triangleright' \{g_{56} \triangleright h_{345}, (g_{56} g_{45} g_{34}) \triangleright h_{123}\}_{\text{pf}}^{-1}. \end{aligned} \quad (\text{E.113})$$

Applying this equality in expression (E.110), we obtain:

$$\begin{aligned}\delta_L(l_{12346}) &= \delta_L((h_{146}g_{46}\triangleright h_{134})\triangleright'\{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1} \\ &\quad (\delta(l_{1346})^{-1}\delta(l_{1356})h_{156}\triangleright'(g_{56}\triangleright\delta(l_{1345}))h_{156}g_{56}\triangleright h_{145})\triangleright'((g_{56}g_{45})\triangleright l_{1234})^{-1}l_{1346}^{-1}l_{1356}h_{156}\triangleright'(g_{56}\triangleright l_{1345}) \\ &\quad h_{156}\triangleright'(g_{56}\triangleright l_{1245})^{-1}(h_{156}g_{56}\triangleright h_{125})\triangleright'(g_{56}\triangleright l_{2345})^{-1}l_{1256}^{-1}h_{126}\triangleright'l_{2356}^{-1}(h_{126}\triangleright'l_{2346})l_{1246}h_{146}\triangleright'(g_{46}\triangleright l_{1234})).\end{aligned}\quad (\text{E.114})$$

Applying (E.100) and identity (2.63), we obtain the equality

$$\begin{aligned}\{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1} &= \{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright((h_{134}^{-1}\triangleright'\delta(l_{1234})^{-1})h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}^{-1} \\ &= (g_{46}\triangleright(h_{134}^{-1}\triangleright'\delta(l_{1234})^{-1}))\triangleright' \\ &\quad \{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright(h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}^{-1} \\ &\quad \{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright(h_{134}^{-1}\triangleright'\delta(l_{1234})^{-1})\}_{\text{pf}}^{-1}.\end{aligned}\quad (\text{E.115})$$

That is, when we expand the term

$$\begin{aligned}\{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright(h_{134}^{-1}\triangleright'\delta(l_{1234})^{-1})\}_{\text{pf}}^{-1} &= (g_{46}\triangleright(h_{134}^{-1}\triangleright'l_{1234}^{-1})) \\ &\quad (h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}))\triangleright'((g_{56}g_{45})\triangleright(h_{134}^{-1}\triangleright'l_{1234})).\end{aligned}\quad (\text{E.116})$$

expression (E.115) is equal to:

$$\begin{aligned}\{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45}g_{34})\triangleright h_{123}\}_{\text{pf}}^{-1} &= g_{46}\triangleright(h_{134}^{-1}\triangleright'\delta(l_{1234})^{-1}) \\ &\quad \{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright(h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}^{-1} \\ &\quad (h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}))\triangleright'((g_{56}g_{45})\triangleright(h_{134}^{-1}\triangleright'l_{1234})).\end{aligned}\quad (\text{E.117})$$

Substituting this result into equation (E.114), the terms containing l_{1234} cancel, and it follows that $\delta_L(l_{12346})$ is:

$$\begin{aligned}\delta_L(l_{12346}) &= \delta_L((h_{146}g_{46}\triangleright h_{134})\triangleright'\{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright(h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}^{-1}l_{1346}^{-1}l_{1356} \\ &\quad h_{156}\triangleright'(g_{56}\triangleright l_{1345})h_{156}\triangleright'(g_{56}\triangleright l_{1245})^{-1}(h_{156}g_{56}\triangleright h_{125})\triangleright'(g_{56}\triangleright l_{2345})^{-1}l_{1256}^{-1}h_{126}\triangleright'l_{2356}^{-1}(h_{126}\triangleright'l_{2346})l_{1246}).\end{aligned}\quad (\text{E.118})$$

From this expression, we see that the integration over l_{1234} is trivial, so the expression on the right-hand side of the Pachner move finally becomes:

$$\begin{aligned}d.s. &= \delta_G(e)\delta_H(e)^2\delta_L(h_{156}\triangleright'(g_{56}\triangleright l_{1245})^{-1}h_{156}\triangleright'(g_{56}\triangleright(h_{125}\triangleright'l_{2345}))^{-1}l_{1256}^{-1}h_{126}\triangleright'l_{2356}^{-1}(h_{126}\triangleright'l_{2346}) \\ &\quad l_{1246}(h_{146}g_{46}\triangleright h_{134})\triangleright'\{h_{346}^{-1}h_{356}(g_{56}\triangleright h_{345}), (g_{56}g_{45})\triangleright(h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}^{-1}l_{1346}^{-1}l_{1356}h_{156}\triangleright'(g_{56}\triangleright l_{1345}).\end{aligned}\quad (\text{E.119})$$

Let us now analyze the left-hand side of the move, i.e., the integral:

$$\int_H dh_{456} \int_{L^3} dl_{1456} dl_{2456} dl_{3456} \delta_G(g_{456}) \delta_H(h_{3456}) \delta_H(h_{2456}) \delta_H(h_{1456}) \delta_L(l_{23456}) \delta_L(l_{13456}) \delta_L(l_{12456}). \quad (\text{E.120})$$

First, we integrate over l_{1456} using the δ -function $\delta_L(l_{13456})$ and obtain:

$$l_{1456} = h_{146} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\} l_{1346}^{-1} (h_{136} \triangleright' l_{3456}) l_{1356} h_{156} \triangleright' (g_{56} \triangleright l_{1345}). \quad (\text{E.121})$$

Then, we integrate over l_{2456} using $\delta_L(l_{23456})$,

$$l_{2456} = h_{246} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{234}\} l_{2346}^{-1} (h_{236} \triangleright' l_{3456}) l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}), \quad (\text{E.122})$$

and over h_{456} using $\delta_H(h_{3456})$:

$$h_{456} = h_{346}^{-1} \delta(l_{3456}) h_{356} (g_{56} \triangleright h_{345}). \quad (\text{E.123})$$

Using equation (E.123), the δ -function $\delta_G(g_{456})$,

$$\delta_G(g_{456}) = \delta_G(\partial(h_{346})^{-1} \partial(h_{356}) g_{56} \triangleright \partial(h_{345}) g_{56} g_{45} g_{46}^{-1}), \quad (\text{E.124})$$

reduces, upon applying identity (9.18) to the triangles (346), (356), and (345), to:

$$\delta_G(g_{456}) = \delta_G(e). \quad (\text{E.125})$$

By a procedure similar to that used on the right-hand side of the move, applying equations (E.121) and (E.122), we find that the δ -functions $\delta_H(h_{1456})$ and $\delta_H(h_{2456})$ are equal to $\delta_H(e)^2$. The remaining $\delta_L(l_{12456})$ is equal to

$$\delta_L(l_{12456}) = \delta_L(l_{1246}^{-1} (h_{126} \triangleright' l_{2456}) l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1245}) l_{1456}^{-1} h_{146} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}). \quad (\text{E.126})$$

Substituting equations (E.121) and (E.122) into the expression for the δ -function $\delta_L(l_{12456})$, we obtain

$$\begin{aligned} \delta_L(l_{12456}) &= \delta_L(l_{1246}^{-1} (h_{126} \triangleright' (h_{246} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{234}\}_{\text{pf}} l_{2346}^{-1} (h_{236} \triangleright' l_{3456}) l_{2356} h_{256} \triangleright' (g_{56} \triangleright l_{2345}))) \\ &\quad l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1245}) h_{156} \triangleright' (g_{56} \triangleright l_{1345})^{-1} l_{1356}^{-1} (h_{136} \triangleright' l_{3456})^{-1} l_{1346} h_{146} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}^{-1} \\ &\quad h_{146} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}), \end{aligned} \quad (\text{E.127})$$

or, after commuting the elements:

$$\begin{aligned} \delta_L(l_{12456}) &= \delta_L((\delta(l_{1246})^{-1} h_{126} h_{246}) \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{234}\}_{\text{pf}} (\delta(l_{1246})^{-1} h_{126} \triangleright \delta(l_{2346})^{-1} h_{126} h_{236}) \triangleright' l_{3456} \\ &\quad l_{1246}^{-1} h_{126} \triangleright' l_{2346}^{-1} h_{126} \triangleright' l_{2356} (h_{126} h_{256}) \triangleright' (g_{56} \triangleright l_{2345})) \\ &\quad l_{1256} h_{156} \triangleright' (g_{56} \triangleright l_{1245}) h_{156} \triangleright' (g_{56} \triangleright l_{1345})^{-1} l_{1356}^{-1} l_{1346} (\delta(l_{1346})^{-1} h_{136}) \triangleright' l_{3456}^{-1} \\ &\quad h_{146} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}^{-1} h_{146} \triangleright \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}). \end{aligned} \quad (\text{E.128})$$

Applying identity (2.68) for the inverse Peiffer lifting $\{h_{456}, (g_{56}g_{45}) \triangleright h_{134}\}_{\text{pf}}^{-1}$, followed by identity (2.63) for three elements, namely,

$$\{h_1, h_2 h_3 h_4\}_{\text{pf}} = \{h_1, h_2\}_{\text{pf}} (\partial(h_1) \triangleright h_2) \triangleright' \{h_1, h_3\}_{\text{pf}} (\partial(h_1) \triangleright (h_2 h_3)) \triangleright' \{h_1, h_4\}_{\text{pf}}, \quad (\text{E.129})$$

we obtain:

$$\begin{aligned} \{h_{456}, (g_{56}g_{45}) \triangleright (h_{134}^{-1} h_{124} h_{234})\}_{\text{pf}} &= \{h_{456}, (g_{56}g_{45}) \triangleright h_{134}^{-1}\}_{\text{pf}} (g_{46} \triangleright h_{134}^{-1}) \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}} \\ &\quad (g_{46} \triangleright (h_{134}^{-1} h_{124})) \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright h_{124}\}_{\text{pf}}. \end{aligned} \quad (\text{E.130})$$

We can then apply this identity in equation (E.128), which yields the equality:

$$\begin{aligned}
\delta_L(l_{12456}) &= \delta_L((\delta(l_{1246})^{-1}h_{126} \triangleright \delta(l_{2346})^{-1}h_{126}h_{236}) \triangleright' l_{3456} \\
&\quad l_{1246}^{-1}h_{126} \triangleright' l_{2346}^{-1}h_{126} \triangleright' l_{2356}(h_{126}h_{256}) \triangleright' (g_{56} \triangleright l_{2345})) \\
&\quad l_{1256}h_{156} \triangleright' (g_{56} \triangleright l_{1245})h_{156} \triangleright' (g_{56} \triangleright l_{1345})^{-1}l_{1356}^{-1}l_{1346}(\delta(l_{1346})^{-1}h_{136}) \triangleright' l_{3456}^{-1} \\
&\quad (h_{146}g_{46} \triangleright h_{134}) \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright (h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}).
\end{aligned} \tag{E.131}$$

Applying equation (E.123) and identity (2.62), by a procedure similar to that used on the right-hand side of the move, the terms containing l_{3456} cancel; hence, the δ -function $\delta_L(l_{12456})$ is:

$$\begin{aligned}
\delta_L(l_{12456}) &= \delta_L(l_{1246}^{-1}h_{126} \triangleright' l_{2346}^{-1}h_{126} \triangleright' l_{2356}(h_{126}h_{256}) \triangleright' (g_{56} \triangleright l_{2345}))l_{1256}h_{156} \triangleright' (g_{56} \triangleright l_{1245}) \\
&\quad h_{156} \triangleright' (g_{56} \triangleright l_{1345})^{-1}l_{1356}^{-1}l_{1346}(h_{146}g_{46} \triangleright h_{134}) \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright (h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}.
\end{aligned} \tag{E.132}$$

We conclude that the integration over l_{3456} is trivial, and hence the left-hand side of the move is finally equal to:

$$\begin{aligned}
l.s. &= \delta_G(e)\delta_H(e)^2\delta_L(h_{126} \triangleright' l_{2346}l_{1246}(h_{146}g_{46} \triangleright h_{134}) \triangleright' \{h_{456}, (g_{56}g_{45}) \triangleright (h_{134}^{-1}h_{124}h_{234})\}_{\text{pf}}}^{-1}l_{1346}^{-1} \\
&\quad l_{1356}h_{156} \triangleright' (g_{56} \triangleright l_{1345})h_{156} \triangleright' (g_{56} \triangleright l_{1245})^{-1}(h_{156}g_{56} \triangleright h_{125}) \triangleright' (g_{56} \triangleright l_{2345})^{-1}l_{1256}^{-1}h_{126} \triangleright' l_{2356}^{-1}).
\end{aligned} \tag{E.133}$$

Expressions (E.119) and (E.126) are identical, and we therefore conclude that the state sum (9.2.1) is invariant under the $3 \leftrightarrow 3$ Pachner move. The number of k -simplices on both sides of the $3 \leftrightarrow 3$ move is the same for every k ; thus, the coefficients in front of the integrals are equal in this case and therefore have no effect.

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Biography

Tijana Radenković was born on 21 March 1992 in Belgrade, where she completed her primary education at the Miloš Crnjanski Primary School and graduated from the Mathematical Grammar School in 2011. In the same year, she enrolled at the Faculty of Physics of the University of Belgrade, in the Theoretical and Experimental Physics program, and graduated on 30 September 2016 with a grade average of 9.33. In 2016, she enrolled in the master's degree program at the Faculty of Physics of the University of Belgrade, which she completed on 27 September 2017 by defending the master's thesis entitled

"Quantum Gravity on Piecewise-Flat Manifolds",

with a grade average of 9.33. The master's thesis focused on a review of established results in the theory of spin-cube models. The spin-cube model is a well-known quantum-gravity model that generalizes spin-foam models in the context of 2-categories. However, the literature lacked a clear review that would facilitate an introduction to this model and the mathematical foundations on which it is based. Tijana's master's thesis reviewed Regge calculus and 2-groups in category theory and then presented the construction of the state sum within the Ponzano–Regge topological model of $3D$ gravity and the Ooguri model, the topological sector of $4D$ gravity, both based on the BF action. It was shown that general relativity is equivalent to the constrained $2BF$ action and can be regarded as a gauge theory for the Poincaré 2-group. The quantization procedure for the topological part of the theory, i.e., the $BFCG$ action, was carried out in a manner similar to that for the BF action. It was shown that the problem present in the spin-foam model is resolved by its categorical generalization: the fundamental variables now label the 3-complex of the dual lattice, and edge lengths in the triangulation become the fundamental degrees of freedom of the theory. The thesis received the Prof. Ljubomir Ćirković Foundation award for the best master's thesis defended at the Faculty of Physics in the 2017/2018 academic year.

In 2017, she began doctoral studies at the Faculty of Physics of the University of Belgrade, in the *Fields, Particles and Gravitation* program. Her grade average was 9.5. She conducted her doctoral research under the supervision of Dr Marko Vojinović. Since April 2018, Tijana has been employed at the Institute of Physics Belgrade as a junior research assistant. During her doctoral studies, her research addressed problems of quantum gravity and its unification with the other fundamental fields. In particular, the formalism of 2-groups and 3-groups provides an excellent platform for the unification of all interactions.

From 2011 to 2019, Tijana worked part-time in the Physics and TEH programs at the Petnica Science Center, where she helped organize seminars and laboratory exercises and delivered lectures. In addition to her long-standing involvement with the Science Center, Tijana gained teaching experience during the 2016/2017 academic year as a physics teacher at the Tenth Belgrade Grammar School and during the 2019/2020 academic year as an acoustics teacher at the Stanković, Vatroslav Lisinski, and Josip Slavenski music schools.